

Egyptian Baccalaureate Mathematics

Student Edition • Open Workbook / Guided Practice • All_Chapters

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Concept 1 | Binomial and Multinomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Binomial

$$\binom{n}{r} a^r b^{n-r}$$

Multinomial

$$\binom{n}{r} \binom{n-r}{s} a^r b^s c^{n-r-s}$$

Exponent check

$$r + (n - r) = n$$

Worked Solutions

Binomial and Multinomial Theorems

Q001 Challenge

For $(2x - 3)^5$, find the coefficient of x^3 and the sum of all coefficients without writing the full expansion:

Solution

1. Use the binomial term rule: $q = 3$ and $r = 2$, so the coefficient is 5C_2 times 72.
2. For the sum of coefficients, evaluate the expression at $x = 1$: -1 .

Correct option: A

Final answer

Coefficient = 720; sum = -1.

Worked Solutions

Binomial and Multinomial Theorems

Q002 Challenge

Find the coefficient of x^2y^3z in $(x + y + z)^6$, then find the coefficient of x^2yz^3 in $(x - 2y + z)^6$:

Solution

1. The first exponents are (2,3,1), so use $6!/(2!3!1!) = 60$.
2. The second exponents are (2,1,3); the factor (-2) is raised to the first power of y, giving -120.

Correct option: A

Final answer

First coefficient = 60; second coefficient = -120.

Worked Solutions

Binomial and Multinomial Theorems

Q003 Challenge

For $(x^2 + x + 1)^4$, find the greatest coefficient and verify it by identifying the coefficient of x^4 :

Solution

1. Expand and collect like powers: the central coefficient is 19.
2. The coefficient of x^4 is also 19; it is therefore the greatest.

Correct option: A

Final answer

Greatest coefficient = 19.

Worked Solutions

Binomial and Multinomial Theorems

Q004 **Written****Expand $(x + 2)^5$ using the Binomial Theorem.****Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 2)^5 = x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$.

Final answer

$$x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$$

Q005 **MCQ****Expand $(3x - 2)^4$ using the Binomial Theorem:****Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3x - 2)^4 = 81x^4 - 216x^3 + 216x^2 - 96x + 16$.

Correct option: D**Final answer**

$$81x^4 - 216x^3 + 216x^2 - 96x + 16$$

Worked Solutions

Binomial and Multinomial Theorems

Q006 Written

Expand $(x + 2)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 2)^6 = x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$.

Final answer

$$x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$$

Q007 MCQ

Expand $(x + 1)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1$.

Correct option: B**Final answer**

$$x^4 + 4x^3 + 6x^2 + 4x + 1$$

Worked Solutions

Binomial and Multinomial Theorems

Q008 MCQ

Expand $(a + b)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$.

Correct option: C**Final answer**

$$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

Q009 Written

Expand $(a - 2b)^4$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a - 2b)^4 = a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$.

Final answer

$$a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$$

Worked Solutions

Binomial and Multinomial Theorems

Q010 MCQ

Expand $(x - 2)^5$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 2)^5 = x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$.

Correct option: A**Final answer**

$$x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$$

Q011 Written

Expand $(3x + 2y)^5$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3x + 2y)^5 = 243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$.

Final answer

$$243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$$

Worked Solutions

Binomial and Multinomial Theorems

Q012 Written

Expand $(2a^2 - 1)^5$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a^2 - 1)^5 = 32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$.

Final answer

$$32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$$

Q013 Written

Expand $(x - 3)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 3)^6 = x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$.

Final answer

$$x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$$

Worked Solutions

Binomial and Multinomial Theorems

Q014 Written

Expand $(2x - 3y)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x - 3y)^6 = 64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$.

Final answer

$$64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$$

Q015 Written

Expand $\left(x + \frac{1}{3}\right)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $\left(x + \frac{1}{3}\right)^6 = x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$.

Final answer

$$x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$$

Worked Solutions

Binomial and Multinomial Theorems

Q016 MCQ

Expand $(x + 3)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 3)^3 = x^3 + 9x^2 + 27x + 27$.

Correct option: B**Final answer**

$$x^3 + 9x^2 + 27x + 27$$

Worked Solutions

Binomial and Multinomial Theorems

Q017 MCQ

Expand $(a + 2b)^5$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a + 2b)^5 = a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5$.

Correct option: D**Final answer**

$$a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5$$

Worked Solutions

Binomial and Multinomial Theorems

Q018 MCQ

Expand $(x + 2)^2$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 2)^2 = x^2 + 4x + 4$.

Correct option: D**Final answer**

$$x^2 + 4x + 4$$

Q019 MCQ

Expand $(a - b)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$.

Correct option: A**Final answer**

$$a^3 - 3a^2b + 3ab^2 - b^3$$

Worked Solutions

Binomial and Multinomial Theorems

Q020 MCQ

Expand $(2x + 1)^5$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x + 1)^5 = 32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$.

Correct option: B**Final answer**

$$32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$$

Q021 Written

Expand $(3a - 2b)^5$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3a - 2b)^5 = 243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$.

Final answer

$$243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$$

Worked Solutions

Binomial and Multinomial Theorems

Q022 MCQ

Expand $(2x - 1)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x - 1)^4 = 16x^4 - 32x^3 + 24x^2 - 8x + 1$.

Correct option: C**Final answer**

$$16x^4 - 32x^3 + 24x^2 - 8x + 1$$

Q023 MCQ

Expand $(x - 3)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 3)^4 = x^4 - 12x^3 + 54x^2 - 108x + 81$.

Correct option: B**Final answer**

$$x^4 - 12x^3 + 54x^2 - 108x + 81$$

Worked Solutions

Binomial and Multinomial Theorems

Q024 MCQ

Expand $(2a + b)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a + b)^3 = 8a^3 + 12a^2b + 6ab^2 + b^3$.

Correct option: C**Final answer**

$$8a^3 + 12a^2b + 6ab^2 + b^3$$

Q025 MCQ

Expand $(2a^2 + 3)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a^2 + 3)^4 = 16a^8 + 96a^6 + 216a^4 + 216a^2 + 81$.

Correct option: C**Final answer**

$$16a^8 + 96a^6 + 216a^4 + 216a^2 + 81$$

Worked Solutions

Binomial and Multinomial Theorems

Q026 MCQ

Expand $(3a^2 - 2)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3a^2 - 2)^3 = 27a^6 - 54a^4 + 36a^2 - 8$.

Correct option: B**Final answer**

$$27a^6 - 54a^4 + 36a^2 - 8$$

Q027 Written

Find the coefficient of a^3b^2 in the expansion of $(a + b)^5$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use 5C_2 times the first factor to power q and the second factor to power r .
3. The required coefficient is 10.

Final answer

10

Worked Solutions

Binomial and Multinomial Theorems

Q028 Written

Find the coefficient of x^4y^2 in the expansion of $(x - 2y)^6$.**Solution**

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 2$.
2. Use $6C2$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 60.

Final answer

60

Q029 Written

Find the coefficient of x^4 in the expansion of $(2x^2 - 3)^7$.**Solution**

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 5$.
2. Use $7C5$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -20412 .

Final answer -20412

Worked Solutions

Binomial and Multinomial Theorems

Q030 Written

Find the coefficient of x^6y in the expansion of $(2x - 3y)^7$.**Solution**

1. The target power uses $q = 6$ copies of the first term, so $r = n - q = 1$.
2. Use $7C1$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -1344 .

Final answer -1344

Q031 MCQ

Find the coefficient of x^2 in the expansion of $(3x - 2)^5$:**Solution**

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 3$.
2. Use $5C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -720 .

Correct option: B**Final answer** -720

Worked Solutions

Binomial and Multinomial Theorems

Q032 Written

Find the coefficient of a^6 in the expansion of $(3a^2 - 2)^8$.

Solution

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 5$.
2. Use $8C5$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -48384 .

Final answer -48384

Q033 MCQ

Find the coefficient of x^4y^4 in the expansion of $(2x - y)^8$:

Solution

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 4$.
2. Use $8C4$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 1120.

Correct option: A

Final answer

1120

Worked Solutions

Binomial and Multinomial Theorems

Q034 MCQ

Find the coefficient of ab^3 in the expansion of $(3a + b)^4$:**Solution**

1. The target power uses $q = 1$ copies of the first term, so $r = n - q = 3$.
2. Use $4C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 12.

Correct option: B**Final answer**

12

Q035 MCQ

Find the coefficient of x^3y^2 in the expansion of $(x - 4y)^5$:**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use $5C2$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 160.

Correct option: C**Final answer**

160

Worked Solutions

Binomial and Multinomial Theorems

Q036 MCQ

Find the coefficient of x^4 in the expansion of $(3x + 1)^5$:**Solution**

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 1$.
2. Use $5C1$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 405.

Correct option: D**Final answer**

405

Q037 Written

Find the coefficient of a in the expansion of $(2a - 3)^4$.**Solution**

1. The target power uses $q = 1$ copies of the first term, so $r = n - q = 3$.
2. Use $4C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -216 .

Final answer -216

Worked Solutions

Binomial and Multinomial Theorems

Q038 Written

Find the coefficient of x^3 in the expansion of $(3x + 2)^5$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use 5C_2 times the first factor to power q and the second factor to power r .
3. The required coefficient is 1080.

Final answer

1080

Q039 Written

Find the coefficient of x^6 in the expansion of $(2x^2 - 1)^6$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 3$.
2. Use 6C_3 times the first factor to power q and the second factor to power r .
3. The required coefficient is -160 .

Final answer -160

Worked Solutions

Binomial and Multinomial Theorems

Q040 MCQ

Find the coefficient of x^4 in the expansion of $(3x^2 - 1)^5$:**Solution**

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 3$.
2. Use $5C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -90 .

Correct option: D**Final answer** -90

Q041 Written

Find the coefficient of x^6 in the expansion of $(3x^2 + 2)^6$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 3$.
2. Use $6C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 4320 .

Final answer 4320

Worked Solutions

Binomial and Multinomial Theorems

Q042 Written

If $(x + 1)^5 = a_0 + a_x + \dots$, find the sum of its coefficients.

Solution

1. Put $x = 1$; the sum of the coefficients is the value of the expansion at $x = 1$.
2. Therefore the required value is 32.

Final answer

32

Q043 MCQ

Find the coefficient of xyz in the expansion of $(x + y + z)^3$.

Solution

1. The target powers are $p = 1$, $q = 1$, $r = 1$, and $p + q + r = 3$.
2. Coefficient = $\frac{3!}{1! \times 1! \times 1!} \times (1)^1 \times (1)^1 \times (1)^1 = 6$.
3. Therefore, the required coefficient is 6.

Correct option: A

Final answer

6

Worked Solutions

Binomial and Multinomial Theorems

Q044 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a + b + c)^8$:**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is 280.

Correct option: C**Final answer**

280

Q045 MCQ

Find the coefficient of a^3bc^2 in the expansion of $(a + b + c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 60$ on the signed factors.
3. The required coefficient is 60.

Correct option: A**Final answer**

60

Worked Solutions

Binomial and Multinomial Theorems

Q046 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + b + c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 90$ on the signed factors.
3. The required coefficient is 90.

Correct option: B**Final answer**

90

Q047 MCQ

Find the coefficient of xy^5z in the expansion of $(x + y + z)^7$:**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 42$ on the signed factors.
3. The required coefficient is 42.

Correct option: A**Final answer**

42

Worked Solutions

Binomial and Multinomial Theorems

Q048 Written

Find the coefficient of x^4y^2z in the expansion of $(x + y + z)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 105$ on the signed factors.
3. The required coefficient is 105.

Final answer

105

Q049 Written

Find the coefficient of $x^2y^3z^2$ in the expansion of $(x + y + z)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 210$ on the signed factors.
3. The required coefficient is 210.

Final answer

210

Worked Solutions

Binomial and Multinomial Theorems

Q050 Written

Find the coefficient of x^6yz^2 in the expansion of $(x + y + z)^9$.**Solution**

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\frac{n!}{p! \times q! \times r!} = 252$ on the signed factors.
3. The required coefficient is 252.

Final answer

252

Q051 Written

Find the coefficient of $x^3y^3z^3$ in the expansion of $(x + y + z)^9$.**Solution**

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\frac{n!}{p! \times q! \times r!} = 1680$ on the signed factors.
3. The required coefficient is 1680.

Final answer

1680

Worked Solutions

Binomial and Multinomial Theorems

Q052 Written

Find the coefficient of $x^2y^4z^3$ in the expansion of $(x + y + z)^9$.**Solution**

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\frac{n!}{p! \times q! \times r!} = 1260$ on the signed factors.
3. The required coefficient is 1260.

Final answer

1260

Q053 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a - 2b + 3c)^8$:**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is -6720 .

Correct option: D**Final answer** -6720

Worked Solutions

Binomial and Multinomial Theorems

Q054 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + 3b - 2c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 90$ on the signed factors.
3. The required coefficient is 3240.

Correct option: C**Final answer**

3240

Q055 MCQ

Find the coefficient of ab^2c^3 in the expansion of $(a + 3b - 2c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 60$ on the signed factors.
3. The required coefficient is -4320 .

Correct option: D**Final answer** -4320

Worked Solutions

Binomial and Multinomial Theorems

Q056 Written

Find the coefficient of $a^2b^3c^2$ in the expansion of $(a + 2b - 3c)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 210$ on the signed factors.
3. The required coefficient is 15120.

Final answer

15120

Q057 Written

Find the coefficient of abc^5 in the expansion of $(a + 2b - 3c)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 42$ on the signed factors.
3. The required coefficient is -20412 .

Final answer -20412

Worked Solutions

Binomial and Multinomial Theorems

Q058 Written

Find the coefficient of a^2b^4c in the expansion of $(a + 2b - 3c)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 105$ on the signed factors.
3. The required coefficient is -5040 .

Final answer -5040

Q059 Written

Find the coefficient of $a^2b^2c^4$ in the expansion of $(2a - b + 3c)^8$.**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 420$ on the signed factors.
3. The required coefficient is 136080.

Final answer

136080

Worked Solutions

Binomial and Multinomial Theorems

Q060 Written

Find the coefficient of a^4bc^3 in the expansion of $(2a - b + 3c)^8$.**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is -120960 .

Final answer -120960

Q061 Written

Find the coefficient of $a^3b^3c^2$ in the expansion of $(2a - b + 3c)^8$.**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 560$ on the signed factors.
3. The required coefficient is -40320 .

Final answer -40320

Worked Solutions

Binomial and Multinomial Theorems

Q062 Written

Find the greatest coefficient in the expansion of $(x^2 + x + 1)^4$.

Solution

1. Expand and collect like powers: $x^8 + 4x^7 + 10x^6 + 16x^5 + 19x^4 + 16x^3 + 10x^2 + 4x + 1$.
2. The greatest coefficient is 19.

Final answer

19

Quiz MCQ 1 Concept Quiz · MCQ

The coefficient of x^4 in $(2x - 3)^5$ is:

Solution

1. $q = 4$ and $r = 1$, so the coefficient is -240 .

Correct option: A

Final answer -240

Worked Solutions

Binomial and Multinomial Theorems

Quiz MCQ 2 [Concept Quiz](#) · [MCQ](#)

The coefficient of $x^2y^2z^2$ in $(x + y + z)^6$ is:

Solution

1. Use $6!/(2!2!2!)$.

Correct option: C

Final answer

90

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

The coefficient of x^6 in $(3x^2 - 2)^6$ is:

Solution

1. Choose the first factor three times.

Correct option: A

Final answer

−4320

Worked Solutions

Binomial and Multinomial Theorems

Quiz MCQ 4 **Concept Quiz · MCQ**

The coefficient of x^5 in $(2x^2 + 1)^4$ is:

Solution

1. All powers of x are even, so the coefficient is zero.

Correct option: A

Final answer

0

Quiz WRITTEN 5 **Concept Quiz · Written**

Expand $(2x - 1)^4$ fully.

Solution

1. The expansion is $16x^4 - 32x^3 + 24x^2 - 8x + 1$.

Final answer

$16x^4 - 32x^3 + 24x^2 - 8x + 1$

Worked Solutions

Binomial and Multinomial Theorems

Quiz WRITTEN 6 Concept Quiz · Written

Find the coefficient of x^4 in $(3x^2 + 2)^5$. Show the exponent check.

Solution

1. The first-term exponent is $q = 2$ and $r = 3$, giving ${}^5C_3 \times 9 \times 8 = 720$.

Final answer

720

Quiz WRITTEN 7 Concept Quiz · Written

Find the coefficient of x^2y^3z in $(x - 2y + z)^6$.

Solution

1. Use the multinomial coefficient and the signed factor; the result is -480 .

Final answer

 -480

Worked Solutions

Binomial and Multinomial Theorems

Quiz WRITTEN 8 Concept Quiz · Written

Find the greatest coefficient in $(x^2 + x + 1)^4$. Show enough work to justify the maximum.

Solution

1. Expand: $x^8 + 4x^7 + 10x^6 + 16x^5 + 19x^4 + 16x^3 + 10x^2 + 4x + 1$; the greatest coefficient is 19.

Final answer

19

Answer key

Use the question number shown on each page.

Q001 A	Q018 D	Q036 D	Q054 C
Q002 A	Q019 A	Q037 -216	Q055 D
Q003 A	Q020 B	Q038 1080	Q056 15120
Q004 $x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$	Q021 $243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$	Q039 -160	Q057 -20412
Q005 D	Q022 C	Q040 D	Q058 -5040
Q006 $x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$	Q023 B	Q041 4320	Q059 136080
Q007 B	Q024 C	Q042 32	Q060 -120960
Q008 C	Q025 C	Q043 A	Q061 -40320
Q009 $a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$	Q026 B	Q044 C	Q062 19
Q010 A	Q027 10	Q045 A	Quiz MCQ 1 A
Q011 $243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$	Q028 60	Q046 B	Quiz MCQ 2 C
Q012 $32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$	Q029 -20412	Q047 A	Quiz MCQ 3 A
Q013 $x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$	Q030 -1344	Q048 105	Quiz MCQ 4 A
Q014 $64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$	Q031 B	Q049 210	Quiz WRITTEN 5 $16x^4 - 32x^3 + 24x^2 - 8x + 1$
Q015 $x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$	Q032 -48384	Q050 252	Quiz WRITTEN 6 720
Q016 B	Q033 A	Q051 1680	Quiz WRITTEN 7 -480
Q017 D	Q034 B	Q052 1260	Quiz WRITTEN 8 19
	Q035 C	Q053 D	

Concept 2 | Polynomial Division

English Student Edition • Focused formulas and guided practice

Formula opener

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder

$$R = P(a) \quad \text{for divisor } x - a$$

Degree check

$$\deg R < \deg B$$

Worked Solutions

Polynomial Division

Q001 Challenge

Find the quotient and the remainder when $2x^3 + x^2 - 7x + 3$ is divided by $x^2 + x - 2$. Then verify the result using $A = B \times Q + R$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 + x^2 - 7x + 3 = (x^2 + x - 2) \times (2x - 1) + (-2x + 1)$.
4. Therefore, quotient $Q = 2x - 1$ and remainder $R = -2x + 1$.

Correct option: B

Final answer

$$Q = 2x - 1, \quad R = -2x + 1$$

Worked Solutions

Polynomial Division

Q002 Challenge

Find the polynomial A when it is divided by $x^2 - 2x + 3$, the quotient is $3x - 1$, and the remainder is $x + 2$. Then divide your answer back to check it:

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x + 3) \times (3x - 1) + (x + 2)$.
3. Expand and collect like powers: $A = 3x^3 - 7x^2 + 12x - 1$.
4. Therefore, $A = 3x^3 - 7x^2 + 12x - 1$.

Correct option: A

Final answer

$$3x^3 - 7x^2 + 12x - 1$$

Worked Solutions

Polynomial Division

Q003 Challenge

When $x^3 + 5x^2 + 7x - 7$ is divided by B, the quotient is $x + 3$ and the remainder is $2x - 4$. Find B. Then verify the degree condition:

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^3+5x^2+7x-7-(2x-4)}{x+3}$.
3. Divide and simplify: $B = x^2 + 2x - 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^3 + 5x^2 + 7x - 7$.

Correct option: D

Final answer

$$x^2 + 2x - 1$$

Worked Solutions

Polynomial Division

Q004 Challenge

A cuboid has volume $2x^3 + 9x^2 + 4x - 15$ and base area $x^2 + 2x - 3$. Write its height in terms of x . State the mathematical check Which of the following is correct?:

Solution

1. Use height = volume $\div$ base area: $h = \frac{2x^3+9x^2+4x-15}{x^2+2x-3}$.
2. Divide the polynomials and check that the remainder is zero.
3. Therefore, $h = 2x + 5$.

Correct option: C

Final answer

$$2x + 5$$

Worked Solutions

Polynomial Division

Q005 Challenge

Find the quotient and the remainder when $3x^4 - x^3 + 2x^2 - 4x + 1$ is divided by $x^2 - 2x + 2$. Decide whether the result is exact and justify from R:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 - x^3 + 2x^2 - 4x + 1 = (x^2 - 2x + 2) \times (3x^2 + 5x + 6) + (-2x - 11)$.
4. Therefore, quotient $Q = 3x^2 + 5x + 6$ and remainder $R = -2x - 11$.

Correct option: B

Final answer

$$Q = 3x^2 + 5x + 6, \quad R = -2x - 11$$

Worked Solutions

Polynomial Division

Q006 Challenge

Find the quotient and the remainder when $x^4 - 3x^2 + 2x + 6$ is divided by $x + 2$. Explain why a zero placeholder is needed:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 3x^2 + 2x + 6 = (x + 2) \times (x^3 - 2x^2 + x) + (6)$.
4. Therefore, quotient $Q = x^3 - 2x^2 + x$ and remainder $R = 6$.

Correct option: A

Final answer

$$Q = x^3 - 2x^2 + x, \quad R = 6$$

Worked Solutions

Polynomial Division

Q007 Written

Find the quotient and the remainder when $x^2 - 5x + 1$ is divided by $x - 2$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^2 - 5x + 1 = (x - 2) \times (x - 3) + (-5)$.
4. Therefore, quotient $Q = x - 3$ and remainder $R = -5$.

Final answer

$$Q = x - 3, \quad R = -5$$

Q008 Written

Find the quotient and the remainder when $x^3 - 4x + 1$ is divided by $x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 4x + 1 = (x - 3) \times (x^2 + 3x + 5) + (16)$.
4. Therefore, quotient $Q = x^2 + 3x + 5$ and remainder $R = 16$.

Final answer

$$Q = x^2 + 3x + 5, \quad R = 16$$

Worked Solutions

Polynomial Division

Q009 MCQ

Find the quotient and the remainder when $3x^3 + 2x^2 - 7x + 5$ is divided by $x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 + 2x^2 - 7x + 5 = (x - 2) \times (3x^2 + 8x + 9) + (23)$.
4. Therefore, quotient $Q = 3x^2 + 8x + 9$ and remainder $R = 23$.

Correct option: D**Final answer**

$$Q = 3x^2 + 8x + 9, \quad R = 23$$

Worked Solutions

Polynomial Division

Q010 MCQ

Find the quotient and the remainder when $x^3 + 6x^2 - 4$ is divided by $2x - 3$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 6x^2 - 4 = (2x - 3) \times \left(\frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8} \right) + \left(\frac{103}{8} \right)$.
4. Therefore, quotient $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}$ and remainder $R = \frac{103}{8}$.

Correct option: B

Final answer

$$Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, \quad R = \frac{103}{8}$$

Worked Solutions

Polynomial Division

Q011 MCQ

Find the quotient and the remainder when $3x^3 - 4x^2 + 2x + 7$ is divided by $x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 - 4x^2 + 2x + 7 = (x - 1) \times (3x^2 - x + 1) + (8)$.
4. Therefore, quotient $Q = 3x^2 - x + 1$ and remainder $R = 8$.

Correct option: B**Final answer**

$$Q = 3x^2 - x + 1, \quad R = 8$$

Q012 MCQ

Find the quotient and the remainder when $2x^4 + x^3 - 5x + 4$ is divided by $x + 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^4 + x^3 - 5x + 4 = (x + 2) \times (2x^3 - 3x^2 + 6x - 17) + (38)$.
4. Therefore, quotient $Q = 2x^3 - 3x^2 + 6x - 17$ and remainder $R = 38$.

Correct option: A**Final answer**

$$Q = 2x^3 - 3x^2 + 6x - 17, \quad R = 38$$

Worked Solutions

Polynomial Division

Q013 MCQ

Find the quotient and the remainder when $4x^4 - 3x^3 + x - 6$ is divided by $x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $4x^4 - 3x^3 + x - 6 = (x - 2) \times (4x^3 + 5x^2 + 10x + 21) + (36)$.
4. Therefore, quotient $Q = 4x^3 + 5x^2 + 10x + 21$ and remainder $R = 36$.

Correct option: D**Final answer**

$$Q = 4x^3 + 5x^2 + 10x + 21, \quad R = 36$$

Worked Solutions

Polynomial Division

Q014 MCQ

Find the quotient and the remainder when $2x^3 - 9x + 1$ is divided by $3x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 9x + 1 = (3x - 2) \times \left(\frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27} \right) + \left(-\frac{119}{27} \right)$.
4. Therefore, quotient $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}$ and remainder $R = -\frac{119}{27}$.

Correct option: B**Final answer**

$$Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = -\frac{119}{27}$$

Worked Solutions

Polynomial Division

Q015 Written

Find the quotient and the remainder when $2x^3 - 5x^2 - 2$ is divided by $2x^2 - x - 1$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 5x^2 - 2 = (2x^2 - x - 1) \times (x - 2) + (-x - 4)$.
4. Therefore, quotient $Q = x - 2$ and remainder $R = -x - 4$.

Final answer

$$Q = x - 2, \quad R = -x - 4$$

Q016 Written

Find the quotient and the remainder when $2x^3 - 13x + 1$ is divided by $x^2 + 2x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 13x + 1 = (x^2 + 2x - 3) \times (2x - 4) + (x - 11)$.
4. Therefore, quotient $Q = 2x - 4$ and remainder $R = x - 11$.

Final answer

$$Q = 2x - 4, \quad R = x - 11$$

Worked Solutions

Polynomial Division

Q017 Written

Find the quotient and the remainder when $2x^3 - 3x^2 + 3x + 4$ is divided by $x^2 - 2x + 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 3x^2 + 3x + 4 = (x^2 - 2x + 3) \times (2x + 1) + (-x + 1)$.
4. Therefore, quotient $Q = 2x + 1$ and remainder $R = -x + 1$.

Final answer

$$Q = 2x + 1, \quad R = -x + 1$$

Q018 Written

Find the quotient and the remainder when $x^3 - 7x + 6$ is divided by $x^2 + 2x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 7x + 6 = (x^2 + 2x - 3) \times (x - 2) + (0)$.
4. Therefore, quotient $Q = x - 2$ and remainder $R = 0$.

Final answer

$$Q = x - 2, \quad R = 0$$

Worked Solutions

Polynomial Division

Q019 MCQ

Find the quotient and the remainder when $x^3 + 2x^2 - 5x + 4$ is divided by $x^2 + x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 2x^2 - 5x + 4 = (x^2 + x - 2) \times (x + 1) + (-4x + 6)$.
4. Therefore, quotient $Q = x + 1$ and remainder $R = -4x + 6$.

Correct option: D**Final answer**

$$Q = x + 1, \quad R = -4x + 6$$

Q020 MCQ

Find the quotient and the remainder when $x^4 - 3x^3 + 2x - 5$ is divided by $x^2 + 2x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 3x^3 + 2x - 5 = (x^2 + 2x - 1) \times (x^2 - 5x + 11) + (-25x + 6)$.
4. Therefore, quotient $Q = x^2 - 5x + 11$ and remainder $R = -25x + 6$.

Correct option: B**Final answer**

$$Q = x^2 - 5x + 11, \quad R = -25x + 6$$

Worked Solutions

Polynomial Division

Q021 MCQ

Find the quotient and the remainder when $2x^3 + x^2 - 4x + 5$ is divided by $x^2 - 2x + 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 + x^2 - 4x + 5 = (x^2 - 2x + 2) \times (2x + 5) + (2x - 5)$.
4. Therefore, quotient $Q = 2x + 5$ and remainder $R = 2x - 5$.

Correct option: D**Final answer**

$$Q = 2x + 5, \quad R = 2x - 5$$

Q022 MCQ

Find the quotient and the remainder when $x^4 - 2x^3 + 5x^2 - x + 3$ is divided by $x^2 + 3x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 2x^3 + 5x^2 - x + 3 = (x^2 + 3x - 1) \times (x^2 - 5x + 21) + (-69x + 24)$.
4. Therefore, quotient $Q = x^2 - 5x + 21$ and remainder $R = -69x + 24$.

Correct option: C**Final answer**

$$Q = x^2 - 5x + 21, \quad R = -69x + 24$$

Worked Solutions

Polynomial Division

Q023 MCQ

Find the quotient and the remainder when $3x^3 - 5x + 2$ is divided by $x^2 - x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 - 5x + 2 = (x^2 - x - 1) \times (3x + 3) + (x + 5)$.
4. Therefore, quotient $Q = 3x + 3$ and remainder $R = x + 5$.

Correct option: B**Final answer**

$$Q = 3x + 3, \quad R = x + 5$$

Q024 MCQ

Find the quotient and the remainder when $2x^4 - x^3 + 4x^2 - 7$ is divided by $x^2 + x + 3$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^4 - x^3 + 4x^2 - 7 = (x^2 + x + 3) \times (2x^2 - 3x + 1) + (8x - 10)$.
4. Therefore, quotient $Q = 2x^2 - 3x + 1$ and remainder $R = 8x - 10$.

Correct option: A**Final answer**

$$Q = 2x^2 - 3x + 1, \quad R = 8x - 10$$

Worked Solutions

Polynomial Division

Q025 Written

Find the quotient and the remainder when $6x^3 - 8x^2 + 5$ is divided by $2x^2 - 2x - 5$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $6x^3 - 8x^2 + 5 = (2x^2 - 2x - 5) \times (3x - 1) + (13x)$.
4. Therefore, quotient $Q = 3x - 1$ and remainder $R = 13x$.

Final answer

$$Q = 3x - 1, \quad R = 13x$$

Q026 Written

Find the quotient and the remainder when $3x^4 - 5x^3 + x - 1$ is divided by $-x^2 + 3x + 2$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 - 5x^3 + x - 1 = (-x^2 + 3x + 2) \times (-3x^2 - 4x - 18) + (63x + 35)$.
4. Therefore, quotient $Q = -3x^2 - 4x - 18$ and remainder $R = 63x + 35$.

Final answer

$$Q = -3x^2 - 4x - 18, \quad R = 63x + 35$$

Worked Solutions

Polynomial Division

Q027 MCQ

Find the quotient and the remainder when $4x^3 - x^2 + 8x - 3$ is divided by $2x^2 + x - 4$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $4x^3 - x^2 + 8x - 3 = (2x^2 + x - 4) \times \left(2x - \frac{3}{2}\right) + \left(\frac{35x}{2} - 9\right)$.
4. Therefore, quotient $Q = 2x - \frac{3}{2}$ and remainder $R = \frac{35x}{2} - 9$.

Correct option: D**Final answer**

$$Q = 2x - \frac{3}{2}, \quad R = \frac{35x}{2} - 9$$

Q028 MCQ

Find the quotient and the remainder when $x^3 - 8x + 2$ is divided by $x^2 - 4$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 8x + 2 = (x^2 - 4) \times (x) + (-4x + 2)$.
4. Therefore, quotient $Q = x$ and remainder $R = -4x + 2$.

Correct option: B**Final answer**

$$Q = x, \quad R = -4x + 2$$

Worked Solutions

Polynomial Division

Q029 MCQ

Find the quotient and the remainder when $5x^3 - 4x^2 + x - 6$ is divided by $2x^2 - x + 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $5x^3 - 4x^2 + x - 6 = (2x^2 - x + 1) \times \left(\frac{5x}{2} - \frac{3}{4}\right) + \left(-\frac{9x}{4} - \frac{21}{4}\right)$.
4. Therefore, quotient $Q = \frac{5x}{2} - \frac{3}{4}$ and remainder $R = -\frac{9x}{4} - \frac{21}{4}$.

Correct option: D**Final answer**

$$Q = \frac{5x}{2} - \frac{3}{4}, \quad R = -\frac{9x}{4} - \frac{21}{4}$$

Worked Solutions

Polynomial Division

Q030 MCQ

Find the quotient and the remainder when $3x^4 + 2x^3 - x^2 + 5$ is divided by $4x^2 + x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 + 2x^3 - x^2 + 5 = (4x^2 + x - 2) \times \left(\frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64} \right) + \left(\frac{37x}{64} + \frac{163}{32} \right)$.
4. Therefore, quotient $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}$ and remainder $R = \frac{37x}{64} + \frac{163}{32}$.

Correct option: C**Final answer**

$$Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, \quad R = \frac{37x}{64} + \frac{163}{32}$$

Worked Solutions

Polynomial Division

Q031 MCQ

Find the quotient and the remainder when $3x^3 + 4x^2 - x + 9$ is divided by $2x^2 - x + 3$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 + 4x^2 - x + 9 = (2x^2 - x + 3) \times \left(\frac{3x}{2} + \frac{11}{4}\right) + \left(-\frac{11x}{4} + \frac{3}{4}\right)$.
4. Therefore, quotient $Q = \frac{3x}{2} + \frac{11}{4}$ and remainder $R = -\frac{11x}{4} + \frac{3}{4}$.

Correct option: D**Final answer**

$$Q = \frac{3x}{2} + \frac{11}{4}, \quad R = -\frac{11x}{4} + \frac{3}{4}$$

Worked Solutions

Polynomial Division

Q032 MCQ

Find the quotient and the remainder when $6x^3 - 5x + 1$ is divided by $2x^2 + 3x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $6x^3 - 5x + 1 = (2x^2 + 3x - 2) \times \left(3x - \frac{9}{2}\right) + \left(\frac{29x}{2} - 8\right)$.
4. Therefore, quotient $Q = 3x - \frac{9}{2}$ and remainder $R = \frac{29x}{2} - 8$.

Correct option: B

Final answer

$$Q = 3x - \frac{9}{2}, \quad R = \frac{29x}{2} - 8$$

Q033 Written

Find the quotient and the remainder when $-9a^2 - 9ax + 4x^2$ is divided by $-3a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-9a^2 - 9ax + 4x^2 = (-3a + x) \times (3a + 4x) + (0)$.
4. Therefore, quotient $Q = 3a + 4x$ and remainder $R = 0$.

Final answer

$$Q = 3a + 4x, \quad R = 0$$

Worked Solutions

Polynomial Division

Q034 Written

Find the quotient and the remainder when $-5a^3 + 5a^2x - 3ax^2 + x^3$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-5a^3 + 5a^2x - 3ax^2 + x^3 = (-2a + x) \times (3a^2 - ax + x^2) + (a^3)$.
4. Therefore, quotient $Q = 3a^2 - ax + x^2$ and remainder $R = a^3$.

Final answer

$$Q = 3a^2 - ax + x^2, \quad R = a^3$$

Q035 Written

Find the quotient and the remainder when $-6a^2 - 5ax + 4x^2$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-6a^2 - 5ax + 4x^2 = (-2a + x) \times (3a + 4x) + (0)$.
4. Therefore, quotient $Q = 3a + 4x$ and remainder $R = 0$.

Final answer

$$Q = 3a + 4x, \quad R = 0$$

Worked Solutions

Polynomial Division

Q036 Written

Find the quotient and the remainder when $-a^3 + a^2x + ax^2 + x^3$ is divided by $2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-a^3 + a^2x + ax^2 + x^3 = (2a + x) \times (3a^2 - ax + x^2) + (-7a^3)$.
4. Therefore, quotient $Q = 3a^2 - ax + x^2$ and remainder $R = -7a^3$.

Final answer

$$Q = 3a^2 - ax + x^2, \quad R = -7a^3$$

Q037 Written

Find the quotient and the remainder when $3a^3 - 2ax^2 + x^3$ is divided by $a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3a^3 - 2ax^2 + x^3 = (a + x) \times (3a^2 - 3ax + x^2) + (0)$.
4. Therefore, quotient $Q = 3a^2 - 3ax + x^2$ and remainder $R = 0$.

Final answer

$$Q = 3a^2 - 3ax + x^2, \quad R = 0$$

Worked Solutions

Polynomial Division

Q038 MCQ

Find the quotient and the remainder when $-2a^3 + 7a^2x - 4ax^2 + x^3$ is divided by $-a + x$. Treat it as a polynomial in x ; other letters are constants:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-2a^3 + 7a^2x - 4ax^2 + x^3 = (-a + x) \times (4a^2 - 3ax + x^2) + (2a^3)$.
4. Therefore, quotient $Q = 4a^2 - 3ax + x^2$ and remainder $R = 2a^3$.

Correct option: D

Final answer

$$Q = 4a^2 - 3ax + x^2, \quad R = 2a^3$$

Q039 MCQ

Find the quotient and the remainder when $x^2 + 3xy - 4y^2$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^2 + 3xy - 4y^2 = (x + y) \times (x + 2y) + (-6y^2)$.
4. Therefore, quotient $Q = x + 2y$ and remainder $R = -6y^2$.

Correct option: B

Final answer

$$Q = x + 2y, \quad R = -6y^2$$

Worked Solutions

Polynomial Division

Q040 Written

Find the quotient and the remainder when $x^3 + 3x^2y - 2y^3$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 3x^2y - 2y^3 = (x + y) \times (x^2 + 2xy - 2y^2) + (0)$.
4. Therefore, quotient $Q = x^2 + 2xy - 2y^2$ and remainder $R = 0$.

Final answer

$$Q = x^2 + 2xy - 2y^2, \quad R = 0$$

Q041 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $x + 3$, and the remainder is $-2x + 5$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x - 1) \times (x + 3) + (-2x + 5)$.
3. Expand and collect like powers: $A = x^3 + x^2 - 9x + 2$.
4. Therefore, $A = x^3 + x^2 - 9x + 2$.

Final answer

$$x^3 + x^2 - 9x + 2$$

Worked Solutions

Polynomial Division

Q042 Written

Find the polynomial **A** when it is divided by $x^2 - 2x - 1$, the quotient is $2x - 3$, and the remainder is $-2x$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x - 1) \times (2x - 3) + (-2x)$.
3. Expand and collect like powers: $A = 2x^3 - 7x^2 + 2x + 3$.
4. Therefore, $A = 2x^3 - 7x^2 + 2x + 3$.

Final answer

$$2x^3 - 7x^2 + 2x + 3$$

Q043 Written

Find the polynomial **A** when it is divided by $x^2 - 3x + 2$, the quotient is $2x + 6$, and the remainder is $2x - 3$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 3x + 2) \times (2x + 6) + (2x - 3)$.
3. Expand and collect like powers: $A = 2x^3 - 12x + 9$.
4. Therefore, $A = 2x^3 - 12x + 9$.

Final answer

$$2x^3 - 12x + 9$$

Worked Solutions

Polynomial Division

Q044 Written

Find the polynomial A when it is divided by $2x^2 - x + 2$, the quotient is $2x - 1$, and the remainder is 2.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (2x^2 - x + 2) \times (2x - 1) + (2)$.
3. Expand and collect like powers: $A = 4x^3 - 4x^2 + 5x$.
4. Therefore, $A = 4x^3 - 4x^2 + 5x$.

Final answer

$$4x^3 - 4x^2 + 5x$$

Q045 Written

Find the polynomial A when it is divided by $2x^2 + x + 1$, the quotient is $3x + 2$, and the remainder is $-2x + 6$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (2x^2 + x + 1) \times (3x + 2) + (-2x + 6)$.
3. Expand and collect like powers: $A = 6x^3 + 7x^2 + 3x + 8$.
4. Therefore, $A = 6x^3 + 7x^2 + 3x + 8$.

Final answer

$$6x^3 + 7x^2 + 3x + 8$$

Worked Solutions

Polynomial Division

Q046 Written

When $4x^3 + 8x - 9$ is divided by B, the quotient is $2x - 3$ and the remainder is $x + 15$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{4x^3+8x-9-(x+15)}{2x-3}$.
3. Divide and simplify: $B = 2x^2 + 3x + 8$.
4. Check by multiplying B by Q and adding R to recover $A = 4x^3 + 8x - 9$.

Final answer

$$2x^2 + 3x + 8$$

Q047 Written

When $x^3 - x^2 + 3x + 1$ is divided by B, the quotient is $x^2 - 2x + 5$ and the remainder is -4 . Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^3-x^2+3x+1-(-4)}{x^2-2x+5}$.
3. Divide and simplify: $B = x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^3 - x^2 + 3x + 1$.

Final answer

$$x + 1$$

Worked Solutions

Polynomial Division

Q048 Written

When $2x^3 + x^2 - 3x + 5$ is divided by B, the quotient is $x + 1$ and the remainder is $-5x + 2$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{2x^3+x^2-3x+5-(-5x+2)}{x+1}$.
3. Divide and simplify: $B = 2x^2 - x + 3$.
4. Check by multiplying B by Q and adding R to recover $A = 2x^3 + x^2 - 3x + 5$.

Final answer

$$2x^2 - x + 3$$

Q049 Written

When $6x^3 + 11x^2 - 2$ is divided by B, the quotient is $2x^2 + 3x - 1$ and the remainder is -1 . Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{6x^3+11x^2-2-(-1)}{2x^2+3x-1}$.
3. Divide and simplify: $B = 3x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = 6x^3 + 11x^2 - 2$.

Final answer

$$3x + 1$$

Worked Solutions

Polynomial Division

Q050 Written

When $2x^3 + 4x^2 + 7$ is divided by B, the quotient is $x + 2$ and the remainder is $3x + 13$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{2x^3+4x^2+7-(3x+13)}{x+2}$.
3. Divide and simplify: $B = 2x^2 - 3$.
4. Check by multiplying B by Q and adding R to recover $A = 2x^3 + 4x^2 + 7$.

Final answer

$$2x^2 - 3$$

Q051 Written

When $x^4 + x^2 + 3x - 1$ is divided by B, the quotient is $x^2 + x + 1$ and the remainder is $3x - 2$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^4+x^2+3x-1-(3x-2)}{x^2+x+1}$.
3. Divide and simplify: $B = x^2 - x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^4 + x^2 + 3x - 1$.

Final answer

$$x^2 - x + 1$$

Worked Solutions

Polynomial Division

Q052 Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Write its height in terms of x .

Solution

1. Use height = volume $\div$ base area: $h = \frac{4x^3 + 4x^2 - 9x - 9}{2x^2 - x - 3}$.
2. Divide the polynomials and check that the remainder is zero.
3. Therefore, $h = 2x + 3$.

Final answer

$$2x + 3$$

Quiz MCQ 1 Concept Quiz · MCQ

For $x^2 - 5x + 1$ divided by $x - 2$, what is R ?:

Solution

1. Long division gives $R = -5$.

Correct option: A

Final answer

$$-5$$

Worked Solutions

Polynomial Division

Quiz MCQ 2 **Concept Quiz · MCQ**

In a polynomial in x , a is treated as... Which of the following is correct?:

Solution

1. Only powers of x determine the division order.

Correct option: B

Final answer

constant

Quiz MCQ 3 **Concept Quiz · MCQ**

To recover B from $A = B \times Q + R$, use... Which of the following is correct?:

Solution

1. Isolate B after moving R to the other side.

Correct option: C

Final answer

$$\frac{A - R}{Q}$$

Worked Solutions

Polynomial Division

Quiz MCQ 4 Concept Quiz · MCQ

If B has degree 2, a valid remainder can have degree...:

Solution

1. The remainder degree must be below 2.

Correct option: C

Final answer

1

Quiz WRITTEN 5 Concept Quiz · Written

Divide $3x^3 - 5x^2 + 2x - 7$ by $x - 2$. State Q, R, and verify $A = B \times Q + R$.

Solution

1. Long division gives $Q = 3x^2 + x + 4$, $R = 1$.
2. Substitution verifies $3x^3 - 5x^2 + 2x - 7 = (x - 2) \times (3x^2 + x + 4) + (1)$.

Final answer

$$Q = 3x^2 + x + 4, \quad R = 1$$

Worked Solutions

Polynomial Division

Quiz WRITTEN 6 Concept Quiz · Written

Recover A when $B = x^2 - x - 2$, $Q = 2x + 3$, $R = -x + 1$.

Solution

1. Use $A = B \times Q + R$.
2. $A = 2x^3 + x^2 - 8x - 5$.

Final answer

$$2x^3 + x^2 - 8x - 5$$

Quiz WRITTEN 7 Concept Quiz · Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Find its height.

Solution

1. Divide volume by base area: $h = 2x + 3$.
2. The remainder is zero, so the height is polynomial.

Final answer

$$2x + 3$$

Worked Solutions

Polynomial Division

Quiz WRITTEN 8 Concept Quiz · Written

Divide $x^4 - 2x^3 + 3x - 1$ by $x^2 - x + 1$ and explain why the remainder is valid.

Solution

1. Long division gives $Q = x^2 - x - 2$ and $R = 2x + 1$.
2. The remainder degree is less than 2, so it is valid even though it is nonzero.

Final answer

$$Q = x^2 - x - 2, \quad R = 2x + 1$$

Answer key

Use the question number shown on each page.

Q001 B	Q016 $Q = 2x - 4$, $R = x - 11$	Q031 D	Q046 $2x^2 + 3x + 8$
Q002 A	Q017 $Q = 2x + 1$, $R = -x + 1$	Q032 B	Q047 $x + 1$
Q003 D	Q018 $Q = x - 2$, $R = 0$	Q033 $Q = 3a + 4x$, $R = 0$	Q048 $2x^2 - x + 3$
Q004 C	Q019 D	Q034 $Q = 3a^2 - ax + x^2$, $R = a^3$	Q049 $3x + 1$
Q005 B	Q020 B	Q035 $Q = 3a + 4x$, $R = 0$	Q050 $2x^2 - 3$
Q006 A	Q021 D	Q036 $Q = 3a^2 - ax + x^2$, $R = -7a^3$	Q051 $x^2 - x + 1$
Q007 $Q = x - 3$, $R = -5$	Q022 C	Q037 $Q = 3a^2 - 3ax + x^2$, $R = 0$	Q052 $2x + 3$
Q008 $Q = x^2 + 3x + 5$, $R = 16$	Q023 B	Q038 D	Quiz MCQ 1 A
Q009 D	Q024 A	Q039 B	Quiz MCQ 2 B
Q010 B	Q025 $Q = 3x - 1$, $R = 13x$	Q040 $Q = x^2 + 2xy - 2y^2$, $R = 0$	Quiz MCQ 3 C
Q011 B	Q026 $Q = -3x^2 - 4x - 18$, $R = 63x + 35$	Q041 $x^3 + x^2 - 9x + 2$	Quiz MCQ 4 C
Q012 A	Q027 D	Q042 $2x^3 - 7x^2 + 2x + 3$	Quiz WRITTEN 5 $Q = 3x^2 + x + 4$, $R = 1$
Q013 D	Q028 B	Q043 $2x^3 - 12x + 9$	Quiz WRITTEN 6 $2x^3 + x^2 - 8x - 5$
Q014 B	Q029 D	Q044 $4x^3 - 4x^2 + 5x$	Quiz WRITTEN 7 $2x + 3$
Q015 $Q = x - 2$, $R = -x - 4$	Q030 C	Q045 $6x^3 + 7x^2 + 3x + 8$	Quiz WRITTEN 8 $Q = x^2 - x - 2$, $R = 2x + 1$

Concept 3 | Rational Expressions

English Student Edition • Focused formulas and guided practice

Formula opener

Restriction

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Worked Solutions

Rational Expressions

Q001 Challenge

After simplifying this transfer chain before substitution, which result is correct for 2():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{2x(x+3)}{(x-3)(x-1)}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{2x(x+3)}{(x-3)(x-1)}$$

Worked Solutions

Rational Expressions

Q002 Challenge

After simplifying this transfer chain before substitution, which result is correct for 3():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3(x-3)(x+2)}{x(x+3)}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{3(x-3)(x+2)}{x(x+3)}$$

Worked Solutions

Rational Expressions

Q003 Challenge

After simplifying this transfer chain before substitution, which result is correct for 4():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8(x-4)(x+1)}{(x-3)(x-2)}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{8(x-4)(x+1)}{(x-3)(x-2)}$$

Worked Solutions

Rational Expressions

Q004 Challenge

After simplifying this transfer chain before substitution, which result is correct for 8():?

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8}{x}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{8}{x}$$

Worked Solutions

Rational Expressions

Q005 Challenge

After simplifying this transfer chain before substitution, which result is correct for 9():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$$

Worked Solutions

Rational Expressions

Q006 Challenge

After simplifying this transfer chain before substitution, which result is correct for 10():?

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$$

Worked Solutions

Rational Expressions

Q007 Written

Simplify $\frac{12ab^5}{9a^3b^2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{4b^3}{3a^2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{4b^3}{3a^2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{4b^3}{3a^2}$$

Q008 Written

Simplify $\frac{6x^2y^3}{20x^3y}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{3y^2}{10x}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{3y^2}{10x}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{3y^2}{10x}$$

Worked Solutions

Rational Expressions

Q009 Written

Simplify $\frac{6a^2bc^2}{8ab^4c^2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{3a}{4b^3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{3a}{4b^3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{3a}{4b^3}$$

Q010 Written

Simplify $\frac{2a^2b}{8a^3b^3}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{1}{4ab^2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{1}{4ab^2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{1}{4ab^2}$$

Worked Solutions

Rational Expressions

Q011 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{12ab^5}{9a^3b^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{4b^3}{a^2}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{4b^3}{a^2}$$

Q012 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6x^2y^3}{20x^3y}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9y^2}{10x}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9y^2}{10x}$$

Worked Solutions

Rational Expressions

Q013 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{6a^2bc^2}{8ab^4c^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3a}{b^3}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{3a}{b^3}$$

Q014 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{2a^2b}{8a^3b^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{1}{2ab^2}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{1}{2ab^2}$$

Worked Solutions

Rational Expressions

Q015 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{18a^3b^2}{24ab^5}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3a^2}{2b^3}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{3a^2}{2b^3}$$

Q016 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{-15x^4y^3}{20x^2y}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= -\frac{9x^2y^2}{4}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$-\frac{9x^2y^2}{4}$$

Worked Solutions

Rational Expressions

Q017 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{45a^2b^4c}{60a^5bc^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3b^3}{a^3c}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{3b^3}{a^3c}$$

Q018 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{14x^5y^2}{21x^2y^4}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{4x^3}{3y^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{4x^3}{3y^2}$$

Worked Solutions

Rational Expressions

Q019 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{32a^4}{48ab^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{2a^3}{b^2}$.
4. Retain the original denominator restrictions.

Correct option: A**Final answer**

$$\frac{2a^3}{b^2}$$

Q020 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{27x^3y^5}{36xy^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= 3x^2y^3$.
4. Retain the original denominator restrictions.

Correct option: B**Final answer**

$$3x^2y^3$$

Worked Solutions

Rational Expressions

Q021 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-42a^2c^3}{56a^5c}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= -\frac{3c^2}{2a^3}$.
4. Retain the original denominator restrictions.

Correct option: C**Final answer**

$$-\frac{3c^2}{2a^3}$$

Q022 Written

Simplify $\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}$.

Solution

1. Factor each polynomial: $\frac{a^3b^2}{3c}$ and $\frac{9c^3}{a^4b^2}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{3c^2}{a}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{3c^2}{a}$$

Worked Solutions

Rational Expressions

Q023 Written

Simplify $\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}$.

Solution

1. Factor each polynomial: $\frac{6bc^3}{5a^2}$ and $\frac{10a^4}{3b^3}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{4a^2c^3}{b^2}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{4a^2c^3}{b^2}$$

Q024 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{12c^2}{a}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{12c^2}{a}$$

Worked Solutions

Rational Expressions

Q025 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8a^2c^3}{b^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{8a^2c^3}{b^2}$$

Q026 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6a^2b}{5c} \times \frac{15c^2}{4ab^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{27ac}{2b^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{27ac}{2b^2}$$

Worked Solutions

Rational Expressions

Q027 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{9x^3y}{14a^2} \times \frac{21a^3}{6xy^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9ax^2}{y}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9ax^2}{y}$$

Q028 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-8ab^2}{3c^2} \times \frac{12c^3}{-5a^2b}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{64bc}{5a}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{64bc}{5a}$$

Worked Solutions

Rational Expressions

Q029 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{10x^2y^3}{9ab} \times \frac{27a^2b^2}{25xy}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{18abxy^2}{5}$.
4. Retain the original denominator restrictions.

Correct option: C**Final answer**

$$\frac{18abxy^2}{5}$$

Q030 Written

Simplify $\frac{x^2-4}{3x^2+7x+2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x-2}{3x+1}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x-2}{3x+1}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x-2}{3x+1}$$

Worked Solutions

Rational Expressions

Q031 Written

Simplify $\frac{2x^2-3x-5}{x^2+3x+2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{2x-5}{x+2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{2x-5}{x+2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{2x-5}{x+2}$$

Q032 Written

Simplify $\frac{x^2+3x}{x^2-9}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x}{x-3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x}{x-3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x}{x-3}$$

Worked Solutions

Rational Expressions

Q033 Written

Simplify $\frac{x^2+x-2}{x^2-5x+4}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x+2}{x-4}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x+2}{x-4}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x+2}{x-4}$$

Q034 Written

Simplify $\frac{2x^2-4xy+2y^2}{x^3+3x^2y-4xy^2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{2(x-y)}{x(x+4y)}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{2(x-y)}{x(x+4y)}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{2(x-y)}{x(x+4y)}$$

Worked Solutions

Rational Expressions

Q035 Written

Simplify $\frac{x^3-1}{x^2+2x-3}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x^2+x+1}{x+3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x^2+x+1}{x+3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x^2 + x + 1}{x + 3}$$

Q036 Written

Simplify $\frac{x^2-11x+24}{x^2-6x-16} \times \frac{x^2+2x}{x^2-6x+9}$.

Solution

1. Factor each polynomial: $\frac{(x-8)(x-3)}{(x-8)(x+2)}$ and $\frac{x(x+2)}{(x-3)^2}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{x}{x-3}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{x}{x-3}$$

Worked Solutions

Rational Expressions

Q037 Written

Simplify $\frac{x+1}{x^2-4} \times \frac{x^2-2x-8}{x^2+x}.$

Solution

1. Factor each polynomial: $\frac{x+1}{(x-2)(x+2)}$ and $\frac{(x-4)(x+2)}{x(x+1)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{x-4}{x(x-2)}.$
4. Retain the original denominator restrictions.

Final answer

$$\frac{x-4}{x(x-2)}$$

Q038 Written

Simplify $\frac{x^2-9}{x^2-2x} \times \frac{x^2-7x+10}{x^2+x-12}.$

Solution

1. Factor each polynomial: $\frac{(x-3)(x+3)}{x(x-2)}$ and $\frac{(x-5)(x-2)}{(x-3)(x+4)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{(x-5)(x+3)}{x(x+4)}.$
4. Retain the original denominator restrictions.

Final answer

$$\frac{(x-5)(x+3)}{x(x+4)}$$

Worked Solutions

Rational Expressions

Q039 Written

Simplify $\frac{x^2+x-6}{2x^2+x-6} \times \frac{x^2+6x+8}{3x^2-5x-2}$.

Solution

1. Factor each polynomial: $\frac{(x-2)(x+3)}{(x+2)(2x-3)}$ and $\frac{(x+2)(x+4)}{(x-2)(3x+1)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{(x+3)(x+4)}{(2x-3)(3x+1)}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$$

Q040 Written

Simplify $\frac{x^2-4x-5}{2x^2+5x+2} \div \frac{x^2-x-20}{2x^2-5x-3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-5)(x+1)}{(x+2)(2x+1)}$ and $\frac{(x-3)(2x+1)}{(x-5)(x+4)}$.
3. Cancel only common factors; the result is $= \frac{(x-3)(x+1)}{(x+2)(x+4)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{(x-3)(x+1)}{(x+2)(x+4)}$$

Worked Solutions

Rational Expressions

Q041 Written

Simplify $\frac{x^2-9}{x^2-x-2} \div \frac{2x^2+5x-3}{x+1}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-3)(x+3)}{(x-2)(x+1)}$ and $\frac{x+1}{(x+3)(2x-1)}$.
3. Cancel only common factors; the result is $= \frac{x-3}{(x-2)(2x-1)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{x-3}{(x-2)(2x-1)}$$

Q042 Written

Simplify $\frac{2xy}{3ab} \div \frac{8x^3y}{9a^2b^3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{2xy}{3ab}$ and $\frac{9a^2b^3}{8x^3y}$.
3. Cancel only common factors; the result is $= \frac{3ab^2}{4x^2}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{3ab^2}{4x^2}$$

Worked Solutions

Rational Expressions

Q043 Written

Simplify $\frac{2x^2-x-10}{x^2-2x-3} \div \frac{x^2+x-2}{x^2-6x+9}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x+2)(2x-5)}{(x-3)(x+1)}$ and $\frac{(x-3)^2}{(x-1)(x+2)}$.
3. Cancel only common factors; the result is $= \frac{(x-3)(2x-5)}{(x-1)(x+1)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{(x-3)(2x-5)}{(x-1)(x+1)}$$

Q044 Written

Simplify $\frac{x^2-3x-10}{2x^2+3x-2} \div \frac{x^2-2x-15}{6x-3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-5)(x+2)}{(x+2)(2x-1)}$ and $\frac{3(2x-1)}{(x-5)(x+3)}$.
3. Cancel only common factors; the result is $= \frac{3}{x+3}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{3}{x+3}$$

Worked Solutions

Rational Expressions

Q045 Written

Simplify $\frac{x^2-y^2}{x^2-2xy+y^2} \div \frac{x^2+xy}{x-y}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-y)(x+y)}{(x-y)^2}$ and $\frac{x-y}{x(x+y)}$.
3. Cancel only common factors; the result is $= \frac{1}{x}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{1}{x}$$

Q046 Written

Simplify $\frac{x^3+8}{x^2-x-6} \times \frac{x^2-6x+9}{x^2-2x+4} \div \frac{2x-6}{x^5-1}$ for $x = \sqrt{123}$.

Solution

1. Factor every polynomial and postpone substitution until the symbolic cancellation is complete.
2. The chain simplifies to $= \frac{(x-1)(x^4+x^3+x^2+x+1)}{2}$.
3. Substitute $x = \sqrt{123}$ only after simplifying: $= -\frac{1}{2} + \frac{15129\sqrt{123}}{2}$.
4. Check all original denominators and divisor-nonzero conditions.

Final answer

$$-\frac{1}{2} + \frac{15129\sqrt{123}}{2}$$

Worked Solutions

Rational Expressions

Q047 Challenge

Simplify $\left(\frac{x+1}{x-2} - \frac{2}{x+3} + \frac{3}{x+3} - \frac{1}{x-2}\right)$; state every excluded value.**Solution**

1. Group like denominators, then combine over the two-factor LCM.
2. The reduced expression is $= \frac{x^2 + 4x - 2}{(x-2)(x+3)}$.
3. The original denominators still control the domain.

Final answer

$$\frac{x^2 + 4x - 2}{(x-2)(x+3)}$$

Worked Solutions

Rational Expressions

Q048 Written

Simplify

$$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= x - 1$.
4. Retain all values excluded by the original denominators.

Final answer

$$x - 1$$

Worked Solutions

Rational Expressions

Q049 MCQ

Simplify

$$\frac{x^2 + 2x + 3}{x + 1} + \frac{2x - 1}{x + 1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x^2 + 4x + 2}{x + 1}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{x^2 + 4x + 2}{x + 1}$$

Worked Solutions

Rational Expressions

Q050 MCQ

Simplify

$$\frac{2x^2 - 5}{x^2 - 4} + \frac{x + 6}{x^2 - 4}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^2 + x + 1}{(x - 2)(x + 2)}$.
5. Keep every value excluded by an original denominator.

Correct option: D**Final answer**

$$\frac{2x^2 + x + 1}{(x - 2)(x + 2)}$$

Worked Solutions

Rational Expressions

Q051 MCQ

Simplify

$$\frac{3x^2 + x - 2}{(x - 2)(x + 1)} + \frac{4x - 3}{(x - 2)(x + 1)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$$

Worked Solutions

Rational Expressions

Q052 MCQ

Simplify

$$\frac{x^3 - 1}{x + 2} + \frac{2x^2 + x + 4}{x + 2};$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x^3 + 2x^2 + x + 3}{x + 2}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{x^3 + 2x^2 + x + 3}{x + 2}$$

Worked Solutions

Rational Expressions

Q053 Written

Simplify

$$\frac{3}{x^2 + x} - \frac{-x + 3}{x^2 + x}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{1}{x + 1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{1}{x + 1}$$

Worked Solutions

Rational Expressions

Q054 MCQ

Simplify

$$\frac{x^2 + 4x + 4}{x + 2} - \frac{x + 4}{x + 2}.$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x(x + 3)}{x + 2}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{x(x + 3)}{x + 2}$$

Worked Solutions

Rational Expressions

Q055 MCQ

Simplify

$$\frac{3x^2 - 2x + 1}{x^2 - 1} - \frac{2x - 5}{x^2 - 1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$$

Worked Solutions

Rational Expressions

Q056 MCQ

Simplify

$$\frac{x^2 - 6x + 8}{(x - 1)(x + 2)} - \frac{x^2 - 3x + 2}{(x - 1)(x + 2)};$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= -\frac{3(x - 2)}{(x - 1)(x + 2)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$-\frac{3(x - 2)}{(x - 1)(x + 2)}$$

Worked Solutions

Rational Expressions

Q057 MCQ

Simplify

$$\frac{2x^3 + x}{x - 3} - \frac{x^2 + 7}{x - 3}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^3 - x^2 + x - 7}{x - 3}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{2x^3 - x^2 + x - 7}{x - 3}$$

Worked Solutions

Rational Expressions

Q058 Written

Simplify

$$\frac{x+4}{x^2-2x} - \frac{3}{x^2-3x+2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x+2}{x(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x+2}{x(x-1)}$$

Q059 Written**Simplify**

$$\frac{x-1}{x^2+4x+3} + \frac{2}{x^2+5x+6}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x}{(x+1)(x+2)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x}{(x+1)(x+2)}$$

Worked Solutions

Rational Expressions

Q060 MCQ

Simplify

$$\frac{x-1}{(x+1)(x+3)} + \frac{2x+1}{(x+3)(x+5)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$$

Worked Solutions

Rational Expressions

Q061 MCQ

Simplify

$$\frac{x+3}{(x+2)(x+4)} + \frac{x-2}{(x+4)(x+6)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$$

Worked Solutions

Rational Expressions

Q062 MCQ

Simplify

$$\frac{3x + 2}{(x + 1)(x + 4)} + \frac{x + 5}{(x + 4)(x + 7)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$.
5. Keep every value excluded by an original denominator.

Correct option: D**Final answer**

$$\frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$$

Worked Solutions

Rational Expressions

Q063 MCQ

Simplify

$$\frac{x+1}{(x+2)(x+5)} + \frac{4x-3}{(x+5)(x+8)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$$

Worked Solutions

Rational Expressions

Q064 MCQ

Simplify

$$\frac{2x + 1}{(x + 3)(x + 6)} + \frac{x + 4}{(x + 6)(x + 9)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$$

Worked Solutions

Rational Expressions

Q065 MCQ

Simplify

$$\frac{x-2}{(x+1)(x+5)} + \frac{3x+1}{(x+5)(x+8)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$$

Worked Solutions

Rational Expressions

Q066 Written

Simplify

$$\frac{x}{x-1} - \frac{2}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x+2}{x+1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x+2}{x+1}$$

Q067 Written**Simplify**

$$\frac{x+2}{x} - \frac{x-2}{x-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3x-2}{x(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3x-2}{x(x-1)}$$

Worked Solutions

Rational Expressions

Q068 **Written****Simplify**

$$\frac{2}{x-1} + \frac{3}{x^2+5x-6}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x+15}{(x-1)(x+6)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x+15}{(x-1)(x+6)}$$

Worked Solutions

Rational Expressions

Q069 MCQ

Simplify

$$\frac{x+2}{(x+1)(x+3)} - \frac{2}{x+1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= -\frac{x+4}{(x+1)(x+3)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$-\frac{x+4}{(x+1)(x+3)}$$

Worked Solutions

Rational Expressions

Q070 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x}{(x-2)(x+2)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x}{(x-2)(x+2)}$$

Q071 Written**Simplify**

$$\frac{x+4}{x^2+2x} - \frac{x+5}{x^2+x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= -\frac{2}{x(x-1)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$-\frac{2}{x(x-1)}$$

Worked Solutions

Rational Expressions

Q072 Written

Simplify

$$\frac{x+8}{x^2+x-2} - \frac{x+5}{x^2-1}$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2}{(x+1)(x+2)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2}{(x+1)(x+2)}$$

Q073 Written**Simplify**

$$\frac{x-3}{x^2+x} + \frac{x}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x^2 - 4x + 3}{x(x-1)(x+1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x^2 - 4x + 3}{x(x-1)(x+1)}$$

Worked Solutions

Rational Expressions

Q074 Written

Simplify

$$\frac{4x}{x^2 - 1} - \frac{x - 1}{x^2 + x}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3x - 1}{x(x - 1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3x - 1}{x(x - 1)}$$

Q075 Written**Simplify**

$$\frac{1}{x^2 + 5x + 6} + \frac{1}{x^2 + 7x + 12}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2}{(x + 2)(x + 4)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2}{(x + 2)(x + 4)}$$

Worked Solutions

Rational Expressions

Q076 Written

Simplify

$$\frac{x-1}{x^2+3x+2} - \frac{x-3}{x^2+4x+3}$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3}{(x+2)(x+3)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3}{(x+2)(x+3)}$$

Worked Solutions

Rational Expressions

Q077 MCQ

A learner combines

 $\frac{2x-1}{(x-1)(x+2)} - \frac{x+4}{x+2}$. Correct the numerator line, then simplify the expression:

Solution

1. Keep one common denominator; never add the denominators.
2. Put the entire second numerator in parentheses before applying the sign.
3. After factoring and reducing, the correct result is $= -\frac{x^2+x-3}{(x-1)(x+2)}$.
4. Check the original denominator restrictions.

Correct option: B

Final answer

$$-\frac{x^2+x-3}{(x-1)(x+2)}$$

Worked Solutions

Rational Expressions

Q078 **Written****Simplify**

$$\frac{x^3 + 8x}{x^2 - x - 2} - \frac{8}{x - 2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x^2 + 2x + 4}{x + 1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x^2 + 2x + 4}{x + 1}$$

Worked Solutions

Rational Expressions

Q079 Challenge

Simplify the single-inner fraction together with the reciprocal-pair quotient; state the full domain.

Solution

1. Identify the two family triggers before manipulating the expression.
2. Simplify each component inside-out, keeping the original restrictions.
3. Combine the reduced components: $= -\frac{x^2 - 4x + 2}{2(x - 2)}$.
4. Audit every original inner and outer denominator.

Final answer

$$-\frac{x^2 - 4x + 2}{2(x - 2)}$$

Worked Solutions

Rational Expressions

Q080 Written

Simplify

$$\frac{1}{1 - \frac{1}{x-1}}.$$

Solution

1. The inner denominator requires $x-1 \neq 0$.
2. Multiply the outer numerator and denominator by $x-1$.
3. The result is $\frac{x-1}{x-2}$.
4. The outer denominator also excludes $x = 2$.

Final answer

$$\frac{x-1}{x-2}$$

Q081 Written**Simplify**

$$\frac{1}{1 - \frac{x}{x+1}}.$$

Solution

1. The inner denominator requires $x+1 \neq 0$.
2. Clear the inner denominator $x+1$ in the outer fraction.
3. The denominator becomes 1, so the result is $x + 1$.
4. Retain $x \neq -1$ from the original expression.

Final answer

$$x + 1$$

Worked Solutions

Rational Expressions

Q082 MCQ

Simplify

$$\frac{1}{1 - \frac{x+2}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 1$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

$$-x - 1$$

Q083 MCQ

Simplify

$$\frac{1}{1 - \frac{x+3}{x+2}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 2$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$-x - 2$$

Worked Solutions

Rational Expressions

Q084 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+1}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+1}{x}$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

$$-\frac{x+1}{x}$$

Q085 MCQ

Simplify

$$\frac{1}{1 - \frac{2x + 3}{x + 1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x + 1}{x + 2}$.
4. State every restriction from the original denominators.

Correct option: C**Final answer**

$$-\frac{x + 1}{x + 2}$$

Worked Solutions

Rational Expressions

Q086 MCQ

Simplify

$$\frac{1}{1 - \frac{3x+1}{x+2}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+2}{2x-1}$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

$$-\frac{x+2}{2x-1}$$

Q087 MCQ

Simplify

$$\frac{1}{1 - \frac{x+4}{x+3}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 3$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$-x - 3$$

Worked Solutions

Rational Expressions

Q088 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+5}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+1}{x+4}$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

$$-\frac{x+1}{x+4}$$

Q089 Written**Simplify**

$$\frac{1}{1-x} + \frac{1}{1+x} \div \left(\frac{1}{1-x} - \frac{1}{1+x} \right).$$

Solution

1. Multiply both outer parts by $(1-x)(1+x)$.
2. The numerator pair becomes 2; the denominator pair becomes $2x$.
3. Reduce to $1/x$.
4. The original inner denominators and the outer denominator give $x \neq -1, 0, 1$.

Final answer

$$1/x$$

Worked Solutions

Rational Expressions

Q090 MCQ

Simplify

$$\frac{1}{x+1} + \frac{1}{x+4} \div \left(\frac{1}{x+1} - \frac{1}{x+4} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= \frac{2x+5}{3}$.
4. State every restriction from the original denominators.

Correct option: A

Final answer

$$\frac{2x+5}{3}$$

Q091 MCQ**Simplify**

$$\frac{1}{x+1} + \frac{1}{x+5} \div \left(\frac{1}{x+1} - \frac{1}{x+5} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= \frac{x+3}{2}$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$\frac{x+3}{2}$$

Worked Solutions

Rational Expressions

Q092 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2} \div \left(\frac{1}{x+2} - \frac{1}{x-2} \right).$$

Solution

1. Multiply numerator and denominator by $(x+2)(x-2)$.
2. The numerator pair becomes $2x$; the denominator pair becomes -4 .
3. The result is $-x/2$.
4. Keep $x \neq -2$ and $x \neq 2$.

Final answer

$$-\frac{x}{2}$$

Q093 MCQ**Simplify**

$$\frac{1}{x+1} + \frac{1}{x+3} \div \left(\frac{1}{x+1} + \frac{1}{x+3} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: C**Final answer**

1

Worked Solutions

Rational Expressions

Q094 MCQ

Simplify

$$\frac{1}{x+2} + \frac{1}{x+5} \div \left(\frac{1}{x+2} + \frac{1}{x+5} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: D

Final answer

1

Q095 MCQ

Simplify

$$\frac{1}{x+3} + \frac{1}{x+6} \div \left(\frac{1}{x+3} + \frac{1}{x+6} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: B**Final answer**

1

Worked Solutions

Rational Expressions

Q096 Written

Simplify

$$\frac{1}{x+y} + \frac{1}{x-y} \div \left(\frac{1}{x+y} - \frac{1}{x-y} \right).$$

Solution

1. Clear both inner denominators using $(x+y)(x-y)$.
2. The numerator pair becomes $2x$; the denominator pair becomes $-2y$.
3. Reduce to $-x/y$.
4. Keep $y \neq 0$ and $x \neq \pm y$.

Final answer

$$-\frac{x}{y}$$

Q097 Written**Simplify**

$$\frac{1}{1 - \frac{1}{1 - \frac{1}{x}}}.$$

Solution

1. Start with the innermost $1 - 1/x$.
2. Substitute that result into the next reciprocal layer.
3. The outer result reduces to $1 - x$.
4. The original denominators exclude $x = 0$ and $x = 1$.

Final answer

$$1 - x$$

Worked Solutions

Rational Expressions

Q098 Written

Simplify

$$\frac{1}{1 - \frac{1}{1 + \frac{1}{x}}}.$$

Solution

1. Simplify $1 + 1/x$ before entering the next layer.
2. Substitute the result into $1 - 1/(...)$.
3. The final result is $x + 1$.
4. Retain $x \neq 0$ and $x \neq -1$.

Final answer

$$x + 1$$

Q099 Written**Simplify**

$$\left(1 + \frac{x-y}{x+y}\right) \div \left(1 - \frac{x-y}{x+y}\right).$$

Solution

1. Let $u = (x - y)/(x + y)$, then combine $1+u$ and $1-u$.
2. Clear $x+y$ in the resulting numerator and denominator.
3. The reduced result is $\frac{x}{y}$.
4. Require $y \neq 0$ and $x + y \neq 0$.

Final answer

$$\frac{x}{y}$$

Worked Solutions

Rational Expressions

Q100 Written

Simplify

$$\left(x - \frac{2}{x-1}\right) \div \left(\frac{x}{x-1} - 2\right).$$

Solution

1. Combine the polynomial pieces over $x-1$ in the numerator and denominator.
2. Form the outer quotient only after both pieces are simplified.
3. Cancel complete factors to obtain $-(x+1)$.
4. The original denominators retain $x \neq 1$ and $x \neq 2$.

Final answer

$$-(x+1)$$

Q101 Written

Simplify

$$\left(\frac{x+1}{x-1} - \frac{x-1}{x+1}\right) \div \left(\frac{x+1}{x-1} + \frac{x-1}{x+1}\right).$$

Solution

1. Use $(x-1)(x+1)$ inside both the numerator and denominator.
2. The difference gives $4x$; the sum gives $2(x^2+1)$.
3. The simplified result is $\frac{2x}{x^2+1}$.
4. The original factors require $x \neq \pm 1$.

Final answer

$$\frac{2x}{x^2+1}$$

Worked Solutions

Rational Expressions

Q102 Written

Simplify

$$\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + x}}}.$$

Solution

1. Translate the three booster stages as an inside-out reciprocal chain.
2. Simplify the innermost stage $1 + 1/(1+x)$.
3. Continuing outward gives $\frac{x+2}{2x+3}$.
4. The stated model condition is $x > 0$.

Final answer

$$\frac{x+2}{2x+3}$$

Quiz MCQ 1 Concept Quiz · MCQ

Simplify $\frac{x^2-4}{3x^2+7x+2}$:

Solution

1. Factor both polynomials and cancel $x+2$.

Correct option: A

Final answer

$$\frac{x-2}{3x+1}$$

Worked Solutions

Rational Expressions

Quiz MCQ 2 [Concept Quiz · MCQ](#)**Simplify**

$$\frac{x+1}{x-2} + \frac{2}{x-2}:$$

Solution

1. The denominators match, so add the numerators only.

Correct option: A

Final answer

$$\frac{x+3}{x-2}$$

Quiz MCQ 3 [Concept Quiz · MCQ](#)**In**

$\frac{3}{x^2+x} - \frac{3-x}{x^2+x}$, the numerator after the minus sign is... Which of the following is correct?:

Solution

1. Parentheses preserve the sign of every term in the second numerator.

Correct option: A

Final answer

$$3 - (3 - x)$$

Worked Solutions

Rational Expressions

Quiz MCQ 4 Concept Quiz · MCQ

Which statement is safe when simplifying a compound fraction?:

Solution

1. A common multiplier preserves the quotient and exposes the structure.

Correct option: A

Final answer

multiply top and bottom by the same inner denominator

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $\frac{x^2-9}{x^2-2x-3} \times \frac{x^2-4}{x^2+x-6}$ and state the original restrictions.

Solution

1. Factor all four polynomials and cancel common factors.
2. Final form: $= \frac{x+2}{x+1}$; retain original denominator exclusions.

Final answer

$$\frac{x+2}{x+1}$$

Worked Solutions

Rational Expressions

Quiz WRITTEN 6 Concept Quiz · Written

Simplify

$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}$ and state the original restriction.

Solution

1. Use the common denominator $x+1$ and combine the numerators.
2. Final form: $= x - 1$; $x \neq -1$ remains excluded.

Final answer

$$x - 1$$

Quiz WRITTEN 7 Concept Quiz · Written

Simplify

$\frac{x^2 - y^2}{x^2 - 2xy + y^2} - \frac{x + y}{x - y}$ in x and y .

Solution

1. Factor $x^2 - y^2$, use the common $(x - y)$ structure, and apply the minus sign to $x + y$.
2. Final form: $= 0$.

Final answer

$$0$$

Worked Solutions

Rational Expressions

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $(x^2 - 16)/(x - 4) = x + 4$ is not an identity for every real x .

Solution

1. The cancellation is valid only when $x - 4 \neq 0$.
2. At $x = 4$ the original expression is undefined, so the graph has a hole.

Final answer

$x \neq 4$; removable hole at $(4, 8)$

Answer key

Use the question number shown on each page.

Q001 A
 Q002 B
 Q003 C
 Q004 D
 Q005 A
 Q006 B
 Q007 $\frac{4b^3}{3a^2}$
 Q008 $\frac{3y^2}{10x}$
 Q009 $\frac{3a}{4b^3}$
 Q010 $\frac{1}{4ab^2}$
 Q011 B
 Q012 A
 Q013 A
 Q014 B
 Q015 A
 Q016 B
 Q017 C
 Q018 D
 Q019 A
 Q020 B
 Q021 C
 Q022 $\frac{3c^2}{a}$
 Q023 $\frac{4a^2c^3}{b^2}$

Q024 C
 Q025 D
 Q026 D
 Q027 A
 Q028 B
 Q029 C
 Q030 $\frac{x-2}{3x+1}$
 Q031 $\frac{2x-5}{x+2}$
 Q032 $\frac{x}{x-3}$
 Q033 $\frac{x+2}{x-4}$
 Q034 $\frac{2(x-y)}{x(x+4y)}$
 Q035 $\frac{x^2+x+1}{x+3}$
 Q036 $\frac{x}{x-3}$
 Q037 $\frac{x-4}{x(x-2)}$
 Q038 $\frac{(x-5)(x+3)}{x(x+4)}$
 Q039 $\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$
 Q040 $\frac{(x-3)(x+1)}{(x+2)(x+4)}$
 Q041 $\frac{x-3}{(x-2)(2x-1)}$
 Q042 $\frac{3ab^2}{4x^2}$

Q043 $\frac{(x-3)(2x-5)}{(x-1)(x+1)}$
 Q044 $\frac{3}{x+3}$
 Q045 $\frac{1}{x}$
 Q046 $-\frac{1}{2} + \frac{15129\sqrt{123}}{2}$
 Q047 $\frac{x^2+4x-2}{(x-2)(x+3)}$
 Q048 $x-1$
 Q049 C
 Q050 D
 Q051 A
 Q052 B
 Q053 $\frac{1}{x+1}$
 Q054 A
 Q055 B
 Q056 C
 Q057 C
 Q058 $\frac{x+2}{x(x-1)}$
 Q059 $\frac{x}{(x+1)(x+2)}$
 Q060 B
 Q061 C
 Q062 D
 Q063 A
 Q064 B
 Q065 C

Q066 $\frac{x+2}{x+1}$
 Q067 $\frac{3x-2}{x(x-1)}$
 Q068 $\frac{2x+15}{(x-1)(x+6)}$
 Q069 C
 Q070 $\frac{2x}{(x-2)(x+2)}$
 Q071 $-\frac{2}{x(x-1)}$
 Q072 $\frac{2}{(x+1)(x+2)}$
 Q073 $\frac{2x^2-4x+3}{x(x-1)(x+1)}$
 Q074 $\frac{3x-1}{x(x-1)}$
 Q075 $\frac{2}{(x+2)(x+4)}$
 Q076 $\frac{3}{(x+2)(x+3)}$
 Q077 B
 Q078 $\frac{x^2+2x+4}{x+1}$
 Q079 $-\frac{x^2-4x+2}{2(x-2)}$
 Q080 $\frac{x-1}{x-2}$
 Q081 $x+1$
 Q082 D
 Q083 A
 Q084 B
 Q085 C

Q086 D

Q087 A

Q088 B

Q089 $1/x$

Q090 A

Q091 A

Q092 $-\frac{x}{2}$

Q093 C

Q094 D

Q095 B

Q096 $-\frac{x}{y}$ Q097 $1 - x$ Q098 $x + 1$ Q099 $\frac{x}{y}$ Q100 $-(x + 1)$ Q101 $\frac{2x}{x^2 + 1}$ Q102 $\frac{x + 2}{2x + 3}$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{x + 2}{x + 1}$ Quiz WRITTEN 6 $x - 1$

Quiz WRITTEN 7 0

Quiz WRITTEN 8 $x \neq 4$; removable hole at (4, 8)

Concept 4 | Identities and Proofs

English Student Edition • Focused formulas and guided practice

Formula opener

Proof route

LHS $\longrightarrow$ RHS

Square

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference

$$a^2 - b^2 = (a - b)(a + b)$$

Worked Solutions

Identities and Proofs

Q001 Challenge

$2x^2 + 3x + 1 = (x + 1)(ax + b) + c$. Find **a, b, c**.

Solution

1. Decide that this is a polynomial identity and expand the product.
2. Match the x^2 , x , and constant coefficients.
3. The constants are $a = 2$, $b = 1$, $c = 0$; substitute back.

Final answer

$$a = 2, b = 1, c = 0$$

Worked Solutions

Identities and Proofs

Q002 Written

$$x^2 + 6 = a(x + 1)^2 + b(x + 1) + c$$

Solution

1. Expand the shifted square: the right side is $ax^2 + (2a+b)x + (a+b+c)$.
2. Match coefficients with $x^2 + 0x + 6$: $a = 1$, $2a + b = 0$, $a + b + c = 6$.
3. Solve in order to obtain $a = 1$, $b = -2$, $c = 7$.
4. Substitution reproduces the original identity.

Final answer

$$a = 1, b = -2, c = 7$$

Q003 Written

$$x^2 - x - 3 = a(x - 1)^2 + b(x - 1) + c$$

Solution

1. Expand and collect the right side as $ax^2 + (-2a+b)x + (a-b+c)$.
2. Match with $x^2 - x - 3$ and solve $a = 1$, $b = 1$, $c = -3$.
3. Check all three coefficients in the original identity.

Final answer

$$a = 1, b = 1, c = -3$$

Worked Solutions

Identities and Proofs

Q004 MCQ

For the identity $x^2 + 6x + 5 = a(x + 2)^2 + b(x + 2) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 1$, $b = 2$, $c = -3$.
4. Substitute back to check.

Correct option: C**Final answer** $(1, 2, -3)$

Q005 MCQ

For the identity $2x^2 + 9x + 8 = a(x + 3)^2 + b(x + 3) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 2$, $b = -3$, $c = -1$.
4. Substitute back to check.

Correct option: C**Final answer** $(2, -3, -1)$

Worked Solutions

Identities and Proofs

Q006 MCQ

For the identity $3x^2 + 28x + 65 = a(x + 4)^2 + b(x + 4) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 3$, $b = 4$, $c = 1$.
4. Substitute back to check.

Correct option: D

Final answer

$(3, 4, 1)$

Q007 MCQ

For the identity $x^2 + 5x + 3 = a(x + 5)^2 + b(x + 5) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 1$, $b = -5$, $c = 3$.
4. Substitute back to check.

Correct option: A

Final answer

$(1, -5, 3)$

Worked Solutions

Identities and Proofs

Q008 Written

$$a(x + 1) - b(x + 5) = 8x$$

Solution

1. Collect coefficients: $(a - b)x + (a - 5b) = 8x$.
2. Solve $a - b = 8$ and $a - 5b = 0$.
3. Hence $a = 10$ and $b = 2$; the constant term checks as zero.

Final answer

$$a = 10, b = 2$$

Q009 MCQ

For the identity $a(x + 1) - b(x + 3) = x - 1$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 2, b = 1$.
4. Check both terms.

Correct option: A

Final answer

$$(2, 1)$$

Worked Solutions

Identities and Proofs

Q010 MCQ

For the identity $a(x + 2) - b(x + 4) = 5x + 14$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 3, b = -2$.
4. Check both terms.

Correct option: B

Final answer

$(3, -2)$

Q011 MCQ

For the identity $a(x + 3) - b(x + 5) = x - 3$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 4, b = 3$.
4. Check both terms.

Correct option: D

Final answer

$(4, 3)$

Worked Solutions

Identities and Proofs

Q012 MCQ

For the identity $a(x + 4) - b(x + 6) = 9x + 44$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 5$, $b = -4$.
4. Check both terms.

Correct option: D

Final answer

$(5, -4)$

Q013 Written

$$(x + a)(b + 3) = 4x + 12$$

Solution

1. Expand as $(b+3)x+a(b+3)$.
2. Match $b + 3 = 4$ and $a(b + 3) = 12$.
3. Therefore $b = 1$ and $a = 3$; c is not present in the printed identity and is not applicable.

Final answer

$a = 3$, $b = 1$

Worked Solutions

Identities and Proofs

Q014 MCQ

For the identity $(x + a)(b + 4) = 3x + 6$, the ordered tuple (a, b) is:

Solution

1. Expand as $(b+d)x+a(b+d)$.
2. Match the coefficient of x and the constant term.
3. Obtain $a = 2$, $b = -1$; no extra parameter is present.
4. Substitute back.

Correct option: B

Final answer $(2, -1)$

Q015 MCQ

For the identity $(x + a)(b + 5) = 5x + 15$, the ordered tuple (a, b) is:

Solution

1. Expand as $(b+d)x+a(b+d)$.
2. Match the coefficient of x and the constant term.
3. Obtain $a = 3$, $b = 0$; no extra parameter is present.
4. Substitute back.

Correct option: C

Final answer $(3, 0)$

Worked Solutions

Identities and Proofs

Q016 MCQ

For the identity $(x + a)(b + 6) = 7x + 28$, the ordered tuple (a, b) is:

Solution

1. Expand as $(b+6)x+a(b+6)$.
2. Match the coefficient of x and the constant term.
3. Obtain $a = 4$, $b = 1$; no extra parameter is present.
4. Substitute back.

Correct option: C

Final answer

$(4, 1)$

Q017 Written

$$2x^2 + 5x + 4 = (x + 1)(ax + b) + c$$

Solution

1. Expand the right side to $ax^2+(a+b)x+(b+c)$.
2. Match $(a, a + b, b + c) = (2, 5, 4)$.
3. Solve $a = 2$, $b = 3$, $c = 1$ and check the constant term.

Final answer

$a = 2$, $b = 3$, $c = 1$

Worked Solutions

Identities and Proofs

Q018 Written

$$(ax + b)(x + 1) = 4x^2 + cx + 1$$

Solution

1. Expand: $ax^2 + (a+b)x + b$.
2. Match the x^2 and constant coefficients to get $a = 4$ and $b = 1$.
3. Then $c = a + b = 5$.

Final answer

$$a = 4, b = 1, c = 5$$

Q019 Written

$$2x^2 - 7x + 8 = (x - 3)(ax + b) + c$$

Solution

1. Expand to $ax^2 + (b-3a)x + (-3b+c)$.
2. Match with $(2, -7, 8)$.
3. Solve $a = 2$, $b = -1$, $c = 5$ and substitute back.

Final answer

$$a = 2, b = -1, c = 5$$

Worked Solutions

Identities and Proofs

Q020 MCQ

For the identity $(ax - 1)(x + 1) + c = x^2 - 3$, the ordered tuple (a, c) is:

Solution

1. Expand the product to $ax^2 + (a+1)x - c$.
2. Match the x^2 , x , and constant coefficients.
3. Use $a = 1$ and $c = -2$, then verify the middle coefficient.
4. Check the full identity.

Correct option: B

Final answer

$(1, -2)$

Q021 MCQ

For the identity $(ax + 2)(x + 2) + c = 2x^2 + 6x + 3$, the ordered tuple (a, c) is:

Solution

1. Expand the product to $ax^2 + (a+2)x + 4 + c$.
2. Match the x^2 , x , and constant coefficients.
3. Use $a = 2$ and $c = -1$, then verify the middle coefficient.
4. Check the full identity.

Correct option: C

Final answer

$(2, -1)$

Worked Solutions

Identities and Proofs

Q022 MCQ

For the identity $(ax + 3)(x + 3) + c = 3x^2 + 12x + 9$, the ordered tuple (a, c) is:

Solution

1. Expand the product to $ax^2 + (ah+b)x + bh + c$.
2. Match the x^2 , x , and constant coefficients.
3. Use $a = 3$ and $c = 0$, then verify the middle coefficient.
4. Check the full identity.

Correct option: B

Final answer

$(3, 0)$

Q023 Written

$$\frac{3x + 1}{(x + 2)(x - 3)} = \frac{a}{x + 2} + \frac{b}{x - 3}$$

Solution

1. Exclude $x = -2$ and $x = 3$ from the original denominator.
2. Multiply by $(x+2)(x-3)$: $3x + 1 = a(x - 3) + b(x + 2)$.
3. Match coefficients: $a + b = 3$ and $-3a + 2b = 1$, giving $a = 1$, $b = 2$.
4. Substitute a and b back into the rational identity.

Final answer

$a = 1$, $b = 2$

Worked Solutions

Identities and Proofs

Q024 Written

$$\frac{x+8}{(x-2)(x+3)} = \frac{a}{x-2} + \frac{b}{x+3}$$

Solution

1. Exclude $x = 2$ and $x = -3$.
2. Clear denominators: $x + 8 = a(x + 3) + b(x - 2)$.
3. Match $a + b = 1$ and $3a - 2b = 8$, giving $a = 2$, $b = -1$.
4. Substitute and verify.

Final answer

$$a = 2, b = -1$$

Q025 Written

$$\frac{1}{(2x-1)(x+1)} = \frac{a}{2x-1} + \frac{b}{x+1}$$

Solution

1. Exclude $x = 1/2$ and $x = -1$.
2. Clear denominators: $1 = a(x + 1) + b(2x - 1)$.
3. Match $a + 2b = 0$ and $a - b = 1$, giving $a = 2/3$, $b = -1/3$.
4. Verify in the original decomposition.

Final answer

$$a = \frac{2}{3}, b = -\frac{1}{3}$$

Worked Solutions

Identities and Proofs

Q026 Written

$$\frac{a}{x-1} + \frac{b}{x-5} = \frac{5x-1}{(x-1)(x-5)}$$

Solution

1. Exclude $x = 1$ and $x = 5$.
2. Clear denominators: $a(x-5) + b(x-1) = 5x-1$.
3. Match $a+b=5$ and $-5a-b=-1$, giving $a=-1$, $b=6$.
4. Check the original rational identity.

Final answer

$$a = -1, b = 6$$

Q027 Written

$$\frac{3x-8}{3x^2+5x-2} = \frac{a}{x+2} + \frac{b}{3x-1}$$

Solution

1. Factor $3x^2 + 5x - 2 = (x+2)(3x-1)$.
2. Exclude $x = -2$ and $x = 1/3$, then clear denominators.
3. Match $3a+b=3$ and $-a+2b=-8$, giving $a=2$, $b=-3$.
4. Check the factored rational identity.

Final answer

$$a = 2, b = -3$$

Worked Solutions

Identities and Proofs

Q028 Written

$$f(t) = \frac{22t + 15}{8t^2 + 10t + 3} = \frac{a}{2t + 1} + \frac{b}{4t + 3}, \quad t > 0$$

Solution

1. Factor ($8t^2 + 10t + 3 = (2t + 1)(4t + 3)$).
2. The stated domain is ($t > 0$); the algebraic exclusions lie outside it.
3. Clear denominators: ($22t + 15 = a(4t + 3) + b(2t + 1)$).
4. Match ($4a + 2b = 22$) and ($3a + b = 15$) to get ($a = 4, b = 3$).

Final answer

$$a = 4, \quad b = 3$$

Q029 Written

Prove that $(x + y)^2 - (x - y)^2 = 4xy$.**Solution**

1. Expand both conjugate squares on the left.
2. Remove the parentheses: $x^2 + 2xy + y^2 - x^2 + 2xy - y^2$.
3. Collect like terms to obtain $4xy$, which is the right-hand side.

Final answer

$$4xy$$

Worked Solutions

Identities and Proofs

Q030 Written

Prove that $(2a + b)^2 - (a + 2b)^2 = 3(a^2 - b^2)$.**Solution**

1. Recognise a difference of squares: $U^2 - V^2 = (U - V)(U + V)$.
2. Set $U = 2a + b$ and $V = a + 2b$; then $U - V = a - b$ and $U + V = 3a + 3b$.
3. Multiply to get $(a - b)(3a + 3b) = 3(a^2 - b^2)$.

Final answer

$$3a^2 - 3b^2$$

Q031 MCQ

For the identity $-(x - 2y + 1)^2 + (x + 2y + 1)^2 = 2y(4x + 4)$, **the value of the simplified left-hand side is:****Solution**

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy + 8y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: A**Final answer**

$$8xy + 8y$$

Worked Solutions

Identities and Proofs

Q032 MCQ

For the identity $-(x - 3y + 2)^2 + (x + 3y + 2)^2 = 3y(4x + 8)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $12xy + 24y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D**Final answer** $12xy + 24y$

Q033 MCQ

For the identity $-(x - y + 3)^2 + (x + y + 3)^2 = y(4x + 12)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $4xy + 12y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D**Final answer** $4xy + 12y$

Worked Solutions

Identities and Proofs

Q034 MCQ

For the identity $-(x - 2y + 4)^2 + (x + 2y + 4)^2 = 2y(4x + 16)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy + 32y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: A

Final answer

$$8xy + 32y$$

Q035 Written

Prove that $(x + y)^2 + (x - y)^2 = 2x^2 + 2y^2$.

Solution

1. Expand both squares on the left.
2. Add $x^2 + 2xy + y^2$ and $x^2 - 2xy + y^2$.
3. The mixed terms cancel, leaving $2x^2 + 2y^2$.

Final answer

$$2x^2 + 2y^2$$

Worked Solutions

Identities and Proofs

Q036 MCQ

For the identity $(x - 2y + 1)^2 + (x + 2y + 1)^2 = 8y^2 + 2(x + 1)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 4x + 8y^2 + 2$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C**Final answer**

$$2x^2 + 4x + 8y^2 + 2$$

Q037 MCQ

For the identity $(x - 3y + 2)^2 + (x + 3y + 2)^2 = 18y^2 + 2(x + 2)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 8x + 18y^2 + 8$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C**Final answer**

$$2x^2 + 8x + 18y^2 + 8$$

Worked Solutions

Identities and Proofs

Q038 MCQ

For the identity $(x - y + 3)^2 + (x + y + 3)^2 = 2y^2 + 2(x + 3)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 12x + 2y^2 + 18$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C**Final answer**

$$2x^2 + 12x + 2y^2 + 18$$

Q039 MCQ

For the identity $(x - 2y + 4)^2 + (x + 2y + 4)^2 = 8y^2 + 2(x + 4)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 16x + 8y^2 + 32$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D**Final answer**

$$2x^2 + 16x + 8y^2 + 32$$

Worked Solutions

Identities and Proofs

Q040 Written

Prove that $(a + 2b)^2 - 4ab = (a - 2b)^2 + 4ab$.**Solution**

1. Expand the left-hand side: $a^2 + 4ab + 4b^2 - 4ab = a^2 + 4b^2$.
2. Expand the right-hand side: $a^2 - 4ab + 4b^2 + 4ab = a^2 + 4b^2$.
3. Both sides simplify to the same expression.

Final answer

$$a^2 + 4b^2$$

Q041 MCQ

For the identity $-(x - 2y + 3)^2 + (x + 2y - 1)^2 = (8x + 8)(y - 1)$, **the value of the simplified left-hand side is:****Solution**

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy - 8x + 8y - 8$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D**Final answer**

$$8xy - 8x + 8y - 8$$

Worked Solutions

Identities and Proofs

Q042 MCQ

For the identity $-(2x - 3y + 5)^2 + (2x + 3y - 1)^2 = (24x + 24)(y - 1)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $24xy - 24x + 24y - 24$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$$24xy - 24x + 24y - 24$$

Q043 Written

Prove that $(x^2 + x + 1)(x^2 - x + 1) = x^4 + x^2 + 1$.

Solution

1. Group the factors as $[(x^2 + 1) - x][(x^2 + 1) + x]$.
2. Use $(U - V)(U + V) = U^2 - V^2$ with $U = x^2 + 1$ and $V = x$.
3. Simplify $(x^2 + 1)^2 - x^2$ to $x^4 + x^2 + 1$.

Final answer

$$x^4 + x^2 + 1$$

Worked Solutions

Identities and Proofs

Q044 Written

Prove that $(a^2 - ab + b^2)(a^2 + ab + b^2) = (a^2 + b^2)^2 - (ab)^2$.

Solution

1. Group the factors as $[(a^2 + b^2) - ab][(a^2 + b^2) + ab]$.
2. Apply $(U - V)(U + V) = U^2 - V^2$ with $U = a^2 + b^2$ and $V = ab$.
3. This gives $(a^2 + b^2)^2 - (ab)^2$, exactly the right-hand side.

Final answer

$$a^4 + a^2b^2 + b^4$$

Q045 Written

Prove that $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$.

Solution

1. Expand the left-hand product to $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$.
2. Expand the first right-hand square: $a^2c^2 - 2abcd + b^2d^2$.
3. Expand the second right-hand square: $a^2d^2 + 2abcd + b^2c^2$.
4. Add the two expansions; the $-2abcd$ and $+2abcd$ terms cancel, leaving the left-hand expression.

Final answer

$$a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$$

Worked Solutions

Identities and Proofs

Q046 Written

Prove that $(x^2 + 1)(y^2 + 1) = (xy + 1)^2 + (x - y)^2$.

Solution

1. Expand the left-hand side to $x^2y^2 + x^2 + y^2 + 1$.
2. Expand the right-hand side: $x^2y^2 + 2xy + 1 + x^2 - 2xy + y^2$.
3. The $+2xy$ and $-2xy$ terms cancel, leaving the same expression as the left.

Final answer

$$x^2y^2 + x^2 + y^2 + 1$$

Q047 Written

Prove that $(a^2 - b^2)(c^2 - d^2) = (ac - bd)^2 - (ad - bc)^2$.

Solution

1. Expand the first right-hand square: $a^2c^2 - 2abcd + b^2d^2$.
2. Expand the second right-hand square: $a^2d^2 - 2abcd + b^2c^2$.
3. Subtract; the mixed terms cancel, giving $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$.
4. Factor the result as $(a^2 - b^2)(c^2 - d^2)$.

Final answer

$$a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$$

Worked Solutions

Identities and Proofs

Q048 Written

Prove that $(a + b)^3 - 3ab(a + b) = a^3 + b^3$.**Solution**

1. Expand $(a + b)^3$ to $a^3 + 3a^2b + 3ab^2 + b^3$.
2. Expand $3ab(a+b)$ to $3a^2b + 3ab^2$.
3. Subtract the matching cross terms; the result is $a^3 + b^3$.

Final answer

$$a^3 + b^3$$

Q049 Written

Prove that $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = a^3 + b^3 + c^3 - 3abc$.**Solution**

1. Distribute $a+b+c$ across the second bracket.
2. Write all products in a fixed order: a^3, b^3, c^3 , then the mixed terms.
3. Each mixed term cancels with its opposite, while $-abc$ appears three times.
4. The remaining expression is $a^3 + b^3 + c^3 - 3abc$.

Final answer

$$a^3 - 3abc + b^3 + c^3$$

Worked Solutions

Identities and Proofs

Q050 Challenge

For the integrated identity $(x^2 + 1)(-(x - 2y + 1)^2 + (x + 2y + 1)^2) = 2y(4x + 4)(x^2 + 1)$, the value of the simplified left-hand side is:

Solution

1. Recognise the cubic or symmetric pattern before expanding.
2. Expand in descending degree and collect like terms.
3. The cross terms cancel, leaving $8x^3y + 8x^2y + 8xy + 8y$.
4. Therefore the proposed equality is proved.

Correct option: B

Final answer

$$8x^3y + 8x^2y + 8xy + 8y$$

Worked Solutions

Identities and Proofs

Q051 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 + 1 = 2(x + y - xy)$.

Solution

1. Use $x + y = 1$, so $(x + y)^2 = 1$.
2. Expand: $x^2 + 2xy + y^2 = 1$, hence $x^2 + y^2 + 1 = 2 - 2xy$.
3. Since $x + y = 1$, $2 - 2xy = 2(x + y - xy)$. Therefore the equality holds.

Final answer

$$x^2 + y^2 + 1 = 2(x + y - xy)$$

Q052 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 = 1 - 2xy$.

Solution

1. Square the condition: $(x + y)^2 = 1$.
2. Expand to $x^2 + 2xy + y^2 = 1$.
3. Rearrange to obtain $x^2 + y^2 = 1 - 2xy$.

Final answer

$$x^2 + y^2 = 1 - 2xy$$

Worked Solutions

Identities and Proofs

Q053 MCQ

Under the condition $x + y = 2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 2$.
2. $y = 2 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$$-2xy + 4$$

Q054 MCQ

Under the condition $x + y = 3$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 3$.
2. $y = 3 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 9$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: D

Final answer

$$-2xy + 9$$

Worked Solutions

Identities and Proofs

Q055 MCQ

Under the condition $x + y = -1$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = -1$.
2. $y = -1 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 1$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$$-2xy + 1$$

Q056 MCQ

Under the condition $x + y = 4$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 4$.
2. $y = 4 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 16$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: A

Final answer

$$-2xy + 16$$

Worked Solutions

Identities and Proofs

Q057 Written

Under the condition $x - y = 1$, prove that $x^2 + y^2 = 2xy + 1$.

Solution

1. Square the condition: $(x - y)^2 = 1$.
2. Expand to $x^2 - 2xy + y^2 = 1$.
3. Rearrange to obtain $x^2 + y^2 = 2xy + 1$.

Final answer

$$x^2 + y^2 = 2xy + 1$$

Q058 MCQ

Under the condition $x - y = 1$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 1$.
2. $x = y + 1$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 1$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: B**Final answer**

$$2xy + 1$$

Worked Solutions

Identities and Proofs

Q059 MCQ

Under the condition $x - y = 2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 2$.
2. $x = y + 2$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: B

Final answer

$$2xy + 4$$

Q060 MCQ

Under the condition $x - y = -2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = -2$.
2. $x = y - 2$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$$2xy + 4$$

Worked Solutions

Identities and Proofs

Q061 MCQ

Under the condition $x - y = 3$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 3$.
2. $x = y + 3$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 9$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: D

Final answer

$$2xy + 9$$

Q062 Written

Under the condition $x + y = 1$, prove that $x^2 - x = y^2 - y$.

Solution

1. Move the right-hand side to the left: $x^2 - y^2 - x + y$.
2. Factor: $x^2 - y^2 - x + y = (x - y)(x + y - 1)$.
3. The condition makes $x + y - 1 = 0$, so the difference is zero.

Final answer

$$x^2 - x = y^2 - y$$

Worked Solutions

Identities and Proofs

Q063 Written

Under the condition $a + b + c = 0$, prove that $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.

Solution

1. From $a + b + c = 0$, write $c = -a - b$.
2. Substitute into the left-hand side: $a^2 + b^2 + (-a - b)^2 = 2a^2 + 2ab + 2b^2$.
3. Substitute into the right-hand side: $-2[ab + b(-a - b) + (-a - b)a] = 2a^2 + 2ab + 2b^2$.
4. Both sides reduce to the same expression, so the equality is proved.

Final answer

$$a^2 + b^2 + c^2 = -2(ab + bc + ca)$$

Q064 Written

Under the condition $a + b + c = 0$, prove that $a^3 + b^3 + c^3 = 3abc$.

Solution

1. Use the identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.
2. The condition gives $a + b + c = 0$, so the product on the right is zero.
3. Therefore $a^3 + b^3 + c^3 - 3abc = 0$, which proves $a^3 + b^3 + c^3 = 3abc$.

Final answer

$$a^3 + b^3 + c^3 = 3abc$$

Worked Solutions

Identities and Proofs

Q065 Written

Under the condition $a + b + c = 0$, prove that $a^2 + ca = b^2 + bc$.

Solution

1. Subtract the right-hand side: $a^2 + ca - b^2 - bc$.
2. Factor by grouping: $(a - b)(a + b) + (a - b)c = (a - b)(a + b + c)$.
3. Since $a + b + c = 0$, the difference is zero; hence the two sides are equal.

Final answer

$$a^2 + ca = b^2 + bc$$

Q066 Written

Under the condition $a + b + c = 0$, prove that $(a + b)(b + c)(c + a) + abc = 0$.

Solution

1. From $a + b + c = 0$, write $a + b = -c$, $b + c = -a$, and $c + a = -b$.
2. Then $(a + b)(b + c)(c + a) = (-c)(-a)(-b) = -abc$.
3. Adding abc gives zero, so the required equality is proved.

Final answer

$$(a + b)(b + c)(c + a) + abc = 0$$

Worked Solutions

Identities and Proofs

Q067 Written

Under the condition $a + b + c = 0$, $abc \neq 0$, prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$.

Solution

1. Because $a + b + c = 0$, $b + c = -a$, $c + a = -b$, and $a + b = -c$.
2. The non-zero condition ensures that all three denominators are non-zero.
3. Therefore each fraction equals -1: $a/(b + c) = b/(c + a) = c/(a + b) = -1$.
4. Their sum is $-1 - 1 - 1 = -3$.

Final answer

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$$

Worked Solutions

Identities and Proofs

Q068 Challenge

Under the condition $x + y = 3$, the expression $x^2 + y^2 + 1$ is equal to:

Solution

1. Use $x + y = 3$ to replace each cyclic denominator.
2. The non-zero condition keeps every original denominator defined.
3. Each term simplifies and the total becomes $-2xy + 10$.
4. Thus the required constant sum is proved.

Correct option: C

Final answer

$$-2xy + 10$$

Worked Solutions

Identities and Proofs

Q069 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$.

Solution

1. Let $a/b = c/d = k$, so $a = bk$ and $c = dk$.
2. Substitute into the left-hand side: $ab/(a^2+b^2) = k/(k^2+1)$.
3. Substitute into the right-hand side: $cd/(c^2+d^2) = k/(k^2+1)$.
4. The two sides are equal.

Final answer

$$\frac{ab}{a^2 + b^2} = \frac{cd}{c^2 + d^2}$$

Q070 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q071 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Q072 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q073 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q074 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q075 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q076 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$.

Solution

1. Let $a = bk$ and $c = dk$.
2. The left-hand side becomes $k(b^2 + d^2)/(b^2 - d^2)$.
3. The right-hand side becomes $k^2(b^2 + d^2)/[k^2(b^2 - d^2)]$, which is the same expression.
4. Hence the equality is proved.

Final answer

$$\frac{ab + cd}{ab - cd} = \frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q077 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q078 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q079 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q080 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q081 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Solution

1. Write $a = bk$ and $c = dk$.
2. Then $(a + b)/(a - b) = (bk + b)/(bk - b) = (k + 1)/(k - 1)$.
3. Similarly $(c + d)/(c - d) = (dk + d)/(dk - d) = (k + 1)/(k - 1)$.
4. The equality follows.

Final answer

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Q082 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: B

Final answer

$$\frac{c+d}{c-d}$$

Worked Solutions

Identities and Proofs

Q083 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: B

Final answer

$$\frac{c+d}{c-d}$$

Q084 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{c+d}{c-d}$$

Worked Solutions

Identities and Proofs

Q085 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{c+d}{c-d}$$

Q086 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$.

Solution

1. Substitute $a = bk$ and $c = dk$.
2. The left-hand side becomes $(3k + k)/(3 + 1) = k$.
3. The right-hand side becomes $(2k - k)/(2 - 1) = k$.
4. Thus both sides are equal.

Final answer

$$\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$$

Worked Solutions

Identities and Proofs

Q087 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{2a-c}{2b-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{2a - c}{2b - d}$$

Q088 Written

Under the condition $x/a = y/b$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b}, \quad \frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$.

Solution

1. Let $x = ak$ and $y = bk$.
2. The left-hand side is $(a^2 + a^2k^2)/(a^2 - a^2k^2) = (1 + k^2)/(1 - k^2)$.
3. The right-hand side reduces to the same value.
4. Hence the equality is proved.

Final answer

$$\frac{a^2 + x^2}{a^2 - x^2} = \frac{b^2 + y^2}{b^2 - y^2}$$

Worked Solutions

Identities and Proofs

Q089 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{a+b}{a-b} = \frac{c+d}{c-d}.$

Solution

1. Use the equal concentration condition to write $a = bk$ and $c = dk$.
2. The first ratio becomes $(k+1)/(k-1)$.
3. The second ratio becomes $(k+1)/(k-1)$.
4. The required equality follows, provided the denominators are non-zero.

Final answer

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Q090 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}.$

Solution

1. Let the common ratio be k , so $x = ak$, $y = bk$ and $z = ck$.
2. Then $x^2 + y^2 + z^2 = k^2(a^2 + b^2 + c^2)$.
3. Also $xy + yz + zx = k^2(ab + bc + ca)$.
4. Both ratios therefore equal k^2 .

Final answer

$$\frac{x^2 + y^2 + z^2}{a^2 + b^2 + c^2} = \frac{xy + yz + zx}{ab + bc + ca}$$

Worked Solutions

Identities and Proofs

Q091 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x+y+z}{a+b+c} = \frac{x}{a}.$

Solution

1. Let the common ratio be k .
2. Then $x + y + z = k(a + b + c)$.
3. Therefore $(x + y + z)/(a + b + c) = k = x/a$.

Final answer

$$\frac{x + y + z}{a + b + c} = \frac{x}{a}$$

Q092 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}.$

Solution

1. Set $x = ak$, $y = bk$ and $z = ck$.
2. The left numerator is $k(a+2b+3c)$, so the left ratio is k .
3. The right ratio is also k because $x + y + z = k(a + b + c)$.
4. The two ratios are equal.

Final answer

$$\frac{x + 2y + 3z}{a + 2b + 3c} = \frac{x + y + z}{a + b + c}$$

Worked Solutions

Identities and Proofs

Q093 Written

Under the condition $x : y : z = 2 : 3 : 5$, prove or find the required expression: $x : y : z = 2 : 3 : 5$, $\frac{2xy+yz+3zx}{x^2+y^2+z^2}$.

Solution

1. Let $x = 2k$, $y = 3k$ and $z = 5k$.
2. The numerator is $(12 + 15 + 30)k^2 = 57k^2$.
3. The denominator is $(4 + 9 + 25)k^2 = 38k^2$.
4. Therefore the value is $57/38$.

Final answer

$$\frac{57}{38}$$

Q094 Written

Under the condition $a : b : c = 2 : 3 : 4$, prove or find the required expression: $a : b : c = 2 : 3 : 4$, $\frac{ab+bc+ca}{a^2+b^2+c^2}$.

Solution

1. Let $a = 2k$, $b = 3k$ and $c = 4k$.
2. The numerator is $(6 + 12 + 8)k^2 = 26k^2$.
3. The denominator is $(4 + 9 + 16)k^2 = 29k^2$.
4. Therefore the value is $26/29$.

Final answer

$$\frac{26}{29}$$

Worked Solutions

Identities and Proofs

Q095 Written

Under the condition $a : b : c = 3 : 5 : 2$, prove or find the required expression: $a : b : c = 3 : 5 : 2$, $\frac{a^2+b^2-c^2}{a^2-b^2+c^2}$.

Solution

1. Let $a = 3k$, $b = 5k$ and $c = 2k$.
2. The numerator is $(9 + 25 - 4)k^2 = 30k^2$.
3. The denominator is $(9 - 25 + 4)k^2 = -12k^2$.
4. Therefore the value is $-5/2$.

Final answer

$$-\frac{5}{2}$$

Q096 Written

Under the condition $x : y : z = 2 : -3 : 4$, prove or find the required expression: $x : y : z = 2 : -3 : 4$, $\frac{x^2+y^2+z^2}{xy+yz+zx}$.

Solution

1. Let $x = 2k$, $y = -3k$ and $z = 4k$.
2. The numerator is $(4 + 9 + 16)k^2 = 29k^2$.
3. The denominator is $(-6 - 12 + 8)k^2 = -10k^2$.
4. Therefore the value is $-29/10$.

Final answer

$$-\frac{29}{10}$$

Worked Solutions

Identities and Proofs

Q097 Written

Under the condition $x/7 = y/3 \neq 0$, prove or find the required expression: $\frac{x}{7} = \frac{y}{3} \neq 0$, $\frac{x^2+5y^2}{x^2+xy+y^2}$.

Solution

1. Let $x = 7k$ and $y = 3k$, where k is non-zero.
2. The numerator is $(49 + 45)k^2 = 94k^2$.
3. The denominator is $(49 + 21 + 9)k^2 = 79k^2$.
4. Therefore the value is $94/79$.

Final answer

$$\frac{94}{79}$$

Q098 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Solution

1. Let $x = ak$, $y = bk$ and $z = ck$.
2. Then $x^2 + y^2 + z^2 = k^2(a^2 + b^2 + c^2)$.
3. Also $ax + by + cz = k(a^2 + b^2 + c^2)$.
4. Both sides equal $k^2(a^2 + b^2 + c^2)^2$.

Final answer

$$(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$$

Worked Solutions

Identities and Proofs

Quiz MCQ 1 **Concept Quiz · MCQ**

If $x^2 + 4x + 1 = (x + 1)(ax + b) + c$, then (a, b, c) is... Which of the following is correct?:

Solution

1. Expand to $ax^2 + (a + b)x + b + c$.
2. Match $a = 1$, $a + b = 4$, $b + c = 1$, so $b = 3$, $c = -2$.

Correct option: A

Final answer

$(1, 3, -2)$

Quiz MCQ 2 **Concept Quiz · MCQ**

For the identity $(x^2 - 2x + 2)(x^2 + 2x + 2) = -4x^2 + (x^2 + 2)^2$, the value of the simplified left-hand side is:

Solution

1. Rewrite the factors as $(U-V)(U+V)$.
2. Apply $U^2 - V^2$ before carrying out any long expansion.
3. Simplify to $x^4 + 4$.
4. This is exactly the stated right-hand side.

Correct option: A

Final answer

$x^4 + 4$

Worked Solutions

Identities and Proofs

Quiz MCQ 3 Concept Quiz · MCQ

Under the condition $a + b + c = 0$, the expression $3a^2 + 3ac$ is equal to:

Solution

1. Move the right-hand side to the left and factor the difference.
2. The factor containing the condition $a + b + c = 0$ is zero.
3. The remaining product is therefore 0, giving $3b^2 + 3bc$.
4. Hence the two sides are equal under the given condition.

Correct option: A

Final answer

$$3b^2 + 3bc$$

Quiz MCQ 4 Concept Quiz · MCQ

Under the condition $x : y : z = 2 : 3 : 5$, the required expression is equal to:

Solution

1. Write the variables as the stated multiples of k.
2. Substitute into the numerator and denominator.
3. Cancel the common factor and simplify to $\frac{3}{2}$.
4. Hence the required value is obtained.

Correct option: B

Final answer

$$\frac{3}{2}$$

Worked Solutions

Identities and Proofs

Quiz WRITTEN 5 Concept Quiz · Written

Find a, b, c if $x^2 - 3x + 5 = (x - 2)(ax + b) + c$.

Solution

1. Expand to $ax^2 + (b - 2a)x - 2b + c$.
2. Match $a = 1$, $b - 2 = -3$, $-2b + c = 5$.
3. Therefore $b = -1$, $c = 3$.

Final answer

$$a = 1, b = -1, c = 3$$

Quiz WRITTEN 6 Concept Quiz · Written

Prove that $(x^2 - 3x + 4)(x^2 + 3x + 4) = -9x^2 + (x^2 + 4)^2$.

Solution

1. Rewrite the factors as $(U - V)(U + V)$.
2. Apply $U^2 - V^2$ before carrying out any long expansion.
3. Simplify to $x^4 - x^2 + 16$.
4. This is exactly the stated right-hand side.

Final answer

$$x^4 - x^2 + 16$$

Worked Solutions

Identities and Proofs

Quiz WRITTEN 7 Concept Quiz · Written

Under the condition $a + b + c = 0$, prove that $4a^2 + 4ac = 4b^2 + 4bc$.

Solution

1. Move the right-hand side to the left and factor the difference.
2. The factor containing the condition $a + b + c = 0$ is zero.
3. The remaining product is therefore 0, giving $4b^2 + 4bc$.
4. Hence the two sides are equal under the given condition.

Final answer

$$4b^2 + 4bc$$

Quiz WRITTEN 8 Concept Quiz · Written

Under the condition $x/a = y/b = z/c$, prove that $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Solution

1. Use the common multiplier from $x/a = y/b = z/c$.
2. Substitute and collect the common factor.
3. Both sides simplify to $(ax + by + cz)^2$.
4. Thus the identity is proved.

Final answer

$$(ax + by + cz)^2$$

Answer key

Use the question number shown on each page.

Q001 $a = 2, b = 1, c = 0$

Q002 $a = 1, b = -2, c = 7$

Q003 $a = 1, b = 1, c = -3$

Q004 C

Q005 C

Q006 D

Q007 A

Q008 $a = 10, b = 2$

Q009 A

Q010 B

Q011 D

Q012 D

Q013 $a = 3, b = 1$

Q014 B

Q015 C

Q016 C

Q017 $a = 2, b = 3, c = 1$

Q018 $a = 4, b = 1, c = 5$

Q019 $a = 2, b = -1, c = 5$

Q020 B

Q021 C

Q022 B

Q023 $a = 1, b = 2$

Q024 $a = 2, b = -1$

Q025 $a = \frac{2}{3}, b = -\frac{1}{3}$

Q026 $a = -1, b = 6$

Q027 $a = 2, b = -3$

Q028 $a = 4, b = 3$

Q029 $4xy$

Q030 $3a^2 - 3b^2$

Q031 A

Q032 D

Q033 D

Q034 A

Q035 $2x^2 + 2y^2$

Q036 C

Q037 C

Q038 C

Q039 D

Q040 $a^2 + 4b^2$

Q041 D

Q042 C

Q043 $x^4 + x^2 + 1$

Q044 $a^4 + a^2b^2 + b^4$

Q045 $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

Q046 $x^2y^2 + x^2 + y^2 + 1$

Q047 $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$

Q048 $a^3 + b^3$

Q049 $a^3 - 3abc + b^3 + c^3$

Q050 B

Q051 $x^2 + y^2 + 1 = 2(x + y - xy)$

Q052 $x^2 + y^2 = 1 - 2xy$

Q053 C

Q054 D

Q055 C

Q056 A

Q057 $x^2 + y^2 = 2xy + 1$

Q058 B

Q059 B

Q060 C

Q061 D

Q062 $x^2 - x = y^2 - y$

Q063 $a^2 + b^2 + c^2 = -2(ab + bc + ca)$

Q064 $a^3 + b^3 + c^3 = 3abc$

Q065 $a^2 + ca = b^2 + bc$

Q066 $(a + b)(b + c)(c + a) + abc = 0$

Q067 $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$

Q068 C

Q069 $\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$

Q070 C

Q071 D

Q072 C

Q073 D

Q074 D

Q075 C

Q076 $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$

Q077 A

Q078 D

Q079 D

Q080 A

Q081 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q082 B

Q083 B

Q084 A

Q085 A

Q086 $\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$

Q087 A

Q088 $\frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$

Q089 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q090 $\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$

Q091 $\frac{x+y+z}{a+b+c} = \frac{x}{a}$

Q092 $\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$

Q093 $\frac{57}{38}$

Q094 $\frac{26}{29}$

Q095 $-\frac{5}{2}$

Q096 $-\frac{29}{10}$

Q097 $\frac{94}{79}$

Q098 $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$

Quiz MCQ 1 A**Quiz MCQ 2** A**Quiz MCQ 3** A**Quiz MCQ 4** B**Quiz WRITTEN 5** $a = 1, b = -1, c = 3$ **Quiz WRITTEN 6** $x^4 - x^2 + 16$ **Quiz WRITTEN 7** $4b^2 + 4bc$ **Quiz WRITTEN 8** $(ax + by + cz)^2$

Concept 5 | Inequalities

English Student Edition • Focused formulas and guided practice

Formula opener

Square rule

$$(x - y)^2 \geq 0$$

Order

$$0 < a < b \Rightarrow a^2 < b^2$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Worked Solutions

Inequalities

Q001 Written

Under the condition $a > b$, $x > y$, prove that $2(ax + by) > (a + b)(x + y)$.

Solution

1. Start with LHS-RHS: $2(ax+by)-(a+b)(x+y)$.
2. Expand and factor: $ax-ay-bx+by=(a-b)(x-y)$.
3. Both factors are positive, so $(a-b)(x-y)>0$; hence the required inequality holds.

Final answer

$$2(ax + by) > (a + b)(x + y)$$

Q002 MCQ

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + y > 2(x + y)$.

Correct option: A

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q003 MCQ

If $x > y$, prove that $3x + 2y > 2x + 3y$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 2y > 2x + 3y$.

Correct option: C

Final answer

$$x - y$$

Q004 MCQ

If $x > y$, prove that $3(x + y) > 2(x + 2y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3(x + y) > 2(x + 2y)$.

Correct option: D

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q005 MCQ

If $x > y$, prove that $3x + 4y > 2x + 5y$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 4y > 2x + 5y$.

Correct option: D

Final answer

$$x - y$$

Q006 MCQ

If $x > y$, prove that $3x + 5y > 2(x + 3y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 5y > 2(x + 3y)$.

Correct option: B

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q007 Written

Under the condition $a > b$, $c > d$, prove that $ac + bd > ad + bc$.

Solution

1. Subtract the right-hand side: $ac + bd - ad - bc$.
2. Factor: $a(c - d) - b(c - d) = (a - b)(c - d)$.
3. Both factors are positive, so the difference is positive; therefore $ac + bd > ad + bc$.

Final answer

$$ac + bd > ad + bc$$

Q008 Written

Under the condition $a > c$, $b > d$, prove that $ab + cd > ad + bc$.

Solution

1. Subtract the right-hand side: $ab + cd - ad - bc$.
2. Factor: $a(b - d) - c(b - d) = (a - c)(b - d)$.
3. Both factors are positive, so the difference is positive; hence $ab + cd > ad + bc$.

Final answer

$$ab + cd > ad + bc$$

Worked Solutions

Inequalities

Q009 MCQ

If $a > b$ and $c > d$, prove that $ac + bd > ad + bc$.

If $a > b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - b)(c - d)$.
2. Factor the difference as $(a - b)(c - d)$.
3. The condition $a > b$ and $c > d$ makes each factor positive.
4. Hence $ac + bd > ad + bc$.

Correct option: D

Final answer

$$(a - b)(c - d)$$

Q010 MCQ

If $a > 2b$ and $c > d$, prove that $ac + 2bd > ad + 2bc$.

If $a > 2b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - 2b)(c - d)$.
2. Factor the difference as $(a - 2b)(c - d)$.
3. The condition $a > 2b$ and $c > d$ makes each factor positive.
4. Hence $ac + 2bd > ad + 2bc$.

Correct option: C

Final answer

$$(a - 2b)(c - d)$$

Worked Solutions

Inequalities

Q011 MCQ

If $a > 3b$ and $c > d$, prove that $ac + 3bd > ad + 3bc$.

If $a > 3b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - 3b)(c - d)$.
2. Factor the difference as $(a - 3b)(c - d)$.
3. The condition $a > 3b$ and $c > d$ makes each factor positive.
4. Hence $ac + 3bd > ad + 3bc$.

Correct option: A

Final answer

$$(a - 3b)(c - d)$$

Q012 Written

Under the condition $x > y$, prove that $3x - 4y > 2x - 3y$.

Solution

1. Subtract the right-hand side: $(3x - 4y) - (2x - 3y)$.
2. Simplify to $x - y$.
3. Since $x > y$, $x - y > 0$; therefore $3x - 4y > 2x - 3y$.

Final answer

$$3x - 4y > 2x - 3y$$

Worked Solutions

Inequalities

Q013 Written

Under the condition $x > 3$, $y > -5$, prove that $xy + 5x > 3y + 15$.

Solution

1. Start with LHS-RHS: $xy+5x-3y-15$.
2. Factor by grouping: $x(y+5) - 3(y+5) = (x-3)(y+5)$.
3. Since $x-3>0$ and $y+5>0$, the difference is positive; hence the inequality holds.

Final answer

$$xy + 5x > 3y + 15$$

Q014 Written

Under the condition $x > 2$, $y > 1$, prove that $xy + 2 > x + 2y$.

Solution

1. Start with LHS-RHS: $xy+2-x-2y$.
2. Factor: $x(y-1) - 2(y-1) = (x-2)(y-1)$.
3. Both factors are positive, so the difference is positive; hence $xy+2>x+2y$.

Final answer

$$xy + 2 > x + 2y$$

Worked Solutions

Inequalities

Q015 Written

Under the condition $a > 1$, $b > 1$, prove that $ab + 1 > a + b$.

Solution

1. Start with LHS-RHS: $ab+1-a-b$.
2. Factor: $ab-a-b+1=(a-1)(b-1)$.
3. Since $a-1>0$ and $b-1>0$, the difference is positive; hence $ab+1>a+b$.

Final answer

$$ab + 1 > a + b$$

Q016 Written

Under the condition $a > b > x > y$, prove that $ax + by > ay + bx$.

Solution

1. Subtract the right-hand side: $ax+by-ay-bx$.
2. Factor: $a(x - y) - b(x - y) = (a - b)(x - y)$.
3. Both factors are positive because $a>b$ and $x>y$; therefore the difference is positive.

Final answer

$$ax + by > ay + bx$$

Worked Solutions

Inequalities

Q017 Written

Under the condition $a > b > c > 0$, $x > y > z > 0$, prove that $ax + by + cz > ay + bz + cx$.

Solution

1. Subtract the right-hand side and regroup: $ax+by+cz-ay-bz-cx$.
2. Rewrite it as $(a-b)(x-y)+(b-c)(x-z)$.
3. Every factor in both products is positive, so the sum is positive; hence $ax+by+cz>ay+bz+cx$.

Final answer

$$ax + by + cz > ay + bz + cx$$

Worked Solutions

Inequalities

Q018 Challenge

If $a > b$ and $x > y$, prove that $2(ax + by) > 2(ay + bx)$.

If $a > b$ and $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $2(a - b)(x - y)$.
2. Factor the difference as $2(a - b)(x - y)$.
3. The condition $a > b$ and $x > y$ makes each factor positive.
4. Hence $2(ax + by) > 2(ay + bx)$.

Correct option: D

Final answer

$$2(a - b)(x - y)$$

Worked Solutions

Inequalities

Q019 Written

Prove that $a^2 + 4b^2 \geq 4ab$.**Solution**

1. Start with LHS - RHS: $a^2 + 4b^2 - 4ab$.
2. Complete the square: $a^2 + 4b^2 - 4ab = (a - 2b)^2 \geq 0$.
3. Therefore $a^2 + 4b^2 \geq 4ab$.
4. Equality holds when $a = 2b$.

Final answer

$$a^2 + 4b^2 \geq 4ab$$

Q020 MCQ

Under a,b real, prove that $a^2 + 4b^2 \geq 4ab$ **select the correct factorisation of LHS - RHS:****Solution**

1. Start with LHS - RHS: $a^2 + 4b^2 - 4ab$.
2. Factor/complete the square to obtain $(a - 2b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 4b^2 \geq 4ab$.

Correct option: D**Final answer**

$$(a - 2b)^2$$

Worked Solutions

Inequalities

Q021 MCQ

Under a, b real, prove that $a^2 + 9b^2 \geq 6ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 9b^2 - 6ab$.
2. Factor/complete the square to obtain $(a - 3b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 9b^2 \geq 6ab$.

Correct option: C**Final answer**

$$(a - 3b)^2$$

Q022 MCQ

Under a, b real, prove that $a^2 + 16b^2 \geq 8ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 16b^2 - 8ab$.
2. Factor/complete the square to obtain $(a - 4b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 16b^2 \geq 8ab$.

Correct option: B**Final answer**

$$(a - 4b)^2$$

Worked Solutions

Inequalities

Q023 MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 25b^2 - 10ab$.
2. Factor/complete the square to obtain $(a - 5b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 25b^2 \geq 10ab$.

Correct option: A

Final answer

$$(a - 5b)^2$$

Q024 MCQ

Under a, b real, prove that $a^2 + 36b^2 \geq 12ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 36b^2 - 12ab$.
2. Factor/complete the square to obtain $(a - 6b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 36b^2 \geq 12ab$.

Correct option: C

Final answer

$$(a - 6b)^2$$

Worked Solutions

Inequalities

Q025 MCQ

Under a, b real, prove that $a^2 + 49b^2 \geq 14ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 49b^2 - 14ab$.
2. Factor/complete the square to obtain $(a - 7b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 49b^2 \geq 14ab$.

Correct option: B**Final answer**

$$(a - 7b)^2$$

Q026 Written

Prove that $a^2 + 5b^2 \geq 4ab$.

Solution

1. Start with LHS - RHS: $a^2 + 5b^2 - 4ab$.
2. Complete the square: $a^2 + 5b^2 - 4ab = (a - 2b)^2 + b^2 \geq 0$.
3. Therefore $a^2 + 5b^2 \geq 4ab$.
4. Equality holds when $a = b = 0$.

Final answer

$$a^2 + 5b^2 \geq 4ab$$

Worked Solutions

Inequalities

Q027 Written

Prove that $9x^2 \geq y(6x - y)$.**Solution**

1. Subtract the right-hand side: $9x^2 - y(6x - y)$.
2. Complete the square: $9x^2 - y(6x - y) = (3x - y)^2 \geq 0$.
3. Therefore $9x^2 \geq y(6x - y)$.
4. Equality holds when $y = 3x$.

Final answer

$$9x^2 \geq y(6x - y)$$

Q028 Written

Prove that $2(x^2 + y^2) \geq (x + y)^2$.**Solution**

1. Subtract the right-hand side: $2(x^2 + y^2) - (x + y)^2$.
2. This is $(x - y)^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $x = y$.

Final answer

$$2(x^2 + y^2) \geq (x + y)^2$$

Worked Solutions

Inequalities

Q029 Written

Prove that $(2x - y)^2 \geq 3(x^2 - y^2)$.**Solution**

1. Subtract the right-hand side: $(2x - y)^2 - 3(x^2 - y^2)$.
2. Simplify to $(x - 2y)^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $x = 2y$.

Final answer

$$(2x - y)^2 \geq 3(x^2 - y^2)$$

Q030 Written

Prove that $(2a + 3b)^2 \geq 12ab$.**Solution**

1. Subtract the right-hand side: $(2a + 3b)^2 - 12ab$.
2. The cross terms cancel, giving $4a^2 + 9b^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $a = b = 0$.

Final answer

$$(2a + 3b)^2 \geq 12ab$$

Worked Solutions

Inequalities

Q031 MCQ

Under x, y real, prove that $4x^2 \geq y(4x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $4x^2 - y(4x - y)$.
2. Factor/complete the square to obtain $(2x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $4x^2 \geq y(4x - y)$.

Correct option: A**Final answer**

$$(2x - y)^2$$

Q032 MCQ

Under x, y real, prove that $9x^2 \geq y(6x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $9x^2 - y(6x - y)$.
2. Factor/complete the square to obtain $(3x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $9x^2 \geq y(6x - y)$.

Correct option: C**Final answer**

$$(3x - y)^2$$

Worked Solutions

Inequalities

Q033 MCQ

Under x, y real, prove that $16x^2 \geq y(8x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $16x^2 - y(8x - y)$.
2. Factor/complete the square to obtain $(4x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $16x^2 \geq y(8x - y)$.

Correct option: D

Final answer

$$(4x - y)^2$$

Q034 MCQ

Under x, y real, prove that $25x^2 \geq y(10x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $25x^2 - y(10x - y)$.
2. Factor/complete the square to obtain $(5x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $25x^2 \geq y(10x - y)$.

Correct option: D

Final answer

$$(5x - y)^2$$

Worked Solutions

Inequalities

Q035 MCQ

Under x, y real, prove that $36x^2 \geq y(12x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $36x^2 - y(12x - y)$.
2. Factor/complete the square to obtain $(6x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $36x^2 \geq y(12x - y)$.

Correct option: A**Final answer**

$$(6x - y)^2$$

Q036 MCQ

Under x, y real, prove that $x^2 + y^2 \geq 2xy$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $x^2 + y^2 - 2xy$.
2. Factor/complete the square to obtain $(x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $x^2 + y^2 \geq 2xy$.

Correct option: C**Final answer**

$$(x - y)^2$$

Worked Solutions

Inequalities

Q037 Written

Prove that $x^2 + y^2 \geq 2(x - y - 1)$.**Solution**

1. Start with LHS - RHS: $x^2 + y^2 - 2(x - y - 1)$.
2. Complete the squares: $x^2 + y^2 - 2(x - y - 1) = (x - 1)^2 + (y + 1)^2 \geq 0$.
3. Therefore $x^2 + y^2 \geq 2(x - y - 1)$.
4. Equality holds when $x = 1$, $y = -1$.

Final answer

$$x^2 + y^2 \geq 2(x - y - 1)$$

Q038 Written

Prove that $x^2 + 2 > 2x$.**Solution**

1. Subtract the right-hand side: $x^2 + 2 - 2x$.
2. Complete the square: $x^2 + 2 - 2x = (x - 1)^2 + 1 > 0$.
3. Therefore $x^2 + 2 > 2x$.
4. There is no equality case.

Final answer

$$x^2 + 2 > 2x$$

Worked Solutions

Inequalities

Q039 Written

Prove that $x^2 + y^2 - 6x + 8y + 25 \geq 0$.**Solution**

1. Complete the squares: $x^2 + y^2 - 6x + 8y + 25 = (x - 3)^2 + (y + 4)^2 \geq 0$.
2. Therefore the inequality holds.
3. Equality holds when $x = 3$, $y = -4$.

Final answer

$$x^2 + y^2 - 6x + 8y + 25 \geq 0$$

Q040 Written

Prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.**Solution**

1. Complete the squares: $x^2 + 2y^2 - 10x + 12y + 43 = (x - 5)^2 + 2(y + 3)^2 \geq 0$.
2. Therefore the expression is never negative.
3. Equality holds when $x = 5$, $y = -3$.

Final answer

$$x^2 + 2y^2 - 10x + 12y + 43 \geq 0$$

Worked Solutions

Inequalities

Q041 Written

Prove that $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$.**Solution**

1. Subtract the right-hand side: $(a^2 + b^2)(x^2 + y^2) - (ax + by)^2$.
2. Use $(a^2 + b^2)(x^2 + y^2) - (ax + by)^2 = (ay - bx)^2 \geq 0$.
3. Therefore the required inequality holds.
4. Equality holds when $ay = bx$.

Final answer

$$(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$$

Worked Solutions

Inequalities

Q042 Challenge

For real x, y , prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Solution

1. Complete both squares: $x^2 + 2y^2 - 10x + 12y + 43 = (x - 5)^2 + 2(y + 3)^2 \geq 0$.
2. Therefore $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.
3. Equality holds when $x = 5$, $y = -3$.

Final answer

0

Worked Solutions

Inequalities

Q043 Written

Prove that $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$.

Solution

1. Both sides are positive. Compare their squares.

$$2. \left(\sqrt{\frac{a^2+b^2}{2}} \right)^2 - \left(\frac{a+b}{2} \right)^2 = \frac{(a-b)^2}{4} \geq 0.$$

3. Hence the required inequality holds.

4. Equality holds when $a = b$.

Final answer

$$\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$$

Worked Solutions

Inequalities

Q044 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-b)^2}{4}$$

Worked Solutions

Inequalities

Q045 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-2b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-2b)^2}{4}$$

Worked Solutions

Inequalities

Q046 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-3b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: C

Final answer

$$\frac{(a-3b)^2}{4}$$

Worked Solutions

Inequalities

Q047 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-4b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-4b)^2}{4}$$

Worked Solutions

Inequalities

Q048 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-5b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-5b)^2}{4}$$

Worked Solutions

Inequalities

Q049 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-6b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.
- Equality occurs when the displayed square is zero.

Correct option: B

Final answer

$$\frac{(a-6b)^2}{4}$$

Worked Solutions

Inequalities

Q050 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-7b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-7b)^2}{4}$$

Worked Solutions

Inequalities

Q051 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-8b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-8b)^2}{4}$$

Worked Solutions

Inequalities

Q052 **Written****Prove that** $\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{2(a+b)})^2 - (\sqrt{a} + \sqrt{b})^2 = (\sqrt{a} - \sqrt{b})^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$$\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$$

Q053 **Written****Prove that** $a + 2 \geq \sqrt{3 + 6a}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(a + 2)^2 - (\sqrt{3 + 6a})^2 = (a - 1)^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = 1$.

Final answer

$$a + 2 \geq \sqrt{3 + 6a}$$

Worked Solutions

Inequalities

Q054 Written

Prove that $\sqrt{2(a^2 + b^2)} \geq a + b$.

Solution

1. Both sides are positive. Compare their squares.
2. $(\sqrt{2(a^2 + b^2)})^2 - (a + b)^2 = (a - b)^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$$\sqrt{2(a^2 + b^2)} \geq a + b$$

Q055 Written

Prove that $\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$.

Solution

1. Both sides are positive. Compare their squares.
2. $\left(\sqrt{\frac{a+b}{2}}\right)^2 - \left(\frac{\sqrt{a} + \sqrt{b}}{2}\right)^2 = \frac{(\sqrt{a} - \sqrt{b})^2}{4} \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$$\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$$

Worked Solutions

Inequalities

Q056 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 2\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

Correct option: B

Final answer

$$2\sqrt{a}\sqrt{b}$$

Q057 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 4\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

Correct option: A

Final answer

$$4\sqrt{a}\sqrt{b}$$

Worked Solutions

Inequalities

Q058 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 6\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

Correct option: C

Final answer

$$6\sqrt{a}\sqrt{b}$$

Q059 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 8\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Correct option: B

Final answer

$$8\sqrt{a}\sqrt{b}$$

Worked Solutions

Inequalities

Q060 Written

Prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{a} + 4\sqrt{b})^2 - (\sqrt{a + 16b})^2 = 8\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$$

Q061 Written

Prove that $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(2\sqrt{a} + \sqrt{b})^2 - (\sqrt{4a + b})^2 = 4\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$$

Worked Solutions

Inequalities

Q062 Written

Prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{a} + \sqrt{b})^2 - (\sqrt{a+b})^2 = 2\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a} + \sqrt{b} > \sqrt{a+b}$$

Q063 Written

Prove that $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a+9b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(2\sqrt{a} + 3\sqrt{b})^2 - (\sqrt{4a+9b})^2 = 12\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$2\sqrt{a} + 3\sqrt{b} > \sqrt{4a+9b}$$

Worked Solutions

Inequalities

Q064 **Written****Prove that** $\sqrt{a-b} > \sqrt{a} - \sqrt{b}$.**Solution**

1. Both sides are positive because $a > b > 0$. Compare their squares.
2. $(\sqrt{a-b})^2 - (\sqrt{a} - \sqrt{b})^2 = 2\sqrt{b}(\sqrt{a} - \sqrt{b}) > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a-b} > \sqrt{a} - \sqrt{b}$$

Q065 **Written****If** $a > b > 0$, **prove that** $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $\left(\sqrt{a} - \frac{b}{2\sqrt{a}}\right)^2 - (\sqrt{a-b})^2 = \frac{b^2}{4a} > 0$.
3. Therefore $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.
4. There is no equality case.

Final answer

$$\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$$

Worked Solutions

Inequalities

Q066 Challenge

For $a, b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Solution

1. Both sides are positive under $a, b > 0$; compare their squares.
2. $(\sqrt{a} + 4\sqrt{b})^2 - (\sqrt{a + 16b})^2 = 8\sqrt{ab} > 0$.
3. Therefore $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Final answer

$$\sqrt{a + 16b}$$

Worked Solutions

Inequalities

Q067 Written

Prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.**Solution**

1. By AM–GM, $\frac{b}{a} + \frac{a}{b} \geq 2\sqrt{\frac{b}{a} \cdot \frac{a}{b}} = 2$.
2. Equality holds when $a = b$.

Final answer

$$\frac{b}{a} + \frac{a}{b} \geq 2$$

Q068 Written

Prove that $\frac{b}{3a} + \frac{12a}{b} \geq 4$.**Solution**

1. Apply AM–GM to the two positive terms: $\frac{b}{3a} + \frac{12a}{b} \geq 2\sqrt{4} = 4$.
2. Equality holds when $b = 6a$.

Final answer

$$\frac{b}{3a} + \frac{12a}{b} \geq 4$$

Worked Solutions

Inequalities

Q069 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{a^2+b^2}{ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{a^2+b^2}{ab} - 2$.
2. Factor/complete the square to obtain $\frac{(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{a^2+b^2}{ab} \geq 2$.

Correct option: A**Final answer**

$$\frac{(a-b)^2}{ab}$$

Q070 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{4a^2 + b^2}{2ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{4a^2 + b^2}{2ab} - 2$.
2. Factor/complete the square to obtain $\frac{(2a - b)^2}{2ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{4a^2 + b^2}{2ab} \geq 2$.

Correct option: C

Final answer

$$\frac{(2a - b)^2}{2ab}$$

Worked Solutions

Inequalities

Q071 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{9a^2 + b^2}{3ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{9a^2 + b^2}{3ab} - 2$.
2. Factor/complete the square to obtain $\frac{(3a - b)^2}{3ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{9a^2 + b^2}{3ab} \geq 2$.

Correct option: B**Final answer**

$$\frac{(3a - b)^2}{3ab}$$

Q072 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{16a^2 + b^2}{4ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{16a^2 + b^2}{4ab} - 2$.
2. Factor/complete the square to obtain $\frac{(4a - b)^2}{4ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{16a^2 + b^2}{4ab} \geq 2$.

Correct option: A

Final answer

$$\frac{(4a - b)^2}{4ab}$$

Worked Solutions

Inequalities

Q073 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{25a^2 + b^2}{5ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{25a^2 + b^2}{5ab} - 2$.
2. Factor/complete the square to obtain $\frac{(5a - b)^2}{5ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{25a^2 + b^2}{5ab} \geq 2$.

Correct option: D**Final answer**

$$\frac{(5a - b)^2}{5ab}$$

Q074 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{36a^2 + b^2}{6ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{36a^2 + b^2}{6ab} - 2$.
2. Factor/complete the square to obtain $\frac{(6a - b)^2}{6ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{36a^2 + b^2}{6ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(6a - b)^2}{6ab}$$

Worked Solutions

Inequalities

Q075 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{49a^2 + b^2}{7ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{49a^2 + b^2}{7ab} - 2$.
2. Factor/complete the square to obtain $\frac{(7a - b)^2}{7ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{49a^2 + b^2}{7ab} \geq 2$.

Correct option: C**Final answer**

$$\frac{(7a - b)^2}{7ab}$$

Q076 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{64a^2 + b^2}{8ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{64a^2 + b^2}{8ab} - 2$.
2. Factor/complete the square to obtain $\frac{(8a - b)^2}{8ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{64a^2 + b^2}{8ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(8a - b)^2}{8ab}$$

Worked Solutions

Inequalities

Q077 Written

Prove that $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$.**Solution**

1. Expand: $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{a}{b} + \frac{b}{a} + 2$.
2. Use $\frac{a}{b} + \frac{b}{a} \geq 2$ to obtain the lower bound 4.
3. Equality holds when $a = b$.

Final answer

$$(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$$

Q078 Written

Prove that $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$.**Solution**

1. Expand to $x^2 + 5 + \frac{4}{x^2}$.
2. Since $x^2 + \frac{4}{x^2} \geq 4$, the product is at least 9.
3. Equality holds when $x = \sqrt{2}$.

Final answer

$$\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$$

Worked Solutions

Inequalities

Q079 Written

Prove that $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$.**Solution**

1. Expand to $10 + \frac{a}{b} + \frac{16b}{a}$.
2. AM–GM gives $\frac{a}{b} + \frac{16b}{a} \geq 8$, so the product is at least 18.
3. Equality holds when $a = 4b$.

Final answer

$$(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$$

Q080 Written

Prove that $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$.**Solution**

1. Expand to $ab + 6 + \frac{9}{ab}$.
2. AM–GM gives $ab + \frac{9}{ab} \geq 6$, so the product is at least 12.
3. Equality holds when $ab = 3$.

Final answer

$$\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$$

Worked Solutions

Inequalities

Q081 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+b)^2}{ab} \geq 4$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+b)^2}{ab} - 4$.
2. Factor/complete the square to obtain $\frac{(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+b)^2}{ab} \geq 4$.

Correct option: D**Final answer**

$$\frac{(a-b)^2}{ab}$$

Q082 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+4b)(4a+b)}{ab} - 25$.
2. Factor/complete the square to obtain $\frac{4(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+4b)(4a+b)}{ab} \geq 25$.

Correct option: B

Final answer

$$\frac{4(a-b)^2}{ab}$$

Worked Solutions

Inequalities

Q083 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+9b)(9a+b)}{ab} \geq 100$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+9b)(9a+b)}{ab} - 100$.
2. Factor/complete the square to obtain $\frac{9(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+9b)(9a+b)}{ab} \geq 100$.

Correct option: C**Final answer**

$$\frac{9(a-b)^2}{ab}$$

Q084 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+16b)(16a+b)}{ab} \geq 289$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+16b)(16a+b)}{ab} - 289$.
2. Factor/complete the square to obtain $\frac{16(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+16b)(16a+b)}{ab} \geq 289$.

Correct option: D

Final answer

$$\frac{16(a-b)^2}{ab}$$

Worked Solutions

Inequalities

Q085 Written

Prove that $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$.**Solution**

1. Expand to $ab + 10 + \frac{9}{ab}$.
2. AM–GM gives $ab + \frac{9}{ab} \geq 6$, so the product is at least 16.
3. Equality holds when $ab = 3$.

Final answer

$$\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$$

Q086 Written

Prove that $a + \frac{4}{a} \geq 4$.**Solution**

1. Apply AM–GM: $a + \frac{4}{a} \geq 2\sqrt{4} = 4$.
2. Equality holds when $a = 2$.

Final answer

$$a + \frac{4}{a} \geq 4$$

Worked Solutions

Inequalities

Q087 Written

Prove that $3ab + \frac{4}{ab} \geq 4\sqrt{3}$.**Solution**

1. Apply AM–GM to $3ab$ and $\frac{4}{ab}$: the sum is at least $2\sqrt{12} = 4\sqrt{3}$.
2. Equality holds when $ab = \frac{2}{\sqrt{3}}$.

Final answer

$$3ab + \frac{4}{ab} \geq 4\sqrt{3}$$

Q088 Written

Prove that $a + b + \frac{9}{a+b} \geq 6$.**Solution**

1. Let $t = a + b > 0$. AM–GM gives $t + \frac{9}{t} \geq 6$.
2. Equality holds when $a + b = 3$.

Final answer

$$a + b + \frac{9}{a+b} \geq 6$$

Worked Solutions

Inequalities

Q089 Written

A rectangle has perimeter 200 cm. Prove that its area $LW \leq 2500 \text{ cm}^2$.

Solution

1. From the perimeter, $L + W = 100$.
2. AM–GM gives $\frac{L+W}{2} \geq \sqrt{LW}$, so $50 \geq \sqrt{LW}$.
3. Therefore $LW \leq 2500$.
4. Equality holds when $L = W = 50$.

Final answer

$$LW \leq 2500$$

Worked Solutions

Inequalities

Q090 Challenge

For $a, b > 0$, prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Solution

1. Put the terms over a positive denominator: $\frac{b}{a} + \frac{a}{b} - 2 = \frac{(a-b)^2}{ab} \geq 0$.
2. Hence $\frac{b}{a} + \frac{a}{b} \geq 2$.
3. Equality holds when $a = b$.

Final answer

2

Worked Solutions

Inequalities

Quiz MCQ 1 **Concept Quiz · MCQ**

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + y > 2(x + y)$.

Correct option: A

Final answer

$$x - y$$

Quiz MCQ 2 **Concept Quiz · MCQ**

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $a^2 + 25b^2 - 10ab$.
2. Factor/complete the square to obtain $(a - 5b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 25b^2 \geq 10ab$.

Correct option: A

Final answer

$$(a - 5b)^2$$

Worked Solutions

Inequalities

Quiz MCQ 3 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-7b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-7b)^2}{4}$$

Worked Solutions

Inequalities

Quiz MCQ 4 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+4b)(4a+b)}{ab} - 25$.
2. Factor/complete the square to obtain $\frac{4(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+4b)(4a+b)}{ab} \geq 25$.

Correct option: B

Final answer

$$\frac{4(a-b)^2}{ab}$$

Quiz WRITTEN 5 Concept Quiz · Written

If $x > y$, prove that $3x + 4y > 2x + 5y$.

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 4y > 2x + 5y$.

Final answer

$$2x + 5y$$

Worked Solutions

Inequalities

Quiz WRITTEN 6 Concept Quiz · Written

Under x, y real, prove that $x^2 + y^2 \geq 2xy$.

Solution

1. Start with LHS - RHS: $x^2 + y^2 - 2xy$.
2. Factor/complete the square to obtain $(x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $x^2 + y^2 \geq 2xy$.

Final answer

$$2xy$$

Quiz WRITTEN 7 Concept Quiz · Written

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 12\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Final answer

$$\sqrt{a + 36b}$$

Worked Solutions

Inequalities

Quiz WRITTEN 8 Concept Quiz · Written

Under $t > 0$, prove that $\frac{t^2+1}{t} \geq 2$.

Solution

1. Start with LHS - RHS: $\frac{t^2+1}{t} - 2$.
2. Factor/complete the square to obtain $\frac{(t-1)^2}{t}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{t^2+1}{t} \geq 2$.

Final answer

2

Answer key

Use the question number shown on each page.

Q001 $2(ax + by) > (a + b)(x + y)$

Q002 A

Q003 C

Q004 D

Q005 D

Q006 B

Q007 $ac + bd > ad + bc$

Q008 $ab + cd > ad + bc$

Q009 D

Q010 C

Q011 A

Q012 $3x - 4y > 2x - 3y$

Q013 $xy + 5x > 3y + 15$

Q014 $xy + 2 > x + 2y$

Q015 $ab + 1 > a + b$

Q016 $ax + by > ay + bx$

Q017 $ax + by + cz > ay + bz + cx$

Q018 D

Q019 $a^2 + 4b^2 \geq 4ab$

Q020 D

Q021 C

Q022 B

Q023 A

Q024 C

Q025 B

Q026 $a^2 + 5b^2 \geq 4ab$

Q027 $9x^2 \geq y(6x - y)$

Q028 $2(x^2 + y^2) \geq (x + y)^2$

Q029 $(2x - y)^2 \geq 3(x^2 - y^2)$

Q030 $(2a + 3b)^2 \geq 12ab$

Q031 A

Q032 C

Q033 D

Q034 D

Q035 A

Q036 C

Q037 $x^2 + y^2 \geq 2(x - y - 1)$

Q038 $x^2 + 2 > 2x$

Q039 $x^2 + y^2 - 6x + 8y + 25 \geq 0$

Q040 $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$

Q041 $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$

Q042 0

Q043 $\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2}$

Q044 A

Q045 A

Q046 C

Q047 A

Q048 A

Q049 B

Q050 A

Q051 A

Q052 $\sqrt{2(a + b)} \geq \sqrt{a} + \sqrt{b}$

Q053 $a + 2 \geq \sqrt{3 + 6a}$

Q054 $\sqrt{2(a^2 + b^2)} \geq a + b$

Q055 $\sqrt{\frac{a + b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$

Q056 B

Q057 A

Q058 C

Q059 B

Q060 $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$

Q061 $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$

Q062 $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$

Q063 $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$

Q064 $\sqrt{a - b} > \sqrt{a} - \sqrt{b}$

Q065 $\sqrt{a - b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$

Q066 $\sqrt{a + 16b}$

Q067 $\frac{b}{a} + \frac{a}{b} \geq 2$

Q068 $\frac{b}{3a} + \frac{12a}{b} \geq 4$

Q069 A

Q070 C

Q071 B

Q072 A

Q073 D

Q074 D

Q075 C

Q076 D

Q077 $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$

Q078 $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$

Q079 $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$

Q080 $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$

Q081 D

Q082 B

Q083 C

Q084 D

Q085 $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$

Q086 $a + \frac{4}{a} \geq 4$

Q087 $3ab + \frac{4}{ab} \geq 4\sqrt{3}$

Q088 $a + b + \frac{9}{a + b} \geq 6$

Q089 $LW \leq 2500$

Q090 2

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $2x + 5y$

Quiz WRITTEN 6 $2xy$

Quiz WRITTEN 7 $\sqrt{a + 36b}$

Quiz WRITTEN 8 2

Concept 1 | Arithmetic Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = a + (n - 1)d$$

Sum

$$S_n = \frac{n}{2}(2a + (n - 1)d)$$

Test

$$a_{n+1} - a_n = d$$

Worked Solutions

Arithmetic Sequences

Q001 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2n - 3$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -1$, $a_2 = 1$, $a_3 = 3$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -1, a_2 = 1, a_3 = 3$$

Q002 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 7n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = -10$, $a_3 = -12$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = -10, a_3 = -12$$

Worked Solutions

Arithmetic Sequences

Q003 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 5n - 8$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 2$, $a_3 = 7$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -3, a_2 = 2, a_3 = 7$$

Q004 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -3n + 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = -5$, $a_3 = -8$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -2, a_2 = -5, a_3 = -8$$

Worked Solutions

Arithmetic Sequences

Q005 Written

When the general term of the sequence $\{a_n\}$ is $a_n = -2n - 4$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = -8$, $a_3 = -10$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = -8, a_3 = -10$$

Q006 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 - 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 0$, $a_2 = 3$, $a_3 = 8$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 0, a_2 = 3, a_3 = 8$$

Worked Solutions

Arithmetic Sequences

Q007 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + n + 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 7$, $a_3 = 13$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 3, a_2 = 7, a_3 = 13$$

Q008 Written

When the general term of the sequence $\{a_n\}$ is $a_n = n^2 + 2n + 4$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 7$, $a_2 = 12$, $a_3 = 19$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 7, a_2 = 12, a_3 = 19$$

Worked Solutions

Arithmetic Sequences

Q009 MCQ

For $a_n = 3n - 5$, the first three terms (a_1, a_2, a_3) are:**Solution**

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = 1$, $a_3 = 4$.
3. Check each value by putting its index back into the formula.

Correct option: B**Final answer** $(-2, 1, 4)$

Q010 MCQ

For $a_n = 4n + 7$, the first three terms (a_1, a_2, a_3) are:**Solution**

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 11$, $a_2 = 15$, $a_3 = 19$.
3. Check each value by putting its index back into the formula.

Correct option: C**Final answer** $(11, 15, 19)$

Worked Solutions

Arithmetic Sequences

Q011 MCQ

For $a_n = n^2 - 4n + 6$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 2$, $a_3 = 3$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(3, 2, 3)

Q012 MCQ

For $a_n = 2n^2 + 3$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 5$, $a_2 = 11$, $a_3 = 21$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(5, 11, 21)

Worked Solutions

Arithmetic Sequences

Q013 MCQ

For $a_n = n^2 + 5n - 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 4$, $a_2 = 10$, $a_3 = 16$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(4, 10, 16)$

Q014 MCQ

For $a_n = 5 - 2n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 1$, $a_3 = -1$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(3, 1, -1)$

Worked Solutions

Arithmetic Sequences

Q015 MCQ

For $a_n = 2n^2 - 7n + 9$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 4$, $a_2 = 3$, $a_3 = 6$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(4, 3, 6)

Q016 MCQ

For $a_n = n^2 - 3n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = -3$, $a_3 = -1$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(-3, -3, -1)

Worked Solutions

Arithmetic Sequences

Q017 MCQ

For $a_n = 3n^2 - 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = 10$, $a_3 = 25$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

(1, 10, 25)

Q018 MCQ

For $a_n = n^2 + 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 6$, $a_3 = 11$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

(3, 6, 11)

Worked Solutions

Arithmetic Sequences

Q019 MCQ

For $a_n = 7n - 10$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 4$, $a_3 = 11$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(-3, 4, 11)$

Q020 MCQ

For $a_n = 2n^2 + n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 9$, $a_3 = 20$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(2, 9, 20)$

Worked Solutions

Arithmetic Sequences

Q021 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 2 \times 3^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 6$, $a_2 = 18$, $a_3 = 54$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 6, a_2 = 18, a_3 = 54$$

Q022 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3^n - 1$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 8$, $a_3 = 26$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 2, a_2 = 8, a_3 = 26$$

Worked Solutions

Arithmetic Sequences

Q023 Written

When the general term of the sequence $\{a_n\}$ is $a_n = (-3)^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -3$, $a_2 = 9$, $a_3 = -27$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -3, a_2 = 9, a_3 = -27$$

Q024 Written

When the general term of the sequence $\{a_n\}$ is $a_n = 3 \times (-2)^n$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -6$, $a_2 = 12$, $a_3 = -24$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = -6, a_2 = 12, a_3 = -24$$

Worked Solutions

Arithmetic Sequences

Q025 MCQ

For $a_n = 2^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 4$, $a_3 = 8$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

$(2, 4, 8)$

Q026 MCQ

For $a_n = (-2)^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -2$, $a_2 = 4$, $a_3 = -8$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

$(-2, 4, -8)$

Worked Solutions

Arithmetic Sequences

Q027 MCQ

For $a_n = 2 \times 3^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 6$, $a_2 = 18$, $a_3 = 54$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

(6, 18, 54)

Q028 MCQ

For $a_n = 5 \times (-3)^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = -15$, $a_2 = 45$, $a_3 = -135$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

(-15, 45, -135)

Worked Solutions

Arithmetic Sequences

Q029 MCQ

For $a_n = 4^n - 1$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 15$, $a_3 = 63$.
3. Check each value by putting its index back into the formula.

Correct option: A

Final answer

(3, 15, 63)

Q030 MCQ

For $a_n = 2 \times 3^{n-1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 6$, $a_3 = 18$.
3. Check each value by putting its index back into the formula.

Correct option: B

Final answer

(2, 6, 18)

Worked Solutions

Arithmetic Sequences

Q031 MCQ

For $a_n = (-1)^{n-1}5^n$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = 5$, $a_2 = -25$, $a_3 = 125$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$(5, -25, 125)$

Q032 MCQ

For $a_n = (-3)^n + 2$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute $(n = 1, 2, 3)$ into the displayed general-term formula.
2. This gives $a_1 = -1$, $a_2 = 11$, $a_3 = -25$.
3. Check each value by putting its index back into the formula.

Correct option: D

Final answer

$(-1, 11, -25)$

Worked Solutions

Arithmetic Sequences

Q033 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{n+1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{2}$, $a_2 = \frac{2}{3}$, $a_3 = \frac{3}{4}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$$

Q034 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n}{2n+1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{3}$, $a_2 = \frac{2}{5}$, $a_3 = \frac{3}{7}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$$

Worked Solutions

Arithmetic Sequences

Q035 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{1}{3n-2}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{1}{4}$, $a_3 = \frac{1}{7}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$$

Q036 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{2n}{n+3}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{2}$, $a_2 = \frac{4}{5}$, $a_3 = 1$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$$

Worked Solutions

Arithmetic Sequences

Q037 Written

When the general term of the sequence $\{a_n\}$ is $a_n = \frac{n+1}{2n-1}$, find the first term through the 3rd term.

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 2$, $a_2 = 1$, $a_3 = \frac{4}{5}$.
3. Check each value by putting its index back into the formula.

Final answer

$$a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$$

Q038 MCQ

For $a_n = \frac{n+1}{n+2}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{2}{3}$, $a_2 = \frac{3}{4}$, $a_3 = \frac{4}{5}$.
3. Check each value by putting its index back into the formula.

Correct option: C

Final answer

$$\left(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}\right)$$

Worked Solutions

Arithmetic Sequences

Q039 MCQ

For $a_n = \frac{2n-1}{3n+1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = \frac{1}{4}$, $a_2 = \frac{3}{7}$, $a_3 = \frac{1}{2}$.
3. Check each value by putting its index back into the formula.

Correct option: D**Final answer**

$$\left(\frac{1}{4}, \frac{3}{7}, \frac{1}{2}\right)$$

Q040 MCQ

For $a_n = \frac{n^2+1}{n+1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{5}{3}$, $a_3 = \frac{5}{2}$.
3. Check each value by putting its index back into the formula.

Correct option: A**Final answer**

$$\left(1, \frac{5}{3}, \frac{5}{2}\right)$$

Worked Solutions

Arithmetic Sequences

Q041 MCQ

For $a_n = \frac{3n}{2n-1}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 3$, $a_2 = 2$, $a_3 = \frac{9}{5}$.
3. Check each value by putting its index back into the formula.

Correct option: B**Final answer** $(3, 2, \frac{9}{5})$

Q042 MCQ

For $a_n = \frac{2n+3}{n+4}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{7}{6}$, $a_3 = \frac{9}{7}$.
3. Check each value by putting its index back into the formula.

Correct option: C**Final answer** $(1, \frac{7}{6}, \frac{9}{7})$

Worked Solutions

Arithmetic Sequences

Q043 MCQ

For $a_n = \frac{n}{3n-2}$, the first three terms (a_1, a_2, a_3) are:

Solution

1. Substitute ($n = 1, 2, 3$) into the displayed general-term formula.
2. This gives $a_1 = 1$, $a_2 = \frac{1}{4}$, $a_3 = \frac{3}{7}$.
3. Check each value by putting its index back into the formula.

Correct option: D**Final answer**

$$(1, \frac{1}{4}, \frac{3}{7})$$

Q044 Written

For the sequence $\{a_n\} : 0, 1, 4, 9, 16, \dots$, deduce the general term.

Solution

1. The terms are (term number $- 1$)², so $a_n = (n - 1)^2$.
2. Checking $n = 1, 2, 3$ gives 0,1,4.

Final answer

$$a_n = (n - 1)^2$$

Worked Solutions

Arithmetic Sequences

Q045 **Written****For the sequence $\{a_n\} : 0, 1, 8, 27, 64, \dots$, deduce the general term.****Solution**

1. The terms are successive cubes starting with 0: $(n - 1)^3$.
2. The formula gives 0,1,8,27,64 for $n = 1, 2, 3, 4, 5$.

Final answer

$$a_n = (n - 1)^3$$

Q046 **Written****For the sequence $\{a_n\} : 1, 4, 9, 16, 25, \dots$, deduce the general term.****Solution**

1. These are the squares $1^2, 2^2, 3^2, \dots$, so $a_n = n^2$.

Final answer

$$a_n = n^2$$

Worked Solutions

Arithmetic Sequences

Q047 **Written**

For the sequence $\{a_n\} : 7^2, 8^2, 9^2, 10^2, 11^2, \dots$, deduce the general term.

Solution

1. The bases are $n + 6$, so $a_n = (n + 6)^2$.

Final answer

$$a_n = (n + 6)^2$$

Q048 **MCQ**

Which formula gives the general term of $\{a_n\} : 1, 8, 27, 64, \dots$?:

Solution

1. Separate the visible pattern: the rule is $a_n = n^3$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B

Final answer

$$n^3$$

Worked Solutions

Arithmetic Sequences

Q049 MCQ

Which formula gives the general term of $\{a_n\}$: 9, 16, 25, 36, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n + 2)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$(n + 2)^2$$

Q050 MCQ

Which formula gives the general term of $\{a_n\}$: -1, 0, 1, 8, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n - 2)^3$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: D**Final answer**

$$(n - 2)^3$$

Worked Solutions

Arithmetic Sequences

Q051 MCQ

Which formula gives the general term of $\{a_n\}$: 36, 49, 64, 81, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n + 5)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: A**Final answer**

$$(n + 5)^2$$

Q052 MCQ

Which formula gives the general term of $\{a_n\}$: 1, 9, 25, 49, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (2n - 1)^2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B**Final answer**

$$(2n - 1)^2$$

Worked Solutions

Arithmetic Sequences

Q053 MCQ

Which formula gives the general term of $\{a_n\} : 0, 1, 16, 81, \dots$?:**Solution**

1. Separate the visible pattern: the rule is $a_n = (n - 1)^4$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$(n - 1)^4$$

Q054 Written

For the sequence $\{a_n\} : 2, 5, 8, 11, 14, \dots$, deduce the general term.**Solution**

1. The common increase is 3 and the first term is 2, so $a_n = 2 + 3(n - 1) = 3n - 1$.

Final answer

$$a_n = 3n - 1$$

Worked Solutions

Arithmetic Sequences

Q055 Written

For the sequence $\{a_n\} : 0, 2, 4, 6, 8, \dots$, deduce the general term.

Solution

1. The common difference is 2 and $a_1=0$, so $a_n=2(n-1)=2n-2$.

Final answer

$$a_n = 2n - 2$$

Q056 Written

For the sequence $\{a_n\} : 0, 3, 6, 9, 12, \dots$, deduce the general term.

Solution

1. The common difference is 3 and $a_1=0$, so $a_n=3(n-1)=3n-3$.

Final answer

$$a_n = 3n - 3$$

Worked Solutions

Arithmetic Sequences

Q057 **Written****For the sequence $\{a_n\}$: 4, 9, 14, 19, 24, ..., deduce the general term.****Solution**

1. The common difference is 5 and $a_1=4$, so $a_n=4+5(n-1)=5n-1$.

Final answer

$$a_n = 5n - 1$$

Q058 **Written****For the sequence $\{a_n\}$: 6, 13, 20, 27, 34, ..., deduce the general term.****Solution**

1. The common difference is 7 and $a_1=6$, so $a_n=6+7(n-1)=7n-1$.

Final answer

$$a_n = 7n - 1$$

Worked Solutions

Arithmetic Sequences

Q059 MCQ

Which formula gives the general term of $\{a_n\}$: 7, 12, 17, 22, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 5n + 2$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: A**Final answer**

$$5n + 2$$

Q060 MCQ

Which formula gives the general term of $\{a_n\}$: -4, -1, 2, 5, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 3n - 7$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: B**Final answer**

$$3n - 7$$

Worked Solutions

Arithmetic Sequences

Q061 MCQ

Which formula gives the general term of $\{a_n\}$: 10, 6, 2, -2, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = 14 - 4n$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: C**Final answer**

$$14 - 4n$$

Q062 MCQ

Which formula gives the general term of $\{a_n\}$: $\frac{1}{2}$, 1, $\frac{3}{2}$, 2, ...?:**Solution**

1. Separate the visible pattern: the rule is $a_n = \frac{n}{2}$.
2. Check $n = 1, 2, 3$ against the displayed terms.

Correct option: D**Final answer**

$$\frac{n}{2}$$

Worked Solutions

Arithmetic Sequences

Q063 Written

For the sequence $\{a_n\} : 2, -4, 6, -8, 10, \dots$, deduce the general term.

Solution

1. The magnitude is $2n$ and the first sign is positive with alternation, so the sign factor is $(-1)^{n-1}$.
2. Therefore $a_n = (-1)^{n-1} 2n$.

Final answer

$$a_n = (-1)^{n-1} 2n$$

Q064 Written

For the sequence $\{a_n\} : -3, 6, -9, 12, -15, \dots$, deduce the general term.

Solution

1. The magnitude is $3n$; the first term is negative and signs alternate, giving $(-1)^n$.
2. Therefore $a_n = (-1)^n 3n$.

Final answer

$$a_n = (-1)^n 3n$$

Worked Solutions

Arithmetic Sequences

Q065 Written

For the sequence $\{a_n\} : -5, 10, -15, 20, -25, \dots$, deduce the general term.

Solution

1. The magnitude is $5n$ and the first sign is negative, so $a_n = (-1)^n 5n$.

Final answer

$$a_n = (-1)^n 5n$$

Q066 Written

For the sequence $\{a_n\} : -1, 4, -7, 10, -13, \dots$, deduce the general term.

Solution

1. The magnitudes 1, 4, 7, 10, 13 are $3n - 2$; the first sign is negative, so $a_n = (-1)^n (3n - 2)$.

Final answer

$$a_n = (-1)^n (3n - 2)$$

Worked Solutions

Arithmetic Sequences

Q067 Written

For the sequence $\{a_n\} : 1, \frac{2}{3}, \frac{3}{5}, \frac{4}{7}, \frac{5}{9}, \dots$, deduce the general term.

Solution

1. The numerator is n and the denominator is the odd number $2n-1$.
2. Therefore $a_n = n/(2n-1)$.

Final answer

$$a_n = \frac{n}{2n-1}$$

Q068 Written

For the sequence $\{a_n\} : 1, \frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \frac{9}{25}, \dots$, deduce the general term.

Solution

1. The numerator is the odd number $2n-1$ and the denominator is n^2 .
2. Therefore $a_n = (2n-1)/n^2$.

Final answer

$$a_n = \frac{2n-1}{n^2}$$

Worked Solutions

Arithmetic Sequences

Q069 Written

For the sequence $\{a_n\} : 1, -\frac{1}{3}, \frac{1}{5}, -\frac{1}{7}, \frac{1}{9}, \dots$, deduce the general term.

Solution

1. The denominator is $2n - 1$ and the signs alternate from a positive first term, so $a_n = (-1)^{n-1}/(2n - 1)$.

Final answer

$$a_n = \frac{(-1)^{n-1}}{2n - 1}$$

Q070 Written

For the sequence $\{a_n\} : \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots$, deduce the general term.

Solution

1. The numerator is n and the denominator is $n+1$, so $a_n = n/(n+1)$.

Final answer

$$a_n = \frac{n}{n + 1}$$

Worked Solutions

Arithmetic Sequences

Q071 Written

For the sequence $\{a_n\} : 1 \times 3, 2 \times 9, 3 \times 27, 4 \times 81, 5 \times 243, \dots$, deduce the general term.

Solution

1. The first factor is n and the second factor is 3^n , so $a_n = n3^n$.

Final answer

$$a_n = n 3^n$$

Q072 Written

For the sequence $\{a_n\} : 1 \times \frac{1}{2}, 2 \times \frac{1}{3}, 3 \times \frac{1}{4}, 4 \times \frac{1}{5}, 5 \times \frac{1}{6}, \dots$, deduce the general term then find the 20th term.

Solution

1. The first factor is the term number n and the second factor is $1/(n+1)$.
2. Hence the general term is $a_n = n/(n+1)$.
3. Substitute $n = 20$: $a_{20} = 20/(20+1) = 20/21$.

Final answer

$$a_n = \frac{n}{n+1}; \quad a_{20} = \frac{20}{21}$$

Worked Solutions

Arithmetic Sequences

Q073 Written

Find the first term and the common difference of the arithmetic sequences: (a) 2, 5, 8, 11, ...; (b) 3, -2, -7, -12, -17,

Solution

1. For (a), $5 - 2 = 8 - 5 = 11 - 8 = 3$, so $a = 2, d = 3$.
2. For (b), each successive difference is -5 , so $a = 3, d = -5$.

Final answer

$$(a) \quad a = 2, d = 3 \quad (b) \quad a = 3, d = -5$$

Q074 Written

Find the first term and common difference of: (a) 1, 5, 9, 13, 17, ...; (b) 23, 19, 15, 11, 7,

Solution

1. (a) The constant increase is (4), so $a = 1, d = 4$.
2. (b) The constant change is -4 , so $a = 23, d = -4$.

Final answer

$$(a) \quad a = 1, d = 4 \quad (b) \quad a = 23, d = -4$$

Worked Solutions

Arithmetic Sequences

Q075 MCQ

The listed terms $\{a_n\} = (4, 9, 14, 19, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 4$.
2. Subtract consecutive terms: $d = 5$.

Correct option: B

Final answer

$$a = 4, d = 5$$

Q076 MCQ

The listed terms $\{a_n\} = (-7, -3, 1, 5, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -7$.
2. Subtract consecutive terms: $d = 4$.

Correct option: A

Final answer

$$a = -7, d = 4$$

Worked Solutions

Arithmetic Sequences

Q077 MCQ

The listed terms $\{a_n\} = (18, 13, 8, 3, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 18$.
2. Subtract consecutive terms: $d = -5$.

Correct option: C

Final answer

$$a = 18, d = -5$$

Q078 MCQ

The listed terms $\{a_n\} = (12, 8, 4, 0, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = 12$.
2. Subtract consecutive terms: $d = -4$.

Correct option: D

Final answer

$$a = 12, d = -4$$

Worked Solutions

Arithmetic Sequences

Q079 Written

Find the first term and common difference of: (a) 3, 5, 7, 9, 11, ...; (b) $-13, -9, -5, -1, 3, \dots$; (c) $-2, 1, 4, 7, 10, \dots$; (d) $10, 7, 4, 1, -2, \dots$; (e) $6, \frac{16}{3}, \frac{14}{3}, 4, \frac{10}{3}, \dots$; (f) $-4, -\frac{9}{2}, -5, -\frac{11}{2}, -6, \dots$

Solution

1. Subtract consecutive terms in each list. The six constant differences are 2, 4, 3, $-3, -\frac{2}{3}, -\frac{1}{2}$, respectively; the first terms are the first entries.

Final answer

(a)(3, 2); (b)($-13, 4$); (c)($-2, 3$); (d)($10, -3$); (e)($6, -\frac{2}{3}$); (f)($-4, -\frac{1}{2}$).

Worked Solutions

Arithmetic Sequences

Q080 MCQ

The listed terms $\{a_n\} = (\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = \frac{1}{2}$.
2. Subtract consecutive terms: $d = \frac{1}{3}$.

Correct option: B**Final answer**

$$a = \frac{1}{2}, d = \frac{1}{3}$$

Q081 MCQ

The listed terms $\{a_n\} = (-\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -\frac{3}{4}$.
2. Subtract consecutive terms: $d = \frac{1}{2}$.

Correct option: A**Final answer**

$$a = -\frac{3}{4}, d = \frac{1}{2}$$

Worked Solutions

Arithmetic Sequences

Q082 MCQ

The listed terms $\{a_n\} = (\frac{5}{3}, 2, \frac{7}{3}, \frac{8}{3}, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = \frac{5}{3}$.
2. Subtract consecutive terms: $d = \frac{1}{3}$.

Correct option: C

Final answer

$$a = \frac{5}{3}, d = \frac{1}{3}$$

Q083 MCQ

The listed terms $\{a_n\} = (-\frac{5}{2}, -1, \frac{1}{2}, 2, \dots)$ form an arithmetic sequence. Find the ordered pair (a, d) of first term and common difference select the correct ordered pair a, d :

Solution

1. The first term is $a = -\frac{5}{2}$.
2. Subtract consecutive terms: $d = \frac{3}{2}$.

Correct option: B

Final answer

$$a = -\frac{5}{2}, d = \frac{3}{2}$$

Worked Solutions

Arithmetic Sequences

Q084 Written

The sequences are arithmetic. Find the common difference and the numbers that fit in (A) and (B): (a) $A, 2, -1, B, \dots$; (b) $13, A, B, -2, \dots$

Solution

- (a) $d = -1 - 2 = -3$; hence $A = 2 - d = 5$ and $B = -1 + d = -4$.
- (b) The change from 13 to -2 covers three equal steps: $3d = -15$, so $d = -5$; therefore $A = 8, B = 3$.

Final answer

$$(a) \quad d = -3, A = 5, B = -4 \quad (b) \quad d = -5, A = 8, B = 3$$

Q085 Written

The sequences are arithmetic. Find the common difference and the missing numbers: (a) $2, 9, A, B, \dots$; (b) $A, 17, B, 5, \dots$

Solution

- (a) $d = 9 - 2 = 7$, so $A = 16, B = 23$.
- (b) $5 - 17 = 2d$, hence $d = -6$; then $A = 23, B = 11$.

Final answer

$$(a) \quad d = 7, A = 16, B = 23 \quad (b) \quad d = -6, A = 23, B = 11$$

Worked Solutions

Arithmetic Sequences

Q086 MCQ

The sequence $\{a_n\} = (6, A, 14, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 10, B = 18$.

Correct option: A

Final answer

$$d = 4, A = 10, B = 18$$

Q087 MCQ

The sequence $\{a_n\} = (A, 5, 9, B, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 1, B = 13$.

Correct option: C

Final answer

$$d = 4, A = 1, B = 13$$

Worked Solutions

Arithmetic Sequences

Q088 MCQ

The sequence $\{a_n\} = (11, A, B, 23, \dots)$ is arithmetic. Determine the common difference and every missing letter select the correct values of d and the missing terms:

Solution

1. Set the equal-step difference (d).
2. Substitute successively to obtain $d = 4, A = 15, B = 19$.

Correct option: B**Final answer**

$$d = 4, A = 15, B = 19$$

Worked Solutions

Arithmetic Sequences

Q089 Written

Find the common difference and missing terms: (a) $8, 15, A, B, \dots$; (b) $7, 13, A, B, \dots$; (c) $A, 9, 4, B, \dots$; (d) $A, 15, B, 9, \dots$; (e) $11, A, B, 20, \dots$; (f) $17, A, B, 5, C, \dots$

Solution

1. Use the equal-step condition. For example, in (d), $9 - 15 = 2d$, so $d = -2$, giving $A = 17, B = 13$. Applying the same calculation to (a)–(f) gives the listed values.

Final answer

(a) $d = 7, A = 22, B = 29$; (b) $d = 6, A = 19, B = 25$; (c) $d = -5, A = 14, B = -1$; (d) $d = -2, A = 17, B = 13$; (e) $d = 3, A = 14, B = 17$; (f) $d = -4, A = 13, B = 9, C = 1$.

Worked Solutions

Arithmetic Sequences

Q090 Written

For each arithmetic sequence, find the general term and the 7th term: (a) first term (4), common difference (3); (b) first term (2), common difference -3 .

Solution

1. (a) $a_n = 4 + 3(n - 1) = 3n + 1$; $a_7 = 22$.
2. (b) $a_n = 2 - 3(n - 1) = 5 - 3n$; $a_7 = -16$.

Final answer

$$(a) \quad a_n = 3n + 1, a_7 = 22 \quad (b) \quad a_n = 5 - 3n, a_7 = -16$$

Q091 Written

For each arithmetic sequence, find the general term and the 7th term: (a) $a = 5, d = 4$; (b) $a = -2, d = 3$; (c) $a = 10, d = -3$; (d) $a = -\frac{1}{2}, d = -\frac{1}{2}$.

Solution

1. Substitute each pair into $a_n = a + (n - 1)d$, then put $n = 7$. The results are $4n + 1, 3n - 5, 13 - 3n, -\frac{n}{2}$, with 7th terms 29, 16, $-8, -\frac{7}{2}$.

Final answer

$$(a) a_n = 4n + 1, a_7 = 29; (b) a_n = 3n - 5, a_7 = 16; (c) a_n = 13 - 3n, a_7 = -8; (d) a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}.$$

Worked Solutions

Arithmetic Sequences

Q092 Written

For the arithmetic sequence a_n with first term (8) and common difference -6 , find the general term and the 20th term.

Solution

1. Use $a_n = a + (n - 1)d$: $a_n = 8 - 6(n - 1) = 14 - 6n$.
2. Substitute $n = 20$: $a_{20} = 14 - 120 = -106$.

Final answer

$$a_n = 8 + (n - 1)(-6) = 14 - 6n; \quad a_{20} = -106$$

Q093 Written

For each arithmetic sequence, find the general term and the 10th term: (a) $a = -7, d = 3$; (b) $a = -2, d = -3$; (c) $a = 7, d = \frac{1}{2}$; (d) $a = \frac{2}{3}, d = -\frac{1}{3}$.

Solution

1. Use $a_n = a + (n - 1)d$ and substitute $n = 10$. Simplification gives the four displayed formulas and terms.

Final answer

$$(a)a_n = 3n - 10, a_{10} = 20; (b)a_n = 1 - 3n, a_{10} = -29; (c)a_n = \frac{n + 13}{2}, a_{10} = \frac{23}{2}; (d)a_n = \frac{3 - n}{3}, a_{10} = -\frac{7}{3}.$$

Worked Solutions

Arithmetic Sequences

Q094 Written

In an arithmetic sequence, $a_3 + a_7 = 38$ and $\frac{a_4}{a_{12}} = \frac{2}{5}$. Find the first term and common difference.

Solution

1. $a_3 + a_7 = (a + 2d) + (a + 6d) = 2a + 8d = 38$, so $a + 4d = 19$.
2. The ratio gives $5(a + 3d) = 2(a + 11d)$, hence $3a = 7d$.
3. Solving yields $d = 3$ and $a = 7$.

Final answer

$$a = 7, d = 3.$$

Q095 Written

Find the general term of the arithmetic sequence whose first term is -5 and whose 17th term is 75.

Solution

1. $75 = -5 + 16d$, so $d = 5$.
2. $a_n = -5 + 5(n - 1) = 5n - 10$.

Final answer

$$a_n = 5n - 10$$

Worked Solutions

Arithmetic Sequences

Q096 Written

The first term is 6, and the 4th term is 18. Find a_n .

Solution

1. $18 = 6 + 3d$, so $d = 4$.
2. $a_n = 6 + 4(n - 1) = 4n + 2$.

Final answer

$$a_n = 4n + 2$$

Q097 Written

The first term is 20, and the 8th term is -1 . Find a_n .

Solution

1. $-1 = 20 + 7d$, so $d = -3$.
2. $a_n = 20 - 3(n - 1) = 23 - 3n$.

Final answer

$$a_n = 23 - 3n$$

Worked Solutions

Arithmetic Sequences

Q098 Written

The first term is -8 , and the 8th term is 55 . Find a_n .

Solution

1. $55 = -8 + 7d$, so $d = 9$.
2. $a_n = -8 + 9(n - 1) = 9n - 17$.

Final answer

$$a_n = 9n - 17$$

Q099 Written

The first term is -18 , and the 10th term is -63 . Find a_n .

Solution

1. $-63 = -18 + 9d$, so $d = -5$.
2. $a_n = -18 - 5(n - 1) = -5n - 13$.

Final answer

$$a_n = -5n - 13$$

Worked Solutions

Arithmetic Sequences

Q100 MCQ

An arithmetic sequence has $a = 4$ and $a_7 = 22$. Select the correct pair (a_n, d) :

Solution

1. $22 = 4 + 6d$, so $d = 3$.
2. $a_n = 4 + 3(n - 1) = 3n + 1$.

Correct option: D

Final answer

$$(a_n, d) = (3n + 1, 3)$$

Q101 MCQ

An arithmetic sequence has $a = -3$ and $a_9 = 29$. Select the correct pair (a_n, d) :

Solution

1. $29 = -3 + 8d$, so $d = 4$.
2. $a_n = -3 + 4(n - 1) = 4n - 7$.

Correct option: B

Final answer

$$(a_n, d) = (4n - 7, 4)$$

Worked Solutions

Arithmetic Sequences

Q102 MCQ

An arithmetic sequence has $a = 12$ and $a_5 = -4$. Select the correct pair (a_n, d) :

Solution

1. $-4 = 12 + 4d$, so $d = -4$.
2. $a_n = 12 - 4(n - 1) = 16 - 4n$.

Correct option: A

Final answer

$$(a_n, d) = (16 - 4n, -4)$$

Q103 MCQ

An arithmetic sequence has $a = 5$ and $a_{11} = 45$. Select the correct pair (a_n, d) :

Solution

1. $45 = 5 + 10d$, so $d = 4$.
2. $a_n = 5 + 4(n - 1) = 4n + 1$.

Correct option: B

Final answer

$$(a_n, d) = (4n + 1, 4)$$

Worked Solutions

Arithmetic Sequences

Q104 Written

Find the general term of the arithmetic sequence whose common difference is 3 and whose 16th term is 81.

Solution

1. $81 = a + 15(3)$, so $a = 36$.
2. $a_n = 36 + 3(n - 1) = 3n + 33$.

Final answer

$$a_n = 3n + 33$$

Q105 Written

The common difference is 2, and the 8th term is 4. Find a_n .

Solution

1. $4 = a + 7(2)$, so $a = -10$.
2. $a_n = -10 + 2(n - 1) = 2n - 12$.

Final answer

$$a_n = 2n - 12$$

Worked Solutions

Arithmetic Sequences

Q106 Written

The common difference is -4 , and the 5th term is -7 . Find a_n .

Solution

1. $-7 = a + 4(-4)$, so $a = 9$.
2. $a_n = 9 - 4(n - 1) = 13 - 4n$.

Final answer

$$a_n = 13 - 4n$$

Q107 Written

The common difference is -7 , and the 6th term is -24 . Find a_n .

Solution

1. $-24 = a + 5(-7)$, so $a = 11$.
2. $a_n = 11 - 7(n - 1) = 18 - 7n$.

Final answer

$$a_n = 18 - 7n$$

Worked Solutions

Arithmetic Sequences

Q108 MCQ

An arithmetic sequence has common difference -2 and $a_4 = 1$. Select the correct pair (a_n, a) :

Solution

1. $1 = a + 3(-2)$, so $a = 7$.
2. $a_n = 7 - 2(n - 1) = 9 - 2n$.

Correct option: A

Final answer

$$(a_n, a) = (9 - 2n, 7)$$

Q109 MCQ

An arithmetic sequence has common difference 5 and $a_6 = 17$. Select the correct pair (a_n, a) :

Solution

1. $17 = a + 5(5)$, so $a = -8$.
2. $a_n = -8 + 5(n - 1) = 5n - 13$.

Correct option: D

Final answer

$$(a_n, a) = (5n - 13, -8)$$

Worked Solutions

Arithmetic Sequences

Q110 MCQ

An arithmetic sequence has common difference 4 and $a_{10} = 38$. Select the correct pair (a_n, a) :

Solution

1. $38 = a + 9(4)$, so $a = 2$.
2. $a_n = 2 + 4(n - 1) = 4n - 2$.

Correct option: C

Final answer

$$(a_n, a) = (4n - 2, 2)$$

Q111 MCQ

An arithmetic sequence has common difference -3 and $a_8 = -18$. Select the correct pair (a_n, a) :

Solution

1. $-18 = a + 7(-3)$, so $a = 3$.
2. $a_n = 3 - 3(n - 1) = 6 - 3n$.

Correct option: D

Final answer

$$(a_n, a) = (6 - 3n, 3)$$

Worked Solutions

Arithmetic Sequences

Q112 Written

Find the general term of the arithmetic sequence $\{a_n\}$ whose 3rd term is 10 and the 8th term is 50.

Solution

1. Use $a_n = a + (n - 1)d$. The data give $10 = a + 2d$ and $50 = a + 7d$.
2. Subtracting gives $5d = 40$, so $d = 8$ and $a = -6$.
3. Therefore $a_n = -6 + 8(n - 1) = 8n - 14$.

Final answer

$$a_n = 8n - 14$$

Q113 Written

Find the general term of the arithmetic sequence whose 6th term is 2 and whose 15th term is -25 .

Solution

1. $2 = a + 5d$ and $-25 = a + 14d$, so $d = -3$, $a = 17$.
2. $a_n = 17 - 3(n - 1) = 20 - 3n$.

Final answer

$$a_n = 20 - 3n$$

Worked Solutions

Arithmetic Sequences

Q114 **Written****The 4th term is 14, and the 10th term is 62. Find a_n .****Solution**

1. $14 = a + 3d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.
2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Q115 **Written****The 3rd term is 30, and the 10th term is 2. Find a_n .****Solution**

1. $30 = a + 2d$ and $2 = a + 9d$, so $d = -4$, $a = 38$.
2. $a_n = 42 - 4n$.

Final answer

$$a_n = 42 - 4n$$

Worked Solutions

Arithmetic Sequences

Q116 Written

The 10th term is 49, and the 23rd term is 10. Find a_n .

Solution

1. $49 = a + 9d$ and $10 = a + 22d$, so $d = -3$, $a = 76$.
2. $a_n = 79 - 3n$.

Final answer

$$a_n = 79 - 3n$$

Q117 Written

The 3rd term is 6, and the 10th term is 62. Find a_n .

Solution

1. $6 = a + 2d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.
2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Worked Solutions

Arithmetic Sequences

Q118 MCQ

An arithmetic sequence has $a_3 = 5$ and $a_9 = 17$. Select the correct pair (a_n, a) :

Solution

1. $5 = a + 2d$ and $17 = a + 8d$, so $d = 2$, $a = 1$.
2. $a_n = 1 + 2(n - 1) = 2n - 1$.

Correct option: C

Final answer

$$(a_n, a) = (2n - 1, 1)$$

Q119 MCQ

An arithmetic sequence has $a_4 = 7$ and $a_{10} = -5$. Select the correct pair (a_n, a) :

Solution

1. $7 = a + 3d$ and $-5 = a + 9d$, so $d = -2$, $a = 13$.
2. $a_n = 13 - 2(n - 1) = 15 - 2n$.

Correct option: C

Final answer

$$(a_n, a) = (15 - 2n, 13)$$

Worked Solutions

Arithmetic Sequences

Q120 MCQ

An arithmetic sequence has $a_7 = 31$ and $a_{12} = 56$. Select the correct pair (a_n, a) :

Solution

1. $31 = a + 6d$ and $56 = a + 11d$, so $d = 5$, $a = 1$.

2. $a_n = 1 + 5(n - 1) = 5n - 4$.

Correct option: B

Final answer

$$(a_n, a) = (5n - 4, 1)$$

Q121 MCQ

An arithmetic sequence has $a_2 = -4$ and $a_8 = 14$. Select the correct pair (a_n, a) :

Solution

1. $-4 = a + d$ and $14 = a + 7d$, so $d = 3$, $a = -7$.

2. $a_n = -7 + 3(n - 1) = 3n - 10$.

Correct option: A

Final answer

$$(a_n, a) = (3n - 10, -7)$$

Worked Solutions

Arithmetic Sequences

Q122 MCQ

An arithmetic sequence has $a_5 = 3$ and $a_{11} = 9$. Select the correct pair (a_n, a) :

Solution

1. $3 = a + 4d$ and $9 = a + 10d$, so $d = \frac{1}{2}$, $a = -\frac{1}{2}$.

2. $a_n = -\frac{1}{2} + \frac{1}{2}(n-1) = \frac{n-2}{2}$.

Correct option: A

Final answer

$$\left(a_n = \frac{n-2}{2}, -\frac{1}{2}\right)$$

Q123 MCQ

An arithmetic sequence has $a_2 = 6$ and $a_9 = 20$. Select the correct pair (a_n, a) :

Solution

1. $6 = a + d$ and $20 = a + 8d$, so $d = 2$, $a = 4$.

2. $a_n = 4 + 2(n-1) = 2n + 2$.

Correct option: B

Final answer

$$(a_n = 2n + 2, 4)$$

Worked Solutions

Arithmetic Sequences

Q124 Written

The common difference is $\frac{2}{5}$, and the 30th term is $\frac{23}{5}$. Find a_n .

Solution

1. $\frac{23}{5} = a + 29(\frac{2}{5})$, so $a = -7$.

2. $a_n = -7 + \frac{2}{5}(n - 1) = \frac{2n - 37}{5}$.

Final answer

$$a_n = \frac{2n - 37}{5}$$

Q125 MCQ

An arithmetic sequence has $d = -\frac{3}{2}$ and $a_5 = -9$. Select the correct pair (a_n, a) :

Solution

1. $-9 = a + 4(-\frac{3}{2})$, so $a = -3$.

2. $a_n = -3 - \frac{3}{2}(n - 1) = -\frac{3(n+1)}{2}$.

Correct option: A

Final answer

$$\left(a_n = -\frac{3(n+1)}{2}, -3 \right)$$

Worked Solutions

Arithmetic Sequences

Q126 MCQ

An arithmetic sequence has $d = \frac{2}{3}$ and $a_{12} = 10$. Select the correct pair (a_n, a) :

Solution

1. $10 = a + 11(\frac{2}{3})$, so $a = \frac{8}{3}$.
2. $a_n = \frac{8}{3} + \frac{2}{3}(n - 1) = \frac{2n+6}{3}$.

Correct option: D

Final answer

$$\left(a_n = \frac{2n+6}{3}, \frac{8}{3}\right)$$

Q127 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, which term is 113?

Solution

1. First find a : $33 = a + 7(4)$, so $a = 5$.
2. Then $a_n = 5 + 4(n - 1) = 4n + 1$.
3. Solve $4n + 1 = 113$: $n = 28$.

Final answer

$n = 28$, the 28th term

Worked Solutions

Arithmetic Sequences

Q128 Written

For the arithmetic sequence with first term 1 and common difference 3, determine which term is 40.

Solution

1. $a_n = 1 + 3(n - 1) = 3n - 2$.
2. $3n - 2 = 40$ gives $n = 14$.

Final answer

$n = 14$, the 14th term

Q129 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , determine which term is -52 .

Solution

1. $-7 = a + 12(-5)$, so $a = 53$ and $a_n = 58 - 5n$.
2. $58 - 5n = -52$ gives $n = 22$.

Final answer

$n = 22$, the 22nd term

Worked Solutions

Arithmetic Sequences

Q130 Written

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , determine which term is -110 .

Solution

1. $-20 = a + 9d$ and $-50 = a + 19d$, so $d = -3$, $a = 7$.
2. $a_n = 10 - 3n$, and $10 - 3n = -110$ gives $n = 40$.

Final answer

$n = 40$, the 40th term

Q131 MCQ

For $a = 3$, $d = 4$, which term of the arithmetic sequence is 59?:

Solution

1. $a_n = 3 + 4(n - 1) = 4n - 1$.
2. $4n - 1 = 59$ gives $n = 15$.

Correct option: C

Final answer

$n = 15$

Worked Solutions

Arithmetic Sequences

Q132 MCQ

For $a = -5, d = 6$, which term of the arithmetic sequence is 55?:

Solution

1. $a_n = -5 + 6(n - 1) = 6n - 11$.
2. $6n - 11 = 55$ gives $n = 11$.

Correct option: D

Final answer

$$n = 11$$

Q133 MCQ

For $a = 20, d = -3$, which term of the arithmetic sequence is -16 ?:

Solution

1. $a_n = 20 - 3(n - 1) = 23 - 3n$.
2. $23 - 3n = -16$ gives $n = 13$.

Correct option: A

Final answer

$$n = 13$$

Worked Solutions

Arithmetic Sequences

Q134 MCQ

For $a = 1$, $d = 5$, which term of the arithmetic sequence is 86?:

Solution

1. $a_n = 1 + 5(n - 1) = 5n - 4$.
2. $5n - 4 = 86$ gives $n = 18$.

Correct option: B

Final answer

$$n = 18$$

Q135 Written

For the arithmetic sequence $\{a_n\}$ with common difference 4 and $a_8 = 33$, find the first term greater than 500.

Solution

1. From the given data, $a_n = 4n + 1$.
2. Require $4n + 1 > 500$, hence $n > 124.75$.
3. The least natural index is $n = 125$.

Final answer

$n = 125$, the 125th term

Worked Solutions

Arithmetic Sequences

Q136 Written

For the arithmetic sequence with first term 1 and common difference 3, find the first term greater than 100.

Solution

1. $a_n = 3n - 2$.
2. $3n - 2 > 100$ gives $n > 34$.
3. The first valid index is $n = 35$.

Final answer

$n = 35$, the 35th term

Q137 MCQ

For $a = 2, d = 7$, find the first term greater than 100:

Solution

1. $a_n = 7n - 5$.
2. $7n - 5 > 100$ gives $n > 15$, so $n = 16$.

Correct option: A

Final answer

$n = 16$

Worked Solutions

Arithmetic Sequences

Q138 MCQ

For $a = -9, d = 4$, find the first term greater than 50:**Solution**

1. $a_n = 4n - 13$.
2. $4n - 13 > 50$ gives $n > 15.75$, so $n = 16$.

Correct option: D**Final answer**

$$n = 16$$

Q139 MCQ

For $a = 6, d = 3$, find the first term greater than 200:**Solution**

1. $a_n = 3n + 3$.
2. $3n + 3 > 200$ gives $n > 65\frac{2}{3}$, so $n = 66$.

Correct option: D**Final answer**

$$n = 66$$

Worked Solutions

Arithmetic Sequences

Q140 MCQ

For $a = -1$, $d = 9$, find the first term greater than 300:

Solution

1. $a_n = 9n - 10$.
2. $9n - 10 > 300$ gives $n > 34\frac{4}{9}$, so $n = 35$.

Correct option: B

Final answer

$$n = 35$$

Q141 Written

For the arithmetic sequence with common difference -5 and 13th term -7 , find the first term smaller than -300 .

Solution

1. $a_n = 58 - 5n$.
2. $58 - 5n < -300$ gives $n > 71.6$.
3. The first valid index is $n = 72$.

Final answer

$n = 72$, the 72nd term

Worked Solutions

Arithmetic Sequences

Q142 Written

For the arithmetic sequence whose 10th term is -20 and 20th term is -50 , find the first term smaller than -500 .

Solution

1. $a_n = 10 - 3n$.
2. $10 - 3n < -500$ gives $n > 170$.
3. The first valid index is $n = 171$.

Final answer

$n = 171$, the 171st term

Q143 MCQ

For $a = 10$, $d = -4$, find the first term smaller than -100 :

Solution

1. $a_n = 14 - 4n$.
2. $14 - 4n < -100$ gives $n > 28.5$, so $n = 29$.

Correct option: B

Final answer

$n = 29$

Worked Solutions

Arithmetic Sequences

Q144 MCQ

For $a = 25, d = -6$, find the first term smaller than -250 :**Solution**

1. $a_n = 25 - 6n$.
2. $25 - 6n < -250$ gives $n > 46\frac{5}{6}$, so $n = 47$.

Correct option: B**Final answer**

$$n = 47$$

Q145 MCQ

For $a = -3, d = -5$, find the first term smaller than -200 :**Solution**

1. $a_n = -3 - 5n$.
2. $-3 - 5n < -200$ gives $n > 40.4$, so $n = 41$.

Correct option: C**Final answer**

$$n = 41$$

Worked Solutions

Arithmetic Sequences

Q146 MCQ

For $a = 4$, $d = 6$, find the first term that is at least 100:

Solution

1. $a_n = 6n - 2$.
2. $6n - 2 \geq 100$ gives $n \geq 17$, so $n = 17$.

Correct option: A

Final answer

$$n = 17$$

Q147 MCQ

For the expression For $a = 5$, $d = 4$, does the sequence contain the term 42?, the required value is:

Solution

1. $a_n = 4n + 1$.
2. $4n + 1 = 42$ gives $n = \frac{41}{4}$, not a natural index.
3. Therefore 42 is not a term.

Correct option: B

Final answer

No term

Worked Solutions

Arithmetic Sequences

Q148 **Written**

When $a, 6, 2a$ form an arithmetic sequence in this order, find a .

Solution

1. $2(6) = a + 2a$.
2. $3a = 12$, so $a = 4$.

Final answer

$$a = 4$$

Q149 **Written**

When the three numbers $x, 6, x^2$ form an arithmetic sequence in this order, find x .

Solution

1. Use $2b = a + c$: $2(6) = x + x^2$.
2. Thus $x^2 + x - 12 = 0 = (x + 4)(x - 3)$.
3. Therefore $x = -4$ or $x = 3$.

Final answer

$$x = -4 \text{ or } x = 3$$

Worked Solutions

Arithmetic Sequences

Q150 **Written****When $x^2, x, -8$ form an arithmetic sequence in this order, find x .****Solution**

1. $2x = x^2 - 8$.
2. $x^2 - 2x - 8 = (x - 4)(x + 2) = 0$.
3. $x = -2$ or $x = 4$.

Final answer

$$x = -2 \text{ or } x = 4$$

Q151 **Written****When $-3, x, x^2$ form an arithmetic sequence in this order, find x .****Solution**

1. $2x = -3 + x^2$.
2. $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$.
3. $x = -1$ or $x = 3$.

Final answer

$$x = -1 \text{ or } x = 3$$

Worked Solutions

Arithmetic Sequences

Q152 Challenge

When $2, x, y, z$ form an arithmetic sequence in this order, and $2x, y + 1, z - y$ also form an arithmetic sequence in this order, find x, y, z .

Solution

1. Let the common difference of $2, x, y, z$ be d : $x = 2 + d$, $y = 2 + 2d$, $z = 2 + 3d$.
2. $2(y + 1) = 2x + (z - y)$ gives $6 + 4d = 4 + 3d$, hence $d = -2$.
3. $(x, y, z) = (0, -2, -4)$.

Final answer

$$x = 0, \quad y = -2, \quad z = -4$$

Worked Solutions

Arithmetic Sequences

Q153 Written

Find the sum (S_7) of a sequence with first term (8), last term (22), and (7) terms.

Solution

1. Use ($S_n = \frac{n}{2}(a + l)$).
2. ($S_7 = \frac{7}{2}(8 + 22) = 105$).

Final answer

105

Q154 Written

Find (S_{10}) for a sequence with first term (7), last term (13), and (10) terms.

Solution

1. ($S_{10} = \frac{10}{2}(7 + 13) = 100$).

Final answer

100

Worked Solutions

Arithmetic Sequences

Q155 Written

Find (S_{10}) for a sequence with first term (3), last term (21), and (10) terms.**Solution**

1. $(S_{10} = \frac{10}{2}(3 + 21) = 120).$

Final answer

120

Q156 Written

Find (S_8) for a sequence with first term (5), last term (33), and (8) terms.**Solution**

1. $(S_8 = \frac{8}{2}(5 + 33) = 152).$

Final answer

152

Worked Solutions

Arithmetic Sequences

Q157 **Written****Find (S_{10}) for a sequence with first term (2), last term (29), and (10) terms.****Solution**

1. $(S_{10} = \frac{10}{2}(2 + 29) = 155).$

Final answer

155

Q158 **MCQ****A sequence has first term 6, last term 30, and 7 terms. Find its sum:****Solution**

1. Use $S_7 = \frac{7}{2}(6 + 30).$
2. Thus $S_7 = 126.$

Correct option: B**Final answer**

126

Worked Solutions

Arithmetic Sequences

Q159 MCQ

A sequence has first term -4 , last term 20 , and 9 terms. Find its sum:

Solution

1. $S_9 = \frac{9}{2}(-4 + 20) = 72.$

Correct option: B

Final answer

72

Q160 MCQ

Find the sum of 10 terms whose first term is 15 and last term is 3 :

Solution

1. $S_{10} = \frac{10}{2}(15 + 3) = 90.$

Correct option: B

Final answer

90

Worked Solutions

Arithmetic Sequences

Q161 MCQ

The first term is 7, the last term is 42, and there are 8 terms. Find S_8 :

Solution

1. $S_8 = \frac{8}{2}(7 + 42) = 196.$

Correct option: B

Final answer

196

Q162 MCQ

Find the sum of 6 terms with first term -12 and last term 8:

Solution

1. $S_6 = \frac{6}{2}(-12 + 8) = -12.$

Correct option: B

Final answer

-12

Worked Solutions

Arithmetic Sequences

Q163 MCQ

A sequence has $S_9 = 72$ and first term -4 . Find its last term:

Solution

1. From $72 = \frac{9}{2}(-4 + l)$, $l = 20$.

Correct option: C

Final answer

20

Q164 Written

Find (S_{13}) when the first term is (20) , the common difference is (-5) , and the number of terms is (13) .

Solution

1. Use $(S_n = \frac{n}{2}\{2a + (n - 1)d\})$.

2. $(S_{13} = \frac{13}{2}\{40 + 12(-5)\} = -130)$.

Final answer

-130

Worked Solutions

Arithmetic Sequences

Q165 **Written****Find (S_{12}) when the first term is (-6) , the common difference is (3) , and there are (12) terms.****Solution**

$$1. (S_{12} = \frac{12}{2}\{-12 + 11(3)\} = 144).$$

Final answer

144

Q166 **Written****Find (S_{15}) when the first term is (10) , the common difference is (-3) , and there are (15) terms.****Solution**

$$1. (S_{15} = \frac{15}{2}\{20 + 14(-3)\} = -165).$$

Final answer

-165

Worked Solutions

Arithmetic Sequences

Q167 Written

Find (S_{11}) when the first term is (-4) , the common difference is (4) , and there are (11) terms.

Solution

1. $(S_{11} = \frac{11}{2}\{-8 + 10(4)\} = 176).$

Final answer

176

Q168 Written

Find (S_{18}) when the first term is (3) , the common difference is (2) , and there are (18) terms.

Solution

1. $(S_{18} = \frac{18}{2}\{6 + 17(2)\} = 360).$

Final answer

360

Worked Solutions

Arithmetic Sequences

Q169 MCQ

For $a = 4, d = 3, n = 12$, find S_{12} :**Solution**

$$1. S_{12} = \frac{12}{2}\{8 + 11(3)\} = 246.$$

Correct option: C**Final answer**

246

Q170 MCQ

For $a = 9, d = -2, n = 10$, find S_{10} :**Solution**

$$1. S_{10} = 5\{18 + 9(-2)\} = 0.$$

Correct option: B**Final answer**

0

Worked Solutions

Arithmetic Sequences

Q171 MCQ

For $a = -5, d = 4, n = 9$, find S_9 :**Solution**

$$1. S_9 = \frac{9}{2}\{-10 + 8(4)\} = 88.$$

Correct option: C**Final answer**

88

Q172 MCQ

For $a = 2, d = -3, n = 15$, find S_{15} :**Solution**

$$1. S_{15} = \frac{15}{2}\{4 + 14(-3)\} = -285.$$

Correct option: B**Final answer**

-285

Worked Solutions

Arithmetic Sequences

Q173 MCQ

For $a = 11, d = 0, n = 8$, find S_8 :**Solution**1. All terms equal 11, so $S_8 = 88$.**Correct option:** C**Final answer**

88

Q174 MCQ

For $a = -3, d = -2, n = 7$, find S_7 :**Solution**1. $S_7 = \frac{7}{2}\{-6 + 6(-2)\} = -54$.**Correct option:** B**Final answer**

-54

Worked Solutions

Arithmetic Sequences

Q175 Written

Find the sum (S) of a sequence with first term (2), common difference (3), and last term (47).**Solution**

1. Find (n) from $(47 = 2 + (n - 1)3)$, giving ($n = 16$).
2. Then $(S = \frac{16}{2}(2 + 47) = 392)$.

Final answer

392

Q176 Written

Find (S) for a sequence with first term (5), common difference (3), and last term (44).**Solution**

1. Solve $(44 = 5 + (n - 1)3)$, so ($n = 14$).
2. $(S = \frac{14}{2}(5 + 44) = 343)$.

Final answer

343

Worked Solutions

Arithmetic Sequences

Q177 Written

Find (S) for a sequence with first term (3), common difference (4), and last term (75).**Solution**

1. $(75 = 3 + (n - 1)4)$ gives $(n = 19)$.
2. $(S = \frac{19}{2}(3 + 75) = 741)$.

Final answer

741

Q178 Written

Find (S) for a sequence with first term (2), common difference (5), and last term (57).**Solution**

1. $(57 = 2 + (n - 1)5)$ gives $(n = 12)$.
2. $(S = \frac{12}{2}(2 + 57) = 354)$.

Final answer

354

Worked Solutions

Arithmetic Sequences

Q179 Written

Find (S) for a sequence with first term (4), common difference (3), and last term (100).

Solution

1. $100 = 4 + (n - 1)3$ gives $(n = 33)$.
2. $S = \frac{33}{2}(4 + 100) = 1716$.

Final answer

1716

Q180 MCQ

A sequence has $a = 5, d = 4, l = 41$. Find its sum:

Solution

1. $41 = 5 + (n - 1)4$ gives $n = 10$.
2. $S = \frac{10}{2}(5 + 41) = 230$.

Correct option: C

Final answer

230

Worked Solutions

Arithmetic Sequences

Q181 MCQ

A sequence has $a = -7, d = 5, l = 48$. Find its sum:**Solution**

1. $48 = -7 + (n - 1)5$ gives $n = 12$.
2. $S = 6(-7 + 48) = 246$.

Correct option: C**Final answer**

246

Q182 MCQ

A sequence has $a = 12, d = 2, l = 50$. Find its sum:**Solution**

1. $50 = 12 + (n - 1)2$ gives $n = 20$.
2. $S = 10(12 + 50) = 620$.

Correct option: C**Final answer**

620

Worked Solutions

Arithmetic Sequences

Q183 MCQ

A sequence has $a = -10, d = 6, l = 50$. Find its sum:**Solution**

1. $50 = -10 + (n - 1)6$ gives $n = 11$.

2. $S = \frac{11}{2}(-10 + 50) = 220$.

Correct option: C**Final answer**

220

Q184 MCQ

A sequence has $a = 18, d = -3, l = -9$. Find its sum:**Solution**

1. Solve $-9 = 18 + (n - 1)(-3)$, giving $n = 10$.

2. $S = \frac{10}{2}(18 - 9) = 45$.

Correct option: C**Final answer**

45

Worked Solutions

Arithmetic Sequences

Q185 MCQ

A sequence has $a = 40, d = -4, l = -12$. Find its sum:

Solution

1. $n = 14$ from $-12 = 40 + (n - 1)(-4)$.
2. $S = 7(40 - 12) = 196$.

Correct option: B

Final answer

196

Q186 Written

For the arithmetic sequence with first term (48) and common difference (-2) , find the sum from the 20th term to the 30th term.

Solution

1. Use $(S_{30} - S_{19})$, since the range is (a_{20}) through (a_{30}) .
2. $(a_{20} = 10, a_{30} = -10)$, so $(11 \times \frac{10 + (-10)}{2} = 0)$.

Final answer

0

Worked Solutions

Arithmetic Sequences

Q187 Written

For the arithmetic sequence with first term (-53) and common difference (4) , find the sum from the 10th term to the 25th term.

Solution

1. $(a_{10} = -53 + 9(4) = -17)$ and $(a_{25} = -53 + 24(4) = 43)$.
2. There are (16) terms, so the sum is $(16 \times \frac{-17+43}{2} = 208)$.

Final answer

208

Q188 MCQ

For $a = 3, d = 4$, find the sum from a_5 to a_{12} :

Solution

1. $a_5 = 19, a_{12} = 47$, with 8 terms.
2. Sum = $8(19 + 47)/2 = 264$.

Correct option: B

Final answer

264

Worked Solutions

Arithmetic Sequences

Q189 MCQ

For $a = 20$, $d = -3$, find the sum from a_4 to a_{10} :**Solution**

1. $a_4 = 11$, $a_{10} = -7$, with 7 terms.
2. Sum = $7(11 - 7)/2 = 14$.

Correct option: C**Final answer**

14

Q190 MCQ

For $a = 60$, $d = -3$, find the sum from a_8 to a_{16} :**Solution**

1. $a_8 = 39$, $a_{16} = 15$, with 9 terms.
2. Sum = $9(39 + 15)/2 = 243$.

Correct option: C**Final answer**

243

Worked Solutions

Arithmetic Sequences

Q191 MCQ

For $a = 1, d = 7$, find the sum from a_3 to a_9 :**Solution**

1. $a_3 = 15, a_9 = 57$, with 7 terms.
2. $\text{Sum} = 7(15 + 57)/2 = 252$.

Correct option: C**Final answer**

252

Q192 MCQ

For $a = 60, d = -3$, find the sum from a_9 to a_{17} :**Solution**

1. $a_9 = 36, a_{17} = 12$, with 9 terms.
2. $\text{Sum} = 9(36 + 12)/2 = 216$.

Correct option: B**Final answer**

216

Worked Solutions

Arithmetic Sequences

Q193 MCQ

For $a = 12$, $d = 5$, find the sum from a_2 to a_8 :

Solution

1. $a_2 = 17$, $a_8 = 47$, with 7 terms.
2. Sum $= 7(17 + 47)/2 = 224$.

Correct option: C

Final answer

224

Q194 Written

For the arithmetic sequence $(90, 84, 78, \dots)$, find the sum from the 10th term to the 20th term.

Solution

1. Here $(a = 90, d = -6)$, so the required sum is $(S_{20} - S_9)$.
2. $(S_{20} = 660)$ and $(S_9 = 594)$, hence the sum is $(660 - 594 = 66)$.

Final answer

66

Worked Solutions

Arithmetic Sequences

Q195 Written

For the arithmetic sequence $(-9, -3, 3, \dots)$, find the sum from the 11th term to the 20th term.

Solution

1. Here $a = -9$, $d = 6$, so use $S_{20} - S_{10}$.
2. $a_{10} = 45$ and $a_{20} = 105$; there are 10 terms.
3. Required sum $= \frac{10(45+105)}{2} = 750$.

Final answer

750

Q196 Written

For the arithmetic sequence $(74, 69, 64, \dots)$, find the sum from the 8th term to the 18th term.

Solution

1. Here $(a = 74, d = -5)$, so $(a_8 = 39)$ and $(a_{18} = -11)$.
2. There are (11) terms; the sum is $(11 \times \frac{39-11}{2} = 154)$.

Final answer

154

Worked Solutions

Arithmetic Sequences

Q197 MCQ

For $a = 4, d = 5$, find the sum from a_6 to a_{13} :**Solution**

1. $a_6 = 29, a_{13} = 64$, with 8 terms.
2. $\text{Sum} = 8(29 + 64)/2 = 372$.

Correct option: C**Final answer**

372

Q198 MCQ

For $a = -25, d = 6$, find the sum from a_5 to a_{11} :**Solution**

1. $a_5 = -1, a_{11} = 35$, with 7 terms.
2. $\text{Sum} = 7(-1 + 35)/2 = 119$.

Correct option: C**Final answer**

119

Worked Solutions

Arithmetic Sequences

Q199 **Written****Find (S_n) for a sequence with first term (1) and common difference (4).****Solution**

1. Substitute $(a = 1, d = 4)$ into $(S_n = \frac{n}{2}\{2a + (n - 1)d\})$.
2. $(S_n = \frac{n}{2}(2 + 4n - 4) = 2n^2 - n)$.

Final answer

$$2n^2 - n$$

Q200 **Written****Find (S_n) for a sequence with first term (2) and common difference (-3) .****Solution**

1. $(S_n = \frac{n}{2}\{4 + (n - 1)(-3)\})$.
2. Therefore $(S_n = \frac{n(7-3n)}{2})$.

Final answer

$$\frac{n(7 - 3n)}{2}$$

Worked Solutions

Arithmetic Sequences

Q201 Written

Find (S_n) for a sequence with first term (0) and common difference (5).**Solution**

1. $(S_n = \frac{n}{2}\{0 + 5(n - 1)\})$.
2. Hence $(S_n = \frac{5n(n-1)}{2})$.

Final answer

$$\frac{5n(n-1)}{2}$$

Q202 Written

Find (S_n) for a sequence with first term (3) and common difference (-4) .**Solution**

1. $(S_n = \frac{n}{2}\{6 - 4(n - 1)\})$.
2. Therefore $(S_n = 5n - 2n^2)$.

Final answer

$$5n - 2n^2$$

Worked Solutions

Arithmetic Sequences

Q203 **Written****Find (S_n) for a sequence with first term (-4) and common difference (-2) .****Solution**

1. $(S_n = \frac{n}{2}\{-8 - 2(n - 1)\})$.
2. Therefore $(S_n = -n^2 - 3n)$.

Final answer

$$-n^2 - 3n$$

Worked Solutions

Arithmetic Sequences

Q204 Challenge

For the arithmetic sequence $(94, 89, 84, \dots)$, find the sum of the odd-numbered terms from the first term to the 11th term.

Solution

1. The odd-numbered terms are $(a_1, a_3, a_5, a_7, a_9, a_{11})$, a new arithmetic sequence with first term (94), common difference $(2(-5) = -10)$, and (6) terms.
2. Its last term is (44), so the sum is $(\frac{6}{2}(94 + 44) = 414)$.

Final answer

414

Worked Solutions

Arithmetic Sequences

Q205 Written

Find the sum of the natural numbers from 1 to 100.**Solution**

1. There are $n = 100$ terms, with $a = 1$ and $l = 100$.
2. $S_{100} = \frac{100}{2}(1 + 100) = 5050$.

Final answer

5050

Q206 Written

Find the sum of the natural numbers from 1 to 200.**Solution**

1. Use $n = 200$, $a = 1$, $l = 200$.
2. $S = \frac{200}{2}(1 + 200) = 20100$.

Final answer

20100

Worked Solutions

Arithmetic Sequences

Q207 **Written****Find the sum of the natural numbers from 1 to 150.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 150$.
2. *Therefore* $S = 11325$.

Final answer

11325

Q208 **Written****Find the sum of the natural numbers from 1 to 70.****Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 70$.
2. *Therefore* $S = 2485$.

Final answer

2485

Worked Solutions

Arithmetic Sequences

Q209 Written

Find the sum of the natural numbers from 1 to 100.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 100$.
2. *Therefore* $S = 5050$.

Final answer

5050

Q210 Written

Find the sum of the natural numbers from 1 to 400.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 400$.
2. *Therefore* $S = 80200$.

Final answer

80200

Worked Solutions

Arithmetic Sequences

Q211 Written

Find the sum of the natural numbers from 1 to 250.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 250$.
2. *Therefore* $S = 31375$.

Final answer

31375

Q212 Written

Find the sum of the natural numbers from 1 to 81.**Solution**

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 81$.
2. *Therefore* $S = 8085$.

Final answer

8085

Worked Solutions

Arithmetic Sequences

Q213 Written

Find the sum of the natural numbers from 1 to 301.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 301$.
2. Therefore $S = 24535$.

Final answer

24535

Q214 Written

Find the sum of the natural numbers from 1 to 99.

Solution

1. Use $S_n = \frac{n}{2}(1 + n)$ with $n = 99$.
2. Therefore $S = 4950$.

Final answer

4950

Worked Solutions

Arithmetic Sequences

Q215 MCQ

Find the sum of the natural numbers from 1 to 30:**Solution**

$$1. S = \frac{30}{2}(1 + 30) = 465.$$

Correct option: B**Final answer**

465

Q216 MCQ

Find the sum of the natural numbers from 1 to 45:**Solution**

$$1. S = \frac{45}{2}(1 + 45) = 1035.$$

Correct option: B**Final answer**

1035

Worked Solutions

Arithmetic Sequences

Q217 MCQ

Find the sum of the natural numbers from 1 to 60:**Solution**

$$1. S = \frac{60}{2}(1 + 60) = 1830.$$

Correct option: B

Final answer

1830

Q218 MCQ

Find the sum of the natural numbers from 1 to 75:**Solution**

$$1. S = \frac{75}{2}(1 + 75) = 2850.$$

Correct option: B

Final answer

2850

Worked Solutions

Arithmetic Sequences

Q219 MCQ

Find the sum of the natural numbers from 1 to 90:**Solution**

$$1. S = \frac{90}{2}(1 + 90) = 4095.$$

Correct option: B**Final answer**

4095

Q220 MCQ

Find the sum of the natural numbers from 1 to 125:**Solution**

$$1. S = \frac{125}{2}(1 + 125) = 7875.$$

Correct option: B**Final answer**

7875

Worked Solutions

Arithmetic Sequences

Q221 MCQ

Find the sum of the natural numbers from 1 to 180:**Solution**

$$1. S = \frac{180}{2}(1 + 180) = 16290.$$

Correct option: B

Final answer

16290

Q222 MCQ

Find the sum of the natural numbers from 1 to 220:**Solution**

$$1. S = \frac{220}{2}(1 + 220) = 24310.$$

Correct option: B

Final answer

24310

Worked Solutions

Arithmetic Sequences

Q223 MCQ

Find the sum of the natural numbers from 1 to 275:**Solution**

$$1. S = \frac{275}{2}(1 + 275) = 37950.$$

Correct option: B**Final answer**

37950

Q224 MCQ

Find the sum of the natural numbers from 1 to 350:**Solution**

$$1. S = \frac{350}{2}(1 + 350) = 61425.$$

Correct option: B**Final answer**

61425

Worked Solutions

Arithmetic Sequences

Q225 **Written****Find the sum of the natural numbers from 101 to 200.****Solution**

1. There are $n = 100$ terms, with $a = 101$ and $l = 200$.
2. $S = \frac{100}{2}(101 + 200) = 15050$.

Final answer

15050

Q226 **Written****Find the sum of the natural numbers from 201 to 300.****Solution**

1. There are 100 terms from 201 to 300.
2. $S = \frac{100}{2}(201 + 300) = 25050$.

Final answer

25050

Worked Solutions

Arithmetic Sequences

Q227 **Written****Find the sum of all two-digit natural numbers.****Solution**

1. Use $10 + 11 + \cdots + 99 : n = 90$.
2. $S = \frac{90}{2}(10 + 99) = 4905$.

Final answer

4905

Q228 **MCQ****A sequence has first term 5, last term 35, and 7 terms. Select its sum:****Solution**

1. $S = \frac{7}{2}(5 + 35) = 140$.

Correct option: B

Final answer

140

Worked Solutions

Arithmetic Sequences

Q229 MCQ

A sequence has first term 12, last term 68, and 9 terms. Select its sum:

Solution

$$1. S = \frac{9}{2}(12 + 68) = 360.$$

Correct option: B

Final answer

360

Q230 MCQ

A sequence has first term -4, last term 20, and 9 terms. Select its sum:

Solution

$$1. S = \frac{9}{2}(-4 + 20) = 72.$$

Correct option: B

Final answer

72

Worked Solutions

Arithmetic Sequences

Q231 MCQ

A sequence has first term 7, last term 42, and 8 terms. Select its sum:

Solution

1. $S = \frac{8}{2}(7 + 42) = 196.$

Correct option: B

Final answer

196

Q232 MCQ

A sequence has first term 15, last term 3, and 10 terms. Select its sum:

Solution

1. $S = \frac{10}{2}(15 + 3) = 90.$

Correct option: B

Final answer

90

Worked Solutions

Arithmetic Sequences

Q233 MCQ

A sequence has first term 10, last term 70, and 13 terms. Select its sum:

Solution

$$1. S = \frac{13}{2}(10 + 70) = 520.$$

Correct option: B

Final answer

520

Q234 MCQ

A sequence has first term 2, last term 57, and 12 terms. Select its sum:

Solution

$$1. S = \frac{12}{2}(2 + 57) = 354.$$

Correct option: B

Final answer

354

Worked Solutions

Arithmetic Sequences

Q235 MCQ

A sequence has first term 25, last term 100, and 16 terms. Select its sum:

Solution

$$1. S = \frac{16}{2}(25 + 100) = 1000.$$

Correct option: B

Final answer

1000

Q236 MCQ

A sequence has first term 101, last term 200, and 100 terms. Select its sum:

Solution

$$1. S = \frac{100}{2}(101 + 200) = 15050.$$

Correct option: B

Final answer

15050

Worked Solutions

Arithmetic Sequences

Q237 MCQ

A sequence has first term 31, last term 91, and 16 terms. Select its sum:

Solution

1. $S = \frac{16}{2}(31 + 91) = 976.$

Correct option: B

Final answer

976

Q238 Written

Find the sum of the multiples of 3 among the two-digit natural numbers.

Solution

1. The sequence is 12, 15, 18, ..., 99, so $a = 12, d = 3, l = 99.$

2. $99 = 12 + (n - 1)3$ gives $n = 30.$

3. $S = \frac{30}{2}(12 + 99) = 1665.$

Final answer

1665

Worked Solutions

Arithmetic Sequences

Q239 Written

Find the sum of all two-digit natural numbers that are multiples of 5.**Solution**

1. The sequence is 10, 15, ..., 95; $n = (95 - 10)/5 + 1 = 18$.
2. $S = \frac{18}{2}(10 + 95) = 945$.

Final answer

945

Q240 Written

Find the sum of the multiples of 4 from 1 to 100.**Solution**

1. The sequence is 4, 8, ..., 100; $n = 25$.
2. $S = \frac{25}{2}(4 + 100) = 1300$.

Final answer

1300

Worked Solutions

Arithmetic Sequences

Q241 **Written****Find the sum of all multiples of 6 from 10 to 100.****Solution**

1. The sequence is 12, 18, ..., 96; $n = 15$.
2. $S = \frac{15}{2}(12 + 96) = 810$.

Final answer

810

Q242 **MCQ****Select the sum of the multiples of 3 from 3 to 99:****Solution**

1. $n = (99 - 3)/3 + 1 = 33$; $S = \frac{33}{2}(3 + 99) = 1683$.

Correct option: B**Final answer**

1683

Worked Solutions

Arithmetic Sequences

Q243 MCQ

Select the sum of the multiples of 4 from 4 to 100:**Solution**

$$1. \ n = (100 - 4)/4 + 1 = 25; S = \frac{25}{2}(4 + 100) = 1300.$$

Correct option: B**Final answer**

1300

Q244 Written

Find the sum of the natural numbers from 1 to 100 that leave a remainder of 1 when divided by 4.**Solution**

$$1. \ \text{The sequence is } 1, 5, \dots, 97; n = 25.$$

$$2. \ S = \frac{25}{2}(1 + 97) = 1225.$$

Final answer

1225

Worked Solutions

Arithmetic Sequences

Q245 **Written****Find the sum of all two-digit natural numbers that leave a remainder of 3 when divided by 5.****Solution**

1. The sequence is 13, 18, ..., 98; $n = (98 - 13)/5 + 1 = 18$.
2. $S = \frac{18}{2}(13 + 98) = 999$.

Final answer

999

Q246 **Written****Find the sum of all two-digit odd natural numbers.****Solution**

1. The sequence is 11, 13, ..., 99; $n = 45$.
2. $S = \frac{45}{2}(11 + 99) = 2475$.

Final answer

2475

Worked Solutions**Arithmetic Sequences****Q247** Written**Find the sum of all natural numbers from 10 to 100 that are not divisible by 6.****Solution**

1. Total 10 to 100 is 5005; multiples of 6 are 12 to 96 with sum 810.
2. Required sum = $5005 - 810 = 4195$.

Final answer

4195

Worked Solutions

Arithmetic Sequences

Q248 Challenge

Find the sum of the numbers from 1 to 100 which are not divisible by 4.

Solution

1. Total $S_{100} = 5050$. Multiples of 4 have sum 1300.
2. Required sum = $5050 - 1300 = 3750$.

Final answer

3750

Worked Solutions

Arithmetic Sequences

Q249 **Written**

In an arithmetic sequence with first term 4 and common difference 3, find how many terms from the first term have sum 50.

Solution

1. $50 = \frac{n}{2}\{2(4) + (n-1)3\} \Rightarrow 3n^2 + 5n - 100 = 0.$
2. The natural root is $n = 5$; $S_5 = 50.$

Final answer

5

Q250 **Written**

Find n when $a = 3, d = 2$ and $S_n = 63.$

Solution

1. $S_n = n(n+2) = 63 \Rightarrow n = 7.$

Final answer

7

Worked Solutions

Arithmetic Sequences

Q251 MCQ

Select the correct option: Find n if $S_n = 48$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 48 \Rightarrow n = 6$.

Correct option: B

Final answer

6

Q252 MCQ

Select the correct option: Find n if $S_n = 80$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 80 \Rightarrow n = 8$.

Correct option: B

Final answer

8

Worked Solutions

Arithmetic Sequences

Q253 MCQ

Select the correct option: Find n if $S_n = 120$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 120 \Rightarrow n = 10.$$

Correct option: B**Final answer**

10

Q254 MCQ

Select the correct option: Find n if $S_n = 168$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 168 \Rightarrow n = 12.$$

Correct option: B**Final answer**

12

Worked Solutions

Arithmetic Sequences

Q255 MCQ

Select the correct option: Find n if $S_n = 255$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 255 \Rightarrow n = 15.$

Correct option: B

Final answer

15

Q256 MCQ

Select the correct option: Find n if $S_n = 360$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 360 \Rightarrow n = 18.$

Correct option: B

Final answer

18

Worked Solutions

Arithmetic Sequences

Q257 MCQ

Select the correct option: Find n if $S_n = 440$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 440 \Rightarrow n = 20.$$

Correct option: B**Final answer**

20

Q258 MCQ

Select the correct option: Find n if $S_n = 624$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 624 \Rightarrow n = 24.$$

Correct option: B**Final answer**

24

Worked Solutions

Arithmetic Sequences

Q259 MCQ

Select the correct option: Find n if $S_n = 840$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 840 \Rightarrow n = 28.$$

Correct option: B

Final answer

28

Q260 MCQ

Select the correct option: Find n if $S_n = 1088$ for $a = 3, d = 2$:**Solution**

$$1. S_n = n(n + 2) = 1088 \Rightarrow n = 32.$$

Correct option: B

Final answer

32

Worked Solutions

Arithmetic Sequences

Q261 MCQ

Select the correct option: Find n if $S_n = 1368$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 1368 \Rightarrow n = 36.$

Correct option: B

Final answer

36

Q262 MCQ

Select the correct option: Find n if $S_n = 1680$ for $a = 3, d = 2$:

Solution

1. $S_n = n(n + 2) = 1680 \Rightarrow n = 40.$

Correct option: B

Final answer

40

Worked Solutions

Arithmetic Sequences

Q263 Written

For first term 17 and common difference -2 , find how many terms have sum -63 .

Solution

1. $S_n = n(18 - n)$.
2. $n(18 - n) = -63 \Rightarrow n^2 - 18n - 63 = 0$; hence $n = 21$.

Final answer

21

Q264 Written

Find n when $a = 24$, $d = -4$ and $S_n = -28$.

Solution

1. $S_n = n(26 - 2n) = -28 \Rightarrow n^2 - 13n - 14 = 0$; $n = 14$.

Final answer

14

Worked Solutions

Arithmetic Sequences

Q265 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 80$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 80$; positive root $n = 5$.

Correct option: C**Final answer**

5

Q266 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 133$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 133$; positive root $n = 7$.

Correct option: C**Final answer**

7

Worked Solutions

Arithmetic Sequences

Q267 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 198$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 198$; positive root $n = 9$.

Correct option: C**Final answer**

9

Q268 MCQ

Select the correct option: For $a = 10, d = 3$, find n when $S_n = 275$:**Solution**

1. $S_n = \frac{n}{2}(17 + 3n) = 275$; positive root $n = 11$.

Correct option: C**Final answer**

11

Worked Solutions

Arithmetic Sequences

Q269 Written

Find the general term of $\{a_n\}$ with first term -60 and sum of the first 15 terms -60 .

Solution

1. $-60 = \frac{15}{2}\{2(-60) + 14d\} \Rightarrow d = 8.$
2. $a_n = -60 + 8(n - 1) = 8n - 68.$

Final answer

$$a_n = 8n - 68$$

Q270 Written

Find the general term with first term 8 and sum of the first 13 terms -130 .

Solution

1. $-130 = \frac{13}{2}(16 + 12d) \Rightarrow d = -3.$
2. $a_n = 8 - 3(n - 1) = 11 - 3n.$

Final answer

$$a_n = 11 - 3n$$

Worked Solutions

Arithmetic Sequences

Q271 Written

Find the general term with first term 4 and $S_6 = 114$.**Solution**

$$1. \frac{6}{2}(8 + 5d) = 114 \Rightarrow d = 6.$$

$$2. a_n = 4 + 6(n - 1) = 6n - 2.$$

Final answer

$$a_n = 6n - 2$$

Q272 Written

Find the general term when $a_3 = 10$ and $S_{10} = 175$.**Solution**

$$1. a + 2d = 10, 2a + 9d = 35 \Rightarrow d = 3, a = 4.$$

$$2. a_n = 4 + 3(n - 1) = 3n + 1.$$

Final answer

$$a_n = 3n + 1$$

Worked Solutions

Arithmetic Sequences

Q273 Written

For first term 65 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Solution

1. $a_n = 65 - 4n$; $a_n > 0 \Rightarrow n < 17.25$.
2. The last positive term is $a_{17} = 1$; $S_{17} = \frac{17}{2}(65 + 1) = 561$.

Final answer

$$n = 17, S = 561$$

Q274 Written

For first term 70 and common difference -4 , find the number of terms giving the maximum sum and the sum.

Solution

1. $a_n = 70 - 4n$; $a_{18} = 2$ is the last positive term.
2. $S_{18} = \frac{18}{2}(70 + 2) = 648$.

Final answer

$$n = 18, S = 648$$

Worked Solutions

Arithmetic Sequences

Q275 Written

For $a = 20, d = -6$, find n giving the maximum sum and the sum.

Solution

1. $a_n = 26 - 6n$; the last positive term is $a_4 = 2$.
2. $S_4 = \frac{4}{2}(20 + 2) = 44$.

Final answer

$$n = 4, S = 44$$

Q276 Written

For $a = 126, d = -13$, find n giving the maximum sum and the sum.

Solution

1. $a_n = 139 - 13n$; the last positive term is $a_{10} = 9$.
2. $S_{10} = \frac{10}{2}(126 + 9) = 675$.

Final answer

$$n = 10, S = 675$$

Worked Solutions

Arithmetic Sequences

Q277 Written

Find the general term where $S_{10} = 10$ and $S_{20} = 40$.**Solution**

1. $2a + 9d = 2, 2a + 19d = 4 \Rightarrow d = \frac{1}{5}, a = \frac{1}{10}.$
2. $a_n = \frac{1}{10} + (n-1)\frac{1}{5} = \frac{2n-1}{10}.$

Final answer

$$a_n = \frac{2n-1}{10}$$

Q278 Written

Find the general term where $S_6 = 3$ and $S_{10} = 65$.**Solution**

1. $2a + 5d = 1, 2a + 9d = 13 \Rightarrow d = 3, a = -7.$
2. $a_n = -7 + 3(n-1) = 3n - 10.$

Final answer

$$a_n = 3n - 10$$

Worked Solutions

Arithmetic Sequences

Q279 Written

Find the general term where $S_{20} = 400$ and $S_{30} = 900$.**Solution**

$$1. \ 2a + 19d = 40, \ 2a + 29d = 60 \Rightarrow d = 2, \ a = 1.$$

$$2. \ a_n = 1 + 2(n - 1) = 2n - 1.$$

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Arithmetic Sequences

Q280 Challenge

The 15th term is 33 and the 50th term is 103. Find the sum of the first 79 terms.

Solution

1. $35d = 70 \Rightarrow d = 2, a = 5.$
2. $S_{79} = \frac{79}{2}\{2(5) + 78(2)\} = 6557.$

Final answer

6557

Worked Solutions

Arithmetic Sequences

Quiz MCQ 1 **Concept Quiz · MCQ**

For $a_n = \frac{2n+1}{n+2}$, what is a_5 ?:

Solution

1. Substitute $n = 5$: $a_5 = (2(5)+1)/(5+2) = 11/7$.

Correct option: A

Final answer

$$\frac{11}{7}$$

Quiz MCQ 2 **Concept Quiz · MCQ**

An arithmetic sequence has $a_3 = 10$ and $a_8 = 30$. Select its general term:

Solution

1. $10 = a + 2d$ and $30 = a + 7d$, so $d = 4$, $a = -2$.

2. $a_n = 4n - 2$.

Correct option: B

Final answer

$$a_n = 4n - 2$$

Worked Solutions

Arithmetic Sequences

Quiz MCQ 3 **Concept Quiz · MCQ**

For $a = 12$, $d = -2$, select the sum of odd-numbered terms through a_9 :

Solution

1. Odd terms are 12, 8, 4, 0, -4 ; their sum is 20.

Correct option: C

Final answer

20

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: If $S_{10} = 10$ and $S_{20} = 40$, find d :

Solution

1. $d = \frac{1}{5}$.

Correct option: B

Final answer

$\frac{1}{5}$

Worked Solutions

Arithmetic Sequences

Quiz WRITTEN 5 Concept Quiz · Written

If $a_n = 3n^2 - 2n + 1$, find a_1, a_2, a_3 .

Solution

1. Substitution gives $a_1=2$, $a_2=9$, and $a_3=22$.

Final answer

$$a_1 = 2, a_2 = 9, a_3 = 22$$

Quiz WRITTEN 6 Concept Quiz · Written

Find a_n for an arithmetic sequence with $a_4 = 14$ and $a_{10} = 62$.

Solution

1. $14 = a + 3d$ and $62 = a + 9d$, so $d = 8$, $a = -10$.

2. $a_n = 8n - 18$.

Final answer

$$a_n = 8n - 18$$

Worked Solutions

Arithmetic Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For $a = 9, d = 4$, find the sum of odd-numbered terms from a_1 to a_{15} .

Solution

1. Selected terms have 8 terms, first 9, last 65.
2. $\text{Sum} = 8(9 + 65)/2 = 296$.

Final answer

296

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term when $S_{20} = 400$ and $S_{30} = 900$.

Solution

1. $d = 2, a = 1$; therefore $a_n = 2n - 1$.

Final answer

$a_n = 2n - 1$

Answer key

Use the question number shown on each page.

Q001 $a_1 = -1, a_2 = 1, a_3 = 3$

Q002 $a_1 = -6, a_2 = -10, a_3 = -12$

Q003 $a_1 = -3, a_2 = 2, a_3 = 7$

Q004 $a_1 = -2, a_2 = -5, a_3 = -8$

Q005 $a_1 = -6, a_2 = -8, a_3 = -10$

Q006 $a_1 = 0, a_2 = 3, a_3 = 8$

Q007 $a_1 = 3, a_2 = 7, a_3 = 13$

Q008 $a_1 = 7, a_2 = 12, a_3 = 19$

Q009 B

Q010 C

Q011 D

Q012 A

Q013 B

Q014 C

Q015 D

Q016 A

Q017 B

Q018 C

Q019 D

Q020 A

Q021 $a_1 = 6, a_2 = 18, a_3 = 54$

Q022 $a_1 = 2, a_2 = 8, a_3 = 26$

Q023 $a_1 = -3, a_2 = 9, a_3 = -27$

Q024 $a_1 = -6, a_2 = 12, a_3 = -24$

Q025 A

Q026 B

Q027 C

Q028 D

Q029 A

Q030 B

Q031 C

Q032 D

Q033 $a_1 = \frac{1}{2}, a_2 = \frac{2}{3}, a_3 = \frac{3}{4}$

Q034 $a_1 = \frac{1}{3}, a_2 = \frac{2}{5}, a_3 = \frac{3}{7}$

Q035 $a_1 = 1, a_2 = \frac{1}{4}, a_3 = \frac{1}{7}$

Q036 $a_1 = \frac{1}{2}, a_2 = \frac{4}{5}, a_3 = 1$

Q037 $a_1 = 2, a_2 = 1, a_3 = \frac{4}{5}$

Q038 C

Q039 D

Q040 A

Q041 B

Q042 C

Q043 D

Q044 $a_n = (n-1)^2$

Q045 $a_n = (n-1)^3$

Q046 $a_n = n^2$

Q047 $a_n = (n+6)^2$

Q048 B

Q049 C

Q050 D

Q051 A

Q052 B

Q053 C

Q054 $a_n = 3n - 1$

Q055 $a_n = 2n - 2$

Q056 $a_n = 3n - 3$

Q057 $a_n = 5n - 1$

Q058 $a_n = 7n - 1$

Q059 A

Q060 B

Q061 C

Q062 D

Q063 $a_n = (-1)^{n-1} 2n$

Q064 $a_n = (-1)^n 3n$

Q065 $a_n = (-1)^n 5n$

Q066 $a_n = (-1)^n (3n - 2)$

Q067 $a_n = \frac{n}{2n-1}$

Q068 $a_n = \frac{2n-1}{n^2}$

Q069 $a_n = \frac{(-1)^{n-1}}{2n-1}$

Q070 $a_n = \frac{n}{n+1}$

Q071 $a_n = n 3^n$

Q072 $a_n = \frac{n}{n+1}; a_{20} = \frac{20}{21}$

Q073 (a) $a = 2, d = 3(b) a = 3, d = -5$

Q074 (a) $a = 1, d = 4(b) a = 23, d = -4$

Q075 B

Q076 A

Q077 C

Q078 D

Q079 (a)(3, 2); (b)(-13, 4); (c)(-2, 3); (d)(10, -3); (e)(6, $-\frac{2}{3}$); (f)(-4, $-\frac{1}{2}$).

Q080 B

Q081 A

Q082 C

Q083 B

Q084 (a) $d = -3, A = 5, B = -4(b) d = -5, A = 8, B = 3$

Q085 (a) $d = 7, A = 16, B = 23(b) d = -6, A = 23, B = 11$

Q086 A

Q087 C

Q088 B

Q089 (a)(d) = 7, A = 22, B = 20; (b)(d) = 6, A = 19, B = 21; (c)(d) = -5, A = 14, B = -1; (d)(d) = -2, A = 17, B = 13; (e)(d) = 3, A = 14, B = 17; (f)(d) = -4, A = 13, B = 9, C = 1.

Q090 (a) $a_n = 3n + 1, a_7 = 22(b) a_n = 5 - 3n, a_7 = -16$

Q091 (a) $a_n = 4n + 1, a_7 = 29$; (b) $a_n = 3n - 5, a_7 = 16$; (c) $a_n = 13 - 3n, a_7 = -8$; (d) $a_n = -\frac{n}{2}, a_7 = -\frac{7}{2}$.

Q092 $a_n = 8 + (n-1)(-6) = 14 - 6n; a_{20} = -106$

Q093 (a) $a_n = 3n - 10, a_{10} = 20$; (b) $a_n = 1 - 3n, a_{10} = -29$; (c) $a_n = \frac{n+13}{2}, a_{10} = \frac{23}{2}$; (d) $a_n = \frac{3-n}{3}, a_{10} = -\frac{7}{3}$.

Q094 $a = 7, d = 3$.

Q095 $a_n = 5n - 10$

Q096 $a_n = 4n + 2$

Q097 $a_n = 23 - 3n$

Q098 $a_n = 9n - 17$

Q099 $a_n = -5n - 13$

Q100 D

Q101 B	Q133 A	Q165 144	Q197 C
Q102 A	Q134 B	Q166 -165	Q198 C
Q103 B	Q135 $n = 125$, the 125th term	Q167 176	Q199 $2n^2 - n$
Q104 $a_n = 3n + 33$	Q136 $n = 35$, the 35th term	Q168 360	Q200 $\frac{n(7-3n)}{2}$
Q105 $a_n = 2n - 12$	Q137 A	Q169 C	Q201 $\frac{5n(n-1)}{2}$
Q106 $a_n = 13 - 4n$	Q138 D	Q170 B	Q202 $5n - 2n^2$
Q107 $a_n = 18 - 7n$	Q139 D	Q171 C	Q203 $-n^2 - 3n$
Q108 A	Q140 B	Q172 B	Q204 414
Q109 D	Q141 $n = 72$, the 72nd term	Q173 C	Q205 5050
Q110 C	Q142 $n = 171$, the 171st term	Q174 B	Q206 20100
Q111 D	Q143 B	Q175 392	Q207 11325
Q112 $a_n = 8n - 14$	Q144 B	Q176 343	Q208 2485
Q113 $a_n = 20 - 3n$	Q145 C	Q177 741	Q209 5050
Q114 $a_n = 8n - 18$	Q146 A	Q178 354	Q210 80200
Q115 $a_n = 42 - 4n$	Q147 B	Q179 1716	Q211 31375
Q116 $a_n = 79 - 3n$	Q148 $a = 4$	Q180 C	Q212 8085
Q117 $a_n = 8n - 18$	Q149 $x = -4$ or $x = 3$	Q181 C	Q213 24535
Q118 C	Q150 $x = -2$ or $x = 4$	Q182 C	Q214 4950
Q119 C	Q151 $x = -1$ or $x = 3$	Q183 C	Q215 B
Q120 B	Q152 $x = 0$, $y = -2$, $z = -4$	Q184 C	Q216 B
Q121 A	Q153 105	Q185 B	Q217 B
Q122 A	Q154 100	Q186 0	Q218 B
Q123 B	Q155 120	Q187 208	Q219 B
Q124 $a_n = \frac{2n-37}{5}$	Q156 152	Q188 B	Q220 B
Q125 A	Q157 155	Q189 C	Q221 B
Q126 D	Q158 B	Q190 C	Q222 B
Q127 $n = 28$, the 28th term	Q159 B	Q191 C	Q223 B
Q128 $n = 14$, the 14th term	Q160 B	Q192 B	Q224 B
Q129 $n = 22$, the 22nd term	Q161 B	Q193 C	Q225 15050
Q130 $n = 40$, the 40th term	Q162 B	Q194 66	Q226 25050
Q131 C	Q163 C	Q195 750	Q227 4905
Q132 D	Q164 -130	Q196 154	

Q228 B	Q244 1225	Q260 B	Q276 $n = 10, S = 675$
Q229 B	Q245 999	Q261 B	Q277 $a_n = \frac{2n-1}{10}$
Q230 B	Q246 2475	Q262 B	Q278 $a_n = 3n - 10$
Q231 B	Q247 4195	Q263 21	Q279 $a_n = 2n - 1$
Q232 B	Q248 3750	Q264 14	Q280 6557
Q233 B	Q249 5	Q265 C	Quiz MCQ 1 A
Q234 B	Q250 7	Q266 C	Quiz MCQ 2 B
Q235 B	Q251 B	Q267 C	Quiz MCQ 3 C
Q236 B	Q252 B	Q268 C	Quiz MCQ 4 B
Q237 B	Q253 B	Q269 $a_n = 8n - 68$	Quiz WRITTEN 5 $a_1 = 2, a_2 = 9, a_3 = 22$
Q238 1665	Q254 B	Q270 $a_n = 11 - 3n$	Quiz WRITTEN 6 $a_n = 8n - 18$
Q239 945	Q255 B	Q271 $a_n = 6n - 2$	Quiz WRITTEN 7 296
Q240 1300	Q256 B	Q272 $a_n = 3n + 1$	Quiz WRITTEN 8 $a_n = 2n - 1$
Q241 810	Q257 B	Q273 $n = 17, S = 561$	
Q242 B	Q258 B	Q274 $n = 18, S = 648$	
Q243 B	Q259 B	Q275 $n = 4, S = 44$	

Concept 2 | Geometric Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1}$$

Infinite sum

$$S_{\infty} = \frac{a}{1-r} \quad (|r| < 1)$$

Worked Solutions

Geometric Sequences

Q001 Written

Find the general term of a geometric sequence whose first term is 2 and common ratio is 3.

Solution

1. Here $a = 2$ and $r = 3$.
2. $a_n = 2 \times 3^{n-1}$

Final answer

$$2 \times 3^{n-1}$$

Q002 Written

Find the general term when the first term is 3 and the common ratio is 4.

Solution

1. $a_n = 3 \times 4^{n-1}$

Final answer

$$3 \times 4^{n-1}$$

Worked Solutions

Geometric Sequences

Q003 **Written**

Find the general term of the geometric sequence in part a. The first term is 5, common ratio 2.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5 \times 2^{n-1}$.

Final answer

$$5 \times 2^{n-1}$$

Q004 **Written**

Find the general term of the geometric sequence in part b. The first term is 2, common ratio 5.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2 \times 5^{n-1}$.

Final answer

$$2 \times 5^{n-1}$$

Worked Solutions

Geometric Sequences

Q005 Written

Find the general term of the geometric sequence in part c. The first term is 7, common ratio 4.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $7 \times 4^{n-1}$.

Final answer

$$7 \times 4^{n-1}$$

Q006 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = 3$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = 3$.

Correct option: A

Final answer

$$a_n = 2(3)^4$$

Worked Solutions

Geometric Sequences

Q007 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = 2$, and $n = 6$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = 2$.

Correct option: A

Final answer

$$a_n = 3(2)^5$$

Q008 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 4$, and $n = 4$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 5$ and $r = 4$.

Correct option: A

Final answer

$$a_n = 5(4)^3$$

Worked Solutions

Geometric Sequences

Q009 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = 5$, and $n = 3$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 4$ and $r = 5$.

Correct option: A

Final answer

$$a_n = 4(5)^2$$

Q010 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 2$, and $n = 7$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 5$ and $r = 2$.

Correct option: A

Final answer

$$a_n = 5(2)^6$$

Worked Solutions

Geometric Sequences

Q011 MCQ

Select the correct option: For first term $a = 7$, common ratio $r = 4$, and $n = 3$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 7$ and $r = 4$.

Correct option: A

Final answer

$$a_n = 7(4)^2$$

Q012 Written

Find the general term when the first term is 5 and the common ratio is -2 .

Solution

1. $a_n = 5(-2)^{n-1}$

Final answer

$$5(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q013 Written

Find the general term of the geometric sequence in part d. The first term is 2, common ratio -3 .

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2(-3)^{n-1}$.

Final answer

$$2(-3)^{n-1}$$

Q014 Written

Find the general term of the geometric sequence in part e. The first term is 3, common ratio -5 .

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3(-5)^{n-1}$.

Final answer

$$3(-5)^{n-1}$$

Worked Solutions

Geometric Sequences

Q015 Written

Find the general term of the geometric sequence in part f. The first term is 4, common ratio -2 .

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $4(-2)^{n-1}$.

Final answer

$$4(-2)^{n-1}$$

Q016 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = -3$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = -3$.

Correct option: A

Final answer

$$a_n = 2(-3)^4$$

Worked Solutions

Geometric Sequences

Q017 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -2$, and $n = 6$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = -2$.

Correct option: A

Final answer

$$a_n = 3(-2)^5$$

Q018 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = -1$, and $n = 4$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 6$ and $r = -1$.

Correct option: A

Final answer

$$a_n = 6(-1)^3$$

Worked Solutions

Geometric Sequences

Q019 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -2$, and $n = 3$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 8$ and $r = -2$.

Correct option: A

Final answer

$$a_n = 8(-2)^2$$

Q020 MCQ

Select the correct option: For first term $a = 9$, common ratio $r = -2$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 9$ and $r = -2$.

Correct option: A

Final answer

$$a_n = 9(-2)^4$$

Worked Solutions

Geometric Sequences

Q021 Written

Find the general term when the first term is 8 and the common ratio is $-\frac{1}{3}$.

Solution

1. $a_n = 8\left(-\frac{1}{3}\right)^{n-1}$

Final answer

$$8\left(-\frac{1}{3}\right)^{n-1}$$

Q022 Written

Find the general term of the geometric sequence in part g. The first term is 4, common ratio $\frac{1}{3}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $4\left(\frac{1}{3}\right)^{n-1}$.

Final answer

$$4\left(\frac{1}{3}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q023 Written

Find the general term of the geometric sequence in part h. The first term is 3, common ratio $-\frac{1}{2}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3\left(-\frac{1}{2}\right)^{n-1}$.

Final answer

$$3\left(-\frac{1}{2}\right)^{n-1}$$

Q024 Written

Find the general term of the geometric sequence in part i. The first term is 6, common ratio $\frac{1}{2}$.

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $6\left(\frac{1}{2}\right)^{n-1}$.

Final answer

$$6\left(\frac{1}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q025 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = \frac{1}{3}$, and $n = 5$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 4$ and $r = \frac{1}{3}$.

Correct option: A

Final answer

$$a_n = 4 \left(\frac{1}{3} \right)^4$$

Q026 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -\frac{1}{2}$, and $n = 6$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 3$ and $r = -\frac{1}{2}$.

Correct option: A

Final answer

$$a_n = 3 \left(-\frac{1}{2} \right)^5$$

Worked Solutions

Geometric Sequences

Q027 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = \frac{1}{2}$, and $n = 4$, which expression gives a_n ?

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 6$ and $r = \frac{1}{2}$.

Correct option: A

Final answer

$$a_n = 6 \left(\frac{1}{2} \right)^3$$

Q028 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = \frac{3}{2}$, and $n = 5$, which expression gives a_n ?

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 2$ and $r = \frac{3}{2}$.

Correct option: A

Final answer

$$a_n = 2 \left(\frac{3}{2} \right)^4$$

Worked Solutions

Geometric Sequences

Q029 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -\frac{3}{2}$, and $n = 4$, which expression gives a_n ?:

Solution

1. Use $a_n = ar^{n-1}$; substitute $a = 8$ and $r = -\frac{3}{2}$.

Correct option: A

Final answer

$$a_n = 8 \left(-\frac{3}{2}\right)^3$$

Q030 Written

Find the general term of 5, 10, 20, 40,

Solution

1. The common ratio is $r = 10/5 = 2$.

2. $a_n = 5 \times 2^{n-1}$

Final answer

$$5 \times 2^{n-1}$$

Worked Solutions

Geometric Sequences

Q031 **Written****Find the general term of 2, 6, 18, 54,****Solution**

1. The common ratio is 3.
2. $a_n = 2 \times 3^{n-1}$

Final answer

$$2 \times 3^{n-1}$$

Q032 **Written****Find the general term of the geometric sequence in part j. The sequence is 3, 6, 12, 24,****Solution**

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3 \times 2^{n-1}$.

Final answer

$$3 \times 2^{n-1}$$

Worked Solutions

Geometric Sequences

Q033 MCQ

Select the correct option: Find the general term of the geometric sequence 3, 6, 12, 24, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$3 \times 2^{n-1}$$

Q034 MCQ

Select the correct option: Find the general term of the geometric sequence 2, 10, 50, 250, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$2 \times 5^{n-1}$$

Worked Solutions

Geometric Sequences

Q035 MCQ

Select the correct option: Find the general term of the geometric sequence 4, 8, 16, 32, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$4 \times 2^{n-1}$$

Q036 MCQ

Select the correct option: Find the general term of the geometric sequence 7, 21, 63, 189, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$7 \times 3^{n-1}$$

Worked Solutions

Geometric Sequences

Q037 MCQ

Select the correct option: Find the general term of the geometric sequence 1, 4, 16, 64, ...:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$4^{n-1}$$

Q038 Written

Find the general term of the geometric sequence in part k. The sequence is 2, -6, 18, -54,

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $2(-3)^{n-1}$.

Final answer

$$2(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q039 Written

Find the general term of the geometric sequence in part I. The sequence is 5, -10, 20, -40, ...

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5(-2)^{n-1}$.

Final answer

$$5(-2)^{n-1}$$

Q040 MCQ

Select the correct option: Find the general term of the geometric sequence 5, -10, 20, -40, ...:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$5(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q041 MCQ

Select the correct option: Find the general term of the geometric sequence $2, -6, 18, -54, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$2(-3)^{n-1}$$

Q042 MCQ

Select the correct option: Find the general term of the geometric sequence $-4, 8, -16, 32, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$-4(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q043 MCQ

Select the correct option: Find the general term of the geometric sequence 6, -18, 54, -162, ...:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$6(-3)^{n-1}$$

Q044 Written

Find the general term of $-4, 6, -9, \frac{27}{2}, \dots$ **Solution**1. The common ratio is $r = 6/(-4) = -3/2$.

2. $a_n = -4\left(-\frac{3}{2}\right)^{n-1}$

Final answer

$$-4\left(-\frac{3}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q045 Written

Find the general term of $-8, 12, -18, 27, \dots$ **Solution**

1. The common ratio is $-3/2$.
2. $a_n = -8\left(-\frac{3}{2}\right)^{n-1}$

Final answer

$$-8\left(-\frac{3}{2}\right)^{n-1}$$

Q046 Written

Find the general term of $2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots$ **Solution**

1. The common ratio is $1/3$.
2. $a_n = 2\left(\frac{1}{3}\right)^{n-1}$

Final answer

$$2\left(\frac{1}{3}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q047 Written

Find the general term of the geometric sequence in part m. The sequence is 48, 24, 12, 6,

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $48\left(\frac{1}{2}\right)^{n-1}$.

Final answer

$$48\left(\frac{1}{2}\right)^{n-1}$$

Q048 Written

Find the general term of the geometric sequence in part n. The sequence is $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \frac{9}{16}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$.

Final answer

$$\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q049 Written

Find the general term of the geometric sequence in part o. The sequence is $8, 16\sqrt{2}, 64, 128\sqrt{2}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $8(2\sqrt{2})^{n-1}$.

Final answer

$$8(2\sqrt{2})^{n-1}$$

Q050 Written

Find the general term of the geometric sequence in part p. The sequence is $3, -\frac{15}{2}, \frac{75}{4}, -\frac{375}{8}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $3\left(-\frac{5}{2}\right)^{n-1}$.

Final answer

$$3\left(-\frac{5}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q051 Written

Find the general term of the geometric sequence in part q. The sequence is $9, -3\sqrt{3}, 3, -\sqrt{3}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$.

Final answer

$$9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$$

Q052 Written

Find the general term of the geometric sequence in part r. The sequence is $5, -\frac{10}{3}, \frac{20}{9}, -\frac{40}{27}, \dots$

Solution

1. Use $a_n = ar^{n-1}$ with the stated first term and ratio.
2. The result is $5\left(-\frac{2}{3}\right)^{n-1}$.

Final answer

$$5\left(-\frac{2}{3}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q053 MCQ

Select the correct option: Find the general term of the geometric sequence $-8, 12, -18, 27, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$-8\left(-\frac{3}{2}\right)^{n-1}$$

Q054 MCQ

Select the correct option: Find the general term of the geometric sequence $48, 24, 12, 6, \dots$:**Solution**1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$48\left(\frac{1}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q055 MCQ

Select the correct option: Find the general term of the geometric sequence $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$\frac{1}{6} \left(\frac{3}{2} \right)^{n-1}$$

Q056 MCQ

Select the correct option: Find the general term of the geometric sequence $8, 16\sqrt{2}, 64, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$8 \left(2\sqrt{2} \right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q057 MCQ

Select the correct option: Find the general term of the geometric sequence $9, -3\sqrt{3}, 3, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$$

Q058 MCQ

Select the correct option: Find the general term of the geometric sequence $3, -\frac{3}{2}, \frac{3}{4}, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$3\left(-\frac{1}{2}\right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q059 MCQ

Select the correct option: Find the general term of the geometric sequence $10, 5, \frac{5}{2}, \dots$:

Solution

1. Divide consecutive terms to obtain r , then use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$10\left(\frac{1}{2}\right)^{n-1}$$

Q060 Written

Find the common ratio and missing values in $A, -18, 6, B, \dots$

Solution

1. $r = 6/(-18) = -\frac{1}{3}$

2. $A = 54, \quad B = 6\left(-\frac{1}{3}\right) = -2$

Final answer

$$r = -\frac{1}{3}, A = 54, B = -2$$

Worked Solutions

Geometric Sequences

Q061 MCQ

Select the correct option: The sequence $A, 12, 6, B, \dots$ is geometric. Find r, A, B :

Solution

1. Divide adjacent known terms to find r , then extend one step in each direction.

Correct option: A

Final answer

$$r = \frac{1}{2}, A = 24, B = 3$$

Q062 Written

Find the common ratio and the missing values in $A, 24, 12, B, \dots$

Solution

1. $24r = 12 \Rightarrow r = \frac{1}{2}$

2. $A\left(\frac{1}{2}\right) = 24 \Rightarrow A = 48, \quad B = 12\left(\frac{1}{2}\right) = 6$

Final answer

$$r = \frac{1}{2}, A = 48, B = 6$$

Worked Solutions

Geometric Sequences

Q063 Written

Find the common ratio and missing values in $A, -3, 9, B$.**Solution**

$$1. \ r = -3, \ A = 1, \ B = -27$$

Final answer

$$r = -3, \ A = 1, \ B = -27$$

Q064 Written

Find the common ratio and missing values in $A, \frac{1}{3}, \frac{1}{9}, B$.**Solution**

$$1. \ r = \frac{1}{3}, \ A = 1, \ B = \frac{1}{27}$$

Final answer

$$r = \frac{1}{3}, \ A = 1, \ B = \frac{1}{27}$$

Worked Solutions

Geometric Sequences

Q065 Written

Find the common ratio and missing values in $A, -\frac{1}{4}, \frac{1}{8}, B$.

Solution

1. $r = -\frac{1}{2}, \quad A = \frac{1}{2}, \quad B = -\frac{1}{16}$

Final answer

$$r = -\frac{1}{2}, A = \frac{1}{2}, B = -\frac{1}{16}$$

Worked Solutions

Geometric Sequences

Q066 Challenge

In an increasing geometric sequence, if $a_3 + a_5 = 80$ and $a_4 = 24$, find a_7 .

Solution

1. $a_3 = \frac{24}{r}$, $a_5 = 24r$, $24\left(r + \frac{1}{r}\right) = 80$

2. $3r^2 - 10r + 3 = 0 \Rightarrow r = 3$ or $\frac{1}{3}$

3. The sequence is increasing, so $r = 3$; hence $a_7 = 24(3)^3 = 648$.

Final answer

$$a_7 = 648$$

Worked Solutions

Geometric Sequences

Q067 Written

Find the general term of a geometric sequence whose 4th term is 24 and 6th term is 96.

Solution

1. $r^2 = 96/24 = 4 \Rightarrow r = \pm 2$
2. $(a, r) = (3, 2)$ or $(-3, -2)$
3. Therefore both real branches are retained.

Final answer

$$a_n = 3(2)^{n-1} \text{ or } -3(-2)^{n-1}$$

Q068 Written

Find the general term when the 3rd term is 18 and the 5th term is 162.

Solution

1. $r^2 = 162/18 = 9 \Rightarrow r = \pm 3$
2. $a = 18/r^2 = 2$

Final answer

$$a_n = 2(3)^{n-1} \text{ or } 2(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q069 Written

Find the requested values. The 2nd term is 6 and the 4th term is 54.

Solution

1. $r^2 = 54/6 = 9 \Rightarrow r = \pm 3$

2. $a = 6/r$

Final answer

$$a_n = 2(3)^{n-1} \text{ or } -2(-3)^{n-1}$$

Q070 Written

Find the requested values. The 3rd term is 27 and the 5th term is 243.

Solution

1. $r^2 = 243/27 = 9 \Rightarrow r = \pm 3$

2. $a = 3$

Final answer

$$a_n = 3(3)^{n-1} \text{ or } 3(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q071 **Written****Find the requested values. The 2nd term is 14 and the 5th term is 112.****Solution**

1. $r^3 = 112/14 = 8 \Rightarrow r = 2$
2. $a = 7$

Final answer

$$a_n = 7(2)^{n-1}$$

Q072 **Written****Find the general term when the 3rd term is 12 and the 6th term is -96 .****Solution**

1. $r^3 = -96/12 = -8 \Rightarrow r = -2$
2. $a = 12/r^2 = 3$
3. $a_n = 3(-2)^{n-1}$

Final answer

$$a_n = 3(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q073 Written

Find the general term when the 2nd term is -6 and the 5th term is 48 .

Solution

1. $r^3 = 48/(-6) = -8 \Rightarrow r = -2$

2. $a = -6/r = 3$

Final answer

$$a_n = 3(-2)^{n-1}$$

Q074 Written

Find the general term when the 3rd term is 12 and the 7th term is 192 .

Solution

1. $r^4 = 192/12 = 16 \Rightarrow r = \pm 2$

2. $a = 12/r^2 = 3$

Final answer

$$a_n = 3(2)^{n-1} \text{ or } 3(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q075 **Written****Find the requested values. The 3rd term is -28 and the 5th term is -112 .****Solution**

1. $r^2 = 4 \Rightarrow r = \pm 2$
2. $a = -28/4 = -7$

Final answer

$$a_n = -7(2)^{n-1} \text{ or } -7(-2)^{n-1}$$

Q076 **Written****Find the requested values. The 4th term is 16 and the 7th term is -128 .****Solution**

1. $r^3 = -8 \Rightarrow r = -2$
2. $a = -2$

Final answer

$$a_n = -2(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q077 Written

Find the requested values. The 3rd term is -20 and the 6th term is 160 .

Solution

1. $r^3 = -8 \Rightarrow r = -2$
2. $a = -5$

Final answer

$$a_n = -5(-2)^{n-1}$$

Q078 Written

Find the requested values. The 2nd term is 3 and the 6th term is 768 .

Solution

1. $r^4 = 768/3 = 256 \Rightarrow r = \pm 4$
2. $a = 3/r$

Final answer

$$a_n = \frac{3}{4}(4)^{n-1} \text{ or } -\frac{3}{4}(-4)^{n-1}$$

Worked Solutions

Geometric Sequences

Q079 Written

Find the requested values. The 3rd term is 9 and the 7th term is $729/16$.

Solution

1. $r^4 = 81/16 \Rightarrow r = \pm \frac{3}{2}$
2. $a = 4$

Final answer

$$a_n = 4\left(\frac{3}{2}\right)^{n-1} \text{ or } 4\left(-\frac{3}{2}\right)^{n-1}$$

Q080 Written

Find the requested values. The 4th term is $2/9$ and the 8th term is 18.

Solution

1. $r^4 = 81 \Rightarrow r = \pm 3$
2. $a = \frac{2/9}{r^3} = \pm \frac{2}{243}$

Final answer

$$a_n = \frac{2}{243}(3)^{n-1} \text{ or } -\frac{2}{243}(-3)^{n-1}$$

Worked Solutions

Geometric Sequences

Q081 MCQ

Select the correct option: For $a = 2$, $r = 3$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 2(3)^{n-1}$$

Q082 MCQ

Select the correct option: For $a = 3$, $r = 2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 3(2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q083 MCQ

Select the correct option: For $a = 5$, $r = 4$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 5(4)^{n-1}$$

Q084 MCQ

Select the correct option: For $a = 4$, $r = 5$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 4(5)^{n-1}$$

Worked Solutions

Geometric Sequences

Q085 MCQ

Select the correct option: For $a = 2$, $r = -3$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 2(-3)^{n-1}$$

Q086 MCQ

Select the correct option: For $a = 3$, $r = -2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 3(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q087 MCQ

Select the correct option: For $a = 6$, $r = -1$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 6(-1)^{n-1}$$

Q088 MCQ

Select the correct option: For $a = 8$, $r = -2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 8(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q089 MCQ

Select the correct option: For $a = 4$, $r = \frac{1}{3}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 4 \left(\frac{1}{3} \right)^{n-1}$$

Q090 MCQ

Select the correct option: For $a = 3$, $r = -\frac{1}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 3 \left(-\frac{1}{2} \right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q091 MCQ

Select the correct option: For $a = 6$, $r = \frac{1}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 6 \left(\frac{1}{2} \right)^{n-1}$$

Q092 MCQ

Select the correct option: For $a = 2$, $r = \frac{3}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 2 \left(\frac{3}{2} \right)^{n-1}$$

Worked Solutions

Geometric Sequences

Q093 MCQ

Select the correct option: For $a = 8$, $r = -\frac{3}{2}$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 8 \left(-\frac{3}{2}\right)^{n-1}$$

Q094 MCQ

Select the correct option: For $a = 5$, $r = 2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 5(2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q095 MCQ

Select the correct option: For $a = 7, r = 4$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 7(4)^{n-1}$$

Q096 MCQ

Select the correct option: For $a = 9, r = -2$, which expression gives the general term a_n :

Solution

1. Use $a_n = ar^{n-1}$.

Correct option: A

Final answer

$$a_n = 9(-2)^{n-1}$$

Worked Solutions

Geometric Sequences

Q097 **Written**

When 2, x , 8 form a geometric sequence in this order, find x .

Solution

1. $x^2 = 2 \times 8 = 16$

2. $x = \pm 4$

Final answer

$x = \pm 4$

Q098 **Written**

Find the requested values. When 1, x , 25 form a geometric sequence, find x .

Solution

1. $x^2 = 25 \Rightarrow x = \pm 5$

Final answer

$x = \pm 5$

Worked Solutions

Geometric Sequences

Q099 **Written****Find the requested values. When $1/2, x, 8$ form a geometric sequence, find x .****Solution**

$$1. x^2 = \frac{1}{2} \times 8 = 4 \Rightarrow x = \pm 2$$

Final answer

$$x = \pm 2$$

Q100 **Written****When $1, x, 12$ form a geometric sequence in this order, find x .****Solution**

$$1. x^2 = 1 \times 12 = 12$$

$$2. x = \pm 2\sqrt{3}$$

Final answer

$$x = \pm 2\sqrt{3}$$

Worked Solutions

Geometric Sequences

Q101 MCQ

Select the correct option: If x is the middle term of the geometric triple $2, x, 8$, find x :

Solution

1. Use $x^2 = (2)(8)$.

Correct option: A

Final answer

± 4

Q102 MCQ

Select the correct option: If x is the middle term of the geometric triple $1, x, 25$, find x :

Solution

1. Use $x^2 = (1)(25)$.

Correct option: A

Final answer

± 5

Worked Solutions

Geometric Sequences

Q103 MCQ

Select the correct option: If x is the middle term of the geometric triple $\frac{1}{2}, x, 8$, find x :

Solution

1. Use $x^2 = (\frac{1}{2})(8)$.

Correct option: A

Final answer

± 2

Q104 MCQ

Select the correct option: If x is the middle term of the geometric triple $3, x, 12$, find x :

Solution

1. Use $x^2 = (3)(12)$.

Correct option: A

Final answer

± 6

Worked Solutions

Geometric Sequences

Q105 MCQ

Select the correct option: If x is the middle term of the geometric triple 4, x , 9, find x :

Solution

1. Use $x^2 = (4)(9)$.

Correct option: A

Final answer

± 6

Q106 MCQ

Select the correct option: If x is the middle term of the geometric triple 2, x , 18, find x :

Solution

1. Use $x^2 = (2)(18)$.

Correct option: A

Final answer

± 6

Worked Solutions

Geometric Sequences

Q107 Written

The numbers $6, a, b$ form an arithmetic sequence and $a, b, 16$ form a geometric sequence. Find a, b .

Solution

1. $2a = 6 + b \Rightarrow b = 2a - 6$
2. $b^2 = 16a \Rightarrow 4a^2 - 40a + 36 = 0$
3. $(a, b) = (9, 12)$ or $(1, -4)$

Final answer

$$(a, b) = (9, 12) \text{ or } (1, -4)$$

Q108 Written

Find the requested values. The numbers $a, b, 2$ form an arithmetic sequence and $a, 2, b$ form a geometric sequence. Find a, b .

Solution

1. $2b = a + 2$
2. $b^2 = 2a$
3. $(a, b) = (2, 2)$ or $(-4, -1)$

Final answer

$$(a, b) = (2, 2) \text{ or } (-4, -1)$$

Worked Solutions

Geometric Sequences

Q109 Written

The numbers $-2, a, b$ form an arithmetic sequence and $a, b, 18$ form a geometric sequence. Find a, b .

Solution

1. $b = 2a + 2$

2. $b^2 = 18a \Rightarrow (a, b) = (2, 6) \text{ or } (\frac{1}{2}, 3)$

Final answer

$$(a, b) = (2, 6) \text{ or } (\frac{1}{2}, 3)$$

Worked Solutions

Geometric Sequences

Q110 Challenge

A planning model gives $-5, a, b$ as an arithmetic sequence, while $a, b, 45$ form a geometric sequence. Find a, b .

Solution

1. $b = 2a + 5$
2. $b^2 = 45a \Rightarrow 4a^2 - 25a + 25 = 0$
3. $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$

Final answer

$$(a, b) = (5, 15) \text{ or } (\frac{5}{4}, \frac{15}{2})$$

Worked Solutions

Geometric Sequences

Q111 Written

Find the sum S of a geometric sequence with first term 4, common ratio 3, and 5 terms.

Solution

1. Use $S_5 = \frac{4(3^5 - 1)}{3 - 1} = 2(3^5 - 1)$.
2. Therefore $S_5 = 484$.

Final answer

484

Q112 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 5 terms.

Solution

1. $S_5 = \frac{3(2^5 - 1)}{2 - 1} = 3(31)$.
2. Therefore $S_5 = 93$.

Final answer

93

Worked Solutions

Geometric Sequences

Q113 Written

Find the sum S of a geometric sequence with first term 5, common ratio 3, and 4 terms.

Solution

1. $S_4 = \frac{5(3^4 - 1)}{3 - 1} = 5(40).$
2. Therefore $S_4 = 200.$

Final answer

200

Q114 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 10 terms.

Solution

1. $S_{10} = \frac{3(2^{10} - 1)}{2 - 1} = 3(1023).$
2. Therefore $S_{10} = 3069.$

Final answer

3069

Worked Solutions

Geometric Sequences

Q115 MCQ

For a geometric sequence with $a = 4$, $r = 3$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{4(1 - (3)^4)}{1 - (3)}$.
2. Therefore $S_4 = 160$.

Correct option: D

Final answer

160

Q116 MCQ

For a geometric sequence with $a = 5$, $r = 2$, and $n = 6$, calculate S_n :

Solution

1. $S_n = \frac{5(1 - (2)^6)}{1 - (2)}$.
2. Therefore $S_6 = 315$.

Correct option: C

Final answer

315

Worked Solutions

Geometric Sequences

Q117 MCQ

For a geometric sequence with $a = 2$, $r = 4$, and $n = 5$, calculate S_n :

Solution

1. $S_n = \frac{2(1 - (4)^5)}{1 - (4)}$.

2. Therefore $S_5 = 682$.

Correct option: B

Final answer

682

Q118 MCQ

For a geometric sequence with $a = 3$, $r = 5$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{3(1 - (5)^4)}{1 - (5)}$.

2. Therefore $S_4 = 468$.

Correct option: A

Final answer

468

Worked Solutions

Geometric Sequences

Q119 MCQ

For a geometric sequence with $a = 7$, $r = 2$, and $n = 5$, calculate S_n :

Solution

1. $S_n = \frac{7(1 - (2)^5)}{1 - (2)}$.
2. Therefore $S_5 = 217$.

Correct option: D

Final answer

217

Q120 MCQ

For a geometric sequence with $a = 6$, $r = 3$, and $n = 3$, calculate S_n :

Solution

1. $S_n = \frac{6(1 - (3)^3)}{1 - (3)}$.
2. Therefore $S_3 = 78$.

Correct option: C

Final answer

78

Worked Solutions

Geometric Sequences

Q121 Written

Find the sum S of a geometric sequence with first term 8, common ratio $\frac{1}{2}$, and 6 terms.

Solution

1. Use $S_6 = \frac{8(1 - (1/2)^6)}{1 - 1/2} = 16\{1 - (1/2)^6\}$.
2. Therefore $S_6 = \frac{63}{4}$.

Final answer

$$\frac{63}{4}$$

Q122 Written

Find the sum S of a geometric sequence with first term 6, common ratio $\frac{1}{2}$, and 6 terms.

Solution

1. $S_6 = \frac{6(1 - (1/2)^6)}{1 - 1/2} = 12\{1 - (1/2)^6\}$.
2. Therefore $S_6 = \frac{189}{16}$.

Final answer

$$\frac{189}{16}$$

Worked Solutions

Geometric Sequences

Q123 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{3}$, and 3 terms.

Solution

$$1. S_3 = \frac{1 - (1/3)^3}{1 - 1/3} = \frac{26}{27} \cdot \frac{3}{2}.$$

$$2. \text{ Therefore } S_3 = \frac{13}{9}.$$

Final answer

$$\frac{13}{9}$$

Q124 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{2}$, and 8 terms.

Solution

$$1. S_8 = \frac{1 - (1/2)^8}{1 - 1/2} = 2\{1 - (1/2)^8\}.$$

$$2. \text{ Therefore } S_8 = \frac{255}{128}.$$

Final answer

$$\frac{255}{128}$$

Worked Solutions

Geometric Sequences

Q125 MCQ

For a geometric sequence with $a = 8$, $r = \frac{1}{2}$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{8(1 - (\frac{1}{2})^4)}{1 - (\frac{1}{2})}$.
2. Therefore $S_4 = 15$.

Correct option: B

Final answer

15

Q126 MCQ

For a geometric sequence with $a = 9$, $r = \frac{1}{3}$, and $n = 3$, calculate S_n :

Solution

1. $S_n = \frac{9(1 - (\frac{1}{3})^3)}{1 - (\frac{1}{3})}$.
2. Therefore $S_3 = 13$.

Correct option: A

Final answer

13

Worked Solutions

Geometric Sequences

Q127 MCQ

For a geometric sequence with $a = 12$, $r = \frac{1}{2}$, and $n = 5$, calculate S_n :

Solution

$$1. S_n = \frac{12(1 - (\frac{1}{2})^5)}{1 - (\frac{1}{2})}.$$

$$2. \text{ Therefore } S_5 = \frac{93}{4}.$$

Correct option: D

Final answer

$$\frac{93}{4}$$

Q128 MCQ

For a geometric sequence with $a = 5$, $r = \frac{1}{4}$, and $n = 3$, calculate S_n :

Solution

$$1. S_n = \frac{5(1 - (\frac{1}{4})^3)}{1 - (\frac{1}{4})}.$$

$$2. \text{ Therefore } S_3 = \frac{105}{16}.$$

Correct option: C

Final answer

$$\frac{105}{16}$$

Worked Solutions

Geometric Sequences

Q129 MCQ

For a geometric sequence with $a = 10$, $r = \frac{1}{2}$, and $n = 6$, calculate S_n :

Solution

$$1. S_n = \frac{10(1 - (\frac{1}{2})^6)}{1 - (\frac{1}{2})}.$$

$$2. \text{ Therefore } S_6 = \frac{315}{16}.$$

Correct option: B

Final answer

$$\frac{315}{16}$$

Q130 MCQ

For a geometric sequence with $a = 4$, $r = \frac{1}{3}$, and $n = 4$, calculate S_n :

Solution

$$1. S_n = \frac{4(1 - (\frac{1}{3})^4)}{1 - (\frac{1}{3})}.$$

$$2. \text{ Therefore } S_4 = \frac{160}{27}.$$

Correct option: A

Final answer

$$\frac{160}{27}$$

Worked Solutions

Geometric Sequences

Q131 Written

Find the sum S of a geometric sequence with first term 6, common ratio -2 , and 4 terms.

Solution

1. $S_4 = \frac{6\{1 - (-2)^4\}}{1 - (-2)} = \frac{6(-15)}{3}.$
2. Therefore $S_4 = -30.$

Final answer

-30

Q132 Written

Find the sum S of a geometric sequence with first term 1, common ratio -3 , and 4 terms.

Solution

1. $S_4 = \frac{1 - (-3)^4}{1 - (-3)} = \frac{-80}{4}.$
2. Therefore $S_4 = -20.$

Final answer

-20

Worked Solutions

Geometric Sequences

Q133 Written

Find the sum S of a geometric sequence with first term 2, common ratio -3 , and 6 terms.

Solution

$$1. S_6 = \frac{2\{1 - (-3)^6\}}{1 - (-3)} = \frac{2(-728)}{4}.$$

$$2. \text{ Therefore } S_6 = -364.$$

Final answer

 -364

Q134 MCQ

For a geometric sequence with $a = 4$, $r = -2$, and $n = 5$, calculate S_n :

Solution

$$1. S_n = \frac{4(1 - (-2)^5)}{1 - (-2)}.$$

$$2. \text{ Therefore } S_5 = 44.$$

Correct option: D

Final answer

44

Worked Solutions

Geometric Sequences

Q135 MCQ

For a geometric sequence with $a = 3$, $r = -3$, and $n = 4$, calculate S_n :

Solution

1. $S_n = \frac{3(1 - (-3)^4)}{1 - (-3)}$.
2. Therefore $S_4 = -60$.

Correct option: C

Final answer

-60

Q136 MCQ

For a geometric sequence with $a = 2$, $r = -1$, and $n = 6$, calculate S_n :

Solution

1. $S_n = \frac{2(1 - (-1)^6)}{1 - (-1)}$.
2. Therefore $S_6 = 0$.

Correct option: B

Final answer

0

Worked Solutions

Geometric Sequences

Q137 MCQ

For a geometric sequence with $a = 5$, $r = -2$, and $n = 6$, calculate S_n :

Solution

1. $S_n = \frac{5(1 - (-2)^6)}{1 - (-2)}$.
2. Therefore $S_6 = -105$.

Correct option: A

Final answer

-105

Q138 MCQ

For a geometric sequence with $a = 7$, $r = -1$, and $n = 5$, calculate S_n :

Solution

1. $S_n = \frac{7(1 - (-1)^5)}{1 - (-1)}$.
2. Therefore $S_5 = 7$.

Correct option: D

Final answer

7

Worked Solutions

Geometric Sequences

Q139 MCQ

For a geometric sequence with $a = 6$, $r = -3$, and $n = 3$, calculate S_n :

Solution

$$1. S_n = \frac{6(1 - (-3)^3)}{1 - (-3)}.$$

$$2. \text{ Therefore } S_3 = 42.$$

Correct option: C

Final answer

42

Q140 Written

Find S_n from the first term to the n -th term when the first term is 3 and the common ratio is -2 .

Solution

$$1. \text{ Here } a = 3 \text{ and } r = -2.$$

$$2. S_n = \frac{3\{1 - (-2)^n\}}{1 - (-2)} = 1 - (-2)^n.$$

Final answer $1 - (-2)^n$

Worked Solutions

Geometric Sequences

Q141 Written

Find S_n from the first term to the n -th term when the first term is 7 and the common ratio is 2.

Solution

1. $S_n = \frac{7(2^n - 1)}{2 - 1}$.
2. Hence $S_n = 7(2^n - 1)$.

Final answer

$$7(2^n - 1)$$

Q142 Written

Find S_n from the first term to the n -th term when the first term is 2 and the common ratio is $\frac{1}{3}$.

Solution

1. $S_n = \frac{2\{1 - (1/3)^n\}}{1 - 1/3}$.
2. Hence $S_n = 3\{1 - (1/3)^n\}$.

Final answer

$$3\{1 - (\frac{1}{3})^n\}$$

Worked Solutions

Geometric Sequences

Q143 Written

Find S_n from the first term to the n -th term when the first term is 4 and the common ratio is 2.

Solution

1. $S_n = \frac{4(2^n - 1)}{2 - 1}$.
2. Hence $S_n = 4(2^n - 1)$.

Final answer

$$4(2^n - 1)$$

Q144 MCQ

For the geometric sequence with $a = 3, r = 2$, find the sum S_n from the first term to the n -th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \frac{a(1 - r^n)}{1 - r}$ to obtain $S_n = 3(2^n - 1)$.

Correct option: C

Final answer

$$3(2^n - 1)$$

Worked Solutions

Geometric Sequences

Q145 MCQ

For the geometric sequence with $a = 5, r = \frac{1}{2}$, find the sum S_n from the first term to the n -th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \frac{a(1 - r^n)}{1 - r}$ to obtain $S_n = 10\{1 - (\frac{1}{2})^n\}$.

Correct option: D**Final answer**

$$10\{1 - (\frac{1}{2})^n\}$$

Q146 MCQ

For the geometric sequence with $a = 4, r = -2$, find the sum S_n from the first term to the n -th term:

Solution

1. Identify a and r from the stated parameters.
2. Substitute into $S_n = \frac{a(1 - r^n)}{1 - r}$ to obtain $S_n = \frac{4\{1 - (-2)^n\}}{3}$.

Correct option: A**Final answer**

$$\frac{4\{1 - (-2)^n\}}{3}$$

Worked Solutions

Geometric Sequences

Q147 **Written****Find S_n from the first term to the n -th term of 2, 6, 18, 54,****Solution**

1. The sequence has $a = 2$ and $r = 3$.
2. $S_n = \frac{2(3^n - 1)}{3 - 1} = 3^n - 1$.

Final answer

$$3^n - 1$$

Q148 **Written****Find S_n from the first term to the n -th term of 2, -6, 18, -54,****Solution**

1. The sequence has $a = 2$ and $r = -3$.
2. $S_n = \frac{2\{1 - (-3)^n\}}{1 - (-3)} = \frac{1 - (-3)^n}{2}$.

Final answer

$$\frac{1 - (-3)^n}{2}$$

Worked Solutions

Geometric Sequences

Q149 Written

Find S_n from the first term to the n -th term of $8, -4, 2, -1, \dots$

Solution

1. The sequence has $a = 8$ and $r = -\frac{1}{2}$.
2. $S_n = \frac{8\{1 - (-1/2)^n\}}{1 + 1/2} = \frac{16}{3}\{1 - (-1/2)^n\}$.

Final answer

$$\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$$

Q150 Written

Find S_n from the first term to the n -th term of $1, -3, 9, -27, \dots$

Solution

1. The sequence has $a = 1$ and $r = -3$.
2. $S_n = \frac{1 - (-3)^n}{1 + 3} = \frac{1 - (-3)^n}{4}$.

Final answer

$$\frac{1 - (-3)^n}{4}$$

Worked Solutions

Geometric Sequences

Q151 Written

Find S_n from the first term to the n -th term when the first term is 16 and the common ratio is $-\frac{1}{2}$.

Solution

1. $S_n = \frac{16\{1 - (-1/2)^n\}}{1 + 1/2}$.
2. Hence $S_n = \frac{32}{3}\{1 - (-1/2)^n\}$.

Final answer

$$\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$$

Q152 Written

Find S_n from the first term to the n -th term of 1, -2, 4, -8,

Solution

1. The sequence has $a = 1$ and $r = -2$.
2. $S_n = \frac{1 - (-2)^n}{1 + 2} = \frac{1 - (-2)^n}{3}$.

Final answer

$$\frac{1 - (-2)^n}{3}$$

Worked Solutions

Geometric Sequences

Q153 MCQ

A culture grows by 3, 9, 27, 81 units. Find the four-stage total:

Solution

1. Recognise the listed stages as a finite geometric sequence and identify its first term and ratio.
2. Apply the finite-GP formula to obtain $S_n = 120$.

Correct option: A

Final answer

120

Worked Solutions

Geometric Sequences

Q154 Challenge

A filtration scale is geometric: $3, 1, \frac{1}{3}, \frac{1}{9}, \dots$ mg/L removed per stage. Calculate the total removed in 4 stages.

Solution

1. Here $a = 3$, $r = \frac{1}{3}$, and $n = 4$.
2. $S_4 = \frac{3\{1 - (1/3)^4\}}{1 - 1/3} = \frac{40}{9}$ mg/L.

Final answer

$$\frac{40}{9}$$

Worked Solutions

Geometric Sequences

Q155 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is 28, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2S_3 = 28$, so $r^2 = 4$; then $a(1 + r + r^2) = 7$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 1$ or $a = \frac{7}{3}$, respectively.

Correct option: D

Final answer

$$(a, r) = (1, 2) \text{ or } (a, r) = \left(\frac{7}{3}, -2\right)$$

Worked Solutions

Geometric Sequences

Q156 MCQ

Given $S_3 = 26$ and the sum from the 3rd to the 5th terms is 234, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 234$, so $r^2 = 9$; then $a(1 + r + r^2) = 26$.
2. Solving in the real numbers gives $r = 3$ or $r = -3$.
3. Back-substitution gives $a = 2$ or $a = \frac{26}{7}$, respectively.

Correct option: C

Final answer

$$(a, r) = (2, 3) \text{ or } (a, r) = \left(\frac{26}{7}, -3\right)$$

Worked Solutions

Geometric Sequences

Q157 MCQ

Given $S_3 = 21$ and the sum from the 3rd to the 5th terms is 84, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 84$, so $r^2 = 4$; then $a(1 + r + r^2) = 21$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 3$ or $a = 7$, respectively.

Correct option: B

Final answer

$$(a, r) = (3, 2) \text{ or } (a, r) = (7, -2)$$

Worked Solutions

Geometric Sequences

Q158 MCQ

Given $S_3 = 28$ and the sum from the 3rd to the 5th terms is 112, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 112$, so $r^2 = 4$; then $a(1 + r + r^2) = 28$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 4$ or $a = \frac{28}{3}$, respectively.

Correct option: A

Final answer

$$(a, r) = (4, 2) \text{ or } (a, r) = \left(\frac{28}{3}, -2\right)$$

Worked Solutions

Geometric Sequences

Q159 MCQ

Given $S_3 = 13$ and the sum from the 3rd to the 5th terms is 117, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 117$, so $r^2 = 9$; then $a(1 + r + r^2) = 13$.
2. Solving in the real numbers gives $r = 3$ or $r = -3$.
3. Back-substitution gives $a = 1$ or $a = \frac{13}{7}$, respectively.

Correct option: D

Final answer

$$(a, r) = (1, 3) \text{ or } (a, r) = \left(\frac{13}{7}, -3\right)$$

Worked Solutions

Geometric Sequences

Q160 MCQ

Given $S_3 = 35$ and the sum from the 3rd to the 5th terms is 140, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 140$, so $r^2 = 4$; then $a(1 + r + r^2) = 35$.
2. Solving in the real numbers gives $r = 2$ or $r = -2$.
3. Back-substitution gives $a = 5$ or $a = \frac{35}{3}$, respectively.

Correct option: C

Final answer

$$(a, r) = (5, 2) \text{ or } (a, r) = \left(\frac{35}{3}, -2\right)$$

Worked Solutions

Geometric Sequences

Q161 MCQ

Given $S_3 = \frac{19}{2}$ and the sum from the 3rd to the 5th terms is $\frac{171}{8}$, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = \frac{171}{8}$, so $r^2 = \frac{9}{4}$; then $a(1 + r + r^2) = \frac{19}{2}$.
2. Solving in the real numbers gives $r = \frac{3}{2}$ or $r = -\frac{3}{2}$.
3. Back-substitution gives $a = 2$ or $a = \frac{38}{7}$, respectively.

Correct option: D**Final answer**

$$(a, r) = \left(2, \frac{3}{2}\right) \text{ or } (a, r) = \left(\frac{38}{7}, -\frac{3}{2}\right)$$

Worked Solutions

Geometric Sequences

Q162 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 4th to the 6th term is -72 .

Solution

1. The 4th+6th block is $r^3 S_3 = -72$, so $r^3 = -8$ and $r = -2$.
2. Then $a(1 - 2 + 4) = 9$, giving $a = 3$.

Final answer

$$(a, r) = (3, -2)$$

Worked Solutions

Geometric Sequences

Q163 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is $\frac{7}{4}$, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = \frac{7}{4}$, so $r^2 = \frac{1}{4}$; then $a(1 + r + r^2) = 7$.
2. Solving in the real numbers gives $r = \frac{1}{2}$ or $r = -\frac{1}{2}$.
3. Back-substitution gives $a = 4$ or $a = \frac{28}{3}$, respectively.

Correct option: C

Final answer

$$(a, r) = \left(4, \frac{1}{2}\right) \text{ or } (a, r) = \left(\frac{28}{3}, -\frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q164 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 35$ and the sum from the 3rd to the 5th term is 140.

Solution

1. Write $S_3 = a(1 + r + r^2) = 35$. The shifted block is $r^2 S_3 = 140$, so $r^2 = 4$ and $r = \pm 2$.
2. For $r = 2$, $7a = 35$, giving $a = 5$. For $r = -2$, $3a = 35$, giving $a = \frac{35}{3}$.

Final answer

$$(a, r) = (5, 2) \text{ or } (a, r) = \left(\frac{35}{3}, -2\right)$$

Q165 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and the sum from the 3rd to the 5th term is 84.

Solution

1. The shifted block is $r^2 S_3 = 84$, so $r^2 = 4$ and $r = \pm 2$.
2. Using $a(1 + r + r^2) = 21$ gives $a = 3$ for $r = 2$ and $a = 7$ for $r = -2$.

Final answer

$$(a, r) = (3, 2) \text{ or } (a, r) = (7, -2)$$

Worked Solutions

Geometric Sequences

Q166 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 3rd to the 5th term is 36.

Solution

1. The shifted block gives $r^2 S_3 = 36$, hence $r = \pm 2$.
2. Substitution into $a(1 + r + r^2) = 9$ gives $a = \frac{9}{7}$ for $r = 2$, or $a = 3$ for $r = -2$.

Final answer

$$(a, r) = \left(\frac{9}{7}, 2\right) \text{ or } (a, r) = (3, -2)$$

Q167 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 78$ and the sum from the 3rd to the 5th term is 702.

Solution

1. The shifted block gives $r^2 S_3 = 702$, so $r^2 = 9$ and $r = \pm 3$.
2. Using $a(1 + r + r^2) = 78$ gives $a = 6$ for $r = 3$, or $a = \frac{78}{7}$ for $r = -3$.

Final answer

$$(a, r) = (6, 3) \text{ or } (a, r) = \left(\frac{78}{7}, -3\right)$$

Worked Solutions

Geometric Sequences

Q168 MCQ

Given $S_3 = 6$ and the sum from the 3rd to the 5th terms is 24, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 24$, so $r^2 = 4$; then $a(1 + r + r^2) = 6$.
2. Solving in the real numbers gives $r = -2$ or $r = 2$.
3. Back-substitution gives $a = 2$ or $a = \frac{6}{7}$, respectively.

Correct option: B

Final answer

$$(a, r) = (2, -2) \text{ or } (a, r) = \left(\frac{6}{7}, 2\right)$$

Worked Solutions

Geometric Sequences

Q169 MCQ

Given $S_3 = 9$ and the sum from the 3rd to the 5th terms is 36, find all real pairs (a, r) :

Solution

1. The shifted block is $r^2 S_3 = 36$, so $r^2 = 4$; then $a(1 + r + r^2) = 9$.
2. Solving in the real numbers gives $r = -2$ or $r = 2$.
3. Back-substitution gives $a = 3$ or $a = \frac{9}{7}$, respectively.

Correct option: A

Final answer

$$(a, r) = (3, -2) \text{ or } (a, r) = \left(\frac{9}{7}, 2\right)$$

Worked Solutions

Geometric Sequences

Q170 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 3 and $S_3 = 13$.

Solution

1. From $ar = 3$, write $a = 3/r$. Then $3/r + 3 + 3r = 13$, so $3r^2 - 10r + 3 = 0$.
2. Hence $r = 3$ or $r = \frac{1}{3}$, giving $a = 1$ or $a = 9$, respectively.

Final answer

$$(a, r) = (1, 3) \text{ or } (a, r) = \left(9, \frac{1}{3}\right)$$

Q171 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 6, and $S_3 = 21$.

Solution

1. From $ar = 6$, $a = 6/r$. Substitution into $S_3 = 21$ gives $6/r + 6 + 6r = 21$, hence $2r^2 - 5r + 2 = 0$.
2. Therefore $r = 2$ or $r = \frac{1}{2}$, giving $a = 3$ or $a = 12$, respectively.

Final answer

$$(a, r) = (3, 2) \text{ or } (a, r) = \left(12, \frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q172 MCQ

The second term is 6 and $S_3 = 21$. Find all real pairs (a, r) :**Solution**

1. Use $ar = 6$ and $a(1 + r + r^2) = 21$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 2$ or $r = \frac{1}{2}$.
3. Back-substitution gives $a = 3$ or $a = 12$, respectively.

Correct option: C**Final answer**

$$(a, r) = (3, 2) \text{ or } (a, r) = \left(12, \frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q173 MCQ

The second term is 6 and $S_3 = 26$. Find all real pairs (a, r) :

Solution

1. Use $ar = 6$ and $a(1 + r + r^2) = 26$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 3$ or $r = \frac{1}{3}$.
3. Back-substitution gives $a = 2$ or $a = 18$, respectively.

Correct option: D

Final answer

$$(a, r) = (2, 3) \text{ or } (a, r) = \left(18, \frac{1}{3}\right)$$

Worked Solutions

Geometric Sequences

Q174 MCQ

The second term is 8 and $S_3 = 28$. Find all real pairs (a, r) :**Solution**

1. Use $ar = 8$ and $a(1 + r + r^2) = 28$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 2$ or $r = \frac{1}{2}$.
3. Back-substitution gives $a = 4$ or $a = 16$, respectively.

Correct option: B**Final answer**

$$(a, r) = (4, 2) \text{ or } (a, r) = \left(16, \frac{1}{2}\right)$$

Worked Solutions

Geometric Sequences

Q175 MCQ

The second term is 3 and $S_3 = 13$. Find all real pairs (a, r) :

Solution

1. Use $ar = 3$ and $a(1 + r + r^2) = 13$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = 3$ or $r = \frac{1}{3}$.
3. Back-substitution gives $a = 1$ or $a = 9$, respectively.

Correct option: C

Final answer

$$(a, r) = (1, 3) \text{ or } (a, r) = \left(9, \frac{1}{3}\right)$$

Worked Solutions

Geometric Sequences

Q176 MCQ

The second term is $\frac{5}{2}$ and $S_3 = \frac{35}{4}$. Find all real pairs (a, r) :

Solution

1. Use $ar = \frac{5}{2}$ and $a(1 + r + r^2) = \frac{35}{4}$; after substitution, solve the quadratic in r .
2. Solving in the real numbers gives $r = \frac{1}{2}$ or $r = 2$.
3. Back-substitution gives $a = 5$ or $a = \frac{5}{4}$, respectively.

Correct option: D**Final answer**

$$(a, r) = \left(5, \frac{1}{2}\right) \text{ or } (a, r) = \left(\frac{5}{4}, 2\right)$$

Worked Solutions

Geometric Sequences

Q177 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 4 and $S_3 = 7$.

Solution

1. The data give $ar^2 = 4$ and $a + ar + ar^2 = 7$. Substitute $a = 4/r^2$: $4 + 4r + 4r^2 = 7r^2$, so $3r^2 - 4r - 4 = 0$.
2. Thus $r = 2$ or $r = -\frac{2}{3}$. Back-substitution gives $a = 1$ or $a = 9$, respectively.

Final answer

$$(a, r) = (1, 2) \text{ or } (a, r) = \left(9, -\frac{2}{3}\right)$$

Q178 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 12, and $S_3 = 9$.

Solution

1. Use $ar^2 = 12$ and $a + ar + ar^2 = 9$. After multiplying by r^2 , $r^2 + 4r + 4 = 0$, so $r = -2$.
2. Then $a = 12/(-2)^2 = 3$.

Final answer

$$(a, r) = (3, -2)$$

Worked Solutions

Geometric Sequences

Q179 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 20, and $S_3 = 35$.

Solution

1. Use $ar^2 = 20$ and $a + ar + ar^2 = 35$. Multiplication by r^2 gives $3r^2 - 4r - 4 = 0$.
2. Thus $r = 2$ or $r = -\frac{2}{3}$, and $a = 5$ or $a = 45$, respectively.

Final answer

$$(a, r) = (5, 2) \text{ or } (a, r) = \left(45, -\frac{2}{3}\right)$$

Q180 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and $S_6 = 189$.

Solution

1. Use $S_6 = S_3 + r^3 S_3$: $189 = 21(1 + r^3)$, hence $r^3 = 8$ and $r = 2$ in the real numbers.
2. Then $a(1 + 2 + 4) = 21$, so $a = 3$.

Final answer

$$(a, r) = (3, 2)$$

Worked Solutions

Geometric Sequences

Q181 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 26$ and $S_6 = 728$.

Solution

1. Since $S_6 = S_3 + r^3 S_3$, $728 = 26(1 + r^3)$, so $r^3 = 27$ and $r = 3$.
2. Then $a(1 + 3 + 9) = 26$, giving $a = 2$.

Final answer

$$(a, r) = (2, 3)$$

Q182 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 65$ and $S_6 = 1820$.

Solution

1. Use $S_6 = S_3(1 + r^3)$: $1820 = 65(1 + r^3)$, so $r^3 = 27$ and $r = 3$.
2. Then $13a = 65$, giving $a = 5$.

Final answer

$$(a, r) = (5, 3)$$

Worked Solutions

Geometric Sequences

Q183 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is 3, and $S_4 = 51$.

Solution

1. Let $a + ar = 3$. Factor $S_4 = (a + ar)(1 + r^2)$, so $3(1 + r^2) = 51$ and $r = \pm 4$.
2. Then $a = 3/(1 + r)$, giving $a = \frac{3}{5}$ for $r = 4$, or $a = -1$ for $r = -4$.

Final answer

$$(a, r) = \left(\frac{3}{5}, 4\right) \text{ or } (a, r) = (-1, -4)$$

Q184 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is -8 , and $S_4 = -80$.

Solution

1. Factor $S_4 = (a + ar)(1 + r^2)$. Thus $-8(1 + r^2) = -80$, so $r = \pm 3$.
2. Using $a(1 + r) = -8$ gives $a = -2$ for $r = 3$, or $a = 4$ for $r = -3$.

Final answer

$$(a, r) = (-2, 3) \text{ or } (a, r) = (4, -3)$$

Worked Solutions

Geometric Sequences

Q185 MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Solution

1. $a_7 = S_7 - S_6 = 64$ and $a_8 = S_8 - S_7 = 128$.

2. Thus $r = a_8/a_7 = 2$, and $a_{10} = a_8 r^2 = 512$.

Correct option: D

Final answer

512

Worked Solutions

Geometric Sequences

Q186 Challenge

If S_n is the sum of the first n terms of a geometric sequence, $S_8 = 255$, $S_6 = 63$, and $a_8 - a_7 = 64$, find a_{10} .

Solution

1. $S_8 - S_6 = a_7 + a_8 = 255 - 63 = 192$. Together with $a_8 - a_7 = 64$, this gives $a_7 = 64$ and $a_8 = 128$.
2. Hence $r = a_8/a_7 = 2$, and $a_{10} = a_8 r^2 = 128(4) = 512$.

Final answer

$$a_7 = 64, a_8 = 128, r = 2, a_{10} = 512$$

Worked Solutions

Geometric Sequences

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Select the correct option: For $a = 3, r = 2$, find a_6 :

Solution

1. $a_6 = 3(2)^5 = 96$

Correct option: B

Final answer

96

Quiz MCQ 2 [Concept Quiz · MCQ](#)

Select the correct option: If 2, x , 8 are consecutive geometric terms, find x :

Solution

1. $x^2 = 16 \Rightarrow x = \pm 4$

Correct option: C

Final answer

± 4

Worked Solutions

Geometric Sequences

Quiz MCQ 3 **Concept Quiz · MCQ**

For 1, 3, 9, ..., find S_n :

Solution

1. $a = 1, r = 3$, so $S_n = (3^n - 1)/2$.

Correct option: B

Final answer

$$\frac{3^n - 1}{2}$$

Quiz MCQ 4 **Concept Quiz · MCQ**

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Solution

1. $a_7 = S_7 - S_6 = 64$, $a_8 = S_8 - S_7 = 128$, so $r = 2$.

2. Therefore $a_{10} = 128(2^2) = 512$.

Correct option: D

Final answer

512

Worked Solutions

Geometric Sequences

Quiz WRITTEN 5 Concept Quiz · Written

Find the general term of $-4, 8, -16, 32, \dots$

Solution

1. The ratio is -2 .
2. $a_n = -4(-2)^{n-1}$

Final answer

$$-4(-2)^{n-1}$$

Quiz WRITTEN 6 Concept Quiz · Written

If $1, x, 25$ are consecutive geometric terms, find x .

Solution

1. $x^2 = 25 \Rightarrow x = \pm 5$

Final answer

$$x = \pm 5$$

Worked Solutions

Geometric Sequences

Quiz WRITTEN 7 Concept Quiz · Written

A four-stage process has outputs 5, 15, 45, 135. Find the total output.

Solution

1. $S_4 = 5(1 + 3 + 9 + 27) = 200.$

Final answer

200

Quiz WRITTEN 8 Concept Quiz · Written

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} .

Solution

1. $a_7 = 64$, $a_8 = 128$, hence $r = 2$.

2. $a_{10} = a_8 r^2 = 512.$

Final answer

512

Answer key

Use the question number shown on each page.

Q001 $2 \times 3^{n-1}$

Q002 $3 \times 4^{n-1}$

Q003 $5 \times 2^{n-1}$

Q004 $2 \times 5^{n-1}$

Q005 $7 \times 4^{n-1}$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 $5(-2)^{n-1}$

Q013 $2(-3)^{n-1}$

Q014 $3(-5)^{n-1}$

Q015 $4(-2)^{n-1}$

Q016 A

Q017 A

Q018 A

Q019 A

Q020 A

Q021 $8\left(-\frac{1}{3}\right)^{n-1}$

Q022 $4\left(\frac{1}{3}\right)^{n-1}$

Q023 $3\left(-\frac{1}{2}\right)^{n-1}$

Q024 $6\left(\frac{1}{2}\right)^{n-1}$

Q025 A

Q026 A

Q027 A

Q028 A

Q029 A

Q030 $5 \times 2^{n-1}$

Q031 $2 \times 3^{n-1}$

Q032 $3 \times 2^{n-1}$

Q033 A

Q034 A

Q035 A

Q036 A

Q037 A

Q038 $2(-3)^{n-1}$

Q039 $5(-2)^{n-1}$

Q040 A

Q041 A

Q042 A

Q043 A

Q044 $-4\left(-\frac{3}{2}\right)^{n-1}$

Q045 $-8\left(-\frac{3}{2}\right)^{n-1}$

Q046 $2\left(\frac{1}{3}\right)^{n-1}$

Q047 $48\left(\frac{1}{2}\right)^{n-1}$

Q048 $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$

Q049 $8(2\sqrt{2})^{n-1}$

Q050 $3\left(-\frac{5}{2}\right)^{n-1}$

Q051 $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

Q052 $5\left(-\frac{2}{3}\right)^{n-1}$

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $r = -\frac{1}{3}, A = 54, B = -2$

Q061 A

Q062 $r = \frac{1}{2}, A = 48, B = 6$

Q063 $r = -3, A = 1, B = -27$

Q064 $r = \frac{1}{3}, A = 1, B = \frac{1}{27}$

Q065 $r = -\frac{1}{2}, A = \frac{1}{2}, B = -\frac{1}{16}$

Q066 $a_7 = 648$

Q067 $a_n = 3(2)^{n-1}$ or $-3(-2)^{n-1}$

Q068 $a_n = 2(3)^{n-1}$ or $2(-3)^{n-1}$

Q069 $a_n = 2(3)^{n-1}$ or $-2(-3)^{n-1}$

Q070 $a_n = 3(3)^{n-1}$ or $3(-3)^{n-1}$

Q071 $a_n = 7(2)^{n-1}$

Q072 $a_n = 3(-2)^{n-1}$

Q073 $a_n = 3(-2)^{n-1}$

Q074 $a_n = 3(2)^{n-1}$ or $3(-2)^{n-1}$

Q075 $a_n = -7(2)^{n-1}$ or $-7(-2)^{n-1}$

Q076 $a_n = -2(-2)^{n-1}$

Q077 $a_n = -5(-2)^{n-1}$

Q078 $a_n = \frac{3}{4}(4)^{n-1}$ or $-\frac{3}{4}(-4)^{n-1}$

Q079 $a_n = 4\left(\frac{3}{2}\right)^{n-1}$ or $4\left(-\frac{3}{2}\right)^{n-1}$

Q080 $a_n = \frac{2}{243}(3)^{n-1}$ or $-\frac{2}{243}(-3)^{n-1}$

Q081 A

Q082 A

Q083 A

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A

Q090 A

Q091 A

Q092 A

Q093 A

Q094 A

Q095 A	Q122 $\frac{189}{16}$	Q148 $\frac{1 - (-3)^n}{2}$	Q171 $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$
Q096 A	Q123 $\frac{13}{9}$	Q149 $\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$	Q172 C
Q097 $x = \pm 4$	Q124 $\frac{255}{128}$	Q150 $\frac{1 - (-3)^n}{4}$	Q173 D
Q098 $x = \pm 5$	Q125 B	Q151 $\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$	Q174 B
Q099 $x = \pm 2$	Q126 A	Q152 $\frac{1 - (-2)^n}{3}$	Q175 C
Q100 $x = \pm 2\sqrt{3}$	Q127 D	Q153 A	Q176 D
Q101 A	Q128 C	Q154 $\frac{40}{9}$	Q177 $(a, r) = (1, 2)$ or $(a, r) = \left(9, -\frac{2}{3}\right)$
Q102 A	Q129 B	Q155 D	Q178 $(a, r) = (3, -2)$
Q103 A	Q130 A	Q156 C	Q179 $(a, r) = (5, 2)$ or $(a, r) = \left(45, -\frac{2}{3}\right)$
Q104 A	Q131 -30	Q157 B	Q180 $(a, r) = (3, 2)$
Q105 A	Q132 -20	Q158 A	Q181 $(a, r) = (2, 3)$
Q106 A	Q133 -364	Q159 D	Q182 $(a, r) = (5, 3)$
Q107 $(a, b) = (9, 12)$ or $(1, -4)$	Q134 D	Q160 C	Q183 $(a, r) = \left(\frac{3}{5}, 4\right)$ or $(a, r) = (-1, -4)$
Q108 $(a, b) = (2, 2)$ or $(-4, -1)$	Q135 C	Q161 D	Q184 $(a, r) = (-2, 3)$ or $(a, r) = (4, -3)$
Q109 $(a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$	Q136 B	Q162 $(a, r) = (3, -2)$	Q185 D
Q110 $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$	Q137 A	Q163 C	Q186 $a_7 = 64, a_8 = 128, r = 2, a_{10} = 512$
Q111 484	Q138 D	Q164 $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$	Quiz MCQ 1 B
Q112 93	Q139 C	Q165 $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$	Quiz MCQ 2 C
Q113 200	Q140 $1 - (-2)^n$	Q166 $(a, r) = \left(\frac{9}{7}, 2\right)$ or $(a, r) = (3, -2)$	Quiz MCQ 3 B
Q114 3069	Q141 $7(2^n - 1)$	Q167 $(a, r) = (6, 3)$ or $(a, r) = \left(\frac{78}{7}, -3\right)$	Quiz MCQ 4 D
Q115 D	Q142 $3\{1 - (\frac{1}{3})^n\}$	Q168 B	Quiz WRITTEN 5 $-4(-2)^{n-1}$
Q116 C	Q143 $4(2^n - 1)$	Q169 A	Quiz WRITTEN 6 $x = \pm 5$
Q117 B	Q144 C	Q170 $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$	Quiz WRITTEN 7 200
Q118 A	Q145 D		Quiz WRITTEN 8 512
Q119 D	Q146 A		
Q120 C	Q147 $3^n - 1$		
Q121 $\frac{63}{4}$			

Concept 3 | Summation

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$\sum (au_k + bv_k) = a \sum u_k + b \sum v_k$$

Standard sums

$$\sum k = \frac{n(n+1)}{2}$$

Telescoping

$$\sum_{k=1}^n (u_k - u_{k+1}) = u_1 - u_{n+1}$$

Worked Solutions

Summation

Q001 Written

Write $\sum_{k=1}^5 2k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 2k = 2 + 4 + 6 + 8 + 10$

Final answer

$$2 + 4 + 6 + 8 + 10$$

Q002 Written

Write $\sum_{k=1}^5 (2k + 1)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 (2k + 1) = 3 + 5 + 7 + 9 + 11$

Final answer

$$3 + 5 + 7 + 9 + 11$$

Worked Solutions

Summation

Q003 Written

Write $\sum_{k=1}^5 3k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 3k = 3 + 6 + 9 + 12 + 15$

Final answer

$$3 + 6 + 9 + 12 + 15$$

Q004 Written

Write $\sum_{k=1}^4 (4k - 2)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (4k - 2) = 2 + 6 + 10 + 14$

Final answer

$$2 + 6 + 10 + 14$$

Worked Solutions

Summation

Q005 Written

Write $\sum_{k=1}^5 (2k - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.
2. $\sum_{k=1}^5 (2k - 3) = -1 + 1 + 3 + 5 + 7$

Final answer

$$-1 + 1 + 3 + 5 + 7$$

Q006 MCQ

Select the correct option: Expand $\sum_{k=1}^3 4k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 8 + 12$$

Worked Solutions

Summation

Q007 MCQ

Select the correct option: Expand $\sum_{k=1}^4 2^k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$2 + 4 + 8 + 16$$

Q008 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (5^k - 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 24 + 124$$

Worked Solutions

Summation

Q009 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (k + 4)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 6 + 7 + 8 + 9$$

Q010 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (2k + 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$7 + 9 + 11 + 13$$

Worked Solutions

Summation

Q011 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2^k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 5 + 9$$

Q012 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 3k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 10 + 18$$

Worked Solutions

Summation

Q013 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (k^2 + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 10 + 17 + 26$$

Q014 Written

Write $\sum_{k=1}^3 (k^2 - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (k^2 - 3) = -2 + 1 + 6$

Final answer

$$-2 + 1 + 6$$

Worked Solutions

Summation

Q015 Written

Write $\sum_{k=1}^3 (3k^2 + 2)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (3k^2 + 2) = 5 + 14 + 29$

Final answer

$$5 + 14 + 29$$

Q016 Written

Write $\sum_{k=1}^4 k^2$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 k^2 = 1 + 4 + 9 + 16$

Final answer

$$1 + 4 + 9 + 16$$

Worked Solutions

Summation

Q017 Written

Write $\sum_{k=1}^3 (2k^2 + 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (2k^2 + 3) = 5 + 11 + 21$

Final answer

$$5 + 11 + 21$$

Q018 Written

Write $\sum_{k=1}^4 (k^2 - 2k)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (k^2 - 2k) = -1 + 0 + 3 + 8$

Final answer

$$-1 + 0 + 3 + 8$$

Worked Solutions

Summation

Q019 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 5 + 7 + 9$$

Q020 MCQ

Select the correct option: Expand $\sum_{k=2}^5 (3k - 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$5 + 8 + 11 + 14$$

Worked Solutions

Summation

Q021 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (2k^2 - 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$-1 + 5 + 15 + 29$$

Q022 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k^2 + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$4 + 13 + 28 + 49$$

Worked Solutions

Summation

Q023 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (k^2 - k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$0 + 2 + 6 + 12$

Q024 MCQ

Select the correct option: Expand $\sum_{k=2}^4 5k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$10 + 15 + 20$

Worked Solutions

Summation

Q025 MCQ

Select the correct option: Expand $\sum_{k=1}^4 3$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 3 + 3 + 3$$

Q026 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (5k + 1)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$6 + 11 + 16 + 21$$

Worked Solutions

Summation

Q027 Written

Write $\sum_{k=1}^4 3^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 3^k = 3 + 9 + 27 + 81$

Final answer

$$3 + 9 + 27 + 81$$

Q028 Written

Write $\sum_{k=1}^3 2$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^3 2 = 2 + 2 + 2$

Final answer

$$2 + 2 + 2$$

Worked Solutions

Summation

Q029 Written

Write $\sum_{k=1}^4 2^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 2^k = 2 + 4 + 8 + 16$

Final answer

$$2 + 4 + 8 + 16$$

Q030 Written

Write $\sum_{k=1}^6 3$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^6 3 = 3 + 3 + 3 + 3 + 3 + 3$

Final answer

$$3 + 3 + 3 + 3 + 3 + 3$$

Worked Solutions

Summation

Q031 Written

Write $\sum_{k=1}^3 3^k$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 3^k = 3 + 9 + 27$

Final answer

$$3 + 9 + 27$$

Q032 Written

Write $\sum_{k=1}^3 (5^k + 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3$.
2. $\sum_{k=1}^3 (5^k + 3) = 8 + 28 + 128$

Final answer

$$8 + 28 + 128$$

Worked Solutions

Summation

Q033 Written

Write $\sum_{k=1}^4 (2^k - 3)$ as a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4$.
2. $\sum_{k=1}^4 (2^k - 3) = -1 + 1 + 5 + 13$

Final answer

$$-1 + 1 + 5 + 13$$

Q034 Written

Write $\sum_{k=1}^4 5$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^4 5 = 5 + 5 + 5 + 5$

Final answer

$$5 + 5 + 5 + 5$$

Worked Solutions

Summation

Q035 Written

Write $\sum_{k=1}^3 3$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^3 3 = 3 + 3 + 3$

Final answer

$$3 + 3 + 3$$

Q036 Written

Write $\sum_{k=1}^5 4$ as a sum of terms.

Solution

1. The summand is constant.
2. $\sum_{k=1}^5 4 = 4 + 4 + 4 + 4 + 4$

Final answer

$$4 + 4 + 4 + 4 + 4$$

Worked Solutions

Summation

Q037 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (k^2 + 2)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$3 + 6 + 11$$

Q038 MCQ

Select the correct option: Expand $\sum_{k=1}^5 7$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$7 + 7 + 7 + 7 + 7$$

Worked Solutions

Summation

Q039 MCQ

Select the correct option: Expand $\sum_{k=3}^6 k^2$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$9 + 16 + 25 + 36$$

Q040 MCQ

Select the correct option: Expand $\sum_{k=1}^3 6^k$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$6 + 36 + 216$$

Worked Solutions

Summation

Q041 MCQ

Select the correct option: Expand $\sum_{k=1}^5 (4k - 3)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$1 + 5 + 9 + 13 + 17$$

Q042 MCQ

Select the correct option: Expand $\sum_{k=1}^4 (3k - 5)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$-2 + 1 + 4 + 7$$

Worked Solutions

Summation

Q043 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (4^k - 2)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$2 + 14 + 62$$

Q044 MCQ

Select the correct option: Expand $\sum_{k=1}^3 (2k^2 - k)$ into a sum of terms:

Solution

1. Substitute each integer value of k in order.

Correct option: A

Final answer

$$1 + 6 + 15$$

Worked Solutions

Summation

Q045 Written

Express $3 + 7 + 11 + \dots + (4n - 1)$ using the summation symbol Σ .**Solution**

1. The general term is $4k - 1$.
2. $\sum_{k=1}^n (4k - 1)$

Final answer

$$\sum_{k=1}^n (4k - 1)$$

Q046 Written

Express $5 + 8 + 11 + \dots + (3n + 2)$ using Σ .**Solution**

1. The general term is $3k + 2$.
2. $\sum_{k=1}^n (3k + 2)$

Final answer

$$\sum_{k=1}^n (3k + 2)$$

Worked Solutions

Summation

Q047 Written

Write the sum of the first n terms of the sequence $a_n = 2n + 1$ using Σ .

Solution

1. Change the term index from n to k .
2. $\sum_{k=1}^n (2k + 1)$

Final answer

$$\sum_{k=1}^n (2k + 1)$$

Q048 Written

Express $1 + 3 + 5 + \dots + (2n - 1)$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (2k - 1)$

Final answer

$$\sum_{k=1}^n (2k - 1)$$

Worked Solutions

Summation

Q049 Written

Write the sum of the first n terms of $a_n = 7n - 1$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (7k - 1)$

Final answer

$$\sum_{k=1}^n (7k - 1)$$

Q050 Written

Write the sum of the first n terms of $a_n = 4n + 5$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (4k + 5)$

Final answer

$$\sum_{k=1}^n (4k + 5)$$

Worked Solutions

Summation

Q051 Written

Write the sum of the first n terms of $a_n = 3n - 2$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (3k - 2)$

Final answer

$$\sum_{k=1}^n (3k - 2)$$

Q052 MCQ

Select the correct option: Express $2 + 5 + 8 + \cdots + (3n - 1)$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n (3k - 1)$$

Worked Solutions

Summation

Q053 MCQ

Select the correct option: Express $7 + 11 + 15 + \dots + 43$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^{10} (4k + 3)$$

Q054 MCQ

Select the correct option: Express $6 + 10 + 14 + \dots + (4n + 2)$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^n (4k + 2)$$

Worked Solutions

Summation

Q055 MCQ

Select the correct option: Express $8 + 13 + 18 + \dots + 103$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^{20} (5k + 3)$$

Q056 MCQ

Select the correct option: Express $11^2 + 12^2 + \dots + 19^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=11}^{19} k^2$$

Worked Solutions

Summation

Q057 MCQ

Select the correct option: Express $2^2 + 3^2 + \dots + 10^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=2}^{10} k^2$$

Q058 MCQ

Select the correct option: Express $1 + 3 + 9 + \dots + 3^{n-1}$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=0}^{n-1} 3^k$$

Worked Solutions

Summation

Q059 MCQ

Select the correct option: Express $2 + 7 + 12 + \dots + (5n - 3)$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n (5k - 3)$$

Q060 Written

Express $6^2 + 7^2 + 8^2 + \dots + 15^2$ using Σ .

Solution

1. The index starts at 6 and ends at 15.

2. $\sum_{k=6}^{15} k^2$

Final answer

$$\sum_{k=6}^{15} k^2$$

Worked Solutions

Summation

Q061 Written

Express $4^2 + 5^2 + 6^2 + \dots + 13^2$ using Σ .**Solution**

1. The index starts at 4 and ends at 13.
2. $\sum_{k=4}^{13} k^2$

Final answer

$$\sum_{k=4}^{13} k^2$$

Q062 Written

Express $1 + 7 + 25 + 79 + \dots + (3^n - 2)$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n (3^k - 2)$

Final answer

$$\sum_{k=1}^n (3^k - 2)$$

Worked Solutions

Summation

Q063 Written

Express $1 + 2 + 4 + 8 + \dots + 2^{n-1}$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^n 2^{k-1}$

Final answer

$$\sum_{k=1}^n 2^{k-1}$$

Q064 Written

Express $7^2 + 8^2 + \dots + 21^2$ using Σ .

Solution

1. Identify the general term and the endpoint indices.
2. $\sum_{k=7}^{21} k^2$

Final answer

$$\sum_{k=7}^{21} k^2$$

Worked Solutions

Summation

Q065 Written

Express $12^2 + 13^2 + \dots + 18^2$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=12}^{18} k^2$

Final answer

$$\sum_{k=12}^{18} k^2$$

Q066 Written

Express $10^2 + 11^2 + \dots + 20^2$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=10}^{20} k^2$

Final answer

$$\sum_{k=10}^{20} k^2$$

Worked Solutions

Summation

Q067 MCQ

Select the correct option: Express $4^2 + 5^2 + \dots + 12^2$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=4}^{12} k^2$$

Q068 MCQ

Select the correct option: Express $3^1 + 3^2 + \dots + 3^n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n 3^k$$

Worked Solutions

Summation

Q069 MCQ

Select the correct option: Express $2 + 4 + 8 + \dots + 2^n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=1}^n 2^k$$

Q070 MCQ

Select the correct option: Express $1 + 2^1 + 2^2 + \dots + 2^{n-1}$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A**Final answer**

$$\sum_{k=0}^{n-1} 2^k$$

Worked Solutions

Summation

Q071 MCQ

Select the correct option: Express $7 + 14 + 21 + \dots + 7n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=1}^n 7k$$

Q072 MCQ

Select the correct option: Express $10 + 15 + 20 + \dots + 5n$ using the summation symbol Σ :**Solution**

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=2}^n 5k$$

Worked Solutions

Summation

Q073 MCQ

Select the correct option: Express $6^2 + 7^2 + \dots + 16^2$ using the summation symbol Σ :

Solution

1. Find the general term and the first/last index.

Correct option: A

Final answer

$$\sum_{k=6}^{16} k^2$$

Q074 Written

Express $2 + 5 + 8 + \dots + 29$ using Σ .

Solution

1. The general term is $3k - 1$, and $29 = 3(10) - 1$.
2. $\sum_{k=1}^{10} (3k - 1)$

Final answer

$$\sum_{k=1}^{10} (3k - 1)$$

Worked Solutions

Summation

Q075 Written

Express $8 + 16 + 24 + \cdots + 56$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^7 8k$

Final answer

$$\sum_{k=1}^7 8k$$

Q076 Written

Express $7 + 13 + 19 + \cdots + 73$ **using** Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{12} (6k + 1)$

Final answer

$$\sum_{k=1}^{12} (6k + 1)$$

Worked Solutions

Summation

Q077 Written

Express $4 + 9 + 14 + \dots + 119$ using Σ .**Solution**

1. Identify the general term and the endpoint indices.
2. $\sum_{k=1}^{24} (5k - 1)$

Final answer

$$\sum_{k=1}^{24} (5k - 1)$$

Worked Solutions

Summation

Q078 Challenge

A green-tech startup installs 1 solar panel in Week 1, 4 in Week 2, 7 in Week 3, and $3n - 2$ in Week n . Write one expression using Σ for the total panels installed from Week 1 through Week n .

Solution

1. The weekly pattern is arithmetic with general term $3k - 2$.
2. Total = $\sum_{k=1}^n (3k - 2)$

Final answer

$$\sum_{k=1}^n (3k - 2)$$

Worked Solutions

Summation

Q079 Written

Find $\sum_{k=1}^n (-3)$.

Solution

1. There are n copies of -3 .
2. $\sum_{k=1}^n (-3) = -3n$

Final answer $-3n$

Q080 Written

Find $\sum_{k=1}^n 3$.

Solution

1. There are n copies of 3.
2. $\sum_{k=1}^n 3 = 3n$

Final answer $3n$

Worked Solutions

Summation

Q081 Written

Find $\sum_{k=1}^n 6$.

Solution

1. $\sum_{k=1}^n 6 = 6n$

Final answer

$6n$

Q082 Written

Find $\sum_{k=1}^n (-4)$.

Solution

1. $\sum_{k=1}^n (-4) = -4n$

Final answer

$-4n$

Worked Solutions

Summation

Q083 Written

Find $\sum_{k=1}^n 1$.

Solution

1. $\sum_{k=1}^n 1 = n$

Final answer

$$n$$

Q084 MCQ

Select the correct option: Find $\sum_{k=1}^n 9$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$9n$$

Worked Solutions

Summation

Q085 MCQ

Select the correct option: Find $\sum_{k=1}^n (-2)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer $-2n$

Q086 MCQ

Select the correct option: Find $\sum_{k=1}^7 4$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

28

Worked Solutions

Summation

Q087 MCQ

Select the correct option: Find $\sum_{k=1}^{12} (-5)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

-60

Q088 MCQ

Select the correct option: Find $\sum_{k=1}^n k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer
$$\frac{n(n+1)}{2}$$

Worked Solutions

Summation

Q089 MCQ

Select the correct option: Find $\sum_{k=1}^n 4k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$2n(n + 1)$$

Q090 MCQ

Select the correct option: Find $\sum_{k=1}^{10} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$55$$

Worked Solutions

Summation

Q091 MCQ

Select the correct option: Find $\sum_{k=1}^8 3k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

108

Q092 MCQ

Select the correct option: Find $\sum_{k=1}^6 5k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

105

Worked Solutions

Summation

Q093 MCQ

Select the correct option: Find $\sum_{k=1}^{20} k$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

210

Q094 MCQ

Select the correct option: Find $\sum_{k=1}^n k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer
$$\frac{n(n+1)(2n+1)}{6}$$

Worked Solutions

Summation

Q095 MCQ

Select the correct option: Find $\sum_{k=1}^n 6k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$n(n+1)(2n+1)$$

Q096 MCQ

Select the correct option: Find $\sum_{k=1}^5 k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$55$$

Worked Solutions

Summation

Q097 MCQ

Select the correct option: Find $\sum_{k=1}^4 2k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

60

Q098 MCQ

Select the correct option: Find $\sum_{k=1}^3 4k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

56

Worked Solutions

Summation

Q099 MCQ

Select the correct option: Find $\sum_{k=1}^7 k^2$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

140

Q100 Written

Find $\sum_{k=1}^n (5k + 4)$.

Solution

1. $\sum (5k + 4) = 5 \sum k + \sum 4$

2. $= \frac{5n(n+1)}{2} + 4n = \frac{n(5n+13)}{2}$

Final answer

$$\frac{n(5n + 13)}{2}$$

Worked Solutions

Summation

Q101 Written

Find $\sum_{k=1}^n (2k - 3)$.

Solution

$$1. \ 2 \sum k - \sum 3 = n(n+1) - 3n$$

$$2. \ = n(n-2)$$

Final answer

$$n(n-2)$$

Q102 Written

Find $\sum_{k=1}^n (4k - 1)$.

Solution

$$1. \ 4 \sum k - \sum 1 = 2n(n+1) - n$$

$$2. \ = n(2n+1)$$

Final answer

$$n(2n+1)$$

Worked Solutions

Summation

Q103 Written

Find $\sum_{k=1}^n (3k + 2)$.

Solution

$$1. \quad 3 \sum k + \sum 2 = \frac{3n(n+1)}{2} + 2n$$

$$2. \quad = \frac{n(3n+7)}{2}$$

Final answer

$$\frac{n(3n+7)}{2}$$

Q104 Written

Find $\sum_{k=1}^n (6k - 3)$.

Solution

$$1. \quad 6 \sum k - \sum 3 = 3n(n+1) - 3n$$

$$2. \quad = 3n^2$$

Final answer

$$3n^2$$

Worked Solutions

Summation

Q105 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k - 1)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(3n + 1)}{2}$$

Q106 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k + 5)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$n(n + 6)$$

Worked Solutions

Summation

Q107 MCQ

Select the correct option: Find $\sum_{k=1}^n (8k - 4)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$4n^2$$

Q108 MCQ

Select the correct option: Find $\sum_{k=1}^n (5k + 3)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(5n + 11)}{2}$$

Worked Solutions

Summation

Q109 Written

Find $\sum_{k=1}^n (k-1)^2$.

Solution

$$1. \sum (k^2 - 2k + 1) = \sum k^2 - 2 \sum k + \sum 1$$

$$2. = \frac{n(n-1)(2n-1)}{6}$$

Final answer

$$\frac{n(n-1)(2n-1)}{6}$$

Q110 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 + k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Summation

Q111 MCQ

Select the correct option: Find $\sum_{k=1}^n (2k^2 - k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(4n-1)}{6}$$

Q112 MCQ

Select the correct option: Find $\sum_{k=1}^n (k^2 - 1)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(2n^2 + 3n - 5)}{6}$$

Worked Solutions

Summation

Q113 MCQ

Select the correct option: Find $\sum_{k=1}^n (3k^2 + 2k)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Q114 MCQ

Select the correct option: Find $\sum_{k=1}^n (4k^2 - 3k + 2)$:

Solution

1. Split the summand if needed, then apply the summation formulas over the displayed limits.

Correct option: A

Final answer

$$\frac{n(8n^2 + 3n + 7)}{6}$$

Worked Solutions

Summation

Q115 Written

Find $1 \cdot 3 + 2 \cdot 8 + 3 \cdot 13 + \cdots + n(5n - 2)$.

Solution

$$\begin{aligned} 1. \quad & \sum_{k=1}^n k(5k - 2) = 5 \sum k^2 - 2 \sum k \\ 2. \quad & = \frac{5n(n+1)(2n+1)}{6} - n(n+1) = \frac{n(n+1)(10n-1)}{6} \end{aligned}$$

Final answer

$$\frac{n(n+1)(10n-1)}{6}$$

Q116 Written

Find $\sum_{k=1}^n (k-1)(3k-1)$.

Solution

$$\begin{aligned} 1. \quad & (k-1)(3k-1) = 3k^2 - 4k + 1 \\ 2. \quad & 3 \sum k^2 - 4 \sum k + \sum 1 = \frac{n(n-1)(2n+1)}{2} \end{aligned}$$

Final answer

$$\frac{n(n-1)(2n+1)}{2}$$

Worked Solutions

Summation

Q117 Written

Find $1 \cdot 3 + 2 \cdot 5 + 3 \cdot 7 + \cdots + n(2n + 1)$.

Solution

$$1. \sum_{k=1}^n k(2k + 1) = 2 \sum k^2 + \sum k$$

$$2. = \frac{n(n+1)(4n+5)}{6}$$

Final answer

$$\frac{n(n+1)(4n+5)}{6}$$

Q118 Written

Find $\sum_{k=1}^n k(k - 2)$.

Solution

$$1. \sum (k^2 - 2k) = \sum k^2 - 2 \sum k$$

$$2. = \frac{n(n+1)(2n-5)}{6}$$

Final answer

$$\frac{n(n+1)(2n-5)}{6}$$

Worked Solutions

Summation

Q119 Written

Find $\sum_{k=1}^n k(2k - 2)$.

Solution

$$\begin{aligned} 1. & \quad 2 \sum k^2 - 2 \sum k \\ 2. & \quad = \frac{2n(n+1)(n-1)}{3} \end{aligned}$$

Final answer

$$\frac{2n(n+1)(n-1)}{3}$$

Q120 Written

Find $1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 + \cdots + n(n+3)$.

Solution

$$\begin{aligned} 1. & \quad \sum_{k=1}^n k(k+3) = \sum k^2 + 3 \sum k \\ 2. & \quad = \frac{n(n+1)(n+5)}{3} \end{aligned}$$

Final answer

$$\frac{n(n+1)(n+5)}{3}$$

Worked Solutions

Summation

Q121 Written

Find $1 \cdot 5 + 2 \cdot 8 + 3 \cdot 11 + \cdots + n(3n + 2)$.**Solution**

$$1. \sum_{k=1}^n k(3k + 2) = 3 \sum k^2 + 2 \sum k$$

$$2. = \frac{n(n+1)(2n+3)}{2}$$

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Q122 Written

Find $2 \cdot 5 + 3 \cdot 6 + 4 \cdot 7 + \cdots + (n+1)(n+4)$.**Solution**

$$1. \sum_{k=1}^n (k+1)(k+4) = \sum (k^2 + 5k + 4)$$

$$2. = \frac{n(n+4)(n+5)}{3}$$

Final answer

$$\frac{n(n+4)(n+5)}{3}$$

Worked Solutions

Summation

Q123 Written

Find $\sum_{k=1}^4 (2k - 1)$.

Solution

1. $2 \sum_{k=1}^4 k - \sum_{k=1}^4 1 = 2(10) - 4$
2. $= 16$

Final answer

16

Q124 Written

Find $\sum_{k=1}^5 (3k - 2)$.

Solution

1. $3 \sum_{k=1}^5 k - 2 \sum_{k=1}^5 1 = 3(15) - 10$
2. $= 35$

Final answer

35

Worked Solutions

Summation

Q125 Written

Find $\sum_{k=1}^{20} (2k - 7)$.

Solution

1. $2 \sum_{k=1}^{20} k - 7 \sum_{k=1}^{20} 1 = 2(210) - 140$

2. $= 280$

Final answer

280

Q126 Written

Find $\sum_{k=1}^8 (4k + 5)$.

Solution

1. $4 \sum_{k=1}^8 k + 5 \sum_{k=1}^8 1 = 4(36) + 40$

2. $= 184$

Final answer

184

Worked Solutions

Summation

Q127 Written

Find $\sum_{k=1}^{10} (2k - 1)$.

Solution

$$\begin{aligned} 1. & 2 \sum_{k=1}^{10} k - \sum_{k=1}^{10} 1 = 2(55) - 10 \\ 2. & = 100 \end{aligned}$$

Final answer

100

Worked Solutions

Summation

Q128 Challenge

Find $\sum_{r=4}^8 (r^2 + r + 5)$.

Solution

1. $\sum_{r=4}^8 (r^2 + r + 5) = \sum_{r=1}^8 (r^2 + r + 5) - \sum_{r=1}^3 (r^2 + r + 5)$
2. $= (204 + 36 + 40) - (14 + 6 + 15) = 245$

Final answer

245

Worked Solutions

Summation

Q129 Written

Find the sum $\sum_{k=1}^n k^2(k+1)$.

Solution

1. Expand: $k^2(k+1) = k^3 + k^2$.
2. Use $\sum k^3 = \frac{n^2(n+1)^2}{4}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $\sum_{k=1}^n k^2(k+1) = \frac{1}{12}n(n+1)(n+2)(3n+1)$

Final answer

$$\frac{n(n+1)(n+2)(3n+1)}{12}$$

Q130 Written

Find the sum $\sum_{k=1}^n k^2(k-2)$.

Solution

1. Expand: $k^2(k-2) = k^3 - 2k^2$.
2. Substitute the formulae for $\sum k^3$ and $\sum k^2$, then factor.
3. $\sum_{k=1}^n k^2(k-2) = \frac{n(n+1)(3n^2-5n-4)}{12}$

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q131 Written

Find the sum $\sum_{k=1}^n k^2(k+2)$.

Solution

1. Expand: $k^2(k+2) = k^3 + 2k^2$.
2. Substitute the standard sums and factor the common $n(n+1)$.
3. $\sum_{k=1}^n k^2(k+2) = \frac{n(n+1)(3n^2 + 11n + 4)}{12}$

Final answer

$$\frac{n(n+1)(3n^2 + 11n + 4)}{12}$$

Q132 Written

Find the sum $\sum_{k=1}^n k^2(k-3)$.

Solution

1. Expand: $k^2(k-3) = k^3 - 3k^2$.
2. Substitute the standard sums and simplify the common factor.
3. $\sum_{k=1}^n k^2(k-3) = \frac{n(n+1)(n^2-3n-2)}{4}$

Final answer

$$\frac{n(n+1)(n^2-3n-2)}{4}$$

Worked Solutions

Summation

Q133 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+1)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n+2)(3n+1)}{12}$$

Q134 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-2)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q135 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+11n+4)}{12}$$

Q136 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2-3n-2)}{4}$$

Worked Solutions

Summation

Q137 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+5n+2)}{4}$$

Q138 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-13n-8)}{12}$$

Worked Solutions

Summation

Q139 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+19n+8)}{12}$$

Q140 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-5n-4)}{12}$$

Worked Solutions

Summation

Q141 Written

Find the sum $\sum_{k=1}^n k(k^2 - 1)$.

Solution

1. Expand: $k(k^2 - 1) = k^3 - k$.
2. Use $\sum k^3$ and $\sum k$, then factor $n^2 + n - 2 = (n - 1)(n + 2)$.
3. $\sum_{k=1}^n k(k^2 - 1) = \frac{n(n + 1)(n - 1)(n + 2)}{4}$

Final answer

$$\frac{n(n + 1)(n - 1)(n + 2)}{4}$$

Q142 Written

Find the sum $\sum_{k=1}^n k(k^2 + 2)$.

Solution

1. Expand: $k(k^2 + 2) = k^3 + 2k$.
2. Use $\sum k^3$ and $\sum k$, then factor.
3. $\sum_{k=1}^n k(k^2 + 2) = \frac{n(n+1)(n^2 + n + 4)}{4}$

Final answer

$$\frac{n(n+1)(n^2 + n + 4)}{4}$$

Worked Solutions

Summation

Q143 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Q144 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2 + n + 4)}{4}$$

Worked Solutions

Summation

Q145 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+6)}{4}$$

Q146 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 2)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-4)}{4}$$

Worked Solutions

Summation

Q147 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 4)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+8)}{4}$$

Q148 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 3)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-6)}{4}$$

Worked Solutions

Summation

Q149 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+23n+10)}{12}$$

Q150 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k-5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2-17n-10)}{12}$$

Worked Solutions

Summation

Q151 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n+10)}{4}$$

Q152 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 5)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2+n-10)}{4}$$

Worked Solutions

Summation

Q153 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2 + 27n + 12)}{12}$$

Q154 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 + 6)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(n^2 + n + 12)}{4}$$

Worked Solutions

Summation

Q155 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k - 6)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(3n^2 - 25n - 12)}{12}$$

Q156 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 6)$:**Solution**

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A**Final answer**

$$\frac{n(n+1)(n^2 + n - 12)}{4}$$

Worked Solutions

Summation

Q157 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+7)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+31n+14)}{12}$$

Q158 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k^2(k+8)$:

Solution

1. Expand the product and use the standard formulae for the powers of k.

Correct option: A

Final answer

$$\frac{n(n+1)(3n^2+35n+16)}{12}$$

Worked Solutions

Summation

Q159 Written

Find the sum $\sum_{k=1}^n 6 \times 3^{k-1}$.

Solution

1. Use $a = 6, r = 3$.
2. $S_n = \frac{6(3^n - 1)}{3 - 1} = 3(3^n - 1)$

Final answer

$$3(3^n - 1)$$

Q160 Written

Find the sum $\sum_{k=1}^n 4 \times 2^{k-1}$.

Solution

1. Use $a = 4, r = 2$.
2. $S_n = \frac{4(2^n - 1)}{2 - 1} = 4(2^n - 1)$

Final answer

$$4(2^n - 1)$$

Worked Solutions

Summation

Q161 Written

Find the sum $\sum_{k=1}^n 2^k$.

Solution

1. Rewrite as $(2 \times 2^{k-1})$.
2. $S_n = \frac{2(2^n - 1)}{2 - 1} = 2(2^n - 1)$

Final answer

$$2(2^n - 1)$$

Q162 Written

Find the sum $\sum_{k=1}^n 2 \times (-3)^{k-1}$.

Solution

1. Here $a = 2$ and $r = -3$.
2. $S_n = \frac{2(1 - (-3)^n)}{1 - (-3)} = \frac{1 - (-3)^n}{2}$

Final answer

$$\frac{1 - (-3)^n}{2}$$

Worked Solutions

Summation

Q163 Written

Find the sum $\sum_{k=1}^n 3 \times (-2)^{k-1}$.

Solution

1. Use $a = 3, r = -2$.
2. $S_n = \frac{3(1-(-2)^n)}{1-(-2)} = 1 - (-2)^n$

Final answer

$$1 - (-2)^n$$

Q164 Written

Find the sum $\sum_{k=1}^n (-5)^k$.

Solution

1. Rewrite as $(-5) \times (-5)^{k-1}$, so $a = -5, r = -5$.
2. $S_n = \frac{-5(1-(-5)^n)}{1-(-5)} = \frac{5((-5)^n - 1)}{6}$

Final answer

$$\frac{5((-5)^n - 1)}{6}$$

Worked Solutions

Summation

Q165 Written

Find the sum $\sum_{k=1}^n 2 \times (-1)^{k-1}$.

Solution

1. Use $a = 2, r = -1$.
2. $S_n = \frac{2(1-(-1)^n)}{2} = 1 - (-1)^n$

Final answer

$$1 - (-1)^n$$

Q166 Written

Find the sum $\sum_{k=1}^n 8 \times (-3)^{k-1}$.

Solution

1. Use $a = 8, r = -3$.
2. $S_n = \frac{8(1-(-3)^n)}{4} = 2(1 - (-3)^n)$

Final answer

$$2(1 - (-3)^n)$$

Worked Solutions

Summation

Q167 Written

Find the sum $\sum_{k=1}^n (-2)^{k-1}$.

Solution

1. Use $(a = 1, r = -2)$.
2. $S_n = \frac{1 - (-2)^n}{1 - (-2)} = \frac{1 - (-2)^n}{3}$

Final answer

$$\frac{1 - (-2)^n}{3}$$

Q168 Written

Find the sum $\sum_{k=1}^n (-3)^k$.

Solution

1. Rewrite as $((-3)(-3)^{k-1})$.
2. $S_n = \frac{-3(1 - (-3)^n)}{4} = \frac{3((-3)^n - 1)}{4}$

Final answer

$$\frac{3((-3)^n - 1)}{4}$$

Worked Solutions

Summation

Q169 Written

Find the sum $\sum_{k=1}^n 4^k$.

Solution

1. Rewrite $4^k = 4 \times 4^{k-1}$, so $a = 4, r = 4$.

2. $S_n = \frac{4(4^n - 1)}{4 - 1} = \frac{4(4^n - 1)}{3}$

Final answer

$$\frac{4(4^n - 1)}{3}$$

Q170 Written

Find the sum $\sum_{k=1}^n 4^{k-1}$.

Solution

1. The first term is 1 and the common ratio is 4.

2. $S_n = \frac{4^n - 1}{4 - 1} = \frac{4^n - 1}{3}$

Final answer

$$\frac{4^n - 1}{3}$$

Worked Solutions

Summation

Q171 Written

Find the sum $\sum_{k=1}^n 3^{k-1}$.

Solution

1. The first term is 1 and the ratio is 3.
2. $S_n = \frac{3^n - 1}{3 - 1} = \frac{3^n - 1}{2}$

Final answer

$$\frac{3^n - 1}{2}$$

Q172 Written

Find the sum $\sum_{k=1}^4 (-2)^{k-1}$.

Solution

1. This is a finite geometric sum with $(a = 1, r = -2, n = 4)$.
2. $S_4 = 1 - 2 + 4 - 8 = -5$

Final answer

$$-5$$

Worked Solutions

Summation

Q173 Written

Find the sum $\sum_{k=1}^3 3^k$.

Solution

1. Use $(a = 3, r = 3, n = 3)$.
2. $S_3 = 3 + 9 + 27 = 39$

Final answer

39

Q174 Written

Find the sum $\sum_{k=1}^5 2^k$.

Solution

1. Use $(a = 2, r = 2, n = 5)$.
2. $S_5 = 2 + 4 + 8 + 16 + 32 = 62$

Final answer

62

Worked Solutions

Summation

Q175 Written

Find the sum $\sum_{k=1}^4 (-3)^k$.

Solution

1. Use $(a = -3, r = -3, n = 4)$.
2. $S_4 = -3 + 9 - 27 + 81 = 60$

Final answer

60

Worked Solutions

Summation

Q176 Challenge

Find the finite sum $\sum_{r=5}^9 (2r + 7)$.

Solution

1. Substitute ($r = 5, 6, 7, 8, 9$): ($17+19+21+23+25$).
2. $17 + 19 + 21 + 23 + 25 = 105$

Final answer

105

Worked Solutions

Summation

Q177 Written

Find $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \cdots + \frac{1}{(n+1)(n+2)}.$

Solution

1. Split each term as $\frac{1}{j} - \frac{1}{j+1}.$
2. $\frac{1}{2} - \frac{1}{n+2} = \frac{n}{2(n+2)}$

Final answer

$$\frac{n}{2(n+2)}$$

Q178 Written

Find $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \cdots + \frac{1}{n(n+1)}.$

Solution

1. Use $\frac{1}{j(j+1)} = \frac{1}{j} - \frac{1}{j+1}.$
2. $1 - \frac{1}{n+1} = \frac{n}{n+1}$

Final answer

$$\frac{n}{n+1}$$

Worked Solutions

Summation

Q179 Written

Find $\frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \cdots + \frac{1}{(n+2)(n+3)}$.

Solution

1. The gap is 1 and the first term starts at 3.

$$2. \frac{1}{3} - \frac{1}{n+3} = \frac{n}{3(n+3)}$$

Final answer

$$\frac{n}{3(n+3)}$$

Q180 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 2} + \cdots + \frac{1}{n(n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{n+1}$$

Worked Solutions

Summation

Q181 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 3} + \cdots + \frac{1}{(n+1)(n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(n+2)}$$

Q182 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 4} + \cdots + \frac{1}{(n+2)(n+3)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3(n+3)}$$

Worked Solutions

Summation

Q183 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 5} + \cdots + \frac{1}{(n+3)(n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4(n+4)}$$

Q184 MCQ

Select the correct option: Find the sum $\frac{1}{6 \times 7} + \cdots + \frac{1}{(n+5)(n+6)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{6(n+6)}$$

Worked Solutions

Summation

Q185 MCQ

Select the correct option: Find the sum $\frac{1}{7 \times 8} + \cdots + \frac{1}{(n+6)(n+7)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{7(n+7)}$$

Q186 MCQ

Select the correct option: Find the sum $\frac{1}{9 \times 10} + \cdots + \frac{1}{(n+8)(n+9)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{9(n+9)}$$

Worked Solutions

Summation

Q187 MCQ

Select the correct option: Find the sum $\frac{1}{10 \times 11} + \cdots + \frac{1}{(n+9)(n+10)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{10(n+10)}$$

Q188 MCQ

Select the correct option: Find the sum $\frac{1}{11 \times 12} + \cdots + \frac{1}{(n+10)(n+11)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{11(n+11)}$$

Worked Solutions

Summation

Q189 Written

Find the following sum: $\frac{1}{2 \times 5} + \frac{1}{5 \times 8} + \frac{1}{8 \times 11} + \cdots + \frac{1}{(3n-1)(3n+2)}$.

Solution

1. Use $\frac{1}{ab} = \frac{1}{b-a} \left(\frac{1}{a} - \frac{1}{b} \right)$ with gap 3.
2. $\frac{1}{3} \left(\frac{1}{2} - \frac{1}{3n+2} \right) = \frac{n}{2(3n+2)}$

Final answer

$$\frac{n}{2(3n+2)}$$

Q190 Written

Find $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \cdots + \frac{1}{(2n-1)(2n+1)}$.

Solution

1. The gap is 2, so factor out $\frac{1}{2}$.
2. $\frac{1}{2} \left(1 - \frac{1}{2n+1} \right) = \frac{n}{2n+1}$

Final answer

$$\frac{n}{2n+1}$$

Worked Solutions

Summation

Q191 Written

Find $\frac{1}{1 \times 3} + \frac{1}{2 \times 4} + \frac{1}{3 \times 5} + \cdots + \frac{1}{n(n+2)}.$

Solution

1. Use the constant gap 2 decomposition; two end terms remain.
2. $\frac{1}{2}(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2})$

Final answer

$$\frac{1}{2}(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2})$$

Q192 Written

Find $\frac{1}{1 \times 4} + \frac{1}{4 \times 7} + \frac{1}{7 \times 10} + \cdots + \frac{1}{(3n-2)(3n+1)}.$

Solution

1. Use gap 3 and the first denominator 1.
2. $\frac{1}{3}(1 - \frac{1}{3n+1}) = \frac{n}{3n+1}$

Final answer

$$\frac{n}{3n+1}$$

Worked Solutions

Summation

Q193 Written

Find $\frac{1}{1 \times 7} + \frac{1}{4 \times 10} + \frac{1}{7 \times 13} + \cdots + \frac{1}{(3n-2)(3n+4)}$.

Solution

1. The two factors differ by 6, while the first factors advance by 3; two terms remain at each end.
2. $\frac{1}{6}(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4})$

Final answer

$$\frac{1}{6}(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4})$$

Q194 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 3} + \cdots + \frac{1}{(2n-1)(2n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2n+1}$$

Worked Solutions

Summation

Q195 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 4} + \cdots + \frac{1}{(3n-2)(3n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3n+1}$$

Q196 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 7} + \cdots + \frac{1}{(3n-2)(3n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{1}{6} \left(1 + \frac{1}{4} - \frac{1}{3n+1} - \frac{1}{3n+4} \right)$$

Worked Solutions

Summation

Q197 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 6} + \cdots + \frac{1}{(4n-2)(4n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(4n+2)}$$

Q198 MCQ

Select the correct option: Find the sum $\frac{1}{3 \times 8} + \cdots + \frac{1}{(5n-2)(5n+3)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{3(5n+3)}$$

Worked Solutions

Summation

Q199 MCQ

Select the correct option: Find the sum $\frac{1}{1 \times 5} + \cdots + \frac{1}{(4n-3)(4n+1)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4n+1}$$

Q200 MCQ

Select the correct option: Find the sum $\frac{1}{2 \times 7} + \cdots + \frac{1}{(5n-3)(5n+2)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{2(5n+2)}$$

Worked Solutions

Summation

Q201 MCQ

Select the correct option: Find the sum $\frac{1}{4 \times 10} + \cdots + \frac{1}{(6n-2)(6n+4)}$:

Solution

1. Split into two fractions, cancel the chain, and retain the first and last terms.

Correct option: A

Final answer

$$\frac{n}{4(6n+4)}$$

Q202 Written

Find $\frac{1}{4 \times 6} + \frac{1}{5 \times 7} + \frac{1}{6 \times 8} + \cdots + \frac{1}{(n+3)(n+5)}$.

Solution

1. Write the general term as $\frac{1}{(j+3)(j+5)}$, gap 2.

2. $\frac{1}{2} \left(\frac{1}{4} + \frac{1}{5} - \frac{1}{n+4} - \frac{1}{n+5} \right)$

Final answer

$$\frac{1}{2} \left(\frac{1}{4} + \frac{1}{5} - \frac{1}{n+4} - \frac{1}{n+5} \right)$$

Worked Solutions

Summation

Q203 Written

Find $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \cdots + \frac{1}{2n(2n+2)}$.

Solution

1. Factor each denominator as $4j(j+1)$.
2. $\frac{1}{4}\left(1 - \frac{1}{n+1}\right) = \frac{n}{4(n+1)}$

Final answer

$$\frac{n}{4(n+1)}$$

Q204 Written

Find the following sum: $\sum_{k=1}^n \frac{1}{\sqrt{k+2} + \sqrt{k+1}}$.

Solution

1. Multiply by the conjugate.
2. $\frac{\sqrt{k+2}-\sqrt{k+1}}{(k+2)-(k+1)} = \sqrt{k+2} - \sqrt{k+1}$
3. $\sum_{k=1}^n (\sqrt{k+2} - \sqrt{k+1}) = \sqrt{n+2} - \sqrt{2}$

Final answer

$$\sqrt{n+2} - \sqrt{2}$$

Worked Solutions

Summation

Q205 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}}.$

Solution

1. Rationalise with $\sqrt{k+1} - \sqrt{k}.$
2. $\sum_{k=1}^n (\sqrt{k+1} - \sqrt{k}) = \sqrt{n+1} - 1$

Final answer

$$\sqrt{n+1} - 1$$

Q206 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{k} + \sqrt{k-1}}.$

Solution

1. Rationalise: the denominator difference is 1.
2. $\sum_{k=1}^n (\sqrt{k} - \sqrt{k-1}) = \sqrt{n}$

Final answer

$$\sqrt{n}$$

Worked Solutions

Summation

Q207 Written

Find $\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \dots + \frac{1}{\sqrt{n} + \sqrt{n+1}}.$

Solution

1. Use the conjugate to obtain $\sqrt{j+1} - \sqrt{j}.$
2. $\sum_{j=1}^n (\sqrt{j+1} - \sqrt{j}) = \sqrt{n+1} - 1$

Final answer

$$\sqrt{n+1} - 1$$

Q208 Written

Find $\sum_{k=1}^n \frac{1}{\sqrt{2k-1} + \sqrt{2k+1}}.$

Solution

1. Rationalise; the square-difference gap is 2.
2. $\frac{1}{2} \sum_{k=1}^n (\sqrt{2k+1} - \sqrt{2k-1}) = \frac{\sqrt{2n+1}-1}{2}$

Final answer

$$\frac{\sqrt{2n+1} - 1}{2}$$

Worked Solutions

Summation

Q209 Written

Find $\frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \cdots + \frac{1}{\sqrt{n+1} + \sqrt{n+2}}.$

Solution

1. Multiply by each conjugate.
2. $\sum_{j=1}^n (\sqrt{j+2} - \sqrt{j+1}) = \sqrt{n+2} - \sqrt{2}$

Final answer

$$\sqrt{n+2} - \sqrt{2}$$

Worked Solutions

Summation

Q210 Challenge

Find the sum, showing your steps: $\sum_{r=1}^{30} \frac{1}{r(r+2)}$.

Solution

1. Use $\frac{1}{r(r+2)} = \frac{1}{2} \left(\frac{1}{r} - \frac{1}{r+2} \right)$.
2. $S = \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{31} - \frac{1}{32} \right)$
3. $S = \frac{1425}{1984}$

Final answer

$$\frac{1425}{1984}$$

Worked Solutions

Summation

Q211 Written

Find the following sum $S = 3 \times 1 + 5 \times 3 + 7 \times 3^2 + 9 \times 3^3 + \cdots + (2n + 1) \times 3^{n-1}$.

Solution

1. The arithmetic coefficient is $2k + 1$ and the geometric ratio is $r = 3$.
2. Write S and $3S$, subtract with aligned powers, and simplify: $-2S = -2n \times 3^n$.
3. $S = n \times 3^n$

Final answer

$$n \times 3^n$$

Q212 Written

Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$.

Solution

1. Write the first sum and multiply by the common ratio.
2. Subtract and keep the final shifted term with its correct sign.
3. $3^n n$

Final answer

$$3^n n$$

Worked Solutions

Summation

Q213 Written

Find the sum $\sum_{k=1}^4 (2k + 1) \times (3)^{k-1}$.

Solution

1. Identify the arithmetic coefficient and geometric ratio.
2. Apply (S-rS) over the finite limit ($n = 4$).
3. 324

Final answer

324

Q214 Written

Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$.

Solution

1. Write the first sum and multiply by the common ratio.
2. Subtract and keep the final shifted term with its correct sign.
3. $\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$

Final answer

$$\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$$

Worked Solutions

Summation

Q215 Written

Find the sum $\sum_{k=1}^6 (2k - 1) \times (4)^{k-1}$.

Solution

1. Identify the arithmetic coefficient and geometric ratio.
2. Apply (S-rS) over the finite limit ($n = 6$).
3. 14109

Final answer

14109

Q216 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer $3^n n$

Worked Solutions

Summation

Q217 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k - 1) \times (4)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{6 \cdot 2^{2n}n - 5 \cdot 2^{2n} + 5}{9}$$

Q218 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 4) \times (2)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$2^n n + 3 \cdot 2^n - 3$$

Worked Solutions

Summation

Q219 Written

Find the following sum $S = 2 \times 1 + 3 \times 3 + 4 \times 3^2 + 5 \times 3^3 + \dots + (n + 1) \times 3^{n-1}$.

Solution

1. Here the coefficient is $k + 1$ and $r = 3$.
2. Subtract $3S$ from S ; the interior coefficients reduce to a geometric sum of 1s.
3. $-2S = \frac{1 - (2n+1) \times 3^n}{2}$
4. $S = \frac{(2n+1) \times 3^n - 1}{4}$

Final answer

$$\frac{(2n + 1) \times 3^n - 1}{4}$$

Q220 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k + 1) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{2 \cdot 3^n n + 3^n - 1}{4}$$

Worked Solutions

Summation

Q221 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k-2) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{2 \cdot 3^n n - 5 \cdot 3^n + 5}{4}$$

Q222 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k-3) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$3^n n - 2 \cdot 3^n + 2$$

Worked Solutions

Summation

Q223 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (2k + 1) \times (-2)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$-\frac{6(-2)^n n + 5(-2)^n - 5}{9}$$

Worked Solutions

Summation

Q224 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (k+2) \times (-3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$-\frac{4(-3)^n n + 9(-3)^n - 9}{16}$$

Worked Solutions

Summation

Q225 Written

Find the following sum $S = 1 \times 1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \cdots + n \times 2^{n-1}$.

Solution

1. Use $a_k = k$ and $r = 2$.
2. Applying $S - rS$ gives $-S = 1 - (n-1) \times 2^n$.
3. $S = (n-1) \times 2^n + 1$

Final answer

$$(n-1) \times 2^n + 1$$

Q226 Written

Find the following sum $S = 1 \times 3 + 3 \times 3^2 + 5 \times 3^3 + 7 \times 3^4 + \cdots + (2n-1) \times 3^n$.

Solution

1. Factor out the leading 3, leaving $\sum_{k=1}^n (2k-1)3^{k-1}$.
2. The $S - rS$ subtraction gives $\sum_{k=1}^n (2k-1)3^{k-1} = (n-1)3^n + 1$.
3. $S = (n-1) \times 3^{n+1} + 3$

Final answer

$$(n-1) \times 3^{n+1} + 3$$

Worked Solutions

Summation

Q227 MCQ

Select the correct option: Find the sum $\sum_{k=1}^n (3k) \times (3)^{k-1}$:

Solution

1. Identify the arithmetic coefficient and the ratio, then apply S-rS.

Correct option: A

Final answer

$$\frac{3(2 \cdot 3^n n - 3^n + 1)}{4}$$

Worked Solutions

Summation

Q228 Challenge

Find the sum of the partial sums $5 + (5 + 9) + (5 + 9 + 13) + \cdots + [5 + 9 + \cdots + (4k + 1)]$.

Solution

1. The j -th inner arithmetic sum is $5 + 9 + \cdots + (4j + 1) = j(2j + 3)$.
2. Sum the partial sums: $\sum_{j=1}^k (2j^2 + 3j)$.
3. $\frac{k(k+1)(4k+11)}{6}$

Final answer

$$\frac{k(k+1)(4k+11)}{6}$$

Worked Solutions

Summation

Q229 Written

For the sequence 3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, ..., find the k -th term.

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 2.
2. $u_i = 2i + 1$, $a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(3) + (k - 1)(2)] = k(k + 2)$

Final answer

$$a_k = k(k + 2)$$

Q230 Written

For the sequence 1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, ..., find the k -th term.

Solution

1. The inner terms form an arithmetic sequence with first term 1 and common difference 2.
2. $u_i = 2i - 1$, $a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(1) + (k - 1)(2)] = k^2$

Final answer

$$a_k = k^2$$

Worked Solutions

Summation

Q231 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, **find the k -th term.**

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 2.
2. $u_i = 2i, \quad a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(2) + (k-1)(2)] = k(k+1)$

Final answer

$$a_k = k(k+1)$$

Q232 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, **find the k -th term.**

Solution

1. The inner terms form an arithmetic sequence with first term 1 and common difference 4.
2. $u_i = 4i - 3, \quad a_k = \sum_{i=1}^k u_i$
3. $a_k = \frac{k}{2} [2(1) + (k-1)(4)] = k(2k-1)$

Final answer

$$a_k = k(2k-1)$$

Worked Solutions

Summation

Q233 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 2.

2. $u_i = 2i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k-1)(2)] = k(k+1)$

Correct option: A

Final answer

$$k(k+1)$$

Q234 MCQ

Select the correct option: For the cumulative sequence $3, 3 + 6, 3 + 6 + 9, 3 + 6 + 9 + 12, \dots$, what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 3.

2. $u_i = 3i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(3) + (k-1)(3)] = \frac{3k(k+1)}{2}$

Correct option: A

Final answer

$$\frac{3k(k+1)}{2}$$

Worked Solutions

Summation

Q235 MCQ

Select the correct option: For the cumulative sequence 4, 4 + 6, 4 + 6 + 8, 4 + 6 + 8 + 10, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 4 and common difference 2.

2. $u_i = 2i + 2, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(4) + (k-1)(2)] = k(k+3)$

Correct option: A

Final answer

$$k(k+3)$$

Q236 MCQ

Select the correct option: For the cumulative sequence 2, 2 + 6, 2 + 6 + 10, 2 + 6 + 10 + 14, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 4.

2. $u_i = 4i - 2, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k-1)(4)] = 2k^2$

Correct option: A

Final answer

$$2k^2$$

Worked Solutions

Summation

Q237 MCQ

Select the correct option: For the cumulative sequence 5, 5 + 8, 5 + 8 + 11, 5 + 8 + 11 + 14, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 5 and common difference 3.

$$2. u_i = 3i + 2, \quad a_k = \sum_{i=1}^k u_i$$

$$3. a_k = \frac{k}{2} [2(5) + (k-1)(3)] = \frac{k(3k+7)}{2}$$

Correct option: A

Final answer

$$\frac{k(3k+7)}{2}$$

Q238 MCQ

Select the correct option: For the cumulative sequence 3, 3 + 8, 3 + 8 + 13, 3 + 8 + 13 + 18, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 3 and common difference 5.

$$2. u_i = 5i - 2, \quad a_k = \sum_{i=1}^k u_i$$

$$3. a_k = \frac{k}{2} [2(3) + (k-1)(5)] = \frac{k(5k+1)}{2}$$

Correct option: A

Final answer

$$\frac{k(5k+1)}{2}$$

Worked Solutions

Summation

Q239 MCQ

Select the correct option: For the cumulative sequence 6, 6 + 8, 6 + 8 + 10, 6 + 8 + 10 + 12, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 6 and common difference 2.

2. $u_i = 2i + 4, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(6) + (k-1)(2)] = k(k+5)$

Correct option: A

Final answer

$$k(k+5)$$

Q240 MCQ

Select the correct option: For the cumulative sequence 4, 4 + 8, 4 + 8 + 12, 4 + 8 + 12 + 16, ..., what is a_k :

Solution

1. The inner terms form an arithmetic sequence with first term 4 and common difference 4.

2. $u_i = 4i, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(4) + (k-1)(4)] = 2k(k+1)$

Correct option: A

Final answer

$$2k(k+1)$$

Worked Solutions

Summation

Q241 Written

For the sequence 3, 3 + 5, 3 + 5 + 7, 3 + 5 + 7 + 9, ..., find the sum from the first term to the n -th term.

Solution

1. First collapse the inner arithmetic sum: $a_k = k(k + 2)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(2n+7)}{6}$

Final answer

$$S_n = \frac{n(n+1)(2n+7)}{6}$$

Q242 Written

For the sequence 1, 1 + 3, 1 + 3 + 5, 1 + 3 + 5 + 7, ..., find the sum from the first term to the n -th term.

Solution

1. First collapse the inner arithmetic sum: $a_k = k^2$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(2n+1)}{6}$

Final answer

$$S_n = \frac{n(n+1)(2n+1)}{6}$$

Worked Solutions

Summation

Q243 Written

For the sequence $1, 1 + 5, 1 + 5 + 9, 1 + 5 + 9 + 13, \dots$, **find the sum from the first term to the n -th term.**

Solution

1. First collapse the inner arithmetic sum: $a_k = k(2k - 1)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(4n-1)}{6}$

Final answer

$$S_n = \frac{n(n+1)(4n-1)}{6}$$

Q244 Written

For the sequence $2, 2 + 4, 2 + 4 + 6, 2 + 4 + 6 + 8, \dots$, **find the sum from the first term to the n -th term.**

Solution

1. First collapse the inner arithmetic sum: $a_k = k(k + 1)$.
2. $S_n = \sum_{k=1}^n a_k$; use $\sum k = \frac{n(n+1)}{2}$ and $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
3. $S_n = \frac{n(n+1)(n+2)}{3}$

Final answer

$$S_n = \frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Summation

Q245 MCQ

Select the correct option: For the cumulative sequence $2, 2 + 5, 2 + 5 + 8, 2 + 5 + 8 + 11, \dots$, find a_5 :

Solution

1. The inner terms form an arithmetic sequence with first term 2 and common difference 3.

2. $u_i = 3i - 1, \quad a_k = \sum_{i=1}^k u_i$

3. $a_k = \frac{k}{2} [2(2) + (k-1)(3)] = \frac{k(3k+1)}{2}$

4. $a_5 = 40$

Correct option: A

Final answer

40

Worked Solutions

Summation

Q246 Challenge

The sum of the first n terms of $\{a_n\}$ is $S_n = 2n^2 + 5n$. Find the general term.

Solution

1. $a_1 = S_1 = 7$
2. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
3. $a_n = (2n^2 + 5n) - [2(n-1)^2 + 5(n-1)] = 4n + 3$
4. The formula also gives $a_1 = 7$, so it is valid for every positive integer n .

Final answer

$$a_n = 4n + 3$$

Worked Solutions

Summation

Q247 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 3)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 4)(n - 1)$
3. $a_n = 2(n - 2)$
4. $a_1 = S_1 = -2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2(n - 2)$$

Worked Solutions

Summation

Q248 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(2n + 5)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(2n+3)$
3. $a_n = 4n + 3$
4. $a_1 = S_1 = 7$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 4n + 3$$

Worked Solutions

Summation

Q249 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)^2$
3. $a_n = 2n - 1$
4. $a_1 = S_1 = 1$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Summation

Q250 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n(n - 4)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 5)(n - 1)$
3. $a_n = 2n - 5$
4. $a_1 = S_1 = -3$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2n - 5$$

Worked Solutions

Summation

Q251 MCQ

Select the correct option: If $S_n = n(n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 1)(n + 1)$
3. $a_n = 2n + 1$
4. $a_1 = S_1 = 3$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 2n + 1$$

Worked Solutions

Summation

Q252 MCQ

Select the correct option: If $S_n = 2n(n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2(n-1)(n+1)$
3. $a_n = 2(2n+1)$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 2(2n + 1)$$

Worked Solutions

Summation

Q253 MCQ

Select the correct option: If $S_n = 3n(n - 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 3(n - 2)(n - 1)$
3. $a_n = 6(n - 1)$
4. $a_1 = S_1 = 0$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 6(n - 1)$$

Worked Solutions

Summation

Q254 MCQ

Select the correct option: If $S_n = 2n(2n - 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2(n-1)(2n-3)$
3. $a_n = 2(4n-3)$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: A

Final answer

$$a_n = 2(4n - 3)$$

Worked Solutions

Summation

Q255 MCQ

Select the correct option: If $S_n = 5n(n + 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 5n(n - 1)$
3. $a_n = 10n$
4. $a_1 = S_1 = 10$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 10n$$

Worked Solutions

Summation

Q256 MCQ

Select the correct option: If $S_n = n(2n + 3)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(2n+1)$
3. $a_n = 4n + 1$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 4n + 1$$

Worked Solutions

Summation

Q257 MCQ

Select the correct option: If $S_n = n(3n + 1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(3n-2)$
3. $a_n = 2(3n-1)$
4. $a_1 = S_1 = 4$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 2(3n - 1)$$

Worked Solutions

Summation

Q258 MCQ

Select the correct option: If $S_n = n(n^2 + 2n + 3)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(n^2 + 2)$
3. $a_n = 3n^2 + n + 2$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: C

Final answer

$$a_n = 3n^2 + n + 2$$

Worked Solutions

Summation

Q259 MCQ

Select the correct option: If $S_n = n(n+1)(2n-1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(2n-3)$
3. $a_n = 2n(3n-2)$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 2n(3n-2)$$

Worked Solutions

Summation

Q260 MCQ

Select the correct option: If $S_n = n(3n^2 - 2n + 4)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(3n^2 - 8n + 9)$
3. $a_n = 9n^2 - 13n + 9$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: A

Final answer

$$a_n = 9n^2 - 13n + 9$$

Worked Solutions

Summation

Q261 MCQ

Select the correct option: If $S_n = n(n^2 - n + 2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-1)(n^2 - 3n + 4)$
3. $a_n = 3n^2 - 5n + 4$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 3n^2 - 5n + 4$$

Worked Solutions

Summation

Q262 MCQ

Select the correct option: If $S_n = n(n+1)(2n+1)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(2n-1)$
3. $a_n = 6n^2$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: D

Final answer

$$a_n = 6n^2$$

Worked Solutions

Summation

Q263 MCQ

Select the correct option: If $S_n = n(n+1)(n+2)$, the general term a_n is:

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n(n-1)(n+1)$
3. $a_n = 3n(n+1)$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Correct option: B

Final answer

$$a_n = 3n(n+1)$$

Worked Solutions

Summation

Q264 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^3 + 2n + 6$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n^3 - 3n^2 + 5n + 3$
3. $a_n = 3(n^2 - n + 1)$
4. $a_1 = S_1 = 9$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 9, \quad a_n = 3(n^2 - n + 1) \quad (n \geq 2)$$

Worked Solutions

Summation

Q265 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = n^2 + 2n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = n^2 - 2$
3. $a_n = 2n + 1$
4. $a_1 = S_1 = 2$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 2, \quad a_n = 2n + 1 \quad (n \geq 2)$$

Worked Solutions

Summation

Q266 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = (n-1)(n^2 + n + 1)$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n-2)(n^2 - n + 1)$
3. $a_n = 3n^2 - 3n + 1$
4. $a_1 = S_1 = 0$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \quad (n \geq 2)$$

Worked Solutions

Summation

Q267 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{5^n - 5}{5}$
3. $a_n = 4 \cdot 5^{n-1}$
4. $a_1 = S_1 = 4$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 4 \cdot 5^{n-1}$$

Worked Solutions

Summation

Q268 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 3^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{3^n - 3}{3}$
3. $a_n = 2 \cdot 3^{n-1}$
4. $a_1 = S_1 = 2$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2 \cdot 3^{n-1}$$

Worked Solutions

Summation

Q269 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 2^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{2^n - 2}{2}$
3. $a_n = 2^{n-1}$
4. $a_1 = S_1 = 1$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 2^{n-1}$$

Worked Solutions

Summation

Q270 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 7^n - 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{7^n - 7}{7}$
3. $a_n = 6 \cdot 7^{n-1}$
4. $a_1 = S_1 = 6$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .

Final answer

$$a_n = 6 \cdot 7^{n-1}$$

Worked Solutions

Summation

Q271 **Written**

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 5^n - 2$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{5^n - 10}{5}$
3. $a_n = 4 \cdot 5^{n-1}$
4. $a_1 = S_1 = 3$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 3, \quad a_n = 4 \cdot 5^{n-1} \quad (n \geq 2)$$

Worked Solutions

Summation

Q272 Written

Find the general term of the sequence $\{a_n\}$ where the sum S_n from the first term to the n -th term is given by $S_n = 4 \cdot 2^n - 3$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 4 \cdot 2^{n-1} - 3$
3. $a_n = 2^{n+1}$
4. $a_1 = S_1 = 5$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 5, \quad a_n = 2^{n+1} \quad (n \geq 2)$$

Worked Solutions

Summation

Q273 MCQ

Select the correct option: If $S_n = n(n + 4)$, find a_5 :**Solution**

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = (n - 1)(n + 3)$
3. $a_n = 2n + 3$
4. $a_1 = S_1 = 5$
5. Substituting $n = 1$ gives the same value, so the formula holds for every positive integer n .
6. $a_5 = 13$

Correct option: B**Final answer**

13

Worked Solutions

Summation

Q274 Challenge

Find the general term of the sequence $\{a_n\}$ where S_n is given by $S_n = 3 \cdot 3^n - 2n + 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 3^n - 2n + 3$
3. $a_n = 2(3^n - 1)$
4. $a_1 = S_1 = 8$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 8, \quad a_n = 2(3^n - 1) \quad (n \geq 2)$$

Worked Solutions

Summation

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: Expand $\sum_{k=1}^4 (3k - 1)$ into a sum of terms:

Solution

1. Substitute $k = 1, 2, 3, 4$.

Correct option: A

Final answer

$$2 + 5 + 8 + 11$$

Quiz MCQ 2 **Concept Quiz · MCQ**

Select the correct option: Find the sum $\sum_{k=1}^n k(k^2 - 1)$:

Solution

1. Expand $k(k^2 - 1) = k^3 - k$.

Correct option: A

Final answer

$$\frac{n(n+1)(n-1)(n+2)}{4}$$

Worked Solutions

Summation

Quiz MCQ 3 Concept Quiz · MCQ

Select the correct option: Find the sum $\sum_{k=1}^n k \times 2^{k-1}$:

Solution

1. Write S and rS, subtract, and isolate S.

Correct option: A

Final answer

$$(n - 1) \times 2^n + 1$$

Worked Solutions

Summation

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: If $S_n = 2^n - 3$, find a_6 :

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = \frac{2^n - 6}{2}$
3. $a_n = 2^{n-1}$
4. $a_1 = S_1 = -1$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.
6. $a_6 = 32$

Correct option: C

Final answer

32

Worked Solutions

Summation

Quiz WRITTEN 5 Concept Quiz · Written

Expand $\sum_{k=1}^5 (k^2 - 2)$ into a sum of terms.

Solution

1. Substitute $k = 1, 2, 3, 4, 5$.

Final answer

$$-1 + 2 + 7 + 14 + 23$$

Quiz WRITTEN 6 Concept Quiz · Written

Find the sum $\sum_{k=1}^n k^2(k + 5)$.

Solution

1. Expand into $(k^3 + 5k^2)$.
2. Use the standard power sums and factor.

Final answer

$$\frac{n(n+1)(3n^2 + 23n + 10)}{12}$$

Worked Solutions

Summation

Quiz WRITTEN 7 Concept Quiz · Written

Find the sum $\sum_{k=1}^n (4k - 1) \times 3^k$.

Solution

1. Show the S-rS alignment (or the inner arithmetic sum for the challenge).
2. $\frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$

Final answer

$$\frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$$

Worked Solutions

Summation

Quiz WRITTEN 8 Concept Quiz · Written

Find the general term of the sequence $\{a_n\}$ when S_n is $2 \cdot 2^n - 2n + 1$.

Solution

1. For $n \geq 2$, use $a_n = S_n - S_{n-1}$.
2. $S_{n-1} = 2^n - 2n + 3$
3. $a_n = 2^n - 2$
4. $a_1 = S_1 = 3$
5. The difference formula does not match a_1 at $n = 1$; keep the first term separately.

Final answer

$$a_1 = 3, \quad a_n = 2^n - 2 \quad (n \geq 2)$$

Answer key

Use the question number shown on each page.

Q001 $2 + 4 + 6 + 8 + 10$

Q002 $3 + 5 + 7 + 9 + 11$

Q003 $3 + 6 + 9 + 12 + 15$

Q004 $2 + 6 + 10 + 14$

Q005 $-1 + 1 + 3 + 5 + 7$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 A

Q013 A

Q014 $-2 + 1 + 6$

Q015 $5 + 14 + 29$

Q016 $1 + 4 + 9 + 16$

Q017 $5 + 11 + 21$

Q018 $-1 + 0 + 3 + 8$

Q019 A

Q020 A

Q021 A

Q022 A

Q023 A

Q024 A

Q025 A

Q026 A

Q027 $3 + 9 + 27 + 81$

Q028 $2 + 2 + 2$

Q029 $2 + 4 + 8 + 16$

Q030 $3 + 3 + 3 + 3 + 3 + 3$

Q031 $3 + 9 + 27$

Q032 $8 + 28 + 128$

Q033 $-1 + 1 + 5 + 13$

Q034 $5 + 5 + 5 + 5$

Q035 $3 + 3 + 3$

Q036 $4 + 4 + 4 + 4 + 4$

Q037 A

Q038 A

Q039 A

Q040 A

Q041 A

Q042 A

Q043 A

Q044 A

Q045 $\sum_{k=1}^n (4k - 1)$

Q046 $\sum_{k=1}^n (3k + 2)$

Q047 $\sum_{k=1}^n (2k + 1)$

Q048 $\sum_{k=1}^n (2k - 1)$

Q049 $\sum_{k=1}^n (7k - 1)$

Q050 $\sum_{k=1}^n (4k + 5)$

Q051 $\sum_{k=1}^n (3k - 2)$

Q052 A

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $\sum_{k=6}^{15} k^2$

Q061 $\sum_{k=4}^{13} k^2$

Q062 $\sum_{k=1}^n (3^k - 2)$

Q063 $\sum_{k=1}^n 2^{k-1}$

Q064 $\sum_{k=7}^{21} k^2$

Q065 $\sum_{k=12}^{18} k^2$

Q066 $\sum_{k=10}^{20} k^2$

Q067 A

Q068 A

Q069 A

Q070 A

Q071 A

Q072 A

Q073 A

Q074 $\sum_{k=1}^{10} (3k - 1)$

Q075 $\sum_{k=1}^7 8k$

Q076 $\sum_{k=1}^{12} (6k + 1)$

Q077 $\sum_{k=1}^{24} (5k - 1)$

Q078 $\sum_{k=1}^n (3k - 2)$

Q079 $-3n$

Q080 $3n$

Q081 $6n$

Q082 $-4n$

Q083 n

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A	Q118 $\frac{n(n+1)(2n-5)}{6}$	Q145 A	Q174 62
Q090 A	Q119 $\frac{2n(n+1)(n-1)}{3}$	Q146 A	Q175 60
Q091 A	Q120 $\frac{n(n+1)(n+5)}{3}$	Q147 A	Q176 105
Q092 A	Q121 $\frac{n(n+1)(2n+3)}{2}$	Q148 A	Q177 $\frac{n}{2(n+2)}$
Q093 A	Q122 $\frac{n(n+4)(n+5)}{3}$	Q149 A	Q178 $\frac{n}{n+1}$
Q094 A	Q123 16	Q150 A	Q179 $\frac{n}{3(n+3)}$
Q095 A	Q124 35	Q151 A	Q180 A
Q096 A	Q125 280	Q152 A	Q181 A
Q097 A	Q126 184	Q153 A	Q182 A
Q098 A	Q127 100	Q154 A	Q183 A
Q099 A	Q128 245	Q155 A	Q184 A
Q100 $\frac{n(5n+13)}{2}$	Q129 $\frac{n(n+1)(n+2)(3n+1)}{12}$	Q156 A	Q185 A
Q101 $n(n-2)$	Q130 $\frac{n(n+1)(3n^2-5n-4)}{12}$	Q157 A	Q186 A
Q102 $n(2n+1)$	Q131 $\frac{n(n+1)(3n^2+11n+4)}{12}$	Q158 A	Q187 A
Q103 $\frac{n(3n+7)}{2}$	Q132 $\frac{n(n+1)(n^2-3n-2)}{4}$	Q159 $3(3^n-1)$	Q188 A
Q104 $3n^2$	Q133 A	Q160 $4(2^n-1)$	Q189 $\frac{n}{2(3n+2)}$
Q105 A	Q134 A	Q161 $2(2^n-1)$	Q190 $\frac{n}{2n+1}$
Q106 A	Q135 A	Q162 $\frac{1-(-3)^n}{2}$	Q191 $\frac{1}{2}(1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2})$
Q107 A	Q136 A	Q163 $1-(-2)^n$	Q192 $\frac{n}{3n+1}$
Q108 A	Q137 A	Q164 $\frac{5((-5)^n-1)}{6}$	Q193 $\frac{1}{6}(1+\frac{1}{4}-\frac{1}{3n+1}-\frac{1}{3n+4})$
Q109 $\frac{n(n-1)(2n-1)}{6}$	Q138 A	Q165 $1-(-1)^n$	Q194 A
Q110 A	Q139 A	Q166 $2(1-(-3)^n)$	Q195 A
Q111 A	Q140 A	Q167 $\frac{1-(-2)^n}{3}$	Q196 A
Q112 A	Q141 $\frac{n(n+1)(n-1)(n+2)}{4}$	Q168 $\frac{3((-3)^n-1)}{4}$	Q197 A
Q113 A	Q142 $\frac{n(n+1)(n^2+n+4)}{4}$	Q169 $\frac{4(4^n-1)}{3}$	Q198 A
Q114 A	Q143 A	Q170 $\frac{4^n-1}{3}$	Q199 A
Q115 $\frac{n(n+1)(10n-1)}{6}$	Q144 A	Q171 $\frac{3^n-1}{2}$	Q200 A
Q116 $\frac{n(n-1)(2n+1)}{2}$		Q172 -5	Q201 A
Q117 $\frac{n(n+1)(4n+5)}{6}$		Q173 39	Q202 $\frac{1}{2}(\frac{1}{4}+\frac{1}{5}-\frac{1}{n+4}-\frac{1}{n+5})$

$$\text{Q203 } \frac{n}{4(n+1)}$$

$$\text{Q204 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q205 } \sqrt{n+1} - 1$$

$$\text{Q206 } \sqrt{n}$$

$$\text{Q207 } \sqrt{n+1} - 1$$

$$\text{Q208 } \frac{\sqrt{2n+1} - 1}{2}$$

$$\text{Q209 } \sqrt{n+2} - \sqrt{2}$$

$$\text{Q210 } \frac{1425}{1984}$$

$$\text{Q211 } n \times 3^n$$

$$\text{Q212 } 3^n n$$

$$\text{Q213 } 324$$

$$\text{Q214 } \frac{6 \cdot 2^{2n} n - 5 \cdot 2^{2n} + 5}{9}$$

$$\text{Q215 } 14109$$

$$\text{Q216 } A$$

$$\text{Q217 } A$$

$$\text{Q218 } A$$

$$\text{Q219 } \frac{(2n+1) \times 3^n - 1}{4}$$

$$\text{Q220 } A$$

$$\text{Q221 } A$$

$$\text{Q222 } A$$

$$\text{Q223 } A$$

$$\text{Q224 } A$$

$$\text{Q225 } (n-1) \times 2^n + 1$$

$$\text{Q226 } (n-1) \times 3^{n+1} + 3$$

$$\text{Q227 } A$$

$$\text{Q228 } \frac{k(k+1)(4k+11)}{6}$$

$$\text{Q229 } a_k = k(k+2)$$

$$\text{Q230 } a_k = k^2$$

$$\text{Q231 } a_k = k(k+1)$$

$$\text{Q232 } a_k = k(2k-1)$$

$$\text{Q233 } A$$

$$\text{Q234 } A$$

$$\text{Q235 } A$$

$$\text{Q236 } A$$

$$\text{Q237 } A$$

$$\text{Q238 } A$$

$$\text{Q239 } A$$

$$\text{Q240 } A$$

$$\text{Q241 } S_n = \frac{n(n+1)(2n+7)}{6}$$

$$\text{Q242 } S_n = \frac{n(n+1)(2n+1)}{6}$$

$$\text{Q243 } S_n = \frac{n(n+1)(4n-1)}{6}$$

$$\text{Q244 } S_n = \frac{n(n+1)(n+2)}{3}$$

$$\text{Q245 } A$$

$$\text{Q246 } a_n = 4n + 3$$

$$\text{Q247 } a_n = 2(n-2)$$

$$\text{Q248 } a_n = 4n + 3$$

$$\text{Q249 } a_n = 2n - 1$$

$$\text{Q250 } a_n = 2n - 5$$

$$\text{Q251 } D$$

$$\text{Q252 } C$$

$$\text{Q253 } B$$

$$\text{Q254 } A$$

$$\text{Q255 } D$$

$$\text{Q256 } C$$

$$\text{Q257 } C$$

$$\text{Q258 } C$$

$$\text{Q259 } B$$

$$\text{Q260 } A$$

$$\text{Q261 } D$$

$$\text{Q262 } D$$

$$\text{Q263 } B$$

$$\text{Q264 } a_1 = 9, \quad a_n = 3(n^2 - n + 1) \quad (n \geq 2)$$

$$\text{Q265 } a_1 = 2, \quad a_n = 2n + 1 \quad (n \geq 2)$$

$$\text{Q266 } a_1 = 0, \quad a_n = 3n^2 - 3n + 1 \quad (n \geq 2)$$

$$\text{Q267 } a_n = 4 \cdot 5^{n-1}$$

$$\text{Q268 } a_n = 2 \cdot 3^{n-1}$$

$$\text{Q269 } a_n = 2^{n-1}$$

$$\text{Q270 } a_n = 6 \cdot 7^{n-1}$$

$$\text{Q271 } a_1 = 3, \quad a_n = 4 \cdot 5^{n-1} \quad (n \geq 2)$$

$$\text{Q272 } a_1 = 5, \quad a_n = 2^{n+1} \quad (n \geq 2)$$

$$\text{Q273 } B$$

$$\text{Q274 } a_1 = 8, \quad a_n = 2(3^n - 1) \quad (n \geq 2)$$

$$\text{Quiz MCQ 1 } A$$

$$\text{Quiz MCQ 2 } A$$

$$\text{Quiz MCQ 3 } A$$

$$\text{Quiz MCQ 4 } C$$

$$\text{Quiz WRITTEN 5 } -1 + 2 + 7 + 14 + 23$$

$$\text{Quiz WRITTEN 6 } \frac{n(n+1)(3n^2 + 23n + 10)}{12}$$

$$\text{Quiz WRITTEN 7 } \frac{3(4 \cdot 3^n n - 3 \cdot 3^n + 3)}{2}$$

$$\text{Quiz WRITTEN 8 } a_1 = 3, \quad a_n = 2^n - 2 \quad (n \geq 2)$$

Concept 4 | Advanced Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Differences

$$d_n = a_{n+1} - a_n$$

Rebuild

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k$$

Grouping

odd/even index $\rightarrow$ separate rules

Worked Solutions

Advanced Sequences

Q001 Written

Find the general term of the sequence $\{2, 5, 12, 23, 38, 57, \dots\}$.

Solution

1. The first differences are 3, 7, 11, 15, 19,
2. Hence $b_k = 4k - 1$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (4k - 1) = 2n^2 - 3n + 3$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = 2n^2 - 3n + 3$$

Worked Solutions

Advanced Sequences

Q002 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, then a_n is:

Solution

1. The first differences are represented by $b_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Correct option: C

Final answer

$$a_n = n^2$$

Q003 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, then a_n is:

Solution

1. The first differences are represented by $b_k = 3k - 2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k - 2) = \frac{3n^2 - 7n + 8}{2}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Correct option: B

Final answer

$$a_n = \frac{3n^2 - 7n + 8}{2}$$

Worked Solutions

Advanced Sequences

Q004 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, then a_n is:

Solution

1. The first differences are represented by $b_k = 6k - 5$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 7 + \sum_{k=1}^{n-1} (6k - 5) = 3n^2 - 8n + 12$.
4. Substituting $n = 1$ gives $a_1 = 7$, matching the given first term.

Correct option: A

Final answer

$$a_n = 3n^2 - 8n + 12$$

Worked Solutions

Advanced Sequences

Q005 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = 2k + 1$, find a_5 :**Solution**

1. The first differences are represented by $b_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.
5. $a_5 = 25$

Correct option: B**Final answer**

25

Worked Solutions

Advanced Sequences

Q006 MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k - 2$, find a_6 :**Solution**

1. The first differences are represented by $b_k = 3k - 2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k - 2) = \frac{3n^2 - 7n + 8}{2}$.
4. Substituting $n = 1$ gives $a_1 = \frac{2}{2}$, matching the given first term.
5. $a_6 = 37$

Correct option: A**Final answer**

37

Worked Solutions

Advanced Sequences

Q007 MCQ

Select the correct option: If $a_1 = 7$ and $b_k = 6k - 5$, find a_6 :**Solution**

1. The first differences are represented by $b_k = 6k - 5$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 7 + \sum_{k=1}^{n-1} (6k - 5) = 3n^2 - 8n + 12$.
4. Substituting $n = 1$ gives $a_1 = 7$, matching the given first term.
5. $a_6 = 72$

Correct option: D**Final answer**

72

Worked Solutions

Advanced Sequences

Q008 Written

Find the general term of the sequence $\{2, 10, 24, 44, 70, 102, \dots\}$.

Solution

1. The first differences are 8, 14, 20, 26, 32,
2. Hence $b_k = 2(3k + 1)$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (2(3k + 1)) = n(3n - 1)$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = n(3n - 1)$$

Worked Solutions

Advanced Sequences

Q009 Written

Find the general term of the sequence $\{1, 6, 15, 28, 45, 66, \dots\}$.

Solution

1. The first differences are 5, 9, 13, 17, 21,
2. Hence $b_k = 4k + 1$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 1 + \sum_{k=1}^{n-1} (4k + 1) = n(2n - 1)$.
5. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = n(2n - 1)$$

Worked Solutions

Advanced Sequences

Q010 MCQ

Select the correct option: If $a_1 = 3$ and $b_k = 5k + 4$, then a_n is:

Solution

1. The first differences are represented by $b_k = 5k + 4$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 3 + \sum_{k=1}^{n-1} (5k + 4) = \frac{(n+1)(5n-2)}{2}$.
4. Substituting $n = 1$ gives $a_1 = 3$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{(n+1)(5n-2)}{2}$$

Q011 MCQ

Select the correct option: If $a_1 = 4$ and $b_k = 7k - 3$, then a_n is:

Solution

1. The first differences are represented by $b_k = 7k - 3$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 4 + \sum_{k=1}^{n-1} (7k - 3) = \frac{7n^2 - 13n + 14}{2}$.
4. Substituting $n = 1$ gives $a_1 = 4$, matching the given first term.

Correct option: D

Final answer

$$a_n = \frac{7n^2 - 13n + 14}{2}$$

Worked Solutions

Advanced Sequences

Q012 MCQ

Select the correct option: If $a_1 = 6$ and $b_k = 2(2k + 3)$, then a_n is:

Solution

1. The first differences are represented by $b_k = 2(2k + 3)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 6 + \sum_{k=1}^{n-1} (2(2k + 3)) = 2n(n + 2)$.
4. Substituting $n = 1$ gives $a_1 = 6$, matching the given first term.

Correct option: B

Final answer

$$a_n = 2n(n + 2)$$

Worked Solutions

Advanced Sequences

Q013 Written

Find the general term of the sequence $\{5, 8, 9, 8, 5, 0, \dots\}$.

Solution

1. The first differences are $3, 1, -1, -3, -5, \dots$
2. Hence $b_k = 5 - 2k$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} (5 - 2k) = -n(n - 6)$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = -n(n - 6)$$

Worked Solutions

Advanced Sequences

Q014 Written

Find the general term of the sequence $\{2, 3, 6, 15, 42, 123, \dots\}$.

Solution

1. The first differences are 1, 3, 9, 27, 81,
2. Hence $b_k = 3^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} (3^{k-1}) = \frac{3^n + 9}{6}$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{3^n + 9}{6}$$

Worked Solutions

Advanced Sequences

Q015 Written

Find the general term of the sequence $\{1, 2, 5, 14, 41, 122, \dots\}$.

Solution

1. The first differences are $1, 3, 9, 27, 81, \dots$
2. Hence $b_k = 3^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 1 + \sum_{k=1}^{n-1} (3^{k-1}) = \frac{3^n + 3}{6}$.
5. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = \frac{3^n + 3}{6}$$

Worked Solutions

Advanced Sequences

Q016 Written

Find the general term of the sequence $\{5, 7, 11, 19, 35, \dots\}$.

Solution

1. The first differences are 2, 4, 8, 16,
2. Hence $b_k = 2^k$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} (2^k) = 2^n + 3$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = 2^n + 3$$

Worked Solutions

Advanced Sequences

Q017 Written

Find the general term of the sequence $\{2, 3, 1, 5, -3, 13, \dots\}$.

Solution

1. The first differences are $1, -2, 4, -8, 16, \dots$
2. Hence $b_k = (-2)^{k-1}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 2 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 14}{6}$.
5. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{(-2)^n + 14}{6}$$

Worked Solutions

Advanced Sequences

Q018 Written

Find the general term of the sequence $\{5, 4, 7, -2, 25, -56, \dots\}$.

Solution

1. The first differences are $-1, 3, -9, 27, -81, \dots$
2. Hence $b_k = \frac{(-3)^k}{3}$.
3. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
4. $a_n = 5 + \sum_{k=1}^{n-1} \left(\frac{(-3)^k}{3}\right) = -\frac{(-3)^n - 57}{12}$.
5. Substituting $n = 1$ gives $a_1 = 5$, matching the given first term.

Final answer

$$a_n = -\frac{(-3)^n - 57}{12}$$

Worked Solutions

Advanced Sequences

Q019 MCQ

Select the correct option: If $a_1 = 1$ and $b_k = k^2$, then a_n is:

Solution

1. The first differences are represented by $b_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (k^2) = \frac{(n+1)(2n^2-5n+6)}{6}$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{(n+1)(2n^2-5n+6)}{6}$$

Worked Solutions

Advanced Sequences

Q020 Challenge

An oncology team records tissue volumes 1, 3, 7, 13, 21, 31, ... in mm^3 . Find the general formula a_n and calculate a_{12} to decide whether the 130 mm^3 threshold is exceeded.

Solution

1. The first differences are represented by $b_k = 2k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k) = n^2 - n + 1$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.
5. $a_{12} = 133$; this is greater than 130, so immediate surgery is required.

Final answer

$$a_n = n^2 - n + 1, \quad a_{12} = 133; \text{ surgery required}$$

Worked Solutions

Advanced Sequences

Q021 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 81$ terms precede the group.
2. First position = $P_9 + 1 = 82$
3. First value = 164

Final answer

164

Q022 Written

For groups with $a_j = 1 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 81$ terms precede the group.
2. First position = $P_9 + 1 = 82$
3. First value = 163

Final answer

163

Worked Solutions

Advanced Sequences

Q023 Written

For groups with $a_j = 1 + (j - 1)1$ and $c_j = n$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 45$ terms precede the group.
2. First position $= P_9 + 1 = 46$
3. First value $= 46$

Final answer

46

Q024 MCQ

Select the correct option: find the first term of group 3:

Solution

1. $P_2 = 3$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 4.

Correct option: B

Final answer

4

Worked Solutions

Advanced Sequences

Q025 MCQ

Select the correct option: find the first term of group 6:**Solution**

1. $P_5 = 25$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 78.

Correct option: C**Final answer**

78

Q026 MCQ

Select the correct option: find the first term of group 9:**Solution**

1. $P_8 = 36$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 37.

Correct option: D**Final answer**

37

Worked Solutions

Advanced Sequences

Q027 MCQ

Select the correct option: find the first term of group 4:**Solution**

1. $P_3 = 9$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 30.

Correct option: B**Final answer**

30

Q028 MCQ

Select the correct option: find the first term of group 7:**Solution**

1. $P_6 = 21$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 22.

Correct option: C**Final answer**

22

Worked Solutions

Advanced Sequences

Q029 MCQ

Select the correct option: find the first term of group 10:**Solution**

1. $P_9 = 81$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 246.

Correct option: D**Final answer**

246

Q030 MCQ

Select the correct option: find the first term of group 4:**Solution**

1. $P_3 = 9$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 20.

Correct option: A**Final answer**

20

Worked Solutions

Advanced Sequences

Q031 MCQ

Select the correct option: find the first term of group 7:**Solution**

1. $P_6 = 27$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 110.

Correct option: B**Final answer**

110

Q032 MCQ

Select the correct option: find the first term of group 10:**Solution**

1. $P_9 = 81$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 164.

Correct option: C**Final answer**

164

Worked Solutions

Advanced Sequences

Q033 MCQ

Select the correct option: find the first term of group 5:**Solution**

1. $P_4 = 14$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 58.

Correct option: A**Final answer**

58

Q034 MCQ

Select the correct option: find the first term of group 8:**Solution**

1. $P_7 = 49$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 100.

Correct option: B**Final answer**

100

Worked Solutions

Advanced Sequences

Q035 **Written**

For groups with $a_j = 1 + (j - 1)3$ and $c_j = 3n - 1$ terms in group j , find the first term of group 10.

Solution

1. $P_9 = 126$ terms precede the group.
2. First position = $P_9 + 1 = 127$
3. First value = 379

Final answer

379

Q036 **MCQ**

Select the correct option: find the first term of group 5:

Solution

1. $P_4 = 14$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 29.

Correct option: D

Final answer

29

Worked Solutions

Advanced Sequences

Q037 MCQ

Select the correct option: find the first term of group 8:**Solution**

1. $P_7 = 28$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 89.

Correct option: A**Final answer**

89

Q038 Written

For groups with $a_j = 2 + (j - 1)2$ and $c_j = 2n - 1$ terms in group j , derive the sum of the terms in group n .**Solution**

1. First value $= 2(n^2 - 2n + 2)$
2. Last value $= 2n^2$
3. Number of terms $= 2n - 1$
4. $S_n = \frac{c_n}{2}(\text{first} + \text{last}) = 2(2n - 1)(n^2 - n + 1)$

Final answer $2(2n - 1)(n^2 - n + 1)$

Worked Solutions

Advanced Sequences

Q039 Written

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value = 163
2. Last value = 199
3. Number of terms = 19
4. $S_{10} = 3439$

Final answer

3439

Q040 Written

For the same grouped sequence, find the sum of the terms in group 10.

Solution

1. First value = 46
2. Last value = 55
3. Number of terms = 10
4. $S_{10} = 505$

Final answer

505

Worked Solutions

Advanced Sequences

Q041 **Written****For the same grouped sequence, find the sum of the terms in group 10.****Solution**

1. First value = 379
2. Last value = 463
3. Number of terms = 29
4. $S_{10} = 12209$

Final answer

12209

Q042 **Written****In the grouped arithmetic sequence, locate the value 88: state its group and its position within that group.****Solution**

1. Sequence position = $(88 - 2)/2 + 1 = 44$
2. $P_6 = 36 < 44 \leq P_7 = 49$
3. Therefore the value is in Group 7, position 8.

Final answer

Group 7, position 8

Worked Solutions

Advanced Sequences

Q043 Written

In the grouped arithmetic sequence, locate the value 101: state its group and its position within that group.

Solution

1. Sequence position $= (101 - 1)/2 + 1 = 51$
2. $P_7 = 49 < 51 \leq P_8 = 64$
3. Therefore the value is in Group 8, position 2.

Final answer

Group 8, position 2

Q044 Written

In the grouped arithmetic sequence, locate the value 124: state its group and its position within that group.

Solution

1. Sequence position $= (124 - 1)/1 + 1 = 124$
2. $P_{15} = 120 < 124 \leq P_{16} = 136$
3. Therefore the value is in Group 16, position 4.

Final answer

Group 16, position 4

Worked Solutions

Advanced Sequences

Q045 **Written**

In the grouped arithmetic sequence, locate the value 100: state its group and its position within that group.

Solution

1. Sequence position $= (100 - 1)/3 + 1 = 34$
2. $P_4 = 26 < 34 \leq P_5 = 40$
3. Therefore the value is in Group 5, position 8.

Final answer

Group 5, position 8

Worked Solutions

Advanced Sequences

Q046 Challenge

For the grouped sequence $2, 6 \mid 10, 14, 18 \mid 22, 26, 30, 34 \mid \dots$, solve: (a) the first term of Group 12; (b) the sum in Group n ; (c) the group and position of 202; (d) the total through Group 12.

Solution

1. $c_n = n + 1, \quad P_n = \frac{n(n+3)}{2}$
2. (a) first term = 310
3. (b) group sum = $2(n+1)(n^2 + 2n - 1)$
4. (c) 202 is in Group 9, position 7.
5. (d) total = 16200

Final answer

(a) 310; (b) $2(n+1)(n^2 + 2n - 1)$; (c) Group 9, position 7; (d) 16200

Worked Solutions

Advanced Sequences

Quiz MCQ 1 Concept Quiz · MCQ

Select the correct option: If $a_1 = 2$ and $b_k = 3k + 1$, then a_n is:

Solution

1. The first differences are represented by $b_k = 3k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k + 1) = \frac{3n^2 - n + 2}{2}$.
4. Substituting $n = 1$ gives $a_1 = \frac{2}{2}$, matching the given first term.

Correct option: A

Final answer

$$a_n = \frac{3n^2 - n + 2}{2}$$

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If $a_1 = 3$ and $b_k = (-2)^{k-1}$, then a_n is:

Solution

1. The first differences are represented by $b_k = (-2)^{k-1}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 3 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 20}{6}$.
4. Substituting $n = 1$ gives $a_1 = 3$, matching the given first term.

Correct option: C

Final answer

$$a_n = \frac{(-2)^n + 20}{6}$$

Worked Solutions

Advanced Sequences

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: find the first term of group 9:

Solution

1. $P_8 = 64$
2. Move to the next sequence position, then substitute in the ungrouped arithmetic sequence.
3. The first value is 195.

Correct option: A

Final answer

195

Quiz MCQ 4 **Concept Quiz · MCQ**

Select the correct option: the value 29 is in:

Solution

1. The value is sequence position 15.
2. It lies in Group 5, position 1.

Correct option: B

Final answer

Group 5, position 1

Worked Solutions

Advanced Sequences

Quiz WRITTEN 5 Concept Quiz · Written

A sequence has $a_1 = 1$ and difference sequence $b_k = 4k - 1$. Find the required general term a_n .

Solution

1. The first differences are represented by $b_k = 4k - 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (4k - 1) = 2n^2 - 3n + 2$.
4. Substituting $n = 1$ gives $a_1 = 1$, matching the given first term.

Final answer

$$a_n = 2n^2 - 3n + 2$$

Quiz WRITTEN 6 Concept Quiz · Written

A sequence has $a_1 = 2$ and difference sequence $b_k = (-2)^{k-1}$. Find the required general term a_n .

Solution

1. The first differences are represented by $b_k = (-2)^{k-1}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} b_k$.
3. $a_n = 2 + \sum_{k=1}^{n-1} ((-2)^{k-1}) = \frac{(-2)^n + 14}{6}$.
4. Substituting $n = 1$ gives $a_1 = 2$, matching the given first term.

Final answer

$$a_n = \frac{(-2)^n + 14}{6}$$

Worked Solutions

Advanced Sequences

Quiz WRITTEN 7 Concept Quiz · Written

For groups with $a_j = 5 + (j - 1)3$ and $c_j = n$ terms in group j , find the first term of group 12.

Solution

1. $P_{11} = 66$ terms precede the group.
2. First position $= P_{11} + 1 = 67$
3. First value $= 203$

Final answer

203

Quiz WRITTEN 8 Concept Quiz · Written

In the grouped arithmetic sequence, locate the value 66: state its group and its position within that group.

Solution

1. Sequence position $= (66 - 2)/4 + 1 = 17$
2. $P_4 = 14 < 17 \leq P_5 = 20$
3. Therefore the value is in Group 5, position 3.

Final answer

Group 5, position 3

Answer key

Use the question number shown on each page.

Q001 $a_n = 2n^2 - 3n + 3$

Q002 C

Q003 B

Q004 A

Q005 B

Q006 A

Q007 D

Q008 $a_n = n(3n - 1)$

Q009 $a_n = n(2n - 1)$

Q010 A

Q011 D

Q012 B

Q013 $a_n = -n(n - 6)$

Q014 $a_n = \frac{3^n + 9}{6}$

Q015 $a_n = \frac{3^n + 3}{6}$

Q016 $a_n = 2^n + 3$

Q017 $a_n = \frac{(-2)^n + 14}{6}$

Q018 $a_n = -\frac{(-3)^n - 57}{12}$

Q019 A

Q020 $a_n = n^2 - n + 1$, $a_{12} = 133$; surgery required

Q021 164

Q022 163

Q023 46

Q024 B

Q025 C

Q026 D

Q027 B

Q028 C

Q029 D

Q030 A

Q031 B

Q032 C

Q033 A

Q034 B

Q035 379

Q036 D

Q037 A

Q038 $2(2n - 1)(n^2 - n + 1)$

Q039 3439

Q040 505

Q041 12209

Q042 Group 7, position 8

Q043 Group 8, position 2

Q044 Group 16, position 4

Q045 Group 5, position 8

Q046 (a) 310; (b) $2(n + 1)(n^2 + 2n - 1)$; (c) Group 9, position 7; (d) 16200

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $a_n = 2n^2 - 3n + 2$

Quiz WRITTEN 6 $a_n = \frac{(-2)^n + 14}{6}$

Quiz WRITTEN 7 203

Quiz WRITTEN 8 Group 5, position 3

Concept 5 | Recurrence and Induction

English Student Edition • Focused formulas and guided practice

Formula opener

Recurrence

$$a_{n+1} = f(a_n)$$

Induction

$$P(1), P(k) \Rightarrow P(k + 1)$$

Divisibility

$$P(k) = mq \Rightarrow P(k + 1) = mq'$$

Worked Solutions

Recurrence and Induction

Q001 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (2)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 8$.
2. $a_3 = 26$.
3. $a_4 = 80$.

Final answer

$$a_2 = 8, a_3 = 26, a_4 = 80$$

Q002 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 2a_n + (3)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -9$.
2. $a_3 = -15$.
3. $a_4 = -27$.

Final answer

$$a_2 = -9, a_3 = -15, a_4 = -27$$

Worked Solutions

Recurrence and Induction

Q003 Written

For the sequence $\{a_n\}$ with $a_1 = -6$ and recurrence $a_{n+1} = 1a_n + (n(n-2))$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -7$.
2. $a_3 = -7$.
3. $a_4 = -4$.

Final answer

$$a_2 = -7, a_3 = -7, a_4 = -4$$

Q004 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = -3$.
2. $a_3 = -5$.
3. $a_4 = -8$.

Final answer

$$a_2 = -3, a_3 = -5, a_4 = -8$$

Worked Solutions

Recurrence and Induction

Q005 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (2n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 4$.
2. $a_3 = 8$.
3. $a_4 = 14$.

Final answer

$$a_2 = 4, a_3 = 8, a_4 = 14$$

Q006 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3^n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 1$.
2. $a_3 = 10$.
3. $a_4 = 37$.

Final answer

$$a_2 = 1, a_3 = 10, a_4 = 37$$

Worked Solutions

Recurrence and Induction

Q007 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The fourth term is:

Solution

1. $a_2 = 9$.
2. $a_3 = 27$.
3. $a_4 = 81$.

Correct option: B

Final answer

81

Q008 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The fourth term is:

Solution

1. $a_2 = 8$.
2. $a_3 = 32$.
3. $a_4 = 128$.

Correct option: D

Final answer

128

Worked Solutions

Recurrence and Induction

Q009 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (n + 1)$. The fourth term is:

Solution

1. $a_2 = 6$.
2. $a_3 = 15$.
3. $a_4 = 34$.

Correct option: A

Final answer

34

Q010 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 3a_n + (2n)$. The fourth term is:

Solution

1. $a_2 = 17$.
2. $a_3 = 55$.
3. $a_4 = 171$.

Correct option: B

Final answer

171

Worked Solutions

Recurrence and Induction

Q011 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 9$.
3. $a_4 = 13$.

Correct option: D

Final answer

13

Q012 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = -2a_n + (5)$. The fourth term is:

Solution

1. $a_2 = -3$.
2. $a_3 = 11$.
3. $a_4 = -17$.

Correct option: A

Final answer

-17

Worked Solutions

Recurrence and Induction

Q013 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -3a_n + (2)$. The fourth term is:

Solution

1. $a_2 = -1$.
2. $a_3 = 5$.
3. $a_4 = -13$.

Correct option: C

Final answer

-13

Q014 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 9$.
3. $a_4 = 18$.

Correct option: D

Final answer

18

Worked Solutions

Recurrence and Induction

Q015 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 2a_n + (1)$. The fourth term is:

Solution

1. $a_2 = 5$.
2. $a_3 = 11$.
3. $a_4 = 23$.

Correct option: C

Final answer

23

Q016 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The fourth term is:

Solution

1. $a_2 = 2$.
2. $a_3 = -1$.
3. $a_4 = -4$.

Correct option: D

Final answer

-4

Worked Solutions

Recurrence and Induction

Q017 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The fourth term is:

Solution

1. $a_2 = 4$.
2. $a_3 = 6$.
3. $a_4 = 9$.

Correct option: B

Final answer

9

Q018 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 2a_n + (-n)$. The fourth term is:

Solution

1. $a_2 = 1$.
2. $a_3 = 0$.
3. $a_4 = -3$.

Correct option: C

Final answer

-3

Worked Solutions

Recurrence and Induction

Q019 Written

For the sequence $\{a_n\}$ with $a_1 = -1$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 5n - 6$.

Final answer

$$a_n = 5n - 6$$

Q020 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 3n - 2$.

Final answer

$$a_n = 3n - 2$$

Worked Solutions

Recurrence and Induction

Q021 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-2)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = -2(n - 3)$.

Final answer

$$a_n = -2(n - 3)$$

Q022 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (6)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 6n - 5$.

Final answer

$$a_n = 6n - 5$$

Worked Solutions

Recurrence and Induction

Q023 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 5n$.

Final answer

$$a_n = 8 - 5n$$

Q024 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = 1a_n + (3)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 3n - 5$.

Final answer

$$a_n = 3n - 5$$

Worked Solutions

Recurrence and Induction

Q025 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (-4)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = -4(n - 2)$.

Final answer

$$a_n = -4(n - 2)$$

Q026 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3^n$.

Correct option: D

Final answer

$$a_n = 3^n$$

Worked Solutions

Recurrence and Induction

Q027 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 4a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2^{2n-1}$.

Correct option: A

Final answer

$$a_n = 2^{2n-1}$$

Q028 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 5a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 4 \cdot 5^{n-1}$.

Correct option: A

Final answer

$$a_n = 4 \cdot 5^{n-1}$$

Worked Solutions

Recurrence and Induction

Q029 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 6$ and recurrence $a_{n+1} = 2a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3 \cdot 2^n$.

Correct option: A

Final answer

$$a_n = 3 \cdot 2^n$$

Q030 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2 \cdot 3^{n-1}$.

Final answer

$$a_n = 2 \cdot 3^{n-1}$$

Worked Solutions

Recurrence and Induction

Q031 Written

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 5a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 3 \cdot 5^{n-1}$.

Final answer

$$a_n = 3 \cdot 5^{n-1}$$

Q032 Written

For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 3a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 4 \cdot 3^{n-1}$.

Final answer

$$a_n = 4 \cdot 3^{n-1}$$

Worked Solutions

Recurrence and Induction

Q033 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 4a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2^{2n-2}$.

Final answer

$$a_n = 2^{2n-2}$$

Q034 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 4n - 3$.

Correct option: B

Final answer

$$a_n = 4n - 3$$

Worked Solutions

Recurrence and Induction

Q035 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 4$ and recurrence $a_{n+1} = 1a_n + (n^2)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.

2. $a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$.

Correct option: B

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 24}{6}$$

Q036 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -4a_n + (0)$. The general term is:

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.

2. $a_n = (-4)^{n-1}$.

Correct option: D

Final answer

$$a_n = (-4)^{n-1}$$

Worked Solutions

Recurrence and Induction

Q037 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (3^n)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{3^{n+7}}{2}$.

Correct option: A

Final answer

$$a_n = \frac{3^n + 7}{2}$$

Q038 Written

For the sequence $\{a_n\}$ with $a_1 = -3$ and recurrence $a_{n+1} = -2a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = \frac{3(-2)^n}{2}$.

Final answer

$$a_n = \frac{3(-2)^n}{2}$$

Worked Solutions

Recurrence and Induction

Q039 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = -5a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = 2(-5)^{n-1}$.

Final answer

$$a_n = 2(-5)^{n-1}$$

Q040 Written

For the sequence $\{a_n\}$ with $a_1 = -2$ and recurrence $a_{n+1} = -4a_n + (0)$. Find the general term.

Solution

1. The rule multiplies by a constant ratio, so the sequence is geometric.
2. $a_n = \frac{(-4)^n}{2}$.

Final answer

$$a_n = \frac{(-4)^n}{2}$$

Worked Solutions

Recurrence and Induction

Q041 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 5$ and recurrence $a_{n+1} = 1a_n + (-3)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 3n$.

Correct option: B**Final answer**

$$a_n = 8 - 3n$$

Q042 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (n)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = \frac{n^2 - n + 6}{2}$.

Correct option: C**Final answer**

$$a_n = \frac{n^2 - n + 6}{2}$$

Worked Solutions

Recurrence and Induction

Q043 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 1a_n + (-5)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 8 - 5n$.

Correct option: B**Final answer**

$$a_n = 8 - 5n$$

Q044 MCQ

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (6)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 2(3n - 2)$.

Correct option: C**Final answer**

$$a_n = 2(3n - 2)$$

Worked Solutions

Recurrence and Induction

Q045 Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 5$.
2. $a_3 = 13$.
3. $a_4 = 36$.

Final answer

$$a_2 = 5, a_3 = 13, a_4 = 36$$

Q046 Written

For the sequence $\{a_n\}$ with $a_1 = 1$ and recurrence $a_{n+1} = -2a_n + (5)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 3$.
2. $a_3 = -1$.
3. $a_4 = 7$.

Final answer

$$a_2 = 3, a_3 = -1, a_4 = 7$$

Worked Solutions

Recurrence and Induction

Q047 **Written**

For the sequence $\{a_n\}$ with $a_1 = 3$ and recurrence $a_{n+1} = 3a_n + (-n)$. Find the 2nd through the 4th terms.

Solution

1. $a_2 = 8$.
2. $a_3 = 22$.
3. $a_4 = 63$.

Final answer

$$a_2 = 8, a_3 = 22, a_4 = 63$$

Worked Solutions

Recurrence and Induction

Q048 Challenge

Solve the third-order recurrence $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ for $n \geq 3$ with $a_0 = 0, a_1 = 1, a_2 = 6$.

Solution

1. $r^3 - 6r^2 + 11r - 6 = (r - 1)(r - 2)(r - 3)$

2. $a_n = A + B2^n + C3^n$

3. $A = \frac{1}{2}, B = -2, C = \frac{3}{2}$

4. $a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$

Final answer

$$a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$$

Worked Solutions

Recurrence and Induction

Q049 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2n - 3)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k - 3$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k - 3) = (n - 2)^2$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = (n - 2)^2$$

Q050 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (3n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (3k) = \frac{3n^2 - 3n + 2}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3n^2 - 3n + 2}{2}$$

Worked Solutions

Recurrence and Induction

Q051 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (2n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (2k) = n^2 - n + 2$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = n^2 - n + 2$$

Q052 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k^2) = \frac{2n^3 - 3n^2 + n + 4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$$

Worked Solutions

Recurrence and Induction

Q053 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (3n + 1)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k + 1) = \frac{3n^2 - n + 2}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3n^2 - n + 2}{2}$$

Q054 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n(2n + 1))$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(2k + 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k(2k + 1)) = \frac{4n^3 - 3n^2 - n + 12}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$$

Worked Solutions

Recurrence and Induction

Q055 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$, then a_n is:**Solution**

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 3 + \sum_{k=1}^{n-1} (k^2) = \frac{2n^3 - 3n^2 + n + 18}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C**Final answer**

$$a_n = \frac{2n^3 - 3n^2 + n + 18}{6}$$

Q056 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$, then a_n is:**Solution**

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k^2) = \frac{2n^3 - 3n^2 + n + 4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A**Final answer**

$$a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$$

Worked Solutions

Recurrence and Induction

Q057 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2 + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k^2 + 1) = \frac{2n^3 - 3n^2 + 7n + 6}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{2n^3 - 3n^2 + 7n + 6}{6}$$

Q058 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(4k - 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 5 + \sum_{k=1}^{n-1} (k(4k - 1)) = \frac{(n+1)(8n^2 - 23n + 30)}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = \frac{(n+1)(8n^2 - 23n + 30)}{6}$$

Worked Solutions

Recurrence and Induction

Q059 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = a_n + (8n)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 8k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (8k) = 2(2n^2 - 2n + 1)$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = 2(2n^2 - 2n + 1)$$

Q060 MCQ

Select the correct option: find a_4 when $a_1 = 3$ and $a_{n+1} = a_n + (n^2)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 3 + \sum_{k=1}^{n-1} (k^2) = \frac{2n^3 - 3n^2 + n + 18}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

17

Worked Solutions

Recurrence and Induction

Q061 MCQ

Select the correct option: find a_5 when $a_1 = 2$ and $a_{n+1} = a_n + (3n^2)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 3k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (3k^2) = \frac{2n^3 - 3n^2 + n + 4}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

92

Q062 MCQ

Select the correct option: find a_3 when $a_1 = 2$ and $a_{n+1} = a_n + (n^2 + 1)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2 + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k^2 + 1) = \frac{2n^3 - 3n^2 + 7n + 6}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

9

Worked Solutions

Recurrence and Induction

Q063 MCQ

Select the correct option: find a_6 when $a_1 = 5$ and $a_{n+1} = a_n + (n(4n - 1))$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k(4k - 1)$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 5 + \sum_{k=1}^{n-1} (k(4k - 1)) = \frac{(n+1)(8n^2 - 23n + 30)}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

210

Q064 MCQ

Select the correct option: find a_2 when $a_1 = 2$ and $a_{n+1} = a_n + (8n)$:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 8k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (8k) = 2(2n^2 - 2n + 1)$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

10

Worked Solutions

Recurrence and Induction

Q065 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (n^2)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = k^2$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (k^2) = \frac{2n^3 - 3n^2 + n + 12}{6}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2n^3 - 3n^2 + n + 12}{6}$$

Q066 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = a_n + (2n + 1)$, then a_n is:

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = 2k + 1$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2k + 1) = n^2$.
4. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = n^2$$

Worked Solutions

Recurrence and Induction

Q067 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 1$ and $a_{n+1} = a_n + (2^n)$.**Solution**

1. The recurrence gives the difference term $a_{k+1} - a_k = 2^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 1 + \sum_{k=1}^{n-1} (2^k) = 2^n - 1$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = 2^n - 1$$

Q068 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 2$ and $a_{n+1} = a_n + (-3^n)$.**Solution**

1. The recurrence gives the difference term $a_{k+1} - a_k = -3^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 2 + \sum_{k=1}^{n-1} (-3^k) = -\frac{3^n - 7}{2}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{3^n - 7}{2}$$

Worked Solutions

Recurrence and Induction

Q069 Written

Find the general term of the sequence $\{a_n\}$ determined by $a_1 = 4$ and $a_{n+1} = a_n + (-4^n)$.

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -4^k$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 4 + \sum_{k=1}^{n-1} (-4^k) = -\frac{4^n - 16}{3}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{4^n - 16}{3}$$

Worked Solutions

Recurrence and Induction

Q070 Challenge

A technology startup has a reserve of 400,000 LE, represented by $a_1 = 4$, and burns $0.04n$ hundred-thousand LE each quarter: $a_{n+1} = a_n - 0.04n$. Find an explicit formula for a_n .

Solution

1. The recurrence gives the difference term $a_{k+1} - a_k = -\frac{k}{25}$.
2. Use $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
3. $a_n = 4 + \sum_{k=1}^{n-1} \left(-\frac{k}{25}\right) = -\frac{n^2 - n - 200}{50}$.
4. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{n^2 - n - 200}{50}$$

Worked Solutions

Recurrence and Induction

Q071 **Written****Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 3a_n + (4)$.****Solution**

1. $c = -2$ from $c = 3c + (4)$.
2. $a_{n+1} - c = 3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 3^n - 2$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = 3^n - 2$$

Worked Solutions

Recurrence and Induction

Q072 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.**Solution**

1. $c = -2$ from $c = 5c + (8)$.
2. $a_{n+1} - c = 5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 5^n - 10}{5}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3 \cdot 5^n - 10}{5}$$

Worked Solutions

Recurrence and Induction

Q073 Written

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = 2a_n + (-1)$.

Solution

1. $c = 1$ from $c = 2c + (-1)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2^n + 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2^n + 2}{2}$$

Worked Solutions

Recurrence and Induction

Q074 **Written****Find the general term of $\{a_n\}$ given $a_1 = -3$ and $a_{n+1} = 4a_n + (3)$.****Solution**

1. $c = -1$ from $c = 4c + (3)$.
2. $a_{n+1} - c = 4(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{4^n + 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{4^n + 2}{2}$$

Worked Solutions

Recurrence and Induction

Q075 Written

Find the general term of $\{a_n\}$ given $a_1 = 3$ and $a_{n+1} = 3a_n + (-2)$.

Solution

1. $c = 1$ from $c = 3c + (-2)$.
2. $a_{n+1} - c = 3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2 \cdot 3^n + 3}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{2 \cdot 3^n + 3}{3}$$

Worked Solutions

Recurrence and Induction

Q076 **Written****Find the general term of $\{a_n\}$ given $a_1 = -2$ and $a_{n+1} = -3a_n + (4)$.****Solution**

1. $c = 1$ from $c = -3c + (4)$.
2. $a_{n+1} - c = -3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = (-3)^n + 1$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = (-3)^n + 1$$

Worked Solutions

Recurrence and Induction

Q077 Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = \frac{1}{3}a_n + (2)$.

Solution

1. $c = 3$ from $c = \frac{1}{3}c + (2)$.
2. $a_{n+1} - c = \frac{1}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 3 \cdot 3^{-n} (3^n - 2)$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = 3 \cdot 3^{-n} (3^n - 2)$$

Worked Solutions

Recurrence and Induction

Q078 **Written**

Find the general term of $\{a_n\}$ given $a_1 = 2$ and $a_{n+1} = \frac{2}{3}a_n + (-\frac{1}{2})$.

Solution

1. $c = -\frac{3}{2}$ from $c = \frac{2}{3}c + (-\frac{1}{2})$.
2. $a_{n+1} - c = \frac{2}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 2 \cdot 3^n)}{4}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 2 \cdot 3^n)}{4}$$

Worked Solutions

Recurrence and Induction

Q079 Written

Find the general term of $\{a_n\}$ given $a_1 = 4$ and $a_{n+1} = -2a_n + (3)$.

Solution

1. $c = 1$ from $c = -2c + (3)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{3(-2)^n - 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = -\frac{3(-2)^n - 2}{2}$$

Worked Solutions

Recurrence and Induction

Q080 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{3}$ from $c = -2c + (5)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2(-2)^n - 5}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = -\frac{2(-2)^n - 5}{3}$$

Worked Solutions

Recurrence and Induction

Q081 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{2}{3}a_n + (-2)$, the general term is:

Solution

1. $c = -6$ from $c = \frac{2}{3}c + (-2)$.
2. $a_{n+1} - c = \frac{2}{3}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 4 \cdot 3^n)}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = \frac{3 \cdot 3^{-n}(7 \cdot 2^n - 4 \cdot 3^n)}{2}$$

Worked Solutions

Recurrence and Induction

Q082 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = 7a_n + (-4)$, the general term is:

Solution

1. $c = \frac{2}{3}$ from $c = 7c + (-4)$.
2. $a_{n+1} - c = 7(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2(2 \cdot 7^n + 7)}{21}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = \frac{2(2 \cdot 7^n + 7)}{21}$$

Worked Solutions

Recurrence and Induction

Q083 MCQ

Select the correct option: If $a_1 = 6$ and $a_{n+1} = 2a_n + (-5)$, the general term is:

Solution

1. $c = 5$ from $c = 2c + (-5)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{2^n + 10}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = \frac{2^n + 10}{2}$$

Worked Solutions

Recurrence and Induction

Q084 MCQ

Select the correct option: If $a_1 = 5$ and $a_{n+1} = -2a_n + (7)$, the general term is:

Solution

1. $c = \frac{7}{3}$ from $c = -2c + (7)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{4(-2)^n - 7}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = -\frac{4(-2)^n - 7}{3}$$

Worked Solutions

Recurrence and Induction

Q085 MCQ

Select the correct option: If $a_1 = 4$ and $a_{n+1} = -5a_n + (-2)$, the general term is:

Solution

1. $c = -\frac{1}{3}$ from $c = -5c + (-2)$.
2. $a_{n+1} - c = -5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{13(-5)^n + 5}{15}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = -\frac{13(-5)^n + 5}{15}$$

Worked Solutions

Recurrence and Induction

Q086 MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -3a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{4}$ from $c = -3c + (5)$.
2. $a_{n+1} - c = -3(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{(-3)^n - 5}{4}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: B

Final answer

$$a_n = -\frac{(-3)^n - 5}{4}$$

Worked Solutions

Recurrence and Induction

Q087 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = \frac{1}{2}a_n + (7)$, the general term is:

Solution

1. $c = 14$ from $c = \frac{1}{2}c + (7)$.
2. $a_{n+1} - c = \frac{1}{2}(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 2 \cdot 2^{-n} (7 \cdot 2^n - 13)$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: C

Final answer

$$a_n = 2 \cdot 2^{-n} (7 \cdot 2^n - 13)$$

Worked Solutions

Recurrence and Induction

Q088 MCQ

Select the correct option: If $a_1 = 1$ and $a_{n+1} = 2a_n + (-1)$, the general term is:

Solution

1. $c = 1$ from $c = 2c + (-1)$.
2. $a_{n+1} - c = 2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = 1$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = 1$$

Worked Solutions

Recurrence and Induction

Q089 MCQ

Select the correct option: If $a_1 = 3$ and $a_{n+1} = -2a_n + (5)$, the general term is:

Solution

1. $c = \frac{5}{3}$ from $c = -2c + (5)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2(-2)^{n-5}}{3}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: D

Final answer

$$a_n = -\frac{2(-2)^n - 5}{3}$$

Worked Solutions

Recurrence and Induction

Q090 Challenge

A sequence satisfies $a_{n+1} = 4a_n - 9$ with $a_1 = 2$. Find the equilibrium value, the explicit formula, and a_5 .

Solution

1. $c = 3$ from $c = 4c + (-9)$.
2. $a_{n+1} - c = 4(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{2^{2n}-12}{4}$.
5. Substitute $n = 1$ to verify the given first term.
6. $a_5 = -253$

Final answer

$$c = 3, \quad a_n = -\frac{2^{2n} - 12}{4}, \quad a_5 = -253$$

Worked Solutions

Recurrence and Induction

Q091 Written

Use mathematical induction to prove $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i = \frac{k(k+1)}{2}$.
3. For $n = k + 1$, add the next term n to the left side and use the hypothesis.
4. $\frac{k(k+1)}{2} + n = \frac{(k+1)(k+2)}{2}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)}{2}$$

Worked Solutions

Recurrence and Induction

Q092 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1)^2 = \frac{n(2n-1)(2n+1)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k (2i - 1)^2 = \frac{k(2k-1)(2k+1)}{3}$.
3. For $n = k + 1$, add the next term $(2k + 1)^2$ to the left side and use the hypothesis.
4. $\frac{k(2k-1)(2k+1)}{3} + (2k + 1)^2 = \frac{(k+1)(2k+1)(2k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(2n-1)(2n+1)}{3}$$

Worked Solutions

Recurrence and Induction

Q093 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q094 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q095 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3 \cdot 4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3 \cdot 4^k = (2 \cdot 2^k - 1)(2 \cdot 2^k + 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$4^n - 1$$

Worked Solutions

Recurrence and Induction

Q096 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 2i + 3 = k(k + 4)$.
3. For $n = k + 1$, add the next term $2k + 5$ to the left side and use the hypothesis.
4. $k(k + 4) + 2k + 5 = (k + 1)(k + 5)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$n(n + 4)$$

Worked Solutions

Recurrence and Induction

Q097 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i^2 + 3i$ is:

Solution

- [1] Base case $n = 1$: both sides equal 5.
- [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 2i^2 + 3i = \frac{k(k+1)(4k+11)}{6}$.
- For $n = k + 1$, add the next term $(k + 1)(2k + 5)$ to the left side and use the hypothesis.
- $\frac{k(k+1)(4k+11)}{6} + (k + 1)(2k + 5) = \frac{(k+1)(k+2)(4k+15)}{6}$, which is the right side at $k + 1$.
- Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$\frac{n(n+1)(4n+11)}{6}$$

Worked Solutions

Recurrence and Induction

Q098 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3i^2 + 2i$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3i^2 + 2i = \frac{k(k+1)(2k+3)}{2}$.
3. For $n = k + 1$, add the next term $(k + 1)(3k + 5)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(2k+3)}{2} + (k + 1)(3k + 5) = \frac{(k+1)(k+2)(2k+5)}{2}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$\frac{n(n+1)(2n+3)}{2}$$

Worked Solutions

Recurrence and Induction

Q099 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 7 \cdot 2^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 7.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 7 \cdot 2^{i-1} = 7(2^k - 1)$.
3. For $n = k + 1$, add the next term $7 \cdot 2^k$ to the left side and use the hypothesis.
4. $7(2^k - 1) + 7 \cdot 2^k = 7(2 \cdot 2^k - 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$7(2^n - 1)$$

Worked Solutions

Recurrence and Induction

Q100 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n i(i+1)$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: C

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q101 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 3 \cdot 4^{i-1}$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3 \cdot 4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3 \cdot 4^k = (2 \cdot 2^k - 1)(2 \cdot 2^k + 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: A

Final answer

$$4^n - 1$$

Worked Solutions

Recurrence and Induction

Q102 MCQ

Select the correct option: the right-hand side equivalent to $\sum_{i=1}^n 2i + 3$ is:

Solution

1. [1] Base case $n = 1$: both sides equal 5.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 2i + 3 = k(k + 4)$.
3. For $n = k + 1$, add the next term $2k + 5$ to the left side and use the hypothesis.
4. $k(k + 4) + 2k + 5 = (k + 1)(k + 5)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Correct option: B

Final answer

$$n(n + 4)$$

Worked Solutions

Recurrence and Induction

Q103 Written

Use mathematical induction to prove $\sum_{i=1}^n (2i - 1) = n^2$.

Solution

1. [1] Base case $n = 1$: both sides equal 1.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k (2i - 1) = k^2$.
3. For $n = k + 1$, add the next term $2k + 1$ to the left side and use the hypothesis.
4. $k^2 + 2k + 1 = (k + 1)^2$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$n^2$$

Worked Solutions

Recurrence and Induction

Q104 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 2.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+1) = \frac{k(k+1)(k+2)}{3}$.
3. For $n = k + 1$, add the next term $(k+1)(k+2)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = \frac{(k+1)(k+2)(k+3)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(n+2)}{3}$$

Worked Solutions

Recurrence and Induction

Q105 Written

Use mathematical induction to prove $\sum_{i=1}^n i(i+2) = \frac{n(n+1)(2n+7)}{6}$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k i(i+2) = \frac{k(k+1)(2k+7)}{6}$.
3. For $n = k + 1$, add the next term $(k+1)(k+3)$ to the left side and use the hypothesis.
4. $\frac{k(k+1)(2k+7)}{6} + (k+1)(k+3) = \frac{(k+1)(k+2)(2k+9)}{6}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n+1)(2n+7)}{6}$$

Worked Solutions

Recurrence and Induction

Q106 Written

Use mathematical induction to prove $\sum_{i=1}^n 3 \cdot 4^{i-1} = 4^n - 1$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k 3 \cdot 4^{i-1} = 4^k - 1$.
3. For $n = k + 1$, add the next term $3 \cdot 4^k$ to the left side and use the hypothesis.
4. $4^k - 1 + 3 \cdot 4^k = (2 \cdot 2^k - 1)(2 \cdot 2^k + 1)$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$4^n - 1$$

Worked Solutions

Recurrence and Induction

Q107 Written

Use mathematical induction to prove $\sum_{i=0}^n 10^i = \frac{10 \cdot 10^n - 1}{9}$.

Solution

1. [1] Base case $n = 1$: both sides equal 11.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=0}^k 10^i = \frac{10 \cdot 10^k - 1}{9}$.
3. For $n = k + 1$, add the next term 10^{k+1} to the left side and use the hypothesis.
4. $\frac{10 \cdot 10^k - 1}{9} + 10^{k+1} = \frac{100 \cdot 10^k - 1}{9}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{10 \cdot 10^n - 1}{9}$$

Worked Solutions

Recurrence and Induction

Q108 Challenge

Use mathematical induction to prove $\sum_{i=1}^n (i^2 + i + 1) = \frac{n(n^2 + 3n + 5)}{3}$.

Solution

1. [1] Base case $n = 1$: both sides equal 3.
2. [2] Assume the statement holds for $n = k$: $\sum_{i=1}^k (i^2 + i + 1) = \frac{k(k^2 + 3k + 5)}{3}$.
3. For $n = k + 1$, add the next term $k^2 + 3k + 3$ to the left side and use the hypothesis.
4. $\frac{k(k^2 + 3k + 5)}{3} + k^2 + 3k + 3 = \frac{(k+1)(k^2 + 5k + 9)}{3}$, which is the right side at $k + 1$.
5. Therefore the equality holds for all natural numbers by mathematical induction.

Final answer

$$\frac{n(n^2 + 3n + 5)}{3}$$

Worked Solutions

Recurrence and Induction

Q109 Written

Prove by mathematical induction that $2^n > 3n$ for $n \geq 4$.

Solution

1. $n = 4$: $2^4 = 16 > 12 = 3 \cdot 4$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^{k+1} > 2(3k + 0)$ after multiplying by the positive base.
4. $2(3k + 0) - (3(k + 1) + 0) = 3(k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 3n$$

Worked Solutions

Recurrence and Induction

Q110 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 3$.

Solution

1. $n = 3$: $3^3 = 27 > 16 = 5 \cdot 3 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 5n + 1$$

Worked Solutions

Recurrence and Induction

Q111 Written

Prove by mathematical induction that $3^n > 2n$ for $n \geq 1$.

Solution

1. $n = 1$: $3^1 = 3 > 2 = 2$.
2. Assume the inequality holds for $n = k$ with $k \geq 1$.
3. $3^k + 1 > 3(2k + 0)$ after multiplying by the positive base.
4. $3(2k + 0) - (2(k + 1) + 0) = 2(2k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 2n$$

Worked Solutions

Recurrence and Induction

Q112 Written

Prove by mathematical induction that $2^n > 1n + 2$ for $n \geq 3$.

Solution

1. $n = 3$: $2^3 = 8 > 5 = 1 \cdot 3 + 2$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $2^k + 1 > 2(1k + 2)$ after multiplying by the positive base.
4. $2(1k + 2) - (1(k + 1) + 2) = k + 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 1n + 2$$

Worked Solutions

Recurrence and Induction

Q113 Written

Prove by mathematical induction that $3^n > 5n + 1$ for $n \geq 4$.

Solution

1. $n = 4$: $3^4 = 81 > 21 = 5 \cdot 4 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 5n + 1$$

Worked Solutions

Recurrence and Induction

Q114 Written

Prove by mathematical induction that $3^n > 5n + 2$ for $n \geq 4$.

Solution

1. $n = 4$: $3^4 = 81 > 22 = 5 \cdot 4 + 2$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 2)$ after multiplying by the positive base.
4. $3(5k + 2) - (5(k + 1) + 2) = 10k - 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$3^n > 5n + 2$$

Worked Solutions

Recurrence and Induction

Q115 Written

Prove by mathematical induction that $4^n > 2n + 3$ for $n \geq 4$.

Solution

1. $n = 4$: $4^4 = 256 > 11 = 11$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(2k + 3)$ after multiplying by the positive base.
4. $4(2k + 3) - (2(k + 1) + 3) = 6k + 7$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$4^n > 2n + 3$$

Worked Solutions

Recurrence and Induction

Q116 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?:**Solution**

1. $n = 3$: $2^3 = 8 > 5 = 5$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $2^k + 1 > 2(1k + 2)$ after multiplying by the positive base.
4. $2(1k + 2) - (1(k + 1) + 2) = k + 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: D**Final answer**

$$2^n > 1n + 2$$

Worked Solutions

Recurrence and Induction

Q117 MCQ

Select the correct option: which inequality is proved for $n \geq 2$?

Solution

1. $n = 2$: $3^2 = 9 > 4 = 4$.
2. Assume the inequality holds for $n = k$ with $k \geq 2$.
3. $3^k + 1 > 3(2k + 0)$ after multiplying by the positive base.
4. $3(2k + 0) - (2(k + 1) + 0) = 2(2k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: A

Final answer

$$3^n > 2n$$

Worked Solutions

Recurrence and Induction

Q118 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Solution**

1. $n = 4$: $3^4 = 81 > 21 = 21$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 1)$ after multiplying by the positive base.
4. $3(5k + 1) - (5(k + 1) + 1) = 10k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B**Final answer**

$$3^n > 5n + 1$$

Worked Solutions

Recurrence and Induction

Q119 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?

Solution

1. $n = 3$: $3^3 = 27 > 9 = 3n$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $3^k + 1 > 3(3k + 0)$ after multiplying by the positive base.
4. $3(3k + 0) - (3(k + 1) + 0) = 3(2k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: D

Final answer

$$3^n > 3n$$

Worked Solutions

Recurrence and Induction

Q120 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:

Solution

1. $n = 4$: $3^4 = 81 > 22 = 22$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $3^k + 1 > 3(5k + 2)$ after multiplying by the positive base.
4. $3(5k + 2) - (5(k + 1) + 2) = 10k - 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: A

Final answer

$$3^n > 5n + 2$$

Worked Solutions

Recurrence and Induction

Q121 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?

Solution

1. $n = 3$: $4^3 = 64 > 10 = 10$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $4^k + 1 > 4(3k + 1)$ after multiplying by the positive base.
4. $4(3k + 1) - (3(k + 1) + 1) = 9k$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B

Final answer

$$4^n > 3n + 1$$

Worked Solutions

Recurrence and Induction

Q122 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?:**Solution**

1. $n = 4$: $4^4 = 256 > 11 = 11$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(2k + 3)$ after multiplying by the positive base.
4. $4(2k + 3) - (2(k + 1) + 3) = 6k + 7$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: D**Final answer**

$$4^n > 2n + 3$$

Worked Solutions

Recurrence and Induction

Q123 MCQ

Select the correct option: which inequality is proved for $n \geq 4$?

Solution

1. $n = 4$: $4^4 = 256 > 17 = 17$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $4^k + 1 > 4(4k + 1)$ after multiplying by the positive base.
4. $4(4k + 1) - (4(k + 1) + 1) = 12k - 1$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: A

Final answer

$$4^n > 4n + 1$$

Worked Solutions

Recurrence and Induction

Q124 Written

Prove by mathematical induction that $2^n > 3n + 1$ for $n \geq 4$.

Solution

1. $n = 4$: $2^4 = 16 > 13 = 3 \cdot 4 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 1)$ after multiplying by the positive base.
4. $2(3k + 1) - (3(k + 1) + 1) = 3k - 2$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 3n + 1$$

Worked Solutions

Recurrence and Induction

Q125 MCQ

Select the correct option: which inequality is proved for $n \geq 3$?

Solution

1. $n = 3$: $4^3 = 64 > 9 = 9$.
2. Assume the inequality holds for $n = k$ with $k \geq 3$.
3. $4^k + 1 > 4(3k + 0)$ after multiplying by the positive base.
4. $4(3k + 0) - (3(k + 1) + 0) = 3(3k - 1)$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B

Final answer

$$4^n > 3n$$

Worked Solutions

Recurrence and Induction

Q126 Challenge

When $h > 0$ and $n \geq 3$, prove $(1 + h)^n > 1 + nh^2$.

Solution

1. $n = 3$: $(1 + h)^3 = 1 + 3h + 3h^2 + h^3 > 1 + 3h^2$ because $h > 0$ and $3h + h^3 > 0$.
2. Assume $(1 + h)^k > 1 + kh^2$ for $k \geq 3$.
3. Multiplying by the positive factor $1 + h$ gives a lower bound $(1 + h)(1 + kh^2)$.
4. $(1 + h)(1 + kh^2) - [1 + (k + 1)h^2] = h(1 - h + kh^2)$. The quadratic $1 - h + kh^2$ is positive for $k \geq 3$ because its discriminant is $1 - 4k < 0$.
5. Therefore the inequality holds for every $n \geq 3$ and $h > 0$.

Final answer

$$(1 + h)^n > 1 + nh^2$$

Worked Solutions

Recurrence and Induction

Q127 Written

Prove that $n^3 + (n + 1)^3 + (n + 2)^3$ is a multiple of 9 for every natural number n , using mathematical induction.

Solution

1. [1] Base case: $n = 1$, $f(1) = 36 = 9(4)$, so the statement holds.
2. [2] Assume $f(k) = 9m$ for an integer m .
3. [3] The induction difference is $f(k + 1) - f(k) = 9(k^2 + 3k + 3) = 9(k^2 + 3k + 3)$.
4. [4] Therefore $f(k + 1) = 9[m + k^2 + 3k + 3]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$9 \mid [n^3 + (n + 1)^3 + (n + 2)^3]$$

Worked Solutions

Recurrence and Induction

Q128 Written

Prove that $n^3 + 2n$ is a multiple of 3 for every natural number n , using mathematical induction.

Solution

1. [1] Base case: $n = 1$, $f(1) = 3 = 3(1)$, so the statement holds.
2. [2] Assume $f(k) = 3m$ for an integer m .
3. [3] The induction difference is $f(k+1) - f(k) = 3(k^2 + k + 1) = 3(k^2 + k + 1)$.
4. [4] Therefore $f(k+1) = 3[m + k^2 + k + 1]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$3 \mid (n^3 + 2n)$$

Worked Solutions

Recurrence and Induction

Q129 Written

Prove by mathematical induction that $n^2 + n$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 2 = 2(1)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+0+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + n)$$

Worked Solutions

Recurrence and Induction

Q130 Written

Prove by mathematical induction that $n^2 + 5n + 8$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 14 = 2(7)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+2+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + 5n + 8)$$

Worked Solutions

Recurrence and Induction

Q131 Written

Prove by mathematical induction that $n^2 + 7n + 12$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 20 = 2(10)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+3+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + 7n + 12)$$

Worked Solutions

Recurrence and Induction

Q132 Written

Prove by mathematical induction that $n^2 + 11n + 32$ is a multiple of 2.

Solution

1. [1] Base case: $n = 1$, $f(1) = 44 = 2(22)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+5+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Final answer

$$2 \mid (n^2 + 11n + 32)$$

Worked Solutions

Recurrence and Induction

Q133 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 2 = 2(1)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k + 1) - f(k) = 2 \prod_{j=1}^1 (k + 0 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k + 1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: A

Final answer

2

Worked Solutions

Recurrence and Induction

Q134 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 5n + 8$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 14 = 2(7)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+2+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

2

Worked Solutions

Recurrence and Induction

Q135 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 7n + 12$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 20 = 2(10)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k + 1) - f(k) = 2 \prod_{j=1}^1 (k + 3 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k + 1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: C

Final answer

2

Worked Solutions

Recurrence and Induction

Q136 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + 11n + 32$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 44 = 2(22)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 2 \prod_{j=1}^1 (k+5+j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k+1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: C

Final answer

2

Worked Solutions

Recurrence and Induction

Q137 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 3n^2 + 2n$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 6 = 6(1)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+0+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: B

Final answer

6

Worked Solutions

Recurrence and Induction

Q138 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 9n^2 + 26n + 30$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 66 = 6(11)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+2+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: A

Final answer

6

Worked Solutions

Recurrence and Induction

Q139 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 12n^2 + 47n + 60$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 120 = 6(20)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+3+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

6

Worked Solutions

Recurrence and Induction

Q140 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^3 + 18n^2 + 107n + 216$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 342 = 6(57)$.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] Consecutive-product shift: $f(k+1) - f(k) = 3 \prod_{j=1}^2 (k+5+j)$.
4. [4] Since the remaining 2 consecutive integers contain a multiple of 2 factorial, the difference is a multiple of 6.
5. Thus $f(k+1) = 6[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

6

Worked Solutions

Recurrence and Induction

Q141 MCQ

Select the correct option: which positive integer is guaranteed to divide $n^2 + n + 2$ for every natural number n ?:

Solution

1. [1] Base case: $n = 1$, $f(1) = 4 = 2(2)$.
2. [2] Assume $f(k) = 2m$ for an integer m .
3. [3] Consecutive-product shift: $f(k + 1) - f(k) = 2 \prod_{j=1}^1 (k + 0 + j)$.
4. [4] Since the remaining 1 consecutive integers contain a multiple of 1 factorial, the difference is a multiple of 2.
5. Thus $f(k + 1) = 2[m + \text{an integer}]$ and the result follows by induction.

Correct option: D

Final answer

2

Worked Solutions

Recurrence and Induction

Q142 Written

Prove by mathematical induction that $2n^3 - 3n^2 + n$ is a multiple of 6.

Solution

1. [1] Base case: $n = 1$, $f(1) = 0 = 6(0)$, so the statement holds.
2. [2] Assume $f(k) = 6m$ for an integer m .
3. [3] The induction difference is $f(k+1) - f(k) = 6k^2 = 6(k^2)$.
4. [4] Therefore $f(k+1) = 6[m + k^2]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$6 \mid (2n^3 - 3n^2 + n)$$

Worked Solutions

Recurrence and Induction

Q143 Written

Prove by mathematical induction that $4n^3 - n$ is a multiple of 3.

Solution

1. [1] Base case: $n = 1$, $f(1) = 3 = 3(1)$, so the statement holds.
2. [2] Assume $f(k) = 3m$ for an integer m .
3. [3] The induction difference is $f(k+1) - f(k) = 3(2k+1)^2 = 3((2k+1)^2)$.
4. [4] Therefore $f(k+1) = 3[m + (2k+1)^2]$ is a multiple of the divisor.
5. The result follows for every natural number by mathematical induction.

Final answer

$$3 \mid (4n^3 - n)$$

Worked Solutions

Recurrence and Induction

Q144 Challenge

For any integer $n \geq 1$, prove that $(n + 1)^n - 1$ is a multiple of n^2 .

Solution

1. Use the binomial theorem on $(n + 1)^n = (1 + n)^n$.
2. $(1 + n)^n - 1 = \binom{n}{1}n + \binom{n}{2}n^2 + \cdots + n^n$.
3. $\binom{n}{1}n = n^2$, and every remaining term also contains n^2 .
4. $(n + 1)^n - 1 = n^2[1 + \binom{n}{2} + \binom{n}{3}n + \cdots + n^{n-2}]$. The bracket is an integer for every integer $n \geq 1$.
5. Hence the expression is a multiple of n^2 .

Final answer

$$n^2 \mid [(n + 1)^n - 1]$$

Worked Solutions

Recurrence and Induction

Q145 Written

For the sequence determined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = 4, \quad a_3 = 5, \quad a_4 = 6.$
2. The pattern suggests $a_n = n + 2.$

Final answer

$$a_2 = 4, \quad a_3 = 5, \quad a_4 = 6; \quad a_n = n + 2$$

Q146 Written

For the sequence determined by $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = -3, \quad a_3 = -5, \quad a_4 = -7.$
2. The pattern suggests $a_n = 1 - 2n.$

Final answer

$$a_2 = -3, \quad a_3 = -5, \quad a_4 = -7; \quad a_n = 1 - 2n$$

Worked Solutions

Recurrence and Induction

Q147 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = 4, \quad a_3 = 9, \quad a_4 = 16.$
2. The pattern suggests $a_n = n^2.$

Final answer

$$a_2 = 4, \quad a_3 = 9, \quad a_4 = 16; \quad a_n = n^2$$

Q148 Written

For the sequence determined by $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, find a_2, a_3, a_4 and estimate the general term.

Solution

1. $a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}.$
2. The pattern suggests $a_n = \frac{2n-1}{n}.$

Final answer

$$a_2 = \frac{3}{2}, \quad a_3 = \frac{5}{3}, \quad a_4 = \frac{7}{4}; \quad a_n = \frac{2n-1}{n}$$

Worked Solutions

Recurrence and Induction

Q149 Written

For a sequence with $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 - 1}{n+1}$, prove by mathematical induction that $a_n = n + 2$.

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = k + 2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = \frac{a_k^2 - 1}{k+1}$: $a_{k+1} = k + 3$.
4. [4] This is $k + 3$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = n + 2$$

Worked Solutions

Recurrence and Induction

Q150 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 2k - 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2k + 1$.
4. [4] This is $2k + 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: D

Final answer

$$a_n = 2n - 1$$

Worked Solutions

Recurrence and Induction

Q151 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 2 = 2$.
2. [2] Assume $a_k = 2k$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k + 1)$.
4. [4] This is $2(k + 1)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 2n$$

Worked Solutions

Recurrence and Induction

Q152 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = 2k + 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2k + 3$.
4. [4] This is $2k + 3$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: B

Final answer

$$a_n = 2n + 1$$

Worked Solutions

Recurrence and Induction

Q153 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 2(k + 1)$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k + 2)$.
4. [4] This is $2(k + 2)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: A

Final answer

$$a_n = 2(n + 1)$$

Worked Solutions

Recurrence and Induction

Q154 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 3k - 2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 1$.
4. [4] This is $3k + 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: D

Final answer

$$a_n = 3n - 2$$

Worked Solutions

Recurrence and Induction

Q155 MCQ

Select the correct option: for $a_1 = 2$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 2 = 2$.
2. [2] Assume $a_k = 3k - 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 2$.
4. [4] This is $3k + 2$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 3n - 1$$

Worked Solutions

Recurrence and Induction

Q156 MCQ

Select the correct option: for $a_1 = 3$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 3 = 3$.
2. [2] Assume $a_k = 3k$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3(k + 1)$.
4. [4] This is $3(k + 1)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: B

Final answer

$$a_n = 3n$$

Worked Solutions

Recurrence and Induction

Q157 MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 3$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 3k + 1$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 3$: $a_{k+1} = 3k + 4$.
4. [4] This is $3k + 4$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: A

Final answer

$$a_n = 3n + 1$$

Worked Solutions

Recurrence and Induction

Q158 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = a_n^2 - n^2 a_n + (n+1)^2$, prove by mathematical induction that $a_n = n^2$.

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = k^2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k^2 - k^2 a_k + (k+1)^2$: $a_{k+1} = (k+1)^2$.
4. [4] This is $(k+1)^2$, the proposed formula with $n = k+1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = n^2$$

Worked Solutions

Recurrence and Induction

Q159 Written

For a sequence with $a_1 = -1$ and $a_{n+1} = a_n^2 + 2na_n - 2$, prove by mathematical induction that $a_n = 1 - 2n$.

Solution

1. [1] Base case: $a_1 = -1 = -1$.
2. [2] Assume $a_k = 1 - 2k$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k^2 + 2ka_k - 2$: $a_{k+1} = -2k - 1$.
4. [4] This is $-2k - 1$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = 1 - 2n$$

Worked Solutions

Recurrence and Induction

Q160 MCQ

Select the correct option: for $a_1 = 1$ and $a_{n+1} = 2a_n + 0$, the general term is:

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = 2^{k-1}$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = 2a_k + 0$: $a_{k+1} = 2^k$.
4. [4] This is 2^k , the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 2^{n-1}$$

Worked Solutions

Recurrence and Induction

Q161 Written

For a sequence with $a_1 = 1$ and $a_{n+1} = \frac{a_n - 4}{a_n - 3}$, prove by mathematical induction that $a_n = \frac{2n-1}{n}$.

Solution

1. [1] Base case: $a_1 = 1 = 1$.
2. [2] Assume $a_k = \frac{2k-1}{k}$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = \frac{a_k - 4}{a_k - 3}$: $a_{k+1} = \frac{2k+1}{k+1}$.
4. [4] This is $\frac{2k+1}{k+1}$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = \frac{2n-1}{n}$$

Worked Solutions

Recurrence and Induction

Q162 Challenge

Three numbers A, B, C in that order form a geometric sequence with common ratio r . If $A, 2B, C$ in that order form an arithmetic sequence, determine the possible values of r .

Solution

1. Let the geometric sequence be A, Ar, Ar^2 .
2. Since $A, 2B, C$ is arithmetic, its middle term is the average of the outer terms: $4B = A + C$.
3. Substitute $B = Ar, C = Ar^2$ and cancel the non-zero scale factor A : $r^2 - 4r + 1 = 0$.
4. $r = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$.

Final answer

$$r = 2 \pm \sqrt{3}$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 1 **Concept Quiz · MCQ**

Select the correct option: For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (4)$. The general term is:

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 2(2n - 1)$.

Correct option: D

Final answer

$$a_n = 2(2n - 1)$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If $a_1 = 2$ and $a_{n+1} = -2a_n + (3)$, the general term is:

Solution

1. $c = 1$ from $c = -2c + (3)$.
2. $a_{n+1} - c = -2(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = -\frac{(-2)^n - 2}{2}$.
5. Substitute $n = 1$ to verify the given first term.

Correct option: A

Final answer

$$a_n = -\frac{(-2)^n - 2}{2}$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 3 **Concept Quiz · MCQ**

Select the correct option: which inequality is proved for $n \geq 4$?

Solution

1. $n = 4$: $2^4 = 16 > 13 = 13$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(3k + 1)$ after multiplying by the positive base.
4. $2(3k + 1) - (3(k + 1) + 1) = 3k - 2$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Correct option: B

Final answer

$$2^n > 3n + 1$$

Worked Solutions

Recurrence and Induction

Quiz MCQ 4 Concept Quiz · MCQ

Select the correct option: for $a_1 = 4$ and $a_{n+1} = a_n + 2$, the general term is:

Solution

1. [1] Base case: $a_1 = 4 = 4$.
2. [2] Assume $a_k = 2(k + 1)$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 2$: $a_{k+1} = 2(k + 2)$.
4. [4] This is $2(k + 2)$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Correct option: C

Final answer

$$a_n = 2(n + 1)$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 5 Concept Quiz · Written

For the sequence $\{a_n\}$ with $a_1 = 2$ and recurrence $a_{n+1} = 1a_n + (5)$. Find the general term.

Solution

1. The rule adds the difference term, so it is arithmetic or a finite sum.
2. $a_n = 5n - 3$.

Final answer

$$a_n = 5n - 3$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 6 Concept Quiz · Written

Find the general term of $\{a_n\}$ given $a_1 = 1$ and $a_{n+1} = 5a_n + (8)$.

Solution

1. $c = -2$ from $c = 5c + (8)$.
2. $a_{n+1} - c = 5(a_n - c)$.
3. Let the shifted sequence be geometric, then undo the shift.
4. $a_n = \frac{3 \cdot 5^n - 10}{5}$.
5. Substitute $n = 1$ to verify the given first term.

Final answer

$$a_n = \frac{3 \cdot 5^n - 10}{5}$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 7 Concept Quiz · Written

Prove by mathematical induction that $2^n > 4n + 1$ for $n \geq 4$.

Solution

1. $n = 4$: $2^4 = 16 > 17 = 4 \cdot 4 + 1$.
2. Assume the inequality holds for $n = k$ with $k \geq 4$.
3. $2^k + 1 > 2(4k + 1)$ after multiplying by the positive base.
4. $2(4k + 1) - (4(k + 1) + 1) = 4k - 3$ is positive for the allowed k .
5. Hence the inequality holds for $k+1$ and therefore for every allowed natural number.

Final answer

$$2^n > 4n + 1$$

Worked Solutions

Recurrence and Induction

Quiz WRITTEN 8 Concept Quiz · Written

For a sequence with $a_1 = 5$ and $a_{n+1} = a_n + 1(2n + 1) + 2$, prove by mathematical induction that $a_n = n^2 + 2n + 2$.

Solution

1. [1] Base case: $a_1 = 5 = 5$.
2. [2] Assume $a_k = k^2 + 2k + 2$ for some positive integer k .
3. [3] Substitute into $a_{k+1} = a_k + 1(2k + 1) + 2$: $a_{k+1} = k^2 + 4k + 5$.
4. [4] This is $k^2 + 4k + 5$, the proposed formula with $n = k + 1$.
5. Therefore the formula holds for every natural number by mathematical induction.

Final answer

$$a_n = n^2 + 2n + 2$$

Answer key

Use the question number shown on each page.

Q001 $a_2 = 8, a_3 = 26, a_4 = 80$

Q002 $a_2 = -9, a_3 = -15, a_4 = -27$

Q003 $a_2 = -7, a_3 = -7, a_4 = -4$

Q004 $a_2 = -3, a_3 = -5, a_4 = -8$

Q005 $a_2 = 4, a_3 = 8, a_4 = 14$

Q006 $a_2 = 1, a_3 = 10, a_4 = 37$

Q007 B

Q008 D

Q009 A

Q010 B

Q011 D

Q012 A

Q013 C

Q014 D

Q015 C

Q016 D

Q017 B

Q018 C

Q019 $a_n = 5n - 6$

Q020 $a_n = 3n - 2$

Q021 $a_n = -2(n - 3)$

Q022 $a_n = 6n - 5$

Q023 $a_n = 8 - 5n$

Q024 $a_n = 3n - 5$

Q025 $a_n = -4(n - 2)$

Q026 D

Q027 A

Q028 A

Q029 A

Q030 $a_n = 2 \cdot 3^{n-1}$

Q031 $a_n = 3 \cdot 5^{n-1}$

Q032 $a_n = 4 \cdot 3^{n-1}$

Q033 $a_n = 2^{2n-2}$

Q034 B

Q035 B

Q036 D

Q037 A

Q038 $a_n = \frac{3(-2)^n}{2}$

Q039 $a_n = 2(-5)^{n-1}$

Q040 $a_n = \frac{(-4)^n}{2}$

Q041 B

Q042 C

Q043 B

Q044 C

Q045 $a_2 = 5, a_3 = 13, a_4 = 36$

Q046 $a_2 = 3, a_3 = -1, a_4 = 7$

Q047 $a_2 = 8, a_3 = 22, a_4 = 63$

Q048 $a_n = -\frac{4 \cdot 2^n - 3 \cdot 3^n - 1}{2}$

Q049 $a_n = (n - 2)^2$

Q050 $a_n = \frac{3n^2 - 3n + 2}{2}$

Q051 $a_n = n^2 - n + 2$

Q052 $a_n = \frac{2n^3 - 3n^2 + n + 4}{2}$

Q053 $a_n = \frac{3n^2 - n + 2}{2}$

Q054 $a_n = \frac{4n^3 - 3n^2 - n + 12}{6}$

Q055 C

Q056 A

Q057 C

Q058 D

Q059 C

Q060 A

Q061 D

Q062 A

Q063 B

Q064 A

Q065 $a_n = \frac{2n^3 - 3n^2 + n + 12}{6}$

Q066 A

Q067 $a_n = 2^n - 1$

Q068 $a_n = -\frac{3^n - 7}{2}$

Q069 $a_n = -\frac{4^n - 16}{3}$

Q070 $a_n = -\frac{n^2 - n - 200}{50}$

Q071 $a_n = 3^n - 2$

Q072 $a_n = \frac{3 \cdot 5^n - 10}{5}$

Q073 $a_n = \frac{2^n + 2}{2}$

Q074 $a_n = -\frac{4^n + 2}{2}$

Q075 $a_n = \frac{2 \cdot 3^n + 3}{3}$

Q076 $a_n = (-3)^n + 1$

Q077 $a_n = 3 \cdot 3^{-n} (3^n - 2)$

Q078 $a_n = \frac{3 \cdot 3^{-n} (7 \cdot 2^n - 2 \cdot 3^n)}{4}$

Q079 $a_n = -\frac{3(-2)^n - 2}{2}$

Q080 B

Q081 C

Q082 D

Q083 B

Q084 C

Q085 D

Q086 B

Q087 C

Q088 D

Q089 D

Q090 $c = 3, a_n = -\frac{2^{2n} - 12}{4}, a_5 = -253$

Q091 $\frac{n(n+1)}{2}$

Q092 $\frac{n(2n-1)(2n+1)}{3}$

Q093 $\frac{n(n+1)(n+2)}{3}$

Q094 A

Q095 B	Q113 $3^n > 5n + 1$	Q133 A	Q152 B
Q096 C	Q114 $3^n > 5n + 2$	Q134 D	Q153 A
Q097 B	Q115 $4^n > 2n + 3$	Q135 C	Q154 D
Q098 C	Q116 D	Q136 C	Q155 C
Q099 A	Q117 A	Q137 B	Q156 B
Q100 C	Q118 B	Q138 A	Q157 A
Q101 A	Q119 D	Q139 D	Q158 $a_n = n^2$
Q102 B	Q120 A	Q140 D	Q159 $a_n = 1 - 2n$
Q103 n^2	Q121 B	Q141 D	Q160 C
Q104 $\frac{n(n+1)(n+2)}{3}$	Q122 D	Q142 $6 \mid (2n^3 - 3n^2 + n)$	Q161 $a_n = \frac{2n-1}{n}$
Q105 $\frac{n(n+1)(2n+7)}{6}$	Q123 A	Q143 $3 \mid (4n^3 - n)$	Q162 $r = 2 \pm \sqrt{3}$
Q106 $4^n - 1$	Q124 $2^n > 3n + 1$	Q144 $n^2 \mid [(n+1)^n - 1]$	Quiz MCQ 1 D
Q107 $\frac{10 \cdot 10^n - 1}{9}$	Q125 B	Q145 $a_2 = 4, a_3 = 5, a_4 = 6; a_n = n + 2$	Quiz MCQ 2 A
Q108 $\frac{n(n^2 + 3n + 5)}{3}$	Q126 $(1+h)^n > 1 + nh^2$	Q146 $a_2 = -3, a_3 = -5, a_4 = -7; a_n = 1 - 2n$	Quiz MCQ 3 B
Q109 $2^n > 3n$	Q127 $9 \mid [n^3 + (n+1)^3 + (n+2)^3]$	Q147 $a_2 = 4, a_3 = 9, a_4 = 16; a_n = n^2$	Quiz MCQ 4 C
Q110 $3^n > 5n + 1$	Q128 $3 \mid (n^3 + 2n)$	Q148 $a_2 = \frac{3}{2}, a_3 = \frac{5}{3}, a_4 = \frac{7}{4}; a_n = \frac{2n-1}{n}$	Quiz WRITTEN 5 $a_n = 5n - 3$
Q111 $3^n > 2n$	Q129 $2 \mid (n^2 + n)$	Q149 $a_n = n + 2$	Quiz WRITTEN 6 $a_n = \frac{3 \cdot 5^n - 10}{5}$
Q112 $2^n > 1n + 2$	Q130 $2 \mid (n^2 + 5n + 8)$	Q150 D	Quiz WRITTEN 7 $2^n > 4n + 1$
	Q131 $2 \mid (n^2 + 7n + 12)$	Q151 C	Quiz WRITTEN 8 $a_n = n^2 + 2n + 2$
	Q132 $2 \mid (n^2 + 11n + 32)$		

Concept 1 | Complex Numbers

English Student Edition • Focused formulas and guided practice

Formula opener

Standard form

$$z = a + bi$$

Conjugate

$$\overline{a + bi} = a - bi$$

Equality

$$a + bi = c + di \Rightarrow a = c, b = d$$

Powers

$$i^4 = 1 \quad i^{4q+r} = i^r$$

Worked Solutions

Complex Numbers

Q001 Written

State the real and imaginary parts of $3 - 2i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(3, -2)$.

Final answer $(3, -2)$

Q002 Written

State the real and imaginary parts of $(4 - \sqrt{5}i)/3$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(4/3, -\sqrt{5}/3)$.

Final answer $(4/3, -\sqrt{5}/3)$

Worked Solutions

Complex Numbers

Q003 Written

State the real and imaginary parts of -8 .

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-8, 0)$.

Final answer

$(-8, 0)$

Q004 Written

State the real and imaginary parts of $2i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, 2)$.

Final answer

$(0, 2)$

Worked Solutions

Complex Numbers

Q005 Written

Find the real values of x and y satisfying $(2 + i)x + (1 + i)y = 2 + 3i$, x, y real.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = -1, y = 4$.

Final answer

$$x = -1, y = 4$$

Q006 Written

State the real and imaginary parts of $-3 + 5i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-3, 5)$.

Final answer

$$(-3, 5)$$

Worked Solutions

Complex Numbers

Q007 Written

State the real and imaginary parts of $(-1 - \sqrt{3}i)/2$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-1/2, -\sqrt{3}/2)$.

Final answer $(-1/2, -\sqrt{3}/2)$

Q008 Written

State the real and imaginary parts of 1.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1, 0)$.

Final answer $(1, 0)$

Worked Solutions

Complex Numbers

Q009 Written

State the real and imaginary parts of $3i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, 3)$.

Final answer $(0, 3)$

Q010 Written

Find the real values of x and y satisfying $(x - 6) + (x + 3y)i = 0$, x, y real.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 6, y = -2$.

Final answer $x = 6, y = -2$

Worked Solutions

Complex Numbers

Q011 Written

Find the real values of x and y satisfying $(3 + 2i)x + (4 - i)y = 6 - 7i$, x, y real.

Solution

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = -2, y = 3$.

Final answer

$$x = -2, y = 3$$

Q012 Written

State the real and imaginary parts of $2 + 5i$.

Solution

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2, 5)$.

Final answer

$$(2, 5)$$

Worked Solutions

Complex Numbers

Q013 Written

State the real and imaginary parts of $-25 + 13i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-25, 13)$.

Final answer $(-25, 13)$

Q014 Written

State the real and imaginary parts of $(2 - \sqrt{5}i)/3$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(2/3, -\sqrt{5}/3)$.

Final answer $(2/3, -\sqrt{5}/3)$

Worked Solutions

Complex Numbers

Q015 **Written****State the real and imaginary parts of $(7 - \sqrt{2}i)/2$.****Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(7/2, -\sqrt{2}/2)$.

Final answer $(7/2, -\sqrt{2}/2)$ Q016 **Written****State the real and imaginary parts of -4 .****Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(-4, 0)$.

Final answer $(-4, 0)$

Worked Solutions

Complex Numbers

Q017 Written

State the real and imaginary parts of $1/4$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(1/4, 0)$.

Final answer $(1/4, 0)$

Q018 Written

State the real and imaginary parts of $\sqrt{3}i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, \sqrt{3})$.

Final answer $(0, \sqrt{3})$

Worked Solutions

Complex Numbers

Q019 Written

State the real and imaginary parts of $-i$.**Solution**

1. Rewrite the number as $(a+bi)$, including a zero coefficient when needed.
2. Read off the real and imaginary coefficients: $(0, -1)$.

Final answer $(0, -1)$

Q020 Written

Find the real values of x and y satisfying $(2x + y) + (3x - y)i = 8 + 7i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 3, y = 2$.

Final answer $x = 3, y = 2$

Worked Solutions

Complex Numbers

Q021 Written

Find the real values of x and y satisfying $(x + 1) - 5i = 4 + (2y + 1)i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 3, y = -3$.

Final answer

$$x = 3, y = -3$$

Q022 Written

Find the real values of x and y satisfying $(x - 2) + (x + y)i = 0$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 2, y = -2$.

Final answer

$$x = 2, y = -2$$

Worked Solutions

Complex Numbers

Q023 Written

Find the real values of x and y satisfying $(3 + 2i)x + (1 - i)y = 7 + 3i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 2, y = 1$.

Final answer

$$x = 2, y = 1$$

Q024 Written

Find the real values of x and y satisfying $(1 - i)(x + yi) = 2 + i$.**Solution**

1. Expand and collect the real and imaginary parts.
2. Equate the two real coefficients to obtain a simultaneous real system.
3. Solving the system gives $x = 1/2, y = 3/2$.

Final answer

$$x = 1/2, y = 3/2$$

Worked Solutions

Complex Numbers

Q025 Written

Solve the complex-number equation and find the exact real values of x and y : $(1 + 2i)(x + i) = x(y + i)$.

Solution

1. Expand: $((1 + 2i)(x + i) = x - 2 + (1 + 2x)i)$.
2. Equate coefficients with $(xy + xi)$: $(x - 2 = xy)$ and $(1 + 2x = x)$.
3. Solving gives $x = -1, y = 3$.

Final answer

$$x = -1, y = 3$$

Q026 Written

Calculate $(3 - 2i) - (2 + 5i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 - 7i$.

Final answer

$$1 - 7i$$

Worked Solutions

Complex Numbers

Q027 Written

Calculate $(i - 2)^2$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $3 - 4i$.

Final answer

$$3 - 4i$$

Q028 Written

Calculate $i^6 + i^{27} + i^{17}$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -1 .

Final answer

$$-1$$

Worked Solutions

Complex Numbers

Q029 **Written****Calculate $(4 + 6i) + (-3 + 2i)$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 + 8i$.

Final answer

$$1 + 8i$$

Q030 **Written****Calculate $(2 + 3i)(1 - 2i)$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $8 - i$.

Final answer

$$8 - i$$

Worked Solutions

Complex Numbers

Q031 **Written****Calculate $i + i^2 + i^3 + i^4$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: 0.

Final answer

0

Q032 **Written****Calculate $(7 + 4i) - (3 + 5i)$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1); reduce powers modulo 4.
2. Collect the real and imaginary parts: $4 - i$.

Final answer $4 - i$

Worked Solutions

Complex Numbers

Q033 Written

Calculate $(-4 + 3i) + (5 - 6i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1 - 3i$.

Final answer $1 - 3i$

Q034 Written

Calculate $(3 - 3i) - (5 - 3i)$ and give the answer in standard form.**Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -2 .

Final answer -2

Worked Solutions

Complex Numbers

Q035 Written

Calculate $(\sqrt{2} + 3i)(\sqrt{2} + i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $-1 + 4\sqrt{2}i$.

Final answer

$$-1 + 4\sqrt{2}i$$

Q036 Written

Calculate $(2 + 3i) + (5 + 8i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $7 + 11i$.

Final answer

$$7 + 11i$$

Worked Solutions

Complex Numbers

Q037 Written

Calculate $(2/5 - 1/2i) - (1/3 + 3/7i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $1/15 - 13/14i$.

Final answer

$$1/15 - 13/14i$$

Q038 Written

Calculate $(2i - 1)^2$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $-3 - 4i$.

Final answer

$$-3 - 4i$$

Worked Solutions

Complex Numbers

Q039 Written

Calculate $(2/i + i)(2/i - i)$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -3 .

Final answer -3

Q040 Written

Calculate $(2 + i)^3 + (2 - i)^3$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: 4.

Final answer 4

Worked Solutions

Complex Numbers

Q041 **Written****Calculate $i^5 + i^3 + i$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: i .

Final answer i Q042 **Written****Calculate $i^9 + i^{14} + i^{11}$ and give the answer in standard form.****Solution**

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: -1 .

Final answer -1

Worked Solutions

Complex Numbers

Q043 Written

Calculate $i^{12} + i^{18} + i^{15}$ and give the answer in standard form.

Solution

1. Expand algebraically and replace (i^2) by (-1) ; reduce powers modulo 4.
2. Collect the real and imaginary parts: $-i$.

Final answer $-i$

Q044 Written

Given x is real, solve $(7x - 7i)(x^2 + xi) = 14x^2i^{40}$.

Solution

1. Use $(i^{40} = 1)$ and $((x - i)(x + i) = x^2 + 1)$.
2. The equation becomes $(7x(x^2 + 1) = 14x^2)$, so $(7x(x - 1)^2 = 0)$.
3. Therefore $x = 0$ or $x = 1$.

Final answer $x = 0$ or $x = 1$

Worked Solutions

Complex Numbers

Q045 Written

State the complex conjugate of $2 + 3i$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $2 - 3i$.

Final answer

$$2 - 3i$$

Q046 Written

State the complex conjugate of $7 - i\sqrt{11}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $7 + i\sqrt{11}$.

Final answer

$$7 + i\sqrt{11}$$

Worked Solutions

Complex Numbers

Q047 **Written****State the complex conjugate of 6.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 6.

Final answer

6

Q048 **Written****State the complex conjugate of $-5i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $5i$.

Final answer $5i$

Worked Solutions

Complex Numbers

Q049 Written

Calculate $(2 + i)/(2 - i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5 + 4/5i$.

Final answer

$$3/5 + 4/5i$$

Q050 Written

Calculate $(3 - i)/(2i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1/2 - 3/2i$.

Final answer

$$-1/2 - 3/2i$$

Worked Solutions

Complex Numbers

Q051 Written

State the complex conjugate of $1 - 4i$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $1 + 4i$.

Final answer

$$1 + 4i$$

Q052 Written

State the complex conjugate of $\sqrt{2} + i\sqrt{10}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{2} - i\sqrt{10}$.

Final answer

$$\sqrt{2} - i\sqrt{10}$$

Worked Solutions

Complex Numbers

Q053 Written

State the complex conjugate of 2.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 2.

Final answer

2

Q054 Written

State the complex conjugate of $-8i$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8i$.

Final answer $8i$

Worked Solutions

Complex Numbers

Q055 Written

Calculate $(3 + i)/(1 + 2i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1 - i$.

Final answer

$$1 - i$$

Q056 Written

Calculate $(1 - i)/i$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-1 - i$.

Final answer

$$-1 - i$$

Worked Solutions

Complex Numbers

Q057 **Written****State the complex conjugate of $3 + 4i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $3 - 4i$.

Final answer

$$3 - 4i$$

Q058 **Written****State the complex conjugate of $9 + 8i$.****Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $9 - 8i$.

Final answer

$$9 - 8i$$

Worked Solutions

Complex Numbers

Q059 Written

State the complex conjugate of $8 + i\sqrt{3}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $8 - i\sqrt{3}$.

Final answer

$$8 - i\sqrt{3}$$

Q060 Written

State the complex conjugate of 5.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is 5.

Final answer

$$5$$

Worked Solutions

Complex Numbers

Q061 Written

State the complex conjugate of -15 .**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is -15 .

Final answer -15

Q062 Written

State the complex conjugate of $\sqrt{7}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $\sqrt{7}$.

Final answer $\sqrt{7}$

Worked Solutions

Complex Numbers

Q063 Written

State the complex conjugate of i .**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i$.

Final answer $-i$

Q064 Written

State the complex conjugate of $-50i$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $50i$.

Final answer $50i$

Worked Solutions

Complex Numbers

Q065 Written

State the complex conjugate of $i\sqrt{6}$.**Solution**

1. Reverse the sign of the imaginary coefficient and leave the real part unchanged.
2. Therefore the conjugate is $-i\sqrt{6}$.

Final answer

$$-i\sqrt{6}$$

Q066 Written

Calculate $(1 + i)/(1 - i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is i .

Final answer

$$i$$

Worked Solutions

Complex Numbers

Q067 Written

Calculate $(3 + i)/(2 + i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $7/5 - 1/5i$.

Final answer

$$7/5 - 1/5i$$

Q068 Written

Calculate $(\sqrt{5} + i)/(\sqrt{5} - i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2/3 + \sqrt{5}/3i$.

Final answer

$$2/3 + \sqrt{5}/3i$$

Worked Solutions

Complex Numbers

Q069 Written

Calculate $5i/(1 + 2i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $2 + i$.

Final answer

$$2 + i$$

Q070 Written

Calculate $5i/(4 + 3i)$ and give the answer in the form $a+bi$.**Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $3/5 + 4/5i$.

Final answer

$$3/5 + 4/5i$$

Worked Solutions

Complex Numbers

Q071 **Written****Calculate $(1 - 3i)/(3 - 2i)$ and give the answer in the form $a+bi$.****Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $9/13 - 7/13i$.

Final answer

$$9/13 - 7/13i$$

Q072 **Written****Calculate $1/i$ and give the answer in the form $a+bi$.****Solution**

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $-i$.

Final answer

$$-i$$

Worked Solutions

Complex Numbers

Q073 Written

Calculate $(2 + i)/(3i)$ and give the answer in the form $a+bi$.

Solution

1. Multiply numerator and denominator by the conjugate of the denominator.
2. The denominator becomes real; simplify the real and imaginary coefficients.
3. The result is $1/3 - 2/3i$.

Final answer

$$1/3 - 2/3i$$

Q074 Written

Simplify exactly: $\left(\frac{i(10-12i)}{(2+i)(-1+4i)}\right)^{2027}$.

Solution

1. Rationalise the base using the conjugate of the denominator.
2. The base simplifies exactly to $(-2 - 144i)/85$.
3. Hence the exact result is $((-2 - 144i)/85)^{2027}$; no expansion of the 2027th power is required.

Final answer

$$((-2 - 144i)/85)^{2027}$$

Worked Solutions

Complex Numbers

Q075 Written

Express the square roots of -6 in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $\pm i\sqrt{6}$.

Final answer

$$\pm i\sqrt{6}$$

Q076 Written

Express $\sqrt{-12}\sqrt{-6}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-6\sqrt{2}$.

Final answer

$$-6\sqrt{2}$$

Worked Solutions

Complex Numbers

Q077 Written

Express $\sqrt{8}/\sqrt{-18}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-(2/3)i$.

Final answer

$$-(2/3)i$$

Q078 Written

Express $\sqrt{-7}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $i\sqrt{7}$.

Final answer

$$i\sqrt{7}$$

Worked Solutions

Complex Numbers

Q079 Written

Express $\sqrt{-27}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $3i\sqrt{3}$.

Final answer

$$3i\sqrt{3}$$

Q080 Written

Express $-\sqrt{-12}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2i\sqrt{3}$.

Final answer

$$-2i\sqrt{3}$$

Worked Solutions

Complex Numbers

Q081 Written

Express the square roots of -12 in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $\pm 2i\sqrt{3}$.

Final answer

$$\pm 2i\sqrt{3}$$

Q082 Written

Express $\sqrt{-3}\sqrt{-6}$ in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $-3\sqrt{2}$.

Final answer

$$-3\sqrt{2}$$

Worked Solutions

Complex Numbers

Q083 Written

Express $\sqrt{27}/\sqrt{-3}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-3i$.

Final answer $-3i$

Q084 Written

Express $(3 + \sqrt{-2})^2$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $7 + 6\sqrt{2}i$.

Final answer $7 + 6\sqrt{2}i$

Worked Solutions

Complex Numbers

Q085 Written

Express $\sqrt{-2}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $i\sqrt{2}$.

Final answer

$$i\sqrt{2}$$

Q086 Written

Express $\sqrt{-9}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $3i$.

Final answer

$$3i$$

Worked Solutions

Complex Numbers

Q087 Written

Express $\sqrt{-45}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $3i\sqrt{5}$.

Final answer

$$3i\sqrt{5}$$

Q088 Written

Express $-\sqrt{-24}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2i\sqrt{6}$.

Final answer

$$-2i\sqrt{6}$$

Worked Solutions

Complex Numbers

Q089 Written

Express $-\sqrt{-36}$ in terms of i .**Solution**

1. Use $(\sqrt{a})^2 = a$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-6i$.

Final answer $-6i$

Q090 Written

Express the square roots of -2 in terms of i .**Solution**

1. Use $(\sqrt{a})^2 = a$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $\pm i\sqrt{2}$.

Final answer $\pm i\sqrt{2}$

Worked Solutions

Complex Numbers

Q091 Written

Express the square roots of -18 in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $\pm 3i\sqrt{2}$.

Final answer

$$\pm 3i\sqrt{2}$$

Q092 Written

Express the square roots of -16 in terms of i .**Solution**

1. Use ($\sqrt{-a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $\pm 4i$.

Final answer

$$\pm 4i$$

Worked Solutions

Complex Numbers

Q093 Written

Express $\sqrt{-2}\sqrt{-6}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2\sqrt{3}$.

Final answer

$$-2\sqrt{3}$$

Q094 Written

Express $\sqrt{-28}\sqrt{-35}$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-14\sqrt{5}$.

Final answer

$$-14\sqrt{5}$$

Worked Solutions

Complex Numbers

Q095 Written

Express $\sqrt{8}/\sqrt{-2}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2i$.

Final answer $-2i$

Q096 Written

Express $\sqrt{27}/\sqrt{-12}$ in terms of i .**Solution**

1. Use $(\sqrt{a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-(3/2)i$.

Final answer $-(3/2)i$

Worked Solutions

Complex Numbers

Q097 Written

Express $\sqrt{-32} + \sqrt{-98} + \sqrt{-8}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $13i\sqrt{2}$.

Final answer

$$13i\sqrt{2}$$

Q098 Written

Express $\sqrt{-98} - \sqrt{-72} + \sqrt{-50}$ in terms of i .**Solution**

1. Use ($\sqrt{a} = i\sqrt{a}$) for ($a > 0$), then simplify the positive radical.
2. Therefore $6i\sqrt{2}$.

Final answer

$$6i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q099 Written

Express $(\sqrt{3} + \sqrt{-5})^2$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore $-2 + 2i\sqrt{15}$.

Final answer

$$-2 + 2i\sqrt{15}$$

Q100 Written

Express $(2 + \sqrt{-3})(2 - \sqrt{-3})$ in terms of i .**Solution**

1. Use $(\sqrt{-a} = i\sqrt{a})$ for $(a > 0)$, then simplify the positive radical.
2. Therefore 7.

Final answer

$$7$$

Worked Solutions

Complex Numbers

Q101 Written

Solve over the complex numbers: $(2x - 8)^2 = -10$.

Solution

1. Take square roots: $(2x - 8 = \pm \sqrt{10})$.
2. Divide by 2 to obtain $x = 4 \pm (\sqrt{10}/2)i$.

Final answer

$$x = 4 \pm (\sqrt{10}/2)i$$

Q102 MCQ

For $2 + 3i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $2 + 3i$ using the relevant family rule.
2. The correct value is (2, 3).

Correct option: C

Final answer

(2, 3)

Worked Solutions

Complex Numbers

Q103 MCQ

For $-4 + 5i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-4 + 5i$ using the relevant family rule.
2. The correct value is $(-4, 5)$.

Correct option: D

Final answer

$(-4, 5)$

Q104 MCQ

For $7 - 2i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $7 - 2i$ using the relevant family rule.
2. The correct value is $(7, -2)$.

Correct option: D

Final answer

$(7, -2)$

Worked Solutions

Complex Numbers

Q105 MCQ

For $(1 - \sqrt{2}i)/3$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $(1 - \sqrt{2}i)/3$ using the relevant family rule.

2. The correct value is $\left(\frac{1}{3}, -\frac{\sqrt{2}}{3}\right)$.

Correct option: B

Final answer

$$\left(\frac{1}{3}, -\frac{\sqrt{2}}{3}\right)$$

Q106 MCQ

For -6 , select the ordered pair (real part, imaginary part):

Solution

1. Simplify -6 using the relevant family rule.

2. The correct value is $(-6, 0)$.

Correct option: D

Final answer

$$(-6, 0)$$

Worked Solutions

Complex Numbers

Q107 MCQ

For $4i$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $4i$ using the relevant family rule.
2. The correct value is $(0, 4)$.

Correct option: A**Final answer** $(0, 4)$

Q108 MCQ

For $-i$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $-i$ using the relevant family rule.
2. The correct value is $(0, -1)$.

Correct option: B**Final answer** $(0, -1)$

Worked Solutions

Complex Numbers

Q109 MCQ

For $\sqrt{5}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{5}i$ using the relevant family rule.
2. The correct value is $(0, \sqrt{5})$.

Correct option: C

Final answer

$(0, \sqrt{5})$

Q110 MCQ

For $3/2 + \sqrt{3}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $3/2 + \sqrt{3}i$ using the relevant family rule.
2. The correct value is $(\frac{3}{2}, \sqrt{3})$.

Correct option: B

Final answer

$(\frac{3}{2}, \sqrt{3})$

Worked Solutions

Complex Numbers

Q111 MCQ

For $-1/4 - 2i$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $-1/4 - 2i$ using the relevant family rule.
2. The correct value is $\left(-\frac{1}{4}, -2\right)$.

Correct option: D**Final answer**

$$\left(-\frac{1}{4}, -2\right)$$

Q112 MCQ

For $5/3$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $5/3$ using the relevant family rule.
2. The correct value is $\left(\frac{5}{3}, 0\right)$.

Correct option: D**Final answer**

$$\left(\frac{5}{3}, 0\right)$$

Worked Solutions

Complex Numbers

Q113 MCQ

For $-\sqrt{7}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-\sqrt{7}i$ using the relevant family rule.
2. The correct value is $(0, -\sqrt{7})$.

Correct option: C

Final answer

$(0, -\sqrt{7})$

Q114 MCQ

For $-9 + \sqrt{2}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $-9 + \sqrt{2}i$ using the relevant family rule.
2. The correct value is $(-9, \sqrt{2})$.

Correct option: D

Final answer

$(-9, \sqrt{2})$

Worked Solutions

Complex Numbers

Q115 MCQ

For $2/5 - 3/4i$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $2/5 - 3/4i$ using the relevant family rule.
2. The correct value is $\left(\frac{2}{5}, -\frac{3}{4}\right)$.

Correct option: B**Final answer**

$$\left(\frac{2}{5}, -\frac{3}{4}\right)$$

Q116 MCQ

For $-\sqrt{11}$, select the ordered pair (real part, imaginary part):**Solution**

1. Simplify $-\sqrt{11}$ using the relevant family rule.
2. The correct value is $(-\sqrt{11}, 0)$.

Correct option: B**Final answer**

$$(-\sqrt{11}, 0)$$

Worked Solutions

Complex Numbers

Q117 MCQ

For $\sqrt{13}i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $\sqrt{13}i$ using the relevant family rule.
2. The correct value is $(0, \sqrt{13})$.

Correct option: C**Final answer** $(0, \sqrt{13})$

Q118 MCQ

For $(2 + i)x + (1 - i)y = 3 + 3i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(2, -1)$.

Correct option: D**Final answer** $(2, -1)$

Worked Solutions

Complex Numbers

Q119 MCQ

For $(1 + 2i)x + (3 - i)y = 5 - 4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1, 2)$.

Correct option: A

Final answer $(-1, 2)$

Q120 MCQ

For $(3 - 2i)x + (1 + i)y = 6 + i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1, 3)$.

Correct option: B

Final answer $(1, 3)$

Worked Solutions

Complex Numbers

Q121 MCQ

For $(2 - i)x + (4 + 2i)y = 4i$, select the ordered pair (x, y) :

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-2, 1)$.

Correct option: B

Final answer

$(-2, 1)$

Q122 MCQ

For $(1 + i)x + (2 + 3i)y = 1$, select the ordered pair (x, y) :

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(3, -1)$.

Correct option: C

Final answer

$(3, -1)$

Worked Solutions

Complex Numbers

Q123 MCQ

For $(4 + i)x + (1 - 2i)y = 2 + 5i$, select the ordered pair (x, y) :

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(1, -2)$.

Correct option: D

Final answer $(1, -2)$

Q124 MCQ

For $(2 + 3i)x + (3 + i)y = -8 - 5i$, select the ordered pair (x, y) :

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives $(-1, -2)$.

Correct option: D

Final answer $(-1, -2)$

Worked Solutions

Complex Numbers

Q125 MCQ

For $(1 - i)x + (2 - i)y = 6 - 4i$, select the ordered pair (x,y):

Solution

1. Expand and equate the real and imaginary coefficients.
2. Solving the real system gives (2, 2).

Correct option: C

Final answer

(2, 2)

Q126 MCQ

For $(3 + 4i) - (1 - 2i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $2 + 6i$.

Correct option: C

Final answer

$2 + 6i$

Worked Solutions

Complex Numbers

Q127 MCQ

For $(-2 + i) + (5 - 3i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $3 - 2i$.

Correct option: C

Final answer

$3 - 2i$

Q128 MCQ

For $(1 + 2i)(3 - i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5 + 5i$.

Correct option: D

Final answer

$5 + 5i$

Worked Solutions

Complex Numbers

Q129 MCQ

For $(\sqrt{3} + i)(\sqrt{3} - i)$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is 4.

Correct option: A

Final answer

4

Q130 MCQ

For $(2 - i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2 - i)^2$.

Correct option: C

Final answer $(2 - i)^2$

Worked Solutions

Complex Numbers

Q131 MCQ

For $(1 + 3i)^2$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-8 + 6i$.

Correct option: A**Final answer** $-8 + 6i$

Q132 MCQ

For i^{37} , select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is i .

Correct option: B**Final answer** i

Worked Solutions

Complex Numbers

Q133 MCQ

For $i^{22} + i^9$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: B**Final answer**

$$-1 + i$$

Q134 MCQ

For $i^{15} - i^6$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $1 - i$.

Correct option: B**Final answer**

$$1 - i$$

Worked Solutions

Complex Numbers

Q135 MCQ

For $(4 - 2i) + (1 + 5i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5 + 3i$.

Correct option: C**Final answer** $5 + 3i$

Q136 MCQ

For $(2 + i)(2 - i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is 5.

Correct option: A**Final answer**

5

Worked Solutions

Complex Numbers

Q137 MCQ

For $(\sqrt{2} + 2i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(\sqrt{2} + 2i)^2$.

Correct option: C**Final answer**

$$(\sqrt{2} + 2i)^2$$

Q138 MCQ

For $i^8 + i^{11} + i^{14}$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-i$.

Correct option: C**Final answer**

$$-i$$

Worked Solutions

Complex Numbers

Q139 MCQ

For $(3 - i)^3$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(3 - i)^3$.

Correct option: D**Final answer**

$$(3 - i)^3$$

Q140 MCQ

For $(1 - 2i)(-2 + i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $5i$.

Correct option: D**Final answer**

$$5i$$

Worked Solutions

Complex Numbers

Q141 MCQ

For $i^5 + i^{10}$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $-1 + i$.

Correct option: D**Final answer**

$$-1 + i$$

Q142 MCQ

For $(5 + 2i) - (3 - 4i)$, select the simplified standard form:**Solution**

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $2 + 6i$.

Correct option: D**Final answer**

$$2 + 6i$$

Worked Solutions

Complex Numbers

Q143 MCQ

For $(2 - 3i)^2$, select the simplified standard form:

Solution

1. Expand and use ($i^2 = -1$), reducing powers of i modulo 4.
2. The simplified value is $(2 - 3i)^2$.

Correct option: C

Final answer

$$(2 - 3i)^2$$

Q144 MCQ

For the expression State the complex conjugate of $3 - 2i$;, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $3 + 2i$.

Correct option: C

Final answer

$$3 + 2i$$

Worked Solutions

Complex Numbers

Q145 MCQ

For the expression State the complex conjugate of $-4 + \sqrt{3}i$; the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-4 - \sqrt{3}i$.

Correct option: A

Final answer

$$-4 - \sqrt{3}i$$

Q146 MCQ

For the expression State the complex conjugate of $7i$; the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-7i$.

Correct option: C

Final answer

$$-7i$$

Worked Solutions

Complex Numbers

Q147 MCQ

For the expression State the complex conjugate of $-\sqrt{5}$, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\sqrt{5}$.

Correct option: C**Final answer**

$$-\sqrt{5}$$

Q148 MCQ

For the expression State the complex conjugate of $1/2 - 3i$, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} + 3i$.

Correct option: B**Final answer**

$$\frac{1}{2} + 3i$$

Worked Solutions

Complex Numbers

Q149 MCQ

For $(2 + i)/(1 - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} + \frac{3i}{2}$.

Correct option: A**Final answer**

$$\frac{1}{2} + \frac{3i}{2}$$

Q150 MCQ

For $(4 - 3i)/(2 + i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $1 - 2i$.

Correct option: A**Final answer**

$$1 - 2i$$

Worked Solutions

Complex Numbers

Q151 MCQ

For $(1 + 2i)/(3 - i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{10} + \frac{7i}{10}$.

Correct option: B

Final answer

$$\frac{1}{10} + \frac{7i}{10}$$

Q152 MCQ

For $5i/(2 - i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $i(2 + i)$.

Correct option: C

Final answer

$$i(2 + i)$$

Worked Solutions

Complex Numbers

Q153 MCQ

For $(3 - i)/(1 + 3i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D**Final answer** $-i$

Q154 MCQ

For $1/i$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-i$.

Correct option: D**Final answer** $-i$

Worked Solutions

Complex Numbers

Q155 MCQ

For $(2 + 3i)/(2 - 3i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13} + \frac{12i}{13}$.

Correct option: B**Final answer**

$$-\frac{5}{13} + \frac{12i}{13}$$

Q156 MCQ

For $(\sqrt{2} + i)/(\sqrt{2} - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{\sqrt{2} + i}{\sqrt{2} - i}$.

Correct option: B**Final answer**

$$\frac{\sqrt{2} + i}{\sqrt{2} - i}$$

Worked Solutions

Complex Numbers

Q157 MCQ

For $(1 - 4i)/(3 + 2i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{5}{13} - \frac{14i}{13}$.

Correct option: D**Final answer**

$$-\frac{5}{13} - \frac{14i}{13}$$

Q158 MCQ

For $7i/(1 - i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{7}{2} + \frac{7i}{2}$.

Correct option: C**Final answer**

$$-\frac{7}{2} + \frac{7i}{2}$$

Worked Solutions

Complex Numbers

Q159 MCQ

For the expression State the complex conjugate of $-2 + 5i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-2 - 5i$.

Correct option: B

Final answer

$$-2 - 5i$$

Q160 MCQ

For the expression State the complex conjugate of $6 - \sqrt{7}i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $6 + \sqrt{7}i$.

Correct option: D

Final answer

$$6 + \sqrt{7}i$$

Worked Solutions

Complex Numbers

Q161 MCQ

For the expression State the complex conjugate of $-i\sqrt{2}$;, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{2}i$.

Correct option: B

Final answer

$$\sqrt{2}i$$

Q162 MCQ

For the expression State the complex conjugate of 9;, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is 9.

Correct option: C

Final answer

$$9$$

Worked Solutions

Complex Numbers

Q163 MCQ

For the expression State the complex conjugate of $-3/4 + 2i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $-\frac{3}{4} - 2i$.

Correct option: A

Final answer

$$-\frac{3}{4} - 2i$$

Q164 MCQ

For $(5 + 2i)/(1 - 2i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{5} + \frac{12i}{5}$.

Correct option: D

Final answer

$$\frac{1}{5} + \frac{12i}{5}$$

Worked Solutions

Complex Numbers

Q165 MCQ

For $(2 - i)/(4 + i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{7}{17} - \frac{6i}{17}$.

Correct option: A**Final answer**

$$\frac{7}{17} - \frac{6i}{17}$$

Q166 MCQ

For $3i/(2 + 3i)$, select the value in the form $a+bi$:**Solution**

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{9}{13} + \frac{6i}{13}$.

Correct option: A**Final answer**

$$\frac{9}{13} + \frac{6i}{13}$$

Worked Solutions

Complex Numbers

Q167 MCQ

For $(\sqrt{3} - i)/(\sqrt{3} + i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{\sqrt{3} - i}{\sqrt{3} + i}$.

Correct option: C

Final answer

$$\frac{\sqrt{3} - i}{\sqrt{3} + i}$$

Q168 MCQ

For $(1 + i)/(2i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{1}{2} - \frac{i}{2}$.

Correct option: A

Final answer

$$\frac{1}{2} - \frac{i}{2}$$

Worked Solutions

Complex Numbers

Q169 MCQ

For the expression State the complex conjugate of $4 - 7i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $4 + 7i$.

Correct option: D

Final answer

$$4 + 7i$$

Q170 MCQ

For the expression State the complex conjugate of $\sqrt{10} + 2i$., the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\sqrt{10} - 2i$.

Correct option: A

Final answer

$$\sqrt{10} - 2i$$

Worked Solutions

Complex Numbers

Q171 MCQ

For the expression State the complex conjugate of $-8i$, the required value is:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $8i$.

Correct option: B

Final answer

$8i$

Q172 MCQ

For $(3 + 2i)/(3 - 2i)$, select the value in the form $a+bi$:

Solution

1. Use the conjugate sign rule; for division, multiply numerator and denominator by the denominator conjugate.
2. The simplified result is $\frac{5}{13} + \frac{12i}{13}$.

Correct option: B

Final answer

$\frac{5}{13} + \frac{12i}{13}$

Worked Solutions

Complex Numbers

Q173 MCQ

For $\sqrt{-5}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{5}$.

Correct option: B**Final answer**

$$i\sqrt{5}$$

Q174 MCQ

For $\sqrt{-20}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{5}$.

Correct option: A**Final answer**

$$2i\sqrt{5}$$

Worked Solutions

Complex Numbers

Q175 MCQ

For $-\sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i\sqrt{2}$.

Correct option: A**Final answer**

$$-3i\sqrt{2}$$

Q176 MCQ

For $\sqrt[4]{-8}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt[4]{-a} = i\sqrt[4]{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{2}$.

Correct option: C**Final answer**

$$\pm 2i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q177 MCQ

For $\sqrt{-45}/\sqrt{-5}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 3.

Correct option: A

Final answer

3

Q178 MCQ

For $\sqrt{-2}\sqrt{-8}$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -4 .

Correct option: B

Final answer -4

Worked Solutions

Complex Numbers

Q179 MCQ

For $\sqrt{12}/\sqrt{-3}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{a} = \sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: C

Final answer $-2i$

Q180 MCQ

For $(2 + \sqrt{-3})^2$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{a} = \sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $1 + 4i\sqrt{3}$.

Correct option: D

Final answer $1 + 4i\sqrt{3}$

Worked Solutions

Complex Numbers

Q181 MCQ

For $\sqrt{-72}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: D

Final answer

$$6i\sqrt{2}$$

Q182 MCQ

For $\sqrt{-50}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 5i\sqrt{2}$.

Correct option: D

Final answer

$$\pm 5i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q183 MCQ

For $\sqrt{-3}\sqrt{-12}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is -6 .

Correct option: A

Final answer -6

Q184 MCQ

For $\sqrt{8}/\sqrt{-2}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-2i$.

Correct option: D

Final answer $-2i$

Worked Solutions

Complex Numbers

Q185 MCQ

For $\sqrt{-32} + \sqrt{-8}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $6i\sqrt{2}$.

Correct option: B

Final answer

$$6i\sqrt{2}$$

Q186 MCQ

For $\sqrt{-98} - \sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $4i\sqrt{2}$.

Correct option: B

Final answer

$$4i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q187 MCQ

For $(\sqrt{5} + \sqrt{-2})^2$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $3 + 2i\sqrt{10}$.

Correct option: A**Final answer**

$$3 + 2i\sqrt{10}$$

Q188 MCQ

For $(3 + \sqrt{-5})(3 - \sqrt{-5})$, select the simplified value in terms of i:

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is 14.

Correct option: D**Final answer**

$$14$$

Worked Solutions

Complex Numbers

Q189 MCQ

For $\sqrt{-7}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $i\sqrt{7}$.

Correct option: C**Final answer**

$$i\sqrt{7}$$

Q190 MCQ

For $\sqrt{-28}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $2i\sqrt{7}$.

Correct option: A**Final answer**

$$2i\sqrt{7}$$

Worked Solutions

Complex Numbers

Q191 MCQ

For $\sqrt{-12}$, select the simplified value in terms of i :

Solution

1. Use $\sqrt{-a} = i\sqrt{a}$, then simplify the positive radical and the sign.
2. Therefore the correct value is $\pm 2i\sqrt{3}$.

Correct option: A

Final answer

$$\pm 2i\sqrt{3}$$

Q192 MCQ

For $\sqrt{27}/\sqrt{-3}$, select the simplified value in terms of i :

Solution

1. Use $\sqrt{-a} = i\sqrt{a}$, then simplify the positive radical and the sign.
2. Therefore the correct value is $-3i$.

Correct option: A

Final answer

$$-3i$$

Worked Solutions

Complex Numbers

Q193 MCQ

For $\sqrt{-2}\sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is -6 .

Correct option: D

Final answer -6

Q194 MCQ

For $\sqrt{-48}/\sqrt{-3}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-4i$.

Correct option: C

Final answer $-4i$

Worked Solutions

Complex Numbers

Q195 MCQ

For $\sqrt{-8} + \sqrt{-18}$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{2}$.

Correct option: C**Final answer**

$$5i\sqrt{2}$$

Q196 MCQ

For $(1 + \sqrt{-2})^2$, select the simplified value in terms of i :

Solution

1. Use (sqrt-a=isqrt a), then simplify the positive radical and the sign.
2. Therefore the correct value is $-1 + 2i\sqrt{2}$.

Correct option: A**Final answer**

$$-1 + 2i\sqrt{2}$$

Worked Solutions

Complex Numbers

Q197 MCQ

For $(4 + \sqrt{-3})(4 - \sqrt{-3})$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is 19.

Correct option: B

Final answer

19

Q198 MCQ

For $\sqrt{-75}$, select the simplified value in terms of i :

Solution

1. Use $(\sqrt{-a} = i\sqrt{a})$, then simplify the positive radical and the sign.
2. Therefore the correct value is $5i\sqrt{3}$.

Correct option: D

Final answer $5i\sqrt{3}$

Worked Solutions

Complex Numbers

Q199 MCQ

Solve for the real values x and y : $(1 + 3i)(x + i) = x(y + i)$:**Solution**

1. Expand and equate real and imaginary parts.
2. The system gives $(x = -1/2)$ and $(y = 7)$.

Correct option: B**Final answer**

$$\left(-\frac{1}{2}, 7\right)$$

Q200 MCQ

Given x is real, solve: $(4x - 4i)(x^2 + xi) = 10x^2i^{32}$:**Solution**

1. Use $(i^{32} = 1)$ and $((x - i)(x + i) = x^2 + 1)$.
2. The real equation is $(4x(x^2 + 1) = 10x^2)$, giving $x = 0, 1/2$, or 2 .

Correct option: D**Final answer**

$$\left\{0, 2, \frac{1}{2}\right\}$$

Worked Solutions

Complex Numbers

Q201 MCQ

Simplify exactly: $(i(8 - 6i)/((1 + 2i)(3 - i)))^{2027}$:

Solution

1. Rationalise the base using the conjugate of the denominator.
2. The exact base simplifies to $((7+i)/5)$. Keep the 2027th power unexpanded.

Correct option: B

Final answer

$$((7 + i)/5)^{2027}$$

Q202 MCQ

Solve over the complex numbers: $(3x - 6)^2 = -8$:

Solution

1. Take square roots: $(3x - 6 = \pm 2i\sqrt{2})$.
2. Divide by 3 to obtain $(x = 2 \pm \frac{2}{3}i\sqrt{2})$.

Correct option: B

Final answer

$$x = 2 \pm (2/3)i\sqrt{2}$$

Worked Solutions

Complex Numbers

Quiz MCQ 1 [Concept Quiz · MCQ](#)

For $4-3i$, select the ordered pair (real part, imaginary part):

Solution

1. Simplify $4 - 3i$ using the relevant family rule.
2. The correct value is $(4, -3)$.

Correct option: A

Final answer

$(4, -3)$

Quiz MCQ 2 [Concept Quiz · MCQ](#)

For $(1+2i)(2-i)$, select the simplified value:

Solution

1. Simplify $(1 + 2i)(2 - i)$ using the relevant family rule.
2. The correct value is $4 + 3i$.

Correct option: A

Final answer

$4 + 3i$

Worked Solutions

Complex Numbers

Quiz MCQ 3 [Concept Quiz · MCQ](#)

For $(3+i)/(1-i)$, select the value in the form $a+bi$:

Solution

1. Simplify $(3+i)/(1-i)$ using the relevant family rule.
2. The correct value is $1+2i$.

Correct option: D

Final answer

$$1 + 2i$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

For $\sqrt{-27}$, select the expression in terms of i :

Solution

1. Simplify $\sqrt{-27}$ using the relevant family rule.
2. The correct value is $3i\sqrt{3}$.

Correct option: A

Final answer

$$3i\sqrt{3}$$

Worked Solutions

Complex Numbers

Quiz WRITTEN 5 Concept Quiz · Written

State the real and imaginary parts of $\frac{5-\sqrt{5}i}{2}$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $(5/2, -\sqrt{5}/2)$

Final answer

$(5/2, -\sqrt{5}/2)$

Quiz WRITTEN 6 Concept Quiz · Written

Solve for real x, y : $(2 + i)x + (1 - i)y = 3 + i$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 4/3, y = 1/3$

Final answer

$x = 4/3, y = 1/3$

Worked Solutions

Complex Numbers

Quiz WRITTEN 7 Concept Quiz · Written

Calculate $i^{18} + i^7 - i^3$ in standard form.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: -1

Final answer

 -1

Quiz WRITTEN 8 Concept Quiz · Written

Solve over the complex numbers: $(x - 3)^2 = -12$.

Solution

1. Apply the relevant standard-form or negative-root rule.
2. Final answer: $x = 3 \pm 2i\sqrt{3}$

Final answer

 $x = 3 \pm 2i\sqrt{3}$

Answer key

Use the question number shown on each page.

Q001 $(3, -2)$	Q027 $3 - 4i$	Q053 2	Q079 $3i\sqrt{3}$
Q002 $(4/3, -\sqrt{5}/3)$	Q028 -1	Q054 $8i$	Q080 $-2i\sqrt{3}$
Q003 $(-8, 0)$	Q029 $1 + 8i$	Q055 $1 - i$	Q081 $\pm 2i\sqrt{3}$
Q004 $(0, 2)$	Q030 $8 - i$	Q056 $-1 - i$	Q082 $-3\sqrt{2}$
Q005 $x = -1, y = 4$	Q031 0	Q057 $3 - 4i$	Q083 $-3i$
Q006 $(-3, 5)$	Q032 $4 - i$	Q058 $9 - 8i$	Q084 $7 + 6\sqrt{2}i$
Q007 $(-1/2, -\sqrt{3}/2)$	Q033 $1 - 3i$	Q059 $8 - i\sqrt{3}$	Q085 $i\sqrt{2}$
Q008 $(1, 0)$	Q034 -2	Q060 5	Q086 $3i$
Q009 $(0, 3)$	Q035 $-1 + 4\sqrt{2}i$	Q061 -15	Q087 $3i\sqrt{5}$
Q010 $x = 6, y = -2$	Q036 $7 + 11i$	Q062 $\sqrt{7}$	Q088 $-2i\sqrt{6}$
Q011 $x = -2, y = 3$	Q037 $1/15 - 13/14i$	Q063 $-i$	Q089 $-6i$
Q012 $(2, 5)$	Q038 $-3 - 4i$	Q064 $50i$	Q090 $\pm i\sqrt{2}$
Q013 $(-25, 13)$	Q039 -3	Q065 $-i\sqrt{6}$	Q091 $\pm 3i\sqrt{2}$
Q014 $(2/3, -\sqrt{5}/3)$	Q040 4	Q066 i	Q092 $\pm 4i$
Q015 $(7/2, -\sqrt{2}/2)$	Q041 i	Q067 $7/5 - 1/5i$	Q093 $-2\sqrt{3}$
Q016 $(-4, 0)$	Q042 -1	Q068 $2/3 + \sqrt{5}/3i$	Q094 $-14\sqrt{5}$
Q017 $(1/4, 0)$	Q043 $-i$	Q069 $2 + i$	Q095 $-2i$
Q018 $(0, \sqrt{3})$	Q044 $x = 0$ or $x = 1$	Q070 $3/5 + 4/5i$	Q096 $-(3/2)i$
Q019 $(0, -1)$	Q045 $2 - 3i$	Q071 $9/13 - 7/13i$	Q097 $13i\sqrt{2}$
Q020 $x = 3, y = 2$	Q046 $7 + i\sqrt{11}$	Q072 $-i$	Q098 $6i\sqrt{2}$
Q021 $x = 3, y = -3$	Q047 6	Q073 $1/3 - 2/3i$	Q099 $-2 + 2i\sqrt{15}$
Q022 $x = 2, y = -2$	Q048 $5i$	Q074 $((-2 - 144i)/85)^{2027}$	Q100 7
Q023 $x = 2, y = 1$	Q049 $3/5 + 4/5i$	Q075 $\pm i\sqrt{6}$	Q101 $x = 4 \pm (\sqrt{10}/2)i$
Q024 $x = 1/2, y = 3/2$	Q050 $-1/2 - 3/2i$	Q076 $-6\sqrt{2}$	Q102 C
Q025 $x = -1, y = 3$	Q051 $1 + 4i$	Q077 $-(2/3)i$	Q103 D
Q026 $1 - 7i$	Q052 $\sqrt{2} - i\sqrt{10}$	Q078 $i\sqrt{7}$	Q104 D

Q105 B	Q132 B	Q159 B	Q186 B
Q106 D	Q133 B	Q160 D	Q187 A
Q107 A	Q134 B	Q161 B	Q188 D
Q108 B	Q135 C	Q162 C	Q189 C
Q109 C	Q136 A	Q163 A	Q190 A
Q110 B	Q137 C	Q164 D	Q191 A
Q111 D	Q138 C	Q165 A	Q192 A
Q112 D	Q139 D	Q166 A	Q193 D
Q113 C	Q140 D	Q167 C	Q194 C
Q114 D	Q141 D	Q168 A	Q195 C
Q115 B	Q142 D	Q169 D	Q196 A
Q116 B	Q143 C	Q170 A	Q197 B
Q117 C	Q144 C	Q171 B	Q198 D
Q118 D	Q145 A	Q172 B	Q199 B
Q119 A	Q146 C	Q173 B	Q200 D
Q120 B	Q147 C	Q174 A	Q201 B
Q121 B	Q148 B	Q175 A	Q202 B
Q122 C	Q149 A	Q176 C	Quiz MCQ 1 A
Q123 D	Q150 A	Q177 A	Quiz MCQ 2 A
Q124 D	Q151 B	Q178 B	Quiz MCQ 3 D
Q125 C	Q152 C	Q179 C	Quiz MCQ 4 A
Q126 C	Q153 D	Q180 D	Quiz WRITTEN 5 $(5/2, -\sqrt{5}/2)$
Q127 C	Q154 D	Q181 D	Quiz WRITTEN 6 $x = 4/3, y = 1/3$
Q128 D	Q155 B	Q182 D	Quiz WRITTEN 7 -1
Q129 A	Q156 B	Q183 A	Quiz WRITTEN 8 $x = 3 \pm 2i\sqrt{3}$
Q130 C	Q157 D	Q184 D	
Q131 A	Q158 C	Q185 B	

Concept 2 | Quadratic Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum/product

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Discriminant

$$\Delta = b^2 - 4ac$$

Worked Solutions

Quadratic Equations

Q001 Written

Solve $x^2 + 8 = 0$.**Solution**

1. $x^2 = -8$
2. $x = \pm\sqrt{-8} = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
3. Therefore the final answer is $x = \pm 2i\sqrt{2}$.

Final answer

$$x = \pm 2i\sqrt{2}$$

Q002 Written

Solve $x^2 + 4 = 0$.**Solution**

1. $x^2 = -4$
2. $x = \pm 2i$
3. Therefore the final answer is $x = \pm 2i$.

Final answer

$$x = \pm 2i$$

Worked Solutions

Quadratic Equations

Q003 Written

Solve $x^2 + 49 = 0$.

Solution

1. $x^2 = -49$
2. Therefore the final answer is $x = \pm 7i$.

Final answer

$$x = \pm 7i$$

Q004 Written

Solve $9x^2 - 4 = 0$.

Solution

1. $(3x - 2)(3x + 2) = 0$
2. Therefore the final answer is $x = \pm \frac{2}{3}$.

Final answer

$$x = \pm \frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q005 Written

Solve $x^2 - 12 = 0$.**Solution**

1. $x = \pm\sqrt{12}$
2. Therefore the final answer is $x = \pm 2\sqrt{3}$.

Final answer

$$x = \pm 2\sqrt{3}$$

Q006 Written

Solve $4x^2 + 3 = 0$.**Solution**

1. $x^2 = -3/4$
2. Therefore the final answer is $x = \pm \frac{i\sqrt{3}}{2}$.

Final answer

$$x = \pm \frac{i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q007 MCQ

Solve $x^2 + 8 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 2i\sqrt{2}$.

Correct option: A**Final answer**

$$x = \pm 2i\sqrt{2}$$

Q008 MCQ

Solve $x^2 + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 2i$.

Correct option: A**Final answer**

$$x = \pm 2i$$

Worked Solutions

Quadratic Equations

Q009 MCQ

Solve $x^2 + 49 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 7i$.

Correct option: C**Final answer**

$$x = \pm 7i$$

Q010 MCQ

Solve $9x^2 - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm \frac{2}{3}$.

Correct option: C**Final answer**

$$x = \pm \frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q011 MCQ

Solve $x^2 - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm 2\sqrt{3}$.

Correct option: B**Final answer**

$$x = \pm 2\sqrt{3}$$

Q012 MCQ

Solve $4x^2 + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm \frac{i\sqrt{3}}{2}$.

Correct option: B**Final answer**

$$x = \pm \frac{i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q013 Written

Solve $(x - 3)^2 = -8$.**Solution**

1. $x - 3 = \pm i\sqrt{8} = \pm 2i\sqrt{2}$
2. Therefore the final answer is $x = 3 \pm 2i\sqrt{2}$.

Final answer

$$x = 3 \pm 2i\sqrt{2}$$

Q014 Written

Solve $(x - 2)^2 = 5$.**Solution**

1. $x - 2 = \pm\sqrt{5}$
2. Therefore the final answer is $x = 2 \pm \sqrt{5}$.

Final answer

$$x = 2 \pm \sqrt{5}$$

Worked Solutions

Quadratic Equations

Q015 Written

Solve $(x + 6)^2 = -6$.**Solution**

1. $x + 6 = \pm i\sqrt{6}$
2. Therefore the final answer is $x = -6 \pm i\sqrt{6}$.

Final answer

$$x = -6 \pm i\sqrt{6}$$

Q016 Written

Solve $(x + 4)^2 = -18$.**Solution**

1. $x + 4 = \pm i\sqrt{18} = \pm 3i\sqrt{2}$
2. Therefore the final answer is $x = -4 \pm 3i\sqrt{2}$.

Final answer

$$x = -4 \pm 3i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q017 Written

Solve $(2x - 3)^2 = 7$.**Solution**

1. $2x - 3 = \pm\sqrt{7}$

2. Therefore the final answer is $x = \frac{3 \pm \sqrt{7}}{2}$.

Final answer

$$x = \frac{3 \pm \sqrt{7}}{2}$$

Q018 MCQ

Solve $(x - 3)^2 = -8$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = 3 \pm 2i\sqrt{2}$.**Correct option:** A**Final answer**

$$x = 3 \pm 2i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q019 MCQ

Solve $(x - 2)^2 = 5$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = 2 \pm \sqrt{5}$.

Correct option: B**Final answer**

$$x = 2 \pm \sqrt{5}$$

Q020 MCQ

Solve $(x + 6)^2 = -6$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = -6 \pm i\sqrt{6}$.

Correct option: C**Final answer**

$$x = -6 \pm i\sqrt{6}$$

Worked Solutions

Quadratic Equations

Q021 MCQ

Solve $(x + 4)^2 = -18$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = -4 \pm 3i\sqrt{2}$.

Correct option: C**Final answer**

$$x = -4 \pm 3i\sqrt{2}$$

Q022 MCQ

Solve $(2x - 3)^2 = 7$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{3 \pm \sqrt{7}}{2}$.

Correct option: D**Final answer**

$$x = \frac{3 \pm \sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q023 Written

Solve $2x^2 - 7x - 9 = 0$.**Solution**

1. $(x + 1)(2x - 9) = 0$
2. $x = -1$ or $x = 9/2$
3. Therefore the final answer is $x = -1, \frac{9}{2}$.

Final answer

$$x = -1, \frac{9}{2}$$

Q024 Written

Solve $2x^2 + 5x - 3 = 0$.**Solution**

1. $(2x - 1)(x + 3) = 0$
2. Therefore the final answer is $x = \frac{1}{2}, -3$.

Final answer

$$x = \frac{1}{2}, -3$$

Worked Solutions

Quadratic Equations

Q025 Written

Solve $2x^2 + 7x = 0$.**Solution**

1. $x(2x + 7) = 0$
2. Therefore the final answer is $x = 0, -\frac{7}{2}$.

Final answer

$$x = 0, -\frac{7}{2}$$

Q026 Written

Solve $2x^2 + 5x - 12 = 0$.**Solution**

1. $(2x - 3)(x + 4) = 0$
2. Therefore the final answer is $x = \frac{3}{2}, -4$.

Final answer

$$x = \frac{3}{2}, -4$$

Worked Solutions

Quadratic Equations

Q027 **Written****Solve** $x^2 - 8x + 15 = 0$.**Solution**

1. $(x - 3)(x - 5) = 0$
2. Therefore the final answer is $x = 3, 5$.

Final answer $x = 3, 5$ Q028 **MCQ****Solve** $2x^2 - 7x - 9 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = -1, \frac{9}{2}$.

Correct option: A**Final answer** $x = -1, \frac{9}{2}$

Worked Solutions

Quadratic Equations

Q029 MCQ

Solve $2x^2 + 5x - 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{1}{2}, -3$.

Correct option: D**Final answer**

$$x = \frac{1}{2}, -3$$

Q030 MCQ

Solve $2x^2 + 7x = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = 0, -\frac{7}{2}$.

Correct option: D**Final answer**

$$x = 0, -\frac{7}{2}$$

Worked Solutions

Quadratic Equations

Q031 MCQ

Solve $2x^2 + 5x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{3}{2}, -4$.

Correct option: A**Final answer**

$$x = \frac{3}{2}, -4$$

Q032 MCQ

Solve $x^2 - 8x + 15 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = 3, 5$.

Correct option: A**Final answer**

$$x = 3, 5$$

Worked Solutions

Quadratic Equations

Q033 Written

Solve $3x^2 - 4x - 1 = 0$.**Solution**

1. $D = 16 + 12 = 28$
2. $x = (4 \pm 2\sqrt{7})/6 = (2 \pm \sqrt{7})/3$
3. Therefore the final answer is $x = \frac{2 \pm \sqrt{7}}{3}$.

Final answer

$$x = \frac{2 \pm \sqrt{7}}{3}$$

Q034 MCQ

Solve $3x^2 - 4x - 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{2 \pm \sqrt{7}}{3}$.

Correct option: B**Final answer**

$$x = \frac{2 \pm \sqrt{7}}{3}$$

Worked Solutions

Quadratic Equations

Q035 Written

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$.

Solution

1. $D = (2\sqrt{6})^2 - 4(2)(11) = -64$

2. $x = (-2\sqrt{6} \pm \sqrt{(-64)})/4 = (-\sqrt{6} \pm 4i)/2$

3. Therefore the final answer is $x = \frac{-\sqrt{6} \pm 4i}{2}$.

Final answer

$$x = \frac{-\sqrt{6} \pm 4i}{2}$$

Q036 Written

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$.

Solution

1. $D = 12 - 40 = -28$

2. $x = (-2\sqrt{3} \pm i\sqrt{28})/4 = (-\sqrt{3} \pm i\sqrt{7})/2$

3. Therefore the final answer is $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q037 Written

Solve $x^2 + x + 1 = 0$.**Solution**

1. $D = 1 - 4 = -3$
2. $x = (-1 \pm i\sqrt{3})/2$
3. Therefore the final answer is $x = \frac{-1 \pm i\sqrt{3}}{2}$.

Final answer

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Q038 Written

Solve $3x^2 - 5x + 3 = 0$.**Solution**

1. $D = 25 - 36 = -11$
2. $x = (5 \pm i\sqrt{11})/6$
3. Therefore the final answer is $x = \frac{5 \pm i\sqrt{11}}{6}$.

Final answer

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

Worked Solutions

Quadratic Equations

Q039 Written

Solve $x^2 - 2\sqrt{3}x + 5 = 0$.**Solution**

1. $D = 12 - 20 = -8$
2. $x = (2\sqrt{3} \pm i\sqrt{8})/2 = \sqrt{3} \pm i\sqrt{2}$
3. Therefore the final answer is $x = \sqrt{3} \pm i\sqrt{2}$.

Final answer

$$x = \sqrt{3} \pm i\sqrt{2}$$

Q040 Written

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$.**Solution**

1. $D = 8 - 12 = -4$
2. $x = (-2\sqrt{2} \pm 2i)/6 = (-\sqrt{2} \pm i)/3$
3. Therefore the final answer is $x = \frac{-\sqrt{2} \pm i}{3}$.

Final answer

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Worked Solutions

Quadratic Equations

Q041 Written

Solve $x^2 - x + 2 = 0$.**Solution**

1. $D = 1 - 8 = -7$
2. $x = (1 \pm i\sqrt{7})/2$
3. Therefore the final answer is $x = \frac{1 \pm i\sqrt{7}}{2}$.

Final answer

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Q042 Written

Solve $3x^2 - \sqrt{5}x + 1 = 0$.**Solution**

1. $D = 5 - 12 = -7$
2. $x = (\sqrt{5} \pm i\sqrt{7})/6$
3. Therefore the final answer is $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$.

Final answer

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Worked Solutions

Quadratic Equations

Q043 **Written****Solve** $6x^2 - 12x + 15 = 0$.**Solution**

1. $D = 144 - 360 = -216$
2. $x = (12 \pm i6\sqrt{6})/12 = 1 \pm i\sqrt{6}/2$
3. Therefore the final answer is $x = 1 \pm \frac{i\sqrt{6}}{2}$.

Final answer

$$x = 1 \pm \frac{i\sqrt{6}}{2}$$

Q044 **MCQ****Solve** $2x^2 + 2\sqrt{6}x + 11 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{-\sqrt{6} \pm 4i}{2}$.

Correct option: B**Final answer**

$$x = \frac{-\sqrt{6} \pm 4i}{2}$$

Worked Solutions

Quadratic Equations

Q045 MCQ

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$.**Correct option:** C**Final answer**

$$x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$$

Q046 MCQ

Solve $x^2 + x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{-1 \pm i\sqrt{3}}{2}$.**Correct option:** D**Final answer**

$$x = \frac{-1 \pm i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q047 MCQ

Solve $3x^2 - 5x + 3 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{5 \pm i\sqrt{11}}{6}$.**Correct option:** B**Final answer**

$$x = \frac{5 \pm i\sqrt{11}}{6}$$

Q048 MCQ

Solve $x^2 - 2\sqrt{3}x + 5 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \sqrt{3} \pm i\sqrt{2}$.**Correct option:** D**Final answer**

$$x = \sqrt{3} \pm i\sqrt{2}$$

Worked Solutions

Quadratic Equations

Q049 MCQ

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{-\sqrt{2} \pm i}{3}$.**Correct option:** B**Final answer**

$$x = \frac{-\sqrt{2} \pm i}{3}$$

Q050 MCQ

Solve $x^2 - x + 2 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{1 \pm i\sqrt{7}}{2}$.**Correct option:** C**Final answer**

$$x = \frac{1 \pm i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q051 MCQ

Solve $3x^2 - \sqrt{5}x + 1 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$.**Correct option:** C**Final answer**

$$x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$$

Q052 MCQ

Solve $6x^2 - 12x + 15 = 0$ **Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x = 1 \pm \frac{i\sqrt{6}}{2}$.**Correct option:** B**Final answer**

$$x = 1 \pm \frac{i\sqrt{6}}{2}$$

Worked Solutions

Quadratic Equations

Q053 Written

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$.**Solution**

1. $D = 4^2 - 4(3)(7) = -68 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q054 Written

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions.**Solution**

1. $D = (-m)^2 - 4(m^2 - 3m - 9) = -3m^2 + 12m + 36$
2. $D \geq 0 \rightarrow m^2 - 4m - 12 \leq 0 \rightarrow (m + 2)(m - 6) \leq 0$
3. Therefore the final answer is $-2 \leq m \leq 6$.

Final answer $-2 \leq m \leq 6$

Worked Solutions

Quadratic Equations

Q055 Written

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = k^2 - 4(3 - k) = (k - 2)(k + 6)$
2. Use $D > 0, D = 0, D < 0$ intervals
3. Therefore the final answer is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Written

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions.

Solution

1. $D = 25 - 48 = -23 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q057 Written

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions.

Solution

1. $D = 49 + 32 = 81 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q058 Written

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions.

Solution

1. $D = 100 - 100 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q059 Written

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$.**Solution**

1. $D = 25 - 12 = 13 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Q060 Written

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$.**Solution**

1. $D = 144 - 144 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q061 Written

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$.**Solution**

1. $D = 64 - 80 = -16 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q062 Written

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$.**Solution**

1. $D = 16 - 16 = 0$
2. Therefore the final answer is One repeated real solution.

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q063 Written

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$.

Solution

1. $D = 9 - 20 = -11 < 0$
2. Therefore the final answer is Two distinct imaginary (non-real complex) solutions.

Final answer

Two distinct imaginary (non-real complex) solutions

Q064 Written

Determine the nature of the solutions of $x^2 - x - 2 = 0$.

Solution

1. $D = 1 + 8 = 9 > 0$
2. Therefore the final answer is Two distinct real solutions.

Final answer

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q065 MCQ

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q066 MCQ

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-2 \leq m \leq 6$.

Correct option: D

Final answer

$$-2 \leq m \leq 6$$

Worked Solutions

Quadratic Equations

Q067 MCQ

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary.

Correct option: C

Final answer

$k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q068 MCQ

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: C

Final answer

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q069 MCQ

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct real solutions.

Correct option: A

Final answer

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q070 MCQ

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is One repeated real solution.

Correct option: A

Final answer

One repeated real solution

Worked Solutions

Quadratic Equations

Q071 MCQ

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct real solutions.

Correct option: C**Final answer**

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q072 MCQ

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is One repeated real solution.

Correct option: C**Final answer**

One repeated real solution

Worked Solutions

Quadratic Equations

Q073 MCQ

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: B**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q074 MCQ

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is One repeated real solution.

Correct option: A**Final answer**

One repeated real solution

Worked Solutions

Quadratic Equations

Q075 MCQ

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct imaginary (non-real complex) solutions.

Correct option: A**Final answer**

Two distinct imaginary (non-real complex) solutions

Worked Solutions

Quadratic Equations

Q076 MCQ

Determine the nature of the solutions of $x^2 - x - 2 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct real solutions.

Correct option: A**Final answer**

Two distinct real solutions

Worked Solutions

Quadratic Equations

Q077 Written

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions.

Solution

1. $D = 4m^2 - 4(m + 2)$
2. $D < 0 \rightarrow m^2 - m - 2 < 0 \rightarrow (m - 2)(m + 1) < 0$
3. Therefore the final answer is $-1 < m < 2$.

Final answer

$$-1 < m < 2$$

Q078 Written

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 - 4(2a - 3) = (a - 2)(a - 6)$
2. Classify by sign of D
3. Therefore the final answer is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Final answer $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions

Quadratic Equations

Q079 Written

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution.

Solution

1. $D = (a + 1)^2 - 16 = 0$
2. $a + 1 = \pm 4$
3. Therefore the final answer is $a = 3$ or $a = -5$.

Final answer $a = 3$ or $a = -5$

Q080 Written

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions.

Solution

1. $D = (m + 6)^2 + 8m = m^2 + 20m + 36 = (m + 2)(m + 18)$
2. $D < 0$
3. Therefore the final answer is $-18 < m < -2$.

Final answer $-18 < m < -2$

Worked Solutions

Quadratic Equations

Q081 Written

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions.

Solution

1. $D = 4m^2 - 4(2m^2 - 3m) = 4m(3 - m)$
2. $D \geq 0$
3. Therefore the final answer is $0 \leq m \leq 3$.

Final answer

$$0 \leq m \leq 3$$

Q082 MCQ

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-1 < m < 2$.

Correct option: C

Final answer

$$-1 < m < 2$$

Worked Solutions

Quadratic Equations

Q083 MCQ

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary.

Correct option: C

Final answer

$a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Worked Solutions

Quadratic Equations

Q084 MCQ

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a = 3$ or $a = -5$.

Correct option: A**Final answer** $a = 3$ or $a = -5$

Q085 MCQ

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-18 < m < -2$.

Correct option: D**Final answer** $-18 < m < -2$

Worked Solutions

Quadratic Equations

Q086 MCQ

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $0 \leq m \leq 3$.

Correct option: C

Final answer

$$0 \leq m \leq 3$$

Q087 Written

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$.

Solution

1. Use $S = -b/a = 6/3 = 2$ and $P = c/a = -5/3$
2. Therefore the final answer is $\alpha + \beta = 2$, $\alpha\beta = -\frac{5}{3}$.

Final answer

$$\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$$

Worked Solutions

Quadratic Equations

Q088 Written

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$.**Solution**

1. $S = -2/4 = -1/2; P = -7/4$
2. Therefore the final answer is $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$.

Final answer

$$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$$

Q089 Written

Find the sum and product of the roots of $x^2 + x + 2 = 0$.**Solution**

1. $S = -1; P = 2$
2. Therefore the final answer is $S = -1, \quad P = 2$.

Final answer

$$S = -1, \quad P = 2$$

Worked Solutions

Quadratic Equations

Q090 Written

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$.**Solution**

1. $S = 7/3; P = -2$
2. Therefore the final answer is $S = \frac{7}{3}, P = -2$.

Final answer

$$S = \frac{7}{3}, P = -2$$

Q091 MCQ

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$.

Correct option: C**Final answer**

$$\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$$

Worked Solutions

Quadratic Equations

Q092 MCQ

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $\alpha + \beta = -\frac{1}{2}$, $\alpha\beta = -\frac{7}{4}$.

Correct option: A**Final answer**

$$\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$$

Q093 MCQ

Find the sum and product of the roots of $x^2 + x + 2 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $S = -1$, $P = 2$.

Correct option: B**Final answer**

$$S = -1, \quad P = 2$$

Worked Solutions

Quadratic Equations

Q094 MCQ

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $S = \frac{7}{3}$, $P = -2$.

Correct option: A

Final answer

$$S = \frac{7}{3}, \quad P = -2$$

Q095 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 3, P = -3/2$
2. $S^2 - 2P = 9 + 3 = 12$
3. Therefore the final answer is 12.

Final answer

12

Worked Solutions

Quadratic Equations

Q096 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = -3/2$
2. $S^3 - 3PS = 27 + 27/2 = 81/2$
3. Therefore the final answer is $\frac{81}{2}$.

Final answer

$$\frac{81}{2}$$

Q097 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3/2$
2. $S^2 - 2P = 4 - 3 = 1$
3. Therefore the final answer is 1.

Final answer

$$1$$

Worked Solutions

Quadratic Equations

Q098 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3/2$
2. $S^3 - 3PS = 8 - 9 = -1$
3. Therefore the final answer is -1 .

Final answer -1

Q099 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = 2, P = 3$
2. $S^2 - 2P = 4 - 6 = -2$
3. Therefore the final answer is -2 .

Final answer -2

Worked Solutions

Quadratic Equations

Q100 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 2, P = 3$
2. $S^3 - 3PS = 8 - 18 = -10$
3. Therefore the final answer is -10 .

Final answer

-10

Q101 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$.

Solution

1. $S = -2, P = -1/3$
2. $S^2 - 2P = 4 + 2/3 = 14/3$
3. Therefore the final answer is $\frac{14}{3}$.

Final answer

$\frac{14}{3}$

Worked Solutions

Quadratic Equations

Q102 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = -2, P = -1/3$
2. $S^3 - 3PS = -8 - 2 = -10$
3. Therefore the final answer is -10 .

Final answer

-10

Q103 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 12.

Correct option: A

Final answer

12

Worked Solutions

Quadratic Equations

Q104 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{81}{2}$.

Correct option: B**Final answer**

$$\frac{81}{2}$$

Q105 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 1.

Correct option: A**Final answer**

$$1$$

Worked Solutions

Quadratic Equations

Q106 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -1 .

Correct option: B

Final answer -1

Q107 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -2 .

Correct option: D

Final answer -2

Worked Solutions

Quadratic Equations

Q108 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -10 .

Correct option: B**Final answer** -10

Q109 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{14}{3}$.

Correct option: B**Final answer** $\frac{14}{3}$

Worked Solutions

Quadratic Equations

Q110 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -10 .

Correct option: D**Final answer** -10

Q111 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$.

Solution

1. $2(\alpha + \beta)/(\alpha\beta) = 2(3)/(-3/2) = -4$
2. Therefore the final answer is -4 .

Final answer -4

Worked Solutions

Quadratic Equations

Q112 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$.

Solution

1. $(\alpha^2 + \beta^2)/(\alpha\beta) = 1/(3/2) = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$$\frac{2}{3}$$

Q113 Written

Find the sum and product of the roots of $5x^2 - 3 = 0$.

Solution

1. $S = 0; P = -3/5$
2. Therefore the final answer is $S = 0, P = -\frac{3}{5}$.

Final answer

$$S = 0, P = -\frac{3}{5}$$

Worked Solutions

Quadratic Equations

Q114 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = 2/3$
2. Therefore the final answer is $\frac{2}{3}$.

Final answer

$$\frac{2}{3}$$

Q115 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Solution

1. $S/P = (-2)/(-1/3) = 6$
2. Therefore the final answer is 6.

Final answer

$$6$$

Worked Solutions

Quadratic Equations

Q116 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is -4 .

Correct option: C**Final answer** -4

Q117 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D**Final answer** $\frac{2}{3}$

Worked Solutions

Quadratic Equations

Q118 MCQ

Find the sum and product of the roots of $5x^2 - 3 = 0$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $S = 0$, $P = -\frac{3}{5}$.

Correct option: D**Final answer**

$$S = 0, \quad P = -\frac{3}{5}$$

Q119 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{2}{3}$.

Correct option: D**Final answer**

$$\frac{2}{3}$$

Worked Solutions

Quadratic Equations

Q120 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 6.

Correct option: D**Final answer**

6

Q121 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$.

Solution

1. $P + S + 1 = -3/2 + 3 + 1 = 5/2$
2. Therefore the final answer is $\frac{5}{2}$.

Final answer $\frac{5}{2}$

Worked Solutions

Quadratic Equations

Q122 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$.

Solution

1. $P - S + 1 = 3/2 - 2 + 1 = 1/2$
2. Therefore the final answer is $\frac{1}{2}$.

Final answer

$$\frac{1}{2}$$

Q123 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$.

Solution

1. $P + 2S + 4 = 3 + 4 + 4 = 11$
2. Therefore the final answer is 11.

Final answer

$$11$$

Worked Solutions

Quadratic Equations

Q124 **Written**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$.

Solution

1. $P - 2S + 4 = -1/3 + 4 + 4 = 23/3$
2. Therefore the final answer is $\frac{23}{3}$.

Final answer

$$\frac{23}{3}$$

Q125 **MCQ**

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{5}{2}$.

Correct option: A**Final answer**

$$\frac{5}{2}$$

Worked Solutions

Quadratic Equations

Q126 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{1}{2}$.

Correct option: C

Final answer

$$\frac{1}{2}$$

Q127 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 11.

Correct option: D

Final answer

$$11$$

Worked Solutions

Quadratic Equations

Q128 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{23}{3}$.

Correct option: D**Final answer**

$$\frac{23}{3}$$

Q129 Written

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots.

Solution

1. Let roots $\alpha, 3\alpha$. $4\alpha = 8 \Rightarrow \alpha = 2$.
2. $k = \alpha(3\alpha) = 6$
3. Therefore the final answer is $k = 6$, $x = 2, 6$.

Final answer

$$k = 6, \quad x = 2, 6$$

Worked Solutions

Quadratic Equations

Q130 Written

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots.

Solution

1. Roots $\alpha, 2\alpha$; $3\alpha = -6 \Rightarrow \alpha = -2$; $m = 2\alpha^2 = 8$
2. Therefore the final answer is $m = 8$, $x = -2, -4$.

Final answer

$$m = 8, \quad x = -2, -4$$

Q131 Written

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots.

Solution

1. Roots $\alpha, 2\alpha$; $2\alpha^2 = 1/8 \Rightarrow \alpha = \pm 1/4$.
2. $m = 8(3\alpha) = \pm 6$.
3. Therefore the final answer is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Final answer

$$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2}) \text{ or } (-6, -\frac{1}{4}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q132 MCQ

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k = 6$, $x = 2, 6$.

Correct option: A**Final answer**

$$k = 6, \quad x = 2, 6$$

Q133 MCQ

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m = 8$, $x = -2, -4$.

Correct option: A**Final answer**

$$m = 8, \quad x = -2, -4$$

Worked Solutions

Quadratic Equations

Q134 MCQ

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$.

Correct option: D

Final answer

$$(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2}) \text{ or } (-6, -\frac{1}{4}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q135 Written

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots.

Solution

1. Let roots be α and $\alpha + 5$.
2. $\alpha(\alpha + 5) = -6 \Rightarrow \alpha = -2$ or -3 .
3. $S = 2\alpha + 5 = m - 1$ gives $m = 2$ or 0 ; obtain the paired roots.
4. Therefore the final answer is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Final answer

$$(m, x_1, x_2) = (2, -2, 3) \text{ or } (0, -3, 2)$$

Q136 Written

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots.

Solution

1. Let roots be α and $\alpha + 1$. $\alpha(\alpha + 1) = 2 \Rightarrow \alpha = 1$ or -2 .
2. $m + 1 = 2\alpha + 1 \Rightarrow m = 2$ or -4 .
3. Therefore the final answer is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Final answer

$$(m, x_1, x_2) = (2, 1, 2) \text{ or } (-4, -2, -1)$$

Worked Solutions

Quadratic Equations

Q137 Written

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots.

Solution

1. Roots $\alpha, \alpha + 4$; $2\alpha + 4 = -6 \rightarrow \alpha = -5$; $m = \alpha(\alpha + 4) = 5$
2. Therefore the final answer is $m = 5$, $x = -5, -1$.

Final answer

$$m = 5, \quad x = -5, -1$$

Q138 Written

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots.

Solution

1. Roots $\alpha, \alpha + 1$; $\alpha(\alpha + 1) = 3/4 \rightarrow \alpha = 1/2$ or $-3/2$.
2. $-m/4 = 2\alpha + 1 \rightarrow m = -8$ or 8 .
3. Therefore the final answer is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Final answer

$$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2}) \text{ or } (8, -\frac{3}{2}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q139 MCQ

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$.

Correct option: D

Final answer

$$(m, x_1, x_2) = (2, -2, 3) \text{ or } (0, -3, 2)$$

Worked Solutions

Quadratic Equations

Q140 MCQ

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$.

Correct option: A

Final answer

$(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Worked Solutions

Quadratic Equations

Q141 MCQ

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m = 5$, $x = -5, -1$.

Correct option: B

Final answer

$$m = 5, \quad x = -5, -1$$

Worked Solutions

Quadratic Equations

Q142 MCQ

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$.

Correct option: B

Final answer

$$(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2}) \text{ or } (8, -\frac{3}{2}, -\frac{1}{2})$$

Worked Solutions

Quadratic Equations

Q143 Written

If the ratio of the roots of $x^2 - 2x + m = 0$ is 2 : 3, find m and the roots.

Solution

- Let roots $2t, 3t$; $5t = 2 \rightarrow t = 2/5$; $m = 6t^2 = 24/25$
- Therefore the final answer is $m = \frac{24}{25}$, $x = \frac{4}{5}, \frac{6}{5}$.

Final answer

$$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$$

Q144 MCQ

If the ratio of the roots of $x^2 - 2x + m = 0$ is 2 : 3, find m and the roots Select the correct answer:

Solution

- Apply the same quadratic method as the given demand.
- The correct value is $m = \frac{24}{25}$, $x = \frac{4}{5}, \frac{6}{5}$.

Correct option: A

Final answer

$$m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$$

Worked Solutions

Quadratic Equations

Q145 Written

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots.

Solution

1. Roots α, α^2 ; $\alpha + \alpha^2 = 6 \Rightarrow \alpha = 2$ or -3 .
2. $k = \alpha^3 = 8$ or -27 .
3. Therefore the final answer is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Final answer

$$(k, x_1, x_2) = (8, 2, 4) \text{ or } (-27, -3, 9)$$

Worked Solutions

Quadratic Equations

Q146 MCQ

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$.

Correct option: B

Final answer

$$(k, x_1, x_2) = (8, 2, 4) \text{ or } (-27, -3, 9)$$

Worked Solutions

Quadratic Equations

Q147 Written

Factorise $2x^2 - 3x + 2$ over the complex numbers.

Solution

1. $D = 9 - 16 = -7$; roots $(3 \pm i\sqrt{7})/4$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$.

Final answer

$$2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$$

Q148 Written

Factorise $x^2 + 2x + 5$ over the complex numbers.

Solution

1. Roots $-1 \pm 2i$
2. Therefore the final answer is $(x + 1 - 2i)(x + 1 + 2i)$.

Final answer

$$(x + 1 - 2i)(x + 1 + 2i)$$

Worked Solutions

Quadratic Equations

Q149 Written

Factorise $2x^2 - 6x + 1$ over the complex numbers.

Solution

1. $D = 28$; roots $(3 \pm \sqrt{7})/2$
2. Keep leading coefficient 2
3. Therefore the final answer is $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$.

Final answer

$$2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$$

Q150 Written

Factorise $x^2 + 4x + 5$ over the complex numbers.

Solution

1. $D = 16 - 20 = -4$; roots $-2 \pm i$
2. Therefore the final answer is $(x + 2 - i)(x + 2 + i)$.

Final answer

$$(x + 2 - i)(x + 2 + i)$$

Worked Solutions

Quadratic Equations

Q151 **Written****Factorise $x^2 + 4x + 2$ over the complex numbers.****Solution**

1. *Roots* $-2 \pm \sqrt{2}$
2. Therefore the final answer is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Final answer

$$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$$

Q152 **Written****Factorise $x^2 - 4x + 7$ over the complex numbers.****Solution**

1. $D = 16 - 28 = -12$; *roots* $2 \pm i\sqrt{3}$
2. Therefore the final answer is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Final answer

$$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$$

Worked Solutions

Quadratic Equations

Q153 Written

Factorise $2x^2 - x + 1$ over the complex numbers.**Solution**

1. $D = 1 - 8 = -7$; roots $(1 \pm i\sqrt{7})/4$

2. Therefore the final answer is $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$.

Final answer

$$2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$$

Q154 Written

Factorise $3x^2 - 5x + 3$ over the complex numbers.**Solution**

1. $D = 25 - 36 = -11$; roots $(5 \pm i\sqrt{11})/6$

2. Therefore the final answer is $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$.

Final answer

$$3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$$

Worked Solutions

Quadratic Equations

Q155 Written

Factorise $2x^2 - 2x - 1$ over the complex numbers.**Solution**

1. $D = 4 + 8 = 12$; roots $(1 \pm \sqrt{3})/2$

2. Therefore the final answer is $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$.

Final answer

$$2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q156 MCQ

Factorise $2x^2 - 3x + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$.

Correct option: C**Final answer**

$$2\left(x - \frac{3 + i\sqrt{7}}{4}\right)\left(x - \frac{3 - i\sqrt{7}}{4}\right)$$

Worked Solutions

Quadratic Equations

Q157 MCQ

Factorise $x^2 + 2x + 5$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x + 1 - 2i)(x + 1 + 2i)$.

Correct option: A**Final answer**

$$(x + 1 - 2i)(x + 1 + 2i)$$

Worked Solutions

Quadratic Equations

Q158 MCQ

Factorise $2x^2 - 6x + 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$.

Correct option: D**Final answer**

$$2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q159 MCQ

Factorise $x^2 + 4x + 5$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x + 2 - i)(x + 2 + i)$.

Correct option: D**Final answer**

$$(x + 2 - i)(x + 2 + i)$$

Q160 MCQ

Factorise $x^2 + 4x + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$.

Correct option: B**Final answer**

$$(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q161 MCQ

Factorise $x^2 - 4x + 7$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$.

Correct option: A**Final answer**

$$(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$$

Worked Solutions

Quadratic Equations

Q162 MCQ

Factorise $2x^2 - x + 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$.**Correct option: C****Final answer**

$$2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$$

Worked Solutions

Quadratic Equations

Q163 MCQ

Factorise $3x^2 - 5x + 3$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$.

Correct option: A**Final answer**

$$3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$$

Worked Solutions

Quadratic Equations

Q164 MCQ

Factorise $2x^2 - 2x - 1$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$.

Correct option: B**Final answer**

$$2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$$

Worked Solutions

Quadratic Equations

Q165 **Written****Factorise $x^4 + 2x^2 - 3$ over the complex numbers.****Solution**

1. $x^4 + 2x^2 - 3 = (x^2 - 1)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$.

Final answer

$$(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$$

Q166 **Written****Factorise $x^4 - x^2 - 6$ over the complex numbers.****Solution**

1. Let $y = x^2$: $y^2 - y - 6 = (y - 3)(y + 2)$
2. Factor $x^2 - 3$ and $x^2 + 2$
3. Therefore the final answer is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$.

Final answer

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$$

Worked Solutions

Quadratic Equations

Q167 Written

Factorise $x^4 - 9$ over the complex numbers.

Solution

1. $x^4 - 9 = (x^2 - 3)(x^2 + 3)$
2. Factor each quadratic
3. Therefore the final answer is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Final answer

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$$

Q168 Written

Factorise $x^4 - 3x^2 + 2$ over the complex numbers.

Solution

1. Let $y = x^2$: $y^2 - 3y + 2 = (y - 1)(y - 2)$
2. Factor $x^2 - 1$ and $x^2 - 2$
3. Therefore the final answer is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Final answer

$$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q169 MCQ

Factorise $x^4 + 2x^2 - 3$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$.

Correct option: B**Final answer**

$$(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$$

Q170 MCQ

Factorise $x^4 - x^2 - 6$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$.

Correct option: B**Final answer**

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$$

Worked Solutions

Quadratic Equations

Q171 MCQ

Factorise $x^4 - 9$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$.

Correct option: D**Final answer**

$$(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$$

Q172 MCQ

Factorise $x^4 - 3x^2 + 2$ over the complex numbers Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$.

Correct option: C**Final answer**

$$(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$$

Worked Solutions

Quadratic Equations

Q173 Written

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots.

Solution

1. $S = 1, P = 1/2$
2. $x^2 - x + 1/2 = 0$; multiply by 2
3. Therefore the final answer is $2x^2 - 2x + 1 = 0$.

Final answer

$$2x^2 - 2x + 1 = 0$$

Q174 Written

Find two numbers whose sum is -1 and product is 2 .

Solution

1. They are roots of $x^2 + x + 2 = 0$
2. $D = -7$
3. Therefore the final answer is $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$.

Final answer

$$\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$$

Worked Solutions

Quadratic Equations

Q175 Written

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$.

Solution

1. $S = 6, P = 13$
2. $x^2 - Sx + P = 0$
3. Therefore the final answer is $x^2 - 6x + 13 = 0$.

Final answer

$$x^2 - 6x + 13 = 0$$

Q176 Written

Find two numbers whose sum and product are both 1.

Solution

1. Roots of $x^2 - x + 1 = 0$
2. $D = -3$
3. Therefore the final answer is $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$.

Final answer

$$\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$$

Worked Solutions

Quadratic Equations

Q177 **Written****Create a quadratic equation with roots -1 and 2 .****Solution**

1. $S = 1, P = -2$
2. Therefore the final answer is $x^2 - x - 2 = 0$.

Final answer

$$x^2 - x - 2 = 0$$

Q178 **Written****Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$.****Solution**

1. $S = 2, P = 17$
2. Therefore the final answer is $x^2 - 2x + 17 = 0$.

Final answer

$$x^2 - 2x + 17 = 0$$

Worked Solutions

Quadratic Equations

Q179 Written

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$.

Solution

1. $S = 10/3, P = 2$
2. $x^2 - (10/3)x + 2 = 0$; multiply by 3
3. Therefore the final answer is $3x^2 - 10x + 6 = 0$.

Final answer

$$3x^2 - 10x + 6 = 0$$

Q180 Written

Find two numbers whose sum is 2 and product is 5.

Solution

1. Roots of $x^2 - 2x + 5 = 0$
2. $D = -16$
3. Therefore the final answer is $1 + 2i, 1 - 2i$.

Final answer

$$1 + 2i, 1 - 2i$$

Worked Solutions

Quadratic Equations

Q181 **Written****Find two numbers whose sum is 4 and product is 2.****Solution**

1. Roots of $x^2 - 4x + 2 = 0$
2. Therefore the final answer is $2 + \sqrt{2}$, $2 - \sqrt{2}$.

Final answer

$$2 + \sqrt{2}, 2 - \sqrt{2}$$

Q182 **MCQ****Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots. Select the correct answer:****Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2x^2 - 2x + 1 = 0$.

Correct option: B**Final answer**

$$2x^2 - 2x + 1 = 0$$

Worked Solutions

Quadratic Equations

Q183 MCQ

Find two numbers whose sum is -1 and product is 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$.**Correct option: B****Final answer**

$$\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$$

Q184 MCQ

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x^2 - 6x + 13 = 0$.**Correct option: B****Final answer**

$$x^2 - 6x + 13 = 0$$

Worked Solutions

Quadratic Equations

Q185 MCQ

Find two numbers whose sum and product are both 1 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$.**Correct option:** A**Final answer**

$$\frac{1 + i\sqrt{3}}{2}, \frac{1 - i\sqrt{3}}{2}$$

Q186 MCQ

Create a quadratic equation with roots -1 and 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.

2. The correct value is $x^2 - x - 2 = 0$.**Correct option:** A**Final answer**

$$x^2 - x - 2 = 0$$

Worked Solutions

Quadratic Equations

Q187 MCQ

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 2x + 17 = 0$.

Correct option: D**Final answer**

$$x^2 - 2x + 17 = 0$$

Q188 MCQ

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$ Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $3x^2 - 10x + 6 = 0$.

Correct option: A**Final answer**

$$3x^2 - 10x + 6 = 0$$

Worked Solutions

Quadratic Equations

Q189 MCQ

Find two numbers whose sum is 2 and product is 5 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $1 + 2i$, $1 - 2i$.

Correct option: B**Final answer** $1 + 2i, 1 - 2i$

Q190 MCQ

Find two numbers whose sum is 4 and product is 2 Select the correct answer:**Solution**

1. Apply the same quadratic method as the given demand.
2. The correct value is $2 + \sqrt{2}$, $2 - \sqrt{2}$.

Correct option: B**Final answer** $2 + \sqrt{2}, 2 - \sqrt{2}$

Worked Solutions

Quadratic Equations

Q191 Written

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$.

Solution

1. $S = -2, P = -4$
2. $\text{Newsum} = S + 4 = 2; \text{newproduct} = P + 2S + 4 = -4$
3. $\text{Equation } x^2 - 2x - 4 = 0$
4. Therefore the final answer is $x^2 - 2x - 4 = 0$.

Final answer

$$x^2 - 2x - 4 = 0$$

Q192 MCQ

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 2x - 4 = 0$.

Correct option: C

Final answer

$$x^2 - 2x - 4 = 0$$

Worked Solutions

Quadratic Equations

Q193 Written

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 .

Solution

1. $S = -1, P = -3$
2. $\text{Newsum} S^2 - 2P = 1 + 6 = 7$; $\text{product} P^2 = 9$
3. Therefore the final answer is $x^2 - 7x + 9 = 0$.

Final answer

$$x^2 - 7x + 9 = 0$$

Q194 Written

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 .

Solution

1. $S = 2, P = 7$
2. $\text{Newsum} = S^2 - 2P = -10$; $\text{product} = P^2 = 49$
3. Therefore the final answer is $x^2 + 10x + 49 = 0$.

Final answer

$$x^2 + 10x + 49 = 0$$

Worked Solutions

Quadratic Equations

Q195 MCQ

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 7x + 9 = 0$.

Correct option: D**Final answer**

$$x^2 - 7x + 9 = 0$$

Q196 MCQ

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 + 10x + 49 = 0$.

Correct option: C**Final answer**

$$x^2 + 10x + 49 = 0$$

Worked Solutions

Quadratic Equations

Q197 Written

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

Solution

1. $S = -3/2, P = 2$
2. $\text{Newsum} = S/P = -3/4$; $\text{product} = 1/P = 1/2$
3. $x^2 + (3/4)x + 1/2 = 0$; multiply by 4
4. Therefore the final answer is $4x^2 + 3x + 2 = 0$.

Final answer

$$4x^2 + 3x + 2 = 0$$

Q198 MCQ

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $4x^2 + 3x + 2 = 0$.

Correct option: A

Final answer

$$4x^2 + 3x + 2 = 0$$

Worked Solutions

Quadratic Equations

Q199 Written

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.

Solution

1. $S = -2, P = 3$
2. $\text{Newsum} = 4S = -8$
3. $\text{Newproduct} = 3(\alpha^2 + \beta^2) + 10P = 3(S^2 - 2P) + 10P = 24$
4. Therefore the final answer is $x^2 + 8x + 24 = 0$.

Final answer

$$x^2 + 8x + 24 = 0$$

Q200 MCQ

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 + 8x + 24 = 0$.

Correct option: D

Final answer

$$x^2 + 8x + 24 = 0$$

Worked Solutions

Quadratic Equations

Q201 Written

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m .

Solution

1. $D > 0 \rightarrow (m - 2)(m + 1) > 0 \rightarrow m < -1 \text{ or } m > 2$
2. $S = -2m > 0 \rightarrow m < 0$
3. $P = m + 2 > 0 \rightarrow m > -2$
4. Intersection : $-2 < m < -1$
5. Therefore the final answer is $-2 < m < -1$.

Final answer

$$-2 < m < -1$$

Worked Solutions

Quadratic Equations

Q202 Written

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m .

Solution

1. $D = 4(4 - m)(m + 1) > 0 \rightarrow -1 < m < 4$
2. $S = 2(1 - m) > 0 \rightarrow m < 1$
3. $P = (2m + 1)(m - 3) > 0 \rightarrow m < -1/2 \text{ or } m > 3$
4. Intersection : $-1 < m < -1/2$
5. Therefore the final answer is $-1 < m < -\frac{1}{2}$.

Final answer

$$-1 < m < -\frac{1}{2}$$

Worked Solutions

Quadratic Equations

Q203 MCQ

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-2 < m < -1$.

Correct option: A**Final answer**

$$-2 < m < -1$$

Q204 MCQ

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-1 < m < -\frac{1}{2}$.

Correct option: B**Final answer**

$$-1 < m < -\frac{1}{2}$$

Worked Solutions

Quadratic Equations

Q205 **Written**

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m .

Solution

1. $D > 0 : 4m^2 - 4(m + 6) > 0 \rightarrow (m - 3)(m + 2) > 0$
2. $S = 2m < 0 \rightarrow m < 0$
3. $P = m + 6 > 0 \rightarrow m > -6$
4. *Intersection* : $-6 < m < -2$
5. Therefore the final answer is $-6 < m < -2$.

Final answer

$$-6 < m < -2$$

Worked Solutions

Quadratic Equations

Q206 Written

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k .

Solution

1. $D = (k - 10)(k + 2) > 0 \rightarrow k < -2 \text{ or } k > 10$
2. $S = -k < 0 \rightarrow k > 0$
3. $P = 2k + 5 > 0 \rightarrow k > -5/2$
4. Intersection $k > 10$
5. Therefore the final answer is $k > 10$.

Final answer

$$k > 10$$

Worked Solutions

Quadratic Equations

Q207 MCQ

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-6 < m < -2$.

Correct option: C**Final answer**

$$-6 < m < -2$$

Q208 MCQ

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k > 10$.

Correct option: B**Final answer**

$$k > 10$$

Worked Solutions

Quadratic Equations

Q209 Written

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m .

Solution

1. Opposite signs if $P = -m - 4 < 0$
2. Therefore the final answer is $m > -4$.

Final answer

$$m > -4$$

Q210 Written

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k .

Solution

1. $P = -k - 3 < 0$
2. Therefore the final answer is $k > -3$.

Final answer

$$k > -3$$

Worked Solutions

Quadratic Equations

Q211 MCQ

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m > -4$.

Correct option: A

Final answer

$$m > -4$$

Q212 MCQ

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $k > -3$.

Correct option: D

Final answer

$$k > -3$$

Worked Solutions

Quadratic Equations

Q213 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = a^2 + 4a = a(a + 4)$
2. Classify sign intervals at -4 and 0
3. Therefore the final answer is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions

Quadratic Equations

Q214 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies.

Solution

1. $D = (k + 2)^2 - 4k^2 = -3k^2 + 4k + 4 = -3(k - 2)(k + 2/3)$
2. Classify D sign
3. Therefore the final answer is $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary.

Final answer

$-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q215 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies.

Solution

1. $D = 4(a - 1)^2 - 4(a^2 - 3a + 4) = 4(a - 3)$
2. Classify by sign of $a - 3$
3. Therefore the final answer is $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary.

Final answer

$a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Worked Solutions

Quadratic Equations

Q216 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$.

Solution

1. $\alpha\beta(\alpha + \beta) = PS = (3/2)(3) = 9/2$
2. Therefore the final answer is $\frac{9}{2}$.

Final answer

$$\frac{9}{2}$$

Worked Solutions

Quadratic Equations

Q217 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$.

Solution

1. $S^2 - 4P = 9 - 6 = 3$
2. Therefore the final answer is 3.

Final answer

3

Worked Solutions

Quadratic Equations

Q218 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. $S = 3, P = 3/2$
2. $S^3 - 3PS = 27 - 27/2 = 27/2$
3. Therefore the final answer is $\frac{27}{2}$.

Final answer

$$\frac{27}{2}$$

Worked Solutions

Quadratic Equations

Q219 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$.

Solution

1. $(\alpha^3 + \beta^3)/(\alpha\beta) = (27/2)/(3/2) = 9$
2. Therefore the final answer is 9.

Final answer

9

Worked Solutions

Quadratic Equations

Q220 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$.

Solution

1. *Subtract equations* : $(a - b)(x - c) = 0$, so *common root* $r = c$.
2. *Substitute* c : $c(c + a + b) = 0$; $c \neq 0 \rightarrow a + b + c = 0$.
3. *The other roots are* $bc/c = b$ and $ca/c = a$. *Their equation is* $x^2 - (a + b)x + ab = x^2 + cx + ab$.
4. Therefore the final answer is Common root c ; other roots a and b .

Final answer

Common root c ; other roots a and b .

Worked Solutions

Quadratic Equations

Q221 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model.

Solution

1. Let $y = x^2$: $6y^2 - 7y - 3 = (2y - 3)(3y + 1)$
2. $x^2 = 3/2$ or $-1/3$; hence $x = \pm\sqrt{6}/2$ or $\pm i\sqrt{3}/3$
3. For real displacement x , only $\pm\sqrt{6}/2$ are admissible
4. Therefore the final answer is $x = \pm\frac{\sqrt{6}}{2}$, $x = \pm\frac{i\sqrt{3}}{3}$; real: $\pm\sqrt{6}/2$.

Final answer

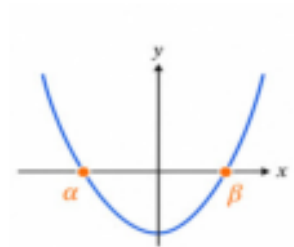
$$x = \pm\frac{\sqrt{6}}{2}, \quad x = \pm\frac{i\sqrt{3}}{3}; \text{ real: } \pm\sqrt{6}/2$$

Worked Solutions

Quadratic Equations

Q222 Challenge

In the opposite figure, α, β are the two roots. If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.



Solution

1. $\text{Newsum} = S/P; \text{newproduct} = 1/P$
2. Equation $x^2 - (S/P)x + 1/P = 0$; multiply by P
3. Therefore the final answer is For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$.

Final answer

For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$

Worked Solutions

Quadratic Equations

Q223 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive.

Solution

1. $D = (m + 1)^2 - 4(m + 4) > 0 \rightarrow m < -3 \text{ or } m > 5$
2. $S = -(m + 1) > 0 \rightarrow m < -1$
3. $P = m + 4 > 0 \rightarrow m > -4$
4. Intersection $(-4, -3)$
5. Therefore the final answer is $-4 < m < -3$.

Final answer

$$-4 < m < -3$$

Worked Solutions

Quadratic Equations

Q224 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative.

Solution

1. $D > 0 \rightarrow m < -3$ or $m > 5$
2. $S < 0 \rightarrow m > -1$
3. $P > 0 \rightarrow m > -4$
4. Intersection $m > 5$
5. Therefore the final answer is $m > 5$.

Final answer

$$m > 5$$

Worked Solutions

Quadratic Equations

Q225 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs.

Solution

1. Opposite signs if $P = m + 4 < 0$
2. Therefore the final answer is $m < -4$.

Final answer

$$m < -4$$

Worked Solutions

Quadratic Equations

Q226 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result.

Solution

1. Set $R = 0$. $D = 12 - 24 = -12$.
2. $x = (-2\sqrt{3} \pm i2\sqrt{3})/6 = (-\sqrt{3} \pm i\sqrt{3})/3$.
3. Because x is real, the complex pair is not admissible; no real equilibrium exists.
4. Therefore the final answer is $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}; \text{ no real equilibrium}$$

Worked Solutions

Quadratic Equations

Q227 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary.

Correct option: B

Final answer

$a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Worked Solutions

Quadratic Equations

Q228 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary.

Correct option: B

Final answer

$-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Worked Solutions

Quadratic Equations

Q229 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary.

Correct option: B

Final answer

$a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Worked Solutions

Quadratic Equations

Q230 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{9}{2}$.

Correct option: B

Final answer

$$\frac{9}{2}$$

Worked Solutions

Quadratic Equations

Q231 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 3.

Correct option: D

Final answer

3

Worked Solutions

Quadratic Equations

Q232 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $\frac{27}{2}$.

Correct option: B

Final answer

$$\frac{27}{2}$$

Worked Solutions

Quadratic Equations

Q233 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 9.

Correct option: A

Final answer

9

Worked Solutions

Quadratic Equations

Q234 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Common root c ; other roots a and b .

Correct option: C

Final answer

Common root c ; other roots a and b .

Worked Solutions

Quadratic Equations

Q235 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$.

Correct option: A

Final answer

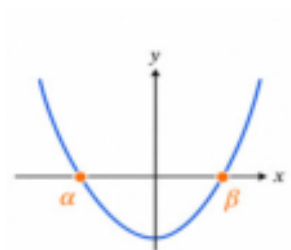
$$x = \pm \frac{\sqrt{6}}{2}, \quad x = \pm \frac{i\sqrt{3}}{3}; \text{ real: } \pm \sqrt{6}/2$$

Worked Solutions

Quadratic Equations

Q236 Challenge

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:



Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$.

Correct option: A

Final answer

For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$

Worked Solutions

Quadratic Equations

Q237 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $-4 < m < -3$.

Correct option: C

Final answer

$$-4 < m < -3$$

Worked Solutions

Quadratic Equations

Q238 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m > 5$.

Correct option: D

Final answer

$$m > 5$$

Worked Solutions

Quadratic Equations

Q239 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $m < -4$.

Correct option: D

Final answer

$$m < -4$$

Worked Solutions

Quadratic Equations

Q240 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result. Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium.

Correct option: D

Final answer

$$x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}; \text{ no real equilibrium}$$

Worked Solutions

Quadratic Equations

Quiz MCQ 1 Concept Quiz · MCQ

Solve $x^2 - 5x + 6 = 0$ **Select the correct answer:**

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 2, 3.

Correct option: B

Final answer

2, 3

Worked Solutions

Quadratic Equations

Quiz MCQ 2 Concept Quiz · MCQ

For $x^2 - 4x + 7 = 0$, determine the nature of the roots Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is Two distinct non-real complex roots.

Correct option: A

Final answer

Two distinct non-real complex roots

Worked Solutions

Quadratic Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

If α, β are roots of $x^2 - 3x + 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is 7.

Correct option: C

Final answer

7

Quiz MCQ 4 **Concept Quiz · MCQ**

Create a monic quadratic with roots $2 + i$ and $2 - i$ Select the correct answer:

Solution

1. Apply the same quadratic method as the given demand.
2. The correct value is $x^2 - 4x + 5 = 0$.

Correct option: A

Final answer

$x^2 - 4x + 5 = 0$

Worked Solutions

Quadratic Equations

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $2x^2 + 3x - 2 = 0$.**Solution**

1. Apply the relevant quadratic identity.
2. Final answer: $x = \frac{1}{2}, -2$

Final answer

$$x = \frac{1}{2}, -2$$

Quiz WRITTEN 6 Concept Quiz · Written**Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary roots.****Solution**

1. Apply the relevant quadratic identity.
2. Final answer: $-1 < m < 2$

Final answer

$$-1 < m < 2$$

Worked Solutions

Quadratic Equations

Quiz WRITTEN 7 Concept Quiz · Written

If α, β are roots of $x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Solution

1. Apply the relevant quadratic identity.
2. Final answer: 28

Final answer

28

Quiz WRITTEN 8 Concept Quiz · Written

If α, β are roots of $x^2 - 2x + 5 = 0$, form the quadratic with roots α^2, β^2 .

Solution

1. Apply the relevant quadratic identity.
2. Final answer: $x^2 + 6x + 25 = 0$

Final answer

$x^2 + 6x + 25 = 0$

Answer key

Use the question number shown on each page.

Q001 $x = \pm 2i\sqrt{2}$

Q002 $x = \pm 2i$

Q003 $x = \pm 7i$

Q004 $x = \pm \frac{2}{3}$

Q005 $x = \pm 2\sqrt{3}$

Q006 $x = \pm \frac{i\sqrt{3}}{2}$

Q007 A

Q008 A

Q009 C

Q010 C

Q011 B

Q012 B

Q013 $x = 3 \pm 2i\sqrt{2}$

Q014 $x = 2 \pm \sqrt{5}$

Q015 $x = -6 \pm i\sqrt{6}$

Q016 $x = -4 \pm 3i\sqrt{2}$

Q017 $x = \frac{3 \pm \sqrt{7}}{2}$

Q018 A

Q019 B

Q020 C

Q021 C

Q022 D

Q023 $x = -1, \frac{9}{2}$

Q024 $x = \frac{1}{2}, -3$

Q025 $x = 0, -\frac{7}{2}$

Q026 $x = \frac{3}{2}, -4$

Q027 $x = 3, 5$

Q028 A

Q029 D

Q030 D

Q031 A

Q032 A

Q033 $x = \frac{2 \pm \sqrt{7}}{3}$

Q034 B

Q035 $x = \frac{-\sqrt{6} \pm 4i}{2}$

Q036 $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

Q037 $x = \frac{-1 \pm i\sqrt{3}}{2}$

Q038 $x = \frac{5 \pm i\sqrt{11}}{6}$

Q039 $x = \sqrt{3} \pm i\sqrt{2}$

Q040 $x = \frac{-\sqrt{2} \pm i}{3}$

Q041 $x = \frac{1 \pm i\sqrt{7}}{2}$

Q042 $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

Q043 $x = 1 \pm \frac{i\sqrt{6}}{2}$

Q044 B

Q045 C

Q046 D

Q047 B

Q048 D

Q049 B

Q050 C

Q051 C

Q052 B

Q053 Two distinct imaginary (non-real complex) solutions

Q054 $-2 \leq m \leq 6$

Q055 $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Two distinct imaginary (non-real complex) solutions

Q057 Two distinct real solutions

Q058 One repeated real solution

Q059 Two distinct real solutions

Q060 One repeated real solution

Q061 Two distinct imaginary (non-real complex) solutions

Q062 One repeated real solution

Q063 Two distinct imaginary (non-real complex) solutions

Q064 Two distinct real solutions

Q065 A

Q066 D

Q067 C

Q068 C

Q069 A

Q070 A

Q071 C

Q072 C

Q073 B

Q074 A

Q075 A

Q076 A

Q077 $-1 < m < 2$

Q078 $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Q079 $a = 3$ or $a = -5$

Q080 $-18 < m < -2$

Q081 $0 \leq m \leq 3$

Q082 C

Q083 C

Q084 A

Q085 D

Q086 C

Q087 $\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$

Q088 $\alpha + \beta = -\frac{1}{2}, \alpha\beta = -\frac{7}{4}$

Q089 $S = -1, P = 2$

Q090 $S = \frac{7}{3}, P = -2$

Q091 C

Q092 A

Q093 B

Q094 A

Q095 12

Q096 $\frac{81}{2}$
 Q097 1
 Q098 -1
 Q099 -2
 Q100 -10
 Q101 $\frac{14}{3}$
 Q102 -10
 Q103 A
 Q104 B
 Q105 A
 Q106 B
 Q107 D
 Q108 B
 Q109 B
 Q110 D
 Q111 -4
 Q112 $\frac{2}{3}$
 Q113 $S = 0, \quad P = -\frac{3}{5}$
 Q114 $\frac{2}{3}$
 Q115 6
 Q116 C
 Q117 D
 Q118 D
 Q119 D
 Q120 D
 Q121 $\frac{5}{2}$
 Q122 $\frac{1}{2}$
 Q123 11
 Q124 $\frac{23}{3}$
 Q125 A

Q126 C
 Q127 D
 Q128 D
 Q129 $k = 6, \quad x = 2, 6$
 Q130 $m = 8, \quad x = -2, -4$
 Q131 $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$
 Q132 A
 Q133 A
 Q134 D
 Q135 $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$
 Q136 $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$
 Q137 $m = 5, \quad x = -5, -1$
 Q138 $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$
 Q139 D
 Q140 A
 Q141 B
 Q142 B
 Q143 $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$
 Q144 A
 Q145 $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$
 Q146 B
 Q147 $2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$
 Q148 $(x+1-2i)(x+1+2i)$
 Q149 $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$
 Q150 $(x+2-i)(x+2+i)$
 Q151 $(x+2-\sqrt{2})(x+2+\sqrt{2})$
 Q152 $(x-2-i\sqrt{3})(x-2+i\sqrt{3})$
 Q153 $2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$
 Q154 $3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$

Q155 $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$
 Q156 C
 Q157 A
 Q158 D
 Q159 D
 Q160 B
 Q161 A
 Q162 C
 Q163 A
 Q164 B
 Q165 $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$
 Q166 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$
 Q167 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{3})(x+i\sqrt{3})$
 Q168 $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$
 Q169 B
 Q170 B
 Q171 D
 Q172 C
 Q173 $2x^2 - 2x + 1 = 0$
 Q174 $\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$
 Q175 $x^2 - 6x + 13 = 0$
 Q176 $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
 Q177 $x^2 - x - 2 = 0$
 Q178 $x^2 - 2x + 17 = 0$
 Q179 $3x^2 - 10x + 6 = 0$
 Q180 $1+2i, 1-2i$
 Q181 $2+\sqrt{2}, 2-\sqrt{2}$
 Q182 B
 Q183 B
 Q184 B

Q185 A
 Q186 A
 Q187 D
 Q188 A
 Q189 B
 Q190 B
 Q191 $x^2 - 2x - 4 = 0$
 Q192 C
 Q193 $x^2 - 7x + 9 = 0$
 Q194 $x^2 + 10x + 49 = 0$
 Q195 D
 Q196 C
 Q197 $4x^2 + 3x + 2 = 0$
 Q198 A
 Q199 $x^2 + 8x + 24 = 0$
 Q200 D
 Q201 $-2 < m < -1$
 Q202 $-1 < m < -\frac{1}{2}$
 Q203 A
 Q204 B
 Q205 $-6 < m < -2$
 Q206 $k > 10$
 Q207 C
 Q208 B
 Q209 $m > -4$
 Q210 $k > -3$
 Q211 A
 Q212 D
 Q213 $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary
 Q214 $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary
 Q215 $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary
 Q216 $\frac{9}{2}$

Q217 3

Q218 $\frac{27}{2}$

Q219 9

Q220 Common root c ; other roots a and b .Q221 $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$ Q222 For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$ Q223 $-4 < m < -3$ Q224 $m > 5$ Q225 $m < -4$ Q226 $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Q227 B

Q228 B

Q229 B

Q230 B

Q231 D

Q232 B

Q233 A

Q234 C

Q235 A

Q236 A

Q237 C

Q238 D

Q239 D

Q240 D

Quiz MCQ 1 B

Quiz MCQ 2 A

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $x = \frac{1}{2}, -2$ Quiz WRITTEN 6 $-1 < m < 2$

Quiz WRITTEN 7 28

Quiz WRITTEN 8 $x^2 + 6x + 25 = 0$

Concept 3 | Polynomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Remainder

$$P(a) = R \text{ when divisor} = x - a$$

Factor

$$P(a) = 0 \Rightarrow (x - a) \mid P(x)$$

Roots

$$P(x) = a \prod (x - r_i)$$

Worked Solutions

Polynomial Theorems

Q001 **Written****Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$.****Solution**

1. $U_{\text{sex}} = -2$
2. $P(-2) = 2$
3. Therefore the final answer is 2.

Final answer

2

Q002 **Written****Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$.****Solution**

1. $U_{\text{sex}} = 1$
2. $P(1) = 6$
3. Therefore the final answer is 6.

Final answer

6

Worked Solutions

Polynomial Theorems

Q003 MCQ

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 2.

Correct option: C**Final answer**

2

Q004 MCQ

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 6.

Correct option: C**Final answer**

6

Worked Solutions

Polynomial Theorems

Q005 Written

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$.**Solution**

1. Use $x = 2/3$
2. Evaluate $P(2/3) = 8$
3. Therefore the final answer is 8.

Final answer

8

Q006 Written

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$.**Solution**

1. Use $x = 1/2$
2. $P(1/2) = -1/8$
3. Therefore the final answer is $-\frac{1}{8}$.

Final answer $-\frac{1}{8}$

Worked Solutions

Polynomial Theorems

Q007 **Written****Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$.****Solution**

1. Use $x = 2/3$
2. $P(2/3) = 0$
3. Therefore the final answer is 0.

Final answer

0

Q008 **Written****Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$.****Solution**

1. Use $x = -1/2$
2. $P(-1/2) = 4$
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions

Polynomial Theorems

Q009 MCQ

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 8.

Correct option: D**Final answer**

8

Q010 MCQ

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $-\frac{1}{8}$.

Correct option: A**Final answer** $-\frac{1}{8}$

Worked Solutions

Polynomial Theorems

Q011 MCQ

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 0.

Correct option: D**Final answer**

0

Q012 MCQ

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 4.

Correct option: A**Final answer**

4

Worked Solutions

Polynomial Theorems

Q013 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a .

Solution

1. $P(-3) = 4$
2. $9a - 35 = 4$, hence $a = 13/3$
3. Therefore the final answer is $a = 13/3$.

Final answer

$$a = 13/3$$

Q014 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a .

Solution

1. $P(3) = 1$
2. $9a + 37 = 1$, hence $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q015 Written

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a .

Solution

1. $P(-1) = -2$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Q016 Written

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8 , find a .

Solution

1. $P(-3) = 8$
2. $a = -16$
3. Therefore the final answer is $a = -16$.

Final answer

$$a = -16$$

Worked Solutions

Polynomial Theorems

Q017 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = 13/3$.

Correct option: B**Final answer**

$$a = 13/3$$

Q018 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -4$.

Correct option: B**Final answer**

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q019 MCQ

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -4$.

Correct option: C**Final answer**

$$a = -4$$

Q020 MCQ

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8 , find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -16$.

Correct option: A**Final answer**

$$a = -16$$

Worked Solutions

Polynomial Theorems

Q021 **Written**

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a .

Solution

1. $P(2) = 0$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Q022 **MCQ**

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -4$.

Correct option: A

Final answer

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q023 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Solution

1. Let $R = ax + b$
2. $a + b = 3$ and $-3a + b = -5$
3. $R = 2x + 1$
4. Therefore the final answer is $2x + 1$.

Final answer

$$2x + 1$$

Q024 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Solution

1. Let $R = ax + b$
2. $2a + b = 3$ and $-3a + b = -7$
3. $R = -10x + 13$
4. Therefore the final answer is $-10x + 13$.

Final answer

$$-10x + 13$$

Worked Solutions

Polynomial Theorems

Q025 **Written**

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$.

Solution

1. Let $R = ax + b$
2. $a + b = 5$ and $-2a + b = -1$
3. $R = 2x + 3$
4. Therefore the final answer is $2x + 3$.

Final answer

$$2x + 3$$

Q026 **Written**

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Solution

1. Let $R = ax + b$
2. $2a + b = 4$ and $-3a + b = -11$
3. $R = -15x + 19$
4. Therefore the final answer is $-15x + 19$.

Final answer

$$-15x + 19$$

Worked Solutions

Polynomial Theorems

Q027 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $2x + 1$.

Correct option: A

Final answer $2x + 1$

Q028 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $-10x + 13$.

Correct option: A

Final answer $-10x + 13$

Worked Solutions

Polynomial Theorems

Q029 MCQ

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $2x + 3$.

Correct option: A

Final answer

$$2x + 3$$

Q030 MCQ

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $-15x + 19$.

Correct option: A

Final answer

$$-15x + 19$$

Worked Solutions

Polynomial Theorems

Q031 Written

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$.

Solution

1. Let $R = ax + b$
2. Use shared roots -2 and 1 : $-2a + b = -8$ and $a + b = 1$
3. $R = 3x - 2$
4. Therefore the final answer is $3x - 2$.

Final answer

$$3x - 2$$

Q032 Written

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$.

Solution

1. Let $R = ax + b$
2. At -1 and 1 : $-a + b = -1$ and $a + b = 1$
3. $R = x$
4. Therefore the final answer is x .

Final answer

$$x$$

Worked Solutions

Polynomial Theorems

Q033 Written

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$.

Solution

1. Use shared roots -1 and 2
2. $-a + b = -4$ and $2a + b = 5$, so $R = 3x - 1$
3. Therefore the final answer is $3x - 1$.

Final answer

$$3x - 1$$

Q034 Written

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$.

Solution

1. Use shared roots 2 and 3
2. $2a + b = 2$ and $3a + b = 9$, so $R = 7x - 12$
3. Therefore the final answer is $7x - 12$.

Final answer

$$7x - 12$$

Worked Solutions

Polynomial Theorems

Q035 MCQ

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $3x - 2$.

Correct option: B

Final answer $3x - 2$

Q036 MCQ

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is x .

Correct option: B

Final answer x

Worked Solutions

Polynomial Theorems

Q037 MCQ

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $3x - 1$.

Correct option: A

Final answer $3x - 1$

Q038 MCQ

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $7x - 12$.

Correct option: D

Final answer $7x - 12$

Worked Solutions

Polynomial Theorems

Q039 Written

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$.

Solution

1. Let $R = ax^2 + bx + c$
2. $R(-1) = -6, R'(-1) = 1, R(2) = 6$
3. Solve to get $R = x^2 + 3x - 4$
4. Therefore the final answer is $x^2 + 3x - 4$.

Final answer

$$x^2 + 3x - 4$$

Q040 MCQ

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x^2 + 3x - 4$.

Correct option: A

Final answer

$$x^2 + 3x - 4$$

Worked Solutions

Polynomial Theorems

Q041 Written

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$.

Solution

1. Use -2 and coefficients 1,0,5,11
2. Bottom row 1,-2,9,-7
3. Therefore the final answer is $Q = x^2 - 2x + 9, R = -7$.

Final answer

$$Q = x^2 - 2x + 9, R = -7$$

Q042 MCQ

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 - 2x + 9, R = -7$.

Correct option: A

Final answer

$$Q = x^2 - 2x + 9, R = -7$$

Worked Solutions

Polynomial Theorems

Q043 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,0,5,-6
2. Bottom row 1,1,6,0
3. Therefore the final answer is $Q = x^2 + x + 6, R = 0$.

Final answer

$$Q = x^2 + x + 6, R = 0$$

Q044 Written

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$.

Solution

1. Use 3 and coefficients 3,-5,-3,-1
2. Bottom row 3,4,9,26
3. Therefore the final answer is $Q = 3x^2 + 4x + 9, R = 26$.

Final answer

$$Q = 3x^2 + 4x + 9, R = 26$$

Worked Solutions

Polynomial Theorems

Q045 Written

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$.

Solution

1. Use 2 and coefficients 1,0,-5,6
2. Bottom row 1,2,-1,4
3. Therefore the final answer is $Q = x^2 + 2x - 1, R = 4$.

Final answer

$$Q = x^2 + 2x - 1, R = 4$$

Q046 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,2,-1,-3
2. Bottom row 1,3,2,-1
3. Therefore the final answer is $Q = x^2 + 3x + 2, R = -1$.

Final answer

$$Q = x^2 + 3x + 2, R = -1$$

Worked Solutions

Polynomial Theorems

Q047 **Written**

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 4,-2,2
2. Bottom row 4,2,4
3. Therefore the final answer is $Q = 4x + 2, R = 4$.

Final answer

$$Q = 4x + 2, R = 4$$

Q048 **MCQ**

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 + x + 6, R = 0$.

Correct option: B

Final answer

$$Q = x^2 + x + 6, R = 0$$

Worked Solutions

Polynomial Theorems

Q049 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = 3x^2 + 4x + 9, R = 26$.

Correct option: D**Final answer**

$$Q = 3x^2 + 4x + 9, R = 26$$

Q050 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 + 2x - 1, R = 4$.

Correct option: D**Final answer**

$$Q = x^2 + 2x - 1, R = 4$$

Worked Solutions

Polynomial Theorems

Q051 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = x^2 + 3x + 2, R = -1$.

Correct option: A

Final answer

$$Q = x^2 + 3x + 2, R = -1$$

Q052 MCQ

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = 4x + 2, R = 4$.

Correct option: B

Final answer

$$Q = 4x + 2, R = 4$$

Worked Solutions

Polynomial Theorems

Q053 Written

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$.

Solution

1. Use -2 and coefficients 3,5,0,-2,-12
2. Bottom row 3,-1,2,-6,0
3. Therefore the final answer is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Final answer

$$Q = 3x^3 - x^2 + 2x - 6, R = 0$$

Q054 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Correct option: D

Final answer

$$Q = 3x^3 - x^2 + 2x - 6, R = 0$$

Worked Solutions

Polynomial Theorems

Q055 Written

Factorise $x^3 + 2x^2 - x - 2$.**Solution**

1. $P(1) = 0$; *divide by* $x - 1$
2. *Quotient* $x^2 + 3x + 2 = (x + 1)(x + 2)$
3. Therefore the final answer is $(x - 1)(x + 1)(x + 2)$.

Final answer

$$(x - 1)(x + 1)(x + 2)$$

Q056 Written

Factorise $x^3 - 2x^2 - 5x + 6$.**Solution**

1. $P(1) = 0$; *quotient* $x^2 - x - 6 = (x - 3)(x + 2)$
2. Completed factorisation
3. Therefore the final answer is $(x - 1)(x - 3)(x + 2)$.

Final answer

$$(x - 1)(x - 3)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q057 Written

Factorise $x^3 - x^2 - 16x - 20$.**Solution**

1. $P(-2) = 0$; divide by $x + 2$ twice
2. Remaining $x - 5$
3. Therefore the final answer is $(x + 2)^2(x - 5)$.

Final answer

$$(x + 2)^2(x - 5)$$

Q058 Written

Factorise $2x^3 - x^2 - 5x - 2$.**Solution**

1. $P(2) = 0$; quotient $2x^2 + 3x + 1$
2. Therefore the final answer is $(x - 2)(2x + 1)(x + 1)$.

Final answer

$$(x - 2)(2x + 1)(x + 1)$$

Worked Solutions

Polynomial Theorems

Q059 Written

Factorise $x^3 - 7x - 6$.**Solution**

1. Test divisor 3 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x - 3)(x + 1)(x + 2)$.

Final answer

$$(x - 3)(x + 1)(x + 2)$$

Q060 Written

Factorise $x^3 + 4x^2 + x - 6$.**Solution**

1. Test divisor 1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x - 1)(x + 2)(x + 3)$.

Final answer

$$(x - 1)(x + 2)(x + 3)$$

Worked Solutions

Polynomial Theorems

Q061 Written

Factorise $x^3 + 6x^2 + 11x + 6$.**Solution**

1. Test divisor -1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x + 1)(x + 2)(x + 3)$.

Final answer

$$(x + 1)(x + 2)(x + 3)$$

Q062 Written

Factorise $x^3 - 3x + 2$.**Solution**

1. Test divisor 1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 1)^2(x + 2)$.

Final answer

$$(x - 1)^2(x + 2)$$

Worked Solutions

Polynomial Theorems

Q063 Written

Factorise $x^3 + x^2 - 8x - 12$.**Solution**

1. Test divisor -2 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 3)(x + 2)^2$.

Final answer

$$(x - 3)(x + 2)^2$$

Q064 Written

Factorise $x^3 - 3x^2 - 9x - 5$.**Solution**

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 5)(x + 1)^2$.

Final answer

$$(x - 5)(x + 1)^2$$

Worked Solutions

Polynomial Theorems

Q065 Written

Factorise $2x^3 - 3x^2 - 3x + 2$.**Solution**

1. Test divisor 2 and divide
2. *Factorquotient* $2x^2 + x - 1$
3. Therefore the final answer is $(x - 2)(x + 1)(2x - 1)$.

Final answer

$$(x - 2)(x + 1)(2x - 1)$$

Q066 Written

Factorise $3x^3 + x^2 - 8x + 4$.**Solution**

1. Test divisor 1 and divide
2. *Factorquotient* $3x^2 + 4x - 4$
3. Therefore the final answer is $(x - 1)(x + 2)(3x - 2)$.

Final answer

$$(x - 1)(x + 2)(3x - 2)$$

Worked Solutions

Polynomial Theorems

Q067 Written

Factorise $2x^3 + 7x^2 + 8x + 3$.**Solution**

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x + 1)^2(2x + 3)$.

Final answer

$$(x + 1)^2(2x + 3)$$

Q068 MCQ

Factorise $x^3 + 2x^2 - x - 2$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 1)(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)(x + 1)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q069 MCQ

Factorise $x^3 - 2x^2 - 5x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x - 3)(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)(x - 3)(x + 2)$$

Q070 MCQ

Factorise $x^3 - x^2 - 16x - 20$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 2)^2(x - 5)$.

Correct option: C**Final answer**

$$(x + 2)^2(x - 5)$$

Worked Solutions

Polynomial Theorems

Q071 MCQ

Factorise $2x^3 - x^2 - 5x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(2x + 1)(x + 1)$.

Correct option: B**Final answer**

$$(x - 2)(2x + 1)(x + 1)$$

Q072 MCQ

Factorise $x^3 - 7x - 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x + 1)(x + 2)$.

Correct option: B**Final answer**

$$(x - 3)(x + 1)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q073 MCQ

Factorise $x^3 + 4x^2 + x - 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 2)(x + 3)$.

Correct option: C**Final answer**

$$(x - 1)(x + 2)(x + 3)$$

Q074 MCQ

Factorise $x^3 + 6x^2 + 11x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 1)(x + 2)(x + 3)$.

Correct option: B**Final answer**

$$(x + 1)(x + 2)(x + 3)$$

Worked Solutions

Polynomial Theorems

Q075 MCQ

Factorise $x^3 - 3x + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)^2(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)^2(x + 2)$$

Q076 MCQ

Factorise $x^3 + x^2 - 8x - 12$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x + 2)^2$.

Correct option: D**Final answer**

$$(x - 3)(x + 2)^2$$

Worked Solutions

Polynomial Theorems

Q077 MCQ

Factorise $x^3 - 3x^2 - 9x - 5$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 5)(x + 1)^2$.

Correct option: D**Final answer**

$$(x - 5)(x + 1)^2$$

Q078 MCQ

Factorise $2x^3 - 3x^2 - 3x + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(x + 1)(2x - 1)$.

Correct option: A**Final answer**

$$(x - 2)(x + 1)(2x - 1)$$

Worked Solutions

Polynomial Theorems

Q079 MCQ

Factorise $3x^3 + x^2 - 8x + 4$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 2)(3x - 2)$.

Correct option: C**Final answer**

$$(x - 1)(x + 2)(3x - 2)$$

Q080 MCQ

Factorise $2x^3 + 7x^2 + 8x + 3$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 1)^2(2x + 3)$.

Correct option: D**Final answer**

$$(x + 1)^2(2x + 3)$$

Worked Solutions

Polynomial Theorems

Q081 Written

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$.**Solution**

1. $P(2) = 0$; divide by $x - 2$ twice
2. Remaining quadratic $x^2 - 2x - 4$
3. Therefore the final answer is $(x - 2)^2(x^2 - 2x - 4)$.

Final answer

$$(x - 2)^2(x^2 - 2x - 4)$$

Q082 Written

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$.**Solution**

1. Test roots and divide successively
2. Four linear factors
3. Therefore the final answer is $(x - 3)(x - 2)(x + 1)(x + 7)$.

Final answer

$$(x - 3)(x - 2)(x + 1)(x + 7)$$

Worked Solutions

Polynomial Theorems

Q083 **Written****Factorise** $x^4 - 4x^3 + 8x^2 - 8x + 3$.**Solution**

1. $P(1) = 0$; *dividetwice*
2. *Remainingquadratic* $x^2 - 2x + 3$
3. Therefore the final answer is $(x - 1)^2(x^2 - 2x + 3)$.

Final answer

$$(x - 1)^2(x^2 - 2x + 3)$$

Q084 **Written****Factorise** $x^4 - 10x^3 + 35x^2 - 50x + 24$.**Solution**

1. Test roots 1,2,3,4 successively
2. Four linear factors
3. Therefore the final answer is $(x - 4)(x - 3)(x - 2)(x - 1)$.

Final answer

$$(x - 4)(x - 3)(x - 2)(x - 1)$$

Worked Solutions

Polynomial Theorems

Q085 Written

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$.**Solution**

1. Test roots 1 and -2 successively
2. Completed factorisation
3. Therefore the final answer is $(x - 5)(x - 1)^2(x + 2)$.

Final answer

$$(x - 5)(x - 1)^2(x + 2)$$

Q086 Written

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$.**Solution**

1. Test roots 2, -2, 4, -3/2
2. Completed factorisation
3. Therefore the final answer is $(x - 4)(x - 2)(x + 2)(2x + 3)$.

Final answer

$$(x - 4)(x - 2)(x + 2)(2x + 3)$$

Worked Solutions

Polynomial Theorems

Q087 **Written****Factorise** $x^4 - 4x^3 - 4x^2 + 13x - 6$.**Solution**

1. Test roots 1,-2
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x + 2)(x^2 - 5x + 3)$.

Final answer

$$(x - 1)(x + 2)(x^2 - 5x + 3)$$

Q088 **Written****Factorise** $x^4 - 4x^3 + 3x^2 + 2x - 6$.**Solution**

1. Test roots 3,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 3)(x + 1)(x^2 - 2x + 2)$.

Final answer

$$(x - 3)(x + 1)(x^2 - 2x + 2)$$

Worked Solutions

Polynomial Theorems

Q089 Written

Factorise $x^4 - x^3 + x^2 + x - 2$.**Solution**

1. Test roots 1,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x + 1)(x^2 - x + 2)$.

Final answer

$$(x - 1)(x + 1)(x^2 - x + 2)$$

Q090 MCQ

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)^2(x^2 - 2x - 4)$.

Correct option: D**Final answer**

$$(x - 2)^2(x^2 - 2x - 4)$$

Worked Solutions

Polynomial Theorems

Q091 MCQ

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x - 2)(x + 1)(x + 7)$.

Correct option: A**Final answer**

$$(x - 3)(x - 2)(x + 1)(x + 7)$$

Q092 MCQ

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)^2(x^2 - 2x + 3)$.

Correct option: D**Final answer**

$$(x - 1)^2(x^2 - 2x + 3)$$

Worked Solutions

Polynomial Theorems

Q093 MCQ

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 4)(x - 3)(x - 2)(x - 1)$.

Correct option: D**Final answer**

$$(x - 4)(x - 3)(x - 2)(x - 1)$$

Q094 MCQ

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 5)(x - 1)^2(x + 2)$.

Correct option: C**Final answer**

$$(x - 5)(x - 1)^2(x + 2)$$

Worked Solutions

Polynomial Theorems

Q095 MCQ

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 4)(x - 2)(x + 2)(2x + 3)$.

Correct option: D**Final answer**

$$(x - 4)(x - 2)(x + 2)(2x + 3)$$

Q096 MCQ

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 2)(x^2 - 5x + 3)$.

Correct option: A**Final answer**

$$(x - 1)(x + 2)(x^2 - 5x + 3)$$

Worked Solutions

Polynomial Theorems

Q097 MCQ

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 3)(x + 1)(x^2 - 2x + 2)$.

Correct option: B**Final answer**

$$(x - 3)(x + 1)(x^2 - 2x + 2)$$

Q098 MCQ

Factorise $x^4 - x^3 + x^2 + x - 2$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x + 1)(x^2 - x + 2)$.

Correct option: C**Final answer**

$$(x - 1)(x + 1)(x^2 - x + 2)$$

Worked Solutions

Polynomial Theorems

Q099 Written

Factorise $2x^3 - 3x^2 - x - 2$.**Solution**

1. $P(2) = 0$; quotient $2x^2 + x + 1$ has discriminant -7
2. Therefore the final answer is $(x - 2)(2x^2 + x + 1)$.

Final answer

$$(x - 2)(2x^2 + x + 1)$$

Q100 Written

Factorise $2x^3 - 7x^2 + 3x + 6$.**Solution**

1. Test divisor 2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 2)(2x^2 - 3x - 3)$.

Final answer

$$(x - 2)(2x^2 - 3x - 3)$$

Worked Solutions

Polynomial Theorems

Q101 Written

Factorise $x^3 + 4x^2 + 2x - 4$.**Solution**

1. Test divisor -2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x + 2)(x^2 + 2x - 2)$.

Final answer

$$(x + 2)(x^2 + 2x - 2)$$

Q102 Written

Factorise $x^3 - 3x^2 + 2$.**Solution**

1. Test divisor 1 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x^2 - 2x - 2)$.

Final answer

$$(x - 1)(x^2 - 2x - 2)$$

Worked Solutions

Polynomial Theorems

Q103 MCQ

Factorise $2x^3 - 3x^2 - x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(2x^2 + x + 1)$.

Correct option: C**Final answer**

$$(x - 2)(2x^2 + x + 1)$$

Q104 MCQ

Factorise $2x^3 - 7x^2 + 3x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 2)(2x^2 - 3x - 3)$.

Correct option: B**Final answer**

$$(x - 2)(2x^2 - 3x - 3)$$

Worked Solutions

Polynomial Theorems

Q105 MCQ

Factorise $x^3 + 4x^2 + 2x - 4$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x + 2)(x^2 + 2x - 2)$.

Correct option: B**Final answer**

$$(x + 2)(x^2 + 2x - 2)$$

Q106 MCQ

Factorise $x^3 - 3x^2 + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $(x - 1)(x^2 - 2x - 2)$.

Correct option: D**Final answer**

$$(x - 1)(x^2 - 2x - 2)$$

Worked Solutions

Polynomial Theorems

Q107 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$.

Solution

1. *Substitute* $x = -1$
2. $(-1)^{57} + 57 = 56$
3. Therefore the final answer is 56.

Final answer

56

Worked Solutions

Polynomial Theorems

Q108 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial.

Solution

1. Use the divisor zeros: $P(-1/2) = 0$ and $P(-2) = 0$.
2. Hence $a = 10$ for $2x + 1$, and $a = -11$ for $x + 2$.
3. Each linear factor corresponds to a polynomial root.

Final answer

$$a = 10 \quad \text{for } 2x + 1$$

$$a = -11 \quad \text{for } x + 2$$

linear factors $\leftrightarrow$ roots

Worked Solutions

Polynomial Theorems

Q109 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Solution

1. $a + b = 0$ and $-3a + b = -4$
2. $R = x - 1$
3. Therefore the final answer is $x - 1$.

Final answer $x - 1$

Worked Solutions

Polynomial Theorems

Q110 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts.

Solution

1. $P = (2x - 1)^2(x^2 - 3)$
2. Remaining roots are $+ / - \sqrt{3}$
3. Therefore the final answer is $x = \sqrt{3}, -\sqrt{3}$.

Final answer

$$x = \sqrt{3}, -\sqrt{3}$$

Worked Solutions

Polynomial Theorems

Q111 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b .

Solution

1. $P(1) = 0$ gives $a + b = -10$
2. $P(-2) = 0$ gives $2a - b = -26$
3. $a = -12, b = 2$
4. Therefore the final answer is $a = -12, b = 2$.

Final answer

$$a = -12, b = 2$$

Worked Solutions

Polynomial Theorems

Q112 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$.

Solution

1. $P = (x - 1)(x + 2)(x^2 - 4x - 6)$
2. Solve quadratic : $x = 2 \pm \sqrt{10}$
3. All four roots satisfy $P = 0$
4. Therefore the final answer is $x = 1, -2, 2 \pm \sqrt{10}$.

Final answer

$$x = 1, -2, 2 \pm \sqrt{10}$$

Worked Solutions

Polynomial Theorems

Q113 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 56.

Correct option: B

Final answer

56

Worked Solutions

Polynomial Theorems

Q114 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.

$$a = 10 \quad \text{for } 2x + 1$$

2. The correct value is $a = -11$ for $x + 2$.

linear factors $\leftrightarrow$ roots

Correct option: D

Final answer

$$a = 10 \quad \text{for } 2x + 1$$

$$a = -11 \quad \text{for } x + 2$$

linear factors $\leftrightarrow$ roots

Worked Solutions

Polynomial Theorems

Q115 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x - 1$.

Correct option: C

Final answer

$x - 1$

Worked Solutions

Polynomial Theorems

Q116 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x = \sqrt{3}, -\sqrt{3}$.

Correct option: A

Final answer

$$x = \sqrt{3}, -\sqrt{3}$$

Worked Solutions

Polynomial Theorems

Q117 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b . Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $a = -12, b = 2$.

Correct option: C

Final answer

$$a = -12, b = 2$$

Worked Solutions

Polynomial Theorems

Q118 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $x = 1, -2, 2 \pm \sqrt{10}$.

Correct option: A

Final answer

$$x = 1, -2, 2 \pm \sqrt{10}$$

Worked Solutions

Polynomial Theorems

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Find the remainder when $P(x) = x^3 - 2x + 5$ is divided by $x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is 9.

Correct option: C

Final answer

9

Quiz MCQ 2 [Concept Quiz · MCQ](#)

If $P(x)$ is divisible by $x + 1$, which statement is true? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $P(-1) = 0$.

Correct option: C

Final answer

$P(-1) = 0$

Worked Solutions

Polynomial Theorems

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The remainder on division by a quadratic divisor has the form $ax + b$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is $ax + b$.

Correct option: B

Final answer

$$ax + b$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

Which method records quotient and remainder for division by $x - a$? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the given demand.
2. The correct value is *Syntheticdivision*.

Correct option: C

Final answer

Syntheticdivision

Worked Solutions

Polynomial Theorems

Quiz WRITTEN 5 Concept Quiz · Written

Find the remainder when $2x^3 + 3x^2 - x + 4$ is divided by $2x + 1$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: 4

Final answer

4

Quiz WRITTEN 6 Concept Quiz · Written

Find the remainder $R = ax + b$ when $P(x)$ has values $P(1) = 3$ and $P(-2) = -6$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: $R = 3x$

Final answer

$R = 3x$

Worked Solutions

Polynomial Theorems

Quiz WRITTEN 7 Concept Quiz · Written**Factorise** $x^3 - 6x^2 + 11x - 6$.**Solution**

1. Apply the relevant polynomial theorem.
2. Final answer: $(x - 1)(x - 2)(x - 3)$

Final answer

$$(x - 1)(x - 2)(x - 3)$$

Quiz WRITTEN 8 Concept Quiz · Written**If** $x - 1$ **and** $x + 2$ **are factors of** $x^4 - 3x^3 + ax^2 + bx + 12$, **find** a, b .**Solution**

1. Apply the relevant polynomial theorem.
2. Final answer: $a = -12, b = 2$

Final answer

$$a = -12, b = 2$$

Answer key

Use the question number shown on each page.

Q001 2	Q027 A	Q053 $Q = 3x^3 - x^2 + 2x - 6, R = 0$	Q079 C
Q002 6	Q028 A	Q054 D	Q080 D
Q003 C	Q029 A	Q055 $(x - 1)(x + 1)(x + 2)$	Q081 $(x - 2)^2(x^2 - 2x - 4)$
Q004 C	Q030 A	Q056 $(x - 1)(x - 3)(x + 2)$	Q082 $(x - 3)(x - 2)(x + 1)(x + 7)$
Q005 8	Q031 $3x - 2$	Q057 $(x + 2)^2(x - 5)$	Q083 $(x - 1)^2(x^2 - 2x + 3)$
Q006 $-\frac{1}{8}$	Q032 x	Q058 $(x - 2)(2x + 1)(x + 1)$	Q084 $(x - 4)(x - 3)(x - 2)(x - 1)$
Q007 0	Q033 $3x - 1$	Q059 $(x - 3)(x + 1)(x + 2)$	Q085 $(x - 5)(x - 1)^2(x + 2)$
Q008 4	Q034 $7x - 12$	Q060 $(x - 1)(x + 2)(x + 3)$	Q086 $(x - 4)(x - 2)(x + 2)(2x + 3)$
Q009 D	Q035 B	Q061 $(x + 1)(x + 2)(x + 3)$	Q087 $(x - 1)(x + 2)(x^2 - 5x + 3)$
Q010 A	Q036 B	Q062 $(x - 1)^2(x + 2)$	Q088 $(x - 3)(x + 1)(x^2 - 2x + 2)$
Q011 D	Q037 A	Q063 $(x - 3)(x + 2)^2$	Q089 $(x - 1)(x + 1)(x^2 - x + 2)$
Q012 A	Q038 D	Q064 $(x - 5)(x + 1)^2$	Q090 D
Q013 $a = 13/3$	Q039 $x^2 + 3x - 4$	Q065 $(x - 2)(x + 1)(2x - 1)$	Q091 A
Q014 $a = -4$	Q040 A	Q066 $(x - 1)(x + 2)(3x - 2)$	Q092 D
Q015 $a = -4$	Q041 $Q = x^2 - 2x + 9, R = -7$	Q067 $(x + 1)^2(2x + 3)$	Q093 D
Q016 $a = -16$	Q042 A	Q068 D	Q094 C
Q017 B	Q043 $Q = x^2 + x + 6, R = 0$	Q069 D	Q095 D
Q018 B	Q044 $Q = 3x^2 + 4x + 9, R = 26$	Q070 C	Q096 A
Q019 C	Q045 $Q = x^2 + 2x - 1, R = 4$	Q071 B	Q097 B
Q020 A	Q046 $Q = x^2 + 3x + 2, R = -1$	Q072 B	Q098 C
Q021 $a = -4$	Q047 $Q = 4x + 2, R = 4$	Q073 C	Q099 $(x - 2)(2x^2 + x + 1)$
Q022 A	Q048 B	Q074 B	Q100 $(x - 2)(2x^2 - 3x - 3)$
Q023 $2x + 1$	Q049 D	Q075 D	Q101 $(x + 2)(x^2 + 2x - 2)$
Q024 $-10x + 13$	Q050 D	Q076 D	Q102 $(x - 1)(x^2 - 2x - 2)$
Q025 $2x + 3$	Q051 A	Q077 D	Q103 C
Q026 $-15x + 19$	Q052 B	Q078 A	Q104 B

Q105 B**Q106** D**Q107** 56

$$a = 10 \quad \text{for } 2x + 1$$

Q108 $a = -11$ for $x + 2$ linear factors $\leftrightarrow$ roots**Q109** $x - 1$

Q110 $x = \sqrt{3}, -\sqrt{3}$

Q111 $a = -12, b = 2$

Q112 $x = 1, -2, 2 \pm \sqrt{10}$

Q113 B**Q114** D**Q115** C**Q116** A**Q117** C**Q118** A**Quiz MCQ 1** C**Quiz MCQ 2** C**Quiz MCQ 3** B**Quiz MCQ 4** C**Quiz WRITTEN 5** 4**Quiz WRITTEN 6** $R = 3x$ **Quiz WRITTEN 7** $(x - 1)(x - 2)(x - 3)$ **Quiz WRITTEN 8** $a = -12, b = 2$

Concept 4 | Higher-Degree Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Factor theorem

$$P(a) = 0 \Rightarrow x - a \text{ is a factor}$$

Degree

$$\deg(PQ) = \deg P + \deg Q$$

Conjugate roots

$$a + bi \Rightarrow a - bi$$

Worked Solutions

Higher-Degree Equations

Q001 Written

$$x^3 - 8 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Q002 Written

$$x^3 = -1$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$.

Final answer

$$x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q003 Written

$$x^3 = 27$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$.

Final answer

$$x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$$

Q004 Written

$$x^3 = -125$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$.

Final answer

$$x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q005 Written

$$8x^3 = 1$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$.

Final answer

$$x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$$

Q006 MCQ

$$x^3 - 8 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Correct option: B

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Worked Solutions

Higher-Degree Equations

Q007 MCQ

 $x^3 = -1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$.

Correct option: D

Final answer

$$x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$$

Q008 MCQ

 $x^3 = 27$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$.

Correct option: B

Final answer

$$x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$$

Worked Solutions

Higher-Degree Equations

Q009 MCQ

 $x^3 = -125$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$.

Correct option: A

Final answer

$$x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$$

Q010 MCQ

 $8x^3 = 1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$.

Correct option: C

Final answer

$$x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$$

Worked Solutions

Higher-Degree Equations

Q011 Written

$$x^3 - 2x^2 - 5x + 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 1, 3$.

Final answer

$$x = -2, 1, 3$$

Q012 Written

$$3x^3 + x^2 - 8x + 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, 2/3, 1$.

Final answer

$$x = -2, 2/3, 1$$

Worked Solutions

Higher-Degree Equations

Q013 Written

$$2x^3 - 3x^2 - x - 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$.

Final answer

$$x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$$

Q014 Written

$$x^3 - x^2 - 16x - 20 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2(\text{repeated}), 5$.

Final answer

$$x = -2(\text{repeated}), 5$$

Worked Solutions

Higher-Degree Equations

Q015 Written

$$x^3 + 6x^2 + 11x + 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, -1$.

Final answer

$$x = -3, -2, -1$$

Q016 Written

$$x^3 + 3x^2 - 4x - 12 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, -2, 2$.

Final answer

$$x = -3, -2, 2$$

Worked Solutions

Higher-Degree Equations

Q017 Written

$$3x^3 - 2x^2 - 3x + 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 2/3, 1$.

Final answer

$$x = -1, 2/3, 1$$

Q018 Written

$$x^3 - x^2 - 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$.

Final answer

$$x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q019 Written

$$x^3 + 4x^2 + 2x - 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Final answer

$$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$$

Q020 Written

$$x^3 - 4x^2 + 9x - 10 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2, 1 - 2i, 1 + 2i$.

Final answer

$$x = 2, 1 - 2i, 1 + 2i$$

Worked Solutions

Higher-Degree Equations

Q021 Written

$$x^3 + x^2 - 8x - 12 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2$ (repeated), 3.

Final answer

$$x = -2(\text{repeated}), 3$$

Q022 Written

$$2x^3 + 7x^2 + 8x + 3 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1$ (repeated), $-3/2$.

Final answer

$$x = -1(\text{repeated}), -3/2$$

Worked Solutions

Higher-Degree Equations

Q023 Written

$$x^3 + 6x^2 + 12x + 8 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2$ (triple).

Final answer

$$x = -2(\text{triple})$$

Q024 MCQ

$$x^3 - 2x^2 - 5x + 6 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, 1, 3$.

Correct option: C

Final answer

$$x = -2, 1, 3$$

Worked Solutions

Higher-Degree Equations

Q025 MCQ

 $3x^3 + x^2 - 8x + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, 2/3, 1$.

Correct option: D

Final answer

$$x = -2, 2/3, 1$$

Q026 MCQ

 $2x^3 - 3x^2 - x - 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$.

Correct option: D

Final answer

$$x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$$

Worked Solutions

Higher-Degree Equations

Q027 MCQ

 $x^3 - x^2 - 16x - 20 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2$ (repeated), 5.

Correct option: B

Final answer $x = -2$ (repeated), 5

Q028 MCQ

 $x^3 + 6x^2 + 11x + 6 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, -2, -1$.

Correct option: C

Final answer $x = -3, -2, -1$

Worked Solutions

Higher-Degree Equations

Q029 MCQ

 $x^3 + 3x^2 - 4x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, -2, 2$.

Correct option: D

Final answer

$$x = -3, -2, 2$$

Q030 MCQ

 $3x^3 - 2x^2 - 3x + 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, 2/3, 1$.

Correct option: D

Final answer

$$x = -1, 2/3, 1$$

Worked Solutions

Higher-Degree Equations

Q031 MCQ

 $x^3 - x^2 - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$.

Correct option: A

Final answer

$$x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q032 MCQ

 $x^3 + 4x^2 + 2x - 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$.

Correct option: D

Final answer

$$x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$$

Worked Solutions

Higher-Degree Equations

Q033 MCQ

 $x^3 - 4x^2 + 9x - 10 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, 1 - 2i, 1 + 2i$.

Correct option: D

Final answer

$$x = 2, 1 - 2i, 1 + 2i$$

Q034 MCQ

 $x^3 + x^2 - 8x - 12 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2(\text{repeated}), 3$.

Correct option: D

Final answer

$$x = -2(\text{repeated}), 3$$

Worked Solutions

Higher-Degree Equations

Q035 MCQ

 $2x^3 + 7x^2 + 8x + 3 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1$ (repeated), $-3/2$.

Correct option: B**Final answer**

$$x = -1(\text{repeated}), -3/2$$

Q036 MCQ

 $x^3 + 6x^2 + 12x + 8 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2$ (triple).

Correct option: D**Final answer**

$$x = -2(\text{triple})$$

Worked Solutions

Higher-Degree Equations

Q037 Written

$$x^4 - 7x^2 - 18 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$.

Final answer

$$x = -3, 3, -\sqrt{2}i, \sqrt{2}i$$

Q038 Written

$$x^4 - 5x^2 + 4 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -2, -1, 1, 2$.

Final answer

$$x = -2, -1, 1, 2$$

Worked Solutions

Higher-Degree Equations

Q039 Written

$$x^4 - 5x^2 - 36 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -2i, 2i$.

Final answer

$$x = -3, 3, -2i, 2i$$

Q040 Written

$$x^4 - 6x^2 - 27 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$.

Final answer

$$x = -3, 3, -\sqrt{3}i, \sqrt{3}i$$

Worked Solutions

Higher-Degree Equations

Q041 Written

$$x^4 - x^2 - 2 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -\sqrt{2}, \sqrt{2}, -i, i$.

Final answer

$$x = -\sqrt{2}, \sqrt{2}, -i, i$$

Q042 MCQ

$$x^4 - 7x^2 - 18 = 0$$
 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$.

Correct option: D

Final answer

$$x = -3, 3, -\sqrt{2}i, \sqrt{2}i$$

Worked Solutions

Higher-Degree Equations

Q043 MCQ

 $x^4 - 5x^2 + 4 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -2, -1, 1, 2$.

Correct option: D

Final answer

$$x = -2, -1, 1, 2$$

Q044 MCQ

 $x^4 - 5x^2 - 36 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, 3, -2i, 2i$.

Correct option: C

Final answer

$$x = -3, 3, -2i, 2i$$

Worked Solutions

Higher-Degree Equations

Q045 MCQ

 $x^4 - 6x^2 - 27 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$.

Correct option: C

Final answer

$$x = -3, 3, -\sqrt{3}i, \sqrt{3}i$$

Q046 MCQ

 $x^4 - x^2 - 2 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -\sqrt{2}, \sqrt{2}, -i, i$.

Correct option: B

Final answer

$$x = -\sqrt{2}, \sqrt{2}, -i, i$$

Worked Solutions

Higher-Degree Equations

Q047 Written

$$x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$.

Final answer

$$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$$

Q048 Written

$$x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3, 4$.

Final answer

$$x = 1, 2, 3, 4$$

Worked Solutions

Higher-Degree Equations

Q049 Written

$$x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = 1, 2, 3(\text{repeated})$.

Final answer

$$x = 1, 2, 3(\text{repeated})$$

Q050 Written

$$x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$$

Solution

1. Factor the polynomial (use a cubic factor or set $u = x^2$ for a biquadratic).
2. Solve each linear or quadratic factor and include the complex roots.
3. Therefore the final answer is $x = -1, 3, 1 - i, 1 + i$.

Final answer

$$x = -1, 3, 1 - i, 1 + i$$

Worked Solutions

Higher-Degree Equations

Q051 MCQ

 $x^4 - 6x^3 + 8x^2 + 8x - 16 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2$ (repeated), $1 - \sqrt{5}$, $1 + \sqrt{5}$.

Correct option: B

Final answer

$$x = 2(\text{repeated}), 1 - \sqrt{5}, 1 + \sqrt{5}$$

Worked Solutions

Higher-Degree Equations

Q052 MCQ

 $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 1, 2, 3, 4$.

Correct option: A

Final answer $x = 1, 2, 3, 4$

Q053 MCQ

 $x^4 - 9x^3 + 29x^2 - 39x + 18 = 0$ Select the correct answer:**Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 1, 2, 3(\text{repeated})$.

Correct option: C

Final answer $x = 1, 2, 3(\text{repeated})$

Worked Solutions

Higher-Degree Equations

Q054 MCQ

 $x^4 - 4x^3 + 3x^2 + 2x - 6 = 0$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, 3, 1 - i, 1 + i$.

Correct option: C

Final answer

$$x = -1, 3, 1 - i, 1 + i$$

Q055 Written

 $x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 1, b = -6$; other roots: -2 .

Final answer

$$a = 1, b = -6; \text{ other roots: } -2$$

Worked Solutions

Higher-Degree Equations

Q056 Written

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a, b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 2, b = 2$; other roots: 1.

Final answer

$a = 2, b = 2$; other roots: 1

Q057 Written

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a, b and the other root.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -7, b = 14$; other roots: 4.

Final answer

$a = -7, b = 14$; other roots: 4

Worked Solutions

Higher-Degree Equations

Q058 MCQ

$x^3 + 4x^2 + ax + b = 0$ has roots -3 and 1; find a,b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 1, b = -6$; other roots: -2 .

Correct option: C

Final answer

$a = 1, b = -6$; other roots: -2

Worked Solutions

Higher-Degree Equations

Q059 MCQ

$x^3 - ax^2 - x + b = 0$ has roots -1 and 2; find a, b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 2, b = 2$; other roots: 1.

Correct option: B

Final answer

$a = 2, b = 2$; other roots: 1

Worked Solutions

Higher-Degree Equations

Q060 MCQ

$x^3 + ax^2 + bx - 8 = 0$ has roots 1 and 2; find a, b and the other root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -7, b = 14$; other roots: 4.

Correct option: C

Final answer

$a = -7, b = 14$; other roots: 4

Worked Solutions

Higher-Degree Equations

Q061 Written

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -3, b = 1$; other roots: $-1, 2 - i$.

Final answer

$a = -3, b = 1$; other roots: $-1, 2 - i$

Q062 Written

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 6$; other roots: $1 - i, 2$.

Final answer

$a = -4, b = 6$; other roots: $1 - i, 2$

Worked Solutions

Higher-Degree Equations

Q063 Written

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -4, b = 14$; other roots: $1 + 3i, 2$.

Final answer

$a = -4, b = 14$; other roots: $1 + 3i, 2$

Q064 Written

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = 0, b = 1$; other roots: $1 - 2i, -2$.

Final answer

$a = 0, b = 1$; other roots: $1 - 2i, -2$

Worked Solutions

Higher-Degree Equations

Q065 MCQ

$x^3 + ax^2 + bx + 5 = 0$ has root $2+i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -3, b = 1$; other roots: $-1, 2 - i$.

Correct option: A

Final answer

$a = -3, b = 1$; other roots: $-1, 2 - i$

Q066 MCQ

$x^3 + ax^2 + bx - 4 = 0$ has root $1+i$; find a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -4, b = 6$; other roots: $1 - i, 2$.

Correct option: B

Final answer

$a = -4, b = 6$; other roots: $1 - i, 2$

Worked Solutions

Higher-Degree Equations

Q067 MCQ

$x^3 + ax^2 + bx - 20 = 0$ has root $1 - 3i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -4, b = 14$; other roots: $1 + 3i, 2$.

Correct option: D

Final answer

$a = -4, b = 14$; other roots: $1 + 3i, 2$

Q068 MCQ

$x^3 + ax^2 + bx + 10 = 0$ has root $1 + 2i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 0, b = 1$; other roots: $1 - 2i, -2$.

Correct option: B

Final answer

$a = 0, b = 1$; other roots: $1 - 2i, -2$

Worked Solutions

Higher-Degree Equations

Q069 Written

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $k = 9$; other roots: 3, 4.

Final answer

$k = 9$; other roots: 3, 4

Q070 MCQ

$x^3 - kx^2 + (3k - 1)x - 24 = 0$ has root 2; find k and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $k = 9$; other roots: 3, 4.

Correct option: B

Final answer

$k = 9$; other roots: 3, 4

Worked Solutions

Higher-Degree Equations

Q071 Written

For an imaginary cube root of unity w , evaluate w^9 .**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q072 Written

For an imaginary cube root of unity w , evaluate w^{2010} .**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Worked Solutions

Higher-Degree Equations

Q073 **Written****For an imaginary cube root of unity w , evaluate w^3 .****Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer

1

Q074 **MCQ****For an imaginary cube root of unity w , evaluate w^9 Select the correct answer:****Solution**

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: A**Final answer**

1

Worked Solutions

Higher-Degree Equations

Q075 MCQ

For an imaginary cube root of unity w , evaluate w^{2010} Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: B**Final answer**

1

Q076 MCQ

For an imaginary cube root of unity w , evaluate w^3 Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: D**Final answer**

1

Worked Solutions

Higher-Degree Equations

Q077 Written

For an imaginary cube root of unity w , evaluate $w^4 + w^5$.**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer -1

Q078 Written

For an imaginary cube root of unity w , evaluate $w^2 + w$.**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer -1

Worked Solutions

Higher-Degree Equations

Q079 **Written**

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q080 **MCQ**

For an imaginary cube root of unity w , evaluate $w^4 + w^5$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: A

Final answer -1

Worked Solutions

Higher-Degree Equations

Q081 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: B**Final answer** -1

Q082 MCQ

For an imaginary cube root of unity w , evaluate $w^2 + w^4 + w^6$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 0 .

Correct option: D**Final answer** 0

Worked Solutions

Higher-Degree Equations

Q083 Written

For an imaginary cube root of unity w , evaluate $w + 1/w$.**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is -1 .

Final answer -1

Q084 Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.**Solution**

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 1.

Final answer 1

Worked Solutions

Higher-Degree Equations

Q085 Written

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 0.

Final answer

0

Q086 MCQ

For an imaginary cube root of unity w , evaluate $w + 1/w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: D

Final answer -1

Worked Solutions

Higher-Degree Equations

Q087 MCQ

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 1.

Correct option: C**Final answer**

1

Q088 MCQ

For an imaginary cube root of unity w , evaluate $1/w + 1/w^2 + 1$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 0.

Correct option: D**Final answer**

0

Worked Solutions

Higher-Degree Equations

Q089 Written

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -4$ or $9/4$.

Final answer

$a = -4$ or $9/4$

Q090 Written

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -3$ or 1 .

Final answer

$a = -3$ or 1

Worked Solutions

Higher-Degree Equations

Q091 **Written** $x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root.**Solution**

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 5$ or 9 .

Final answer $a = 5$ or 9 Q092 **Written** $x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root.**Solution**

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -6, -2$, or 10 .

Final answer $a = -6, -2$, or 10

Worked Solutions

Higher-Degree Equations

Q093 Written

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = -7, -3$, or 2 .

Final answer $a = -7, -3$, or 2

Q094 MCQ

$x^3 - 2x^2 + (a - 3)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -4$ or $9/4$.

Correct option: D

Final answer $a = -4$ or $9/4$

Worked Solutions

Higher-Degree Equations

Q095 MCQ

$x^3 - x^2 + (a - 2)x + a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -3$ or 1 .

Correct option: D

Final answer

$a = -3$ or 1

Q096 MCQ

$x^3 - 7x^2 + (a + 6)x - a = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 5$ or 9 .

Correct option: B

Final answer

$a = 5$ or 9

Worked Solutions

Higher-Degree Equations

Q097 MCQ

$x^3 - (a + 1)x^2 + (3a + 5)x - 2a - 5 = 0$ Find all real parameter values a producing a repeated root. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -6, -2$, or 10 .

Correct option: A**Final answer** $a = -6, -2$, or 10

Q098 MCQ

$x^3 + (2a - 1)x^2 - 3(a - 2)x + a - 6 = 0$ Find all real parameter values a producing a repeated root. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -7, -3$, or 2 .

Correct option: A**Final answer** $a = -7, -3$, or 2

Worked Solutions

Higher-Degree Equations

Q099 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots.

Solution

1. Use the known root and its conjugate when coefficients are real.
2. Compare coefficients (or use Vieta) to determine the unknown coefficients and final root.
3. Therefore the final answer is $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$.

Final answer

$a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

Worked Solutions

Higher-Degree Equations

Q100 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion.

Solution

1. Use ($w^3 = 1$) and reduce powers modulo 3.
2. Use ($1 + w + w^2 = 0$) to simplify the remaining expression.
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions

Higher-Degree Equations

Q101 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root.

Solution

1. Factor a guaranteed linear factor or compute the cubic discriminant.
2. For the remaining quadratic impose ($D = 0$), then retain the real parameter values.
3. Therefore the final answer is $a = 2/3, 2$, or 6 .

Final answer

$a = 2/3, 2$, or 6

Worked Solutions

Higher-Degree Equations

Q102 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$.

Solution

1. Set inflow equal to outflow: $(x^4 + x^2 + x = x^3 + 2)$.
2. Move all terms to one side.
3. Therefore the final answer is $x^4 - x^3 + x^2 + x - 2 = 0$.

Final answer

$$x^4 - x^3 + x^2 + x - 2 = 0$$

Worked Solutions

Higher-Degree Equations

Q103 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ **for the stable pressure levels.**

Solution

1. Factor $(x^4 - x^3 + x^2 + x - 2)$ and solve the remaining quadratic.
2. Apply the stated physical restriction ($x \geq 0$) only after listing the algebraic roots.
3. Therefore the final answer is $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$.

Final answer

$$x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q104 Challenge

Classify the stable pressure solutions as real or complex.

Solution

1. Separate the real roots from the non-real conjugate pair.

Both types occur: real $x = -1, 1$

2. Therefore the final answer is non-real conjugates $x = (1 \pm \sqrt{7}i)/2$.

if $x \geq 0 : x = 1$

Final answer

Both types occur: real $x = -1, 1$

non-real conjugates $x = (1 \pm \sqrt{7}i)/2$

if $x \geq 0 : x = 1$

Worked Solutions

Higher-Degree Equations

Q105 Challenge

$x^3 + ax + b = 0$ has root $1 + \sqrt{2}i$; find real a, b and the other roots Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$.

Correct option: C

Final answer

$a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$

Worked Solutions

Higher-Degree Equations

Q106 Challenge

For an imaginary cube root of unity w , evaluate $(1 - w + w^2)(1 + w - w^2)$ without direct expansion. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is 4.

Correct option: D

Final answer

4

Worked Solutions

Higher-Degree Equations

Q107 Challenge

$x^3 + (a - 1)x^2 + (a - 3)x - 2a + 3 = 0$ Find all real parameter values a producing a repeated root Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $a = 2/3, 2$, or 6 .

Correct option: D

Final answer

$a = 2/3, 2$, or 6

Worked Solutions

Higher-Degree Equations

Q108 Challenge

Write the balance equation: inflow $x^4 + x^2 + x$ equals outflow $x^3 + 2$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x^4 - x^3 + x^2 + x - 2 = 0$.

Correct option: A

Final answer

$$x^4 - x^3 + x^2 + x - 2 = 0$$

Worked Solutions

Higher-Degree Equations

Q109 Challenge

Solve $x^4 - x^3 + x^2 + x - 2 = 0$ **for the stable pressure levels Select the correct answer:**

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$.

Correct option: D

Final answer

$$x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$$

Worked Solutions

Higher-Degree Equations

Q110 Challenge

Classify the stable pressure solutions as real or complex Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.

Both types occur: real $x = -1, 1$

2. The correct value is non-real conjugates $x = (1 \pm \sqrt{7}i)/2$.

if $x \geq 0 : x = 1$

Correct option: B

Final answer

Both types occur: real $x = -1, 1$

non-real conjugates $x = (1 \pm \sqrt{7}i)/2$

if $x \geq 0 : x = 1$

Worked Solutions

Higher-Degree Equations

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $x^3 - 8 = 0$. Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$.

Correct option: D

Final answer

$$x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$$

Quiz MCQ 2 **Concept Quiz · MCQ**

Are $1 + 2i$ and $1 - 2i$ roots of $x^3 - 3x^2 + 4x - 2 = 0$? Which root must also occur? Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $1 - 2i$.

Correct option: D

Final answer

$$1 - 2i$$

Worked Solutions

Higher-Degree Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

For an imaginary cube root of unity w , evaluate $w^2 + w$ Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is -1 .

Correct option: B

Final answer

-1

Quiz MCQ 4 **Concept Quiz · MCQ**

For the remaining quadratic factor, a repeated root requires Select the correct answer:

Solution

1. Apply the same higher-degree equation strategy as the given demand.
2. The correct value is $D = 0$.

Correct option: B

Final answer

$D = 0$

Worked Solutions

Higher-Degree Equations

Quiz WRITTEN 5 Concept Quiz · Written

$$\text{Solve } x^4 - 5x^2 + 4 = 0.$$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $x = -2, -1, 1, 2$

Final answer

$$x = -2, -1, 1, 2$$

Quiz WRITTEN 6 Concept Quiz · Written

$$\text{For } x^3 + ax^2 + bx + 10 = 0 \text{ with root } 1 + 2i, \text{ find } a, b \text{ and the other roots.}$$

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = 0, b = 1$; other roots: $1 - 2i, -2$

Final answer

$$a = 0, b = 1; \text{ other roots: } 1 - 2i, -2$$

Worked Solutions

Higher-Degree Equations

Quiz WRITTEN 7 Concept Quiz · Written

For an imaginary cube root of unity w , evaluate $(1 + w)(1 + w^2)$.

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: 1

Final answer

1

Quiz WRITTEN 8 Concept Quiz · Written

Find all real a for $x^3 - 2x^2 + (a - 3)x + a = 0$ to have a repeated root.

Solution

1. Apply the relevant higher-degree equation strategy.
2. Final answer: $a = -4$ or $9/4$

Final answer

$a = -4$ or $9/4$

Answer key

Use the question number shown on each page.

Q001 $x = 2, -1 - \sqrt{3}i, -1 + \sqrt{3}i$	Q027 B	Q053 C	Q079 0
Q002 $x = -1, (1 - \sqrt{3}i)/2, (1 + \sqrt{3}i)/2$	Q028 C	Q054 C	Q080 A
Q003 $x = 3, (-3 - 3\sqrt{3}i)/2, (-3 + 3\sqrt{3}i)/2$	Q029 D	Q055 $a = 1, b = -6$; other roots: -2	Q081 B
Q004 $x = -5, (5 - 5\sqrt{3}i)/2, (5 + 5\sqrt{3}i)/2$	Q030 D	Q056 $a = 2, b = 2$; other roots: 1	Q082 D
Q005 $x = 1/2, (-1 - \sqrt{3}i)/4, (-1 + \sqrt{3}i)/4$	Q031 A	Q057 $a = -7, b = 14$; other roots: 4	Q083 -1
Q006 B	Q032 D	Q058 C	Q084 1
Q007 D	Q033 D	Q059 B	Q085 0
Q008 B	Q034 D	Q060 C	Q086 D
Q009 A	Q035 B	Q061 $a = -3, b = 1$; other roots: $-1, 2 - i$	Q087 C
Q010 C	Q036 D	Q062 $a = -4, b = 6$; other roots: $1 - i, 2$	Q088 D
Q011 $x = -2, 1, 3$	Q037 $x = -3, 3, -\sqrt{2}i, \sqrt{2}i$	Q063 $a = -4, b = 14$; other roots: $1 + 3i, 2$	Q089 $a = -4$ or $9/4$
Q012 $x = -2, 2/3, 1$	Q038 $x = -2, -1, 1, 2$	Q064 $a = 0, b = 1$; other roots: $1 - 2i, -2$	Q090 $a = -3$ or 1
Q013 $x = 2, (-1 - \sqrt{7}i)/4, (-1 + \sqrt{7}i)/4$	Q039 $x = -3, 3, -2i, 2i$	Q065 A	Q091 $a = 5$ or 9
Q014 $x = -2$ (repeated), 5	Q040 $x = -3, 3, -\sqrt{3}i, \sqrt{3}i$	Q066 B	Q092 $a = -6, -2$, or 10
Q015 $x = -3, -2, -1$	Q041 $x = -\sqrt{2}, \sqrt{2}, -i, i$	Q067 D	Q093 $a = -7, -3$, or 2
Q016 $x = -3, -2, 2$	Q042 D	Q068 B	Q094 D
Q017 $x = -1, 2/3, 1$	Q043 D	Q069 $k = 9$; other roots: $3, 4$	Q095 D
Q018 $x = 2, (-1 - \sqrt{7}i)/2, (-1 + \sqrt{7}i)/2$	Q044 C	Q070 B	Q096 B
Q019 $x = -2, -1 - \sqrt{3}, -1 + \sqrt{3}$	Q045 C	Q071 1	Q097 A
Q020 $x = 2, 1 - 2i, 1 + 2i$	Q046 B	Q072 1	Q098 A
Q021 $x = -2$ (repeated), 3	Q047 $x = 2$ (repeated), $1 - \sqrt{5}, 1 + \sqrt{5}$	Q073 1	Q099 $a = -1, b = 6$; other roots: $1 - \sqrt{2}i, -2$
Q022 $x = -1$ (repeated), $-3/2$	Q048 $x = 1, 2, 3, 4$	Q074 A	Q100 4
Q023 $x = -2$ (triple)	Q049 $x = 1, 2, 3$ (repeated)	Q075 B	Q101 $a = 2/3, 2$, or 6
Q024 C	Q050 $x = -1, 3, 1 - i, 1 + i$	Q076 D	Q102 $x^4 - x^3 + x^2 + x - 2 = 0$
Q025 D	Q051 B	Q077 -1	Q103 $x = -1, 1, (1 - \sqrt{7}i)/2, (1 + \sqrt{7}i)/2$
Q026 D	Q052 A	Q078 -1	

Both types occur: real $x = -1, 1$

Q104 non-real conjugates $x = (1 \pm \sqrt{7}i)/2$
if $x \geq 0 : x = 1$

Q105 C

Q106 D

Q107 D

Q108 A

Q109 D

Q110 B

Quiz MCQ 1 D

Quiz MCQ 2 D

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $x = -2, -1, 1, 2$

Quiz WRITTEN 6 $a = 0, b = 1$; other roots: $1 - 2i, -2$

Quiz WRITTEN 7 1

Quiz WRITTEN 8 $a = -4$ or $9/4$

Concept 1 | Coordinate Geometry Basics

English Student Edition • Focused formulas and guided practice

Formula opener

Distance

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Worked Solutions

Coordinate Geometry Basics

Q001 **Written****A (−4), B (8): distance AB****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 12.
3. Final answer: 12.

Final answer

12

Q002 **Written****A (−5), B (7): distance AB****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 12.
3. Final answer: 12.

Final answer

12

Worked Solutions

Coordinate Geometry Basics

Q003 Written

For two points A (-2) and B (6) , find the distance AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 8.
3. Final answer: 8.

Final answer

8

Q004 Written

For two points A (4) and B (10) , find the distance AB.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 6.
3. Final answer: 6.

Final answer

6

Worked Solutions

Coordinate Geometry Basics

Q005 MCQ

Find a (-4) , B (8) : distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 12.

Correct option: C**Final answer**

12

Q006 MCQ

Find a (-5) , B (7) : distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 12.

Correct option: C**Final answer**

12

Worked Solutions

Coordinate Geometry Basics

Q007 MCQ

For two points A (-2) and B (6) , find the distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 8.

Correct option: C**Final answer**

8

Q008 MCQ

For two points A (4) and B (10) , find the distance AB:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 6.

Correct option: C**Final answer**

6

Worked Solutions

Coordinate Geometry Basics

Q009 Written

A (−4), B (8): internal point C in ratio 2:1**Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 4.
3. Final answer: 4.

Final answer

4

Q010 Written

A (−5), B (7): internal point C in ratio 2:1**Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 3.
3. Final answer: 3.

Final answer

3

Worked Solutions

Coordinate Geometry Basics

Q011 Written

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 4.
3. Final answer: 4.

Final answer

4

Q012 Written

For two points A (4) and B (10) , find the coordinate of point C dividing segment AB internally in the ratio 2:1.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 8.
3. Final answer: 8.

Final answer

8

Worked Solutions

Coordinate Geometry Basics

Q013 MCQ

Find a (-4) , B (8) : internal point C in ratio 2:1:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 4.

Correct option: B**Final answer**

4

Q014 MCQ

Find a (-5) , B (7) : internal point C in ratio 2:1:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 3.

Correct option: C**Final answer**

3

Worked Solutions

Coordinate Geometry Basics

Q015 MCQ

For two points A (-2) and B (6) , find the coordinate of point C dividing segment AB internally in the ratio 3:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 4.

Correct option: B

Final answer

4

Q016 MCQ

For two points A (4) and B (10) , find the coordinate of point C dividing segment AB internally in the ratio 2:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 8.

Correct option: B

Final answer

8

Worked Solutions

Coordinate Geometry Basics

Q017 Written

A (−4), B (8): external point D in ratio 2:3**Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -28 .
3. Final answer: -28 .

Final answer -28

Q018 Written

A (−5), B (7): external point D in ratio 2:5**Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives -13 .
3. Final answer: -13 .

Final answer -13

Worked Solutions

Coordinate Geometry Basics

Q019 **Written**

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 18.
3. Final answer: 18.

Final answer

18

Q020 **Written**

For two points A (4) and B (10) , find the coordinate of point D dividing segment AB externally in the ratio 1:3.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 1.
3. Final answer: 1.

Final answer

1

Worked Solutions

Coordinate Geometry Basics

Q021 MCQ

Find a (-4) , B (8) : external point D in ratio 2:3:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -28 .

Correct option: A**Final answer** -28

Q022 MCQ

Find a (-5) , B (7) : external point D in ratio 2:5:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -13 .

Correct option: B**Final answer** -13

Worked Solutions

Coordinate Geometry Basics

Q023 MCQ

For two points A (-2) and B (6) , find the coordinate of point D dividing segment AB externally in the ratio 5:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 18.

Correct option: A**Final answer**

18

Q024 MCQ

For two points A (4) and B (10) , find the coordinate of point D dividing segment AB externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 1.

Correct option: B**Final answer**

1

Worked Solutions

Coordinate Geometry Basics

Q025 **Written****A (−4), B (8): midpoint M****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 2.
3. Final answer: 2.

Final answer

2

Q026 **Written****A (−5), B (7): midpoint M****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 1.
3. Final answer: 1.

Final answer

1

Worked Solutions

Coordinate Geometry Basics

Q027 **Written****For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB.****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 2.
3. Final answer: 2.

Final answer

2

Q028 **Written****For two points A (4) and B (10) , find the coordinate of midpoint M of segment AB.****Solution**

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives 7.
3. Final answer: 7.

Final answer

7

Worked Solutions

Coordinate Geometry Basics

Q029 MCQ

Find a (-4) , B (8) : midpoint M:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 2.

Correct option: C**Final answer**

2

Q030 MCQ

Find a (-5) , B (7) : midpoint M:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 1.

Correct option: A**Final answer**

1

Worked Solutions

Coordinate Geometry Basics

Q031 MCQ

For two points A (-2) and B (6) , find the coordinate of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 2.

Correct option: D

Final answer

2

Q032 MCQ

For two points A (4) and B (10) , find the coordinate of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 7.

Correct option: A

Final answer

7

Worked Solutions**Coordinate Geometry Basics****Q033 Challenge****Pods A (-8) and B (-3) : fibre-optic tether length****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is 5.

Final answer

5

Worked Solutions

Coordinate Geometry Basics

Q034 Challenge**Pods A (-8) , B (-3) : internal sensor point in ratio 1:4****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -7 .

Final answer -7

Worked Solutions

Coordinate Geometry Basics

Q035 Challenge**Pods A (-8) , B (-3) : external outpost in ratio 3:4****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -23 .

Final answer -23

Worked Solutions**Coordinate Geometry Basics****Q036 Challenge****Pods A (−8), B (−3): midpoint supply hub****Solution**

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $-11/2$.

Final answer $-11/2$

Worked Solutions

Coordinate Geometry Basics

Q037 Challenge

Find pods A (-8) and B (-3) : fibre-optic tether length:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: A

Final answer

5

Worked Solutions

Coordinate Geometry Basics

Q038 Challenge

Find pods A (-8) , B (-3) : internal sensor point in ratio 1:4:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -7 .

Correct option: C

Final answer

-7

Worked Solutions

Coordinate Geometry Basics

Q039 Challenge

Find pods A (-8) , B (-3) : external outpost in ratio 3:4:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -23 .

Correct option: B

Final answer

-23

Worked Solutions

Coordinate Geometry Basics

Q040 Challenge**Find pods A (−8), B (−3): midpoint supply hub:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $-11/2$.

Correct option: C**Final answer** $-11/2$

Worked Solutions

Coordinate Geometry Basics

Q041 **Written****Point a has coordinates (2,3): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*Q042 **Written****Point b has coordinates (2,-3): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*

Worked Solutions

Coordinate Geometry Basics

Q043 **Written****Point c has coordinates $(-2,3)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*Q044 **Written****Point d has coordinates $(-2,-3)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*

Worked Solutions

Coordinate Geometry Basics

Q045 **Written****Point a has coordinates $(-4,2)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*Q046 **Written****Point b has coordinates $(1,2)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*

Worked Solutions

Coordinate Geometry Basics

Q047 **Written****Point c has coordinates (3,-2): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*Q048 **Written****Point d has coordinates (-3,-1): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*

Worked Solutions

Coordinate Geometry Basics

Q049 **Written****Point a has coordinates $(-3,1)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*Q050 **Written****Point b has coordinates $(2,-8)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*

Worked Solutions

Coordinate Geometry Basics

Q051 **Written****Point c has coordinates (4,4): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*Q052 **Written****Point d has coordinates (-6,-4): determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*

Worked Solutions

Coordinate Geometry Basics

Q053 Written

Point e has coordinates $(-2,6)$: determine its quadrant**Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *II*.

Final answer*II*

Q054 Written

Point f has coordinates $(3,-3)$: determine its quadrant**Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *IV*.

Final answer*IV*

Worked Solutions

Coordinate Geometry Basics

Q055 **Written****Point g has coordinates $(-5,-7)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *III*.

Final answer*III*Q056 **Written****Point h has coordinates $(1,4)$: determine its quadrant****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is *I*.

Final answer*I*

Worked Solutions

Coordinate Geometry Basics

Q057 MCQ

Point a has coordinates (2,3): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: C**Final answer***I*

Q058 MCQ

Point b has coordinates (2,-3): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: B**Final answer***IV*

Worked Solutions

Coordinate Geometry Basics

Q059 MCQ

Point c has coordinates $(-2,3)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: A**Final answer***II*

Q060 MCQ

Point d has coordinates $(-2,-3)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: C**Final answer***III*

Worked Solutions

Coordinate Geometry Basics

Q061 MCQ

Point a has coordinates $(-4,2)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: A**Final answer***II*

Q062 MCQ

Point b has coordinates $(1,2)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: C**Final answer***I*

Worked Solutions

Coordinate Geometry Basics

Q063 MCQ

Point c has coordinates (3,-2): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: B**Final answer***IV*

Q064 MCQ

Point d has coordinates (-3,-1): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: A**Final answer***III*

Worked Solutions

Coordinate Geometry Basics

Q065 MCQ

Point a has coordinates $(-3,1)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: B**Final answer***II*

Q066 MCQ

Point b has coordinates $(2,-8)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: A**Final answer***IV*

Worked Solutions

Coordinate Geometry Basics

Q067 MCQ

Point c has coordinates (4,4): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: A**Final answer***I*

Q068 MCQ

Point d has coordinates (-6,-4): determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: C**Final answer***III*

Worked Solutions

Coordinate Geometry Basics

Q069 MCQ

Point e has coordinates $(-2,6)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: B**Final answer***II*

Q070 MCQ

Point f has coordinates $(3,-3)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *IV*.

Correct option: C**Final answer***IV*

Worked Solutions

Coordinate Geometry Basics

Q071 MCQ

Point g has coordinates $(-5, -7)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *III*.

Correct option: C**Final answer***III*

Q072 MCQ

Point h has coordinates $(1, 4)$: determine its quadrant:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *I*.

Correct option: D**Final answer***I*

Worked Solutions

Coordinate Geometry Basics

Q073 Written

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-3, -2)$.

Final answer

$(-3, -2)$

Q074 Written

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1, -2)$.

Final answer

$(-1, -2)$

Worked Solutions

Coordinate Geometry Basics

Q075 Written

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-4, 2)$.

Final answer

$(-4, 2)$

Q076 Written

For point A $(1, -2)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1, 2)$.

Final answer

$(1, 2)$

Worked Solutions

Coordinate Geometry Basics

Q077 **Written**

For point A $(-3, 4)$, find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-3, -4)$.

Final answer $(-3, -4)$ Q078 **MCQ**

For point A $(-3, 2)$, find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3, -2)$.

Correct option: C**Final answer** $(-3, -2)$

Worked Solutions

Coordinate Geometry Basics

Q079 MCQ

For point A $(-1, 2)$, find the coordinates of point P symmetric with respect to the x-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1, -2)$.

Correct option: D**Final answer** $(-1, -2)$

Q080 MCQ

For point A $(-4, -2)$, find the coordinates of point P symmetric with respect to the x-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-4, 2)$.

Correct option: C**Final answer** $(-4, 2)$

Worked Solutions

Coordinate Geometry Basics

Q081 MCQ

For point A (1, -2), find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (1, 2).

Correct option: D

Final answer

(1, 2)

Q082 MCQ

For point A (-3, 4), find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-3, -4).

Correct option: C

Final answer

(-3, -4)

Worked Solutions

Coordinate Geometry Basics

Q083 Written

For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, 2)$.

Final answer

$(3, 2)$

Q084 Written

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1, 2)$.

Final answer

$(1, 2)$

Worked Solutions

Coordinate Geometry Basics

Q085 Written

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(4, -2)$.

Final answer

$(4, -2)$

Q086 Written

For point A $(1, -2)$, find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1, -2)$.

Final answer

$(-1, -2)$

Worked Solutions

Coordinate Geometry Basics

Q087 **Written****For point A $(-3, 4)$, find the coordinates of point Q symmetric with respect to the y-axis.****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, 4)$.

Final answer $(3, 4)$ Q088 **MCQ****For point A $(-3, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 2)$.

Correct option: C**Final answer** $(3, 2)$

Worked Solutions

Coordinate Geometry Basics

Q089 MCQ

For point A $(-1, 2)$, find the coordinates of point Q symmetric with respect to the y-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1, 2)$.

Correct option: C**Final answer** $(1, 2)$

Q090 MCQ

For point A $(-4, -2)$, find the coordinates of point Q symmetric with respect to the y-axis:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4, -2)$.

Correct option: A**Final answer** $(4, -2)$

Worked Solutions

Coordinate Geometry Basics

Q091 MCQ

For point A (1, -2), find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-1, -2).

Correct option: A

Final answer

(-1, -2)

Q092 MCQ

For point A (-3, 4), find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (3, 4).

Correct option: A

Final answer

(3, 4)

Worked Solutions

Coordinate Geometry Basics

Q093 **Written****For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin.****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, -2)$.

Final answer $(3, -2)$ Q094 **Written****For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin.****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(1, -2)$.

Final answer $(1, -2)$

Worked Solutions

Coordinate Geometry Basics

Q095 Written

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(4, 2)$.

Final answer

$(4, 2)$

Q096 Written

For point A $(1, -2)$, find the coordinates of point R symmetric with respect to the origin.

Solution

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(-1, 2)$.

Final answer

$(-1, 2)$

Worked Solutions

Coordinate Geometry Basics

Q097 **Written****For point A $(-3, 4)$, find the coordinates of point R symmetric with respect to the origin.****Solution**

1. Read the coordinate signs for a quadrant, or change the required coordinate signs for the stated reflection.
2. The transformed point/quadrant is $(3, -4)$.

Final answer $(3, -4)$ Q098 **MCQ****For point A $(-3, 2)$, find the coordinates of point R symmetric with respect to the origin:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, -2)$.

Correct option: B**Final answer** $(3, -2)$

Worked Solutions

Coordinate Geometry Basics

Q099 MCQ

For point A $(-1, 2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(1, -2)$.

Correct option: B

Final answer

$(1, -2)$

Q100 MCQ

For point A $(-4, -2)$, find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4, 2)$.

Correct option: B

Final answer

$(4, 2)$

Worked Solutions

Coordinate Geometry Basics

Q101 MCQ

For point A (1, -2), find the coordinates of point R symmetric with respect to the origin:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-1, 2).

Correct option: A**Final answer**

(-1, 2)

Q102 MCQ

For point A (-3, 4), find the coordinates of point R symmetric with respect to the origin:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (3, -4).

Correct option: D**Final answer**

(3, -4)

Worked Solutions

Coordinate Geometry Basics

Q103 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(-3, -4)$.

Final answer

$(-3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q104 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(3, 4)$.

Final answer

$(3, 4)$

Worked Solutions

Coordinate Geometry Basics

Q105 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin.

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(3, -4)$.

Final answer

$(3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q106 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point P symmetric with respect to the x-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-3, -4)$.

Correct option: A

Final answer

$(-3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q107 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point Q symmetric with respect to the y-axis:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 4)$.

Correct option: B

Final answer

$(3, 4)$

Worked Solutions

Coordinate Geometry Basics

Q108 Challenge

A robotic welding arm is programmed at point A $(-3, 4)$ on a metal sheet. To weld identical support brackets, the arm must mirror this point: find the coordinates of point R symmetric with respect to the origin:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, -4)$.

Correct option: C

Final answer

$(3, -4)$

Worked Solutions

Coordinate Geometry Basics

Q109 Written

Distance between A (2, 6) and B (5, 10)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives 5.

Final answer

5

Q110 Written

Distance between A (4, 3) and B (1, 2)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $\sqrt{10}$.

Final answer $\sqrt{10}$

Worked Solutions

Coordinate Geometry Basics

Q111 Written

Distance between A($\sqrt{3}$, -1) and B (0, 1)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $\sqrt{7}$.

Final answer

$$\sqrt{7}$$

Q112 Written

Distance between A (1, 2) and B (11, 7)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $5\sqrt{5}$.

Final answer

$$5\sqrt{5}$$

Worked Solutions

Coordinate Geometry Basics

Q113 Written

Distance between A (3, -1) and B (-1, 3)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $4\sqrt{2}$.

Final answer

$$4\sqrt{2}$$

Q114 MCQ

Find distance between A (2, 6) and B (5, 10):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: C**Final answer**

$$5$$

Worked Solutions

Coordinate Geometry Basics

Q115 MCQ

Find distance between A (4, 3) and B (1, 2):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: A**Final answer**

$$\sqrt{10}$$

Q116 MCQ

Find distance between A($\sqrt{3}$, -1) and B (0, 1):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $\sqrt{7}$.

Correct option: C**Final answer**

$$\sqrt{7}$$

Worked Solutions

Coordinate Geometry Basics

Q117 MCQ

Find distance between A (1, 2) and B (11, 7):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $5\sqrt{5}$.

Correct option: B**Final answer**

$$5\sqrt{5}$$

Q118 MCQ

Find distance between A (3, -1) and B (-1, 3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $4\sqrt{2}$.

Correct option: A**Final answer**

$$4\sqrt{2}$$

Worked Solutions

Coordinate Geometry Basics

Q119 Written

A (3, -2), **B** (x, 1), **distance 5: find x****Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, 7$.

Final answer

$$x = -1, 7$$

Q120 Written

A (-1, 3), **B** (x, 1), **distance $\sqrt{5}$: find x****Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -2, 0$.

Final answer

$$x = -2, 0$$

Worked Solutions

Coordinate Geometry Basics

Q121 Written

A (3, -1), B (x, 2), distance 5: find x

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, 7$.

Final answer

$$x = -1, 7$$

Q122 Written

A (1, -3), B (-2, y), distance $\sqrt{13}$: find y

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $y = -5, -1$.

Final answer

$$y = -5, -1$$

Worked Solutions

Coordinate Geometry Basics

Q123 **Written****A** $(-4, -2)$, **B** $(x, 3)$, **distance** $5\sqrt{5}$: **find** **x****Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -14, 6$.

Final answer

$$x = -14, 6$$

Q124 **MCQ****A** $(3, -2)$, **B** $(x, 1)$, **distance** **5**: **find** **x**:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -1, 7$.

Correct option: B**Final answer**

$$x = -1, 7$$

Worked Solutions

Coordinate Geometry Basics

Q125 MCQ

A $(-1, 3)$, **B** $(x, 1)$, **distance** $\sqrt{5}$: **find x:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -2, 0$.

Correct option: C**Final answer**

$$x = -2, 0$$

Q126 MCQ

A $(3, -1)$, **B** $(x, 2)$, **distance** 5: **find x:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -1, 7$.

Correct option: B**Final answer**

$$x = -1, 7$$

Worked Solutions

Coordinate Geometry Basics

Q127 MCQ

A (1, -3), **B** (-2, y), **distance** $\sqrt{13}$: **find** y :**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $y = -5, -1$.

Correct option: A**Final answer**

$$y = -5, -1$$

Q128 MCQ

A (-4, -2), **B** (x , 3), **distance** $5\sqrt{5}$: **find** x :**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -14, 6$.

Correct option: D**Final answer**

$$x = -14, 6$$

Worked Solutions

Coordinate Geometry Basics

Q129 Written

Distance from origin to A (2, -4)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $2\sqrt{5}$.

Final answer

$$2\sqrt{5}$$

Q130 Written

Distance from origin to A (-3, 4)**Solution**

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives 5.

Final answer

$$5$$

Worked Solutions

Coordinate Geometry Basics

Q131 MCQ

Find distance from origin to A (2, -4):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $2\sqrt{5}$.

Correct option: C**Final answer**

$$2\sqrt{5}$$

Q132 MCQ

Find distance from origin to A (-3, 4):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: C**Final answer**

$$5$$

Worked Solutions

Coordinate Geometry Basics

Q133 Challenge

Distance between $(x, 7)$ and $(3x-1, -5)$ is 13: find x

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $x = -2, 3$.

Final answer

$$x = -2, 3$$

Worked Solutions

Coordinate Geometry Basics

Q134 Challenge

Distance between $(x, 7)$ and $(3x - 1, -5)$ is 13: find x :

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $x = -2, 3$.

Correct option: C

Final answer

$$x = -2, 3$$

Worked Solutions

Coordinate Geometry Basics

Q135 **Written****Point on the x-axis equidistant from A $(-2, 1)$ and B $(3, 4)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2, 0)$.

Final answer $(2, 0)$ Q136 **Written****Point on the y-axis equidistant from A $(4, 2)$ and B $(1, -1)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(0, 3)$.

Final answer $(0, 3)$

Worked Solutions

Coordinate Geometry Basics

Q137 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(4, 3)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(2, 0)$.

Final answer $(2, 0)$ Q138 **Written****Point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(0, 5/6)$.

Final answer $(0, 5/6)$

Worked Solutions

Coordinate Geometry Basics

Q139 **Written****Point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$** **Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(17/2, 0)$.

Final answer $(17/2, 0)$ Q140 **MCQ****Find point on the x-axis equidistant from A $(-2, 1)$ and B $(3, 4)$:****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, 0)$.

Correct option: C**Final answer** $(2, 0)$

Worked Solutions

Coordinate Geometry Basics

Q141 MCQ

Find point on the y-axis equidistant from A (4, 2) and B (1, -1):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (0, 3).

Correct option: A**Final answer**

(0, 3)

Q142 MCQ

Find point on the x-axis equidistant from A (-1, 2) and B (4, 3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, 0).

Correct option: C**Final answer**

(2, 0)

Worked Solutions

Coordinate Geometry Basics

Q143 MCQ

Find point on the y-axis equidistant from A $(-5, 2)$ and B $(3, 5)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, 5/6)$.

Correct option: C**Final answer** $(0, 5/6)$

Q144 MCQ

Find point on the x-axis equidistant from A $(-1, 2)$ and B $(3, 8)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(17/2, 0)$.

Correct option: B**Final answer** $(17/2, 0)$

Worked Solutions

Coordinate Geometry Basics

Q145 **Written****Point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(13/8, 9/4)$.

Final answer $(13/8, 9/4)$ Q146 **Written****Point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are $(-4, -11)$.

Final answer $(-4, -11)$

Worked Solutions

Coordinate Geometry Basics

Q147 **Written****Point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (2, 1).

Final answer

(2, 1)

Q148 **Written****Point on $y = x - 1$ equidistant from A (-1, 5) and B (7, -1)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (3, 2).

Final answer

(3, 2)

Worked Solutions

Coordinate Geometry Basics

Q149 **Written****Point on $y = -x + 1$ equidistant from A (1, 2) and B (−3, 6)****Solution**

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (−2, 3).

Final answer

(−2, 3)

Q150 **MCQ****Find point on $y = 2x - 1$ equidistant from A (1, 4) and B (3, 1):****Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (13/8, 9/4).

Correct option: B**Final answer**

(13/8, 9/4)

Worked Solutions

Coordinate Geometry Basics

Q151 MCQ

Find point on $y = 3x + 1$ equidistant from A (4, -5) and B (2, -3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-4, -11).

Correct option: C**Final answer**

(-4, -11)

Q152 MCQ

Find point on $y = 2x - 3$ equidistant from A (1, 4) and B (3, -2):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, 1).

Correct option: A**Final answer**

(2, 1)

Worked Solutions

Coordinate Geometry Basics

Q153 MCQ

Find point on $y = x - 1$ equidistant from A $(-1, 5)$ and B $(7, -1)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 2)$.

Correct option: C**Final answer** $(3, 2)$

Q154 MCQ

Find point on $y = -x + 1$ equidistant from A $(1, 2)$ and B $(-3, 6)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2, 3)$.

Correct option: B**Final answer** $(-2, 3)$

Worked Solutions

Coordinate Geometry Basics

Q155 Challenge

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $3\sqrt{5}$.

Final answer

$$3\sqrt{5}$$

Worked Solutions

Coordinate Geometry Basics

Q156 Challenge

A (2, 1), B (4, 5), C (3, y) with CA=CB and D (x, 0) with DA=DB: find CD:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $3\sqrt{5}$.

Correct option: A

Final answer

$$3\sqrt{5}$$

Worked Solutions

Coordinate Geometry Basics

Q157 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(4/3, 11/3)$.

Final answer $(4/3, 11/3)$

Worked Solutions

Coordinate Geometry Basics

Q158 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-15/4, 13/4)$.

Final answer $(-15/4, 13/4)$

Worked Solutions

Coordinate Geometry Basics

Q159 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (2, -9/5).

Final answer

(2, -9/5)

Worked Solutions

Coordinate Geometry Basics

Q160 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(3, 3)$.

Final answer $(3, 3)$

Worked Solutions

Coordinate Geometry Basics

Q161 Written

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-2/3, -5/3)$.

Final answer

$(-2/3, -5/3)$

Worked Solutions

Coordinate Geometry Basics

Q162 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4/3, 11/3)$.

Correct option: C

Final answer

$(4/3, 11/3)$

Worked Solutions

Coordinate Geometry Basics

Q163 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point P dividing segment AC internally in the ratio 3:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-15/4, 13/4)$.

Correct option: C

Final answer

$(-15/4, 13/4)$

Worked Solutions

Coordinate Geometry Basics

Q164 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point P dividing segment AB internally in the ratio 4:1:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, -9/5).

Correct option: C

Final answer

(2, -9/5)

Worked Solutions

Coordinate Geometry Basics

Q165 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point P dividing segment BC internally in the ratio 3:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 3)$.

Correct option: C

Final answer

$(3, 3)$

Worked Solutions

Coordinate Geometry Basics

Q166 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point P dividing segment BA internally in the ratio 1:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2/3, -5/3)$.

Correct option: A

Final answer

$(-2/3, -5/3)$

Worked Solutions

Coordinate Geometry Basics

Q167 Written

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, 19/2)$.

Final answer

$(2, 19/2)$

Worked Solutions

Coordinate Geometry Basics

Q168 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(11, -18)$.

Final answer $(11, -18)$

Worked Solutions

Coordinate Geometry Basics

Q169 Written

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (10, -15).

Final answer

(10, -15)

Worked Solutions

Coordinate Geometry Basics

Q170 Written

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-8, -11)$.

Final answer

$(-8, -11)$

Worked Solutions

Coordinate Geometry Basics

Q171 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0, -15/2)$.

Final answer $(0, -15/2)$

Worked Solutions

Coordinate Geometry Basics

Q172 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, 19/2)$.

Correct option: B

Final answer

$(2, 19/2)$

Worked Solutions

Coordinate Geometry Basics

Q173 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(11, -18)$.

Correct option: C

Final answer

$(11, -18)$

Worked Solutions

Coordinate Geometry Basics

Q174 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of point Q dividing segment BC externally in the ratio 2:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (10, -15).

Correct option: B**Final answer**

(10, -15)

Worked Solutions

Coordinate Geometry Basics

Q175 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of point Q dividing segment CA externally in the ratio 3:2:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8, -11)$.

Correct option: B

Final answer

$(-8, -11)$

Worked Solutions

Coordinate Geometry Basics

Q176 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of point Q dividing segment BC externally in the ratio 1:3:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, -15/2)$.

Correct option: C**Final answer** $(0, -15/2)$

Worked Solutions

Coordinate Geometry Basics

Q177 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(3/2, -1/2)$.

Final answer $(3/2, -1/2)$

Worked Solutions

Coordinate Geometry Basics

Q178 **Written**

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, 2)$.

Final answer $(2, 2)$

Worked Solutions

Coordinate Geometry Basics

Q179 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-1, 0)$.

Final answer $(-1, 0)$

Worked Solutions

Coordinate Geometry Basics

Q180 **Written**

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(-2, 1)$.

Final answer $(-2, 1)$

Worked Solutions

Coordinate Geometry Basics

Q181 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (0, 3).

Final answer

(0, 3)

Worked Solutions

Coordinate Geometry Basics

Q182 **Written**

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2/3, 1)$.

Final answer $(2/3, 1)$

Worked Solutions

Coordinate Geometry Basics

Q183 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, -1)$.

Final answer $(2, -1)$

Worked Solutions

Coordinate Geometry Basics

Q184 **Written**

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(5/3, 5/3)$.

Final answer $(5/3, 5/3)$

Worked Solutions

Coordinate Geometry Basics

Q185 Written

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(2, -5/2)$.

Final answer

$(2, -5/2)$

Worked Solutions

Coordinate Geometry Basics

Q186 **Written**

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC.

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are $(0, 0)$.

Final answer $(0, 0)$

Worked Solutions

Coordinate Geometry Basics

Q187 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of midpoint M of segment CA:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3/2, -1/2)$.

Correct option: A

Final answer

$(3/2, -1/2)$

Worked Solutions

Coordinate Geometry Basics

Q188 MCQ

Given A $(-2, -3)$, B $(3, 7)$, and C $(5, 2)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, 2)$.

Correct option: C

Final answer

$(2, 2)$

Worked Solutions

Coordinate Geometry Basics

Q189 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-1, 0)$.

Correct option: B

Final answer

$(-1, 0)$

Worked Solutions

Coordinate Geometry Basics

Q190 MCQ

Given A $(-3, 4)$, B $(1, -4)$, and C $(-4, 3)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-2, 1)$.

Correct option: A

Final answer

$(-2, 1)$

Worked Solutions

Coordinate Geometry Basics

Q191 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of midpoint M of segment CA:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (0, 3).

Correct option: A

Final answer

(0, 3)

Worked Solutions

Coordinate Geometry Basics

Q192 MCQ

Given A (2, 3), B (2, -3), and C (-2, 3), find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2/3, 1).

Correct option: D

Final answer

(2/3, 1)

Worked Solutions

Coordinate Geometry Basics

Q193 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of midpoint M of segment AB:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, -1)$.

Correct option: B

Final answer

$(2, -1)$

Worked Solutions

Coordinate Geometry Basics

Q194 MCQ

Given A $(-2, 1)$, B $(6, -3)$, and C $(1, 7)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(5/3, 5/3)$.

Correct option: A

Final answer

$(5/3, 5/3)$

Worked Solutions

Coordinate Geometry Basics

Q195 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of midpoint M of segment BC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(2, -5/2)$.

Correct option: B

Final answer

$(2, -5/2)$

Worked Solutions

Coordinate Geometry Basics

Q196 MCQ

Given A $(-4, 5)$, B $(1, -5)$, and C $(3, 0)$, find the coordinates of centroid G of triangle ABC:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, 0)$.

Correct option: C

Final answer

$(0, 0)$

Worked Solutions

Coordinate Geometry Basics

Q197 Challenge

A $(-2, 3)$, **B** $(6, 1)$, **C** $(0, y)$, **CA=CB: find the centroid G**

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is $(4/3, -2/3)$.

Final answer

$(4/3, -2/3)$

Worked Solutions

Coordinate Geometry Basics

Q198 Challenge

A $(-2, 3)$, **B** $(6, 1)$, **C** $(0, y)$, **CA=CB: find the centroid G:**

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(4/3, -2/3)$.

Correct option: C

Final answer

$(4/3, -2/3)$

Worked Solutions

Coordinate Geometry Basics

Q199 **Written****Point with A $(-3, 2)$ symmetric about centre $(2, 6)$** **Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(7, 10)$.

Final answer $(7, 10)$ Q200 **Written****Point with A $(1, -2)$ symmetric about centre $(2, 3)$** **Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(3, 8)$.

Final answer $(3, 8)$

Worked Solutions

Coordinate Geometry Basics

Q201 **Written****Point with A (3, 2) symmetric about centre (-1, 5)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-5, 8)$.

Final answer $(-5, 8)$ Q202 **Written****Point with A $(-3, 1)$ symmetric about centre $(2, 0)$** **Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(7, -1)$.

Final answer $(7, -1)$

Worked Solutions

Coordinate Geometry Basics

Q203 **Written****Point with A (7, 4) symmetric about centre (-3, 5)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-13, 6)$.

Final answer $(-13, 6)$ Q204 **Written****Point with A (4, -8) symmetric about centre (-3, 2)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-10, 12)$.

Final answer $(-10, 12)$

Worked Solutions

Coordinate Geometry Basics

Q205 **Written****Point with A (0, -4) symmetric about centre (-4, 1)****Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(-8, 6)$.

Final answer $(-8, 6)$ Q206 **Written****Point with A $(-4, 2)$ symmetric about centre $(-1, -3)$** **Solution**

1. The centre is the midpoint, so the unknown point is twice the centre minus the known point.
2. Therefore the symmetric point/result is $(2, -8)$.

Final answer $(2, -8)$

Worked Solutions

Coordinate Geometry Basics

Q207 MCQ

Find point with A $(-3, 2)$ symmetric about centre $(2, 6)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(7, 10)$.

Correct option: C**Final answer** $(7, 10)$

Q208 MCQ

Find point with A $(1, -2)$ symmetric about centre $(2, 3)$:**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(3, 8)$.

Correct option: B**Final answer** $(3, 8)$

Worked Solutions

Coordinate Geometry Basics

Q209 MCQ

Find point with A (3, 2) symmetric about centre (-1, 5):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-5, 8).

Correct option: A**Final answer**

(-5, 8)

Q210 MCQ

Find point with A (-3, 1) symmetric about centre (2, 0):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (7, -1).

Correct option: C**Final answer**

(7, -1)

Worked Solutions

Coordinate Geometry Basics

Q211 MCQ

Find point with A (7, 4) symmetric about centre (-3, 5):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-13, 6)$.

Correct option: C**Final answer** $(-13, 6)$

Q212 MCQ

Find point with A (4, -8) symmetric about centre (-3, 2):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-10, 12)$.

Correct option: C**Final answer** $(-10, 12)$

Worked Solutions

Coordinate Geometry Basics

Q213 MCQ

Find point with A (0, -4) symmetric about centre (-4, 1):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (-8, 6).

Correct option: B**Final answer**

(-8, 6)

Q214 MCQ

Find point with A (-4, 2) symmetric about centre (-1, -3):**Solution**

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is (2, -8).

Correct option: A**Final answer**

(2, -8)

Worked Solutions

Coordinate Geometry Basics

Q215 Challenge

B (m , 5), **C** (4, n) symmetric about **A** (1, -1): find $m+n$

Solution

1. Translate each coordinate condition, solve the shortest linked route, and check the final distance or centroid.
2. The checked result is -9 .

Final answer

-9

Worked Solutions

Coordinate Geometry Basics

Q216 Challenge

B (m , 5), **C** (4, n) symmetric about **A** (1, -1): find $m+n$:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is -9 .

Correct option: C

Final answer

-9

Worked Solutions

Coordinate Geometry Basics

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Point A ($-3, 4$) lies in which quadrant?:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is *II*.

Correct option: A

Final answer

II

Quiz MCQ 2 [Concept Quiz · MCQ](#)

Find the distance between A ($1, 2$) and B ($4, 6$):

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is 5.

Correct option: B

Final answer

5

Worked Solutions

Coordinate Geometry Basics

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Point P divides the segment from A $(-2, 3)$ to B $(6, 1)$ internally in ratio 1:3. Find P:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(0, 5/2)$.

Correct option: C

Final answer

$(0, 5/2)$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

Point B is symmetric to A $(2, -1)$ about P $(-3, 4)$. Find B:

Solution

1. Use the same coordinate-geometry route as the paired written demand.
2. The correct option is $(-8, 9)$.

Correct option: C

Final answer

$(-8, 9)$

Worked Solutions

Coordinate Geometry Basics

Quiz WRITTEN 5 Concept Quiz · Written

Find the point on $y = 2x + 1$ equidistant from A (0, 2) and B (4, 0).

Solution

1. Parameterise the required point on its axis or line, then equate the two squared distances.
2. The parameter and coordinates are (1, 3).

Final answer

(1, 3)

Quiz WRITTEN 6 Concept Quiz · Written

For points A (−5) and B (7), find the point dividing AB externally in ratio 2:5.

Solution

1. Use the number-line distance, internal, external, or midpoint formula as requested.
2. Substitution gives −13.
3. Final answer: −13.

Final answer

−13

Worked Solutions

Coordinate Geometry Basics

Quiz WRITTEN 7 Concept Quiz · Written

If the distance between A (3, -1) and B (x, 2) is 5, find x.

Solution

1. Use $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$. Square both sides when the distance is given.
2. Solving and simplifying gives $x = -1, 7$.

Final answer

$$x = -1, 7$$

Quiz WRITTEN 8 Concept Quiz · Written

Find the centroid of A (-2, 1), B (6, -3), C (1, 7).

Solution

1. Apply the internal/external coordinate weights to both coordinates; for a centroid, average the three vertices.
2. The required coordinates are (5/3, 5/3).

Final answer

$$(5/3, 5/3)$$

Answer key

Use the question number shown on each page.

Q001 12	Q027 2	Q053 <i>II</i>	Q079 D
Q002 12	Q028 7	Q054 <i>IV</i>	Q080 C
Q003 8	Q029 C	Q055 <i>III</i>	Q081 D
Q004 6	Q030 A	Q056 <i>I</i>	Q082 C
Q005 C	Q031 D	Q057 C	Q083 (3, 2)
Q006 C	Q032 A	Q058 B	Q084 (1, 2)
Q007 C	Q033 5	Q059 A	Q085 (4, -2)
Q008 C	Q034 -7	Q060 C	Q086 (-1, -2)
Q009 4	Q035 -23	Q061 A	Q087 (3, 4)
Q010 3	Q036 -11/2	Q062 C	Q088 C
Q011 4	Q037 A	Q063 B	Q089 C
Q012 8	Q038 C	Q064 A	Q090 A
Q013 B	Q039 B	Q065 B	Q091 A
Q014 C	Q040 C	Q066 A	Q092 A
Q015 B	Q041 <i>I</i>	Q067 A	Q093 (3, -2)
Q016 B	Q042 <i>IV</i>	Q068 C	Q094 (1, -2)
Q017 -28	Q043 <i>II</i>	Q069 B	Q095 (4, 2)
Q018 -13	Q044 <i>III</i>	Q070 C	Q096 (-1, 2)
Q019 18	Q045 <i>II</i>	Q071 C	Q097 (3, -4)
Q020 1	Q046 <i>I</i>	Q072 D	Q098 B
Q021 A	Q047 <i>IV</i>	Q073 (-3, -2)	Q099 B
Q022 B	Q048 <i>III</i>	Q074 (-1, -2)	Q100 B
Q023 A	Q049 <i>II</i>	Q075 (-4, 2)	Q101 A
Q024 B	Q050 <i>IV</i>	Q076 (1, 2)	Q102 D
Q025 2	Q051 <i>I</i>	Q077 (-3, -4)	Q103 (-3, -4)
Q026 1	Q052 <i>III</i>	Q078 C	Q104 (3, 4)

Q105 (3, -4)

Q106 A

Q107 B

Q108 C

Q109 5

Q110 $\sqrt{10}$ Q111 $\sqrt{7}$ Q112 $5\sqrt{5}$ Q113 $4\sqrt{2}$

Q114 C

Q115 A

Q116 C

Q117 B

Q118 A

Q119 $x = -1, 7$ Q120 $x = -2, 0$ Q121 $x = -1, 7$ Q122 $y = -5, -1$ Q123 $x = -14, 6$

Q124 B

Q125 C

Q126 B

Q127 A

Q128 D

Q129 $2\sqrt{5}$

Q130 5

Q131 C

Q132 C

Q133 $x = -2, 3$

Q134 C

Q135 (2, 0)

Q136 (0, 3)

Q137 (2, 0)

Q138 (0, $5/6$)

Q139 (17/2, 0)

Q140 C

Q141 A

Q142 C

Q143 C

Q144 B

Q145 (13/8, 9/4)

Q146 (-4, -11)

Q147 (2, 1)

Q148 (3, 2)

Q149 (-2, 3)

Q150 B

Q151 C

Q152 A

Q153 C

Q154 B

Q155 $3\sqrt{5}$

Q156 A

Q157 (4/3, 11/3)

Q158 (-15/4, 13/4)

Q159 (2, -9/5)

Q160 (3, 3)

Q161 (-2/3, -5/3)

Q162 C

Q163 C

Q164 C

Q165 C

Q166 A

Q167 (2, 19/2)

Q168 (11, -18)

Q169 (10, -15)

Q170 (-8, -11)

Q171 (0, -15/2)

Q172 B

Q173 C

Q174 B

Q175 B

Q176 C

Q177 (3/2, -1/2)

Q178 (2, 2)

Q179 (-1, 0)

Q180 (-2, 1)

Q181 (0, 3)

Q182 (2/3, 1)

Q183 (2, -1)

Q184 (5/3, 5/3)

Q185 (2, -5/2)

Q186 (0, 0)

Q187 A

Q188 C

Q189 B

Q190 A

Q191 A

Q192 D

Q193 B

Q194 A

Q195 B

Q196 C

Q197 (4/3, -2/3)

Q198 C

Q199 (7, 10)

Q200 (3, 8)

Q201 (-5, 8)

Q202 (7, -1)

Q203 (-13, 6)

Q204 (-10, 12)

Q205 (-8, 6)

Q206 (2, -8)

Q207 C

Q208 B

Q209 A

Q210 C

Q211 C

Q212 C

Q213 B

Q214 A

Q215 -9

Q216 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 C

Quiz WRITTEN 5 (1, 3)

Quiz WRITTEN 6 -13

Quiz WRITTEN 7 $x = -1, 7$

Quiz WRITTEN 8 (5/3, 5/3)

Concept 2 | Triangles and Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Line

$$y - y_1 = m(x - x_1)$$

Perpendicular

$$m_1 m_2 = -1$$

Area

$$A = \frac{1}{2}bh$$

Worked Solutions

Triangles and Lines

Q001 Written

Determine the shape of triangle ABC with A (3, 2), B (−1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *equilateral*.

Final answer*equilateral*

Worked Solutions

Triangles and Lines

Q002 **Written**

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$.

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *equilateral*.

Final answer*equilateral*

Worked Solutions

Triangles and Lines

Q003 MCQ

Determine the shape of triangle ABC with A (3, 2), B (−1, 0), C $(1 + \sqrt{3}, 1 - 2\sqrt{3})$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *equilateral*.

Correct option: D

Final answer

equilateral

Worked Solutions

Triangles and Lines

Q004 MCQ

Determine the shape of triangle ABC with A (2, 2), B (2, 0), C $(2 + \sqrt{3}, 1)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *equilateral*.

Correct option: C

Final answer*equilateral*

Q005 Written

Determine the shape of triangle ABC with A (−1, 2), B (1, 1), C (0, 4).

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *isosceles right – angled; right angle at A*.

Final answer*isosceles right – angled; right angle at A*

Worked Solutions

Triangles and Lines

Q006 Written

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1).**Solution**

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *isosceles right – angled; right angle at A*.

Final answer*isosceles right – angled; right angle at A*

Q007 MCQ

Determine the shape of triangle ABC with A (−1, 2), B (1, 1), C (0, 4):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *isosceles right – angled; right angle at A*.

Correct option: B**Final answer***isosceles right – angled; right angle at A*

Worked Solutions

Triangles and Lines

Q008 MCQ

Determine the shape of triangle ABC with A (0, 2), B (−1, −2), C (4, 1):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *isosceles right – angled; right angle at A*.

Correct option: D**Final answer***isosceles right – angled; right angle at A*

Q009 Written

Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4).**Solution**

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *scalene right – angled; right angle at A*.

Final answer*scalene right – angled; right angle at A*

Worked Solutions

Triangles and Lines

Q010 MCQ

Determine the shape of triangle ABC with A (−3, 4), B (1, 7), C (3, −4):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *scalene right – angled; right angle at A*.

Correct option: D**Final answer***scalene right – angled; right angle at A*

Q011 Written

Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2).**Solution**

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *isosceles right – angled; right angle at B*.

Final answer*isosceles right – angled; right angle at B*

Worked Solutions

Triangles and Lines

Q012 Written

Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4).

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *isosceles right – angled; right angle at B.*

Final answer

isosceles right – angled; right angle at B

Q013 Written

Determine the shape of triangle ABC with A (−1, 1), B (4, 2), C (1, 4).

Solution

1. Find the three squared side lengths.
2. Compare equality and $a^2 + b^2 = c^2$, taking c as the longest side.
3. Final classification: *isosceles right – angled; right angle at C.*

Final answer

isosceles right – angled; right angle at C

Worked Solutions

Triangles and Lines

Q014 MCQ

Determine the shape of triangle ABC with A (4, 9), B (0, 6), C (3, 2):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *isosceles right – angled; right angle at B*.

Correct option: B**Final answer***isosceles right – angled; right angle at B*

Q015 MCQ

Determine the shape of triangle ABC with A (1, 2), B (5, 8), C (11, 4):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *isosceles right – angled; right angle at B*.

Correct option: C**Final answer***isosceles right – angled; right angle at B*

Worked Solutions

Triangles and Lines

Q016 MCQ

Determine the shape of triangle ABC with A $(-1, 1)$, B $(4, 2)$, C $(1, 4)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *isosceles right – angled; right angle at C*.

Correct option: B**Final answer***isosceles right – angled; right angle at C*

Worked Solutions

Triangles and Lines

Q017 Challenge

If A (1, 2), B ($2k$, k), C (4, 5) and ABC is isosceles and right-angled at B, find k .

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $k = 2$.

Final answer

$$k = 2$$

Worked Solutions

Triangles and Lines

Q018 Challenge

If A (1, 2), B (2k, k), C (4, 5) and ABC is isosceles and right-angled at B, find k:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $k = 2$.

Correct option: C

Final answer

$$k = 2$$

Worked Solutions

Triangles and Lines

Q019 Written

Find the equation of the line through (3,-4) parallel to the x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -4$.

Final answer

$$y = -4$$

Q020 Written

Find the equation of the line through (-3,8) parallel to the y-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$$x = -3$$

Worked Solutions

Triangles and Lines

Q021 Written

Find the equation of the line through (2,6) parallel to the x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 6$.

Final answer

$$y = 6$$

Q022 Written

Find the equation of the line through (3,5) perpendicular to the x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 3$.

Final answer

$$x = 3$$

Worked Solutions

Triangles and Lines

Q023 Written

Find the equation of the line through $(-4,5)$ parallel to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 5$.

Final answer

$$y = 5$$

Q024 Written

Find the equation of the line through $(2,1)$ parallel to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 1$.

Final answer

$$y = 1$$

Worked Solutions

Triangles and Lines

Q025 Written

Find the equation of the line through $(-4, -7)$ parallel to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -7$.

Final answer

$$y = -7$$

Q026 Written

Find the equation of the line through $(1, -2)$ perpendicular to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 1$.

Final answer

$$x = 1$$

Worked Solutions

Triangles and Lines

Q027 Written

Find the equation of the line through $(-3, -4)$ perpendicular to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$$x = -3$$

Q028 Written

Find the equation of the line through $(3, -5)$ perpendicular to x-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 3$.

Final answer

$$x = 3$$

Worked Solutions

Triangles and Lines

Q029 Written

Find the equation of the line through (2,-7) parallel to y-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 2$.

Final answer

$$x = 2$$

Q030 Written

Find the equation of the line through (4,3) parallel to y-axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 4$.

Final answer

$$x = 4$$

Worked Solutions

Triangles and Lines

Q031 Written

Find the equation of the line through $(-6, -2)$ parallel to y -axis.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -6$.

Final answer

$$x = -6$$

Q032 MCQ

Find the equation of the line through $(3, -4)$ parallel to the x -axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -4$.

Correct option: B**Final answer**

$$y = -4$$

Worked Solutions

Triangles and Lines

Q033 MCQ

Find the equation of the line through $(-3,8)$ parallel to the y-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: C**Final answer**

$$x = -3$$

Q034 MCQ

Find the equation of the line through $(2,6)$ parallel to the x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 6$.

Correct option: D**Final answer**

$$y = 6$$

Worked Solutions

Triangles and Lines

Q035 MCQ

Find the equation of the line through (3,5) perpendicular to the x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 3$.

Correct option: D**Final answer**

$$x = 3$$

Q036 MCQ

Find the equation of the line through (-4,5) parallel to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 5$.

Correct option: B**Final answer**

$$y = 5$$

Worked Solutions

Triangles and Lines

Q037 MCQ

Find the equation of the line through (2,1) parallel to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 1$.

Correct option: D**Final answer**

$$y = 1$$

Q038 MCQ

Find the equation of the line through (-4,-7) parallel to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -7$.

Correct option: B**Final answer**

$$y = -7$$

Worked Solutions

Triangles and Lines

Q039 MCQ

Find the equation of the line through (1,-2) perpendicular to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 1$.

Correct option: C**Final answer**

$$x = 1$$

Q040 MCQ

Find the equation of the line through (-3,-4) perpendicular to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: D**Final answer**

$$x = -3$$

Worked Solutions

Triangles and Lines

Q041 MCQ

Find the equation of the line through (3,-5) perpendicular to x-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 3$.

Correct option: D**Final answer**

$$x = 3$$

Q042 MCQ

Find the equation of the line through (2,-7) parallel to y-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 2$.

Correct option: B**Final answer**

$$x = 2$$

Worked Solutions

Triangles and Lines

Q043 MCQ

Find the equation of the line through (4,3) parallel to y-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 4$.

Correct option: D**Final answer**

$$x = 4$$

Q044 MCQ

Find the equation of the line through (-6,-2) parallel to y-axis:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -6$.

Correct option: A**Final answer**

$$x = -6$$

Worked Solutions

Triangles and Lines

Q045 Written

Find the equation of the line through $(-3,1)$ and $(-3,2)$.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -3$.

Final answer

$$x = -3$$

Q046 Written

Find the equation of the line through $(-2,3)$ and $(-5,3)$.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3$.

Final answer

$$y = 3$$

Worked Solutions

Triangles and Lines

Q047 Written

Find the equation of the line through (1,-1) and (4,0).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$).

Final answer

$$y = x/3 - 4/3 \text{ (equivalently } x - 3y - 4 = 0 \text{)}$$

Q048 Written

Find the equation of the line through (-2,-2) and (2,-2).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2$.

Final answer

$$y = -2$$

Worked Solutions

Triangles and Lines

Q049 Written

Find the equation of the line through $(-4,2)$ and $(-4,8)$.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -4$.

Final answer

$$x = -4$$

Q050 Written

Find the equation of the line through $(-2,-6)$ and $(-4,1)$.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $7x + 2y + 26 = 0$.

Final answer

$$7x + 2y + 26 = 0$$

Worked Solutions

Triangles and Lines

Q051 Written

Find the equation of the line through (4,0) and (4,3).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = 4$.

Final answer

$$x = 4$$

Q052 Written

Find the equation of the line through (-1,2) and (-1,6).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -1$.

Final answer

$$x = -1$$

Worked Solutions

Triangles and Lines

Q053 Written

Find the equation of the line through $(-5,-3)$ and $(-5,2)$.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $x = -5$.

Final answer

$$x = -5$$

Q054 Written

Find the equation of the line through $(-2,3)$ and $(-3,3)$.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3$.

Final answer

$$y = 3$$

Worked Solutions

Triangles and Lines

Q055 Written

Find the equation of the line through (4,-2) and (1,-2).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2$.

Final answer

$$y = -2$$

Q056 Written

Find the equation of the line through (-3,6) and (2,6).**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 6$.

Final answer

$$y = 6$$

Worked Solutions

Triangles and Lines

Q057 **Written****Find the equation of the line through (1,2) and (3,10).****Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 4x - 2$.

Final answer

$$y = 4x - 2$$

Q058 **Written****Find the equation of the line through (-1,-8) and (4,2).****Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 2x - 6$.

Final answer

$$y = 2x - 6$$

Worked Solutions

Triangles and Lines

Q059 MCQ

Find the equation of the line through $(-3,1)$ and $(-3,2)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -3$.

Correct option: B**Final answer**

$$x = -3$$

Q060 MCQ

Find the equation of the line through $(-2,3)$ and $(-5,3)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3$.

Correct option: C**Final answer**

$$y = 3$$

Worked Solutions

Triangles and Lines

Q061 MCQ

Find the equation of the line through (1,-1) and (4,0):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$).

Correct option: D**Final answer**

$$y = x/3 - 4/3 \text{ (equivalently } x - 3y - 4 = 0 \text{)}$$

Q062 MCQ

Find the equation of the line through (-2,-2) and (2,-2):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2$.

Correct option: A**Final answer**

$$y = -2$$

Worked Solutions

Triangles and Lines

Q063 MCQ

Find the equation of the line through $(-4,2)$ and $(-4,8)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -4$.

Correct option: B**Final answer**

$$x = -4$$

Q064 MCQ

Find the equation of the line through $(-2,-6)$ and $(-4,1)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $7x + 2y + 26 = 0$.

Correct option: D**Final answer**

$$7x + 2y + 26 = 0$$

Worked Solutions

Triangles and Lines

Q065 MCQ

Find the equation of the line through (4,0) and (4,3):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = 4$.

Correct option: C**Final answer**

$$x = 4$$

Q066 MCQ

Find the equation of the line through (-1,2) and (-1,6):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -1$.

Correct option: A**Final answer**

$$x = -1$$

Worked Solutions

Triangles and Lines

Q067 MCQ

Find the equation of the line through $(-5,-3)$ and $(-5,2)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x = -5$.

Correct option: C**Final answer**

$$x = -5$$

Q068 MCQ

Find the equation of the line through $(-2,3)$ and $(-3,3)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3$.

Correct option: D**Final answer**

$$y = 3$$

Worked Solutions

Triangles and Lines

Q069 MCQ

Find the equation of the line through (4,-2) and (1,-2):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2$.

Correct option: D**Final answer**

$$y = -2$$

Q070 MCQ

Find the equation of the line through (-3,6) and (2,6):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 6$.

Correct option: C**Final answer**

$$y = 6$$

Worked Solutions

Triangles and Lines

Q071 MCQ

Find the equation of the line through (1,2) and (3,10):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 4x - 2$.

Correct option: D**Final answer**

$$y = 4x - 2$$

Q072 MCQ

Find the equation of the line through (-1,-8) and (4,2):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2x - 6$.

Correct option: D**Final answer**

$$y = 2x - 6$$

Worked Solutions

Triangles and Lines

Q073 Written

Find the equation of the line through (2,-1) with gradient 3.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 3x - 7$.

Final answer

$$y = 3x - 7$$

Q074 Written

Find the equation of the line through (-3,4) with gradient 2.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = 2x + 10$.

Final answer

$$y = 2x + 10$$

Worked Solutions

Triangles and Lines

Q075 Written

Find the equation of the line through (-1,5) with gradient -1.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -x + 4$.

Final answer

$$y = -x + 4$$

Q076 Written

Find the equation of the line through (-3,-8) with gradient 1.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = x - 5$.

Final answer

$$y = x - 5$$

Worked Solutions

Triangles and Lines

Q077 Written

Find the equation of the line through (2,-1) with gradient -2.**Solution**

1. Recognise $x = k$ or $y = k$; otherwise calculate m .
2. Use $y - y_1 = m(x - x_1)$ and simplify.
3. The line is $y = -2x + 3$.

Final answer

$$y = -2x + 3$$

Q078 MCQ

Find the equation of the line through (2,-1) with gradient 3:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3x - 7$.

Correct option: D**Final answer**

$$y = 3x - 7$$

Worked Solutions

Triangles and Lines

Q079 MCQ

Find the equation of the line through (-3,4) with gradient 2:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2x + 10$.

Correct option: C**Final answer**

$$y = 2x + 10$$

Q080 MCQ

Find the equation of the line through (-1,5) with gradient -1:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -x + 4$.

Correct option: D**Final answer**

$$y = -x + 4$$

Worked Solutions

Triangles and Lines

Q081 MCQ

Find the equation of the line through $(-3, -8)$ with gradient 1:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = x - 5$.

Correct option: B**Final answer**

$$y = x - 5$$

Q082 MCQ

Find the equation of the line through $(2, -1)$ with gradient -2:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = -2x + 3$.

Correct option: D**Final answer**

$$y = -2x + 3$$

Worked Solutions

Triangles and Lines

Q083 Challenge

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $y = 2$.

Final answer

$$y = 2$$

Worked Solutions

Triangles and Lines

Q084 Challenge

Sensors A $(-4, 2)$ and B $(9, 2)$ have the same height. Find the straight-line boundary through A and B:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 2$.

Correct option: D

Final answer

$$y = 2$$

Worked Solutions

Triangles and Lines

Q085 **Written**

A (2, 3), B (4, -1), and C ($a - 1$, $2 - a$) are collinear. Find a .

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 7$.

Final answer

$$a = 7$$

Q086 **Written**

A (2, 5), B (4, 9), and C (-1 , a) are collinear. Find a .

Solution

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = -1$.

Final answer

$$a = -1$$

Worked Solutions

Triangles and Lines

Q087 **Written****A (1, -1), B (-2, -3), and C (4, a) are collinear. Find a.****Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 1$.

Final answer

$$a = 1$$

Q088 **Written****A (1, 2), B (3, 1), and C (a, -1) are collinear. Find a.****Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 7$.

Final answer

$$a = 7$$

Worked Solutions

Triangles and Lines

Q089 **Written****A** $(-3, -9)$, **B** $(5, 7)$, and **C** $(a + 3, 3a - 2)$ are collinear. Find a .**Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $a = 5$.

Final answer

$$a = 5$$

Q090 **MCQ****A** $(2, 3)$, **B** $(4, -1)$, and **C** $(a - 1, 2 - a)$ are collinear. Find a :**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 7$.

Correct option: D**Final answer**

$$a = 7$$

Worked Solutions

Triangles and Lines

Q091 MCQ

A (2, 5), B (4, 9), and C (-1, a) are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -1$.

Correct option: D**Final answer**

$$a = -1$$

Q092 MCQ

A (1, -1), B (-2, -3), and C (4, a) are collinear. Find a:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 1$.

Correct option: C**Final answer**

$$a = 1$$

Worked Solutions

Triangles and Lines

Q093 MCQ

A (1, 2), **B** (3, 1), and **C** (a , -1) are collinear. Find a :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 7$.

Correct option: D**Final answer**

$$a = 7$$

Q094 MCQ

A (-3 , -9), **B** (5, 7), and **C** ($a + 3$, $3a - 2$) are collinear. Find a :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 5$.

Correct option: D**Final answer**

$$a = 5$$

Worked Solutions

Triangles and Lines

Q095 Written

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1,3)$.**Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + y - 2 = 0$.

Final answer

$$x + y - 2 = 0$$

Q096 Written

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through $(5,6)$.**Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + 3y - 23 = 0$.

Final answer

$$x + 3y - 23 = 0$$

Worked Solutions

Triangles and Lines

Q097 Written

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1, 2)$.**Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $x + y - 1 = 0$.

Final answer

$$x + y - 1 = 0$$

Q098 Written

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3, -1)$.**Solution**

1. Find the relevant line or simultaneous intersection first.
2. Impose the collinearity or fixed-point condition and simplify.
3. The checked result is $2x - 5y + 1 = 0$.

Final answer

$$2x - 5y + 1 = 0$$

Worked Solutions

Triangles and Lines

Q099 MCQ

Find the line through the intersection of $2x - 3y + 1 = 0$ and $x + 2y - 3 = 0$, passing through $(-1,3)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + y - 2 = 0$.

Correct option: B**Final answer**

$$x + y - 2 = 0$$

Q100 MCQ

Find the line through the intersection of $x + 2y - 10 = 0$ and $2x + 3y - 7 = 0$, passing through $(5,6)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + 3y - 23 = 0$.

Correct option: A**Final answer**

$$x + 3y - 23 = 0$$

Worked Solutions

Triangles and Lines

Q101 MCQ

Find the line through the intersection of $3x + 5y - 1 = 0$ and $7x + 8y - 6 = 0$, passing through $(-1,2)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + y - 1 = 0$.

Correct option: B**Final answer**

$$x + y - 1 = 0$$

Q102 MCQ

Find the line through the intersection of $x + 2y - 4 = 0$ and $2x - y - 3 = 0$, passing through $(-3,-1)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - 5y + 1 = 0$.

Correct option: D**Final answer**

$$2x - 5y + 1 = 0$$

Worked Solutions

Triangles and Lines

Q103 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $3x + 13y = 0$.

Final answer

$$3x + 13y = 0$$

Worked Solutions

Triangles and Lines

Q104 Challenge

The intersection of $6x + 5y + 7 = 0$ and $3x - 2y + 5 = 0$ is joined to the origin. Find the equation of that trajectory:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x + 13y = 0$.

Correct option: C

Final answer

$$3x + 13y = 0$$

Worked Solutions

Triangles and Lines

Q105 **Written****Find the gradient of line $y = 2x + 3$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$$m = 2$$

Q106 **Written****Find the gradient of line $y = -3x + 2$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$$m = -3$$

Worked Solutions

Triangles and Lines

Q107 Written

Find the gradient of line $x - 3y + 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1/3$.

Final answer

$$m = 1/3$$

Q108 Written

Find the gradient of line $2x - y - 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$$m = 2$$

Worked Solutions

Triangles and Lines

Q109 Written

Find the gradient of line $x + 3y + 3 = 0$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/3$.

Final answer

$$m = -1/3$$

Q110 Written

Find the gradient of line $y = 3x + 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$$m = 3$$

Worked Solutions

Triangles and Lines

Q111 Written

Find the gradient of line $y = 4x - 1$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 4$.

Final answer

$$m = 4$$

Q112 Written

Find the gradient of line $y = -4x - 7$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -4$.

Final answer

$$m = -4$$

Worked Solutions

Triangles and Lines

Q113 Written

Find the gradient of line $6x - 2y = 5$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$$m = 3$$

Q114 Written

Find the gradient of line $8x + 2y - 3 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -4$.

Final answer

$$m = -4$$

Worked Solutions

Triangles and Lines

Q115 Written

Find the gradient of line $x + 4y - 4 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/4$.

Final answer

$$m = -1/4$$

Q116 Written

Find the gradient of line $y = 3x - 1$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3$.

Final answer

$$m = 3$$

Worked Solutions

Triangles and Lines

Q117 Written

Find the gradient of line $y = -3x + 4$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$$m = -3$$

Q118 Written

Find the gradient of line $y = x - 3$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1$.

Final answer

$$m = 1$$

Worked Solutions

Triangles and Lines

Q119 Written

Find the gradient of line $6x + 2y + 5 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -3$.

Final answer

$$m = -3$$

Q120 Written

Find the gradient of line $2x - 2y - 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 1$.

Final answer

$$m = 1$$

Worked Solutions

Triangles and Lines

Q121 Written

Find the gradient of line $x + 3y - 2 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/3$.

Final answer

$$m = -1/3$$

Q122 Written

Find the gradient of line $y = 2x - 1$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 2$.

Final answer

$$m = 2$$

Worked Solutions

Triangles and Lines

Q123 Written

Find the gradient of line $y = -x + 4$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1$.

Final answer

$$m = -1$$

Q124 Written

Find the gradient of line $y = 3x/2 + 6$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = 3/2$.

Final answer

$$m = 3/2$$

Worked Solutions

Triangles and Lines

Q125 Written

Find the gradient of line $x + y + 3 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1$.

Final answer

$$m = -1$$

Q126 Written

Find the gradient of line $2x + 3y + 8 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -2/3$.

Final answer

$$m = -2/3$$

Worked Solutions

Triangles and Lines

Q127 Written

Find the gradient of line $3x + 6y - 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $m = -1/2$.

Final answer

$$m = -1/2$$

Q128 MCQ

Find the gradient of line $y = 2x + 3$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: D**Final answer**

$$m = 2$$

Worked Solutions

Triangles and Lines

Q129 MCQ

Find the gradient of line $y = -3x + 2$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: C**Final answer**

$$m = -3$$

Q130 MCQ

Find the gradient of line $x - 3y + 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1/3$.

Correct option: A**Final answer**

$$m = 1/3$$

Worked Solutions

Triangles and Lines

Q131 MCQ

Find the gradient of line $2x - y - 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: B**Final answer**

$$m = 2$$

Q132 MCQ

Find the gradient of line $x + 3y + 3 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/3$.

Correct option: A**Final answer**

$$m = -1/3$$

Worked Solutions

Triangles and Lines

Q133 MCQ

Find the gradient of line $y = 3x + 1$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: A**Final answer**

$$m = 3$$

Q134 MCQ

Find the gradient of line $y = 4x - 1$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 4$.

Correct option: B**Final answer**

$$m = 4$$

Worked Solutions

Triangles and Lines

Q135 MCQ

Find the gradient of line $y = -4x - 7$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -4$.

Correct option: D**Final answer**

$$m = -4$$

Q136 MCQ

Find the gradient of line $6x - 2y = 5$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: A**Final answer**

$$m = 3$$

Worked Solutions

Triangles and Lines

Q137 MCQ

Find the gradient of line $8x + 2y - 3 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -4$.

Correct option: C**Final answer**

$$m = -4$$

Q138 MCQ

Find the gradient of line $x + 4y - 4 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/4$.

Correct option: C**Final answer**

$$m = -1/4$$

Worked Solutions

Triangles and Lines

Q139 MCQ

Find the gradient of line $y = 3x - 1$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3$.

Correct option: D**Final answer**

$$m = 3$$

Q140 MCQ

Find the gradient of line $y = -3x + 4$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: A**Final answer**

$$m = -3$$

Worked Solutions

Triangles and Lines

Q141 MCQ

Find the gradient of line $y = x - 3$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1$.

Correct option: D**Final answer**

$$m = 1$$

Q142 MCQ

Find the gradient of line $6x + 2y + 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -3$.

Correct option: C**Final answer**

$$m = -3$$

Worked Solutions

Triangles and Lines

Q143 MCQ

Find the gradient of line $2x - 2y - 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 1$.

Correct option: C**Final answer**

$$m = 1$$

Q144 MCQ

Find the gradient of line $x + 3y - 2 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/3$.

Correct option: A**Final answer**

$$m = -1/3$$

Worked Solutions

Triangles and Lines

Q145 MCQ

Find the gradient of line $y = 2x - 1$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 2$.

Correct option: D**Final answer**

$$m = 2$$

Q146 MCQ

Find the gradient of line $y = -x + 4$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1$.

Correct option: A**Final answer**

$$m = -1$$

Worked Solutions

Triangles and Lines

Q147 MCQ

Find the gradient of line $y = 3x/2 + 6$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = 3/2$.

Correct option: B**Final answer**

$$m = 3/2$$

Q148 MCQ

Find the gradient of line $x + y + 3 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1$.

Correct option: D**Final answer**

$$m = -1$$

Worked Solutions

Triangles and Lines

Q149 MCQ

Find the gradient of line $2x + 3y + 8 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -2/3$.

Correct option: B**Final answer**

$$m = -2/3$$

Q150 MCQ

Find the gradient of line $3x + 6y - 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m = -1/2$.

Correct option: B**Final answer**

$$m = -1/2$$

Worked Solutions

Triangles and Lines

Q151 **Written****Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x - 2y - 14 = 0$.

Final answer

$$3x - 2y - 14 = 0$$

Q152 **Written****Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + y + 1 = 0$.

Final answer

$$2x + y + 1 = 0$$

Worked Solutions

Triangles and Lines

Q153 **Written****Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x - y - 4 = 0$.

Final answer

$$2x - y - 4 = 0$$

Q154 **Written****Find the equation of the line through (-1,2) parallel to $2x + 3y = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 3y - 4 = 0$.

Final answer

$$2x + 3y - 4 = 0$$

Worked Solutions

Triangles and Lines

Q155 Written

Find the equation of the line through (3,-1) parallel to $3x + 2y + 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x + 2y - 7 = 0$.

Final answer

$$3x + 2y - 7 = 0$$

Q156 MCQ

Find the line through (2,-4) parallel to $3x - 2y + 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 14 = 0$.

Correct option: C**Final answer**

$$3x - 2y - 14 = 0$$

Worked Solutions

Triangles and Lines

Q157 MCQ

Find the line through (1,-3) parallel to $6x + 3y - 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + y + 1 = 0$.

Correct option: D**Final answer**

$$2x + y + 1 = 0$$

Q158 MCQ

Find the equation of the line through (1,-2) parallel to $2x - y + 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y - 4 = 0$.

Correct option: C**Final answer**

$$2x - y - 4 = 0$$

Worked Solutions

Triangles and Lines

Q159 MCQ

Find the equation of the line through $(-1,2)$ parallel to $2x + 3y = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 3y - 4 = 0$.

Correct option: D**Final answer**

$$2x + 3y - 4 = 0$$

Q160 MCQ

Find the equation of the line through $(3,-1)$ parallel to $3x + 2y + 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x + 2y - 7 = 0$.

Correct option: D**Final answer**

$$3x + 2y - 7 = 0$$

Worked Solutions

Triangles and Lines

Q161 **Written****Find the line through $(-2,5)$ perpendicular to $3x + 5y + 1 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $5x - 3y + 25 = 0$.

Final answer

$$5x - 3y + 25 = 0$$

Q162 **Written****Find the line through $(2,1)$ perpendicular to $2x + 3y + 4 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $3x - 2y - 4 = 0$.

Final answer

$$3x - 2y - 4 = 0$$

Worked Solutions

Triangles and Lines

Q163 Written

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $x - 2y - 12 = 0$.

Final answer

$$x - 2y - 12 = 0$$

Q164 Written

Find the equation of the line through (-1,3) perpendicular to $5x - 2y + 3 = 0$.**Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 5y - 13 = 0$.

Final answer

$$2x + 5y - 13 = 0$$

Worked Solutions

Triangles and Lines

Q165 **Written****Find the equation of the line through $(-2, -5)$ perpendicular to $3x - 2y - 7 = 0$.****Solution**

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $2x + 3y + 19 = 0$.

Final answer

$$2x + 3y + 19 = 0$$

Q166 **MCQ****Find the line through $(-2, 5)$ perpendicular to $3x + 5y + 1 = 0$:****Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $5x - 3y + 25 = 0$.

Correct option: D**Final answer**

$$5x - 3y + 25 = 0$$

Worked Solutions

Triangles and Lines

Q167 MCQ

Find the line through (2,1) perpendicular to $2x + 3y + 4 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 4 = 0$.

Correct option: D**Final answer**

$$3x - 2y - 4 = 0$$

Q168 MCQ

Find the equation of the line through (2,-5) perpendicular to $2x + y - 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x - 2y - 12 = 0$.

Correct option: B**Final answer**

$$x - 2y - 12 = 0$$

Worked Solutions

Triangles and Lines

Q169 MCQ

Find the equation of the line through $(-1,3)$ perpendicular to $5x - 2y + 3 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 5y - 13 = 0$.

Correct option: A**Final answer**

$$2x + 5y - 13 = 0$$

Q170 MCQ

Find the equation of the line through $(-2,-5)$ perpendicular to $3x - 2y - 7 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x + 3y + 19 = 0$.

Correct option: D**Final answer**

$$2x + 3y + 19 = 0$$

Worked Solutions

Triangles and Lines

Q171 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k .

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $m/k = 2$ ($k = 3, m = 6$).

Final answer

$$m/k = 2 \text{ (} k = 3, m = 6 \text{)}$$

Worked Solutions

Triangles and Lines

Q172 Challenge

If $L1 : 2x + 3y - 10 = 0$ is perpendicular to $L2 : kx - 2y + 5 = 0$ and parallel to $L3 : 4x + my - 1 = 0$, find m/k :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $m/k = 2$ ($k = 3$, $m = 6$).

Correct option: D

Final answer

$$m/k = 2 \ (k = 3, \ m = 6)$$

Worked Solutions

Triangles and Lines

Q173 Written

For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB.**Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $x + 3y - 18 = 0$.

Final answer

$$x + 3y - 18 = 0$$

Q174 Written

For A (-1, 2), B (3, -4), find the equation of the perpendicular bisector of AB.**Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - 3y - 5 = 0$.

Final answer

$$2x - 3y - 5 = 0$$

Worked Solutions

Triangles and Lines

Q175 **Written****Find the perpendicular bisector of segment A $(-2, 3)$, B $(2, 1)$.****Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - y + 2 = 0$.

Final answer

$$2x - y + 2 = 0$$

Q176 **Written****Find the perpendicular bisector of segment A $(-1, 2)$, B $(5, -2)$.****Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $3x - 2y - 6 = 0$.

Final answer

$$3x - 2y - 6 = 0$$

Worked Solutions

Triangles and Lines

Q177 **Written****Find the perpendicular bisector of segment A (0, 6), B (4, 4).****Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $2x - y + 1 = 0$.

Final answer

$$2x - y + 1 = 0$$

Q178 **MCQ****For A (5, 11), B (1, -1), find the equation of the perpendicular bisector of AB:****Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $x + 3y - 18 = 0$.

Correct option: A**Final answer**

$$x + 3y - 18 = 0$$

Worked Solutions

Triangles and Lines

Q179 MCQ

For A $(-1, 2)$, B $(3, -4)$, find the equation of the perpendicular bisector of AB:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - 3y - 5 = 0$.

Correct option: C**Final answer**

$$2x - 3y - 5 = 0$$

Q180 MCQ

Find the perpendicular bisector of segment A $(-2, 3)$, B $(2, 1)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y + 2 = 0$.

Correct option: C**Final answer**

$$2x - y + 2 = 0$$

Worked Solutions

Triangles and Lines

Q181 MCQ

Find the perpendicular bisector of segment A $(-1, 2)$, B $(5, -2)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3x - 2y - 6 = 0$.

Correct option: D**Final answer**

$$3x - 2y - 6 = 0$$

Q182 MCQ

Find the perpendicular bisector of segment A $(0, 6)$, B $(4, 4)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $2x - y + 1 = 0$.

Correct option: A**Final answer**

$$2x - y + 1 = 0$$

Worked Solutions

Triangles and Lines

Q183 **Written****Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$.****Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (2, 1)$.

Final answer

$$B = (2, 1)$$

Q184 **Written****Find the point B symmetric to A $(0, 4)$ with respect to $2x - y - 1 = 0$.****Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (4, 2)$.

Final answer

$$B = (4, 2)$$

Worked Solutions

Triangles and Lines

Q185 Written

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$.**Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (3, 6)$.

Final answer

$$B = (3, 6)$$

Q186 Written

Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$.**Solution**

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $B = (-2, 2)$.

Final answer

$$B = (-2, 2)$$

Worked Solutions

Triangles and Lines

Q187 MCQ

Find the point B symmetric to A $(-4, 3)$ with respect to $3x - y + 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (2, 1)$.

Correct option: A**Final answer**

$$B = (2, 1)$$

Q188 MCQ

Find the point B symmetric to A $(0, 4)$ with respect to $2x - y - 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (4, 2)$.

Correct option: A**Final answer**

$$B = (4, 2)$$

Worked Solutions

Triangles and Lines

Q189 MCQ

Find the point B symmetric to A (1, 2) with respect to $x + 2y - 10 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (3, 6)$.

Correct option: C**Final answer**

$$B = (3, 6)$$

Q190 MCQ

Find the point B symmetric to A (4, -1) with respect to $4x - 2y - 3 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (-2, 2)$.

Correct option: C**Final answer**

$$B = (-2, 2)$$

Worked Solutions

Triangles and Lines

Q191 Challenge

Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $B = (-3, 4)$.

Final answer

$$B = (-3, 4)$$

Worked Solutions

Triangles and Lines

Q192 Challenge

Reflect A (5, 0) across the targeting line $y = 2x$. Find the receiver point B:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $B = (-3, 4)$.

Correct option: C

Final answer

$$B = (-3, 4)$$

Worked Solutions

Triangles and Lines

Q193 Written

Find the distance from $(-2,1)$ to $3x - 2y + 5 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $3/\sqrt{13} = 3\sqrt{13}/13$.

Final answer

$$3/\sqrt{13} = 3\sqrt{13}/13$$

Q194 Written

Find the distance from $(3,-2)$ to $x + 2y - 4 = 0$.

Solution

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{5}$.

Final answer

$$\sqrt{5}$$

Worked Solutions

Triangles and Lines

Q195 Written

Find the perpendicular distance from $(-3,2)$ to $2x - 3y + 6 = 0$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $6/\sqrt{13} = 6\sqrt{13}/13$.

Final answer

$$6/\sqrt{13} = 6\sqrt{13}/13$$

Q196 Written

Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 6.

Final answer

6

Worked Solutions

Triangles and Lines

Q197 **Written****Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$.****Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $4\sqrt{5}$.

Final answer

$$4\sqrt{5}$$

Q198 **Written****Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$.****Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 1.

Final answer

$$1$$

Worked Solutions

Triangles and Lines

Q199 Written

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{13}$.

Final answer $\sqrt{13}$

Q200 Written

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: 10.

Final answer

10

Worked Solutions

Triangles and Lines

Q201 Written

Find the perpendicular distance from (1,2) to $y = 3x + 1$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}/5$.

Final answer

$$\sqrt{10}/5$$

Q202 MCQ

Find the distance from (-2,1) to $3x - 2y + 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $3/\sqrt{13} = 3\sqrt{13}/13$.

Correct option: A**Final answer**

$$3/\sqrt{13} = 3\sqrt{13}/13$$

Worked Solutions

Triangles and Lines

Q203 MCQ

Find the distance from (3,-2) to $x + 2y - 4 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{5}$.

Correct option: D**Final answer**

$$\sqrt{5}$$

Q204 MCQ

Find the perpendicular distance from (-3,2) to $2x - 3y + 6 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $6/\sqrt{13} = 6\sqrt{13}/13$.

Correct option: D**Final answer**

$$6/\sqrt{13} = 6\sqrt{13}/13$$

Worked Solutions

Triangles and Lines

Q205 MCQ

Find the perpendicular distance from $(-3,7)$ to $12x - 5y = 7$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 6.

Correct option: B**Final answer**

6

Q206 MCQ

Find the perpendicular distance from $(-8,1)$ to $y = 2x - 3$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $4\sqrt{5}$.

Correct option: D**Final answer** $4\sqrt{5}$

Worked Solutions

Triangles and Lines

Q207 MCQ

Find the perpendicular distance from the origin to $3x - 4y - 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 1.

Correct option: D**Final answer**

1

Q208 MCQ

Find the perpendicular distance from (2,3) to $3x + 2y + 1 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{13}$.

Correct option: A**Final answer** $\sqrt{13}$

Worked Solutions

Triangles and Lines

Q209 MCQ

Find the perpendicular distance from (13,4) to $3x + 4y - 5 = 0$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 10.

Correct option: B**Final answer**

10

Q210 MCQ

Find the perpendicular distance from (1,2) to $y = 3x + 1$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}/5$.

Correct option: D**Final answer** $\sqrt{10}/5$

Worked Solutions

Triangles and Lines

Q211 Written

Find the perpendicular length from the origin to $3x - y = 5$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}/2$.

Final answer

$$\sqrt{10}/2$$

Q212 Written

Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$.**Solution**

1. Rewrite the line as $ax + by + c = 0$.
2. Substitute the point into $d = |ax_1 + by_1 + c|/\sqrt{a^2 + b^2}$.
3. Distance: $\sqrt{10}$.

Final answer

$$\sqrt{10}$$

Worked Solutions

Triangles and Lines

Q213 MCQ

Find the perpendicular length from the origin to $3x - y = 5$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}/2$.

Correct option: D**Final answer**

$$\sqrt{10}/2$$

Q214 MCQ

Find the perpendicular length from $(-1,5)$ to $y = 3x - 2$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: B**Final answer**

$$\sqrt{10}$$

Worked Solutions

Triangles and Lines

Q215 Challenge

Find the shortest distance from P (−1, 5) to the radiation line $y = x/3 + 2$.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: $\sqrt{10}$.

Final answer

$$\sqrt{10}$$

Worked Solutions

Triangles and Lines

Q216 Challenge

Find the shortest distance from P $(-1, 5)$ to the radiation line $y = x/3 + 2$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $\sqrt{10}$.

Correct option: A

Final answer

$\sqrt{10}$

Worked Solutions

Triangles and Lines

Q217 **Written****Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1).****Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 10.

Final answer

10

Q218 **Written****Find the area for A (1, 5), B (−3, −3), C (3, −6).****Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 30.

Final answer

30

Worked Solutions

Triangles and Lines

Q219 **Written****Find the area for A** $(-2, 1)$, **B** $(2, -1)$, **C** $(0, 4)$.**Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 8.

Final answer

8

Q220 **Written****Find the area for A** $(0, -6)$, **B** $(2, -2)$, **C** $(1, 5)$.**Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 9.

Final answer

9

Worked Solutions

Triangles and Lines

Q221 **Written****Find the area for A** $(-2, 4)$, **B** $(-3, -5)$, **C** $(5, -1)$.**Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 34.

Final answer

34

Q222 **Written****Find the area for A** $(2, 5)$, **B** $(0, 1)$, **C** $(4, 2)$.**Solution**

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 7.

Final answer

7

Worked Solutions

Triangles and Lines

Q223 MCQ

Find the area of triangle ABC with A (1, 1), B (3, 7), C (−3, −1):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 10.

Correct option: D**Final answer**

10

Q224 MCQ

Find the area for A (1, 5), B (−3, −3), C (3, −6):**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 30.

Correct option: C**Final answer**

30

Worked Solutions

Triangles and Lines

Q225 MCQ

Find the area for A $(-2, 1)$, B $(2, -1)$, C $(0, 4)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 8.

Correct option: A**Final answer**

8

Q226 MCQ

Find the area for A $(0, -6)$, B $(2, -2)$, C $(1, 5)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 9.

Correct option: B**Final answer**

9

Worked Solutions

Triangles and Lines

Q227 MCQ

Find the area for A $(-2, 4)$, B $(-3, -5)$, C $(5, -1)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 34.

Correct option: B**Final answer**

34

Q228 MCQ

Find the area for A $(2, 5)$, B $(0, 1)$, C $(4, 2)$:**Solution**

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 7.

Correct option: D**Final answer**

7

Worked Solutions

Triangles and Lines

Q229 Challenge

Find the area of the wound triangle A $(-2, 1)$, **B** $(1, 5)$, **C** $(-21, -21)$.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: 5.

Final answer

5

Worked Solutions

Triangles and Lines

Q230 Challenge

Find the area of the wound triangle A $(-2, 1)$, B $(1, 5)$, C $(-21, -21)$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 5.

Correct option: D

Final answer

5

Worked Solutions

Triangles and Lines

Q231 **Written**

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(1/2, -3/2)$.

Final answer $(1/2, -3/2)$ Q232 **Written**

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(-2, 3)$.

Final answer $(-2, 3)$

Worked Solutions

Triangles and Lines

Q233 Written

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(3, -3)$.

Final answer $(3, -3)$

Q234 Written

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(1, 1)$.

Final answer $(1, 1)$

Worked Solutions

Triangles and Lines

Q235 **Written**

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it.

Solution

1. Collect the coefficient of k and the constant part.
2. Set both parts to zero and solve the resulting simultaneous equations.
3. Fixed point: $(-3, 1)$.

Final answer $(-3, 1)$ Q236 **MCQ**

The line $(k - 4)x + (k + 2)y + k + 5 = 0$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(1/2, -3/2)$.

Correct option: D

Final answer $(1/2, -3/2)$

Worked Solutions

Triangles and Lines

Q237 MCQ

The line $(2k + 1)x - (k + 2)y + 7k + 8 = 0$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(-2, 3)$.

Correct option: B**Final answer** $(-2, 3)$

Q238 MCQ

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(3, -3)$.

Correct option: D**Final answer** $(3, -3)$

Worked Solutions

Triangles and Lines

Q239 MCQ

The line $(k + 3)x - (2k + 1)y + k - 2 = 0$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is (1, 1).

Correct option: D**Final answer**

(1, 1)

Q240 MCQ

The line $(k + 3)x + (3k + 2)y + 7 = 0$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $(-3, 1)$.

Correct option: A**Final answer** $(-3, 1)$

Worked Solutions

Triangles and Lines

Q241 Challenge

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it.

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: (4, 1).

Final answer

(4, 1)

Worked Solutions

Triangles and Lines

Q242 Challenge

The line $-(k - 2)x + (k - 3)y = -3k + 5$ passes through a fixed point for every k . Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is (4, 1).

Correct option: A

Final answer

(4, 1)

Worked Solutions

Triangles and Lines

Q243 Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -4, -3, \text{ or } 6$.

Final answer

$$a = -4, -3, \text{ or } 6$$

Q244 Written

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a .

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -1, -1/2, \text{ or } 1$.

Final answer

$$a = -1, -1/2, \text{ or } 1$$

Worked Solutions

Triangles and Lines

Q245 Written

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a .

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = 11/10$, $3/2$, or $-9/2$.

Final answer

$$a = 11/10, 3/2, \text{ or } -9/2$$

Q246 Written

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a .

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = 1$, $-3/2$, or 2 .

Final answer

$$a = 1, -3/2, \text{ or } 2$$

Worked Solutions

Triangles and Lines

Q247 MCQ

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -4, -3, \text{ or } 6$.

Correct option: A

Final answer $a = -4, -3, \text{ or } 6$

Q248 MCQ

For $x - 2y + 8 = 0$, $x + y - 1 = 0$, and $ax + y - 5 = 0$ not to form a triangle, find a :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = -1, -1/2, \text{ or } 1$.

Correct option: A

Final answer $a = -1, -1/2, \text{ or } 1$

Worked Solutions

Triangles and Lines

Q249 MCQ

For $x - 2y + 1 = 0$, $3x + 2y = 6$, and $ax - 3y + 2 = 0$ not to form a triangle, find a :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 11/10$, $3/2$, or $-9/2$.

Correct option: C**Final answer** $a = 11/10, 3/2, \text{ or } -9/2$

Q250 MCQ

For $x - y = -1$, $3x + 2y = 12$, and $ax - y = a - 1$ not to form a triangle, find a :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $a = 1$, $-3/2$, or 2 .

Correct option: D**Final answer** $a = 1, -3/2, \text{ or } 2$

Worked Solutions

Triangles and Lines

Q251 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m .

Solution

1. Translate the geometry condition into equations.
2. Solve, simplify, and check the defining condition.
3. Final result: 15.

Final answer

15

Worked Solutions

Triangles and Lines

Q252 Challenge

For $x + 2y - 1 = 0$, $3x + y + 2 = 0$, and $mx + 4y - 3 = 0$ not to form a triangle, find the sum of all possible m :

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 15.

Correct option: A

Final answer

15

Worked Solutions

Triangles and Lines

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Determine the shape of triangle A (0, 0), B (3, 0), C (0, 4):

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is *scalene right – angled*.

Correct option: C

Final answer

scalene right – angled

Quiz MCQ 2 [Concept Quiz · MCQ](#)

Find the equation of the line through (2,-1) with gradient 3:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is $y = 3x - 7$.

Correct option: B

Final answer

$y = 3x - 7$

Worked Solutions

Triangles and Lines

Quiz MCQ 3 **Concept Quiz · MCQ**

Find the perpendicular distance from (1,2) to $3x + 4y - 10 = 0$:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is 1/5.

Correct option: D

Final answer

1/5

Quiz MCQ 4 **Concept Quiz · MCQ**

The line $(k + 2)x + ky - 6 = 0$ passes through a fixed point. Find it:

Solution

1. Use the same geometry/line route as the paired written demand.
2. The correct option is (3, -3).

Correct option: C

Final answer

(3, -3)

Worked Solutions

Triangles and Lines

Quiz WRITTEN 5 Concept Quiz · Written

Find the area of triangle A (0, 0), B (4, 0), C (0, 3).

Solution

1. Choose a convenient base and find its perpendicular height.
2. Use $A = \frac{1}{2} \text{ base} \times \text{height}$.
3. Area: 6.

Final answer

6

Quiz WRITTEN 6 Concept Quiz · Written

Find the perpendicular bisector of the segment joining A (1, 2) and B (5, 4).

Solution

1. Find the midpoint and the gradient of the given segment.
2. Use the negative reciprocal gradient for the perpendicular bisector or reflection line.
3. The checked result is $y = -2x + 9$.

Final answer

$$y = -2x + 9$$

Worked Solutions

Triangles and Lines

Quiz WRITTEN 7 Concept Quiz · Written

Find the equation of the line through (1,2) perpendicular to $y = 3x - 1$.

Solution

1. Write each line in gradient form where possible.
2. Use equal gradients for parallel lines and negative reciprocal gradients for perpendicular lines.
3. The required result is $y = -x/3 + 7/3$.

Final answer

$$y = -x/3 + 7/3$$

Quiz WRITTEN 8 Concept Quiz · Written

For $x - y + 1 = 0$, $2x + y + 2 = 0$, and $ax + 3y - 4 = 0$ not to form a triangle, find a .

Solution

1. Check parallel pairs first by comparing gradients.
2. For the remaining case, test concurrence at the common intersection.
3. The parameter result is $a = -4, -3, \text{ or } 6$.

Final answer

$$a = -4, -3, \text{ or } 6$$

Answer key

Use the question number shown on each page.

Q001 <i>equilateral</i>	Q027 $x = -3$	Q053 $x = -5$	Q079 C
Q002 <i>equilateral</i>	Q028 $x = 3$	Q054 $y = 3$	Q080 D
Q003 D	Q029 $x = 2$	Q055 $y = -2$	Q081 B
Q004 C	Q030 $x = 4$	Q056 $y = 6$	Q082 D
Q005 <i>isosceles right – angled; right angle at A</i>	Q031 $x = -6$	Q057 $y = 4x - 2$	Q083 $y = 2$
Q006 <i>isosceles right – angled; right angle at A</i>	Q032 B	Q058 $y = 2x - 6$	Q084 D
Q007 B	Q033 C	Q059 B	Q085 $a = 7$
Q008 D	Q034 D	Q060 C	Q086 $a = -1$
Q009 <i>scalene right – angled; right angle at A</i>	Q035 D	Q061 D	Q087 $a = 1$
Q010 D	Q036 B	Q062 A	Q088 $a = 7$
Q011 <i>isosceles right – angled; right angle at B</i>	Q037 D	Q063 B	Q089 $a = 5$
Q012 <i>isosceles right – angled; right angle at B</i>	Q038 B	Q064 D	Q090 D
Q013 <i>isosceles right – angled; right angle at A</i>	Q039 C	Q065 C	Q091 D
Q014 B	Q040 D	Q066 A	Q092 C
Q015 C	Q041 D	Q067 C	Q093 D
Q016 B	Q042 B	Q068 D	Q094 D
Q017 $k = 2$	Q043 D	Q069 D	Q095 $x + y - 2 = 0$
Q018 C	Q044 A	Q070 C	Q096 $x + 3y - 23 = 0$
Q019 $y = -4$	Q045 $x = -3$	Q071 D	Q097 $x + y - 1 = 0$
Q020 $x = -3$	Q046 $y = 3$	Q072 D	Q098 $2x - 5y + 1 = 0$
Q021 $y = 6$	Q047 $y = x/3 - 4/3$ (equivalently $x - 3y - 4 = 0$)	Q073 $y = 3x - 7$	Q099 B
Q022 $x = 3$	Q048 $y = -2$	Q074 $y = 2x + 10$	Q100 A
Q023 $y = 5$	Q049 $x = -4$	Q075 $y = -x + 4$	Q101 B
Q024 $y = 1$	Q050 $7x + 2y + 26 = 0$	Q076 $y = x - 5$	Q102 D
Q025 $y = -7$	Q051 $x = 4$	Q077 $y = -2x + 3$	Q103 $3x + 13y = 0$
Q026 $x = 1$	Q052 $x = -1$	Q078 D	Q104 C

Q105 $m = 2$	Q137 C	Q169 A	Q201 $\sqrt{10}/5$
Q106 $m = -3$	Q138 C	Q170 D	Q202 A
Q107 $m = 1/3$	Q139 D	Q171 $m/k = 2 (k = 3, m = 6)$	Q203 D
Q108 $m = 2$	Q140 A	Q172 D	Q204 D
Q109 $m = -1/3$	Q141 D	Q173 $x + 3y - 18 = 0$	Q205 B
Q110 $m = 3$	Q142 C	Q174 $2x - 3y - 5 = 0$	Q206 D
Q111 $m = 4$	Q143 C	Q175 $2x - y + 2 = 0$	Q207 D
Q112 $m = -4$	Q144 A	Q176 $3x - 2y - 6 = 0$	Q208 A
Q113 $m = 3$	Q145 D	Q177 $2x - y + 1 = 0$	Q209 B
Q114 $m = -4$	Q146 A	Q178 A	Q210 D
Q115 $m = -1/4$	Q147 B	Q179 C	Q211 $\sqrt{10}/2$
Q116 $m = 3$	Q148 D	Q180 C	Q212 $\sqrt{10}$
Q117 $m = -3$	Q149 B	Q181 D	Q213 D
Q118 $m = 1$	Q150 B	Q182 A	Q214 B
Q119 $m = -3$	Q151 $3x - 2y - 14 = 0$	Q183 $B = (2, 1)$	Q215 $\sqrt{10}$
Q120 $m = 1$	Q152 $2x + y + 1 = 0$	Q184 $B = (4, 2)$	Q216 A
Q121 $m = -1/3$	Q153 $2x - y - 4 = 0$	Q185 $B = (3, 6)$	Q217 10
Q122 $m = 2$	Q154 $2x + 3y - 4 = 0$	Q186 $B = (-2, 2)$	Q218 30
Q123 $m = -1$	Q155 $3x + 2y - 7 = 0$	Q187 A	Q219 8
Q124 $m = 3/2$	Q156 C	Q188 A	Q220 9
Q125 $m = -1$	Q157 D	Q189 C	Q221 34
Q126 $m = -2/3$	Q158 C	Q190 C	Q222 7
Q127 $m = -1/2$	Q159 D	Q191 $B = (-3, 4)$	Q223 D
Q128 D	Q160 D	Q192 C	Q224 C
Q129 C	Q161 $5x - 3y + 25 = 0$	Q193 $3/\sqrt{13} = 3\sqrt{13}/13$	Q225 A
Q130 A	Q162 $3x - 2y - 4 = 0$	Q194 $\sqrt{5}$	Q226 B
Q131 B	Q163 $x - 2y - 12 = 0$	Q195 $6/\sqrt{13} = 6\sqrt{13}/13$	Q227 B
Q132 A	Q164 $2x + 5y - 13 = 0$	Q196 6	Q228 D
Q133 A	Q165 $2x + 3y + 19 = 0$	Q197 $4\sqrt{5}$	Q229 5
Q134 B	Q166 D	Q198 1	Q230 D
Q135 D	Q167 D	Q199 $\sqrt{13}$	Q231 $(1/2, -3/2)$
Q136 A	Q168 B	Q200 10	Q232 $(-2, 3)$

Q233 (3, -3)

Q234 (1, 1)

Q235 (-3, 1)

Q236 D

Q237 B

Q238 D

Q239 D

Q240 A

Q241 (4, 1)

Q242 A

Q243 $a = -4, -3, \text{ or } 6$ Q244 $a = -1, -1/2, \text{ or } 1$ Q245 $a = 11/10, 3/2, \text{ or } -9/2$ Q246 $a = 1, -3/2, \text{ or } 2$

Q247 A

Q248 A

Q249 C

Q250 D

Q251 15

Q252 A

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 D

Quiz MCQ 4 C

Quiz WRITTEN 5 6

Quiz WRITTEN 6 $y = -2x + 9$ Quiz WRITTEN 7 $y = -x/3 + 7/3$ Quiz WRITTEN 8 $a = -4, -3, \text{ or } 6$

Concept 3 | Circles

English Student Edition • Focused formulas and guided practice

Formula opener

Circle

$$(x - a)^2 + (y - b)^2 = r^2$$

General form

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Tangent

$$xx_1 + yy_1 = r^2$$

Worked Solutions

Circles

Q001 Written

Find the equation of the circle with centre (3,-2) and radius 5.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x - 3)^2 + (y + 2)^2 = 25$.

Final answer

$$(x - 3)^2 + (y + 2)^2 = 25$$

Q002 Written

Find the equation of the circle with centre (-3,1) and radius 2.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x + 3)^2 + (y - 1)^2 = 4$.

Final answer

$$(x + 3)^2 + (y - 1)^2 = 4$$

Worked Solutions

Circles

Q003 Written

Find the equation of the circle with centre the origin and radius 4.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $x^2 + y^2 = 16$.

Final answer

$$x^2 + y^2 = 16$$

Q004 Written

Find the equation of the circle with centre (3,4), radius 6.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x - 3)^2 + (y - 4)^2 = 36$.

Final answer

$$(x - 3)^2 + (y - 4)^2 = 36$$

Worked Solutions

Circles

Q005 Written

Find the equation of the circle with centre $(-2, 6)$, radius $2\sqrt{3}$.**Solution**

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x + 2)^2 + (y - 6)^2 = 12$.

Final answer

$$(x + 2)^2 + (y - 6)^2 = 12$$

Q006 Written

Find the equation of the circle with centre the origin, radius 5.**Solution**

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $x^2 + y^2 = 25$.

Final answer

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q007 MCQ

Find the equation of the circle with centre (3,-2) and radius 5:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y + 2)^2 = 25$.

Correct option: A**Final answer**

$$(x - 3)^2 + (y + 2)^2 = 25$$

Q008 MCQ

Find the equation of the circle with centre (-3,1) and radius 2:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y - 1)^2 = 4$.

Correct option: A**Final answer**

$$(x + 3)^2 + (y - 1)^2 = 4$$

Worked Solutions

Circles

Q009 MCQ

Find the equation of the circle with centre the origin and radius 4:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 16$.

Correct option: D**Final answer**

$$x^2 + y^2 = 16$$

Q010 MCQ

Find the equation of the circle with centre (3,4), radius 6:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y - 4)^2 = 36$.

Correct option: B**Final answer**

$$(x - 3)^2 + (y - 4)^2 = 36$$

Worked Solutions

Circles

Q011 MCQ

Find the equation of the circle with centre $(-2,6)$, radius $2\sqrt{3}$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 2)^2 + (y - 6)^2 = 12$.

Correct option: B**Final answer**

$$(x + 2)^2 + (y - 6)^2 = 12$$

Q012 MCQ

Find the equation of the circle with centre the origin, radius 5:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 25$.

Correct option: A**Final answer**

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q013 Written

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(3, -1)$, radius 2.

Final answer

circle; centre $(3, -1)$, radius 2

Q014 Written

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(2, 3)$, radius 3.

Final answer

circle; centre $(2, 3)$, radius 3

Worked Solutions

Circles

Q015 Written

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(2, -4)$, radius 4.

Final answer

circle; centre $(2, -4)$, radius 4

Q016 Written

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(2, 0)$, radius 2.

Final answer

circle; centre $(2, 0)$, radius 2

Worked Solutions

Circles

Q017 Written

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(-1, 2)$, radius 5.

Final answer

circle; centre $(-1, 2)$, radius 5

Q018 Written

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents.

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: circle; centre $(-2, 3)$, radius $\sqrt{13}$.

Final answer

circle; centre $(-2, 3)$, radius $\sqrt{13}$

Worked Solutions

Circles

Q019 MCQ

Determine what kind of figure $x^2 + y^2 - 6x + 2y + 6 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(3, -1)$, radius 2.

Correct option: A**Final answer**circle; centre $(3, -1)$, radius 2

Worked Solutions

Circles

Q020 MCQ

Determine what kind of figure $x^2 + y^2 - 4x - 6y + 4 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(2, 3)$, radius 3.

Correct option: C**Final answer**circle; centre $(2, 3)$, radius 3

Worked Solutions

Circles

Q021 MCQ

Determine what kind of figure $x^2 + y^2 - 4x + 8y + 4 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(2, -4)$, radius 4.

Correct option: D**Final answer**circle; centre $(2, -4)$, radius 4

Worked Solutions

Circles

Q022 MCQ

Determine what kind of figure $x^2 + y^2 - 4x = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(2, 0)$, radius 2.

Correct option: C**Final answer**circle; centre $(2, 0)$, radius 2

Worked Solutions

Circles

Q023 MCQ

Determine what kind of figure $x^2 + y^2 + 2x - 4y - 20 = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(-1, 2)$, radius 5.

Correct option: C**Final answer**circle; centre $(-1, 2)$, radius 5

Worked Solutions

Circles

Q024 MCQ

Determine what kind of figure $x^2 + y^2 + 4x - 6y = 0$ represents:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is circle; centre $(-2, 3)$, radius $\sqrt{13}$.

Correct option: C**Final answer**circle; centre $(-2, 3)$, radius $\sqrt{13}$

Worked Solutions

Circles

Q025 Challenge

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$.

Final answer

centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$

Worked Solutions

Circles

Q026 Challenge

A circle passes through A (2, 5), B (8, 1), and has its centre on $x + y = 8$. Find its standard equation, centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$.

Correct option: B

Final answer

centre (5, 3), $\text{radius}^2 = 13$; $(x - 5)^2 + (y - 3)^2 = 13$

Worked Solutions

Circles

Q027 Written

Find the equation of the circle with centre (2,3) passing through (4,6).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 2)^2 + (y - 3)^2 = 13$.

Final answer

$$(x - 2)^2 + (y - 3)^2 = 13$$

Q028 Written

Find the equation of the circle with centre (-3,1) passing through (4,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 3)^2 + (y - 1)^2 = 58$.

Final answer

$$(x + 3)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q029 Written

Find the equation of the circle with centre (1,-3) through (-3,0).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 1)^2 + (y + 3)^2 = 25$.

Final answer

$$(x - 1)^2 + (y + 3)^2 = 25$$

Q030 Written

Find the equation of the circle with centre (-4,1) through (3,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 4)^2 + (y - 1)^2 = 58$.

Final answer

$$(x + 4)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q031 Written

Find the equation of the circle with centre (2,-1) through the origin.**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 2)^2 + (y + 1)^2 = 5$.

Final answer

$$(x - 2)^2 + (y + 1)^2 = 5$$

Q032 Written

Find the equation of the circle with centre the origin through (4,-3).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 = 25$.

Final answer

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q033 MCQ

Find the equation of the circle with centre (2,3) passing through (4,6):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y - 3)^2 = 13$.

Correct option: C**Final answer**

$$(x - 2)^2 + (y - 3)^2 = 13$$

Q034 MCQ

Find the equation of the circle with centre (-3,1) passing through (4,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y - 1)^2 = 58$.

Correct option: A**Final answer**

$$(x + 3)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q035 MCQ

Find the equation of the circle with centre (1,-3) through (-3,0):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y + 3)^2 = 25$.

Correct option: D**Final answer**

$$(x - 1)^2 + (y + 3)^2 = 25$$

Q036 MCQ

Find the equation of the circle with centre (-4,1) through (3,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 4)^2 + (y - 1)^2 = 58$.

Correct option: A**Final answer**

$$(x + 4)^2 + (y - 1)^2 = 58$$

Worked Solutions

Circles

Q037 MCQ

Find the equation of the circle with centre (2,-1) through the origin:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y + 1)^2 = 5$.

Correct option: B**Final answer**

$$(x - 2)^2 + (y + 1)^2 = 5$$

Q038 MCQ

Find the equation of the circle with centre the origin through (4,-3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 = 25$.

Correct option: A**Final answer**

$$x^2 + y^2 = 25$$

Worked Solutions

Circles

Q039 Written

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 2)^2 + (y + 1)^2 = 17$.

Final answer

$$(x - 2)^2 + (y + 1)^2 = 17$$

Q040 Written

Find the equation of the circle with diameter endpoints (3,4), (5,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 4)^2 + (y - 1)^2 = 10$.

Final answer

$$(x - 4)^2 + (y - 1)^2 = 10$$

Worked Solutions

Circles

Q041 Written

Find the equation of the circle with diameter endpoints (4,7), (-2,-1).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x - 1)^2 + (y - 3)^2 = 25$.

Final answer

$$(x - 1)^2 + (y - 3)^2 = 25$$

Q042 Written

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 1)^2 + (y + 2)^2 = 25$.

Final answer

$$(x + 1)^2 + (y + 2)^2 = 25$$

Worked Solutions

Circles

Q043 Written

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$.**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $(x + 2)^2 + (y - 1)^2 = 2$.

Final answer

$$(x + 2)^2 + (y - 1)^2 = 2$$

Q044 Written

Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$.**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + (y - 2)^2 = 25$.

Final answer

$$x^2 + (y - 2)^2 = 25$$

Worked Solutions

Circles

Q045 MCQ

Find the equation of the circle with diameter endpoints A (1, 3), B (3, -5):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y + 1)^2 = 17$.

Correct option: A**Final answer**

$$(x - 2)^2 + (y + 1)^2 = 17$$

Q046 MCQ

Find the equation of the circle with diameter endpoints (3,4), (5,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 4)^2 + (y - 1)^2 = 10$.

Correct option: C**Final answer**

$$(x - 4)^2 + (y - 1)^2 = 10$$

Worked Solutions

Circles

Q047 MCQ

Find the equation of the circle with diameter endpoints (4,7), (-2,-1):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y - 3)^2 = 25$.

Correct option: C**Final answer**

$$(x - 1)^2 + (y - 3)^2 = 25$$

Q048 MCQ

Find the equation of the circle with diameter endpoints (4,-2), (-6,-2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 2)^2 = 25$.

Correct option: A**Final answer**

$$(x + 1)^2 + (y + 2)^2 = 25$$

Worked Solutions

Circles

Q049 MCQ

Find the equation of the circle with diameter endpoints $(-3,0)$, $(-1,2)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 2)^2 + (y - 1)^2 = 2$.

Correct option: B**Final answer**

$$(x + 2)^2 + (y - 1)^2 = 2$$

Q050 MCQ

Find the equation of the circle with diameter endpoints $(-3,6)$, $(3,-2)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + (y - 2)^2 = 25$.

Correct option: C**Final answer**

$$x^2 + (y - 2)^2 = 25$$

Worked Solutions

Circles

Q051 Written

Find the equation of the circle passing through (1,1), (2,-1), and (3,2).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 5x - y + 4 = 0$.

Final answer

$$x^2 + y^2 - 5x - y + 4 = 0$$

Q052 Written

Find the equation of the circle passing through (0,1), (-1,2), and (3,1).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 3x - 7y + 6 = 0$.

Final answer

$$x^2 + y^2 - 3x - 7y + 6 = 0$$

Worked Solutions

Circles

Q053 Written

Find the equation of the circle with through (0,0), (-1,1), (2,4).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 2x - 4y = 0$.

Final answer

$$x^2 + y^2 - 2x - 4y = 0$$

Q054 Written

Find the equation of the circle with through (-4,6), (0,8), (3,-1).**Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 6y - 16 = 0$.

Final answer

$$x^2 + y^2 - 6y - 16 = 0$$

Worked Solutions

Circles

Q055 **Written****Find the equation of the circle with through (1,1), (5,-1), (-3,-7).****Solution**

1. For a diameter, find its midpoint and radius; for known points, substitute each point.
2. Solve the resulting centre/radius or general-form equations and verify every point.
3. The required circle is $x^2 + y^2 - 2x + 8y - 8 = 0$.

Final answer

$$x^2 + y^2 - 2x + 8y - 8 = 0$$

Q056 **MCQ****Find the equation of the circle passing through (1,1), (2,-1), and (3,2):****Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 5x - y + 4 = 0$.

Correct option: A**Final answer**

$$x^2 + y^2 - 5x - y + 4 = 0$$

Worked Solutions

Circles

Q057 MCQ

Find the equation of the circle passing through (0,1), (-1,2), and (3,1):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 3x - 7y + 6 = 0$.

Correct option: B**Final answer**

$$x^2 + y^2 - 3x - 7y + 6 = 0$$

Q058 MCQ

Find the equation of the circle with through (0,0), (-1,1), (2,4):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 2x - 4y = 0$.

Correct option: D**Final answer**

$$x^2 + y^2 - 2x - 4y = 0$$

Worked Solutions

Circles

Q059 MCQ

Find the equation of the circle with through $(-4,6)$, $(0,8)$, $(3,-1)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 6y - 16 = 0$.

Correct option: D**Final answer**

$$x^2 + y^2 - 6y - 16 = 0$$

Q060 MCQ

Find the equation of the circle with through $(1,1)$, $(5,-1)$, $(-3,-7)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 2x + 8y - 8 = 0$.

Correct option: D**Final answer**

$$x^2 + y^2 - 2x + 8y - 8 = 0$$

Worked Solutions

Circles

Q061 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$.

Final answer

$$x^2 + y^2 - 6x - 6y + 8 = 0; \text{ centre } (3, 3), \text{ radius } \sqrt{10}$$

Worked Solutions

Circles

Q062 Challenge

Find the circle through (0,2), (2,0), and (4,6), giving its centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$.

Correct option: A

Final answer

$$x^2 + y^2 - 6x - 6y + 8 = 0; \text{ centre } (3, 3), \text{ radius } \sqrt{10}$$

Worked Solutions

Circles

Q063 Written

Find the circle with centre (4,-2) tangent to the x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 4)^2 + (y + 2)^2 = 4$.

Final answer

$$(x - 4)^2 + (y + 2)^2 = 4$$

Q064 Written

Find the circle with centre (-2,3) tangent to the y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 2)^2 + (y - 3)^2 = 4$.

Final answer

$$(x + 2)^2 + (y - 3)^2 = 4$$

Worked Solutions

Circles

Q065 Written

Find the circle with centre $(-1,4)$ tangent to x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y - 4)^2 = 16$.

Final answer

$$(x + 1)^2 + (y - 4)^2 = 16$$

Q066 Written

Find the circle with centre $(3,-4)$ tangent to x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 3)^2 + (y + 4)^2 = 16$.

Final answer

$$(x - 3)^2 + (y + 4)^2 = 16$$

Worked Solutions

Circles

Q067 Written

Find the circle with centre (5,-4) tangent to x-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 5)^2 + (y + 4)^2 = 16$.

Final answer

$$(x - 5)^2 + (y + 4)^2 = 16$$

Q068 Written

Find the circle with centre (-1,6) tangent to y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y - 6)^2 = 1$.

Final answer

$$(x + 1)^2 + (y - 6)^2 = 1$$

Worked Solutions

Circles

Q069 Written

Find the circle with centre (2,-5) tangent to y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 2)^2 + (y + 5)^2 = 4$.

Final answer

$$(x - 2)^2 + (y + 5)^2 = 4$$

Q070 Written

Find the circle with centre (-3,-4) tangent to y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 3)^2 + (y + 4)^2 = 9$.

Final answer

$$(x + 3)^2 + (y + 4)^2 = 9$$

Worked Solutions

Circles

Q071 MCQ

Find the circle with centre (4,-2) tangent to the x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 4)^2 + (y + 2)^2 = 4$.

Correct option: B**Final answer**

$$(x - 4)^2 + (y + 2)^2 = 4$$

Worked Solutions

Circles

Q072 MCQ

Find the circle with centre $(-2,3)$ tangent to the y -axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 2)^2 + (y - 3)^2 = 4$.

Correct option: B**Final answer**

$$(x + 2)^2 + (y - 3)^2 = 4$$

Worked Solutions

Circles

Q073 MCQ

Find the circle with centre $(-1,4)$ tangent to x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y - 4)^2 = 16$.

Correct option: B**Final answer**

$$(x + 1)^2 + (y - 4)^2 = 16$$

Worked Solutions

Circles

Q074 MCQ

Find the circle with centre (3,-4) tangent to x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y + 4)^2 = 16$.

Correct option: D**Final answer**

$$(x - 3)^2 + (y + 4)^2 = 16$$

Worked Solutions

Circles

Q075 MCQ

Find the circle with centre (5,-4) tangent to x-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 5)^2 + (y + 4)^2 = 16$.

Correct option: D**Final answer**

$$(x - 5)^2 + (y + 4)^2 = 16$$

Worked Solutions

Circles

Q076 MCQ

Find the circle with centre $(-1,6)$ tangent to y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y - 6)^2 = 1$.

Correct option: D**Final answer**

$$(x + 1)^2 + (y - 6)^2 = 1$$

Worked Solutions

Circles

Q077 MCQ

Find the circle with centre (2,-5) tangent to y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y + 5)^2 = 4$.

Correct option: B**Final answer**

$$(x - 2)^2 + (y + 5)^2 = 4$$

Worked Solutions

Circles

Q078 MCQ

Find the circle with centre $(-3,-4)$ tangent to y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y + 4)^2 = 9$.

Correct option: C**Final answer**

$$(x + 3)^2 + (y + 4)^2 = 9$$

Worked Solutions

Circles

Q079 Written

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$.

Final answer

$$(x + 1)^2 + (y - 1)^2 = 1 \text{ or } (x + 5)^2 + (y - 5)^2 = 25$$

Q080 Written

Find the circles through $(1,2)$ tangent to both axes.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$.

Final answer

$$(x - 1)^2 + (y - 1)^2 = 1 \text{ or } (x - 5)^2 + (y - 5)^2 = 25$$

Worked Solutions

Circles

Q081 **Written****Find the circle with through $(-1, -2)$ tangent to both axes.****Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Final answer

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Q082 **Written****Find the circle with through $(-2, -1)$ tangent to both axes.****Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Final answer

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Worked Solutions

Circles

Q083 Written

Find the circle with through (8,-1) tangent to both axes.**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$.

Final answer

$$(x - 5)^2 + (y + 5)^2 = 25 \text{ or } (x - 13)^2 + (y + 13)^2 = 169$$

Worked Solutions

Circles

Q084 MCQ

Find the circles through $(-2,1)$ tangent to both the x-axis and y-axis:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$.

Correct option: C**Final answer**

$$(x + 1)^2 + (y - 1)^2 = 1 \text{ or } (x + 5)^2 + (y - 5)^2 = 25$$

Worked Solutions

Circles

Q085 MCQ

Find the circles through (1,2) tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$.

Correct option: D**Final answer**

$$(x - 1)^2 + (y - 1)^2 = 1 \text{ or } (x - 5)^2 + (y - 5)^2 = 25$$

Worked Solutions

Circles

Q086 MCQ

Find the circle with through $(-1,-2)$ tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Correct option: C**Final answer**

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Worked Solutions

Circles

Q087 MCQ

Find the circle with through $(-2,-1)$ tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$.

Correct option: C**Final answer**

$$(x + 1)^2 + (y + 1)^2 = 1 \text{ or } (x + 5)^2 + (y + 5)^2 = 25$$

Worked Solutions

Circles

Q088 MCQ

Find the circle with through (8,-1) tangent to both axes:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$.

Correct option: B**Final answer**

$$(x - 5)^2 + (y + 5)^2 = 25 \text{ or } (x - 13)^2 + (y + 13)^2 = 169$$

Worked Solutions

Circles

Q089 Written

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3).**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 1)^2 + (y + 1)^2 = 25$.

Final answer

$$(x - 1)^2 + (y + 1)^2 = 25$$

Q090 Written

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2).**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$.

Final answer

$$(x + 5/2)^2 + (y - 5/2)^2 = 25/2$$

Worked Solutions

Circles

Q091 Written

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4).**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 2)^2 + (y - 3)^2 = 2$.

Final answer

$$(x - 2)^2 + (y - 3)^2 = 2$$

Q092 Written

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3).**Solution**

1. Translate tangency to an equal-distance condition from the centre to the relevant axis or line.
2. Intersect the centre locus with the stated constraint and reject non-geometric radii.
3. The checked tangency result is $(x - 4)^2 + (y - 2)^2 = 26$.

Final answer

$$(x - 4)^2 + (y - 2)^2 = 26$$

Worked Solutions

Circles

Q093 MCQ

Find the circle whose centre is on $y = -x$ and which passes through (5,2) and (-2,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 1)^2 + (y + 1)^2 = 25$.

Correct option: D**Final answer**

$$(x - 1)^2 + (y + 1)^2 = 25$$

Worked Solutions

Circles

Q094 MCQ

Find the circle whose centre is on $y = x + 5$ and which passes through the origin and (1,2):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$.

Correct option: C**Final answer**

$$(x + 5/2)^2 + (y - 5/2)^2 = 25/2$$

Worked Solutions

Circles

Q095 MCQ

Find the circle with centre on $y = x + 1$ through (3,4) and (1,4):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 2)^2 + (y - 3)^2 = 2$.

Correct option: C**Final answer**

$$(x - 2)^2 + (y - 3)^2 = 2$$

Worked Solutions

Circles

Q096 MCQ

Find the circle with centre on $y = 2x - 6$ through (3,7) and (-1,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 4)^2 + (y - 2)^2 = 26$.

Correct option: D**Final answer**

$$(x - 4)^2 + (y - 2)^2 = 26$$

Worked Solutions

Circles

Q097 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$.

Final answer

$$(x + 1)^2 + y^2 = 5; \text{ centre } (-1, 0), \text{ radius } \sqrt{5}$$

Worked Solutions

Circles

Q098 Challenge

A circle centre is constrained to the x-axis and passes through P1(-2,2), P2(1,1). Find its standard equation, centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + y^2 = 5$; centre $(-1, 0)$, radius $\sqrt{5}$.

Correct option: C

Final answer

$$(x + 1)^2 + y^2 = 5; \text{ centre } (-1, 0), \text{ radius } \sqrt{5}$$

Worked Solutions

Circles

Q099 Written

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle.**Solution**

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -1$ or $k > 3$.

Final answer

$$k < -1 \text{ or } k > 3$$

Q100 Written

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle.**Solution**

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -\sqrt{6}$ or $k > \sqrt{6}$.

Final answer

$$k < -\sqrt{6} \text{ or } k > \sqrt{6}$$

Worked Solutions

Circles

Q101 Written

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < 5/4$.

Final answer

$$k < 5/4$$

Q102 Written

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $k < -1/5$ or $k > 3$.

Final answer

$$k < -1/5 \text{ or } k > 3$$

Worked Solutions

Circles

Q103 Written

Find the range of k for $x^2 + y^2 + 2kx - 2(k + 3)y + 4k^2 + 1 = 0$ to represent a circle.

Solution

1. Complete the square and isolate the parameter-dependent radius squared.
2. Impose $r^2 > 0$ for a genuine real circle (and $r^2 = 0$ only if a point circle is allowed).
3. The admissible parameter result is $-1 < k < 4$.

Final answer

$$-1 < k < 4$$

Worked Solutions

Circles

Q104 MCQ

Find the range of k for $x^2 + y^2 - 2kx + 2ky + 4k + 6 = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -1$ or $k > 3$.

Correct option: C**Final answer**

$$k < -1 \text{ or } k > 3$$

Worked Solutions

Circles

Q105 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 4ky + 4k^2 + 6 = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -\sqrt{6}$ or $k > \sqrt{6}$.

Correct option: B**Final answer**

$$k < -\sqrt{6} \text{ or } k > \sqrt{6}$$

Worked Solutions

Circles

Q106 MCQ

Find the range of k for $x^2 + y^2 + 2x - y + k = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < 5/4$.

Correct option: C**Final answer**

$$k < 5/4$$

Worked Solutions

Circles

Q107 MCQ

Find the range of k for $x^2 + y^2 + 6kx - 2ky + 28k + 6 = 0$ to represent a circle:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k < -1/5$ or $k > 3$.

Correct option: B**Final answer**

$$k < -1/5 \text{ or } k > 3$$

Worked Solutions

Circles

Q108 MCQ

Find the range of k for $x^2 + y^2 + 2kx - 2(k + 3)y + 4k^2 + 1 = 0$ to represent a circle:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-1 < k < 4$.

Correct option: D

Final answer

$$-1 < k < 4$$

Worked Solutions

Circles

Q109 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k -range giving a positive radius.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $0 < k < 4$.

Final answer

$$0 < k < 4$$

Worked Solutions

Circles

Q110 Challenge

An acoustic-safety circle is $x^2 + y^2 + 6x - 2ky + 2k^2 - 4k + 9 = 0$. Find the exact k -range giving a positive radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < k < 4$.

Correct option: B

Final answer

$$0 < k < 4$$

Worked Solutions

Circles

Q111 Written

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(1, -1)$ and $(-1, 3)$.

Final answer

$(1, -1)$ and $(-1, 3)$

Q112 Written

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(0, 2)$ and $(2, 0)$.

Final answer

$(0, 2)$ and $(2, 0)$

Worked Solutions

Circles

Q113 Written

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(3, -1)$.

Final answer $(3, -1)$

Q114 Written

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(-2, 1)$.

Final answer $(-2, 1)$

Worked Solutions

Circles

Q115 Written

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(-2, 1)$ and $(4, -1)$.

Final answer

$(-2, 1)$ and $(4, -1)$

Worked Solutions

Circles

Q116 MCQ

Find the coordinates of the intersections of $(x + 2)^2 + y^2 = 10$ and $y = -2x + 1$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(1, -1)$ and $(-1, 3)$.

Correct option: C**Final answer** $(1, -1)$ and $(-1, 3)$

Worked Solutions

Circles

Q117 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 4$ and $x + y = 2$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is (0, 2) and (2, 0).

Correct option: D**Final answer**

(0, 2) and (2, 0)

Worked Solutions

Circles

Q118 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 10$ with $y = 3x - 10$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(3, -1)$.

Correct option: C**Final answer** $(3, -1)$

Worked Solutions

Circles

Q119 MCQ

Find the coordinates of the intersections of $x^2 + y^2 = 5$ with $2x - y + 5 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-2, 1)$.

Correct option: D**Final answer** $(-2, 1)$

Worked Solutions

Circles

Q120 MCQ

Find the coordinates of the intersections of $(x - 1)^2 + y^2 = 10$ with $x + 3y - 1 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-2, 1)$ and $(4, -1)$.

Correct option: B**Final answer** $(-2, 1)$ and $(4, -1)$

Worked Solutions

Circles

Q121 Written

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 0 points.

Final answer

0 points

Q122 Written

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$.**Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 2 points : (1, -3) and (3, 1).

Final answer

2 points : (1, -3) and (3, 1)

Worked Solutions

Circles

Q123 **Written****Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$.****Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$.

Final answer2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$ Q124 **Written****Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$.****Solution**

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 0 points.

Final answer

0 points

Worked Solutions

Circles

Q125 Written

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: 1 point : $(-2, 0)$.

Final answer

1 point : $(-2, 0)$

Worked Solutions

Circles

Q126 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ and $x - y + 3 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0 points.

Correct option: D**Final answer**

0 points

Worked Solutions

Circles

Q127 MCQ

Find the number of intersections of $x^2 + y^2 = 10$ and $y = 2x - 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2 points : $(1, -3)$ and $(3, 1)$.

Correct option: A**Final answer**2 points : $(1, -3)$ and $(3, 1)$

Worked Solutions

Circles

Q128 MCQ

Find the number of intersections of $x^2 + y^2 = 5$ with $y = x + 2$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$.

Correct option: C**Final answer**2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$

Worked Solutions

Circles

Q129 MCQ

Find the number of intersections of $x^2 + y^2 = 2$ with $2x + 3y - 6 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0 points.

Correct option: C**Final answer**

0 points

Worked Solutions

Circles

Q130 MCQ

Find the number of intersections of $(x + 1)^2 + (y - 2)^2 = 5$ with $x + 2y + 2 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 1 point : $(-2, 0)$.

Correct option: D**Final answer**1 point : $(-2, 0)$

Worked Solutions

Circles

Q131 Written

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $-5 < k < 5$.

Final answer

$$-5 < k < 5$$

Q132 Written

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r = \sqrt{5}$.

Final answer

$$r = \sqrt{5}$$

Worked Solutions

Circles

Q133 Written

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $m = \pm\sqrt{3}$.

Final answer

$$m = \pm\sqrt{3}$$

Q134 Written

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $0 < r < \sqrt{10}$.

Final answer

$$0 < r < \sqrt{10}$$

Worked Solutions

Circles

Q135 Written

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $m = \pm 4$.

Final answer

$$m = \pm 4$$

Q136 Written

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $n < -10$ or $n > 10$.

Final answer

$$n < -10 \text{ or } n > 10$$

Worked Solutions

Circles

Q137 **Written**

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $-5 \leq m \leq 5$.

Final answer

$$-5 \leq m \leq 5$$

Q138 **Written**

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r .

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r = \sqrt{2}$.

Final answer

$$r = \sqrt{2}$$

Worked Solutions

Circles

Q139 Written

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $r > \sqrt{17}$.

Final answer

$$r > \sqrt{17}$$

Worked Solutions

Circles

Q140 MCQ

If $x^2 + y^2 = 5$ and $y = 2x - k$ intersect at two distinct points, find the range of k :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5 < k < 5$.

Correct option: D

Final answer

$$-5 < k < 5$$

Worked Solutions

Circles

Q141 MCQ

If $x^2 + y^2 = r^2$ and $2x + y - 5 = 0$ are tangent, find r :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r = \sqrt{5}$.

Correct option: B

Final answer

$$r = \sqrt{5}$$

Worked Solutions

Circles

Q142 MCQ

If $x^2 + y^2 = 4$ and $y = mx + 4$ are tangent, find m :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $m = \pm\sqrt{3}$.

Correct option: B

Final answer

$$m = \pm\sqrt{3}$$

Worked Solutions

Circles

Q143 MCQ

If $x^2 + y^2 = r^2$ and $x - 3y - 10 = 0$ have no intersections, find the positive-radius range:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < r < \sqrt{10}$.

Correct option: B

Final answer

$$0 < r < \sqrt{10}$$

Worked Solutions

Circles

Q144 MCQ

If $x^2 + y^2 = 8$ and $y = x + m$ are tangent, find m :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $m = \pm 4$.

Correct option: C

Final answer

$$m = \pm 4$$

Worked Solutions

Circles

Q145 MCQ

If $x^2 + y^2 = 10$ and $y = 3x + n$ have no intersections, find n :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $n < -10$ or $n > 10$.

Correct option: A

Final answer

$$n < -10 \text{ or } n > 10$$

Worked Solutions

Circles

Q146 MCQ

If $x^2 + y^2 = 5$ and $y = 2x + m$ have points of intersection, find the range of m :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5 \leq m \leq 5$.

Correct option: C

Final answer

$$-5 \leq m \leq 5$$

Worked Solutions

Circles

Q147 MCQ

If $x^2 + y^2 = r^2$ and $y = x + 2$ are tangent, find r :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r = \sqrt{2}$.

Correct option: B

Final answer

$$r = \sqrt{2}$$

Worked Solutions

Circles

Q148 MCQ

If $x^2 + y^2 = r^2$ and $4x - y + 17 = 0$ intersect at two distinct points, find the positive-radius range:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $r > \sqrt{17}$.

Correct option: D

Final answer

$$r > \sqrt{17}$$

Worked Solutions

Circles

Q149 Challenge

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r .

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $0 < r < 2$.

Final answer

$$0 < r < 2$$

Worked Solutions

Circles

Q150 Challenge

A radar circle $x^2 + y^2 = r^2$ must have no intersection with $3x + 4y - 10 = 0$. Find the positive range of r :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $0 < r < 2$.

Correct option: D

Final answer

$$0 < r < 2$$

Worked Solutions

Circles

Q151 Written

Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$.**Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $\sqrt{14}$.

Final answer $\sqrt{14}$

Q152 Written

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$.**Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: 4.

Final answer

4

Worked Solutions

Circles

Q153 **Written****Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$.****Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $2\sqrt{5}$.

Final answer

$$2\sqrt{5}$$

Q154 **Written****Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$.****Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $3\sqrt{2}$.

Final answer

$$3\sqrt{2}$$

Worked Solutions

Circles

Q155 **Written****Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$.****Solution**

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: 2.

Final answer

2

Q156 **MCQ****Find the chord length cut off by $x^2 + y^2 = 4$ on $x + y - 1 = 0$:****Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $\sqrt{14}$.

Correct option: A**Final answer** $\sqrt{14}$

Worked Solutions

Circles

Q157 MCQ

Find the chord length cut off by $(x - 2)^2 + (y + 1)^2 = 9$ on $x + 2y - 5 = 0$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 4.

Correct option: C**Final answer**

4

Q158 MCQ

Find the chord length cut off by $x^2 + y^2 = 10$ on $2x - y = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\sqrt{5}$.

Correct option: D**Final answer** $2\sqrt{5}$

Worked Solutions

Circles

Q159 MCQ

Find the chord length cut off by $(x - 2)^2 + (y - 1)^2 = 5$ on $x + y = 4$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $3\sqrt{2}$.

Correct option: B**Final answer** $3\sqrt{2}$

Q160 MCQ

Find the chord length cut off by $x^2 + y^2 = 3$ on $x - y = 2$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is 2.

Correct option: D**Final answer**

2

Worked Solutions

Circles

Q161 Written

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm 2$.

Final answer

$$k = \pm 2$$

Q162 Written

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm\sqrt{31}$.

Final answer

$$k = \pm\sqrt{31}$$

Worked Solutions

Circles

Q163 Written

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = \pm 2$.

Final answer

$$k = \pm 2$$

Q164 Written

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$.

Solution

1. Find the distance d from the centre to the chord line.
2. Use $L = 2\sqrt{r^2 - d^2}$ and simplify.
3. Chord length: $k = 0$ or $k = -2$.

Final answer

$$k = 0 \text{ or } k = -2$$

Worked Solutions

Circles

Q165 MCQ

For $x^2 + y^2 = 3$ and $x - y + k = 0$, find k when the chord length is 2:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm 2$.

Correct option: C

Final answer

$$k = \pm 2$$

Q166 MCQ

For $x^2 + y^2 = 16$ and $y = x + k$, find k when the chord length is $\sqrt{2}$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm\sqrt{31}$.

Correct option: D

Final answer

$$k = \pm\sqrt{31}$$

Worked Solutions

Circles

Q167 MCQ

For $x^2 + y^2 = 10$ and $y = x + k$, find k when the chord length is $4\sqrt{2}$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = \pm 2$.

Correct option: A

Final answer

$$k = \pm 2$$

Worked Solutions

Circles

Q168 MCQ

For $(x - 1)^2 + y^2 = 5$ and $y = x + k$, find k when the chord length is $3\sqrt{2}$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = 0$ or $k = -2$.

Correct option: B

Final answer

$$k = 0 \text{ or } k = -2$$

Worked Solutions

Circles

Q169 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $k = -1$ or $k = 7$.

Final answer

$$k = -1 \text{ or } k = 7$$

Worked Solutions

Circles

Q170 Challenge

For $x^2 + y^2 + 4x - 2y - 7 = 0$ and $y = x + k$, find k when the chord length is 4:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $k = -1$ or $k = 7$.

Correct option: A

Final answer

$$k = -1 \text{ or } k = 7$$

Worked Solutions

Circles

Q171 Written

Find the tangent to $x^2 + y^2 = 25$ at (4,3).**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $4x + 3y = 25$.

Final answer

$$4x + 3y = 25$$

Q172 Written

Find the tangent to $x^2 + y^2 = 34$ at (5,3).**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $5x + 3y = 34$.

Final answer

$$5x + 3y = 34$$

Worked Solutions

Circles

Q173 **Written****Find the tangents $x^2 + y^2 = 10$ at (3,-1).****Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $3x - y = 10$.

Final answer

$$3x - y = 10$$

Q174 **Written****Find the tangents $x^2 + y^2 = 29$ at (-5,-2).****Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-5x - 2y = 29$.

Final answer

$$-5x - 2y = 29$$

Worked Solutions

Circles

Q175 Written

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-2x + y = 5$.

Final answer

$$-2x + y = 5$$

Worked Solutions

Circles

Q176 MCQ

Find the tangent to $x^2 + y^2 = 25$ at (4,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $4x + 3y = 25$.

Correct option: A**Final answer**

$$4x + 3y = 25$$

Worked Solutions

Circles

Q177 MCQ

Find the tangent to $x^2 + y^2 = 34$ at (5,3):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $5x + 3y = 34$.

Correct option: C**Final answer**

$$5x + 3y = 34$$

Worked Solutions

Circles

Q178 MCQ

Find the tangents $x^2 + y^2 = 10$ at (3,-1):**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $3x - y = 10$.

Correct option: A**Final answer**

$$3x - y = 10$$

Worked Solutions

Circles

Q179 MCQ

Find the tangents $x^2 + y^2 = 29$ at $(-5, -2)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-5x - 2y = 29$.

Correct option: C**Final answer**

$$-5x - 2y = 29$$

Worked Solutions

Circles

Q180 MCQ

Find the tangents $x^2 + y^2 = 5$ at $(-2,1)$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-2x + y = 5$.

Correct option: A**Final answer**

$$-2x + y = 5$$

Worked Solutions

Circles

Q181 Written

Find the tangents from (1,3) to $x^2 + y^2 = 5$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $2x + y = 5$ and $-x + 2y = 5$.

Final answer

$$2x + y = 5 \text{ and } -x + 2y = 5$$

Q182 Written

Find the tangents from (-2,4) to $x^2 + y^2 = 10$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-3x + y = 10$ and $x + 3y = 10$.

Final answer

$$-3x + y = 10 \text{ and } x + 3y = 10$$

Worked Solutions

Circles

Q183 **Written****Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$.****Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-2x + y = 5$ and $x + 2y = 5$.

Final answer

$$-2x + y = 5 \text{ and } x + 2y = 5$$

Q184 **Written****Find the tangents from $(2,1)$ to $x^2 + y^2 = 1$.****Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $y = 1$ and $4x - 3y = 5$.

Final answer

$$y = 1 \text{ and } 4x - 3y = 5$$

Worked Solutions

Circles

Q185 Written

Find the tangents from (2,6) to $x^2 + y^2 = 20$.**Solution**

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $-x + 2y = 10$ and $2x + y = 10$.

Final answer

$$-x + 2y = 10 \text{ and } 2x + y = 10$$

Worked Solutions

Circles

Q186 MCQ

Find the tangents from (1,3) to $x^2 + y^2 = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2x + y = 5$ and $-x + 2y = 5$.

Correct option: D**Final answer**

$$2x + y = 5 \text{ and } -x + 2y = 5$$

Worked Solutions

Circles

Q187 MCQ

Find the tangents from $(-2,4)$ to $x^2 + y^2 = 10$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-3x + y = 10$ and $x + 3y = 10$.

Correct option: A**Final answer**

$$-3x + y = 10 \text{ and } x + 3y = 10$$

Worked Solutions

Circles

Q188 MCQ

Find the tangents from $(-1,3)$ to $x^2 + y^2 = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-2x + y = 5$ and $x + 2y = 5$.

Correct option: D**Final answer**

$$-2x + y = 5 \text{ and } x + 2y = 5$$

Worked Solutions

Circles

Q189 MCQ

Find the tangents from (2,1) to $x^2 + y^2 = 1$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $y = 1$ and $4x - 3y = 5$.

Correct option: B**Final answer**

$$y = 1 \text{ and } 4x - 3y = 5$$

Worked Solutions

Circles

Q190 MCQ

Find the tangents from (2,6) to $x^2 + y^2 = 20$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-x + 2y = 10$ and $2x + y = 10$.

Correct option: C**Final answer**

$$-x + 2y = 10 \text{ and } 2x + y = 10$$

Worked Solutions

Circles

Q191 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$.

Final answer

$-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$

Worked Solutions

Circles

Q192 Challenge

A drone at (2,6) needs tangents to $x^2 + y^2 = 20$. Find both sightline equations and their exact contact points:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $-x + 2y = 10$ at $(-2, 4)$; $2x + y = 10$ at $(4, 2)$.

Correct option: B

Final answer

$$-x + 2y = 10 \text{ at } (-2, 4); 2x + y = 10 \text{ at } (4, 2)$$

Worked Solutions

Circles

Q193 Written

Classify $x^2 + y^2 = 4$ **and** $(x - 4)^2 + y^2 = 9$.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: intersect at 2 points.

Final answer

intersect at 2 points

Q194 Written

Classify $x^2 + y^2 = 9$ **and** $(x - 3)^2 + (y - 4)^2 = 4$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: B : touch externally.

Final answer B : touch externally

Worked Solutions

Circles

Q195 **Written****Classify** $(x - 1)^2 + y^2 = 5^2$ **and** $x^2 + y^2 + 2x - 6y = 3$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: E : intersect at 2 points.

Final answer E : intersect at 2 pointsQ196 **Written****Classify** $x^2 + y^2 = 4$ **and** $(x - 1)^2 + y^2 = 1$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: C : touch internally.

Final answer C : touch internally

Worked Solutions

Circles

Q197 Written

Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: C : touch internally.

Final answer C : touch internally

Q198 Written

Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: E : intersect at 2 points.

Final answer E : intersect at 2 points

Worked Solutions

Circles

Q199 **Written****Classify** $x^2 + (y - 3)^2 = 45$ **and** $x^2 + y^2 + 2x - 2y = 3$ **from A-E.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: D : one is inside the other .

Final answer D : one is inside the other

Worked Solutions

Circles

Q200 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 4)^2 + y^2 = 9$, the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is intersect at 2 points.

Correct option: C**Final answer**

intersect at 2 points

Worked Solutions

Circles

Q201 MCQ

For the expression Classify $x^2 + y^2 = 9$ and $(x - 3)^2 + (y - 4)^2 = 4$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is B : touch externally.

Correct option: D

Final answer

B : touch externally

Worked Solutions

Circles

Q202 MCQ

For the expression Classify $(x - 1)^2 + y^2 = 5^2$ and $x^2 + y^2 + 2x - 6y = 3$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is E : intersect at 2 points.

Correct option: B

Final answer

E : intersect at 2 points

Worked Solutions

Circles

Q203 MCQ

For the expression Classify $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 1$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is C : touch internally.

Correct option: A

Final answer

C : touch internally

Worked Solutions

Circles

Q204 MCQ

For the expression Classify $x^2 + y^2 = 18$ and $(x + 2)^2 + (y - 2)^2 = 2$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is C : touch internally.

Correct option: D**Final answer**

C : touch internally

Worked Solutions

Circles

Q205 MCQ

For the expression Classify $(x + 3)^2 + y^2 = 16$ and $x^2 + y^2 - 2x - 8y - 8 = 0$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is E : intersect at 2 points.

Correct option: B

Final answer

E : intersect at 2 points

Worked Solutions

Circles

Q206 MCQ

For the expression Classify $x^2 + (y - 3)^2 = 45$ and $x^2 + y^2 + 2x - 2y = 3$ from A-E., the required value is:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is D : one is inside the other .

Correct option: C

Final answer

D : one is inside the other

Worked Solutions

Circles

Q207 **Written****Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $(x - 3)^2 + (y - 1)^2 = 45$.

Final answer

$$(x - 3)^2 + (y - 1)^2 = 45$$

Q208 **Written****Find the circle with centre (-3,-9) touching $(x + 6)^2 + (y + 5)^2 = 1$ externally.****Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $(x + 3)^2 + (y + 9)^2 = 16$.

Final answer

$$(x + 3)^2 + (y + 9)^2 = 16$$

Worked Solutions

Circles

Q209 Written

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally.**Solution**

1. Find the centre distance D and both radii.
2. Compare D with $r_1 + r_2$ and $|r_1 - r_2|$.
3. Relative position: $(x - 3)^2 + (y + 9)^2 = 144$.

Final answer

$$(x - 3)^2 + (y + 9)^2 = 144$$

Worked Solutions

Circles

Q210 MCQ

Find the circle with centre (3,1) touching $(x - 1)^2 + (y + 3)^2 = 5$ internally:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y - 1)^2 = 45$.

Correct option: A**Final answer**

$$(x - 3)^2 + (y - 1)^2 = 45$$

Worked Solutions

Circles

Q211 MCQ

Find the circle with centre $(-3, -9)$ touching $(x + 6)^2 + (y + 5)^2 = 1$ externally:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 3)^2 + (y + 9)^2 = 16$.

Correct option: D**Final answer**

$$(x + 3)^2 + (y + 9)^2 = 16$$

Worked Solutions

Circles

Q212 MCQ

Find the circle with centre (3,-9) touching $(x - 3)^2 + y^2 = 9$ internally:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x - 3)^2 + (y + 9)^2 = 144$.

Correct option: A**Final answer**

$$(x - 3)^2 + (y + 9)^2 = 144$$

Worked Solutions

Circles

Q213 Challenge

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(x + 1)^2 + (y + 2)^2 = 9$.

Final answer

$$(x + 1)^2 + (y + 2)^2 = 9$$

Worked Solutions

Circles

Q214 Challenge

Find the circle with centre $(-1, -2)$ touching $x^2 + y^2 - 4x - 4y + 4 = 0$ externally:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(x + 1)^2 + (y + 2)^2 = 9$.

Correct option: D

Final answer

$$(x + 1)^2 + (y + 2)^2 = 9$$

Worked Solutions

Circles

Q215 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$).

Final answer

$$2x - 4y - 5 = 0 \text{ (equivalently } y = x/2 - 5/4)$$

Q216 Written

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $x + y - 1 = 0$.

Final answer

$$x + y - 1 = 0$$

Worked Solutions

Circles

Q217 Written

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $12x - 6y - 7 = 0$.

Final answer

$$12x - 6y - 7 = 0$$

Q218 Written

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $x = 1/2$.

Final answer

$$x = 1/2$$

Worked Solutions

Circles

Q219 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $x^2 + y^2 + 2x - 4y = 9$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$).

Correct option: A**Final answer**

$$2x - 4y - 5 = 0 \text{ (equivalently } y = x/2 - 5/4 \text{)}$$

Worked Solutions

Circles

Q220 MCQ

Find the line through the intersections of $x^2 + y^2 = 6$ and $x^2 + y^2 + 3x + 3y = 9$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x + y - 1 = 0$.

Correct option: C**Final answer**

$$x + y - 1 = 0$$

Worked Solutions

Circles

Q221 MCQ

Find the line through the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 - 12x + 6y = 3$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $12x - 6y - 7 = 0$.

Correct option: B**Final answer**

$$12x - 6y - 7 = 0$$

Worked Solutions

Circles

Q222 MCQ

Find the line through the intersections of $x^2 + y^2 = 4$ and $(x - 1)^2 + y^2 = 4$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x = 1/2$.

Correct option: D**Final answer**

$$x = 1/2$$

Worked Solutions

Circles

Q223 Written

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $(-3, -1)$ and $(-1, 3)$.

Final answer $(-3, -1)$ and $(-1, 3)$

Q224 Written

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$.**Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: $(0, 2)$ and $(8/5, -6/5)$.

Final answer $(0, 2)$ and $(8/5, -6/5)$

Worked Solutions

Circles

Q225 **Written****Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$.****Solution**

1. Subtract the two circle equations to obtain the radical axis (common chord line).
2. Solve that line with either circle and verify the resulting points in both equations.
3. Intersections: (1, 2) and (2, 1).

Final answer

(1, 2) and (2, 1)

Worked Solutions

Circles

Q226 MCQ

Find the intersections of $x^2 + y^2 = 10$ and $x^2 + y^2 + 2x - y = 5$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-3, -1)$ and $(-1, 3)$.

Correct option: B**Final answer** $(-3, -1)$ and $(-1, 3)$

Worked Solutions

Circles

Q227 MCQ

Find the intersections of $x^2 + y^2 = 4$ and $(x - 4)^2 + (y - 2)^2 = 16$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(0, 2)$ and $(8/5, -6/5)$.

Correct option: C**Final answer** $(0, 2)$ and $(8/5, -6/5)$

Worked Solutions

Circles

Q228 MCQ

Find the intersections of $x^2 + y^2 = 5$ and $(x - 2)^2 + (y - 2)^2 = 1$:**Solution**

1. Use the same circle strategy as the paired written demand.
2. The correct option is (1, 2) and (2, 1).

Correct option: B**Final answer**

(1, 2) and (2, 1)

Worked Solutions

Circles

Q229 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points.

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $(-4, 3)$ and $(3, -4)$.

Final answer

$(-4, 3)$ and $(3, -4)$

Worked Solutions

Circles

Q230 Challenge

Two radar circles are $x^2 + y^2 = 25$ and $(x + 5)^2 + (y + 5)^2 = 65$. Find their intersection points:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $(-4, 3)$ and $(3, -4)$.

Correct option: C

Final answer

$(-4, 3)$ and $(3, -4)$

Worked Solutions

Circles

Q231 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 2x + 6y - 15 = 0$.

Final answer

$$x^2 + y^2 + 2x + 6y - 15 = 0$$

Q232 Written

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin.

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 4x + 2y = 0$.

Final answer

$$x^2 + y^2 + 4x + 2y = 0$$

Worked Solutions

Circles

Q233 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 10x + 20y - 39 = 0$.

Final answer

$$x^2 + y^2 + 10x + 20y - 39 = 0$$

Q234 Written

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through (-3,2).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 + 24x + 32y - 5 = 0$.

Final answer

$$x^2 + y^2 + 24x + 32y - 5 = 0$$

Worked Solutions

Circles

Q235 Written

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through $(4,0)$.

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $x^2 + y^2 - 10x - 6y + 24 = 0$.

Final answer

$$x^2 + y^2 - 10x - 6y + 24 = 0$$

Worked Solutions

Circles

Q236 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 5 = 0$ and $x^2 + y^2 = 5$, passing through (3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 2x + 6y - 15 = 0$.

Correct option: D

Final answer

$$x^2 + y^2 + 2x + 6y - 15 = 0$$

Worked Solutions

Circles

Q237 MCQ

Find the circle through the intersections of $x^2 + y^2 - 4x - 2y - 8 = 0$ and $x^2 + y^2 = 4$, passing through the origin:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 4x + 2y = 0$.

Correct option: A

Final answer

$$x^2 + y^2 + 4x + 2y = 0$$

Worked Solutions

Circles

Q238 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 4y + 3 = 0$ and $x^2 + y^2 = 4$, passing through (3,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 10x + 20y - 39 = 0$.

Correct option: D

Final answer

$$x^2 + y^2 + 10x + 20y - 39 = 0$$

Worked Solutions

Circles

Q239 MCQ

Find the circle through the intersections of $x^2 + y^2 - 6x - 8y = 0$ and $x^2 + y^2 - 1 = 0$, passing through $(-3,2)$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 24x + 32y - 5 = 0$.

Correct option: C

Final answer

$$x^2 + y^2 + 24x + 32y - 5 = 0$$

Worked Solutions

Circles

Q240 MCQ

Find the circle through the intersections of $x^2 + y^2 - 2x - 2y = 0$ and $x^2 + y^2 - 6x - 4y + 12 = 0$, passing through (4,0):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 - 10x - 6y + 24 = 0$.

Correct option: B

Final answer

$$x^2 + y^2 - 10x - 6y + 24 = 0$$

Worked Solutions

Circles

Q241 Challenge

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2).

Solution

1. Translate the combined circle/line conditions into equations.
2. Solve, classify the geometry, and verify every stated point or tangent.
3. Final checked result: $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$.

Final answer

$$x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$$

Worked Solutions

Circles

Q242 Challenge

Find the circle through the intersections of $x^2 + y^2 - 2x - 6y + 6 = 0$ and $x^2 + y^2 + 4x - 2y - 11 = 0$, passing through (2,2):

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$.

Correct option: A

Final answer

$$x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$$

Worked Solutions

Circles

Quiz MCQ 1 **Concept Quiz · MCQ**

For the circle $(x - 2)^2 + (y + 3)^2 = 25$, determine its centre and radius:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is centre $(2, -3)$, radius 5.

Correct option: B

Final answer

centre $(2, -3)$, radius 5

Worked Solutions

Circles

Quiz MCQ 2 **Concept Quiz · MCQ**

Determine the substitution discriminant when a line meets a circle at exactly one real point:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is 0.

Correct option: D

Final answer

0

Worked Solutions

Circles

Quiz MCQ 3 **Concept Quiz · MCQ**

Determine the chord length for radius r and centre-to-chord distance d :

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is $2\sqrt{r^2 - d^2}$.

Correct option: A

Final answer

$$2\sqrt{r^2 - d^2}$$

Worked Solutions

Circles

Quiz MCQ 4 **Concept Quiz · MCQ**

Determine the relative position of two non-overlapping circles when $D = r_1 + r_2$:

Solution

1. Use the same circle strategy as the paired written demand.
2. The correct option is externally tangent.

Correct option: D

Final answer

externally tangent

Worked Solutions

Circles

Quiz WRITTEN 5 Concept Quiz · Written

Find the circle with centre (1,-2) passing through (4,2).

Solution

1. Read the centre and radius, or complete the square in the given equation.
2. Use $(x - h)^2 + (y - k)^2 = r^2$ and compare coefficients when needed.
3. Circle: $(x - 1)^2 + (y + 2)^2 = 25$.

Final answer

$$(x - 1)^2 + (y + 2)^2 = 25$$

Quiz WRITTEN 6 Concept Quiz · Written

Find the intersections of $x^2 + y^2 = 25$ and $y = 3$.

Solution

1. Substitute the line equation into the circle equation, or compute the centre-to-line distance.
2. Use the discriminant (or d versus r) to classify two points, tangency, or no real point.
3. Intersection result: $(4, 3)$, $(-4, 3)$.

Final answer

$$(4, 3), (-4, 3)$$

Worked Solutions

Circles

Quiz WRITTEN 7 Concept Quiz · Written

Find the tangent to $x^2 + y^2 = 25$ at (3,4).

Solution

1. Confirm the contact point lies on the circle.
2. Use $(x - h)(x_1 - h) + (y - k)(y_1 - k) = r^2$ and rearrange.
3. Tangent: $3x + 4y = 25$.

Final answer

$$3x + 4y = 25$$

Quiz WRITTEN 8 Concept Quiz · Written

The circles $S_1 = x^2 + y^2 - 5 = 0$ and $S_2 = x^2 + y^2 - 2x - 2y - 4 = 0$ meet. Find the circle $S_1 + \lambda S_2 = 0$ passing through (1,1).

Solution

1. Use $S_1 + \lambda S_2 = 0$ for the pencil of circles through the common points.
2. Substitute the extra point or tangent condition to determine the parameter, then normalise.
3. Required circle: $\lambda = -1/2$; $x^2 + y^2 + 2x + 2y - 6 = 0$.

Final answer

$$\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$$

Answer key

Use the question number shown on each page.

Q001 $(x - 3)^2 + (y + 2)^2 = 25$	Q027 $(x - 2)^2 + (y - 3)^2 = 13$	Q053 $x^2 + y^2 - 2x - 4y = 0$	Q079 $(x + 1)^2 + (y - 1)^2 = 1$ or $(x + 5)^2 + (y - 5)^2 = 25$
Q002 $(x + 3)^2 + (y - 1)^2 = 4$	Q028 $(x + 3)^2 + (y - 1)^2 = 58$	Q054 $x^2 + y^2 - 6y - 16 = 0$	Q080 $(x - 1)^2 + (y - 1)^2 = 1$ or $(x - 5)^2 + (y - 5)^2 = 25$
Q003 $x^2 + y^2 = 16$	Q029 $(x - 1)^2 + (y + 3)^2 = 25$	Q055 $x^2 + y^2 - 2x + 8y - 8 = 0$	Q081 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$
Q004 $(x - 3)^2 + (y - 4)^2 = 36$	Q030 $(x + 4)^2 + (y - 1)^2 = 58$	Q056 A	Q082 $(x + 1)^2 + (y + 1)^2 = 1$ or $(x + 5)^2 + (y + 5)^2 = 25$
Q005 $(x + 2)^2 + (y - 6)^2 = 12$	Q031 $(x - 2)^2 + (y + 1)^2 = 5$	Q057 B	Q083 $(x - 5)^2 + (y + 5)^2 = 25$ or $(x - 13)^2 + (y + 13)^2 = 169$
Q006 $x^2 + y^2 = 25$	Q032 $x^2 + y^2 = 25$	Q058 D	Q084 C
Q007 A	Q033 C	Q059 D	Q085 D
Q008 A	Q034 A	Q060 D	Q086 C
Q009 D	Q035 D	Q061 $x^2 + y^2 - 6x - 6y + 8 = 0$; centre (3, 3), radius $\sqrt{10}$	Q087 C
Q010 B	Q036 A	Q062 A	Q088 B
Q011 B	Q037 B	Q063 $(x - 4)^2 + (y + 2)^2 = 4$	Q089 $(x - 1)^2 + (y + 1)^2 = 25$
Q012 A	Q038 A	Q064 $(x + 2)^2 + (y - 3)^2 = 4$	Q090 $(x + 5/2)^2 + (y - 5/2)^2 = 25/2$
Q013 circle; centre (3, -1), radius 2	Q039 $(x - 2)^2 + (y + 1)^2 = 17$	Q065 $(x + 1)^2 + (y - 4)^2 = 16$	Q091 $(x - 2)^2 + (y - 3)^2 = 2$
Q014 circle; centre (2, 3), radius 3	Q040 $(x - 4)^2 + (y - 1)^2 = 10$	Q066 $(x - 3)^2 + (y + 4)^2 = 16$	Q092 $(x - 4)^2 + (y - 2)^2 = 26$
Q015 circle; centre (2, -4), radius 4	Q041 $(x - 1)^2 + (y - 3)^2 = 25$	Q067 $(x - 5)^2 + (y + 4)^2 = 16$	Q093 D
Q016 circle; centre (2, 0), radius 2	Q042 $(x + 1)^2 + (y + 2)^2 = 25$	Q068 $(x + 1)^2 + (y - 6)^2 = 1$	Q094 C
Q017 circle; centre (-1, 2), radius 5	Q043 $(x + 2)^2 + (y - 1)^2 = 2$	Q069 $(x - 2)^2 + (y + 5)^2 = 4$	Q095 C
Q018 circle; centre (-2, 3), radius $\sqrt{13}$	Q044 $x^2 + (y - 2)^2 = 25$	Q070 $(x + 3)^2 + (y + 4)^2 = 9$	Q096 D
Q019 A	Q045 A	Q071 B	Q097 $(x + 1)^2 + y^2 = 5$; centre (-1, 0), radius $\sqrt{5}$
Q020 C	Q046 C	Q072 B	Q098 C
Q021 D	Q047 C	Q073 B	Q099 $k < -1$ or $k > 3$
Q022 C	Q048 A	Q074 D	Q100 $k < -\sqrt{6}$ or $k > \sqrt{6}$
Q023 C	Q049 B	Q075 D	Q101 $k < 5/4$
Q024 C	Q050 C	Q076 D	Q102 $k < -1/5$ or $k > 3$
Q025 centre (5, 3), radius ² = 13; $(x - 5)^2 + (y - 3)^2 = 13$	Q051 $x^2 + y^2 - 5x - y + 4 = 0$	Q077 B	Q103 $-1 < k < 4$
Q026 B	Q052 $x^2 + y^2 - 3x - 7y + 6 = 0$	Q078 C	Q104 C

Q105 B	Q137 $-5 \leq m \leq 5$	Q169 $k = -1$ or $k = 7$	Q201 D
Q106 C	Q138 $r = \sqrt{2}$	Q170 A	Q202 B
Q107 B	Q139 $r > \sqrt{17}$	Q171 $4x + 3y = 25$	Q203 A
Q108 D	Q140 D	Q172 $5x + 3y = 34$	Q204 D
Q109 $0 < k < 4$	Q141 B	Q173 $3x - y = 10$	Q205 B
Q110 B	Q142 B	Q174 $-5x - 2y = 29$	Q206 C
Q111 (1, -1) and (-1, 3)	Q143 B	Q175 $-2x + y = 5$	Q207 $(x - 3)^2 + (y - 1)^2 = 45$
Q112 (0, 2) and (2, 0)	Q144 C	Q176 A	Q208 $(x + 3)^2 + (y + 9)^2 = 16$
Q113 (3, -1)	Q145 A	Q177 C	Q209 $(x - 3)^2 + (y + 9)^2 = 144$
Q114 (-2, 1)	Q146 C	Q178 A	Q210 A
Q115 (-2, 1) and (4, -1)	Q147 B	Q179 C	Q211 D
Q116 C	Q148 D	Q180 A	Q212 A
Q117 D	Q149 $0 < r < 2$	Q181 $2x + y = 5$ and $-x + 2y = 5$	Q213 $(x + 1)^2 + (y + 2)^2 = 9$
Q118 C	Q150 D	Q182 $-3x + y = 10$ and $x + 3y = 10$	Q214 D
Q119 D	Q151 $\sqrt{14}$	Q183 $-2x + y = 5$ and $x + 2y = 5$	Q215 $2x - 4y - 5 = 0$ (equivalently $y = x/2 - 5/4$)
Q120 B	Q152 4	Q184 $y = 1$ and $4x - 3y = 5$	Q216 $x + y - 1 = 0$
Q121 0 points	Q153 $2\sqrt{5}$	Q185 $-x + 2y = 10$ and $2x + y = 10$	Q217 $12x - 6y - 7 = 0$
Q122 2 points : (1, -3) and (3, 1)	Q154 $3\sqrt{2}$	Q186 D	Q218 $x = 1/2$
Q123 2 points : $(-1 + \sqrt{6}/2, 1 + \sqrt{6}/2)$ and $(-1 - \sqrt{6}/2, 1 - \sqrt{6}/2)$	Q155 2	Q187 A	Q219 A
Q124 0 points	Q156 A	Q188 D	Q220 C
Q125 1 point : (-2, 0)	Q157 C	Q189 B	Q221 B
Q126 D	Q158 D	Q190 C	Q222 D
Q127 A	Q159 B	Q191 $-x + 2y = 10$ at (-2, 4); $2x + y = 10$ at (4, 0)	Q223 (-3, -1) and (-1, 3)
Q128 C	Q160 D	Q192 B	Q224 (0, 2) and $(8/5, -6/5)$
Q129 C	Q161 $k = \pm 2$	Q193 intersect at 2 points	Q225 (1, 2) and (2, 1)
Q130 D	Q162 $k = \pm\sqrt{31}$	Q194 B : touch externally	Q226 B
Q131 $-5 < k < 5$	Q163 $k = \pm 2$	Q195 E : intersect at 2 points	Q227 C
Q132 $r = \sqrt{5}$	Q164 $k = 0$ or $k = -2$	Q196 C : touch internally	Q228 B
Q133 $m = \pm\sqrt{3}$	Q165 C	Q197 C : touch internally	Q229 (-4, 3) and (3, -4)
Q134 $0 < r < \sqrt{10}$	Q166 D	Q198 E : intersect at 2 points	Q230 C
Q135 $m = \pm 4$	Q167 A	Q199 D : one is inside the other	Q231 $x^2 + y^2 + 2x + 6y - 15 = 0$
Q136 $n < -10$ or $n > 10$	Q168 B	Q200 C	Q232 $x^2 + y^2 + 4x + 2y = 0$

Q233 $x^2 + y^2 + 10x + 20y - 39 = 0$

Q234 $x^2 + y^2 + 24x + 32y - 5 = 0$

Q235 $x^2 + y^2 - 10x - 6y + 24 = 0$

Q236 D**Q237** A**Q238** D**Q239** C**Q240** B

Q241 $x^2 + y^2 + 2x - (10/3)y - 16/3 = 0$

Q242 A**Quiz MCQ 1** B**Quiz MCQ 2** D**Quiz MCQ 3** A**Quiz MCQ 4** D

Quiz WRITTEN 5 $(x - 1)^2 + (y + 2)^2 = 25$

Quiz WRITTEN 6 $(4, 3), (-4, 3)$

Quiz WRITTEN 7 $3x + 4y = 25$

Quiz WRITTEN 8 $\lambda = -1/2; x^2 + y^2 + 2x + 2y - 6 = 0$

Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$S = r\theta$$

Trig

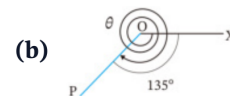
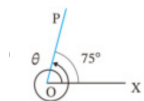
$$\sin \theta = \frac{y}{r}, \cos \theta = \frac{x}{r}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q001 Written

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75° ; (b) two full clockwise revolutions and a further 135° .



Solution

1. Add $360^\circ + 75^\circ$ for (a).
2. Use $360^\circ \times (-2) - 135^\circ$ for (b).

Final answer

(a) 435° (b) -855°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q002 Written

Draw the terminal side for 300° and -450° .

Solution

1. Reduce each angle by complete turns while keeping the sign of rotation.
2. 300° ends in quadrant IV; -450° is coterminal with -90° .

Final answer

300° : QIV -450° : negative y – axis

Q003 Written

Express 420° and -210° as $\alpha + 360^\circ \times n$, where n is an integer and $0^\circ \leq \alpha < 360^\circ$.

Solution

1. Choose the coterminal angle in the interval $0^\circ \leq \alpha < 360^\circ$.
2. Write the full family by adding $360^\circ \times n$.

Final answer

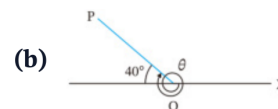
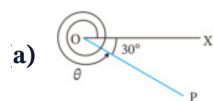
$60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q004 Written

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x -axis; (b) a terminal side 40° above the negative x -axis, measured counter-clockwise from the positive x -axis).



2 Draw the terminal side for the following angles.

Solution

1. Clockwise rotation is negative, so (a) is -30° .
2. For (b), $180^\circ - 40^\circ = 140^\circ$.

Final answer

(a) -30° (b) 140°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q005 Written

Draw the terminal side for 500° and -120° .

Solution

1. Subtract or add complete turns: $500^\circ - 360^\circ = 140^\circ$; -120° is already in QIII.

Final answer

500° : *QI* -120° : *QIII*

Q006 Written

Express 520° and -140° as $\alpha + 360^\circ \times n$.

Solution

1. Reduce each angle to the principal representative in $[0^\circ, 360^\circ)$.
2. Attach $360^\circ n$.

Final answer

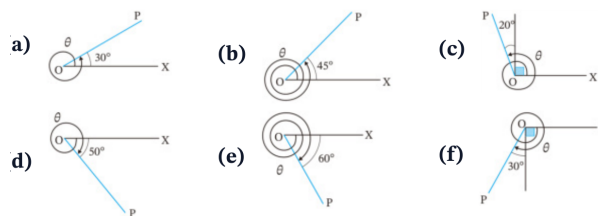
$160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q007 Written

Find the six diagram measures θ shown in the given figure (a)–(f).



2 Draw the terminal side for the following angles.

Solution

1. Read the marked acute angle and the direction of each complete turn.
2. Use $360^\circ \pm$ marked angle and $90^\circ \pm 20^\circ$ where shown.

Final answer

30° ; 405° ; 110° ; 310° ; 300° ; 240°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q008 Written

Draw the terminal sides for 470° , 780° , 225° , -380° , -990° , -200° .

Solution

1. Reduce each angle modulo 360° , preserving clockwise direction for negative angles.
2. Sketch the ray from the positive x -axis in the resulting quadrant.

Final answer

QII; QII; QIII; QI; QII; QII

Q009 Written

Express 610° , 920° , 840° , -430° , -960° , -20° as $\alpha + 360^\circ \times n$.

Solution

1. Find each representative α in $[0^\circ, 360^\circ)$.
2. The integer coefficient of 360° is absorbed into arbitrary integer n .

Final answer

$250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q010 Written

Express 129720° and -2592100° as $\alpha + 360^\circ \times n$, with $0^\circ \leq \alpha < 360^\circ$.

Solution

1. Divide by 360° and retain the remainder in the required interval.
2. Write each result as $\alpha + 360^\circ n$.

Final answer

$$120^\circ + 360^\circ n \quad ; \quad 260^\circ + 360^\circ n$$

Q011 MCQ

The principal angle coterminal with 765° is:

Solution

$$1. \quad 765^\circ - 720^\circ = 45^\circ$$

Correct option: A

Final answer

$$45^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q012 MCQ

The principal angle coterminal with -410° is:**Solution**

$$1. -410^\circ + 720^\circ = 310^\circ$$

Correct option: C

Final answer 310°

Q013 MCQ

The terminal side of 540° lies on the:**Solution**

$$1. 540^\circ - 360^\circ = 180^\circ$$

Correct option: B

Final answernegative x -axis

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q014 MCQ

A positive angle of 315° terminates in:**Solution**1. $315^\circ \in \text{QIV}$

Correct option: D

Final answer

QIV

Q015 MCQ

Which is the general-angle form for -75° ?**Solution**1. $-75^\circ + 360^\circ = 285^\circ$

Correct option: B

Final answer $285^\circ + 360^\circ n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q016 MCQ

The direction of a negative angle is:**Solution**

1. Negative rotation is clockwise.

Correct option: B

Final answer

clockwise

Q017 MCQ

The terminal side of 1080° is the:**Solution**1. $1080^\circ = 3 \times 360^\circ$

Correct option: A

Final answerpositive x -axis

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q018 MCQ

For $\alpha = 0^\circ$, the general-angle family is:**Solution**

1. Substitute $\alpha = 0^\circ$ in $\alpha + 360^\circ n$.

Correct option: A**Final answer** $360^\circ n$

Q019 MCQ

The principal angle for -990° is:**Solution**

1. $-990^\circ + 1080^\circ = 90^\circ$

Correct option: C**Final answer** 90°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q020 MCQ

A ray (P) is 25° above the negative x -axis. Its standard-position angle is:

Solution

1. $180^\circ - 25^\circ = 155^\circ$

Correct option: B

Final answer

155°

Q021 Written

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S .

Solution

1. Multiply degrees by $\pi/180$, and radians by $180/\pi$.
2. Use $L = r\theta$ and $S = \frac{1}{2}r^2\theta$ with θ in radians.

Final answer

$\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q022 Written

Complete the standard-angle degree/radian table, then: convert $288^\circ, -400^\circ$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Solution

1. Use the conversion multipliers and the standard table.
2. For a sector, substitute directly in $L = r\theta$ and $S = \frac{1}{2}r^2\theta$.

Final answer

$$\frac{8\pi}{5}, -\frac{20\pi}{9}; 330^\circ, -108^\circ; L = \frac{9\pi}{2}, S = \frac{27\pi}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q023 Written

Convert $108^\circ, 36^\circ, -240^\circ, -225^\circ, 420^\circ, 400^\circ, -570^\circ, -2700^\circ$ to radians.**Solution**

1. Multiply every degree measure by $\pi/180$.
2. Reduce each fraction to lowest terms.

Final answer

$$\frac{3\pi}{5}, \frac{\pi}{5}, -\frac{4\pi}{3}, -\frac{5\pi}{4}, \frac{7\pi}{3}, \frac{20\pi}{9}, -\frac{19\pi}{6}, -15\pi$$

Q024 Written

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ to degrees.**Solution**

1. Multiply every radian measure by $180^\circ/\pi$.
2. Keep the sign and simplify.

Final answer

$$210^\circ, 144^\circ, 120^\circ, 240^\circ, -225^\circ, -90^\circ, -990^\circ, -70^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q025 Written

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Solution

1. Apply $L = r\theta$.
2. Apply $S = \frac{1}{2}r^2\theta$ to each sector.

Final answer

(a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 Written

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Solution

1. Use the standard radian-angle table and the sign of rotation.
2. Sketch each terminal ray from the positive x -axis.

Final answer

QI; QI; QII; QII; QIII; QIII; QIV; QIV

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q027 Written

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Solution

1. The sector arc becomes the cone base circumference: $2\pi R = 9(4\pi/3)$, so $R = 6$.
2. The slant height is 9, hence $h = \sqrt{9^2 - 6^2} = 3\sqrt{5}$.
3. Use $V = \frac{1}{3}\pi R^2 h$.

Final answer

$$h = 3\sqrt{5} \text{ cm}; \quad V = 36\pi\sqrt{5} \text{ cm}^3$$

Q028 MCQ

Convert 150° to radians:

Solution

1. $150 \times \pi/180 = 5\pi/6$

Correct option: A

Final answer

$$\frac{5\pi}{6}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:

Solution

1. $-5\pi/6 \times 180/\pi = -150^\circ$

Correct option: B

Final answer

-150°

Q030 MCQ

For $r = 5, \theta = \frac{2\pi}{3}$, the arc length is:

Solution

1. $L = r\theta = 5(2\pi/3)$

Correct option: B

Final answer

$\frac{10\pi}{3}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q031 MCQ

For $r = 4$, $\theta = \frac{\pi}{2}$, the sector area is:

Solution

1. $S = \frac{1}{2}(4^2)(\pi/2) = 4\pi$

Correct option: B

Final answer

4π

Q032 MCQ

The radian measure of a full turn is:

Solution

1. A full circumference subtends 2π radians.

Correct option: B

Final answer

2π

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Solution

1. $7\pi/4 \times 180/\pi = 315^\circ$

Correct option: C

Final answer

315°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Solution

1. $-11\pi/6 + 2\pi = \pi/6$

Correct option: A

Final answer

$\frac{\pi}{6}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q035 MCQ

For $r = 3$, $\theta = \pi$, the sector area is:

Solution

1. $S = \frac{1}{2}(3^2)\pi = 9\pi/2$

Correct option: D

Final answer

$$\frac{9\pi}{2}$$

Q036 MCQ

The angle $\frac{5\pi}{3}$ lies in:

Solution

1. $5\pi/3 = 300^\circ$

Correct option: D

Final answer

QIV

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q037 Written

Convert 252° to radians.**Solution**

$$1. 252\pi/180 = 7\pi/5.$$

Final answer

$$\frac{7\pi}{5}$$

Q038 Written

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S .**Solution**

$$1. L = 8(3\pi/4) = 6\pi.$$

$$2. S = \frac{1}{2}(8^2)(3\pi/4) = 24\pi.$$

Final answer

$$L = 6\pi, \quad S = 24\pi$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q039 Written

If $\theta = \frac{4\pi}{3}$, find $\sin \theta$, $\cos \theta$, $\tan \theta$.

Solution

1. The terminal side is in QIII with reference angle $\pi/3$.
2. Use the signs of x, y on the unit circle.

Final answer

$$-\frac{\sqrt{3}}{2}, -\frac{1}{2}, \sqrt{3}$$

Q040 Written

Find the exact values (write undefined where appropriate): $\sin \frac{7\pi}{4}$, $\cos \frac{\pi}{3}$, $\tan \frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

Solution

1. Locate each angle on the unit circle.
2. Read x, y , then use $\sin \theta = y$, $\cos \theta = x$, $\tan \theta = y/x$.

Final answer

$$-\frac{\sqrt{2}}{2}; \frac{1}{2}; \frac{1}{\sqrt{3}}; -\frac{\sqrt{3}}{2}; -1; \text{undefined}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q041 Written

Find the exact values of the eighteen trigonometric functions listed in the given exercise (a)–(r), including negative and undefined cases.

Find the values of the following trigonometric functions. If the value is undefined, write “undefined”.

(a) $\sin \frac{\pi}{3}$

(b) $\sin \frac{7}{6}\pi$

(c) $\sin \pi$

(d) $\cos \frac{5}{4}\pi$

(e) $\cos \frac{5}{3}\pi$

(f) $\cos \frac{7}{6}\pi$

(g) $\tan \frac{2}{3}\pi$

(h) $\tan \frac{5}{4}\pi$

(i) $\tan \pi$

(j) $\sin \left(-\frac{\pi}{6}\right)$

(k) $\sin \left(-\frac{5}{4}\pi\right)$

(l) $\sin \left(-\frac{5}{3}\pi\right)$

(m) $\cos \left(-\frac{\pi}{4}\right)$

(n) $\cos \left(-\frac{5}{6}\pi\right)$

(o) $\cos \left(-\frac{4}{3}\pi\right)$

(p) $\tan \left(-\frac{4}{3}\pi\right)$

(q) $\tan \left(-\frac{11}{6}\pi\right)$

(r) $\tan \left(-\frac{3}{2}\pi\right)$

Solution

1. Reduce each angle to a standard reference angle.
2. Apply quadrant signs and the unit-circle exact values; tangent is undefined when $x = 0$.

Final answer

$\frac{\sqrt{3}}{2}; -\frac{1}{2}; 0; -\frac{\sqrt{2}}{2}; \frac{1}{2}; -\frac{\sqrt{3}}{2}; -\sqrt{3}; 1; 0; -\frac{1}{2}; \frac{\sqrt{2}}{2}; \frac{\sqrt{3}}{2}; \frac{\sqrt{2}}{2}; -\frac{\sqrt{3}}{2}; -\frac{1}{2}; -\sqrt{3}; \frac{1}{\sqrt{3}}; \text{undefined}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q042 Written

The panel height is $h(t) = 3 \sin(\frac{2\pi}{3}t - \frac{\pi}{4}) + 1.5$. Find the exact height at $t = 0$.

Solution

1. Substitute $t = 0$: $h(0) = 3 \sin(-\pi/4) + 1.5$.
2. Use $\sin(-\pi/4) = -\sqrt{2}/2$.

Final answer

$$\frac{3(1 - \sqrt{2})}{2} \text{ m}$$

Q043 MCQ

$\sin \frac{\pi}{6}$ equals:

Solution

1. Use the QI special-angle value.

Correct option: B

Final answer

$$\frac{1}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q044 MCQ

 $\cos \frac{2\pi}{3}$ equals:**Solution**1. QII has negative cosine; reference angle $\pi/3$.

Correct option: B

Final answer

$$-\frac{1}{2}$$

Q045 MCQ

 $\tan \frac{3\pi}{4}$ equals:**Solution**1. QII reference angle $\pi/4$.

Correct option: A

Final answer

$$-1$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q046 MCQ

 $\sin\left(-\frac{\pi}{2}\right)$ equals:**Solution**1. The terminal point is $(0, -1)$.

Correct option: A

Final answer

-1

Q047 MCQ

 $\tan \frac{\pi}{2}$ is:**Solution**1. $\tan = y/x$ and $x = 0$.

Correct option: D

Final answer

undefined

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q048 MCQ

 $\cos \frac{5\pi}{3}$ equals:**Solution**1. $5\pi/3$ has reference angle $\pi/3$ in QIV.

Correct option: D

Final answer

$$\frac{1}{2}$$

Q049 MCQ

If $P(x, y)$ lies on the unit circle, then $\sin \theta$ is:**Solution**1. By definition, $\sin \theta = y$.

Correct option: B

Final answer

$$y$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q050 MCQ

 $\tan\left(-\frac{3\pi}{4}\right)$ equals:**Solution**1. Tangent is odd: $\tan(-3\pi/4) = -\tan(3\pi/4) = 1$.

Correct option: C

Final answer

1

Q051 MCQ

 $\sin \frac{11\pi}{6}$ equals:**Solution**1. QIV reference angle $\pi/6$, so sine is negative.

Correct option: A

Final answer $-\frac{1}{2}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q052 Written

Find $\sin \frac{5\pi}{6}, \cos \frac{4\pi}{3}, \tan \frac{7\pi}{4}$.**Solution**

1. Use reference angles $\pi/6, \pi/3, \pi/4$ and quadrant signs.

Final answer

$$\frac{1}{2}, -\frac{1}{2}, -1$$

Q053 Written

Find $\sin(-\frac{5\pi}{6}), \cos(-\frac{7\pi}{6}), \tan(-\frac{\pi}{3})$.**Solution**

1. Reduce negative angles and apply odd/even symmetry.

Final answer

$$-\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\sqrt{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q054 Written

For $P = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$ on the unit circle, find the corresponding $\sin \theta$, $\cos \theta$, $\tan \theta$.

Solution

1. Read x, y directly, then compute y/x .

Final answer

$$\frac{\sqrt{3}}{2}, -\frac{1}{2}, -\sqrt{3}$$

Q055 Written

Find the exact value of $3 \sin \frac{3\pi}{2} - 2 \cos \pi$.

Solution

1. $3(-1) - 2(-1) = -1$.

Final answer

$$-1$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q056 Written

State the two standard angles where $\tan \theta$ is undefined in $[0, 2\pi)$.**Solution**

1. Tangent is undefined when the unit-circle x -coordinate is zero.

Final answer

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Q057 Written

Solve for θ , $0 \leq \theta < 2\pi$: (a) $\sin \theta = -\frac{1}{2}$; (b) $\cos \theta = \frac{1}{\sqrt{2}}$; (c) $\tan \theta = -\sqrt{3}$.**Solution**

1. Draw the unit circle and the corresponding horizontal/vertical/oblique line.
2. Use the reference angle and select the quadrants with the required sign.

Final answer

$$(a) \frac{7\pi}{6}, \frac{11\pi}{6}; (b) \frac{\pi}{4}, \frac{7\pi}{4}; (c) \frac{2\pi}{3}, \frac{5\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q058 Written

Solve the six trigonometric equations in the given Try set for $0 \leq \theta < 2\pi$: $\sin \theta = 1/\sqrt{2}$, $\cos \theta = -\sqrt{3}/2$, $\tan \theta = 1/\sqrt{3}$, $\sin \theta = 0$, $\cos \theta = 1/2$, $\tan \theta = -1$.

Solution

1. Find the reference angle from the special triangle.
2. Use the sign of the function to choose the correct quadrants.

Final answer

$$\frac{\pi}{4}, \frac{3\pi}{4}; \frac{5\pi}{6}, \frac{7\pi}{6}; \frac{\pi}{6}, \frac{7\pi}{6}; 0, \pi; \frac{\pi}{3}, \frac{5\pi}{3}; \frac{3\pi}{4}, \frac{7\pi}{4}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q059 Written

Solve the sixteen equations in the given exercise set (a)–(p) for $0 \leq \theta < 2\pi$.

Solution

1. Apply the same unit-circle/reference-angle method to each equation.
2. Remember that sine/cosine can have two solutions, while $\sin \theta = \pm 1$ and $\cos \theta = \pm 1$ have one in the interval.

Final answer

$\frac{\pi}{6}, \frac{5\pi}{6}; \frac{\pi}{3}, \frac{2\pi}{3}; \frac{\pi}{2}, \frac{3\pi}{2}; 0; 0, \pi; \frac{\pi}{3}, \frac{4\pi}{3}; \frac{\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}; \frac{3\pi}{4}, \frac{5\pi}{4}; \frac{2\pi}{3}, \frac{4\pi}{3}; \frac{5\pi}{6}, \frac{11\pi}{6}; \frac{\pi}{4}, \frac{5\pi}{4}; \frac{4\pi}{3}, \frac{5\pi}{3}; \frac{3\pi}{2}; \pi; \frac{\pi}{6}, \frac{11\pi}{6}$

Q060 Written

Solve $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta = \sqrt{3}$ for $0 \leq \theta < 360^\circ$.

Solution

1. Let $u = \tan \theta$ and factor $u^2 + (1 - \sqrt{3})u - \sqrt{3} = 0$.
2. The two tangent values are 1 and $-\sqrt{3}$; list their angles in $[0, 360^\circ)$.

Final answer

$60^\circ, 135^\circ, 240^\circ, 315^\circ$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = \frac{1}{2}$ gives:

Solution

1. Sine is positive in QI and QII with reference angle $\pi/6$.

Correct option: A

Final answer

$$\frac{\pi}{6}, \frac{5\pi}{6}$$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = -\frac{1}{2}$ gives:

Solution

1. Cosine is negative in QII and QIII.

Correct option: B

Final answer

$$\frac{2\pi}{3}, \frac{4\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = 1$ gives:**Solution**

1. Tangent is positive in QI and QIII.

Correct option: A**Final answer**

$$\frac{\pi}{4}, \frac{5\pi}{4}$$

Q064 MCQ

The solutions of $\sin \theta = 0$ in $[0, 2\pi)$ are:**Solution**

1. The unit-circle y-coordinate is zero on the x-axis.

Correct option: A**Final answer**

$$0, \pi$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q065 MCQ

If $\cos \theta = 1$, then in $[0, 2\pi)$, θ is:

Solution

1. The point $(1, 0)$ corresponds to $\theta = 0$.

Correct option: A

Final answer

0

Q066 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = -\sqrt{3}$ gives:

Solution

1. Tangent is negative in QII/QIV; reference angle $\pi/3$.

Correct option: B

Final answer

$\frac{2\pi}{3}, \frac{5\pi}{3}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = -1$ gives:**Solution**

1. The lowest unit-circle point has y-coordinate -1 .

Correct option: B**Final answer**

$$\frac{3\pi}{2}$$

Q068 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = 0$ gives:**Solution**

1. Cosine is the x-coordinate, zero on the y-axis.

Correct option: B**Final answer**

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q069 MCQ

The equation $\tan \theta = 0$ has solutions in $[0, 2\pi)$:**Solution**1. Tangent is zero when $y = 0$ and $x \neq 0$.

Correct option: A

Final answer $0, \pi$

Q070 Written

Solve $\cos \theta = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.**Solution**1. Cosine is positive in QI/QIV with reference angle $\pi/6$.**Final answer** $\frac{\pi}{6}, \frac{11\pi}{6}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q071 Written

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Sine is negative in QIII/QIV with reference angle $\pi/3$.

Final answer

$$\frac{4\pi}{3}, \frac{5\pi}{3}$$

Q072 Written

Solve $\tan \theta = \frac{1}{\sqrt{3}}$ for $0 \leq \theta < 2\pi$.

Solution

1. Tangent is positive in QI/QIII with reference angle $\pi/6$.

Final answer

$$\frac{\pi}{6}, \frac{7\pi}{6}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q073 Written

Solve $2 \cos \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.**Solution**1. Rearrange to $\cos \theta = 1/2$, then choose QI/QIV.**Final answer**

$$\frac{\pi}{3}, \frac{5\pi}{3}$$

Q074 Written

Solve $2 \sin \theta + \sqrt{2} = 0$ for $0 \leq \theta < 2\pi$.**Solution**1. Rearrange to $\sin \theta = -\sqrt{2}/2$, then choose QIII/QIV.**Final answer**

$$\frac{5\pi}{4}, \frac{7\pi}{4}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 1 **Concept Quiz · MCQ**

The radian measure of 240° is:

Solution

1. $240\pi/180 = 4\pi/3$.

Correct option: B

Final answer

$$\frac{4\pi}{3}$$

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\sin \theta = -\frac{\sqrt{3}}{2}$, $0 \leq \theta < 2\pi$, then θ is:

Solution

1. Sine is negative in QIII/QIV with reference angle $\pi/3$.

Correct option: B

Final answer

$$\frac{4\pi}{3}, \frac{5\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 3 Concept Quiz · MCQ

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Solution

1. $S = \frac{1}{2}(6^2)(\pi/3) = 6\pi$.

Correct option: C

Final answer

$$6\pi$$

Quiz MCQ 4 Concept Quiz · MCQ

The general-angle family for -300° is:

Solution

1. Add 360° to obtain the principal angle 60° .

Correct option: A

Final answer

$$60^\circ + 360^\circ n$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

Solve $\cos \theta = -\frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Cosine is negative in QII/QIII with reference angle $\pi/4$.

Final answer

$$\frac{3\pi}{4}, \frac{5\pi}{4}$$

Quiz WRITTEN 6 Concept Quiz · Written

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Solution

1. Multiply by $180/\pi$: -390° . Add 720° for the principal representative 330° .

Final answer

$$-390^\circ; 330^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Solution

1. $L = 10(2\pi/5) = 4\pi$.
2. $S = \frac{1}{2}(10^2)(2\pi/5) = 20\pi$.

Final answer

$$L = 4\pi, \quad S = 20\pi$$

Quiz WRITTEN 8 Concept Quiz · Written

Evaluate $3 \sin \frac{7\pi}{6} - 2 \cos \frac{4\pi}{3}$.

Solution

1. $3(-1/2) - 2(-1/2) = -1/2$.

Final answer

$$-\frac{1}{2}$$

Answer key

Use the question number shown on each page.

Q001 (a) 435° (b) -855°

Q002 300° : QIV

-450° : *negative y - axis*

Q003 $60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Q004 (a) -30° (b) 140°

Q005 500° : QI -120° : QIII

Q006 $160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Q007 30° ; 405° ; 110° ; 310° ; 300° ; 240°

Q008 QII; QII; QIII; QI; QII; QII

Q009 $250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Q010 $120^\circ + 360^\circ n$; $260^\circ + 360^\circ n$

Q011 A

Q012 C

Q013 B

Q014 D

Q015 B

Q016 B

Q017 A

Q018 A

Q019 C

Q020 B

Q021 $\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Q022 $\frac{8\pi}{5}$, $-\frac{20\pi}{9}$; 330° , -108° ; $L = \frac{9\pi}{2}$, $S = \frac{27\pi}{2}$

Q023 $\frac{3\pi}{5}$, $\frac{\pi}{5}$, $-\frac{4\pi}{3}$, $-\frac{5\pi}{4}$, $\frac{7\pi}{3}$, $\frac{20\pi}{9}$, $-\frac{19\pi}{6}$, -15π

Q024 210° , 144° , 120° , 240° , -225° , -90° , -990° , -70°

Q025 (a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 QI; QI; QII; QII; QIII; QIII; QIV; QIV

Q027 $h = 3\sqrt{5}$ cm; $V = 36\pi\sqrt{5}$ cm³

Q028 A

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 A

Q035 D

Q036 D

Q037 $\frac{7\pi}{5}$

Q038 $L = 6\pi$, $S = 24\pi$

Q039 $-\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $\sqrt{3}$

Q040 $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $\frac{1}{\sqrt{3}}$; $-\frac{\sqrt{3}}{2}$; -1 ; *undefined*

Q041 $\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; 0 ; $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $-\frac{\sqrt{3}}{2}$; $-\sqrt{3}$; 1 ; 0 ; $-\frac{1}{2}$; $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q042 $\frac{3(1-\sqrt{2})}{2}$ m

Q043 B

Q044 B

Q045 A

Q046 A

Q047 D

Q048 D

Q049 B

Q050 C

Q051 A

Q052 $\frac{1}{2}$, $-\frac{1}{2}$, -1

Q053 $-\frac{1}{2}$, $-\frac{\sqrt{3}}{2}$, $-\sqrt{3}$

Q054 $\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $-\sqrt{3}$

Q055 -1

Q056 $\frac{\pi}{2}$, $\frac{3\pi}{2}$

Q057 (a) $\frac{7\pi}{6}$, $\frac{11\pi}{6}$; (b) $\frac{\pi}{4}$, $\frac{7\pi}{4}$; (c) $\frac{2\pi}{3}$, $\frac{5\pi}{3}$

Q058 $\frac{\pi}{4}$, $\frac{3\pi}{4}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{\pi}{6}$, $\frac{7\pi}{6}$; 0 , π ; $\frac{\pi}{3}$, $\frac{5\pi}{3}$; $\frac{3\pi}{4}$, $\frac{7\pi}{4}$

Q059 $\frac{\pi}{6}$, $\frac{5\pi}{6}$; $\frac{\pi}{3}$, $\frac{2\pi}{3}$; $\frac{\pi}{2}$, $\frac{3\pi}{2}$; 0 , π ; $\frac{4\pi}{3}$, $\frac{2\pi}{3}$; $\frac{\pi}{4}$, $\frac{5\pi}{4}$; $\frac{3\pi}{4}$, $\frac{5\pi}{4}$; $\frac{2\pi}{3}$, $\frac{4\pi}{3}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{\pi}{6}$, $\frac{5\pi}{6}$; $\frac{\pi}{4}$, $\frac{3\pi}{4}$; $\frac{5\pi}{3}$, $\frac{4\pi}{3}$; $\frac{3\pi}{2}$, $\frac{1\pi}{2}$

Q060 60° , 135° , 240° , 315°

Q061 $\frac{1}{2}$, $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q062 B

Q063 A

Q064 A

Q065 A

Q066 B

Q067 B

Q068 B

Q069 A

Q070 $\frac{\pi}{6}$, $\frac{11\pi}{6}$

Q071 $\frac{4\pi}{3}$, $\frac{5\pi}{3}$

Q072 $\frac{\pi}{6}$, $\frac{7\pi}{6}$

Q073 $\frac{\pi}{3}$, $\frac{5\pi}{3}$

Q074 $\frac{5\pi}{4}$, $\frac{7\pi}{4}$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{3\pi}{4}$, $\frac{5\pi}{4}$

Quiz WRITTEN 6 -390° ; 330°

Quiz WRITTEN 7 $L = 4\pi$, $S = 20\pi$

Quiz WRITTEN 8 $-\frac{1}{2}$

Concept 2 | Trigonometric Identities and Expression Evaluation

English Student Edition • Focused formulas and guided practice

Formula opener

Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Tangent

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Secant

$$1 + \tan^2 \theta = \sec^2 \theta$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q001 Written

Given (a) $\sin \theta = -\frac{1}{5}$ in QIII and (b) $\tan \theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

Solution

1. Use $\sin^2 \theta + \cos^2 \theta = 1$, then choose the sign from the quadrant.
2. Use $\tan \theta = \sin \theta / \cos \theta$ (or $1 + \tan^2 \theta = 1 / \cos^2 \theta$).

Final answer

$$(a) \cos \theta = -\frac{2\sqrt{6}}{5}, \tan \theta = \frac{\sqrt{6}}{12}; \quad (b) \cos \theta = \frac{\sqrt{7}}{7}, \sin \theta = -\frac{\sqrt{42}}{7}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q002 Written

Given one value and the quadrant, find the other two: (a) $\cos \theta = -1/\sqrt{3}$, QIII; (b) $\tan \theta = -2$, QIV; (c) $\sin \theta = 3/5$, QII; (d) $\tan \theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

Solution

1. Find the missing squared value from a Pythagorean identity.
2. Select signs using the stated quadrant, then form the remaining ratio.

Final answer

$$(a) \sin \theta = -\frac{\sqrt{6}}{3}, \tan \theta = \sqrt{2}; (b) \cos \theta = \frac{\sqrt{5}}{5}, \sin \theta = -\frac{2\sqrt{5}}{5}; (c) \cos \theta = -\frac{4}{5}, \tan \theta = -\frac{3}{4}; (d) \sin \theta = -\frac{\sqrt{21}}{7}, \cos \theta = -\frac{2\sqrt{7}}{7}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q003 Written

For each of the twelve numbered cases in the given Exercise, one of $\sin \theta$, $\cos \theta$, $\tan \theta$ and a quadrant/interval is given. Find the other two values.

Solution

1. Square the known ratio and use $\sin^2 \theta + \cos^2 \theta = 1$ or $1 + \tan^2 \theta = 1/\cos^2 \theta$.
2. Apply the quadrant sign before simplifying the final ratio.

Final answer

$$(a) \left(-\frac{\sqrt{7}}{4}, \frac{3\sqrt{7}}{7}\right) (b) \left(-\frac{\sqrt{5}}{5}, -\frac{1}{2}\right) (c) \left(\frac{\sqrt{2}}{4}, -\frac{\sqrt{14}}{4}\right) (d) \left(-\frac{\sqrt{5}}{3}, -\frac{2\sqrt{5}}{5}\right) (e) \left(-\frac{\sqrt{15}}{4}, -\sqrt{15}\right) (f) \left(\frac{1}{3}, -\frac{2\sqrt{2}}{3}\right) (g) \left(-\frac{3\sqrt{10}}{10}, \frac{\sqrt{10}}{10}\right) (h) \left(-\frac{\sqrt{5}}{5}, -2\right) (i) \left(-\frac{12}{13}, -\frac{12}{5}\right) (j) \left(\frac{12}{13}, -\frac{5}{12}\right) (k) \left(-\frac{3}{5}, \frac{3}{4}\right) (l) \left(\frac{2\sqrt{5}}{5}, -\frac{\sqrt{5}}{5}\right)$$

Q004 Written

Using trigonometric identities, find the other two values given $\tan \theta + \cot \theta = 12$ and $\pi < \theta < 2\pi$.

Solution

1. Use $\tan \theta + \cot \theta = (\sin^2 \theta + \cos^2 \theta)/(\sin \theta \cos \theta)$ to obtain $\sin \theta \cos \theta = 1/12$.
2. The interval gives QIII or QIV; take the pair consistent with the stated signs.

Final answer

$$\sin \theta = -\frac{1}{2}, \quad \cos \theta = -\frac{\sqrt{3}}{2} \quad \text{or the swapped pair}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q005 Written

Given $\sin \theta = \frac{5}{13}$ **in QII, find** $\cos \theta$ **and** $\tan \theta$.

Solution

1. Use $\cos^2 \theta = 1 - 25/169$ and QII signs.

Final answer

$$-\frac{12}{13}, -\frac{5}{12}$$

Q006 Written

Given $\cos \theta = -\frac{3}{5}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Solution

1. Use the Pythagorean identity and QIII signs.

Final answer

$$-\frac{4}{5}, \frac{4}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q007 Written

Given $\tan \theta = \frac{3}{4}$ **in QI, find** $\sin \theta$ **and** $\cos \theta$.

Solution

1. Use a 3-4-5 triangle with positive signs.

Final answer

$$\frac{3}{5}, \frac{4}{5}$$

Q008 Written

Given $\tan \theta = -\frac{5}{12}$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$.

Solution

1. Use a 5-12-13 triangle and QIV signs.

Final answer

$$-\frac{5}{13}, \frac{12}{13}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q009 Written

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ **in QIII, find** $\cos \theta$ **and** $\tan \theta$.

Solution

1. The reference angle is $\pi/4$; both sine and cosine are negative in QIII.

Final answer

$$-\frac{\sqrt{2}}{2}, 1$$

Q010 MCQ

If $\sin \theta = -\frac{3}{5}$ **in QIII, $\cos \theta$ is:**

Solution

1. QIII makes cosine negative.

Correct option: B

Final answer

$$-\frac{4}{5}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q011 MCQ

If $\tan \theta = 2$ in QI, $\sin \theta$ is:**Solution**

1. Use a 1-2- $\sqrt{5}$ triangle.

Correct option: A

Final answer

$$\frac{2\sqrt{5}}{5}$$

Q012 MCQ

If $\cos \theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan \theta$ is:**Solution**

1. The reference angle is $\pi/4$ and tangent is negative in QIV.

Correct option: B

Final answer

$$-1$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q013 MCQ

For $\tan \theta = -\sqrt{3}$ in QII, $\sin \theta$ is:

Solution

1. Reference angle $\pi/3$, sine positive in QII.

Correct option: C

Final answer

$$\frac{\sqrt{3}}{2}$$

Q014 MCQ

If $\sin \theta = \frac{7}{25}$ in QI, $\tan \theta$ is:

Solution

1. The triangle is 7-24-25.

Correct option: A

Final answer

$$\frac{7}{24}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q015 Written

Given $\sin \theta + \cos \theta = \frac{1}{2}$, **with** θ **in QII**, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Solution

1. Square the sum: $1 + 2 \sin \theta \cos \theta = 1/4$.
2. Use $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$, then choose the QII sign for (c).

Final answer

$$(a) -\frac{3}{8} \quad (b) \frac{11}{16} \quad (c) \frac{\sqrt{7}}{2}$$

Q016 Written

Given $\sin \theta + \cos \theta = \frac{1}{3}$, **with** θ **in QII**, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Solution

1. Square the given sum to find the product.
2. Use the sum-of-cubes identity and the quadrant sign.

Final answer

$$(a) -\frac{4}{9} \quad (b) \frac{37}{27} \quad (c) \frac{\sqrt{17}}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q017 Written

Solve the three numbered Exercise problems: (1) given $\sin \theta - \cos \theta = \sqrt{3}/2$ in QI; (2) given $\sin \theta + \cos \theta = \sqrt{3}/2$ in QIV; (3) given $\sin \theta - \cos \theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

Solution

1. Square the given sum or difference to obtain the product.
2. Use the sum/difference of cubes and select the sign from the quadrant.

Final answer

$$(1) \sin \theta \cos \theta = \frac{1}{8}, \sin^3 \theta - \cos^3 \theta = \frac{7\sqrt{3}}{16}, \sin \theta + \cos \theta = \frac{\sqrt{5}}{2}; (2) -\frac{1}{8}, \frac{9\sqrt{3}}{16}, -\frac{\sqrt{5}}{2}; (3) \frac{3}{8}, \frac{11}{16}, -\frac{\sqrt{7}}{2}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q018 Written

Given $\sin \theta - \cos \theta = 0.5$ **and** θ **in QIII**, **evaluate the exact value of** $\sin^6 \theta - \cos^6 \theta$.

Solution

1. Find $\sin \theta \cos \theta = 3/8$, then $\sin^3 \theta - \cos^3 \theta = 11/16$ and $\sin^3 \theta + \cos^3 \theta = -5\sqrt{7}/16$.
2. Multiply the last two expressions.

Final answer

$$-\frac{55\sqrt{7}}{256}$$

Q019 Written

Given $\sin \theta + \cos \theta = \frac{2}{3}$ **in QII**, **find** $\sin \theta \cos \theta$ **and** $\sin \theta - \cos \theta$.

Solution

1. Square the sum and use QII signs.

Final answer

$$-\frac{5}{18}, \frac{\sqrt{14}}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q020 Written

Given $\sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$ **in QI, find** $\sin \theta \cos \theta$ **and** $\sin^3 \theta - \cos^3 \theta$.

Solution

1. Use $(s - c)^2 = 1 - 2sc$, then the difference-of-cubes identity.

Final answer

$$-\frac{1}{8}, \frac{7\sqrt{5}}{16}$$

Q021 Written

Given $\sin \theta + \cos \theta = -\frac{1}{2}$ **in QIII, find** $\sin^3 \theta + \cos^3 \theta$.

Solution

1. Find $sc = -3/8$, then apply the sum-of-cubes identity.

Final answer

$$-\frac{11}{16}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q022 Written

Given $\sin \theta - \cos \theta = \frac{1}{3}$ **in QIV, find** $\sin \theta + \cos \theta$.

Solution

1. Find $sc = 4/9$, so the square of the sum is $17/9$; use QIV sign.

Final answer

$$\frac{\sqrt{11}}{3}$$

Q023 Written

Given $\sin \theta - \cos \theta = 1$ **in QIII, find** $\sin^3 \theta - \cos^3 \theta$.

Solution

1. The product is zero and the difference-of-cubes factor is $1 + sc$.

Final answer

$$\frac{3}{2}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q024 MCQ

If $s + c = \frac{1}{2}$, then sc equals:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$$-\frac{3}{8}$$

Q025 MCQ

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

Solution

1. Use $(s + c)^3 - 3sc(s + c)$.

Correct option: B

Final answer

$$\frac{11}{16}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q026 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

Solution

1. The sum is negative in QIII.

Correct option: B

Final answer

$$-\frac{\sqrt{7}}{2}$$

Q027 MCQ

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

Solution

1. Use $(s - c)^2 = 1 - 2sc$.

Correct option: A

Final answer

$$\frac{1}{8}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q028 MCQ

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

Solution

1. Use the two cubic factors and multiply.

Correct option: A

Final answer

$$-\frac{55\sqrt{7}}{256}$$

Q029 Written

Prove $\frac{\sin \theta}{1 + \cos \theta} + \frac{1}{\tan \theta} = \frac{1}{\sin \theta}$.

Solution

1. Rewrite $1/\tan \theta = \cos \theta/\sin \theta$.
2. Use the common denominator $(1 + \cos \theta)\sin \theta$, then $\sin^2 \theta + \cos^2 \theta = 1$.

Final answer

$$\text{LHS} = \text{RHS}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q030 Written

Prove (a) $\frac{1-\tan \theta}{1+\tan \theta} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$; **(b)** $\frac{\cos \theta}{1-\sin \theta} - \tan \theta = \frac{1}{\cos \theta}$.

Solution

1. For (a), multiply numerator and denominator by $\cos \theta$.
2. For (b), multiply the first fraction by $(1 + \sin \theta)/(1 + \sin \theta)$, then simplify.

Final answer

Both identities verified

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q031 Written

Prove the six identities in the given Exercise: (a) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$; (b) $\tan \theta + 1/\tan \theta = 1/(\sin \theta \cos \theta)$; (c) $\cos \theta/(1 - \sin \theta) - 1/\cos \theta = \tan \theta$; (d) $\tan \theta/\sin \theta - \sin \theta/\tan \theta = \sin \theta \tan \theta$; (e) $(1 + \sin \theta)/\cos \theta + \cos \theta/(1 + \sin \theta) = 2/\cos \theta$; (f) $(\tan^2 \theta - \sin^2 \theta)/\sin^2 \theta = \tan^2 \theta$.

Solution

1. Replace tangent by $\sin \theta / \cos \theta$ where useful.
2. Combine fractions and use $\sin^2 \theta + \cos^2 \theta = 1$.

Final answer

LHS = RHS for (a)+(f)

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q032 Written

Prove that, for all defined values of x , $\frac{1+\sin x-\cos x}{1+\sin x+\cos x} + \frac{1+\sin x+\cos x}{1+\sin x-\cos x} = 2 \csc x$.

Solution

1. Set $A = 1 + \sin(x)$ and $(B = \cos(x))$, combine the two fractions, and simplify the numerator to $(2(1 + \sin(x)))$.
2. Use $(1 - \sin^2 x = \cos^2 x)$ in the denominator to obtain $(2 / \sin(x))$.

Final answer

$$\text{LHS} = 2 \csc x$$

Q033 Written

Prove $\frac{1+\sin \theta}{\cos \theta} = \frac{\cos \theta}{1-\sin \theta}$.

Solution

1. Cross-multiply and use $1 - \sin^2 \theta = \cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q034 **Written**

Prove $\frac{1}{1+\cos \theta} + \frac{1}{1-\cos \theta} = 2 \sec^2 \theta.$

Solution

1. Combine over $1 - \cos^2 \theta = \sin^2 \theta$, then rewrite.

Final answer

Identity proved

Q035 **Written**

Prove $\frac{\tan \theta}{1+\tan \theta} = \frac{\sin \theta}{\sin \theta + \cos \theta}.$

Solution

1. Multiply numerator and denominator on the left by $\cos \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q036 Written

Prove $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}.$

Solution

1. Multiply top and bottom by $1 + \cos \theta$, then use the Pythagorean identity.

Final answer

Identity proved

Q037 Written

Prove $\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}.$

Solution

1. Rationalise the left denominator and cancel $\cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q038 MCQ

The first move in proving a trig identity is usually to:

Solution

1. Keep one side fixed and transform the other.

Correct option: B

Final answer

Simplify one side

Q039 MCQ

The identity used to replace $1 - \sin^2 \theta$ is:

Solution

1. From $\sin^2 \theta + \cos^2 \theta = 1$.

Correct option: A

Final answer

$\cos^2 \theta$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q040 MCQ

$\tan \theta$ should be rewritten as:

Solution

1. Use the quotient identity.

Correct option: B

Final answer $\sin \theta / \cos \theta$

Q041 MCQ

To add two trig fractions, first find a:

Solution

1. Fraction addition requires a common denominator.

Correct option: B

Final answer

common denominator

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q042 MCQ

The result of a valid identity proof is:**Solution**

1. The two expressions are shown equal.

Correct option: B**Final answer**

LHS=RHS

Q043 MCQ

If $\cos \theta = -\frac{12}{13}$ in QII, $\sin \theta$ is:**Solution**

1. Use the 5-12-13 triangle and the QII sign.

Correct option: A**Final answer** $\frac{5}{13}$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q044 MCQ

If $\tan \theta = -\frac{1}{2}$ in QII, $\cos \theta$ is:

Solution

1. Cosine is negative in QII.

Correct option: A

Final answer

$$-\frac{2\sqrt{5}}{5}$$

Q045 MCQ

If $\sin \theta = \frac{8}{17}$ in QIV, $\tan \theta$ is:

Solution

1. The 8-15-17 triangle has negative sine in QIV.

Correct option: B

Final answer

$$-\frac{8}{15}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q046 MCQ

If $\tan \theta = \sqrt{3}$ and θ is in QIII, $\sin \theta$ is:

Solution

1. Reference angle $\pi/3$, with both ratios negative in QIII.

Correct option: D

Final answer

$$-\frac{\sqrt{3}}{2}$$

Q047 MCQ

If $s + c = 1$, then sc equals:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$$0$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q048 MCQ

If $s - c = \sqrt{2}$, then sc equals:

Solution

1. Use $(s - c)^2 = 1 - 2sc$.

Correct option: A

Final answer

$$-\frac{1}{2}$$

Q049 MCQ

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

Solution

1. Use $(s + c)^3 - 3sc(s + c)$.

Correct option: A

Final answer

$$\frac{19}{27}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q050 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

Solution

1. Use $(s - c)(1 + sc)$ with $sc = 3/8$.

Correct option: A

Final answer

$$\frac{11}{16}$$

Q051 MCQ

Which identity converts $\frac{1}{\tan \theta}$ into sine and cosine?

Solution

1. Take the reciprocal of tangent.

Correct option: B

Final answer

$$\frac{\cos \theta}{\sin \theta}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q052 MCQ

The identity $1 - \cos^2 \theta$ simplifies to:**Solution**

1. Rearrange the Pythagorean identity.

Correct option: A**Final answer**

$$\sin^2 \theta$$

Q053 MCQ

When proving an identity, which side should normally be transformed?**Solution**

1. Keep one side fixed and simplify the other to it.

Correct option: B**Final answer**

One side only

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q054 MCQ

After combining trig fractions, the next step is usually to use:

Solution

1. Replace $1 - \sin^2\theta$ or $1 - \cos^2\theta$ using the identity.

Correct option: B

Final answer

The Pythagorean identity

Quiz MCQ 1 Concept Quiz · MCQ

If $\sin \theta + \cos \theta = \frac{1}{2}$, then $\sin \theta \cos \theta$ is:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$-\frac{3}{8}$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz MCQ 2 [Concept Quiz · MCQ](#)

If $\tan \theta = \frac{3}{4}$ in QI, then $\cos \theta$ is:

Solution

1. Use a 3-4-5 triangle.

Correct option: B

Final answer

$$\frac{4}{5}$$

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Which identity is always valid where defined?

Solution

1. Use the fundamental quotient identity.

Correct option: B

Final answer

$$1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz MCQ 4 Concept Quiz · MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

Solution

1. Square the difference and use the QIII sign.

Correct option: B

Final answer

$$-\frac{\sqrt{7}}{2}$$

Quiz WRITTEN 5 Concept Quiz · Written

Given $\cos \theta = -\frac{5}{13}$ in QIII, find $\sin \theta$ and $\tan \theta$.

Solution

1. Use the 5-12-13 triangle and QIII signs.

Final answer

$$-\frac{12}{13}, \frac{12}{5}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 6 Concept Quiz · Written

Given $s + c = \frac{2}{3}$, **find** sc and $s^3 + c^3$.

Solution

1. Square the sum to find sc , then apply the sum-of-cubes identity.

Final answer

$$-\frac{5}{18}, \frac{19}{27}$$

Quiz WRITTEN 7 Concept Quiz · Written

Prove $\frac{1-\sin \theta}{\cos \theta} = \frac{\cos \theta}{1+\sin \theta}$.

Solution

1. Cross-multiply and use $1 - \sin^2 \theta = \cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 8 Concept Quiz · Written

Given $\tan \theta = -2$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$, **then verify** $1 + \tan^2 \theta = 1/\cos^2 \theta$.

Solution

1. Use a 1-2- $\sqrt{5}$ triangle and QIV signs.
2. Substitute into both sides of the identity.

Final answer

$$\sin \theta = -\frac{2\sqrt{5}}{5}, \cos \theta = \frac{\sqrt{5}}{5}; 6 = 6$$

Answer key

Use the question number shown on each page.

Q001 (a) $\cos \theta = -\frac{2\sqrt{6}}{5}$, $\tan \theta = \frac{\sqrt{6}}{12}$; (b) $\cos \theta = \frac{\sqrt{7}}{7}$, $\sin \theta = -\frac{\sqrt{42}}{7}$ **Q015** (a) $-\frac{3}{8}$ (b) $\frac{11}{16}$ (c) $\frac{\sqrt{7}}{2}$

Q002 (a) $\sin \theta = -\frac{\sqrt{6}}{3}$, $\tan \theta = \sqrt{2}$; (b) $\cos \theta = \frac{\sqrt{5}}{5}$, $\sin \theta = -\frac{2\sqrt{5}}{5}$; (c) $\cos \theta = -\frac{4}{5}$, $\tan \theta = -\frac{3}{4}$; (d) $\sin \theta = -\frac{\sqrt{21}}{7}$, $\cos \theta = -\frac{2\sqrt{7}}{7}$

Q003 $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (b) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (c) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (d) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$ **Q016** (a) $-\frac{4}{9}$ (b) $\frac{37}{27}$ (c) $\frac{\sqrt{17}}{3}$

Q004 $\sin \theta = -\frac{1}{2}$, $\cos \theta = -\frac{\sqrt{3}}{2}$ or the swapped pair **Q017** (1) $\sin \theta \cos \theta = \frac{1}{8}$, $\sin^2 \theta - \cos^2 \theta = \frac{7\sqrt{3}}{16}$, $\sin \theta + \cos \theta = \frac{\sqrt{5}}{2}$; (2) $-\frac{1}{8}$, $\frac{9\sqrt{3}}{16}$, $-\frac{\sqrt{5}}{2}$; (3) $\frac{3}{8}$, $\frac{11}{16}$ **Q031** LHS = RHS for (a)+(f)

Q005 $-\frac{12}{13}$, $-\frac{5}{12}$

Q018 $-\frac{55\sqrt{7}}{256}$

Q006 $-\frac{4}{5}$, $\frac{4}{3}$

Q019 $-\frac{5}{18}$, $\frac{\sqrt{14}}{3}$

Q007 $\frac{3}{5}$, $\frac{4}{5}$

Q020 $-\frac{1}{8}$, $\frac{7\sqrt{5}}{16}$

Q008 $-\frac{5}{13}$, $\frac{12}{13}$

Q021 $-\frac{11}{16}$

Q009 $-\frac{\sqrt{2}}{2}$, 1

Q022 $\frac{\sqrt{11}}{3}$

Q010 B

Q023 $\frac{3}{2}$

Q011 A

Q024 A

Q012 B

Q025 B

Q013 C

Q026 B

Q014 A

Q027 A

Q028 A

Q029 LHS = RHS

Q030 Both identities verified

Q031 LHS = RHS for (a)+(f)

Q032 LHS = $2 \csc x$

Q033 Identity proved

Q034 Identity proved

Q035 Identity proved

Q036 Identity proved

Q037 Identity proved

Q038 B

Q039 A

Q040 B

Q041 B

Q042 B

Q043 A

Q044 A

Q045 B

Q046 D

Q047 A

Q048 A

Q049 A

Q050 A

Q051 B

Q052 A

Q053 B

Q054 B

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $-\frac{12}{13}$, $\frac{12}{5}$

Quiz WRITTEN 6 $-\frac{5}{18}$, $\frac{19}{27}$

Quiz WRITTEN 7 Identity proved

Quiz WRITTEN 8 $\sin \theta = -\frac{2\sqrt{5}}{5}$, $\cos \theta = \frac{\sqrt{5}}{5}$; $6 = 6$

Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Amplitude

$$y = a \sin(bx + c) + d \Rightarrow |a|$$

Period

$$T = \frac{2\pi}{|b|}$$

Range

$$d - |a| \leq y \leq d + |a|$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q001 Written

(1) Express $\sin \frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin \theta + \sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi) + \sin(\theta + \frac{3\pi}{2})$.

Solution

1. Reduce the angle to its reference angle and keep the quadrant sign.
2. Use the quarter-turn identities: $\sin(\theta + \frac{\pi}{2}) = \cos \theta$, $\sin(\theta + \pi) = -\sin \theta$, and $\sin(\theta + \frac{3\pi}{2}) = -\cos \theta$.

Final answer

(1) $-\sin \frac{\pi}{5}$ (2) 0

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q002 Written

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin \frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2} - \theta) \sin(\pi - \theta) - \sin(\frac{3\pi}{2} + \theta) \cos(\pi - \theta)$.

Solution

1. Use evenness of cosine and the reference-angle rules.
2. Replace shifted functions by $\sin \theta$ or $\cos \theta$, then simplify.

Final answer

(1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q003 Written

Solve all three given demands: (1) express $\cos \frac{17\pi}{5}$, $\cos \frac{5\pi}{7}$, $\sin(-\frac{7\pi}{8})$, $\sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin \theta + \cos(\pi + \theta) - \cos(\pi - \theta) + \sin(\pi + \theta)$, (b) $\sin(\theta + \frac{\pi}{2}) \cos(\theta + \pi) + \sin(-\theta) \cos(\frac{\pi}{2} - \theta)$.

Solution

1. Reduce each angle to an acute reference angle and apply the sign in its quadrant.
2. For (3), use the quarter-turn and half-turn identities before collecting terms.

Final answer

$$(1) -\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8} \quad (2) -\frac{1}{\sqrt{3}}, -1 \quad (3a) 0, (3b) -1$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q004 Written

Simplify $\sin(\frac{3\pi}{2} - \theta) \sec(\pi + \theta) - \tan(\theta - \frac{\pi}{2}) \cot(\frac{5\pi}{2} + \theta)$.**Solution**

1. Use $\sin(\frac{3\pi}{2} - \theta) = -\cos \theta$, $\sec(\pi + \theta) = -\sec \theta$, $\tan(\theta - \frac{\pi}{2}) = -\cot \theta$, and $\cot(\frac{5\pi}{2} + \theta) = -\tan \theta$.
2. Both products equal 1, so their difference is 0.

Final answer

0

Q005 Written

Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, **and** $y = \tan \theta$, **and state each period and range.****Solution**

1. Mark one standard cycle, then repeat it periodically.
2. For tangent, draw vertical asymptotes at $\theta = \frac{\pi}{2} + k\pi$.

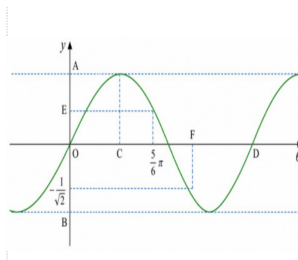
Final answer $\sin, \cos : T = 2\pi, -1 \leq y \leq 1; \quad \tan : T = \pi, y \in \mathbb{R}$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q006 Written

For the given graph of $y = \sin \theta$, determine the labelled scales A to F .



Solution

1. Read the extrema and the x-axis scale from the standard sine cycle.
2. Use $\sin \frac{5\pi}{6} = \frac{1}{2}$ and the descending branch to locate F.

Final answer

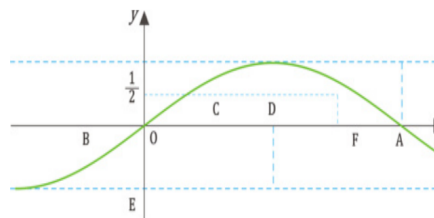
$$A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q007 Written

(1) Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, $y = \tan \theta$, and state their periods. (2) For the given graph of $y = \cos \theta$, determine the labelled scales A to F .



Solution

1. Use the standard parent graph and mark quarter-period points.
2. The vertical values are the cosine range $[-1, 1]$; the horizontal values follow the 2π cycle.

Final answer

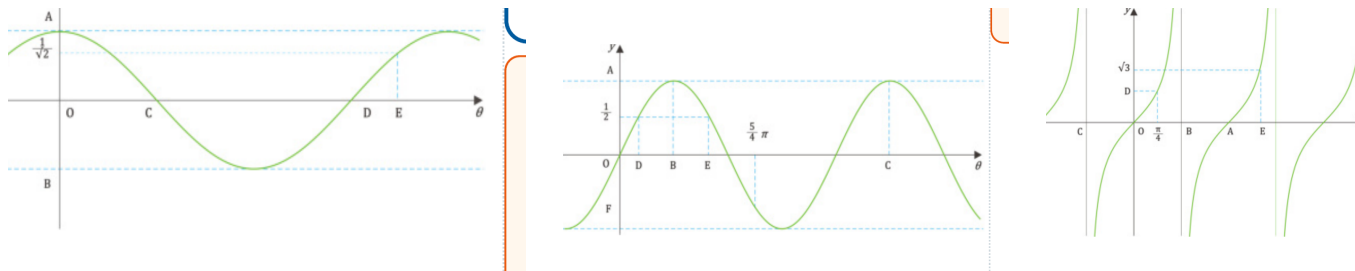
(1) $T_{\sin} = 2\pi$, $T_{\cos} = 2\pi$, $T_{\tan} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q008 Written

Complete all four given graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the given cosine graph; read the labelled scales on the given sine graph; and read the labelled scales on the given tangent graph.



Solution

1. Draw axes and the standard cycle first.
2. Read each label from an intercept, extremum, asymptote or quarter-period tick; do not change the graph scale.

Final answer

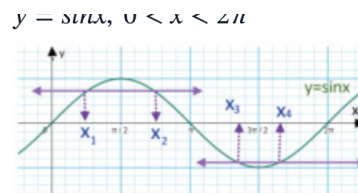
Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q009 Written

The given graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



Then find: $x_1 + x_2 + x_3 + x_4$

Solution

1. Identify the y-level associated with each marked point and use the sine cycle on $(0, 2\pi)$.
2. Add the four x-coordinates only after they have been read from the same horizontal scale.

Final answer

Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q010 Written

Draw the graphs of $y = 2 \cos \theta$ and $y = \tan \frac{\theta}{2}$, and find their periods.

Solution

1. The coefficient 2 outside cosine doubles the y-values but not the period.
2. For $\tan(\theta/2)$, multiply the parent tangent period by the reciprocal of $1/2$.

Final answer

$$y = 2 \cos \theta : T = 2\pi, -2 \leq y \leq 2; \quad y = \tan \frac{\theta}{2} : T = 2\pi$$

Q011 Written

Draw the graphs of $y = 2 \sin \theta$, $y = \tan 2\theta$, and $y = \cos \frac{\theta}{2}$, and find their periods.

Solution

1. Use the vertical factor for amplitude/range and the reciprocal of the coefficient of θ for horizontal scale.
2. For tangent, the parent period is π ; for sine/cosine it is 2π .

Final answer

$$T = 2\pi, \frac{\pi}{2}, 4\pi \text{ respectively; ranges } [-2, 2], \mathbb{R}, [-1, 1]$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q012 Written

Draw the nine given graphs and find each period: (a) $3 \sin \theta$, (b) $3 \cos \theta$, (c) $\frac{1}{2} \cos \theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin \frac{\theta}{2}$, (h) $\cos \frac{\theta}{3}$, (i) $\tan \frac{\theta}{3}$.

Solution

1. Read the vertical coefficient as the amplitude/range factor.
2. Divide the parent period by the absolute coefficient of θ .

Final answer

(a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q013 Written

Draw $y = -2|\sin(3x)|$ and find its period.**Solution**

1. The absolute value makes the sine pattern repeat after π in its input, so the x-period is $\pi/3$.
2. The negative outside factor reflects the nonnegative $|\sin(3x)|$ graph below the x-axis.

Final answer

$$T = \frac{\pi}{3}, \quad 0 \geq y \geq -2$$

Q014 Written

Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.**Solution**

1. Start the standard sine cycle at $\theta = \pi/6$.
2. A horizontal shift does not change the parent period.

Final answer

$$\text{shift right by } \frac{\pi}{6}, \quad T = 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q015 **Written**

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

Solution

1. Read $\theta - p$ as a shift right by p and $\theta + p$ as a shift left by p .
2. Keep the parent period unchanged.

Final answer

Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q016 Written

Draw all given graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; **cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; **tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

Solution

1. Translate the parent graph horizontally by the signed amount inside the bracket.
2. Use the same vertical range and asymptote spacing as the parent function.

Final answer

All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q017 Written

Draw $y = 3 \cos |x + \frac{\pi}{3}|$ and find its period.

Solution

1. Use $\cos |u| = \cos(u)$, then read the horizontal shift and vertical scale.
2. The period of cosine remains 2π .

Final answer

$$y = 3 \cos(x + \frac{\pi}{3}) \text{ since cosine is even; } T = 2\pi$$

Q018 Written

Draw the graphs and find their periods: (1) $y = 2 \cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

Solution

1. Factor the coefficient of θ from the bracket before reading the shift.
2. Apply shift, horizontal scale, then vertical scale.

Final answer

$$(1) y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2; \quad (2) y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3 \sin \frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2 \cos(2\theta - \frac{\pi}{3})$.

Solution

1. Identify the coefficient b , factor the angle, then read the shift q/b .
2. Use the parent period divided by $|b|$ and the outside amplitude $|a|$.

Final answer

$$(a) T = 4\pi, -3 \leq y \leq 3; \quad (b) T = \frac{\pi}{2}; \quad (c) T = \pi, -2 \leq y \leq 2$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q020 Written

Draw all nine given graphs and find their periods: (a) $2 \cos \frac{\theta}{2}$, (b) $3 \sin 2\theta$, (c) $2 \cos 2\theta$, (d) $2 \cos(\theta - \frac{\pi}{4})$, (e) $2 \sin(\theta - \frac{\pi}{3})$, (f) $\tan(2\theta - \frac{\pi}{3})$, (g) $3 \cos(2\theta - \frac{\pi}{2})$, (h) $3 \sin(2\theta - \frac{2\pi}{3})$, (i) $2 \cos(\frac{\theta}{2} - \frac{\pi}{4})$.

Solution

1. Rewrite every angle as $b(\theta - q/b)$.
2. Apply the transformations from the inside out and record period, range and shift.

Final answer

Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q021 Written

Draw $y = 3 \sin |2x - \pi|$ and find its period.**Solution**

1. The absolute value makes the sine pattern repeat after π in its input; with coefficient 2 the x-period is $\pi/2$.
2. The graph is nonnegative with maximum 3.

Final answer

$$y = 3|\sin(2x - \pi)|, \quad T = \frac{\pi}{2}, \quad 0 \leq y \leq 3$$

Q022 MCQ

 $\sin(\theta + \pi)$ equals:**Solution**

1. A half-turn changes the sign of sine.

Correct option: B**Final answer**

$$-\sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q023 MCQ

 $\cos(\theta + \pi)$ equals:**Solution**

1. A half-turn changes the sign of cosine.

Correct option: B

Final answer

$$-\cos \theta$$

Q024 MCQ

 $\sin\left(\frac{\pi}{2} - \theta\right)$ equals:**Solution**

1. Use the complementary-angle identity.

Correct option: C

Final answer

$$\cos \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q025 MCQ

 $\cos(-\theta)$ equals:**Solution**

1. Cosine is even.

Correct option: A

Final answer

$$\cos \theta$$

Q026 MCQ

 $\sin(-\theta)$ equals:**Solution**

1. Sine is odd.

Correct option: B

Final answer

$$-\sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q027 MCQ

 $\cos(\pi - \theta)$ equals:**Solution**

1. Use the supplementary-angle identity.

Correct option: B

Final answer

$$-\cos \theta$$

Q028 MCQ

 $\sin(\frac{3\pi}{2} + \theta)$ equals:**Solution**

1. Shift by three quarter-turns.

Correct option: D

Final answer

$$-\cos \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q029 MCQ

 $\tan(\theta - \frac{\pi}{2})$ equals:**Solution**

1. Use the quarter-turn tangent identity.

Correct option: D

Final answer $-\cot \theta$

Q030 MCQ

The period of $y = \sin \theta$ is:**Solution**

1. One sine cycle spans 2π .

Correct option: C

Final answer 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q031 MCQ

The range of $y = \cos \theta$ is:**Solution**

1. Cosine is bounded by -1 and 1.

Correct option: B

Final answer

$$-1 \leq y \leq 1$$

Q032 MCQ

The period of $y = \tan \theta$ is:**Solution**1. Tangent repeats every π .

Correct option: B

Final answer

$$\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q033 MCQ

A vertical asymptote of $y = \tan \theta$ occurs at:**Solution**

1. Cosine is zero at odd multiples of $\pi/2$.

Correct option: B**Final answer**

$$\theta = \frac{\pi}{2} + k\pi$$

Q034 MCQ

The maximum value of $\sin \theta$ is:**Solution**

1. Read the top of the standard sine graph.

Correct option: C**Final answer**

1

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q035 MCQ

The x-coordinate of the first positive sine maximum is:**Solution**1. Sine first reaches 1 at $\pi/2$.**Correct option:** B**Final answer** $\pi/2$

Q036 MCQ

At $\theta = 0$, $\cos \theta$ equals:**Solution**

1. The cosine graph starts at its maximum.

Correct option: C**Final answer**

1

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q037 MCQ

At $\theta = 0$, $\tan \theta$ equals:**Solution**

1. Tangent passes through the origin.

Correct option: B**Final answer**

0

Q038 MCQ

The range of $y = 3 \sin \theta$ is:**Solution**

1. Multiply the sine range by 3.

Correct option: B**Final answer** $-3 \leq y \leq 3$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q039 MCQ

The period of $y = \sin 4\theta$ is:**Solution**1. Use $2\pi/|b|$.**Correct option:** B**Final answer** $\pi/2$

Q040 MCQ

The period of $y = \cos(\theta/3)$ is:**Solution**1. The coefficient is $1/3$, so multiply the parent period by 3.**Correct option:** D**Final answer** 6π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q041 MCQ

The period of $y = \tan 5\theta$ is:**Solution**1. Use $\pi/|b|$.**Correct option:** A**Final answer**

$$\pi/5$$

Q042 MCQ

The range of $y = \frac{1}{2} \cos \theta$ is:**Solution**

1. The outside factor halves the cosine values.

Correct option: C**Final answer**

$$-\frac{1}{2} \leq y \leq \frac{1}{2}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q043 MCQ

The graph of $y = -2 \sin \theta$ is the parent sine graph:**Solution**

1. The negative sign reflects; 2 stretches.

Correct option: B**Final answer**

reflected and stretched vertically

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:**Solution**1. Multiply the tangent period by the reciprocal of $1/2$.**Correct option:** C**Final answer** 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q045 MCQ

The period of $y = \sin(3\theta)$ is:**Solution**

1. Use $2\pi/3$.

Correct option: B**Final answer** $2\pi/3$

Q046 MCQ

The graph of $y = \sin(\theta - \pi/4)$ is shifted:**Solution**

1. $\theta - p$ means shift right by p .

Correct option: B**Final answer**right by $\pi/4$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q047 MCQ

The graph of $y = \cos(\theta + \pi/3)$ is shifted:**Solution**1. $\theta + p$ means shift left by p .

Correct option: B

Final answerleft by $\pi/3$

Q048 MCQ

A horizontal shift changes the sine period from 2π to:**Solution**

1. Translation does not change period.

Correct option: B

Final answer 2π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q049 MCQ

A horizontal shift changes the tangent period from π to:**Solution**

1. Translation does not change period.

Correct option: B**Final answer** π

Q050 MCQ

The graph of $y = \sin(\theta + \pi/2)$ equals:**Solution**

1. Use the quarter-turn identity.

Correct option: C**Final answer** $\cos \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q051 MCQ

The graph of $y = \cos(\theta - \pi)$ equals:**Solution**1. A π shift reverses cosine.

Correct option: B

Final answer

$$-\cos \theta$$

Q052 MCQ

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:**Solution**1. Set $\theta - \pi/2 = 0$.

Correct option: B

Final answer

$$\theta = \pi/2$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q053 MCQ

Since cosine is even, $\cos |x|$ equals:**Solution**

1. Evenness removes the absolute value inside cosine.

Correct option: A**Final answer** $\cos x$

Q054 MCQ

In $y = a \sin(b\theta - q)$, the horizontal shift is:**Solution**

1. Factor b from the angle first.

Correct option: B**Final answer** q/b

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q055 MCQ

The period of $y = 2 \cos(3\theta - \pi)$ is:**Solution**

1. The coefficient of θ is 3.

Correct option: B**Final answer** $2\pi/3$

Q056 MCQ

The range of $y = 4 \sin(2\theta - 1)$ is:**Solution**

1. The outside factor controls the range.

Correct option: C**Final answer** $[-4, 4]$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q057 MCQ

For $y = \tan(2\theta - \pi/3)$, the shift is:**Solution**

1. Rewrite $2(\theta - \pi/6)$.

Correct option: B**Final answer**right by $\pi/6$

Q058 MCQ

The period of $y = 3 \sin(\theta/2)$ is:**Solution**

1. The coefficient is $1/2$.

Correct option: D**Final answer** 4π

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:**Solution**

1. Factor b and read q/b first.

Correct option: B

Final answer

horizontal shift

Q060 MCQ

The period of $y = \tan(4\theta + q)$ is:**Solution**1. Tangent period is $\pi/|4|$.

Correct option: A

Final answer $\pi/4$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q061 MCQ

The range of $y = 3 \sin |2x - \pi|$ is:**Solution**

1. The absolute sine is nonnegative and the factor is 3.

Correct option: B**Final answer**

$$0 \leq y \leq 3$$

Q062 Written

Express $\cos \frac{11\pi}{6}$ as a function of an acute angle.**Solution**

1. Use the QIV reference angle and cosine's positive sign.

Final answer

$$\cos \frac{\pi}{6}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q063 Written

Simplify $\sin(\theta + \pi) - \cos(\frac{\pi}{2} - \theta)$.**Solution**

1. Replace the shifted functions by $-\sin \theta$ and $\sin \theta$.

Final answer

$$-\sin \theta - \sin \theta = -2 \sin \theta$$

Q064 Written

Simplify $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.**Solution**

1. Use the half-turn and complementary-angle identities.

Final answer

$$-\cos \theta + \sin \theta$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q065 Written

Evaluate $\tan(-\frac{3\pi}{4})$ using an acute reference angle.**Solution**

1. The reference angle is $\pi/4$ and tangent is positive in QIII.

Final answer

1

Q066 Written

State the period and range of $y = \cos \theta$, then list its key points on one cycle.**Solution**

1. Read the standard cosine cycle at quarter-period intervals.

Final answer

$$T = 2\pi, -1 \leq y \leq 1; (0, 1), (\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan \theta$ on $[-\pi, \pi]$.**Solution**

1. Tangent is undefined at odd multiples of $\pi/2$.

Final answer

$$T = \pi; \theta = \pm \frac{\pi}{2}$$

Q068 Written

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.**Solution**

1. Sine crosses the x-axis at integer multiples of π .

Final answer

$$0, \pi, 2\pi$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q069 Written

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Solution

1. Cosine is zero at odd quarter-turns.

Final answer

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Q070 Written

Find the period and range of $y = 4 \cos \theta$.

Solution

1. The outside factor changes the range, not the period.

Final answer

$$T = 2\pi, -4 \leq y \leq 4$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q071 Written

Find the period and range of $y = \sin(5\theta)$.**Solution**

1. Divide the sine period by 5.

Final answer

$$T = \frac{2\pi}{5}, -1 \leq y \leq 1$$

Q072 Written

Find the period and range of $y = \tan(\theta/4)$.**Solution**

1. Multiply the parent tangent period by 4.

Final answer

$$T = 4\pi, y \in \mathbb{R}$$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q073 Written

Describe the transformations from $y = \cos \theta$ to $y = -\frac{1}{2} \cos(2\theta)$.

Solution

1. Read the inside coefficient and outside factor separately.

Final answer

horizontal compression by $1/2$, reflection in x-axis and vertical scale $1/2$; $T = \pi$

Q074 Written

Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.

Solution

1. $\theta + p$ shifts left and does not alter period.

Final answer

left by $\frac{\pi}{6}$, $T = 2\pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q075 Written

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.**Solution**

1. θ -p shifts right by p.

Final answerright by $\frac{\pi}{2}$, $T = 2\pi$

Q076 Written

Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.**Solution**

1. Set the bracket equal to zero and retain tangent's period.

Final answer $\theta = -\frac{\pi}{4}$; $T = \pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q077 Written

For $y = 2 \sin(4\theta - \pi)$, state the shift, period and range.

Solution

- Factor 4 from the angle: $4(\theta - \pi/4)$.

Final answer

right by $\frac{\pi}{4}$, $T = \frac{\pi}{2}$, $-2 \leq y \leq 2$

Q078 Written

For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.

Solution

- Rewrite the angle as $\frac{1}{3}(\theta + \pi/2)$.

Final answer

left by $\frac{\pi}{2}$, $T = 6\pi$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Q079 Written

For $y = 3 \tan(2\theta + \pi)$, state the shift and period.**Solution**

1. Rewrite as $3 \tan 2(\theta + \pi/2)$.

Final answerleft by $\frac{\pi}{2}$, $T = \frac{\pi}{2}$

Q080 Written

For $y = 2 \cos(\theta/2 - \pi/8)$, state the shift, period and range.**Solution**

1. Rewrite the angle as $\frac{1}{2}(\theta - \pi/4)$.

Final answerright by $\frac{\pi}{4}$, $T = 4\pi$, $-2 \leq y \leq 2$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz MCQ 1 **Concept Quiz · MCQ**

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:

Solution

1. Use $2\pi/2$.

Correct option: B

Final answer

π

Quiz MCQ 2 **Concept Quiz · MCQ**

$\sin(\theta + \pi)$ equals:

Solution

1. Use the half-turn identity.

Correct option: B

Final answer

$-\sin \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz MCQ 3 Concept Quiz · MCQ

The range of $y = 2 \cos \theta$ is:

Solution

1. Multiply the cosine range by 2.

Correct option: B

Final answer

$$-2 \leq y \leq 2$$

Quiz MCQ 4 Concept Quiz · MCQ

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Solution

1. $\theta - p$ shifts right by p .

Correct option: B

Final answer

right by $\frac{\pi}{4}$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

For $y = 3 \sin(2\theta - \pi)$, state the horizontal shift, period and range.

Solution

1. Rewrite as $3 \sin 2(\theta - \pi/2)$.

Final answer

right by $\frac{\pi}{2}$, $T = \pi$, $-3 \leq y \leq 3$

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Solution

1. Use the quarter-turn and half-turn identities.

Final answer

$\cos \theta - \sin \theta$

Worked Solutions

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.

Solution

1. Shift the parent tangent graph right by $\pi/3$; shift its asymptotes with it.

Final answer

$$T = \pi; \theta = \frac{5\pi}{6} + k\pi$$

Quiz WRITTEN 8 Concept Quiz · Written

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Solution

1. Absolute sine has input period π ; the coefficient 3 gives $\pi/3$, and -2 reflects and scales.

Final answer

$$T = \frac{\pi}{3}, -2 \leq y \leq 0; \text{ reflected below the x-axis}$$

Answer key

Use the question number shown on each page.

- Q001** (1) $-\sin \frac{\pi}{5}$ (2) 0
- Q002** (1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$
- Q003** (1) $-\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) $-\frac{1}{\sqrt{3}}$
- Q004** 0
- Q005** $\sin, \cos$: $T = 2\pi, -1 \leq y \leq 1$; $\tan$: $T = \pi, y \in \mathbb{R}$
- Q006** $A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$
- Q007** (1) $T_{\max} = 2\pi, T_{\min} = 2\pi, T_{\max} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph
- Q008** Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.
- Q009** Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.
- Q010** $y = 2 \cos \theta$: $T = 2\pi, -2 \leq y \leq 2$; $y = \tan \frac{\theta}{2}$: $T = 2\pi$
- Q011** $T = 2\pi, \frac{\pi}{2}, 4\pi$ respectively; ranges $[-2, 2], \mathbb{R}, [-1, 1]$
- Q012** (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient
- Q013** $T = \frac{\pi}{3}, 0 \geq y \geq -2$
- Q014** shift right by $\frac{\pi}{6}, T = 2\pi$
- Q015** Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.
- Q016** All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.
- Q017** $y = 3 \cos(x + \frac{\pi}{3})$ since cosine is even; $T = 2\pi$
- Q018** (1) $y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2$; (2) $y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$
- Q019** (a) $T = 4\pi, -3 \leq y \leq 3$; (b) $T = \frac{\pi}{2}$; (c) $T = \pi, -2 \leq y \leq 2$
- Q020** Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.
- Q021** $y = 3|\sin(2x - \pi)|, T = \frac{\pi}{2}, 0 \leq y \leq 3$
- Q022** B
- Q023** B
- Q024** C
- Q025** A
- Q026** B
- Q027** B
- Q028** D
- Q029** D
- Q030** C
- Q031** B
- Q032** B
- Q033** B
- Q034** C
- Q035** B
- Q036** C
- Q037** B
- Q038** B
- Q039** B
- Q040** D
- Q041** A
- Q042** C
- Q043** B
- Q044** C
- Q045** B
- Q046** B
- Q047** B
- Q048** B
- Q049** B
- Q050** C
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Q055** B
- Q056** C
- Q057** B
- Q058** D
- Q059** B
- Q060** A
- Q061** B
- Q062** $\cos \frac{\pi}{6}$
- Q063** $-\sin \theta - \sin \theta = -2 \sin \theta$
- Q064** $-\cos \theta + \sin \theta$
- Q065** 1
- Q066** $T = 2\pi, -1 \leq y \leq 1$; (0, 1), $(\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$
- Q067** $T = \pi; \theta = \pm \frac{\pi}{2}$
- Q068** 0, $\pi, 2\pi$
- Q069** $\frac{\pi}{2}, \frac{3\pi}{2}$
- Q070** $T = 2\pi, -4 \leq y \leq 4$
- Q071** $T = \frac{2\pi}{5}, -1 \leq y \leq 1$
- Q072** $T = 4\pi, y \in \mathbb{R}$
- Q073** horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T = \pi$
- Q074** left by $\frac{\pi}{6}, T = 2\pi$
- Q075** right by $\frac{\pi}{2}, T = 2\pi$
- Q076** $\theta = -\frac{\pi}{4}; T = \pi$
- Q077** right by $\frac{\pi}{4}, T = \frac{\pi}{2}, -2 \leq y \leq 2$
- Q078** left by $\frac{\pi}{2}, T = 6\pi$
- Q079** left by $\frac{\pi}{2}, T = \frac{\pi}{2}$
- Q080** right by $\frac{\pi}{4}, T = 4\pi, -2 \leq y \leq 2$
- Quiz MCQ 1** B
- Quiz MCQ 2** B
- Quiz MCQ 3** B

Quiz MCQ 4 B**Quiz WRITTEN 6** $\cos \theta - \sin \theta$ **Quiz WRITTEN 7** $T = \pi; \theta = \frac{5\pi}{6} + k\pi$ **Quiz WRITTEN 8** $T = \frac{\pi}{3}, -2 \leq y \leq 0$; reflected below the**Quiz WRITTEN 5** right by $\frac{\pi}{2}$, $T = \pi, -3 \leq y \leq 3$

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Sine

$$\sin x = \sin \alpha$$

Cosine

$$\cos x = \cos \alpha$$

Tangent

$$\tan x = \tan \alpha$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q001 Written

Solve (a) $-2 \cos \theta = \sqrt{2}$ **and (b)** $\tan \theta + \sqrt{3} = 0$, giving the general solution for (b).

Solution

1. Rewrite as $\cos \theta = -\frac{\sqrt{2}}{2}$ and use QII/QIII.
2. For tangent, the reference angle is $\pi/3$ and the period is π .

Final answer

$$(a) \theta = \frac{3\pi}{4}, \frac{5\pi}{4} \quad (b) \theta = \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}$$

Q002 Written

Solve (a) $2 \cos \theta - 1 = 0$ **and (b)** $\tan \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

Solution

1. Use the unit circle for cosine and include both turns in the stated interval.
2. For tangent, repeat the $\pi/4$ reference angle after one period π .

Final answer

$$(a) \theta = \frac{\pi}{3}, \frac{5\pi}{3} \quad (b) \theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q003 Written

Solve all given equations on $0 \leq \theta \leq 2\pi$: (a) $2 \sin \theta + 1 = 0$, (b) $2 \sin \theta = \sqrt{3}$, (c) $3 \sin \theta + 3 = 0$, (d) $\sqrt{2} \cos \theta + 1 = 0$, (e) $2 \cos \theta = \sqrt{3}$, (f) $2 \cos \theta - \sqrt{2} = 0$, (g) $\tan \theta + 1 = 0$, (h) $\sqrt{3} \tan \theta = 1$, (i) $\sqrt{3} \tan \theta = 3$, **and give general solutions for** (j) $2 \sin \theta + \sqrt{3} = 0$, (k) $\cos \theta + 1 = 0$, (l) $\sqrt{3} \tan \theta + 3 = 0$.

Solution

1. Put each equation into a single-function form, find the reference angle, then select the quadrants from the sign.
2. Use period 2π for sine/cosine and π for tangent; reject the endpoint 2π when the interval is half-open.

Final answer

(a) $\frac{7\pi}{6}, \frac{11\pi}{6}$ (b) $\frac{\pi}{3}, \frac{2\pi}{3}$ (c) $\frac{3\pi}{2}$ (d) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (e) $\frac{\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{4}, \frac{7\pi}{4}$ (g) $\frac{3\pi}{4}, \frac{7\pi}{4}$ (h) $\frac{\pi}{6}, \frac{7\pi}{6}$ (i) $\frac{\pi}{3}, \frac{4\pi}{3}$ (j) $\frac{4\pi}{3} + 2n\pi, \frac{5\pi}{3} + 2n\pi$ (k) $\pi + 2n\pi$ (l) $\frac{2\pi}{3} + n\pi$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q004 Written

Solve $\tan^2 \theta - (2 + \sqrt{3}) \tan \theta + 2\sqrt{3} = 0$ **and give the general solution.**

Solution

1. Factor the quadratic as $(\tan \theta - 2)(\tan \theta - \sqrt{3}) = 0$.
2. Apply the tangent period π to both roots.

Final answer

$$\tan \theta = 2 \text{ or } \tan \theta = \sqrt{3}; \quad \theta = \arctan(2) + n\pi \text{ or } \theta = \frac{\pi}{3} + n\pi$$

Q005 Written

Solve for $0 \leq \theta < 2\pi$: **(a)** $2 \cos \theta > -\sqrt{3}$, **(b)** $\tan \theta \leq \frac{1}{\sqrt{3}}$.

Solution

1. Solve the boundary equation first, then shade the correct arcs on the unit circle.
2. Tangent is undefined at odd multiples of $\pi/2$; those points are never included.

Final answer

$$(a) 0 \leq \theta < \frac{5\pi}{6} \text{ or } \frac{7\pi}{6} < \theta < 2\pi \quad (b) 0 \leq \theta \leq \frac{\pi}{6} \text{ or } \frac{\pi}{2} < \theta \leq \frac{7\pi}{6} \text{ or } \frac{3\pi}{2} < \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q006 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin \theta \leq -1$, (b) $\cos \theta > \frac{1}{2}$, (c) $\tan \theta > 1$.

Solution

1. Mark the equality angles and choose the arcs where the function has the required sign.
2. Open endpoints correspond to strict inequalities and to tangent asymptotes.

Final answer

$$(a) \frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6} \quad (b) 0 \leq \theta < \frac{\pi}{3} \text{ or } \frac{5\pi}{3} < \theta < 2\pi \quad (c) \frac{\pi}{4} < \theta < \frac{\pi}{2} \text{ or } \frac{5\pi}{4} < \theta < \frac{3\pi}{2}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q007 Written

Solve all given inequalities for $0 \leq \theta < 2\pi$: (a) $\sin \theta > \frac{\sqrt{3}}{2}$, (b) $\sin \theta - \frac{\sqrt{2}}{2} > 0$, (c) $2 \sin \theta + \sqrt{3} \leq 0$, (d) $\cos \theta > 0$, (e) $\cos \theta - \frac{1}{2} \leq 0$, (f) $2 \cos \theta + \sqrt{2} \leq 0$, (g) $\tan \theta \leq -\frac{1}{\sqrt{3}}$, (h) $\tan \theta + \sqrt{3} > 0$, (i) $\tan \theta + 1 \geq 0$.

Solution

- For each item, solve the equality, draw the relevant horizontal line, and select the arcs satisfying the sign.
- Keep the distinction between open and closed endpoints and remove all tangent asymptotes.

Final answer

$$(a) \left(\frac{\pi}{3}, \frac{2\pi}{3}\right) (b) \left(\frac{\pi}{4}, \frac{3\pi}{4}\right) (c) \left[\frac{4\pi}{3}, \frac{5\pi}{3}\right] (d) \left[0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right) (e) \left[\frac{\pi}{3}, \frac{5\pi}{3}\right] (f) \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right] (g) \left(\frac{\pi}{2}, \frac{5\pi}{6}\right] \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right] (h) \left[0, \frac{\pi}{2}\right) \cup \left(\frac{2\pi}{3}, \frac{3\pi}{2}\right) \cup \left(\frac{5\pi}{3}, 2\pi\right) (i) \left[0, \frac{\pi}{2}\right) \cup \left[\frac{3\pi}{4}, \frac{3\pi}{2}\right) \cup \left[\frac{7\pi}{4}, 2\pi\right)$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q008 Written

Solve $2 \sin^2 \theta - \cos \theta - 1 > 0$ for $0 \leq \theta < 2\pi$.**Solution**

1. Use $\sin^2 \theta = 1 - \cos^2 \theta$ to obtain $(1 - 2 \cos \theta)(1 + \cos \theta) > 0$.
2. Since $-1 \leq \cos \theta \leq 1$, the solution is $-1 < \cos \theta < 1/2$; exclude $\theta = \pi$ where the product is zero.

Final answer

$$\frac{\pi}{3} < \theta < \pi \text{ or } \pi < \theta < \frac{5\pi}{3}$$

Q009 Written

Solve for $0 \leq \theta < 2\pi$: **(a)** $2 \cos^2 \theta - 3 \sin \theta = 0$, **(b)** $2 \sin^2 \theta - 3 \cos \theta - 3 \leq 0$.**Solution**

1. For (a), replace $\cos^2 \theta$ and solve $2s^2 + 3s - 2 = 0$; reject $s = -2$.
2. For (b), replace $\sin^2 \theta$, solve $(2c + 1)(c + 1) \geq 0$, and intersect with $[-1, 1]$.

Final answer

$$(a) \theta = \frac{\pi}{6}, \frac{5\pi}{6} \quad (b) 0 \leq \theta \leq \frac{2\pi}{3} \text{ or } \theta = \pi \text{ or } \frac{4\pi}{3} \leq \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q010 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta - \sin \theta = 0$, (b) $2 \cos^2 \theta - \sin \theta = 2$, (c) $2 \sin^2 \theta - 4 < 5 \cos \theta$.

Solution

1. Set $s = \sin \theta$ or $c = \cos \theta$ after using $s^2 + c^2 = 1$.
2. Solve the resulting quadratic, reject impossible trig values, then read the unit-circle intervals.

Final answer

$$(a) \theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi \quad (b) \theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6} \quad (c) 0 \leq \theta < \frac{2\pi}{3} \text{ or } \frac{4\pi}{3} < \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q011 Written

Solve all given equations and inequalities for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta + \sin \theta - 1 = 0$, (b) $2 \sin^2 \theta + 3 \cos \theta = 0$, (c) $2 \sin^2 \theta - \cos \theta = 2$, (d) $2 \cos^2 \theta - 5 \cos \theta - 3 = 0$, (e) $2 \cos^2 \theta + \sin \theta - 1 = 0$, (f) $2 \cos^2 \theta + 3 \sin \theta = 3$, (g) $2 \cos^2 \theta + 3 \sin \theta \leq 0$, (h) $2 \sin^2 \theta + \cos \theta > 1$, (i) $2 \cos^2 \theta - 4 \leq 5 \sin \theta$.

Solution

1. Convert to one trig function, solve the quadratic, and apply $-1 \leq \sin \theta, \cos \theta \leq 1$.
2. For inequalities, use the sign chart in the trig variable and then translate the accepted interval(s) to θ .

Final answer

(a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ (b) $\frac{2\pi}{3}, \frac{4\pi}{3}$ (c) $\frac{2\pi}{3}, \frac{\pi}{2}, \frac{4\pi}{3}, \frac{3\pi}{2}$ (d) $\frac{2\pi}{3}, \frac{4\pi}{3}$ (e) $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$ (g) $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (h) $0 < \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ (i) $0 \leq \theta \leq \frac{7\pi}{6}$ or $\frac{11\pi}{6} \leq \theta < 2\pi$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q012 Written

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ for $0 \leq \theta < 2\pi$.

Solution

1. Subtract 1 and simplify the numerator to $\sin\theta(1 - \sin\theta)$.
2. The denominator is positive but undefined at $\theta = \pi/2$; hence $0 \leq \sin\theta \leq 1$ with that point removed.

Final answer

$$0 \leq \theta < \frac{\pi}{2} \text{ or } \frac{\pi}{2} < \theta \leq \pi$$

Q013 Written

Solve for $0 \leq \theta < 2\pi$: **(a)** $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, **(b)** $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

Solution

1. Let $t = \theta + \alpha$, write the range of t , solve there, then subtract α .
2. For the inequality, shade one complete revolution in the shifted t-range and preserve strict endpoints.

Final answer

$$(a) \theta = \frac{\pi}{2}, \frac{11\pi}{6} \quad (b) 0 \leq \theta < \pi \text{ or } \frac{5\pi}{3} < \theta < 2\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q014 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, (b) $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, (c) $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$.

Solution

1. Find each shifted argument's exact range before selecting the unit-circle arcs.
2. For tangent, exclude asymptotes automatically; for strict cosine inequality, endpoints are open.

Final answer

$$(a) \theta = 0, \frac{2\pi}{3} \quad (b) \theta = \pi, \frac{13\pi}{12} \quad (c) \frac{\pi}{2} < \theta < \pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q015 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$, (b) $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$, (c) $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$, (d) $\tan(\theta + \frac{\pi}{4}) = -1$, (e) $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$, (f) $\tan(\theta + \frac{\pi}{6}) = 1$, (g) $\sin(\theta + \frac{\pi}{6}) \leq \frac{\sqrt{3}}{2}$, (h) $\sin(\theta + \frac{\pi}{4}) \geq \frac{\sqrt{2}}{2}$, (i) $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$.

Solution

1. Set $t = \theta + \alpha$, record its interval, solve the standard trig relation in that interval, then subtract α .
2. Check every endpoint against the original strict/non-strict sign and the domain $0 \leq \theta < 2\pi$.

Final answer

(a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, 2\pi$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{5\pi}{6}, \frac{11\pi}{6}$ (f) $\pi, 2\pi$ (g) $[0, \frac{\pi}{6}] \cup [\frac{\pi}{2}, 2\pi)$ (h) $[0, \frac{\pi}{2}]$ (i) $\frac{4\pi}{3} < \theta < 2\pi$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q016 Written

Solve $4 \sin^2(\theta + \frac{\pi}{6}) \geq 1$ for $0 \leq \theta < 2\pi$.**Solution**

1. Let $t = \theta + \pi/6$; then $|\sin(t)| \geq 1/2$.
2. On the shifted range $[\pi/6, 13\pi/6)$, the accepted arcs give the stated closed θ -intervals.

Final answer

$$0 \leq \theta \leq \frac{2\pi}{3} \text{ or } \pi \leq \theta \leq \frac{5\pi}{3}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q017 Written

A student divides $\tan x \times \cos x = \frac{1}{2}$ by $\cos x$, obtaining $\tan x = \frac{1}{2 \cos x}$. Evaluate this strategy and propose a better substitution on $[0^\circ, 360^\circ[$.

Solution

1. Replace $\tan(x)$ by $\sin(x)/\cos(x)$ before cancelling; the original tangent already excludes $\cos(x) = 0$.
2. Solve the resulting sine equation on the stated degree interval.

Final answer

Dividing by $\cos(x)$ separates the interacting factors and is not helpful; use $\tan(x) \cos(x) = \sin(x)$ (with $\cos(x) \neq 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30^\circ, 150^\circ$.

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q018 MCQ

The period used for a general tangent solution is:**Solution**

1. Tangent repeats after π .

Correct option: B**Final answer** π

Q019 MCQ

The solutions of $2 \cos \theta = 1$ on $[0, 2\pi]$ are:**Solution**

1. Cosine is $1/2$ in QI and QIV.

Correct option: B**Final answer** $\pi/3, 5\pi/3$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q020 MCQ

Which root is impossible when solving $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$?

Solution

1. A sine value must lie in $[-1, 1]$.

Correct option: B

Final answer

$$\sin \theta = -2$$

Q021 MCQ

The quadratic challenge factors into roots:

Solution

1. Use $(u - 2)(u - \sqrt{3})$ with $u = \tan \theta$.

Correct option: B

Final answer

$$2, \sqrt{3}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q022 MCQ

For $2 \cos \theta > -\sqrt{3}$, the excluded arc is:

Solution

1. Cosine is at most $-\sqrt{3}/2$ on the central QII arc.

Correct option: B

Final answer

$[5\pi/6, 7\pi/6]$

Q023 MCQ

The inequality $\tan \theta > 1$ on $[0, 2\pi)$ holds on:

Solution

1. Use the increasing tangent branches and omit asymptotes.

Correct option: A

Final answer

$(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q024 MCQ

A closed endpoint in a trig inequality is used when the sign is:**Solution**

1. Equality is included for $\leq$ or $\geq$.

Correct option: C**Final answer**

greater than or equal to

Q025 MCQ

The 5-14 challenge reduces to which cosine condition?**Solution**

1. Factor the expression after replacing $\sin^2\theta$.

Correct option: B**Final answer** $-1 < \cos \theta < 1/2$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q026 MCQ

When a quadratic gives $\sin \theta = -2$, we:**Solution**

1. Sine cannot be outside $[-1, 1]$.

Correct option: B

Final answer

reject it

Q027 MCQ

The first step for $2 \cos^2 \theta - 3 \sin \theta = 0$ is to use:**Solution**

1. Rewrite in one trig function.

Correct option: A

Final answer

$$\sin^2 \theta + \cos^2 \theta = 1$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q028 MCQ

The solution of $2 \sin^2 \theta + \sin \theta - 1 = 0$ includes:**Solution**

1. The valid root $\sin \theta = 1/2$ gives $\pi/6, 5\pi/6$.

Correct option: A**Final answer** $\pi/6$

Q029 MCQ

In the rational challenge, which angle must be excluded?**Solution**

1. The denominator $\cos^2 \theta$ is zero there.

Correct option: C**Final answer** $\pi/2$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q030 MCQ

After setting $t = \theta + \pi/3$ with $0 \leq \theta < 2\pi$, the t-range is:

Solution

1. Add $\pi/3$ to both domain endpoints.

Correct option: B

Final answer

$[\pi/3, 7\pi/3)$

Q031 MCQ

The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ include:

Solution

1. Set the shifted argument to $\pi/3$.

Correct option: A

Final answer

$\pi/12$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q032 MCQ

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ -solution is:

Solution

1. The shifted sine is at least $\sqrt{2}/2$ from $\pi/4$ to $3\pi/4$.

Correct option: A

Final answer

$$[0, \pi/2]$$

Q033 MCQ

The 5-16 challenge is equivalent to:

Solution

1. Divide by 4 and take the square-root inequality.

Correct option: A

Final answer

$$|\sin t| \geq 1/2$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Q034 MCQ

In the Reflect problem, $\tan x \cos x$ simplifies to:**Solution**1. Use $\tan(x) = \sin(x)/\cos(x)$ where defined.

Correct option: B

Final answer $\sin x$

Quiz MCQ 1 Concept Quiz · MCQ

The general solution of $\tan \theta = -1$ is:**Solution**1. Tangent has period π .

Correct option: B

Final answer

$$\theta = -\frac{\pi}{4} + n\pi$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 2 Concept Quiz · MCQ

The solution of $\cos \theta > 0$ on $[0, 2\pi)$ is:

Solution

1. Cosine is positive in QI and QIV.

Correct option: B

Final answer

$$[0, \pi/2) \cup (3\pi/2, 2\pi)$$

Quiz MCQ 3 Concept Quiz · MCQ

The valid sine root of $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$ is:

Solution

1. The other algebraic root is outside the sine range.

Correct option: B

Final answer

$$\sin \theta = \frac{1}{2}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 4 **Concept Quiz · MCQ**

The first operation in a shifted-argument problem is to:

Solution

1. The t-range prevents losing or adding solutions.

Correct option: B

Final answer

set t equal to the whole argument and find its range

Quiz WRITTEN 5 **Concept Quiz · Written**

Solve $2 \cos \theta = \sqrt{3}$ **for** $0 \leq \theta < 2\pi$.

Solution

1. Cosine is $\sqrt{3}/2$ in QI and QIV.

Final answer

$$\theta = \frac{\pi}{6}, \frac{11\pi}{6}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\tan \theta \leq -1$ for $0 \leq \theta < 2\pi$.**Solution**

1. Use the two tangent branches and include the equality angles.

Final answer

$$\frac{\pi}{2} < \theta \leq \frac{3\pi}{4} \text{ or } \frac{3\pi}{2} < \theta \leq \frac{7\pi}{4}$$

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $2 \sin^2 \theta + \sin \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.**Solution**

1. Factor to $(2 \sin \theta - 1)(\sin \theta + 1) = 0$, then use the unit circle.

Final answer

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

Worked Solutions

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 8 Concept Quiz · Written

Solve $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ **for** $0 \leq \theta < 2\pi$.

Solution

1. Set $t = \theta + \pi/3$, solve $\cos(t) > 1/2$ in $[\pi/3, 7\pi/3)$, then subtract.

Final answer

$$\frac{4\pi}{3} < \theta < 2\pi$$

Answer key

Use the question number shown on each page.

- Q001** (a) $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$ (b) $\theta = \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}$ **Q012** $0 \leq \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \leq \pi$ **Q020** B
- Q002** (a) $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (b) $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ **Q013** (a) $\theta = \frac{\pi}{2}, \frac{11\pi}{6}$ (b) $0 \leq \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$ **Q021** B
- Q003** (a) $\frac{3\pi}{4}, \frac{11\pi}{4}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (d) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q014** (a) $\theta = 0, \frac{2\pi}{3}$ (b) $\theta = \pi, \frac{13\pi}{12}$ (c) $\frac{\pi}{2} < \theta < \pi$ **Q022** B
- Q004** $\tan \theta = 2$ or $\tan \theta = \sqrt{3}$; $\theta = \arctan(2) + n\pi$ or $\theta = \frac{\pi}{3} + n\pi$ **Q015** (a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, \frac{5\pi}{12}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q023** A
- Q005** (a) $0 \leq \theta < \frac{5\pi}{6}$ or $\frac{7\pi}{6} < \theta < 2\pi$ (b) $0 \leq \theta \leq \frac{\pi}{6}$ or $\frac{\pi}{2} < \theta \leq \frac{7\pi}{6}$ or $\frac{3\pi}{2} < \theta < 2\pi$ **Q016** $0 \leq \theta \leq \frac{2\pi}{3}$ or $\pi \leq \theta \leq \frac{5\pi}{3}$ **Q024** C
- Q006** (a) $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (b) $0 \leq \theta < \frac{\pi}{3}$ or $\frac{5\pi}{3} < \theta < 2\pi$ (c) $\frac{\pi}{4} < \theta < \frac{\pi}{2}$ or $\frac{5\pi}{4} < \theta < \frac{3\pi}{2}$ **Q017** Dividing by $\cos(x)$ separates the interacting factors and is not helpful, use $\tan(x) \cos(x) = \sin(x)$ (with $\cos(x) \neq 0$), so $\sin(x) = \frac{1}{2}$ and $x = 30^\circ, 150^\circ$. **Q025** B
- Q007** (a) $\frac{\pi}{2}, \frac{3\pi}{2}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{\pi}{2}, \frac{3\pi}{2}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q026** B
- Q008** $\frac{\pi}{3} < \theta < \pi$ or $\pi < \theta < \frac{5\pi}{3}$ **Q027** A
- Q009** (a) $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ (b) $0 \leq \theta \leq \frac{2\pi}{3}$ or $\theta = \pi$ or $\frac{4\pi}{3} \leq \theta < 2\pi$ **Q028** A
- Q010** (a) $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$ (b) $\theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $0 \leq \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ **Q029** C
- Q011** (a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (b) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (d) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (e) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (g) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (h) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (i) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (j) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (k) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (l) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (m) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (n) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (o) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (p) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (q) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (r) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (s) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (t) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (u) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (v) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (w) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (x) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (y) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (z) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ **Q030** B
- Q018** B **Q031** A
- Q019** B

Q032 A

Q033 A

Q034 B

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $\theta = \frac{\pi}{6}, \frac{11\pi}{6}$

Quiz WRITTEN 6 $\frac{\pi}{2} < \theta \leq \frac{3\pi}{4}$ or $\frac{3\pi}{2} < \theta \leq \frac{7\pi}{4}$

Quiz WRITTEN 7 $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

Quiz WRITTEN 8 $\frac{4\pi}{3} < \theta < 2\pi$

Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

Worked Solutions

Advanced Theorems and Applications

Q001 Written

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Solution

1. Put $t = \cos\theta$: $y = -t^2 + t + 2 = -(t - \frac{1}{2})^2 + \frac{9}{4}$.
2. Check the vertex and the endpoints $t = -1, 1$, then return to θ .

Final answer

$$\max y = \frac{9}{4} \text{ at } \theta = \frac{\pi}{3}, \frac{5\pi}{3}; \quad \min y = 0 \text{ at } \theta = \pi$$

Q002 Written

Find the maximum and minimum of $y = \cos^2\theta - \sin\theta$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Solution

1. Put $t = \sin\theta$: $y = 1 - t^2 - t = -(t + \frac{1}{2})^2 + \frac{5}{4}$.
2. The endpoint $t = 1$ gives the minimum -1 .

Final answer

$$\max y = \frac{5}{4} \text{ at } \theta = \frac{7\pi}{6}, \frac{11\pi}{6}; \quad \min y = -1 \text{ at } \theta = \frac{\pi}{2}$$

Worked Solutions

Advanced Theorems and Applications

Q003 Written

Solve the four given maximum/minimum problems on $0 \leq \theta < 2\pi$, including $y = \sin^2 \theta + \sin \theta - 1$, $y = \cos^2 \theta - \cos \theta + 1$, $y = 2 \sin \theta + \cos^2 \theta - 1$, and $y = -2 \sin^2 \theta - 2 \cos \theta + 1$. Include the given challenge $y = \cos^2 \theta - \sin \theta$.

Solution

1. Choose the single trig variable that makes the expression quadratic.
2. The vertex is valid only when it lies in $[-1, 1]$; convert every extremal t back to the corresponding angles.

Final answer

See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.

Worked Solutions

Advanced Theorems and Applications

Q004 Written

Use addition theorems to find $\sin 75^\circ$ and $\cos(\pi/12)$.

Solution

1. Rewrite $75^\circ = 45^\circ + 30^\circ$ and $\pi/12 = 45^\circ - 30^\circ$.
2. Apply the sine-sum and cosine-difference identities.

Final answer

$$\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Q005 Written

Given $\sin \alpha = \frac{\sqrt{3}}{2}$ in QII and $\cos \beta = -\frac{3}{5}$ in QIII, find $\sin(\alpha + \beta)$ and $\cos(\alpha - \beta)$.

Solution

1. The quadrant signs give $\cos \alpha = -1/2$ and $\sin \beta = -4/5$.
2. Substitute in the addition identities without a calculator.

Final answer

$$\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}; \quad \cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$$

Worked Solutions

Advanced Theorems and Applications

Q006 Written

Complete the given addition-theorem exercises *exact values, quadrant – controlled α, β , and the three – angle challenge* ($\sin \alpha = 0.6, \cos \beta = 0.8, \tan \gamma = 0.75$).

Solution

1. Convert radians to degrees only when it clarifies the reference angle.
2. Recover the missing sine/cosine value from the quadrant before applying the sum or difference identity.

Final answer

See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.

Q007 Written

Find $\tan 105^\circ$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.

Solution

1. Use $105^\circ = 45^\circ + 60^\circ$ and the tangent addition theorem.
2. For the lines, use $m_1 = 2, m_2 = -3$ in $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$.

Final answer

$$\tan 105^\circ = -2 + \sqrt{3}; \quad \theta = \frac{\pi}{4}$$

Worked Solutions

Advanced Theorems and Applications

Q008 Written

Find the angle between each given pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

Solution

1. Convert each line to slope-intercept form.
2. Use the acute angle convention $0 < \theta < \pi/2$, replacing θ_1 by $\pi - \theta_1$ when necessary.

Final answer

The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.

Worked Solutions

Advanced Theorems and Applications

Q009 **Written**

Find the exact angles for all given line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan \theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y -axis.

Solution

1. Solve the two linear equations produced by the tangent formula for the unknown slope.
2. Use the y -intercepts and the line intersection to obtain each triangle area.

Final answer

See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.

Worked Solutions

Advanced Theorems and Applications

Q010 Written

Given $\sin \alpha = 1/3$ **with** α **in** QII, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Solution

1. The quadrant gives $\cos \alpha = -2\sqrt{2}/3$.
2. Apply $\sin 2\alpha = 2 \sin \alpha \cos \alpha$, $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$, then divide.

Final answer

$$\sin 2\alpha = -\frac{4\sqrt{2}}{9}, \quad \cos 2\alpha = \frac{7}{9}, \quad \tan 2\alpha = -\frac{4\sqrt{2}}{7}$$

Q011 Written

Given $\sin \alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Solution

1. The interval places α in QIII, so $\cos \alpha = -\sqrt{5}/3$.
2. Substitute into the three double-angle formulas.

Final answer

$$\sin 2\alpha = \frac{4\sqrt{5}}{9}, \quad \cos 2\alpha = \frac{1}{9}, \quad \tan 2\alpha = 4\sqrt{5}$$

Worked Solutions

Advanced Theorems and Applications

Q012 Written

Solve the given double-angle exercises for given sine/cosine values and the challenge $\cos \theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Solution

1. Determine the missing trig value from the interval before using the identities.
2. For 3θ , use $4\cos^3\theta - 3\cos\theta$; for 2θ , use the tangent double-angle identity.

Final answer

For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.

Q013 Written

Given $\cos \alpha = 3/5, 0 < \alpha < \pi$, find $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, and evaluate $\cos(5\pi/8)$.

Solution

1. Use the half-angle formulas and choose signs from $0 < \alpha/2 < \pi/2$.
2. For $(5\pi/8)$, use $\cos(\pi - 3\pi/8)$.

Final answer

$$\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}, \quad \cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}, \quad \tan \frac{\alpha}{2} = \frac{1}{2}, \quad \cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$$

Worked Solutions

Advanced Theorems and Applications

Q014 Written

Given $\cos \alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Solution

1. Since $\alpha/2$ is in QI, both half-angle roots are positive.
2. Use $3\pi/8 = \pi/2 - \pi/8$ or the half-angle identity.

Final answer

$$\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}, \quad \cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}, \quad \tan \frac{\alpha}{2} = \sqrt{2}, \quad \sin \frac{3\pi}{8} = \frac{\sqrt{2 + \sqrt{2}}}{2}$$

Q015 Written

Solve the given half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}$, $4\pi < A < 5\pi$, $\cos B = -\frac{\sqrt{5}}{3}$, $5\pi < B < 6\pi$.

Solution

1. Halve the interval first; its quadrant controls every sign.
2. Apply the half-angle formula again for $(A/4)$ after finding $(\cos(A/2))$.

Final answer

Use the signed half-angle formulas; for the challenge, $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$, $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.

Worked Solutions

Advanced Theorems and Applications

Q016 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin \theta = 0$, $b \cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Solution

1. Use $\sin 2\theta = 2 \sin \theta \cos \theta$ and factor.
2. Use $\cos 2\theta = 2 \cos^2 \theta - 1$, solve the quadratic in $c = \cos \theta$, and map the interval.

Final answer

$$a \ \theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}; \quad b \ \frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$$

Q017 Written

Solve the given equation $\sin 2\theta + \cos \theta = 0$ **on** $0 \leq \theta < 2\pi$.

Solution

1. Factor $\cos \theta (2 \sin \theta + 1) = 0$.
2. List the two cosine zeros and the two angles with $\sin \theta = -1/2$.

Final answer

$$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$$

Worked Solutions

Advanced Theorems and Applications

Q018 Written

Solve all given double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Solution

1. Choose the identity that leaves one trig function only.
2. For inequalities, use a sign chart and include or exclude boundary points exactly as stated.

Final answer

See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.

Q019 Written

Rewrite $\sin \theta - \sqrt{3} \cos \theta$ as $r \sin(\theta + \alpha)$, $r > 0$, $-\pi < \alpha < \pi$.

Solution

1. Plot $P(1, -\sqrt{3})$, find $r = OP$, then read the angle α .

Final answer

$$r = 2, \quad \alpha = -\frac{\pi}{3}, \quad \sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$$

Worked Solutions

Advanced Theorems and Applications

Q020 Written

Rewrite the given expressions $\sin \theta - \cos \theta$, $\sin \theta + \cos \theta$, $3 \sin \theta - \sqrt{3} \cos \theta$, and $3 \sin \theta - \cos \theta$ in harmonic-addition form.

Solution

1. For $a \sin \theta + b \cos \theta$, use $r = \sqrt{a^2 + b^2}$, $\cos \alpha = a/r$, $\sin \alpha = b/r$.

Final answer

$$\sqrt{2} \sin \theta - \frac{\pi}{4}, \sqrt{2} \sin \theta + \frac{\pi}{4}, 2\sqrt{3} \sin \theta - \frac{\pi}{6}, \sqrt{10} \sin \theta - \arctan \frac{1}{3}$$

Q021 Written

Rewrite all given harmonic-addition exercises in the form $r \sin(\theta + \alpha)$, including $2 \sin \theta + 2 \cos \theta$ and the challenge $\sin \theta - \sqrt{3} \cos \theta$.

Solution

1. Keep the sign of the cosine coefficient when locating P(a,b).
2. Substitute the obtained (r,a) and expand once to verify both coefficients.

Final answer

See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.

Worked Solutions

Advanced Theorems and Applications

Q022 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin \theta - \cos \theta = 1$, $b \sin \theta + \cos \theta < 3/2$.**Solution**

1. Synthesize each left-hand side and account for the shifted angle range.
2. The maximum of $\sin \theta + \cos \theta$ is $\sqrt{2} < 3/2$, so the second inequality is always true.

Final answer

$$a \theta = \frac{\pi}{2}, \pi; \quad b \ 0 \leq \theta < 2\pi$$

Q023 Written

Solve $\sin \theta \leq \cos \theta$ on $0 \leq \theta < 2\pi$.**Solution**

1. Rewrite as $\sqrt{2} \sin(\theta - \pi/4) \leq 0$.
2. Read the accepted half-turns and retain the equality endpoints.

Final answer

$$0 \leq \theta \leq \frac{\pi}{4} \quad \text{or} \quad \frac{5\pi}{4} \leq \theta < 2\pi$$

Worked Solutions

Advanced Theorems and Applications

Q024 Written

Solve all given harmonic-addition equations and inequalities on $0 \leq \theta < 2\pi$, including $\sin \theta + \cos \theta + 1 = 0$ and $\sin \theta - \cos \theta + 3/2 > 0$.

Solution

1. After synthesis set $t = \theta + \alpha$ and write the exact t-range.
2. Use one full revolution in t, then subtract α and intersect with the original domain.

Final answer

See the numbered solution; synthesize first, then solve in the shifted variable over its full range.

Q025 Written

Find the maximum and minimum of $y = 2 \sin(\theta - \pi/3)$, and of $y = 2 \cos 2\theta + 4 \cos \theta + 1$, $0 \leq \theta < 2\pi$.

Solution

1. Use $-1 \leq \sin(t) \leq 1$ for the first function.
2. For the second, set $t = \cos \theta$ and complete the square after the double-angle substitution.

Final answer

First: max = 2, min = -2. Second: max = 7 at $\theta = 0$; min = -2 at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

Worked Solutions

Advanced Theorems and Applications

Q026 Written

Find the maximum and minimum of the given functions $y = -\frac{4}{3}\sin^2\theta + \cos\theta$ and $y = \cos 2\theta - 2\cos\theta$.

Solution

1. Reduce double angles or $\sin^2\theta$ to one variable.
2. Check the vertex and both endpoints of the allowed interval.

Final answer

For the first, $\max = 1$, $\min = -\frac{73}{48}$; for the second, use $t = \cos\theta$ and the endpoint/vertex test.

Q027 Written

Solve all given maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Solution

1. Use the same transformation rules as for equations and inequalities.
2. Never report a vertex outside $[-1, 1]$; compare it with both endpoints.

Final answer

See the numbered solution: synthesize $a\sin\theta + b\cos\theta$ or substitute $t = \cos\theta$, then test the full allowed range.

Worked Solutions

Advanced Theorems and Applications

Q028 MCQ

The substitution for $\sin^2 \theta + \cos \theta + 1$ should be:**Solution**1. Use $\sin^2 \theta = 1 - t^2$.

Correct option: B

Final answer $t = \cos \theta$

Q029 MCQ

The maximum of $\sin^2 \theta + \cos \theta + 1$ is:**Solution**

1. Complete the square.

Correct option: C

Final answer $9/4$

Worked Solutions

Advanced Theorems and Applications

Q030 MCQ

An extremal vertex is valid only if t lies in:**Solution**

1. Trig sine and cosine values are bounded.

Correct option: B**Final answer** $[-1, 1]$

Q031 MCQ

The identity for $\sin(a+b)$ is:**Solution**

1. Use the addition theorem.

Correct option: A**Final answer** $\sin a \cos b + \cos a \sin b$

Worked Solutions

Advanced Theorems and Applications

Q032 MCQ

sin 75 degrees equals:**Solution**

1. Rewrite 75 as 45+30.

Correct option: B

Final answer

$$\sqrt{6} + \sqrt{2}/4$$

Q033 MCQ

A missing trig value is selected using:**Solution**

1. The quadrant fixes the sign.

Correct option: A

Final answer

the quadrant

Worked Solutions

Advanced Theorems and Applications

Q034 MCQ

The slope of $y=mx+k$ is:**Solution**

1. The coefficient of x is the slope.

Correct option: B**Final answer** m

Q035 MCQ

The angle formula uses denominator:**Solution**

1. Use the tangent difference theorem.

Correct option: A**Final answer** $1 + m_1 m_2$

Worked Solutions

Advanced Theorems and Applications

Q036 MCQ

For an acute angle, if $\tan \theta_1$ is negative we use:

Solution

1. Choose the acute representative.

Correct option: B

Final answer

$\pi - \theta_1$

Q037 MCQ

$\sin(2a)$ equals:

Solution

1. Double-angle identity.

Correct option: B

Final answer

$2 \sin a \cos a$

Worked Solutions

Advanced Theorems and Applications

Q038 MCQ

cos(2a) can be written as:**Solution**

1. Use the standard formula.

Correct option: A**Final answer**

$$\cos^2 a - \sin^2 a$$

Q039 MCQ

For theta in QIV, sin theta is:**Solution**

1. Read the quadrant.

Correct option: B**Final answer**

negative

Worked Solutions

Advanced Theorems and Applications

Q040 MCQ

The half-angle formula for $\sin(a/2)$ uses:**Solution**1. Choose the sign from $a/2$.

Correct option: A

Final answer $\sqrt{(1 - \cos a)/2}$

Q041 MCQ

The sign of a half-angle root is chosen from:**Solution**

1. Halving changes the interval.

Correct option: A

Final answer

the original quadrant interval

Worked Solutions

Advanced Theorems and Applications

Q042 MCQ

If $\cos a = -1/3$ and $0 < a < \pi$, $\cos(a/2)$ is:

Solution

1. $a/2$ is in QI.

Correct option: B

Final answer

positive

Q043 MCQ

For double-angle equations, first rewrite in:

Solution

1. Use a double-angle identity.

Correct option: A

Final answer

one trig function

Worked Solutions

Advanced Theorems and Applications

Q044 MCQ

The factorisation of $\sin(2\theta) + \sin\theta$ is:**Solution**1. Factor after substituting $\sin(2\theta)$.

Correct option: A

Final answer $\sin\theta(2\cos\theta + 1)$

Q045 MCQ

A tangent asymptote is always:**Solution**

1. Tangent is undefined there.

Correct option: B

Final answer

excluded

Worked Solutions

Advanced Theorems and Applications

Q046 MCQ

For a $\sin \theta + b \cos \theta$, r is:**Solution**

1. It is the length OP.

Correct option: B**Final answer**

$$\sqrt{a^2 + b^2}$$

Q047 MCQ

The harmonic target form is:**Solution**

1. Match the coefficients.

Correct option: A**Final answer**

$$r \sin(\theta + a)$$

Worked Solutions

Advanced Theorems and Applications

Q048 MCQ

The angle a is located from point:**Solution**

1. The coefficient point determines direction.

Correct option: A

Final answer $P(a, b)$

Q049 MCQ

After synthesis, t is usually:**Solution**

1. Track the shifted range.

Correct option: A

Final answer $\theta + a$

Worked Solutions

Advanced Theorems and Applications

Q050 MCQ

If $\max(\sin \theta + \cos \theta) = \sqrt{2}$, then $\sin \theta + \cos \theta < 3/2$ is:

Solution

1. $\sqrt{2} < 3/2$.

Correct option: B

Final answer

always true

Q051 MCQ

In a strict inequality, equality endpoints are:

Solution

1. Use open endpoints.

Correct option: B

Final answer

excluded

Worked Solutions

Advanced Theorems and Applications

Q052 MCQ

The range of $2\sin(\theta - \pi/3)$ is:**Solution**

1. Multiply the sine range by 2.

Correct option: B

Final answer $[-2, 2]$

Q053 MCQ

For a double-angle maximum problem, use:**Solution**1. Reduce to $t = \sin \theta$ or $\cos \theta$.

Correct option: A

Final answer

the double-angle formula then a bounded variable

Worked Solutions

Advanced Theorems and Applications

Q054 MCQ

A vertex outside $[-1,1]$ is:**Solution**

1. It is not attainable by sine/cosine.

Correct option: B**Final answer**

rejected

Quiz MCQ 1 Concept Quiz · MCQ

The maximum of $2 \sin(\theta - \pi/3)$ is:**Solution**

1. Use the sine range.

Correct option: D**Final answer**

2

Worked Solutions

Advanced Theorems and Applications

Quiz MCQ 2 [Concept Quiz · MCQ](#)

The addition-theorem value $\sin 75^\circ$ is:

Solution

1. Use 45 degrees plus 30 degrees.

Correct option: A

Final answer

$$\sqrt{6} + \sqrt{2}/4$$

Quiz MCQ 3 [Concept Quiz · MCQ](#)

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Solution

1. Complete the square.

Correct option: A

Final answer

$$-2$$

Worked Solutions

Advanced Theorems and Applications

Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Solution

1. The tangent magnitude is 1.

Correct option: B

Final answer

$$\pi/4$$

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the maximum and minimum of $y = \sin^2 \theta + \cos \theta + 1$ on $0 \leq \theta < 2\pi$.

Solution

1. Let $t = \cos \theta$ and complete the square.

Final answer

$$\max = \frac{9}{4}, \min = 0$$

Worked Solutions

Advanced Theorems and Applications

Quiz WRITTEN 6 Concept Quiz · Written

Find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$ **when** $\sin \alpha = 1/3$ **and** α **is in** QII.

Solution

1. Use the quadrant to select $\cos \alpha$.

Final answer

$$-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$$

Quiz WRITTEN 7 Concept Quiz · Written

Rewrite $3 \sin \theta - \sqrt{3} \cos \theta$ **as** $r \sin(\theta + \alpha)$.

Solution

1. Find r and the coefficient angle from $P(3, -\sqrt{3})$.

Final answer

$$2\sqrt{3} \sin \theta - \frac{\pi}{6}$$

Worked Solutions

Advanced Theorems and Applications

Quiz WRITTEN 8 **Concept Quiz · Written****Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.**Solution**1. Synthesize to $\sqrt{2} \sin(\theta - \pi/4) \leq 0$.**Final answer**

$$[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$$

Answer key

Use the question number shown on each page.

- Q001** $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q002** $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- Q003** See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.
- Q004** $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$
- Q005** $\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$
- Q006** See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.
- Q007** $\tan 105^\circ = -2 + \sqrt{3}$; $\theta = \frac{\pi}{4}$
- Q008** The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.
- Q009** See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.
- Q010** $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$
- Q011** $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$
- Q012** For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.
- Q013** $\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan \frac{\alpha}{2} = \frac{1}{2}$, $\cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$
- Q014** $\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan \frac{\alpha}{2} = \sqrt{2}$, $\sin \frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$
- Q015** Use the signed half-angle formulas; for the challenge,
 $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$,
 $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.
- Q016** $a \theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; $b \frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$
- Q017** $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$
- Q018** See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.
- Q019** $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$
- Q020** $\sqrt{2} \sin \theta - \frac{\pi}{4}$, $\sqrt{2} \sin \theta + \frac{\pi}{4}$, $2\sqrt{3} \sin \theta - \frac{\pi}{6}$, $\sqrt{10} \sin \theta - \arctan \frac{1}{3}$
- Q021** See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.
- Q022** $a \theta = \frac{\pi}{2}, \pi$; $b \ 0 \leq \theta < 2\pi$
- Q023** $0 \leq \theta \leq \frac{\pi}{4}$ or $\frac{5\pi}{4} \leq \theta < 2\pi$
- Q024** See the numbered solution; synthesize first, then solve in the shifted variable over its full range.
- Q025** First: $\max = 2$, $\min = -2$. Second: $\max = 7$ at $\theta = 0$; $\min = -2$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q026** For the first, $\max = 1$, $\min = -\frac{73}{48}$; for the second, use $t = \cos \theta$ and the endpoint/vertex test.
- Q027** See the numbered solution: synthesize $a \sin \theta + b \cos \theta$ or substitute $t = \cos \theta$, then test the full allowed range.
- Q028** B
- Q029** C
- Q030** B
- Q031** A
- Q032** B
- Q033** A
- Q034** B
- Q035** A
- Q036** B
- Q037** B
- Q038** A
- Q039** B
- Q040** A
- Q041** A
- Q042** B
- Q043** A
- Q044** A
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** A
- Q050** B
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Quiz MCQ 1** D
- Quiz MCQ 2** A
- Quiz MCQ 3** A
- Quiz MCQ 4** B
- Quiz WRITTEN 5** $\max = \frac{9}{4}$, $\min = 0$
- Quiz WRITTEN 6** $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$
- Quiz WRITTEN 7** $2\sqrt{3} \sin \theta - \frac{\pi}{6}$
- Quiz WRITTEN 8** $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$

Concept 1 | Laws of Exponents and Roots

English Student Edition • Focused formulas and guided practice

Formula opener

Product

$$a^m a^n = a^{m+n}$$

Power

$$(a^m)^n = a^{mn}$$

Root

$$a^{1/n} = \sqrt[n]{a}$$

Worked Solutions

Laws of Exponents and Roots

Q001 Written

Find: (a) 10^0 , (b) 2^{-3} , (c) $(0.2)^{-2}$.**Solution**

1. Use $a^0 = 1$ and $a^{-n} = 1/a^n$; convert $0.2 = 1/5$.

Final answer

$$1, \frac{1}{8}, 25$$

Q002 Written

Let $a \neq 0, b \neq 0$. Simplify: (a) $a^{-5}a^8$, (b) $a^5 \div a^{-3}$, (c) $(a^{-1}b)^2$, (d) $(2^2)^3 \div 2^{-2} \times 2^{-3}$.**Solution**

1. Add exponents when multiplying, subtract when dividing, and multiply exponents for a power of a power.

Final answer

$$a^3, a^8, \frac{b^2}{a^2}, 32$$

Worked Solutions

Laws of Exponents and Roots

Q003 **Written****Find:** (a) 5^0 , (b) 3^{-2} , (c) $(0.25)^{-2}$.**Solution**1. Apply the zero and reciprocal-power laws; $0.25 = 1/4$.**Final answer** $1, \frac{1}{9}, 16$

Worked Solutions

Laws of Exponents and Roots

Q004 **Written**

Simplify the eight given expressions: a^5a^4 , $a^3 \div a^{-2}$, $(a^{-2}b^4)^{-1}$, $2^3 \times 2^{-7}$, $5^{-4} \div 5^{-6}$, $(a^3b^{-1})^3(a^2b^{-3})^{-2}$, $4^{-5}4^2 \div 4^{-2}$, and $2^4 \div (2^{-1})^3 \times 2^{-2}$.

Solution

1. Collect the exponent of each common base; distribute a negative outer power over every factor.

Final answer

$$a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$$

Worked Solutions

Laws of Exponents and Roots

Q005 Written

Find the nine values: $3^0, 11^0, (2/3)^0, 4^{-3}, 5^{-3}, 6^{-2}, (1/3)^{-3}, (3/5)^{-2}, (0.04)^{-2}$.**Solution**

1. Convert terminating decimals to fractions before applying the reciprocal rule.

Final answer $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Worked Solutions

Laws of Exponents and Roots

Q006 Written

Simplify the twelve same-base expressions in given Question 2: $a^2a^4, a^{-2}a^5, a^{-3}a^{-4}, a^3 \div a^5, a^8 \div a^{-3}, a^{-6} \div a^{-2}, (a^{-1})^3, (a^3b^4)^{-2}, (a^{-3}b^2)^{-3}, 3^53^{-3}, 4^44^{-3}, 5^55^{-5}$.

Solution

1. Use one exponent line for each common base and leave negative powers in reciprocal form when useful.

Final answer

$$a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, 4, 1$$

Worked Solutions

Laws of Exponents and Roots

Q007 Written

Simplify given Question 3: $5^4 \div 5^2$, $3^2 \div 3^{-2}$, $2^{-4} \div 2^{-5}$, $(a^{-2}b)^3(a^2b^{-1})^2$, $(a^{-3}b)^{-3}(a^4b^2)^{-2}$, $(2a^3b^{-1})^3(-3ab^{-2})^{-3}$, $3^23^{-1} \div 3^3$, $5^{-4}5^5 \div 5^{-2}$, $2^42^{-8} \div 2^{-6}$, $5^3(5^{-2})^2 \div 5^{-4}$, $3^5 \div (3^{-2})^{-3} \times 3$, and $(a/2)^3a^{-4} \div (2a)^{-2}$.

Solution

1. Distribute powers first, then combine exponents base by base; keep coefficients separate from variables.

Final answer

$$25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}$$

Worked Solutions

Laws of Exponents and Roots

Q008 Written

Simplify completely with positive exponents: $\left(\frac{2a^2}{b^{-3}}\right)^{-2} (4a^{-3}b^2)^3 \div \left(\frac{a}{2b}\right)^{-4} \div (a^{-2}b^2)$.

Solution

1. Rewrite the first factor as $(2a^2b^3)^{-2}$; distribute powers and combine all a- and b-exponents.

Final answer

$$\frac{1}{a^7b^6}$$

Q009 Written

Find $\sqrt[3]{64}$ and $\sqrt[3]{1/27}$.

Solution

1. Recognise $64 = 4^3$ and $1/27 = (1/3)^3$.

Final answer

$$4, \frac{1}{3}$$

Worked Solutions

Laws of Exponents and Roots

Q010 Written

Perform: $\sqrt[3]{12}\sqrt[3]{18}$, $\sqrt[3]{250}/\sqrt[3]{2}$, $(\sqrt[4]{8})^2$, $\sqrt[3]{\sqrt{729}}$, and $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Solution

1. Combine equal-index roots, factor perfect powers, then collect like cube-root terms.

Final answer $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 Written

Find $\sqrt[4]{81}$ and $\sqrt[4]{1/16}$.

Solution

1. Use perfect fourth powers.

Final answer $3, \frac{1}{2}$

Worked Solutions

Laws of Exponents and Roots

Q012 Written

Perform: $\sqrt[4]{8}\sqrt[4]{32}$, $\sqrt[4]{48}/\sqrt[4]{3}$, $(\sqrt[6]{100})^3$, $\sqrt[3]{\sqrt[3]{64}}$, and $\sqrt[3]{81} - \sqrt[3]{3} + \sqrt[3]{24}$.

Solution

- Combine roots with the same index and extract perfect powers; leave the nested radical exact.

Final answer

4, 2, 10, $\sqrt[3]{4}$, $4\sqrt[3]{3}$

Q013 Written

Find: $\sqrt[3]{216}$, $\sqrt[4]{625}$, $\sqrt[5]{243}$, $\sqrt[3]{1/64}$, $\sqrt[3]{1/125}$, $\sqrt[4]{1/81}$, $\sqrt[3]{1}$, $\sqrt[5]{100000}$, $\sqrt[4]{0.0001}$.

Solution

- Prime-factorise each radicand and match the index to a perfect power.

Final answer

6, 5, 3, $\frac{1}{4}$, $\frac{1}{5}$, $\frac{1}{3}$, 1, 10, 0.1

Worked Solutions

Laws of Exponents and Roots

Q014 Written

Perform the fifteen **given** calculations involving products, quotients, powers and sums of roots (including $\sqrt[3]{2}\sqrt[3]{4}$, $\sqrt[4]{27}\sqrt[4]{243}$, nested roots and the final three like-radical sums).

Solution

1. Convert every radicand to prime factors, combine equal-index roots, and collect only like radicals.

Final answer

$2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Worked Solutions

Laws of Exponents and Roots

Q015 Written

Perform $\sqrt[3]{\sqrt[4]{4096}} \div \sqrt{\sqrt[6]{4096}}$.

Solution

1. Since $4096 = 2^{12}$, both numerator and denominator equal 2.

Final answer

1

Q016 Written

Express in the form a^r : $\sqrt{5}$, $\sqrt[5]{2^4}$, and $1/\sqrt[5]{3^3}$.

Solution

1. Use $\sqrt[n]{a^m} = a^{m/n}$ and move a radical denominator to a negative exponent.

Final answer

$5^{1/2}$, $2^{4/5}$, $3^{-3/5}$

Worked Solutions

Laws of Exponents and Roots

Q017 Written

Find: $27^{2/3}$, $64^{-3/2}$, $(25/4)^{-3/2}$, and $[(16/9)^{-3/4}]^{2/3}$.

Solution

1. Rewrite each base as a perfect power, multiply fractional exponents, and apply negative powers last.

Final answer

$$9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$$

Q018 Written

Express in the form a^r : $\sqrt{a}$, $\sqrt[4]{a^3}$, $\sqrt[6]{a^5}$, $\sqrt[3]{2^2}$, $1/\sqrt{a}$, $1/\sqrt[3]{a^2}$, and $1/\sqrt[4]{5^3}$.

Solution

1. Index over radicand exponent gives the rational power; reciprocals change the sign.

Final answer

$$a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$$

Worked Solutions

Laws of Exponents and Roots

Q019 Written

Find $81^{3/4}$, $16^{-3/2}$, $(25/4)^{-3/2}$, **and** $[(27/125)^{1/2}]^{-4/3}$.**Solution**

1. Prime-factorise the bases before multiplying the fractional exponents.

Final answer

$$27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$$

Q020 Written

Express the nine given expressions in the form a^r , including roots of a and numerical bases 2, 3, 5.**Solution**

1. Translate each root and reciprocal directly into a rational exponent.

Final answer

$$2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$$

Worked Solutions

Laws of Exponents and Roots

Q021 Written

Evaluate the twelve given expressions with rational powers: $25^{3/2}$, $16^{3/4}$, $64^{2/3}$, $9^{-3/2}$, $125^{-2/3}$, $81^{-3/4}$, $(1/8)^{-1/3}$, $(125/8)^{-2/3}$, $(16/81)^{-3/4}$, $[(1/2)^{-1/3}]^{-2}$, $[(1/2)^{-1/3}]^{-2}$, $[(1/2)^{-1/3}]^{-2}$

Solution

1. Express every base as a power of a small integer, then multiply exponents exactly.

Final answer

$125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Worked Solutions

Laws of Exponents and Roots

Q022 Written

Simplify $\left[\left(\left(1000/27\right)^{-2/3}\right)^{7/15}\right]^{-15/14}$.

Solution

1. Since $1000/27 = (10/3)^3$, the inner power is $(10/3)^{-2}$; multiply the remaining exponents.

Final answer

$$\frac{10}{3}$$

Q023 Written

Perform: (a) $2^{2/3}2^{1/2} \div 2^{7/6}$, (b) $6^{1/2}36^{1/4}$, (c) $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$, (d) $\sqrt{2} \div \sqrt[5]{32} \times \sqrt[3]{16}$.

Solution

1. Make the bases the same, then add/subtract rational exponents.

Final answer

1, 6, 6, 2

Worked Solutions

Laws of Exponents and Roots

Q024 Written

Perform: $5^{1/2}5^{1/3} \div 5^{-1/6}$, $4^{5/6}2^{4/3}$, $\sqrt{7}\sqrt[3]{7}\sqrt[6]{7}$, and $\sqrt[4]{27}\sqrt{27} \div \sqrt[4]{3}$.

Solution

1. Rewrite powers of 4 and 27 with bases 2 and 3, then combine exponents.

Final answer

5, 8, 7, 9

Q025 Written

For $a > 0, b > 0$, perform: $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ and $\sqrt[3]{a}\sqrt{ab} \div \sqrt[6]{a^2/b^3}$.

Solution

1. Translate every radical to exponents of a and b, then collect like bases.

Final answer

$a^{1/3}$, $b\sqrt{a}$

Worked Solutions

Laws of Exponents and Roots

Q026 Written

Perform the twelve numerical calculations in given Question 1, including rational powers of 2, 3 and 5 and mixed radical products/quotients.

Solution

1. Make all bases equal before adding and subtracting rational exponents; extract perfect powers.

Final answer

$$1, 9, 5, 9, \sqrt{5}, 4, 5, 3, 2, 1, 8, \sqrt[3]{3}$$

Q027 Written

For $a > 0, b > 0$, simplify the six given expressions built from $\sqrt{a}, \sqrt[3]{a}, \sqrt[4]{a}, \sqrt[6]{a}$ and mixed a,b radicals.

Solution

1. Convert all radicals to rational powers and combine exponents separately for a and b.

Final answer

$$a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$$

Worked Solutions

Laws of Exponents and Roots

Q028 Written

Simplify the ministry expression $\left(\frac{(243/32)^{-2/5}(9/4)^{-1}}{(125/64)^{1/3}(0.004)^{-1/2}} \right)^{-3/2}$.

Solution

1. The printed decimal is $0.004 = 1/250$, so the exact result is not rational despite the prompt's wording; retain the exact radical form.

Final answer

$$\left(\frac{2025\sqrt{10}}{64} \right)^{3/2}$$

Q029 Written

Assess the statement $\sqrt{a^2} = a$ **for all real a, using positive and negative examples.**

Solution

1. For $a = 3$ the value is 3; for $a = -3$ the value is 3, not -3.

Final answer

Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Worked Solutions

Laws of Exponents and Roots

Q030 MCQ

The value of 2^{-3} is:**Solution**

1. Take the reciprocal of 2^3 .

Correct option: B**Final answer** $1/8$

Q031 MCQ

When multiplying equal bases, we:**Solution**

1. Use $a^m a^n = a^{m+n}$.

Correct option: B**Final answer**

add exponents

Worked Solutions

Laws of Exponents and Roots

Q032 MCQ

The expression $a^5 \div a^{-3}$ simplifies to:

Solution

1. Subtract: $5 - (-3) = 8$.

Correct option: B

Final answer

a^8

Q033 MCQ

A negative exponent means:

Solution

1. Use $a^{-n} = 1/a^n$.

Correct option: B

Final answer

take the reciprocal power

Worked Solutions

Laws of Exponents and Roots

Q034 MCQ

The value of $(0.04)^{-2}$ is:**Solution**1. $0.04 = 1/25$, so the reciprocal square is 25^2 .

Correct option: C

Final answer

625

Q035 MCQ

The exponent of a in $(a^{-2}b)^3(a^2b^{-1})^2$ is:**Solution**1. Compute $-6 + 4 = -2$.

Correct option: A

Final answer

-2

Worked Solutions

Laws of Exponents and Roots

Q036 MCQ

The positive-exponent form of the 6-1 challenge is:**Solution**1. Combine the final powers as $a^{-7}b^{-6}$.

Correct option: B

Final answer

$$1/(a^7b^6)$$

Q037 MCQ

The value of $3^5 \div (3^{-2})^{-3} \times 3$ is:**Solution**1. The exponent is $5 - 6 + 1 = 0$.

Correct option: B

Final answer

$$1$$

Worked Solutions

Laws of Exponents and Roots

Q038 MCQ

 $\sqrt[3]{64}$ equals:**Solution**1. Since $4^3 = 64$.

Correct option: B

Final answer

4

Q039 MCQ

 $\sqrt[3]{12}\sqrt[3]{18}$ equals:**Solution**1. Combine to $\sqrt[3]{216} = 6$.

Correct option: C

Final answer

6

Worked Solutions

Laws of Exponents and Roots

Q040 MCQ

The fourth root of $1/16$ is:**Solution**

1. $(1/2)^4 = 1/16$.

Correct option: A**Final answer**

$$1/2$$

Q041 MCQ

The result of $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$ is:**Solution**

1. Collect $3\sqrt[3]{2} - 2\sqrt[3]{2} + \sqrt[3]{2}$.

Correct option: B**Final answer**

$$2\sqrt[3]{2}$$

Worked Solutions

Laws of Exponents and Roots

Q042 MCQ

 $\sqrt[4]{625}$ equals:

Solution

1. $5^4 = 625$.

Correct option: B

Final answer

5

Q043 MCQ

When combining roots with the same index, multiply:

Solution

1. Use $\sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$.

Correct option: B

Final answer

the radicands

Worked Solutions

Laws of Exponents and Roots

Q044 MCQ

The roots challenge evaluates to:**Solution**

1. Both nested roots are 2.

Correct option: B**Final answer**

1

Q045 MCQ

 $\sqrt[5]{2^4}$ in rational-index form is:**Solution**

1. Index becomes the denominator.

Correct option: A**Final answer** $2^{4/5}$

Worked Solutions

Laws of Exponents and Roots

Q046 MCQ

The value of $27^{2/3}$ is:**Solution**

1. $(3^3)^{2/3} = 3^2$.

Correct option: C**Final answer**

9

Q047 MCQ

A reciprocal square root is written as:**Solution**

1. A denominator changes the exponent sign.

Correct option: B**Final answer** $a^{-1/2}$

Worked Solutions

Laws of Exponents and Roots

Q048 MCQ

The value of $81^{-3/4}$ is:**Solution**1. $81^{3/4} = 27$, then take the reciprocal.

Correct option: C

Final answer

1/27

Q049 MCQ

The exponent of a in $\sqrt[4]{a^3}$ is:**Solution**

1. Radical index is the denominator.

Correct option: A

Final answer

3/4

Worked Solutions

Laws of Exponents and Roots

Q050 MCQ

The value of $[(64/27)^{-4/3}]^{-1/2}$ is:**Solution**

1. Use $64/27 = (4/3)^3$ and multiply exponents.

Correct option: C

Final answer

16/9

Q051 MCQ

The 6-3 challenge simplifies to:**Solution**

1. Collect the three rational exponents.

Correct option: B

Final answer

10/3

Worked Solutions

Laws of Exponents and Roots

Q052 MCQ

 $2^{2/3}2^{1/2} \div 2^{7/6}$ equals:**Solution**1. The exponent is $4/6 + 3/6 - 7/6 = 0$.

Correct option: B

Final answer

1

Q053 MCQ

 $4^{5/6}2^{4/3}$ equals:**Solution**1. Rewrite $4 = 2^2$, giving exponent 3.

Correct option: C

Final answer

8

Worked Solutions

Laws of Exponents and Roots

Q054 MCQ

 $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ equals:**Solution**1. Compute $3/2 - 1/3 - 5/6 = 1/3$.

Correct option: A

Final answer

$$a^{1/3}$$

Q055 MCQ

The numerical result of $\sqrt[3]{6}\sqrt{6}\sqrt[6]{6}$ is:**Solution**

1. The exponents add to 1.

Correct option: C

Final answer

$$6$$

Worked Solutions

Laws of Exponents and Roots

Q056 MCQ

The given expression $\sqrt{a^3} \div \sqrt[3]{a} \div \sqrt[6]{a^5}$ tests:

Solution

1. Translate roots before combining.

Correct option: B

Final answer

combining rational exponents

Q057 MCQ

The printed decimal in the 6-4 challenge is:

Solution

1. Retain the ministry's printed value exactly.

Correct option: B

Final answer

0.004

Worked Solutions

Laws of Exponents and Roots

Q058 MCQ

For real a , $\sqrt{a^2}$ equals:

Solution

1. The principal square root is nonnegative.

Correct option: C

Final answer

$|a|$

Quiz MCQ 1 Concept Quiz · MCQ

Which law gives $a^m a^n = a^{m+n}$?

Solution

1. Add exponents for multiplication.

Correct option: A

Final answer

same-base multiplication

Worked Solutions

Laws of Exponents and Roots

Quiz MCQ 2 [Concept Quiz · MCQ](#)

The value of $\sqrt[3]{250}/\sqrt[3]{2}$ is:

Solution

1. Combine to $\sqrt[3]{125}$.

Correct option: B

Final answer

5

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The value of $64^{-3/2}$ is:

Solution

1. $64^{3/2} = 512$, then take the reciprocal.

Correct option: C

Final answer

1/512

Worked Solutions

Laws of Exponents and Roots

Quiz MCQ 4 Concept Quiz · MCQ

For a negative real a , $\sqrt{a^2}$ is:

Solution

1. The principal root is $|a| = -a$ when a is negative.

Correct option: B

Final answer

$$-a$$

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $a^{-3}a^7 \div a^2$.

Solution

1. The exponent is $-3 + 7 - 2 = 2$.

Final answer

$$a^2$$

Worked Solutions

Laws of Exponents and Roots

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sqrt[3]{54} - \sqrt[3]{16} + \sqrt[3]{2}$.

Solution

1. Extract $3\sqrt[3]{2} - 2\sqrt[3]{2} + \sqrt[3]{2}$.

Final answer

$$2\sqrt[3]{2}$$

Quiz WRITTEN 7 Concept Quiz · Written

Evaluate $125^{-2/3}$.

Solution

1. $125^{2/3} = 25$, then take the reciprocal.

Final answer

$$1/25$$

Worked Solutions

Laws of Exponents and Roots

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $\sqrt{a^2} = a$ is not true for every real a .

Solution

1. Use one positive and one negative counterexample.

Final answer

$\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Answer key

Use the question number shown on each page.

Q001 $1, \frac{1}{8}, 25$

Q002 $a^3, a^8, \frac{b^2}{a^2}, 32$

Q003 $1, \frac{1}{9}, 16$

Q004 $a^9, a^5, a^2b^{-4}, \frac{1}{16}, 25, a^5b^3, \frac{1}{4}, 32$

Q005 $1, 1, 1, \frac{1}{64}, \frac{1}{125}, \frac{1}{36}, 27, \frac{25}{9}, 625$

Q006 $a^6, a^3, a^{-7}, a^{-2}, a^{11}, a^{-4}, a^{-3}, a^{-6}b^{-8}, a^9b^{-6}, 9, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q007 $25, 81, 2, \frac{b}{a^2}, \frac{a}{b^7}, -\frac{8a^6b^3}{27}, \frac{1}{9}, 125, 4, 125, 1, \frac{a}{2}, \frac{10}{3}$

Q008 $\frac{1}{a^7b^6}$

Q009 $4, \frac{1}{3}$

Q010 $6, 5, 2, 3, 2\sqrt[3]{2}$

Q011 $3, \frac{1}{2}$

Q012 $4, 2, 10, \sqrt[3]{4}, 4\sqrt[3]{3}$

Q013 $6, 5, 3, \frac{1}{4}, \frac{1}{5}, \frac{1}{3}, 1, 10, 0.1$

Q014 $2, 4, 9, \sqrt[4]{27}, \frac{1}{4}, \frac{1}{5}, 7, 3, 4, 9, 4, \sqrt[3]{9}, \sqrt[3]{3}, 0, 0$

Q015 1

Q016 $5^{1/2}, 2^{4/5}, 3^{-3/5}$

Q017 $9, \frac{1}{512}, \frac{8}{125}, \frac{3}{4}$

Q018 $a^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q019 $27, \frac{1}{64}, \frac{8}{125}, \frac{25}{9}$

Q020 $2^{1/2}, 6^{1/2}, 7^{1/2}, a^{3/4}, a^{5/6}, 2^{2/3}, a^{-1/2}, a^{-2/3}, 5^{-3/4}$

Q021 $125, 8, 16, \frac{1}{27}, \frac{1}{25}, \frac{1}{27}, 2, \frac{4}{25}, \frac{27}{8}, 5, \frac{10}{9}, \frac{16}{9}$

Q022 $\frac{10}{3}$

Q023 1, 6, 6, 2

Q024 5, 8, 7, 9

Q025 $a^{1/3}, b\sqrt{a}$

Q026 1, 9, 5, 9, $\sqrt{5}$, 4, 5, 3, 2, 1, 8, $\sqrt[3]{3}$

Q027 $a^{1/3}, a^{7/6}, a^4, a^{2/3}b^{5/6}, a^{1/2}b^3, a^{13/6}b^{-5/6}$

Q028 $\left(\frac{2025\sqrt{10}}{64}\right)^{3/2}$

Q029 Sometimes: $\sqrt{a^2} = |a|$; it equals a only when $a \geq 0$.

Q030 B

Q031 B

Q032 B

Q033 B

Q034 C

Q035 A

Q036 B

Q037 B

Q038 B

Q039 C

Q040 A

Q041 B

Q042 B

Q043 B

Q044 B

Q045 A

Q046 C

Q047 B

Q048 C

Q049 A

Q050 C

Q051 B

Q052 B

Q053 C

Q054 A

Q055 C

Q056 B

Q057 B

Q058 C

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 B

Quiz WRITTEN 5 a^2

Quiz WRITTEN 6 $2\sqrt[3]{2}$

Quiz WRITTEN 7 $1/25$

Quiz WRITTEN 8 $\sqrt{a^2} = |a|$; e.g. $a = -3$ gives $3 \neq -3$

Concept 2 | Exponential Functions and Their Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Exponential

$$y = ab^x$$

Growth

$$A = A_0(1 + r)^t$$

Decay

$$A = A_0(1 - r)^t$$

Worked Solutions

Exponential Functions and Their Applications

Q001 Written

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1, 0, 1$.

Solution

1. Plot $x = -1, 0, 1$. A base > 1 gives growth; a base between 0 and 1 gives decay. The curve stays above the x-axis.

Final answer

Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.

Q002 Written

Draw $y = 2^x$ and $y = (1/2)^x$.

Solution

1. Use $(-1, 1/2), (0, 1), (1, 2)$ for 2^x and $(-1, 2), (0, 1), (1, 1/2)$ for its reciprocal-base graph.

Final answer

2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.

Worked Solutions

Exponential Functions and Their Applications

Q003 Written

Draw the six given graphs $y = 4^x$, $y = 10^x$, $y = (2/3)^x$, $y = (1/4)^x$, $y = (1/10)^x$, $y = (3/2)^x$.

Solution

1. For each graph mark $x = -1, 0, 1$, classify the base, and sketch a curve that never meets the x-axis.

Final answer

Bases 4, 10, 3/2: strictly increasing. Bases 2/3, 1/4, 1/10: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.

Q004 Written

Draw $y = \pi^x$ for $-1 \leq x \leq 2$, marking the key points.

Solution

1. Because $\pi > 1$, the curve increases. Restrict the drawing to the interval $[-1, 2]$ and keep $y > 0$.

Final answer

Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.

Worked Solutions

Exponential Functions and Their Applications

Q005 Written

Order the given sets 0.7^{-3} , 0.7^{-1} , 0.7^2 and $5/8$, $5/64$, $6/16$ using inequality signs.

Solution

1. For a base $0 < a < 1$, larger exponents give smaller values. For the second set rewrite with a common base 2 and compare exponents.

Final answer

$$0.7^2 < 0.7^{-1} < 0.7^{-3}; \text{ and } 5/64 < 5/8 < 6/16.$$

Q006 Written

Order the three displayed Try sets: 5^0 , 5^{-4} , $5^{3/2}$; $(1/3)^7$, $(1/3)^2$, $(1/3)^{-5}$; and 3^0 , 3^2 , 3^{-2} .

Solution

1. Make the base common. Keep exponent order for base > 1 and reverse it for $0 < a < 1$.

Final answer

$$5^{-4} < 5^0 < 5^{3/2}; (1/3)^7 < (1/3)^2 < (1/3)^{-5}; 3^{-2} < 3^0 < 3^2.$$

Worked Solutions**Exponential Functions and Their Applications****Q007** Written

Order the given sets with bases 3, 2, $\frac{3}{2}$, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.

Solution

1. The nine given comparisons are solved by one rule: rewrite, state whether the base is above or below one, then write the ordered chain.

Final answer

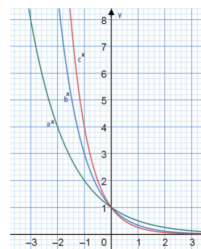
Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.

Worked Solutions

Exponential Functions and Their Applications

Q008 Written

Order $(\sqrt{2})^5$, $(\sqrt[3]{4})^3$, $(\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a , b , c .



Solution

1. Convert radicals to rational powers before ordering; the opposite graph reverses the order when its base lies between 0 and 1.

Final answer

Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.

Worked Solutions

Exponential Functions and Their Applications

Q009 Written

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, and $8 \geq 2^{2x-1}$.

Solution

1. Rewrite with a common base > 1 , compare exponents, and reverse an inequality only when the chosen base is between 0 and 1.

Final answer

$x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.

Q010 Written

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, **and the reciprocal-base inequalities.**

Solution

Use the same two-step method: common base, then solve the exponent equation or inequality with correct sign control.

Final answer

Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x .

Worked Solutions

Exponential Functions and Their Applications

Q011 Written

Solve all 24 given equations and inequalities in Exercise (a)–(x), including the 2, 3, 4, 5, 8, 9, 25, 27, 32, 64, 81, 125 common-base forms.

Solution

1. For each subpart: factor numerical constants into one base, write the exponent comparison, state $\text{base} > 1$ or $0 < \text{base} < 1$, and solve.

Final answer

Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control.

Q012 Written

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

Solution

1. Use base $4 > 1$, solve the quadratic inequality, then apply the physical time restriction $x \geq 0$.

Final answer

$x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.

Worked Solutions

Exponential Functions and Their Applications

Q013 Written

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

Solution

1. Set $t = a^x > 0$, solve the quadratic in t , discard non-positive roots, then convert back to x .

Final answer

First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.

Q014 Written

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

Solution

1. The substitution changes the exponential problem to a quadratic; the positivity condition on t is mandatory before returning to x .

Final answer

Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.

Worked Solutions

Exponential Functions and Their Applications

Q015 Written

Solve the twelve given items (a)–(l), each reducible to a quadratic in 2^x or 3^x .

Solution

1. Write $t > 0$ beside the substitution, solve the quadratic, reject $t \leq 0$, and only then use monotonicity of the base to recover x .

Final answer

For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .

Q016 Written

Solve completely for real x : $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \leq 0$.

Solution

1. Factor the quadratic as $(t - 1)(t - 4) \leq 0$, convert the bounds with the decreasing base $1/2$, then solve the resulting quadratic inequalities.

Final answer

Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.

Worked Solutions

Exponential Functions and Their Applications

Q017 Written

For $-1 \leq x \leq 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^x$, with the corresponding x -values.

Solution

1. Find the exact t -range first, compare the vertex with both endpoints, then convert t back to x .

Final answer

Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.

Q018 Written

For $-1 \leq x \leq 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x -values.

Solution

1. The quadratic has vertex $t = -1$, outside the positive t -range, so compare the two endpoints.

Final answer

Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.

Worked Solutions

Exponential Functions and Their Applications

Q019 Written

Solve the three given extrema models: $-3 \leq x \leq 2$, $y = 2 \cdot 9^x - 12 \cdot 3^x + 7$; $-1 \leq x \leq 3$, $y = -4 + 2^{2x} - 3^{x+2}$; **and the third displayed interval/model.**

Solution

1. A correct solution must include the t-range; an unconstrained quadratic vertex is not accepted unless it lies in that range.

Final answer

In each case set $t = a^x$, write the exact t-range, and compare the quadratic vertex and endpoints; report both extreme values and their x-locations.

Q020 Written

For $-1 \leq x \leq 2$, **optimise** $y = 2 \cdot 4^x - 2^{x+2} + 11$.

Solution

1. Use the interval $1/2 \leq t \leq 4$, evaluate the vertex $t = 1$, and compare both endpoints.

Final answer

Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.

Worked Solutions

Exponential Functions and Their Applications

Q021 **Written**

For $f(x) = a^x$, $g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

Solution

1. AM-GM gives $a^x + a^{-x} \geq 2$, with equality at $a^x = 1$, i.e. $x = 0$; the two terms move in opposite directions away from zero.

Final answer

h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.

Q022 **MCQ**

Which base gives a strictly decreasing exponential graph?

Solution

1. A base between 0 and 1 gives decay.

Correct option: C

Final answer

$1/2$

Worked Solutions

Exponential Functions and Their Applications

Q023 MCQ

The horizontal asymptote of $y = a^x$ is:**Solution**

1. Exponential values stay positive and approach zero.

Correct option: B**Final answer**

$$y = 0$$

Q024 MCQ

The range of $y = 5^x$ is:**Solution**

1. Every exponential output is positive.

Correct option: C**Final answer**

$$(0, \infty)$$

Worked Solutions

Exponential Functions and Their Applications

Q025 MCQ

For $y = \pi^x$ on $[-1, 2]$, which point is on the graph?**Solution**1. At $x = 1$, $y = \pi$.

Correct option: B

Final answer $(1, \pi)$

Q026 MCQ

For $0 < a < 1$ and $r < s$, which is true?**Solution**

1. The exponent order reverses below base 1.

Correct option: B

Final answer $a^r > a^s$

Worked Solutions

Exponential Functions and Their Applications

Q027 MCQ

The first step when bases differ is to:**Solution**

1. Rewrite all terms using one base.

Correct option: B**Final answer**

make the bases equal

Q028 MCQ

Which is largest? 3^{-2} , 3^0 , 3^2 **Solution**

1. Base 3 is greater than 1.

Correct option: C**Final answer** 3^2

Worked Solutions

Exponential Functions and Their Applications

Q029 MCQ

The challenge term $(\sqrt[3]{4})^3$ equals:

Solution

1. $(2^{2/3})^3 = 2^2$.

Correct option: B

Final answer

$$2^2$$

Q030 MCQ

Solving $4^{x-1} = 8^{x+3}$ gives:

Solution

1. Use $2^{2x-2} = 2^{3x+9}$.

Correct option: A

Final answer

$$-11$$

Worked Solutions

Exponential Functions and Their Applications

Q031 MCQ

For $0 < a < 1$, an exponential inequality requires:**Solution**

1. Decreasing functions reverse inequalities.

Correct option: B**Final answer**

a reversed exponent order

Q032 MCQ

The solution of $3^x = 81$ is:**Solution**

1. $81 = 3^4$.

Correct option: C**Final answer**

4

Worked Solutions

Exponential Functions and Their Applications

Q033 MCQ

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

Solution

1. Solve $(x - 2)(x + 1) > 0$, then use $x \geq 0$.

Correct option: B

Final answer

$$x > 2$$

Q034 MCQ

In substitution methods, $t = a^x$ must satisfy:

Solution

1. Every positive base power is positive.

Correct option: C

Final answer

$$t > 0$$

Worked Solutions

Exponential Functions and Their Applications

Q035 MCQ

The positive root of $t^2 - 3t - 16 = 0$ is:**Solution**1. Discard the negative root because $t > 0$.**Correct option:** B**Final answer**

$$(3 + \sqrt{73})/2$$

Q036 MCQ

The challenge quadratic in t is:**Solution**1. Set $t = (1/2)^{x^2 - 2x}$.**Correct option:** B**Final answer**

$$t^2 - 5t + 4$$

Worked Solutions

Exponential Functions and Their Applications

Q037 MCQ

After solving an exponential quadratic, reject roots with:**Solution**1. A power a^x cannot be zero or negative.

Correct option: C

Final answer

$$t \leq 0$$

Q038 MCQ

For extrema, the first required range is the range of:**Solution**

1. The quadratic is constrained by the exact t-range.

Correct option: B

Final answer

$$t = a^x$$

Worked Solutions

Exponential Functions and Their Applications

Q039 MCQ

The minimum of $y = t^2 - 4t$ at $t = 2$ is:**Solution**1. Complete the square: $(t - 2)^2 - 4$.

Correct option: B

Final answer

-4

Q040 MCQ

The C06-K02 challenge has minimum value:**Solution**1. $y = 2(t - 1)^2 + 9$.

Correct option: A

Final answer $9atx = 0$

Worked Solutions

Exponential Functions and Their Applications

Q041 MCQ

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:

Solution

1. AM–GM gives a minimum of 2 at $x = 0$.

Correct option: C

Final answer

2

Q042 MCQ

At which value of x does $a^x + a^{-x}$ attain its minimum?

Solution

1. Equality in AM–GM occurs when $a^x = 1$.

Correct option: B

Final answer

$x = 0$

Worked Solutions

Exponential Functions and Their Applications

Quiz MCQ 1 **Concept Quiz · MCQ**

Which statement about $y = a^x$ is always true for $a > 0$, $a \neq 1$?

Solution

1. The domain is all real numbers and the range is positive.

Correct option: A

Final answer

range $(0, \infty)$

Quiz MCQ 2 **Concept Quiz · MCQ**

For base 0.5, increasing the exponent makes the value:

Solution

1. The function is decreasing.

Correct option: B

Final answer

smaller

Worked Solutions

Exponential Functions and Their Applications

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The positive-substitution condition is:

Solution

1. An exponential power is always positive.

Correct option: C

Final answer

$$t > 0$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

For $y = 2(t - 1)^2 + 9$, the minimum is:

Solution

1. The square is zero at $t = 1$.

Correct option: C

Final answer

$$9$$

Worked Solutions

Exponential Functions and Their Applications

Quiz WRITTEN 5 Concept Quiz · Written

Sketch $y = (1/4)^x$ and state its domain, range, and asymptote.

Solution

1. Use the three points and the base $0 < 1/4 < 1$.

Final answer

Decreasing; $D = \mathbb{R}, R = (0, \infty), y = 0$

Quiz WRITTEN 6 Concept Quiz · Written

Order $2^{-3}, 2^0, 2^{5/2}$.

Solution

1. Base 2 is greater than 1, so preserve exponent order.

Final answer

$$2^{-3} < 2^0 < 2^{5/2}$$

Worked Solutions

Exponential Functions and Their Applications

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $4^x - 5 \cdot 2^x + 4 = 0$.**Solution**1. Let $t = 2^x > 0$: $t^2 - 5t + 4 = (t - 1)(t - 4) = 0$.**Final answer**

$$x = 0 \text{ or } x = 2$$

Quiz WRITTEN 8 Concept Quiz · Written**Find the minimum of** $a^x + a^{-x}$ **for** $a > 1$.**Solution**1. By AM-GM, $a^x + a^{-x} \geq 2$, with equality when $a^x = 1$.**Final answer**

$$2 \text{ at } x = 0$$

Answer key

Use the question number shown on each page.

- Q001** Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.
- Q002** 2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.
- Q003** Bases 4, 10, $3/2$: strictly increasing. Bases $2/3$, $1/4$, $1/10$: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.
- Q004** Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.
- Q005** $0.7^2 < 0.7^{-1} < 0.7^{-3}$; and $5/64 < 5/8 < 6/16$.
- Q006** $5^{-4} < 5^0 < 5^{3/2}$; $(1/3)^7 < (1/3)^2 < (1/3)^{-5}$; $3^{-2} < 3^0 < 3^2$.
- Q007** Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.
- Q008** Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.
- Q009** $x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.
- Q010** Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x .
- Q011** Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control.
- Q012** $x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.
- Q013** First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.
- Q014** Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.
- Q015** For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .
- Q016** Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.
- Q017** Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.
- Q018** Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.
- Q019** In each case set $t = a^x$, write the exact t -range, and compare the quadratic vertex and endpoints; report both extreme values and their x -locations.
- Q020** Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.
- Q021** h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.
- Q022** C
- Q023** B
- Q024** C
- Q025** B
- Q026** B
- Q027** B
- Q028** C
- Q029** B
- Q030** A
- Q031** B
- Q032** C
- Q033** B
- Q034** C
- Q035** B
- Q036** B
- Q037** C
- Q038** B
- Q039** B
- Q040** A
- Quiz MCQ 1** A
- Quiz MCQ 2** B
- Quiz MCQ 3** C
- Quiz MCQ 4** C
- Quiz WRITTEN 5** Decreasing; $D = \mathbb{R}, R = (0, \infty), y =$
- Quiz WRITTEN 6** $2^{-3} < 2^0 < 2^{5/2}$
- Quiz WRITTEN 7** $x = 0$ or $x = 2$
- Quiz WRITTEN 8** $2atx = 0$

Concept 3 | Introduction to Logarithms and Their Properties

English Student Edition • Focused formulas and guided practice

Formula opener

Definition

$$\log_b a = c \iff b^c = a$$

Product

$$\log_b(xy) = \log_b x + \log_b y$$

Power

$$\log_b(x^k) = k \log_b x$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q001 **Written****Convert the given forms $2^5 = 32$ and $\log_3 9 = 2$ to the opposite notation.****Solution**

1. Use $a^p = M \iff \log_a M = p$; swap the last two numbers.

Final answer

$$\log_2 32 = 5; 3^2 = 9$$

Q002 **Written****Convert $2^6 = 64$, $4^{-2} = 1/16$, and the two remaining Try expressions into logarithmic or exponential form.****Solution**

1. Identify base, exponent, and argument; then interchange the exponent and argument positions.

Final answer

$$\log_2 64 = 6; \log_4(1/16) = -2, \text{ with the same swap rule for the remaining items.}$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q003 Written

Convert all Exercise items (a)+(l): the first six exponential statements to logarithms and the last six logarithmic statements to exponentials.

Solution

1. Check $a > 0$, $a \neq 1$, $M > 0$, then write the equivalent form without changing the values.

Final answer

Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.

Q004 Written

If $\log_5 25 = x$, evaluate $\log_{1/2} x$ without a calculator.

Solution

1. First find x from $5^x = 25$, then recognise $2 = (1/2)^{-1}$.

Final answer

$x = 2$, so $\log_{1/2} 2 = -1$.

Worked Solutions

Introduction to Logarithms and Their Properties

Q005 Written

Find the six direct logarithmic values shown in Warm Up, including $\log_{10} 1, \log_5 5, \log_2 8, \log_2 16, \log_5(1/5), \log_7(49^{3/2})$.

Solution

1. Rewrite each argument as a power of its base; the exponent is the logarithm.

Final answer

Values: 0, 1, 3, 4, -1 , $3/2$.

Q006 Written

Find the six Try values $\log_5 1, \log_3 3, \log_4 4, \log_3 27, \log_2(1/4), \log_5(1/125)$.

Solution

1. Use $\log_a 1 = 0$, $\log_a a = 1$, and the power definition.

Final answer

0, 1, 1, 3, -2 , -3 .

Worked Solutions

Introduction to Logarithms and Their Properties

Q007 Written

Find all 18 Exercise values (a)+(r), including reciprocal arguments and fractional powers.

Solution

1. Do not approximate. For reciprocals, use a negative exponent; for roots, use a fractional exponent.

Final answer

Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.

Q008 Written

Put the printed logarithmic product $\log_{a^3}(a^2)$ into simplest form.

Solution

1. Since $a^2 = (a^3)^{2/3}$, the logarithm is $2/3$.

Final answer

$2/3$.

Worked Solutions

Introduction to Logarithms and Their Properties

Q009 Written

Perform $\log_6 2 + \log_6 3$ **and** $\log_2 30 - \log_2 3 - \log_2 5$.**Solution**

1. Combine products for addition and quotients for subtraction, then express the result as a base power.

Final answer

1 and 1.

Q010 Written

Perform the six Try calculations using product, quotient, and power laws, including $4\log_5 3 - 2\log_5 15 - \log_5 45$.**Solution**

1. Move coefficients into exponents first, combine logs, simplify the argument, and read the final exponent.

Final answerEach expression reduces to one logarithm and then an integer; the displayed worked example gives -3 .

Worked Solutions

Introduction to Logarithms and Their Properties

Q011 Written

Perform all 18 given expressions (a)+(r), combining same-base logarithms exactly.**Solution**

1. Apply the power law before product/quotient combination; simplify the final argument as a power of the base.

Final answer

Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.

Q012 Written

Simplify the fractional expression $\log_5 75 + 3 \log_5 \sqrt{2} - 2 \log_5 \sqrt{6}$ to one reduced fraction.**Solution**

1. Move coefficients inside, combine the argument, and reduce it to a power of 5.

Final answer

1.

Worked Solutions

Introduction to Logarithms and Their Properties

Q013 Written

Given $\log_{10} 2 = a$ **and** $\log_{10} 3 = b$, **express** $\log_{10} 24, \log_{10}(27/4), \log_{10} 45, \log_{10}(24/75)$ **in terms of** a, b .

Solution

1. Prime-factorise the argument, use $5 = 10/2$ when required, then apply product, quotient, and power laws.

Final answer

$3a + b, 2 - 2a + 3b, 2a + 1 - a, b + 2 - 2a.$

Q014 Written

Given $\log_{10} 2 = a, \log_{10} 3 = b$, **express the four Try arguments** $72, 3^8/10, 15, 12$ **in terms of** a, b .

Solution

1. Write every integer as $2^m 3^n 5^p$, replace $\log_{10} 2, \log_{10} 3$, and convert the 5-factor using $1 - a$.

Final answer

Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.

Worked Solutions

Introduction to Logarithms and Their Properties

Q015 Written

Express all 12 Exercise logarithms in terms of $a = \log_{10} 2$ and $b = \log_{10} 3$, including fractions and reciprocal arguments.

Solution

1. Keep numerator and denominator factors separate until the final subtraction.

Final answer

Each result is a linear combination of $a, b, 1$ obtained from prime factors.

Q016 Written

If $\log_c 3 = x, \log_c 2 = y$, write $\log_c 12, \log_c(3/8), \log_c \sqrt{6}, \log_2 3, \log_{3b}(2^b c)$ in terms of x, y .

Solution

1. Factorise arguments, use the logarithm laws, and use change of base for the last two forms.

Final answer

$x + 2y, x - 3y, (x + y)/2, x/y, (1 + by)/(bx)$.

Worked Solutions

Introduction to Logarithms and Their Properties

Q017 Written

Find $\log_{16} 64$ and $\log_5(1/125)$ by changing to bases 2 and 5.

Solution

1. Use $\log_a b = \log_c b / \log_c a$, choosing a base that turns both numbers into powers.

Final answer

$3/2$ and -3 .

Q018 Written

Evaluate the five Try logarithms using a suitable common base.

Solution

1. Choose the smallest convenient base; do not use decimal approximations when exact powers are available.

Final answer

Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.

Worked Solutions

Introduction to Logarithms and Their Properties

Q019 Written

Find all 15 Exercise values using change of base, including the mixed-base and fractional-argument items.

Solution

1. Convert numerator and denominator to the same base, then cancel common factors.

Final answer

All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.

Q020 Written

Find $\log_9(\sqrt{3}/4\sqrt{2})$ exactly.

Solution

1. Apply change of base to 2, then express the radical denominator as a power.

Final answer

$$\frac{\log_2 3 - 5/2}{2 \log_2 3}.$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q021 **Written****Solve the three warm-up expressions using a common base, including $\log_4 16 \cdot \log_9 3$.****Solution**

1. Make all logarithms use base 2 (or 3), then cancel adjacent change-of-base factors.

Final answer

Use reciprocal and power identities; the worked values are exact constants.

Q022 **Written****Solve the four Try products of logarithms after making the bases the same.****Solution**

1. Treat each logarithm as a change-of-base fraction and cancel before multiplying.

Final answer

Each product collapses to a rational number after change of base and cancellation.

Worked Solutions

Introduction to Logarithms and Their Properties

Q023 Written

Simplify all 15 Exercise products and sums of logarithms by common-base conversion.**Solution**

1. Make the base 2, 3, or 5 consistently, simplify inside parentheses, then multiply.

Final answer

Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.

Q024 Written

Simplify $\log_x(\sqrt{y}) \log_{y^2}(z^3) \log_{\sqrt{z}}(x^4)$ for $x > 1$.**Solution**

1. Convert each factor to a common natural-log ratio; exponents contribute $1/2, 3/2, 8$, whose product is 12.

Final answer

12.

Worked Solutions

Introduction to Logarithms and Their Properties

Q025 Written

Compare $f(x) = \log_4(x^2)$ and $g(x) = 2\log_4 x$ across the real line.**Solution**

1. Use $\log_4(x^2) = 2\log_4 |x|$; negative x belongs to f 's domain but not g 's.

Final answer

They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.

Q026 Written

Draw $y = \log_4 x$, state its positional relationship with $y = 4^x$, and draw $y = \log_{1/4} x$.**Solution**

1. Plot $y = -1, 0, 1$ in the exponential counterpart, then reflect in $y = x$.

Final answer

Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.

Worked Solutions

Introduction to Logarithms and Their Properties

Q027 **Written****State graph relationships and draw the three Try logarithmic functions with bases 2, 5, and $1/3$.****Solution**

1. Use the domain $x > 0$, vertical asymptote $x = 0$, and inverse reflection.

Final answer

Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.

Q028 **Written****Complete both Exercise groups: state positional relationships and draw the six requested logarithmic graphs.****Solution**

1. Every graph passes through $(1, 0)$; do not draw it across the y-axis.

Final answer

Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.

Worked Solutions

Introduction to Logarithms and Their Properties

Q029 **Written**

For $f(x) = \log_5(x - 2) + 1$ and $g(x) = 5^{x-1} + 2$, state their exact geometric relationship and line of symmetry.

Solution

1. Translate the standard inverse pair by the same horizontal/vertical shifts and identify the resulting symmetry line.

Final answer

They are inverse-reflected curves with symmetry line $y = x + 1$.

Q030 **Written**

Order the displayed logarithmic sets, including $\log_2 3$, $\log_2 5$, $2 \log_2(1/2)$, $\log_{1/2} 3$, $\log_{1/2} 5$.

Solution

1. Compare arguments only after the bases are the same and the base condition is stated.

Final answer

Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.

Worked Solutions

Introduction to Logarithms and Their Properties

Q031 **Written****Order the four Try sets of logarithms using monotonicity and change of base.****Solution**

1. Rewrite $k \log_a M$ as $\log_a (M^k)$, then compare the arguments.

Final answer

Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.

Q032 **Written****Order all 12 Exercise sets (a)+(l), including negative coefficients and reciprocal bases.****Solution**

1. Move coefficients inside first, then compare arguments with the correct direction.

Final answer

Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.

Worked Solutions

Introduction to Logarithms and Their Properties

Q033 Written

If $\log_c(3^2/2) = x$ and $\log_c(9) = y$, write the relation between x and y for $0 < c < 1$ and $c > 1$.

Solution

1. Express both logs with a common base and state the monotonicity case before writing the inequality.

Final answer

Use the same argument comparison; the relation reverses when the base crosses 1.

Q034 MCQ

$\log_2 32$ equals:

Solution

1. Since $2^5 = 32$.

Correct option: C

Final answer

5

Worked Solutions

Introduction to Logarithms and Their Properties

Q035 MCQ

 $\log_{1/2} 2$ equals:**Solution**

1. $(1/2)^{-1} = 2.$

Correct option: B

Final answer

-1

Q036 MCQ

 $\log_5(1/125)$ equals:**Solution**

1. $1/125 = 5^{-3}.$

Correct option: A

Final answer

-3

Worked Solutions

Introduction to Logarithms and Their Properties

Q037 MCQ

 $\log_7 49^{3/2}$ equals:**Solution**1. Write $49^{3/2} = 7^3$.

Correct option: C

Final answer $3/2$

Q038 MCQ

 $\log_a M + \log_a N$ equals:**Solution**

1. Use the product law.

Correct option: B

Final answer $\log_a(MN)$

Worked Solutions

Introduction to Logarithms and Their Properties

Q039 MCQ

The coefficient $3 \log_5 2$ becomes:**Solution**1. Use $k \log_a M = \log_a (M^k)$.**Correct option:** B**Final answer**

$$\log_5 2^3$$

Q040 MCQ

If $a = \log_{10} 2$ and $b = \log_{10} 3$, $\log_{10} 24$ is:**Solution**1. $24 = 2^3 \cdot 3$.**Correct option:** C**Final answer**

$$3a + b$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q041 MCQ

$\log_c \sqrt{6}$ in terms of $x = \log_c 3, y = \log_c 2$ is:

Solution

1. Use $\sqrt{6} = 6^{1/2}$.

Correct option: B

Final answer

$$(x + y)/2$$

Q042 MCQ

$\log_{16} 64$ equals:

Solution

1. $16 = 2^4, 64 = 2^6$.

Correct option: B

Final answer

$$3/2$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q043 MCQ

Change of base gives $\log_a b =$:**Solution**

1. Use the quotient of common-base logs.

Correct option: B**Final answer**

$$\log_c b / \log_c a$$

Q044 MCQ

 $\log_4 16 \cdot \log_9 3$ equals:**Solution**1. Values are $2 \cdot 1/2$.**Correct option:** A**Final answer**

$$1/2$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q045 MCQ

The challenge product of three logarithms equals:**Solution**

1. Multiply exponent factors after change of base.

Correct option: C**Final answer**

12

Q046 MCQ

The domain of $y = \log_a x$ is:**Solution**

1. The argument must be positive.

Correct option: C**Final answer** $x > 0$

Worked Solutions

Introduction to Logarithms and Their Properties

Q047 MCQ

The graph of $y = \log_a x$ is symmetric with $y = a^x$ about:

Solution

1. Inverse graphs reflect in $y = x$.

Correct option: C

Final answer

$$y = x$$

Q048 MCQ

For $a > 1$ and $M < N$, which is true?

Solution

1. The logarithm is increasing.

Correct option: B

Final answer

$$\log_a M < \log_a N$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q049 MCQ

For $0 < a < 1$ and $M < N$, which is true?**Solution**

1. The logarithm is decreasing.

Correct option: A**Final answer**

$$\log_a M > \log_a N$$

Q050 MCQ

The vertical asymptote of $y = \log_a x$ is:**Solution**1. Logarithmic graphs require $x > 0$.**Correct option:** A**Final answer**

$$x = 0$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q051 MCQ

A negative coefficient $-2 \log_a M$ can be written as:

Solution

1. Move the coefficient into the exponent.

Correct option: A

Final answer

$$\log_a M^{-2}$$

Q052 MCQ

The base of $\log_a M$ is:

Solution

1. The subscript is the base.

Correct option: B

Final answer

$$a$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q053 MCQ

The argument of $\log_a M$ is:**Solution**

1. The number inside the logarithm is the argument.

Correct option: B**Final answer** M

Q054 MCQ

 $\log_2(1/8)$ equals:**Solution**

1. $1/8 = 2^{-3}$.

Correct option: A**Final answer** -3

Worked Solutions

Introduction to Logarithms and Their Properties

Q055 MCQ

 $\log_4 4$ equals:**Solution**

1. The argument equals the base.

Correct option: B

Final answer

1

Q056 MCQ

 $\log_3 1$ equals:**Solution**

1. Every allowed base to power zero is 1.

Correct option: B

Final answer

0

Worked Solutions

Introduction to Logarithms and Their Properties

Q057 MCQ

 $\log_a M - \log_a N$ equals:**Solution**

1. Use the quotient law.

Correct option: B**Final answer**

$$\log_a(M/N)$$

Q058 MCQ

 $2\log_3 5$ equals:**Solution**

1. Move 2 into the argument as an exponent.

Correct option: B**Final answer**

$$\log_3 25$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q059 MCQ

If $a = \log_{10} 2$, $b = \log_{10} 3$, $\log_{10} 12$ is:

Solution

1. $12 = 2^2 \cdot 3$.

Correct option: B

Final answer

$$2a + b$$

Q060 MCQ

$\log_{10} 5$ in terms of $a = \log_{10} 2$ is:

Solution

1. $5 = 10/2$.

Correct option: B

Final answer

$$1 - a$$

Worked Solutions

Introduction to Logarithms and Their Properties

Q061 MCQ

 $\log_8 32$ equals:**Solution**1. $8 = 2^3, 32 = 2^5$.

Correct option: B

Final answer

5/3

Q062 MCQ

 $\log_{25} 125$ equals:**Solution**1. $25 = 5^2, 125 = 5^3$.

Correct option: B

Final answer

3/2

Worked Solutions

Introduction to Logarithms and Their Properties

Q063 MCQ

 $\log_a b \cdot \log_b a$ equals:**Solution**

1. Change of base makes reciprocal factors.

Correct option: B

Final answer

1

Q064 MCQ

 $\log_2 8 \cdot \log_8 2$ equals:**Solution**

1. The factors are reciprocals.

Correct option: B

Final answer

1

Worked Solutions

Introduction to Logarithms and Their Properties

Q065 MCQ

The best common base for 16 and 64 is:**Solution**

1. Both numbers are powers of 2.

Correct option: A**Final answer**

2

Q066 MCQ

The range of $y = \log_a x$ is:**Solution**

1. Every real exponent is attained.

Correct option: B**Final answer** $\mathbb{R}$

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz MCQ 1 **Concept Quiz · MCQ**

The logarithmic equivalent of $a^p = M$ is:

Solution

1. Swap the exponent and argument.

Correct option: A

Final answer

$$\log_a M = p$$

Quiz MCQ 2 **Concept Quiz · MCQ**

$\log_2 8 + \log_2 4$ equals:

Solution

1. Combine to $\log_2 32$.

Correct option: C

Final answer

5

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz MCQ 3 Concept Quiz · MCQ

$\log_{16} 64$ equals:

Solution

1. Use base 2 exponents 6/4.

Correct option: B

Final answer

$$3/2$$

Quiz MCQ 4 Concept Quiz · MCQ

For $a > 1$, $M < N$ implies:

Solution

1. The logarithmic function is increasing.

Correct option: B

Final answer

$$\log_a M < \log_a N$$

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz WRITTEN 5 Concept Quiz · Written**Find** $\log_3(1/27)$.**Solution**

1. $1/27 = 3^{-3}$.

Final answer -3 **Quiz WRITTEN 6** Concept Quiz · Written**Simplify** $\log_5 25 + \log_5 5$.**Solution**

1. Combine to $\log_5 125 = 3$.

Final answer 3

Worked Solutions

Introduction to Logarithms and Their Properties

Quiz WRITTEN 7 Concept Quiz · Written**Find $\log_{16} 64$ using change of base.****Solution**

1. $\log_2 64 / \log_2 16 = 6/4 = 3/2.$

Final answer

$$3/2$$

Quiz WRITTEN 8 Concept Quiz · Written**State the domain and range of $y = \log_a x$.****Solution**

1. The argument is positive and every real exponent is attained.

Final answer

Domain $x > 0$; range all real numbers

Answer key

Use the question number shown on each page.

- Q001** $\log_2 32 = 5$; $3^2 = 9$
- Q002** $\log_2 64 = 6$; $\log_4 (1/16) = -2$, with the same swap rule for the remaining items.
- Q003** Each item follows $a^p = M \iff \log_a M = p$, preserving the printed bases and signs.
- Q004** $x = 2$, so $\log_{1/2} 2 = -1$.
- Q005** Values: 0, 1, 3, 4, -1, 3/2.
- Q006** 0, 1, 1, 3, -2, -3.
- Q007** Convert each argument to a power of its base; the answers are the corresponding integers or rational exponents.
- Q008** 2/3.
- Q009** 1 and 1.
- Q010** Each expression reduces to one logarithm and then an integer; the displayed worked example gives -3.
- Q011** Every item is reduced by $\log_a M + \log_a N = \log_a(MN)$, $\log_a M - \log_a N = \log_a(M/N)$, and $k \log_a M = \log_a(M^k)$.
- Q012** 1.
- Q013** $3a + b$, $2 - 2a + 3b$, $2a + 1 - a$, $b + 2 - 2a$.
- Q014** Use prime factorisation and the same linear combinations of a, b produced by the logarithm laws.
- Q015** Each result is a linear combination of $a, b, 1$ obtained from prime factors.
- Q016** $x + 2y$, $x - 3y$, $(x + y)/2$, x/y , $(1 + by)/(bx)$.
- Q017** 3/2 and -3.
- Q018** Rewrite powers of 2, 3, or 5 and apply the change-of-base quotient exactly.
- Q019** All values are exact rational numbers obtained from $\log_a b = \log_c b / \log_c a$.
- Q020** $\frac{\log_2 3 - 5/2}{2 \log_2 3}$.
- Q021** Use reciprocal and power identities; the worked values are exact constants.
- Q022** Each product collapses to a rational number after change of base and cancellation.
- Q023** Use $\log_a b \log_b a = 1$ and the power law; every printed item reduces exactly.
- Q024** 12.
- Q025** They agree only for $x > 0$; f is defined for $x \neq 0$, while g requires $x > 0$.
- Q026** Both logarithmic graphs have domain $x > 0$, range $\mathbb{R}$, asymptote $x = 0$; $y = \log_4 x$ increases and is the reflection of $y = 4^x$ in $y = x$, while $\log_{1/4} x$ decreases.
- Q027** Base > 1 gives increasing curves; base between 0 and 1 gives decreasing curves; each is inverse-symmetric to its exponential.
- Q028** Use inverse symmetry, domain $x > 0$, range $\mathbb{R}$, and increasing/decreasing classification from the base.
- Q029** They are inverse-reflected curves with symmetry line $y = x + 1$.
- Q030** Make bases equal, turn coefficients into powers, and reverse order for base $0 < a < 1$.
- Q031** Each chain follows argument order for base > 1 and reverses it for base $0 < a < 1$.
- Q032** Convert every term to one base-log form and apply monotonicity; negative coefficients become reciprocal arguments.
- Q033** Use the same argument comparison; the relation reverses when the base crosses 1.
- Q034** C
- Q035** B
- Q036** A
- Q037** C
- Q038** B
- Q039** B
- Q040** C
- Q041** B
- Q042** B
- Q043** B
- Q044** A
- Q045** C
- Q046** C
- Q047** C
- Q048** B
- Q049** A
- Q050** A
- Q051** A
- Q052** B
- Q053** B
- Q054** A
- Q055** B
- Q056** B
- Q057** B
- Q058** B
- Q059** B
- Q060** B
- Q061** B
- Q062** B
- Q063** B
- Q064** B
- Q065** A

Q066 B**Quiz MCQ 1** A**Quiz MCQ 2** C**Quiz MCQ 3** B**Quiz MCQ 4** B**Quiz WRITTEN 5** -3 **Quiz WRITTEN 6** 3**Quiz WRITTEN 7** $\frac{3}{2}$ **Quiz WRITTEN 8** Domain $x > 0$; range all real numbers

Concept 4 | Applications of Logarithms

English Student Edition • Focused formulas and guided practice

Formula opener

Change base

$$\log_b a = \frac{\ln a}{\ln b}$$

Domain

$$x > 0$$

Inverse

$$b^{\log_b x} = x$$

Worked Solutions

Applications of Logarithms

Q001 Written

Solve $\log_2 x = 4$.**Solution**

1. Set the argument equal to 2^4 ; hence $x = 16$.

Final answer

16

Q002 Written

Solve $\log_2(2x - 1) = \log_2 3$.**Solution**

1. Domain: $2x - 1 > 0$. Equal arguments give $2x - 1 = 3$, so $x = 2$.

Final answer $x = 2$

Worked Solutions

Applications of Logarithms

Q003 Written

Solve all 12 given logarithmic equations, including same-base, product, quotient, and shifted-argument forms.

Solution

1. Write every argument condition, combine logs, solve the resulting algebraic equation, and verify each candidate.

Final answer

Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.

Q004 Written

Solve $\log_3(x + 2) - \log_9((x - 2)^2) = 1$ completely.

Solution

1. Domain gives $x > 2$. Convert $\log_9$ to base 3, solve $(x + 2)^2 / (x - 2)^2 = 9$, and retain only $x = 4$.

Final answer

$x = 4$

Worked Solutions

Applications of Logarithms

Q005 Written

Solve $\log_3 x < \log_3 81$ **and** $\log_{1/2}(x + 2) + \log_{1/2}(x - 5) > -3$.**Solution**

1. State the domain, preserve the sign for base 3, reverse it for base $1/2$, then intersect.

Final answer

$$0 < x < 81; \quad 5 < x < 6$$

Q006 Written

Solve the six given logarithmic inequalities using base monotonicity and argument conditions.**Solution**

1. For $a > 1$ keep the inequality direction; for $0 < a < 1$ reverse it before solving.

Final answer

Each solution is the intersection of the argument domain with the monotonicity interval.

Worked Solutions

Applications of Logarithms

Q007 Written

Solve all 24 given logarithmic inequalities, including shifted arguments, coefficients, and rational arguments.

Solution

1. Never omit the positive-argument condition; it can remove one branch of an algebraic solution.

Final answer

Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.

Q008 Written

Solve the given challenge inequality involving a reciprocal-base logarithmic sum and state the complete interval.

Solution

1. Combine the logarithms, solve the rational inequality, and retain only the common domain.

Final answer

Reverse the comparison after converting to a base below one, then intersect with both argument conditions.

Worked Solutions

Applications of Logarithms

Q009 Written

Solve $5(\log_2 x)^2 - 16\log_2 x + 3 < 0$.**Solution**

1. Let $t = \log_2 x$: $(5t - 1)(t - 3) < 0$, so $1/5 < t < 3$, then back-substitute.

Final answer

$$2^{1/5} < x < 8$$

Q010 Written

Solve the three given substitution problems: two quadratic logarithmic inequalities/equations and one exponential-log value.**Solution**

1. The substitution turns the problem into a quadratic; only then return to the original logarithm.

Final answerSubstitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.

Worked Solutions

Applications of Logarithms

Q011 Written

Solve the given groups: three equations, three inequalities, and three exact logarithmic values.

Solution

1. Record the domain before setting $t = \log_a x$, and verify every returned value.

Final answer

All groups follow the same substitution, domain, solve, and back-substitution sequence.

Q012 Written

Solve $(\log_3 x)^2 - \log_3 x - 6 < 0$ for the positive wave-intensity multiplier x .

Solution

1. Let $t = \log_3 x$: $(t - 3)(t + 2) < 0$, so $-2 < t < 3$, hence $3^{-2} < x < 3^3$.

Final answer

$1/9 < x < 27$

Worked Solutions

Applications of Logarithms

Q013 Written

For $y = -(\log_3 x)^2 + 4 \log_3 x + 3$, $1 \leq x \leq 27$, find maximum/minimum values and the corresponding x .

Solution

1. Let $t = \log_3 x$, so $0 \leq t \leq 3$; complete the square $y = 7 - (t - 2)^2$.

Final answer

Maximum 7 at $x = 9$; minimum 3 at $x = 1$.

Q014 Written

Find the maximum and minimum of the given logarithmic quadratic on its stated interval, with the corresponding values of x .

Solution

1. Convert the x -interval to a t -interval before reading the quadratic range.

Final answer

Use the vertex in $t = \log_a x$ and compare it with both endpoints.

Worked Solutions

Applications of Logarithms

Q015 Written

Solve the three given extrema problems, each with a different base and interval.

Solution

1. The maximum/minimum in t must be translated back to the exact $x = a^t$.

Final answer

For every item, map the domain to t , locate the vertex, and check endpoints.

Q016 Written

For $y = -(\log_2 x)^2 + 2\log_2 x + 4$, $1 \leq x \leq 8$, find the maximum and minimum health ratings and their concentrations.

Solution

1. Here $0 \leq t \leq 3$ and $y = 5 - (t - 1)^2$; compare $t = 0, 1, 3$.

Final answer

Maximum 5 at $x = 2$; minimum 4 at $x = 1$.

Worked Solutions

Applications of Logarithms

Q017 Written

Using $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, determine the digits of 6^{30} and the first non-zero decimal place of $(2/3)^{50}$.

Solution

1. Take common logs: $30(\log 2 + \log 3) = 23.343$ and $50(\log 2 - \log 3) = -8.805$.

Final answer

24 digits; 9th decimal place.

Q018 Written

Complete the given digit-count and first-non-zero-decimal-place sets for the displayed powers and fractions.

Solution

1. Use the larger absolute exponent for a number between two negative powers of ten.

Final answer

Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.

Worked Solutions

Applications of Logarithms

Q019 Written

Using the printed common logs, determine the number of digits and the leading digit of 12^{50} .

Solution

1. Compute $50 \log_{10} 12 = 50(2 \log 2 + \log 3) = 53.965$; then use the characteristic and mantissa.

Final answer

55 digits; leading digit 1.

Q020 Written

Solve $\log_x(x+2) \leq \log_x(4-x)$ for all real x .

Solution

1. Domain: $0 < x < 4, x \neq 1$. For $0 < x < 1$ reverse the sign and for $1 < x < 4$ keep it; both cases are impossible.

Final answer

No real solution.

Worked Solutions

Applications of Logarithms

Q021 MCQ

The argument condition for $\log_3(x - 5)$ is:**Solution**

1. A logarithm argument must be positive.

Correct option: C**Final answer**

$$x > 5$$

Q022 MCQ

If $\log_2(2x - 1) = \log_2 3$, then $x =$:**Solution**

1. Equate same-base arguments.

Correct option: B**Final answer**

$$2$$

Worked Solutions

Applications of Logarithms

Q023 MCQ

The valid solution of the challenge equation is:**Solution**

1. The domain $x > 2$ rejects the other algebraic root.

Correct option: C**Final answer**

4

Q024 MCQ

For same-base logarithms, equality of logs gives:**Solution**

1. The logarithm is one-to-one on its domain.

Correct option: A**Final answer**

arguments equal

Worked Solutions

Applications of Logarithms

Q025 MCQ

For base $a > 1$, $\log_a M < \log_a N$ means:**Solution**

1. The logarithm is increasing.

Correct option: B**Final answer**

$$M < N$$

Q026 MCQ

For base $0 < a < 1$, the inequality direction is:**Solution**

1. A reciprocal-base logarithm is decreasing.

Correct option: B**Final answer**

reversed

Worked Solutions

Applications of Logarithms

Q027 MCQ

The domain of $\log_{1/2}(x - 5)$ requires:**Solution**

1. The argument must be positive.

Correct option: C

Final answer

$$x > 5$$

Q028 MCQ

The warm-up intersection is:**Solution**1. Intersect the algebraic interval with $x > 5$.

Correct option: B

Final answer

$$5 < x < 6$$

Worked Solutions

Applications of Logarithms

Q029 MCQ

The best substitution for a quadratic logarithmic equation is:**Solution**

1. It turns the logarithm into a single variable.

Correct option: B**Final answer**

$$t = \log_a x$$

Q030 MCQ

The inequality $5t^2 - 16t + 3 < 0$ gives:**Solution**

1. Factor and use the sign chart.

Correct option: B**Final answer**

$$1/5 < t < 3$$

Worked Solutions

Applications of Logarithms

Q031 MCQ

The substitution variable must satisfy:**Solution**

1. A logarithm value can be any real number.

Correct option: B**Final answer**

$$t \in \mathbb{R}$$

Q032 MCQ

The challenge interval for $(\log_3 x)^2 - \log_3 x - 6 < 0$ is:**Solution**

1. Convert $-2 < t < 3$ back with base 3.

Correct option: B**Final answer**

$$1/9 < x < 27$$

Worked Solutions

Applications of Logarithms

Q033 MCQ

For $y = -(t - 2)^2 + 7$, the maximum occurs at:

Solution

1. The vertex gives the maximum.

Correct option: C

Final answer

$$t = 2$$

Q034 MCQ

The correct first step for extrema is:

Solution

1. Then map the domain to a t-range.

Correct option: B

Final answer

$$\text{set } t = \log_a x$$

Worked Solutions

Applications of Logarithms

Q035 MCQ

For $1 \leq x \leq 27$, $t = \log_3 x$ lies in:**Solution**1. Apply $\log_3$ to the endpoints.

Correct option: B

Final answer $[0, 3]$

Q036 MCQ

The water-tank challenge maximum health rating is:**Solution**1. The vertex is at $t = 1$, so $x = 2$.

Correct option: C

Final answer

5

Worked Solutions

Applications of Logarithms

Q037 MCQ

The number of digits of N is determined by:**Solution**

1. Use the characteristic of the common logarithm.

Correct option: B**Final answer**

$$[\log_{10} N] + 1$$

Q038 MCQ

If $-9 < \log_{10} N < -8$, the first non-zero digit is at:**Solution**

1. Use the larger absolute negative exponent.

Correct option: B**Final answer** 9^{th}

Worked Solutions

Applications of Logarithms

Q039 MCQ

The common logarithm base is:**Solution**

1. A common logarithm has base 10.

Correct option: C**Final answer**

10

Q040 MCQ

The solution set of $\log_x(x + 2) \leq \log_x(4 - x)$ is:**Solution**

1. The increasing and decreasing cases both contradict their domains.

Correct option: D**Final answer**

empty

Worked Solutions

Applications of Logarithms

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Solve $\log_5(x - 1) = 2$.

Solution

1. $x - 1 = 25$.

Correct option: A

Final answer

$$x = 26$$

Quiz MCQ 2 [Concept Quiz · MCQ](#)

For $0 < a < 1$ **and** $M < N$, **compare** $\log_a M$ **and** $\log_a N$.

Solution

1. The function is decreasing.

Correct option: B

Final answer

$$\log_a M > \log_a N$$

Worked Solutions

Applications of Logarithms

Quiz MCQ 3 Concept Quiz · MCQ

Solve $4(\log_2 x)^2 - 5\log_2 x + 1 = 0$.

Solution

1. Let $t = \log_2 x$: $(4t - 1)(t - 1) = 0$.

Correct option: C

Final answer

$$x = 2, \sqrt{2}$$

Quiz MCQ 4 Concept Quiz · MCQ

Find the maximum of $y = -(\log_3 x)^2 + 2\log_3 x + 4$ **on** $1 \leq x \leq 27$.

Solution

1. Let $t = \log_3 x \in [0, 3]$; vertex at $t = 1$.

Correct option: D

Final answer

$$5 \text{ at } x = 3$$

Worked Solutions

Applications of Logarithms

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\log_3(x + 1) = 2$.**Solution**

1. $x + 1 = 9$.

Final answer

$$x = 8$$

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\log_2 x \geq 3$.**Solution**

1. Base 2 is increasing.

Final answer

$$x \geq 8$$

Worked Solutions

Applications of Logarithms

Quiz WRITTEN 7 Concept Quiz · Written**Solve** $(\log_5 x)^2 - 4 < 0$.**Solution**

1. $-2 < \log_5 x < 2$.

Final answer

$$1/25 < x < 25$$

Quiz WRITTEN 8 Concept Quiz · Written**How many digits does 6^{20} have using $\log_{10} 2 = 0.3010, \log_{10} 3 = 0.4771$?****Solution**

1. $20(0.3010 + 0.4771) = 15.562$.

Final answer

16 digits

Answer key

Use the question number shown on each page.

- Q001 16
- Q002 $x = 2$
- Q003 Apply the argument condition first; then equate same-base arguments and reject any extraneous roots.
- Q004 $x = 4$
- Q005 $0 < x < 81$; $5 < x < 6$
- Q006 Each solution is the intersection of the argument domain with the monotonicity interval.
- Q007 Use argument conditions, convert coefficients by the power law, then solve and intersect the intervals.
- Q008 Reverse the comparison after converting to a base below one, then intersect with both argument conditions.
- Q009 $2^{1/5} < x < 8$
- Q010 Substitute $t = \log_a x$, solve in t , enforce $x > 0$, and back-substitute.
- Q011 All groups follow the same substitution, domain, solve, and back-substitution sequence.
- Q012 $1/9 < x < 27$
- Q013 Maximum 7 at $x = 9$; minimum 3 at $x = 1$.
- Q014 Use the vertex in $t = \log_a x$ and compare it with both endpoints.
- Q015 For every item, map the domain to t , locate the vertex, and check endpoints.
- Q016 Maximum 5 at $x = 2$; minimum 4 at $x = 1$.
- Q017 24 digits; 9th decimal place.
- Q018 Use $n - 1 < \log_{10} N < n$ for digits and $-n < \log_{10} N < -n + 1$ for the first non-zero place.
- Q019 55 digits; leading digit 1.
- Q020 No real solution.
- Q021 C
- Q022 B
- Q023 C
- Q024 A
- Q025 B
- Q026 B
- Q027 C
- Q028 B
- Q029 B
- Q030 B
- Q031 B
- Q032 B
- Q033 C
- Q034 B
- Q035 B
- Q036 C
- Q037 B
- Q038 B
- Q039 C
- Q040 D
- Quiz MCQ 1 A
- Quiz MCQ 2 B
- Quiz MCQ 3 C
- Quiz MCQ 4 D
- Quiz WRITTEN 5 $x = 8$
- Quiz WRITTEN 6 $x \geq 8$
- Quiz WRITTEN 7 $1/25 < x < 25$
- Quiz WRITTEN 8 16 digits

Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

English Student Edition • Focused formulas and guided practice

Formula opener

Derivative

$$\frac{d}{dx} x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q001 Written

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3 , and from 3 to $3 + h$.

Solution

1. Use $[f(b) - f(a)]/(b - a)$. For the h-form, factor $f(3 + h) - f(3) = h(8 + h)$.

Final answer

4; $8 + h$

Q002 Written

Find the average rate of change for the three given Try functions on their stated intervals.

Solution

1. Evaluate both endpoint values, subtract, and divide by the change in x; cancel h in the variable-endpoint case.

Final answer

2; $-5/3$; $h - 6$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q003 Written

Find the average rate of change for all nine given Exercise functions, including fixed and h-form intervals.

Solution

1. Keep the endpoint order consistent and simplify before substituting numerical values.

Final answer

Apply $[f(b) - f(a)]/(b - a)$ to each printed item exactly.

Q004 Written

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

Solution

1. With $A = \pi d^2/4$, compute $[A(18) - A(10)]/(18 - 10) = 7\pi$.

Final answer

$7\pi \text{ cm}^2/\text{cm}$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q005 Written

Find the given limits by direct substitution and by simplifying the indeterminate fraction before substitution.

Solution

1. Direct substitution is valid when defined; otherwise factor/cancel first, then put the limiting value into the simplified expression.

Final answer

−2; 6

Q006 Written

Find the three given Try limits, including polynomial and h-quotient forms.

Solution

1. Test direct substitution, identify 0/0, simplify, and substitute again.

Final answer

Substitute after removing any common factor; polynomial limits are obtained directly.

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q007 Written

Find all nine given Exercise limits, including x-limits and h-limits.

Solution

1. For a polynomial use continuity; for a quotient, factor before taking the limit.

Final answer

Each limit is obtained by direct substitution or cancellation of the factor causing 0/0.

Q008 Written

If $\lim_{x \rightarrow 3} (x^2 - 5x + k)/(x - 3) = L$ is finite, find Lk .

Solution

1. Set the numerator to zero at 3: $k = 6$. Then cancel $x - 3$, giving $L = 1$, so $Lk = 6$.

Final answer

6

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q009 Written

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

Solution

1. Use $\lim_{h \rightarrow 0} [f(-2 + h) - f(-2)]/h = \lim(-12 + 3h) = -12$.

Final answer

$$f'(-2) = -12$$

Q010 Written

Find the differential coefficients at the given values for the two given Try functions using the definition.

Solution

1. Expand the shifted polynomial fully before cancelling the common h.

Final answer

Substitute a into $[f(a + h) - f(a)]/h$, cancel h, and let h tend to zero.

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q011 Written

Find the differential coefficients for all six given Exercise functions at their stated values.

Solution

1. The resulting constant is the instantaneous slope at the given input.

Final answer

Use the definition at each specified point and simplify exactly.

Q012 Written

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

Solution

1. The differential coefficient is $s'(t) = 9.8t$; hence $s'(10) = 98$.

Final answer

98 m s^{-1}

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q013 Written

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.**Solution**

1. Expand $f(x + h) - f(x) = 2xh + h^2 + 3h$, divide by h , and let h tend to zero.

Final answer

$$f'(x) = 2x + 3$$

Q014 Written

Differentiate the three given Try functions using the definition.**Solution**

1. Do not use a memorised rule; retain the h -limit structure until the cancellation is visible.

Final answer

Expand each $f(x + h) - f(x)$, cancel h , and take the limit.

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q015 **Written****Differentiate all nine given Exercise functions using the definition.****Solution**

1. Expand, collect the coefficient of h , divide by h , then set $h = 0$.

Final answerApply $f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)]/h$ to every printed polynomial.Q016 **Written****A regular hexagon has side length x . Write its area and differentiate it using the definition.****Solution**

1. Substitute the area formula into the difference quotient; the derivative of the quadratic factor is $3\sqrt{3}x$.

Final answer

$$A = \frac{3\sqrt{3}}{2}x^2, \quad A'(x) = 3\sqrt{3}x$$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q017 Written

If $f(x) = 5$ for every x except that it is undefined at $x=2$, does $\lim_{x \rightarrow 2} f(x)$ exist?

Solution

1. A limit depends on nearby values, not on whether the function is defined at the point itself.

Final answer

Yes; the limit is 5.

Q018 MCQ

The average rate of change from a to b is:

Solution

1. Use the secant slope.

Correct option: B

Final answer

$$[f(b) - f(a)] / (b - a)$$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q019 MCQ

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

Solution

1. Evaluate the two endpoints.

Correct option: C

Final answer

4

Q020 MCQ

The average area-rate for diameter 10 to 18 is:

Solution

1. Use $A = \pi d^2/4$.

Correct option: A

Final answer

7π

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q021 MCQ

The h-form average rate is found by:**Solution**

1. Simplify the difference quotient.

Correct option: B**Final answer**

cancelling the common h

Q022 MCQ

A direct-substitution limit is valid when:**Solution**

1. Use continuity where defined.

Correct option: A**Final answer**

the expression is defined

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q023 MCQ

The 0/0 form requires:**Solution**

1. Remove the common vanishing factor.

Correct option: A**Final answer**

factor/cancel first

Q024 MCQ

In the finite-limit challenge, k equals:**Solution**

1. The numerator must vanish at $x = 3$.

Correct option: C**Final answer**

6

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q025 MCQ

For a polynomial, the limit equals:**Solution**

1. Polynomials are continuous.

Correct option: A**Final answer**

its value at the point

Q026 MCQ

The definition of the differential coefficient at a is:**Solution**

1. Use the difference quotient.

Correct option: A**Final answer** $\lim [f(a+h)-f(a)]/h$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q027 MCQ

For $f(x) = 3x^2$, $f'(-2)$ is:**Solution**

1. Expand and cancel h.

Correct option: A

Final answer

-12

Q028 MCQ

The stone's velocity at 10 s is:**Solution**1. Use $s'(t) = 9.8t$.

Correct option: B

Final answer

98

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q029 MCQ

After cancelling h , take the limit as:**Solution**

1. The definition uses h approaching zero.

Correct option: A**Final answer**

$$h \rightarrow 0$$

Q030 MCQ

The derivative function is:**Solution**

1. Associate the coefficient with each x .

Correct option: B**Final answer**the differential coefficient for every x

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q031 MCQ

The derivative of $x^2 + 3x$ from the definition is:**Solution**

1. Expand the difference quotient.

Correct option: B

Final answer

$$2x + 3$$

Q032 MCQ

The area of a regular hexagon of side x is:**Solution**

1. Use the standard hexagon area formula.

Correct option: B

Final answer

$$(3\sqrt{3}/2)x^2$$

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Q033 MCQ

A function need not be defined at a for its limit to:**Solution**

1. Nearby values determine a limit.

Correct option: A**Final answer**

exist

Q034 MCQ

The derivative of a constant function is:**Solution**

1. The difference quotient of a constant is zero.

Correct option: A**Final answer**

0

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Find the average rate of change of x^2 from 1 to 3.

Solution

1. $[9 - 1]/2 = 4$.

Correct option: A

Final answer

4

Quiz MCQ 2 [Concept Quiz · MCQ](#)

Find $\lim_{x \rightarrow 2}(x^2 + 1)$.

Solution

1. Direct substitution gives 5.

Correct option: B

Final answer

5

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Find $f'(x)$ for $f(x) = x^2$ using the definition.

Solution

1. Expand $(x + h)^2 - x^2$, cancel h , and let h tend to zero.

Correct option: C

Final answer

$2x$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

If $s(t) = 4.9t^2$, find the velocity at $t=5$.

Solution

1. $s'(t) = 9.8t$.

Correct option: D

Final answer

49

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 5 Concept Quiz · Written

Find the average rate of change of $3x + 1$ from 2 to 7.

Solution

1. A linear function has constant slope 3.

Final answer

3

Quiz WRITTEN 6 Concept Quiz · Written

Evaluate $\lim_{h \rightarrow 0}(6 + 2h)$.

Solution

1. Substitute $h = 0$.

Final answer

6

Worked Solutions

Fundamentals of Calculus (Limits and Derivatives)

Quiz WRITTEN 7 Concept Quiz · Written

Find the differential coefficient of $2x^2$ at $x=3$ from the definition.

Solution

1. The difference quotient simplifies to $4x + 2h$, giving 12.

Final answer

12

Quiz WRITTEN 8 Concept Quiz · Written

State $\lim_{x \rightarrow 2} f(x)$ when $f(x)=5$ for all x except $x=2$ where it is undefined.

Solution

1. Nearby values are constantly 5.

Final answer

5

Answer key

Use the question number shown on each page.

Q001 4; $8 + h$	Q009 $f'(-2) = -12$	Q017 Yes; the limit is 5.	Q030 B
Q002 2; $-5/3$; $h - 6$	Q010 Substitute a into $[f(a + h) - f(a)]/h$	Q018 B	Q031 B
Q003 Apply $[f(b) - f(a)]/(b - a)$ to each printed item exactly.	cancel h , and let h tend to zero.	Q019 C	Q032 B
Q004 $7\pi \text{ cm}^2/\text{cm}$	Q011 Use the definition at each specified point and simplify exactly.	Q020 A	Q033 A
Q005 -2 ; 6	Q012 98 m s^{-1}	Q021 B	Q034 A
Q006 Substitute after removing any common factor; polynomial limits are obtained directly.	Q013 $f'(x) = 2x + 3$	Q022 A	Quiz MCQ 1 A
Q007 Each limit is obtained by direct substitution or cancellation of the factor causing $0/0$.	Q014 Expand each $f(x + h) - f(x)$, cancel h and take the limit.	Q023 A	Quiz MCQ 2 B
Q008 6	Q015 Apply $f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)]/h$ to every printed polynomial.	Q024 C	Quiz MCQ 3 C
	Q016 $A = \frac{3\sqrt{3}}{2}x^2$, $A'(x) = 3\sqrt{3}x$	Q025 A	Quiz MCQ 4 D
		Q026 A	Quiz WRITTEN 5 3
		Q027 A	Quiz WRITTEN 6 6
		Q028 B	Quiz WRITTEN 7 12
		Q029 A	Quiz WRITTEN 8 5

Concept 2 | Differentiation and Tangent Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

Worked Solutions

Differentiation and Tangent Lines

Q001 Written

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ with respect to (r), and find $f'(-1)$ for $f(x) = x^3 - 3x + 1$.

Solution

1. Use $(x^n)' = nx^{n-1}$. For a composite cubic, expand or use the chain rule. Treat (h) as constant in dV/dr , then substitute $x = -1$ into $3x^2 - 3$.

Final answer

$$2x - 3; 6(2x + 1)^2; 2\pi rh; 0$$

Q002 Written

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2)$, $f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

Solution

1. Differentiate each term, keeping non-differentiated variables constant. For the values, first form $f'(x) = 12x^2 - 6x$, then substitute.

Final answer

$$3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 126$$

Worked Solutions

Differentiation and Tangent Lines

Q003 **Written**

Differentiate every given Exercise function 18 polynomial/product forms, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.

Solution

1. Bring down the exponent and subtract one. Expand products before differentiating when no chain rule is being used; then substitute the stated input.

Final answer

Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Worked Solutions

Differentiation and Tangent Lines

Q004 Written

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$, find $f'(1) + f'(-1)$.

Solution

1. Write $f(x) = x^{5/3} + x^{-2}$, so $f'(x) = \frac{5}{3}x^{2/3} - 2x^{-3}$. Hence $f'(1) = -\frac{1}{3}$, $f'(-1) = \frac{11}{3}$, and the sum is $\frac{10}{3}$.

Final answer

$$\frac{10}{3}$$

Q005 Written

Find the quadratic $f(x) = ax^2 + bx + c$ satisfying $f'(0) = -2$, $f'(2) = 2$, $f(1) = 2$.

Solution

1. Since $f'(x) = 2ax + b$, $b = -2$, $4a + b = 2$, and $a + b + c = 2$. Thus $a = 1$, $b = -2$, $c = 3$.

Final answer

$$f(x) = x^2 - 2x + 3$$

Worked Solutions

Differentiation and Tangent Lines

Q006 Written

Find the quadratic satisfying $f'(0) = 2$, $f'(1) = 4$, $f(2) = 6$.**Solution**

1. From $2ax + b$: $b = 2$, $2a + b = 4$, and $4a + 2b + c = 6$, giving $a = 1$, $b = 2$, $c = 0$.

Final answer

$$f(x) = x^2 + 2x$$

Q007 Written

Find the four quadratics satisfying the four given triples of derivative and function conditions.**Solution**

1. For each item set $f(x) = ax^2 + bx + c$, use $f'(x) = 2ax + b$, substitute the two derivative conditions and one function condition, then solve the three linear equations.

Final answer

$$x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x + \frac{3}{2}$$

Worked Solutions

Differentiation and Tangent Lines

Q008 Written

Find the quadratic ($f(x)$) satisfying $f'(2) + f'(4) = 0$, $f(3) = 0$, $f(1) = 12$.

Solution

1. With $f'(x) = 2ax + b$, $12a + 2b = 0$, $9a + 3b + c = 0$, and $a + b + c = 12$. Solving gives $a = 3$, $b = -18$, $c = 27$.

Final answer

$$f(x) = 3x^2 - 18x + 27$$

Q009 Written

Find the tangent line to $y = x^2 - 4x + 1$ at $(3, -2)$.

Solution

1. The gradient is $f'(3) = 2$. Use $y + 2 = 2(x - 3)$, hence $y = 2x - 8$.

Final answer

$$y = 2x - 8$$

Worked Solutions

Differentiation and Tangent Lines

Q010 Written

Find the tangent line for the three given functions at their stated points: $y = x^2 + 3$ at $(-2, 7)$, $y = x^2 - 3x + 2$ at $((1, 0))$, and $y = -x^3 + 2x + 4$ at $(-2, 8)$.

Solution

1. Differentiate, evaluate the gradient at the given x-coordinate, and use $y - y_1 = m(x - x_1)$ for each point.

Final answer

$$y = -4x - 1; y = -x + 1; y = -10x - 12$$

Q011 Written

Find the tangent line for all nine given Exercise functions at their given points.

Solution

1. For each item compute $m = f'(a)$, insert $((a, f(a)))$ into $y - f(a) = m(x - a)$, then rearrange.

Final answer

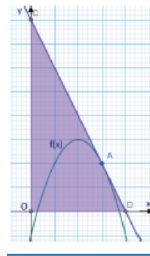
$$y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = x + 1$$

Worked Solutions

Differentiation and Tangent Lines

Q012 Written

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



Solution

1. The tangent is $y = -2x + 8$, so $D = (4, 0)$ and $C = (0, 8)$. Therefore area $= \frac{1}{2}(4)(8) = 16$. The ministry graph is retained beside the demand.

Final answer

16 square units

Worked Solutions

Differentiation and Tangent Lines

Q013 Written

Solve the two given Warm Up demands: tangents from $C(0,2)$ to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

Solution

1. Let the point of tangency be $((a, f(a)))$. Equate the tangent line through the external point, or set $(f'(a))$ equal to the required gradient, then solve for (a) .

Final answer

$$y = -7x + 2, y = x + 2; \quad y = 3x + 15, y = 3x - \frac{95}{27}$$

Q014 Written

Solve the two given Try demands: tangents from $C(1, -1)$ to $y = x^2 - x + 3$, and tangents to $y = x^3 + 2x - 3$ with gradient 5.

Solution

1. Use $((a, f(a)))$ and $m = f'(a)$. The first condition gives $a = 3, -1$; the second gives $a = 1, -1$. Substitute each point into the tangent formula.

Final answer

$$y = 5x - 6, y = -3x + 2; \quad y = 5x - 5, y = 5x - 1$$

Worked Solutions

Differentiation and Tangent Lines

Q015 **Written****Solve all six given Exercise demands involving external points or prescribed gradients.****Solution**

1. Let the contact point be $((a, f(a)))$, form $y - f(a) = f'(a)(x - a)$, and impose the external point or gradient condition. Retain every real value of (a) .

Final answer

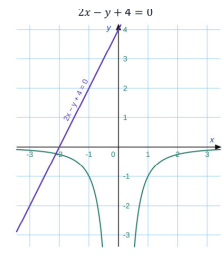
$$y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 8$$

Worked Solutions

Differentiation and Tangent Lines

Q016 Written

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



Solution

1. The given line has gradient 2. Since $f'(x) = 2/x^3$, $2/a^3 = 2$ gives $a = 1$. The tangent at $(1, -1)$ is $y + 1 = 2(x - 1)$, so $y = 2x - 3$. The ministry graph is retained beside the demand.

Final answer

$$y = 2x - 3$$

Worked Solutions

Differentiation and Tangent Lines

Q017 MCQ

The derivative of x^n is:**Solution**

1. Bring down n and subtract 1 from the exponent.

Correct option: B**Final answer**

$$nx^{n-1}$$

Q018 MCQ

The derivative of $V = \pi r^2 h$ with respect to r is:**Solution**

1. Treat h as a constant.

Correct option: B**Final answer**

$$2\pi rh$$

Worked Solutions

Differentiation and Tangent Lines

Q019 MCQ

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:

Solution

1. $f'(x) = 3x^2 - 3$.

Correct option: C

Final answer

0

Q020 MCQ

The challenge function can be written as:

Solution

1. Convert the cube root and reciprocal to powers.

Correct option: A

Final answer

$$x^{5/3} + x^{-2}$$

Worked Solutions

Differentiation and Tangent Lines

Q021 MCQ

For $f(x) = ax^2 + bx + c$, $(f'(x))$ is:**Solution**

1. Differentiate the quadratic.

Correct option: B**Final answer**

$$2ax + b$$

Q022 MCQ

The condition $f'(0) = -2$ gives:**Solution**

1. Substitute $x = 0$ into $2ax + b$.

Correct option: B**Final answer**

$$b = -2$$

Worked Solutions

Differentiation and Tangent Lines

Q023 MCQ

The warm-up quadratic is:**Solution**

1. Solve the three coefficient equations.

Correct option: A

Final answer

$$x^2 - 2x + 3$$

Q024 MCQ

The challenge coefficient a is:**Solution**1. Use $f'(2) + f'(4) = 0$, then the two function values.

Correct option: C

Final answer

3

Worked Solutions

Differentiation and Tangent Lines

Q025 MCQ

The tangent gradient at $x=a$ is:**Solution**

1. The derivative gives the instantaneous gradient.

Correct option: C**Final answer**

$$f'(a)$$

Q026 MCQ

The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:**Solution**

1. Use gradient 2 and point $3, -2$.

Correct option: B**Final answer**

$$y = 2x - 8$$

Worked Solutions

Differentiation and Tangent Lines

Q027 MCQ

In the triangle challenge, the x-intercept of the tangent is:**Solution**1. The tangent is $y = -2x + 8$.

Correct option: C

Final answer

4

Q028 MCQ

The area of triangle (ODC) in the challenge is:**Solution**

1. Use one half base times height.

Correct option: C

Final answer

16

Worked Solutions

Differentiation and Tangent Lines

Q029 MCQ

When the contact point is not given, set it to:**Solution**

1. Use an unknown contact x-coordinate.

Correct option: B**Final answer** $(a, f(a))$

Q030 MCQ

The gradient of $2x - y + 4 = 0$ is:**Solution**

1. Rearrange to $y = 2x + 4$.

Correct option: D**Final answer**

2

Worked Solutions

Differentiation and Tangent Lines

Q031 MCQ

For $f(x) = -1/x^2$, $(f'(x))$ is:**Solution**1. Write $f = -x^{-2}$ and differentiate.

Correct option: B

Final answer

$$2/x^3$$

Q032 MCQ

The parallel tangent in the challenge is:**Solution**1. The contact point is $a = 1$.

Correct option: B

Final answer

$$y = 2x - 3$$

Worked Solutions

Differentiation and Tangent Lines

Quiz WRITTEN 1 Concept Quiz · Written

Differentiate $3x^3 - 2x$.**Solution**

1. Apply the power rule.

Final answer

$$9x^2 - 2$$

Quiz WRITTEN 2 Concept Quiz · Written

Find the quadratic with $f'(0) = 1, f'(1) = 3, f(1) = 2$.**Solution**

1. Use $f'(x) = 2ax + b$ and solve for (a,b,c).

Final answer

$$x^2 + x$$

Worked Solutions

Differentiation and Tangent Lines

Quiz WRITTEN 3 Concept Quiz · Written

Find the tangent to $y = x^2$ at $((2,4))$.

Solution

1. Gradient $f'(2) = 4$, then use point-slope form.

Final answer

$$y = 4x - 4$$

Quiz WRITTEN 4 Concept Quiz · Written

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

Solution

1. Set $f'(a) = 1$, giving $a = 1$.

Final answer

$$y = x - 4$$

Worked Solutions

Differentiation and Tangent Lines

Quiz MCQ 5 **Concept Quiz · MCQ**

The derivative of a constant is:

Solution

1. A constant has zero rate of change.

Correct option: A

Final answer

0

Quiz MCQ 6 **Concept Quiz · MCQ**

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

Solution

1. There are three unknown coefficients.

Correct option: B

Final answer

two derivatives and one function value

Worked Solutions

Differentiation and Tangent Lines

Quiz MCQ 7 [Concept Quiz · MCQ](#)

The tangent equation in point-slope form is:

Solution

1. Use the given point and tangent gradient.

Correct option: A

Final answer

$$y - y_1 = mx - x_1$$

Quiz MCQ 8 [Concept Quiz · MCQ](#)

A tangent parallel to $y = 3x - 1$ has gradient:

Solution

1. Parallel lines have equal gradients.

Correct option: C

Final answer

3

Answer key

Use the question number shown on each page.

Q001 $2x - 3; 6(2x + 1)^2; 2\pi rh; 0$

Q008 $f(x) = 3x^2 - 18x + 27$

Q019 C

Q002 $3, -6x + 8, 2x + 3, 6x^2 - 8x + 6, 6(2x - 3)^2, 4x^2 + x - 1; 2\pi r, v - gt; 36, 20$

Q009 $y = 2x - 8$

Q020 A

Q003 Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Q010 $y = -4x - 1; y = -x + 1; y = -10x - 12$

Q021 B

Q011 $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = -4x - 1$

Q022 B

Q012 16 square units

Q023 A

Q013 $y = -7x + 2, y = x + 2; y = 3x + 15, y = 3x - \frac{85}{27}$

Q024 C

Q014 $y = 5x - 6, y = -3x + 2; y = 5x - 5, y = 5x - 1$

Q025 C

Q004 $\frac{10}{3}$

Q015 $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 4$

Q026 B

Q005 $f(x) = x^2 - 2x + 3$

Q016 $y = 2x - 3$

Q027 C

Q006 $f(x) = x^2 + 2x$

Q007 $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x - \frac{3}{2}$

Q017 B

Q028 C

Q018 B

Q029 B

Q030 D

Q031 B

Q032 B

Quiz WRITTEN 1 $9x^2 - 2$

Quiz WRITTEN 2 $x^2 + x$

Quiz WRITTEN 3 $y = 4x - 4$

Quiz WRITTEN 4 $y = x - 4$

Quiz MCQ 5 A

Quiz MCQ 6 B

Quiz MCQ 7 A

Quiz MCQ 8 C

Concept 3 | Applications of Differentiation: Polynomial Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Stationary

$$f'(x) = 0$$

Second derivative

$$f''(x) > 0 \Rightarrow \text{minimum}$$

Rate

$$\frac{dy}{dx}$$

Eng Eslam Ahmed | 01120009622

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q001 Written

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

Solution

1. Differentiate, solve $f'(x) = 0$, make a sign table, and substitute each stationary x-value into $f(x)$.

Final answer

Use the derivative sign table: for $x^3 - 6x^2 + 16$, local maximum 16 at $x = 0$ and local minimum -16 at $x = 4$; classify the other two from their stationary points.

Q002 Written

Find the extreme values and sketch the three given Try cubics.

Solution

1. Record the sign of f' from left to right and label each change $+$ $\rightarrow$ $-$ as a local maximum and $-$ $\rightarrow$ $+$ as a local minimum.

Final answer

Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q003 Written

Find the extreme values and draw the graphs for all nine given cubic functions in Exercise.

Solution

1. Use exact stationary coordinates and keep the graph's increasing/decreasing intervals consistent with the sign table.

Final answer

Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.

Q004 Written

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

Solution

1. $P'(x) = 3[(x - a)^2 - 1]$, so the critical points are $a - 1$ and $a + 1$. The derivative changes $+$ $\rightarrow$ $-$ then $-$ $\rightarrow$ $+$.

Final answer

Local maximum at $x = a - 1$; local minimum at $x = a + 1$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q005 **Written**

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

Solution

1. Compare $f(-3)$, $f(4)$ with the stationary values $f(-2) = -11$ and $f(2) = 21$.

Final answer

Maximum 21 at $x = 2$; minimum -11 at $x = -2$.

Q006 **Written**

Find the maximum and minimum values of the two given Try cubics on their stated closed intervals.

Solution

1. A local extreme is not automatically absolute: compare it with both domain endpoints.

Final answer

Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q007 Written

Find the maximum and minimum values of all six given cubics on their given closed intervals.

Solution

1. State the extreme value together with its x-coordinate and retain only points in the specified domain.

Final answer

For each function, compare the endpoint values with the values at all critical x-values lying in the interval.

Q008 Written

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

Solution

1. Critical points are $x = \pm\sqrt{2}$, with values $\mp 4\sqrt{2}$; compare these with $f(-3) = -9$ and $f(3) = 9$.

Final answer

Absolute minimum -9 at $x = -3$; absolute maximum 9 at $x = 3$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q009 Written

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a , b , and the local minimum.

Solution

1. Use $f'(2) = 0$ to get $a = 12$, then $f(2) = 9$ to get $b = -7$. The other stationary point is $x = -2$.

Final answer

$a = 12, b = -7$; local minimum -23 at $x = -2$.

Q010 Written

Solve the two given Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

Solution

1. Differentiate first, use each stated extreme point, then evaluate the remaining extreme value from the resulting cubic.

Final answer

Use $f'(a) = 0$, $f(a) = M$, and the second stationary condition to form three simultaneous equations in the coefficients.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q011 Written

Solve all six given Exercise problems asking for cubic coefficients from extrema.

Solution

1. Do not skip the condition $f'(a) = 0$: it identifies the stationary point before the function-value equation is used.

Final answer

Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.

Q012 Written

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a, b, c .

Solution

1. The derivative roots are $-2, 4$, so $f'(x) = 3(x + 2)(x - 4) = 3x^2 - 6x - 24$. Thus $a = -3, b = -24$. Substitution of $f(-2) = 178$ gives $c = 150$.

Final answer

$a = -3, b = -24, c = 150$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q013 Written

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k .

Solution

1. The derivative is $f'(x) = 3x^2 - 6kx + k$. No two distinct stationary roots means $D = 36k^2 - 12k \leq 0$, hence $0 \leq k \leq \frac{1}{3}$.

Final answer

$$0 \leq k \leq \frac{1}{3}.$$

Q014 Written

Determine the parameter ranges in the given Try problems for a cubic to have no extreme values or to have both extremes.

Solution

1. Differentiate, calculate the quadratic discriminant, and solve the resulting parameter inequality exactly.

Final answer

Use $D \leq 0$ for no extremes and $D > 0$ for two distinct stationary points.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q015 Written

Determine the parameter ranges for all six given Exercise cubics according to whether they have extreme values.

Solution

1. Remember: $D > 0$ gives two distinct real stationary roots; $D \leq 0$ gives no pair of distinct extremes.

Final answer

Apply the derivative-discriminant criterion separately to each printed cubic.

Q016 Written

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, find the exact k -range that guarantees both a local maximum and minimum.

Solution

1. $f'(x) = 3x^2 + 6kx + 12k$ has two distinct real roots when $36k^2 - 144k > 0$, i.e. $k(k - 4) > 0$.

Final answer

$k < 0$ or $k > 4$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q017 Written

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k .

Solution

1. For a positive leading coefficient, monotonic increase requires the derivative quadratic to have no two distinct real roots: $D = 24k^2 - 60 \leq 0$.

Final answer

$$-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}.$$

Q018 Written

Find the parameter ranges in the two given Try problems for the cubic to be monotonically increasing.

Solution

1. Monotonicity is checked from $f'(x) \geq 0$ for every real x ; use the discriminant condition for the quadratic derivative.

Final answer

Set the derivative discriminant $D \leq 0$, with the derivative leading coefficient positive.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q019 Written

Find the parameter ranges in all six given Exercise problems for monotonic increase.

Solution

1. Complete the discriminant test and include equality when the derivative has a repeated zero.

Final answer

Differentiate each cubic and impose that its quadratic derivative never becomes negative.

Q020 Written

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k + 6)x + 10$, find the k-range for monotonic increase on $x \in [1, \infty)$.

Solution

1. $f'(x) = x^2 - 2kx + k + 6$. If $k \leq 1$, its minimum on $[1, \infty)$ is $7 - k \geq 0$; if $k > 1$, the vertex condition gives $k + 6 - k^2 \geq 0$, hence $k \leq 3$.

Final answer

$k \leq 3$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q021 Written

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.**Solution**

1. The stationary values are $f(1) = 1 > 0$ and $f(3) = -3 < 0$, so the cubic crosses the x-axis three times.

Final answer

Three distinct real roots.

Q022 Written

Find the number of distinct real roots for the three given Try equations.**Solution**

1. Draw $y = f(x)$, find its critical points, and compare the local maximum/minimum with zero.

Final answer

Use the cubic graph: count x-axis intersections from the stationary points and end behavior.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q023 Written

Find the number of distinct real roots for all nine given cubic equations.

Solution

1. Differentiate, locate the turning points, and use their signs relative to zero to decide between one, two, or three distinct roots.

Final answer

Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.

Q024 Written

Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.

Solution

1. The derivative is $-3(h - 2)(h - 6)$; both stationary values are positive, so the graph crosses the axis only once.

Final answer

One distinct real root.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q025 Written

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a .

Solution

1. Rewrite as $x^3 - 3x^2 - 9x = a$. The local values are 5 and -27 ; a horizontal line intersects the cubic three times strictly between them.

Final answer

$$-27 < a < 5.$$

Q026 Written

Solve the two given Try parameter problems for three or two distinct real roots.

Solution

1. Strict inequalities give three roots; equality at a stationary value produces two distinct roots and must be handled separately.

Final answer

Rewrite each equation as $g(x) = a$, then compare the parameter line with the local maximum and minimum of g .

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q027 **Written****Solve all four given Exercise parameter problems for the required number of distinct roots.****Solution**

1. Keep the endpoint cases: tangency at an extreme value counts as two distinct roots, not three.

Final answer

Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.

Q028 **Written****For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.****Solution**

1. Set $g(x) = x^3 - 6x^2 + 9x$, whose extrema are 0 and 4. The horizontal level is $2a^2 - 7a + 2$; require $0 < 2a^2 - 7a + 2 < 4$.

Final answer

$$\frac{7 - \sqrt{65}}{4} < a < \frac{7 - \sqrt{33}}{4} \text{ or } \frac{7 + \sqrt{33}}{4} < a < \frac{7 + \sqrt{65}}{4}.$$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q029 **Written****Prove the given inequalities using an auxiliary cubic and a derivative sign table.****Solution**

1. Differentiate the auxiliary function, make the sign table on the domain, and substitute the minimum point or endpoint.

Final answer

Set $f(x) = \text{LHS} - \text{RHS}$, prove $f(x) \geq 0$ on the stated domain, and identify its minimum.

Q030 **Written****Prove all three given cubic inequalities on their stated domains.****Solution**

1. The proof is complete only after the derivative sign table and the minimum value are stated.

Final answer

Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q031 Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ for every $x \in [-1, 3]$.

Solution

1. The maximum of $x^3 - 3x^2 - 3x + 3$ on the interval is $4\sqrt{2} - 2$ at $x = 1 - \sqrt{2}$. Hence $k + 4\sqrt{2} - 2 \leq 0$.

Final answer

$k \leq 2 - 4\sqrt{2}$; the largest value is $2 - 4\sqrt{2}$.

Q032 Written

Find the extreme values and sketch the two given quartic functions.

Solution

1. Differentiate the quartic, use a sign table for the cubic derivative, and substitute the critical values.

Final answer

For $3x^4 + 4x^3 - 12x^2$: local maximum 0 at $x = 0$, local minima -32 at $x = -2$ and -5 at $x = 1$. For $-x^4 - 4x^3 - 8$: maximum 19 at $x = -3$; $x = 0$ is stationary but not an extreme.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q033 Written

Find the extreme values and graphs for all six given quartic functions.

Solution

1. Repeated derivative roots may be stationary without being extrema; check the sign change before labelling a point.

Final answer

Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.

Q034 Written

For $f(x) = x^4 - 4x^3 + k$, find the exact k -range for exactly two distinct real roots.

Solution

1. The global minimum of $x^4 - 4x^3 + k$ is $k - 27$ at $x = 3$. Exactly two crossings require this minimum to be negative.

Final answer

$k < 27$.

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q035 MCQ

At a local maximum, the sign of f' changes:

Solution

1. Use the first derivative sign change.

Correct option: B

Final answer

$+to-$

Q036 MCQ

For $P'(x) = 3[(x - a)^2 - 1]$, the local maximum occurs at:

Solution

1. The derivative changes + to - at $a - 1$.

Correct option: A

Final answer

$a - 1$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q037 MCQ

Absolute extrema on a closed interval require comparing:**Solution**

1. Closed intervals include both.

Correct option: C**Final answer**

stationary points and endpoints

Q038 MCQ

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:**Solution**

1. Compare the endpoints with the critical values.

Correct option: B**Final answer**

9

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q039 MCQ

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

Solution

1. Set $f'(2) = -12 + a = 0$.

Correct option: C

Final answer

12

Q040 MCQ

The derivative roots in the challenge are:

Solution

1. Use the two turning-point x -values.

Correct option: B

Final answer

-2 and 4

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q041 MCQ

No two distinct extrema for a cubic means the derivative discriminant is:**Solution**

1. A repeated root or no real roots gives no pair.

Correct option: C**Final answer**

$$D \leq 0$$

Q042 MCQ

The challenge range for both extrema is:**Solution**

1. Require $36k^2 - 144k > 0$.

Correct option: B**Final answer**

$$k < 0 \text{ or } k > 4$$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q043 MCQ

For a positive-leading cubic to be monotone increasing, its derivative must be:

Solution

1. Use $f'(x) \geq 0$.

Correct option: B

Final answer

non-negative everywhere

Q044 MCQ

The challenge domain in 7-13 is:

Solution

1. Use the restricted domain in the vertex test.

Correct option: B

Final answer

[1,infinity)

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q045 MCQ

Three distinct roots occur when the x-axis lies:**Solution**

1. Then the graph crosses the axis three times.

Correct option: B**Final answer**

between the extrema

Q046 MCQ

The 7-14 challenge has how many distinct real roots?**Solution**

1. Both turning values are above zero.

Correct option: B**Final answer**

1

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q047 MCQ

Three intersections with $y = a$ require the level to lie:**Solution**

1. Tangency is an endpoint case.

Correct option: A

Final answer

strictly between the extrema

Q048 MCQ

The 7-15 warm-up range is:**Solution**

1. Use the two local values.

Correct option: A

Final answer $-27 < a < 5$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q049 MCQ

To prove $LHS \geq RHS$, define:**Solution**

1. Then prove $f \geq 0$.

Correct option: B**Final answer**

LHS-RHS

Q050 MCQ

The largest k in the challenge is:**Solution**

1. Bound k by the maximum of the k -free part.

Correct option: A**Final answer** $2-4\sqrt{2}$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q051 MCQ

A repeated root of a derivative may be:**Solution**

1. Check the sign change.

Correct option: C**Final answer**

stationary but not an extreme

Q052 MCQ

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require:**Solution**

1. The global minimum must be below zero.

Correct option: B**Final answer** $k < 27$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q053 MCQ

A local minimum is identified by the change:**Solution**

1. A negative derivative followed by a positive derivative gives a local minimum.

Correct option: B**Final answer** $-to+$

Q054 MCQ

For $y = x^3 - 6x^2 + 16$, the critical x-values are:**Solution**

1. Solve $3x^2 - 12x = 0$.

Correct option: A**Final answer**

0 and 4

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q055 MCQ

The derivative sign table is used to determine:**Solution**

1. Signs of (f') control monotonicity.

Correct option: B**Final answer**

increase/decrease and extrema

Q056 MCQ

On a closed interval, an endpoint can be:**Solution**

1. Compare both endpoints with stationary values.

Correct option: B**Final answer**

an absolute extreme

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q057 MCQ

For $y = -x^3 + 12x + 5$, the stationary x-values are:

Solution

1. Solve $-3x^2 + 12 = 0$.

Correct option: A

Final answer

-2 and 2

Q058 MCQ

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at:

Solution

1. Compare all candidates; $f(-3) = -9$.

Correct option: A

Final answer

$x = -3$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q059 MCQ

To determine a cubic from an extreme value, use both $f(a) = M$ and:

Solution

1. An extreme point is stationary.

Correct option: B

Final answer

$$f'(a) = 0$$

Q060 MCQ

In the warm-up $f(x) = -x^3 + 12x - 7$, the local minimum occurs at:

Solution

1. The derivative changes from negative to positive at -2.

Correct option: A

Final answer

$$x = -2$$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q061 MCQ

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires:

Solution

1. Use $D = 12k(3k - 1) \leq 0$.

Correct option: A

Final answer

$$0 \leq k \leq 1/3$$

Q062 MCQ

Two distinct stationary points occur when the derivative discriminant is:

Solution

1. A positive discriminant gives two distinct real derivative roots.

Correct option: C

Final answer

$$D > 0$$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q063 MCQ

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

Solution

1. The quadratic has no two distinct real zeros.

Correct option: A

Final answer

non-negative for all x

Q064 MCQ

In the 7-13 challenge, monotonicity is required on:

Solution

1. Use the operational interval.

Correct option: A

Final answer

[1,infinity)

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q065 MCQ

A cubic with both stationary values above the x-axis has:**Solution**

1. The graph crosses the x-axis once.

Correct option: B**Final answer**

one distinct real root

Q066 MCQ

To count cubic roots, compare the extrema with:**Solution**

1. Roots are x-axis intersections.

Correct option: B**Final answer**

the x-axis

Worked Solutions

Applications of Differentiation: Polynomial Functions

Q067 MCQ

At an extremal parameter level, a cubic and a horizontal line have:

Solution

1. One contact is tangent and the other crossing remains.

Correct option: B

Final answer

two distinct intersections

Q068 MCQ

For $x^4 - 4x^3 + k$, the global minimum occurs at:

Solution

1. The derivative is $4x^2(x - 3)$.

Correct option: C

Final answer

$x = 3$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 1 Concept Quiz · Written

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

Solution

1. Use $f'(x) = 3x(x - 2)$ and a sign table.

Final answer

Maximum 1 at $x = 0$; minimum -3 at $x = 2$

Quiz WRITTEN 2 Concept Quiz · Written

Find a, b if $f(x) = -x^3 + ax + b$ has a local maximum 4 at $x=1$.

Solution

1. Use $f'(1) = 0$ and $f(1) = 4$.

Final answer

$a = 3, b = 2$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Quiz WRITTEN 3 Concept Quiz · Written

Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.

Solution

1. Set the derivative discriminant ≤ 0 .

Final answer

$$-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$$

Quiz WRITTEN 4 Concept Quiz · Written

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ on $[-1, 3]$.

Solution

1. Use the maximum of the k-free cubic on the interval.

Final answer

$$2 - 4\sqrt{2}$$

Worked Solutions

Applications of Differentiation: Polynomial Functions

Quiz MCQ 5 **Concept Quiz · MCQ**

At a local minimum, f' changes:

Solution

1. Use the first derivative sign change.

Correct option: B

Final answer

$-to+$

Quiz MCQ 6 **Concept Quiz · MCQ**

A cubic with a positive-leading derivative and $D < 0$ is:

Solution

1. The derivative quadratic stays positive.

Correct option: A

Final answer

always increasing

Worked Solutions

Applications of Differentiation: Polynomial Functions

Quiz MCQ 7 [Concept Quiz · MCQ](#)

The 7-15 challenge requires the level to satisfy:

Solution

1. Three distinct intersections lie strictly between extrema.

Correct option: B

Final answer

$$0 < t < 4$$

Quiz MCQ 8 [Concept Quiz · MCQ](#)

The global minimum of $x^4 - 4x^3 + k$ is:

Solution

1. Evaluate at the quartic's global minimum $x = 3$.

Correct option: A

Final answer

$$k - 27$$

Answer key

Use the question number shown on each page.

- Q001** Use the derivative sign table: for $x^3 - 6x^2 + 16$, local maximum 16 at $x = 0$ and local minimum -16 at $x = 4$; classify the other two from their stationary points.
- Q002** Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.
- Q003** Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.
- Q004** Local maximum at $x = a - 1$; local minimum at $x = a + 1$.
- Q005** Maximum 21 at $x = 2$; minimum -11 at $x = -2$.
- Q006** Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.
- Q007** For each function, compare the endpoint values with the values at all critical x-values lying in the interval.
- Q008** Absolute minimum -9 at $x = -3$; absolute maximum 9 at $x = 3$.
- Q009** $a = 12, b = -7$; local minimum -23 at $x = -2$.
- Q010** Use $f'(a) = 0, f(a) = M$, and the second stationary condition to form three simultaneous equations in the coefficients.
- Q011** Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.
- Q012** $a = -3, b = -24, c = 150$.
- Q013** $0 \leq k \leq \frac{1}{3}$.
- Q014** Use $D \leq 0$ for no extremes and $D > 0$ for two distinct stationary points.
- Q015** Apply the derivative-discriminant criterion separately to each printed cubic.
- Q016** $k < 0$ or $k > 4$.
- Q017** $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$.
- Q018** Set the derivative discriminant $D \leq 0$, with the derivative leading coefficient positive.
- Q019** Differentiate each cubic and impose that its quadratic derivative never becomes negative.
- Q020** $k \leq 3$.
- Q021** Three distinct real roots.
- Q022** Use the cubic graph: count x-axis intersections from the stationary points and end behavior.
- Q023** Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.
- Q024** One distinct real root.
- Q025** $-27 < a < 5$.
- Q026** Rewrite each equation as $g(x) = a$, then compare the parameter line with the local maximum and minimum of g .
- Q027** Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.
- Q028** $\frac{7 - \sqrt{65}}{4} < a < \frac{7 - \sqrt{33}}{4}$ or $\frac{7 + \sqrt{33}}{4} < a < \frac{7 + \sqrt{65}}{4}$.
- Q029** Set $f(x) = \text{LHS} - \text{RHS}$, prove $f(x) \geq 0$ on the stated domain, and identify its minimum.
- Q030** Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.
- Q031** $k \leq 2 - 4\sqrt{2}$; the largest value is $2 - 4\sqrt{2}$.
- Q032** For $3x^4 + 4x^3 - 12x^2$: local maximum 0 at $x = 0$, local minima -32 at $x = -2$ and -5 at $x = 1$. For $-x^4 - 4x^3 - 8$: maximum 19 at $x = -3$; $x = 0$ is stationary but not an extreme.
- Q033** Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.
- Q034** $k < 27$.
- Q035** B
- Q036** A
- Q037** C
- Q038** B
- Q039** C
- Q040** B
- Q041** C
- Q042** B
- Q043** B
- Q044** B
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** B
- Q050** A
- Q051** C
- Q052** B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 A

Q058 A

Q059 B

Q060 A

Q061 A

Q062 C

Q063 A

Q064 A

Q065 B

Q066 B

Q067 B

Q068 C

Quiz WRITTEN 1 Maximum 1 at $x = 0$; minimum -3 at
 $x = 2$

Quiz WRITTEN 2 $a = 3, b = 2$ Quiz WRITTEN 3 $-\frac{\sqrt{10}}{2} \leq k \leq \frac{\sqrt{10}}{2}$ Quiz WRITTEN 4 $2 - 4\sqrt{2}$

Quiz MCQ 5 B

Quiz MCQ 6 A

Quiz MCQ 7 B

Quiz MCQ 8 A

Concept 4 | Integration: Definite Integrals and Applications

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Formula opener

Antiderivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Definite

$$\int_a^b f(x) dx = F(b) - F(a)$$

Area

$$A = \int_a^b |f(x)| dx$$

Worked Solutions

Integration: Definite Integrals and Applications

Q001 Written

Find the two indefinite integrals in the Warm Up.

Solution

1. Use $\int x^n dx = \frac{x^{n+1}}{n+1} + C$, then differentiate your result to check.

Final answer

Integrate term by term and add C to each primitive.

Q002 Written

Find the six indefinite integrals in the given Try section, expanding a squared bracket first when needed.

Solution

1. For $(x - 1)^2$, write $x^2 - 2x + 1$ before applying the power rule.

Final answer

Expand products or powers, collect like powers, then integrate every term.

Worked Solutions

Integration: Definite Integrals and Applications

Q003 Written

Solve the first ten given indefinite-integral exercises (constants, linear and quadratic polynomials).

Solution

1. A constant integrates to a constant times x ; combine the terms only after integration.

Final answer

Apply linearity and the power rule; retain the constant of integration.

Q004 Written

Solve the remaining eight given integrals, including products and squared brackets.

Solution

1. Do not integrate a product factor-by-factor; expand first, then use the power rule.

Final answer

Expand each product or square, simplify, and integrate term by term.

Worked Solutions

Integration: Definite Integrals and Applications

Q005 **Written****Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.****Solution**

1. Integrate $F'(x)$, substitute $x = 1$, and solve the resulting equation for C .

Final answer

$$F(x) = x^3 + 4x^2 - x + 3.$$

Q006 **Written****Find the two given Try functions from their derivatives and one given value of F .****Solution**

1. The initial value fixes the otherwise undetermined constant of integration.

Final answerIntegrate the derivative and use the given point to determine C .

Worked Solutions

Integration: Definite Integrals and Applications

Q007 **Written****Find $F(x)$ for all six given Exercise conditions.****Solution**

1. Keep the derivative and the initial condition as two separate equations before solving for C .

Final answer

For each item, integrate $F'(x)$, then impose the stated value of F .

Q008 **Written****Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.****Solution**

1. The gradient is $f'(x)$; integrate it and use $f(3) = 2$ to determine C .

Final answer

$$f(x) = x^3 - x^2 + x - 19.$$

Worked Solutions

Integration: Definite Integrals and Applications

Q009 Written

Solve the given Try tangent-gradient problem through $(1, 1)$.**Solution**

1. Replace the gradient by $f'(x)$, integrate, then substitute the known coordinate.

Final answer

Integrate the given gradient and use the point on the curve to determine the constant.

Q010 Written

Solve the three given tangent-gradient exercises, finding each function from its point and gradient.**Solution**

1. A point (x_0, y_0) gives $f(x_0) = y_0$, which fixes C .

Final answer

Integrate the polynomial gradient and apply the given point condition in each case.

Worked Solutions

Integration: Definite Integrals and Applications

Q011 Written

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

Solution

1. Write $[F(x)]_a^b = F(b) - F(a)$ before substituting the limits.

Final answer

Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.

Q012 Written

Evaluate the six given Try definite integrals, simplifying the integrand before applying the limits.

Solution

1. Linearity allows sums and differences of integrals to be combined before integration.

Final answer

Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.

Worked Solutions

Integration: Definite Integrals and Applications

Q013 Written

Evaluate the first twelve given Exercise definite integrals, including reversed and equal limits.

Solution

1. Check the order of the limits before evaluating the antiderivative.

Final answer

Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.

Q014 Written

Evaluate the remaining twelve given Exercise integrals, including sums, differences and products.

Solution

1. Combine only integrals with the same limits; otherwise split or evaluate separately.

Final answer

Use linearity, expand products, then apply the endpoint-substitution rule.

Worked Solutions

Integration: Definite Integrals and Applications

Q015 Written

Use the four properties of definite integrals to simplify the three Warm Up expressions.

Solution

1. Choose the property that makes the integrand or limits match a known form.

Final answer

Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.

Q016 Written

Solve the four given Try definite-integral transformations.

Solution

1. For $(x - a)(x - b)$, first make the coefficient of x equal to 1.

Final answer

Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.

Worked Solutions

Integration: Definite Integrals and Applications

Q017 **Written****Solve the first six given Exercise questions using interval splitting and reversal.****Solution**

1. A minus sign before an integral can be absorbed by reversing its limits.

Final answer

Rewrite each expression until adjacent intervals have matching integrands and compatible limits.

Q018 **Written****Solve the final six given Exercise questions using the factorized definite-integral identity.****Solution**

1. Identify (a,b) from the two linear factors and account for every constant multiplier.

Final answerScale to $\int_a^b (x - a)(x - b) dx = -\frac{(b-a)^3}{6}$.

Worked Solutions

Integration: Definite Integrals and Applications

Q019 Written

Find $f(x)$ from the given equation $f(x) = x^2 + \int_0^1 f(t) dt$.

Solution

1. Replace the definite integral by a constant, substitute the resulting $f(t)$ back into it, and solve.

Final answer

Let $A = \int_0^1 f(t) dt$; then $A = \frac{1}{3} + A$, so $A = -\frac{1}{3}$ and $f(x) = x^2 - \frac{1}{3}$.

Q020 Written

Solve the two given Try integral equations for $f(x)$.

Solution

1. The unknown integral is a number independent of x ; determine it before writing the final function.

Final answer

Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.

Worked Solutions

Integration: Definite Integrals and Applications

Q021 **Written****Find $f(x)$ for all five given integral equations.****Solution**

1. Check the final function by integrating it over the stated interval and substituting back.

Final answer

Let the definite integral be A , substitute the expression for $f(t)$, solve for A , then recover $f(x)$.

Q022 **Written****Differentiate $\int_3^x (2t^2 + 3t - 5) dt$ with respect to x .****Solution**

1. By the Fundamental Theorem, $\frac{d}{dx} \int_a^x f(t) dt = f(x)$.

Final answer

$2x^2 + 3x - 5$.

Worked Solutions

Integration: Definite Integrals and Applications

Q023 Written

Solve the two given Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

Solution

1. A constant lower limit disappears under differentiation; it is recovered from the original equation.

Final answer

Replace t by x when differentiating, then use equal limits or the given condition to determine a .

Q024 Written

Differentiate the three given expressions with respect to x .

Solution

1. After differentiation, replace t with the variable upper limit x .

Final answer

Apply the variable-limit Fundamental Theorem directly to each integrand.

Worked Solutions

Integration: Definite Integrals and Applications

Q025 **Written****Find $f(x)$ and a for the three given equations with a variable upper limit.****Solution**

1. Use $0 = \int_a^x f(t) dt$ in the original equation after finding $f(x)$.

Final answer

Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a .

Q026 **MCQ****The constant of integration is required because:****Solution**

1. Differentiation removes additive constants.

Correct option: B**Final answer**

primitives differ by a constant

Worked Solutions

Integration: Definite Integrals and Applications

Q027 MCQ

The integral of x^n for $n \neq -1$ is:**Solution**

1. Use the power rule for integration.

Correct option: B

Final answer

$$x^{(n+1)/(n+1)} + C$$

Q028 MCQ

Before integrating $(x - 1)^2$, first:**Solution**1. Write $x^2 - 2x + 1$.

Correct option: B

Final answer

expand it

Worked Solutions

Integration: Definite Integrals and Applications

Q029 MCQ

The integral of $3x^2 - 2x + 1$ is:**Solution**

1. Integrate term by term.

Correct option: A**Final answer**

$$x^3 - x^2 + x + C$$

Q030 MCQ

To find $F(x)$ from $F'(x)$, you should first:**Solution**

1. Integration reverses differentiation.

Correct option: B**Final answer**integrate $F'x$

Worked Solutions

Integration: Definite Integrals and Applications

Q031 MCQ

The condition $F(1) = 7$ determines:**Solution**

1. Use the given function value after integrating.

Correct option: C

Final answer

C

Q032 MCQ

If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:**Solution**1. Integrate and use $F(0) = 1$.

Correct option: A

Final answer $3x^2 - 2x + 1$

Worked Solutions

Integration: Definite Integrals and Applications

Q033 MCQ

The gradient of the tangent at (x, y) is:**Solution**

1. By definition, the derivative is the tangent gradient.

Correct option: B**Final answer**

$$f'x$$

Q034 MCQ

A point (x_0, y_0) on $y = f(x)$ gives:**Solution**

1. Use the point as an initial condition.

Correct option: B**Final answer**

$$f(x_0) = y_0$$

Worked Solutions

Integration: Definite Integrals and Applications

Q035 MCQ

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

Solution

1. Integrate each term.

Correct option: B

Final answer

$$x^3 - x^2 + x + C$$

Q036 MCQ

The Fundamental Theorem gives $\int_a^b f(x)dx =$:

Solution

1. Evaluate an antiderivative at the upper minus lower limit.

Correct option: B

Final answer

$$Fb - Fa$$

Worked Solutions

Integration: Definite Integrals and Applications

Q037 MCQ

When the limits are equal, $\int_a^a f(x)dx$ equals:**Solution**

1. There is no interval of accumulation.

Correct option: C**Final answer**

0

Q038 MCQ

A reversed definite integral satisfies:**Solution**

1. Reversing the limits changes the sign.

Correct option: B**Final answer**

negative reversed integral

Worked Solutions

Integration: Definite Integrals and Applications

Q039 MCQ

To evaluate a definite integral of a product, first:**Solution**

1. Then apply the power rule and endpoint substitution.

Correct option: B

Final answer

expand the product

Q040 MCQ

Adjacent integrals with the same integrand can be:**Solution**

1. Use interval additivity.

Correct option: B

Final answer

combined over the joined interval

Worked Solutions

Integration: Definite Integrals and Applications

Q041 MCQ

To use $\int_{\alpha}^{\beta} (x - \alpha)(x - \beta)dx$, the factors must have:

Solution

1. Match each factor to its endpoint.

Correct option: B

Final answer

roots a and b

Q042 MCQ

The identity for the factored integral is:

Solution

1. Use the signed cubic interval length.

Correct option: B

Final answer

$-(b - a)^3/6$

Worked Solutions

Integration: Definite Integrals and Applications

Q043 MCQ

A minus sign before an integral can be removed by:

Solution

1. $-\int_a^b f = \int_b^a f.$

Correct option: B

Final answer

reversing its limits

Q044 MCQ

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

Solution

1. The limits contain no x .

Correct option: B

Final answer

a constant

Worked Solutions

Integration: Definite Integrals and Applications

Q045 MCQ

The best first step for an integral equation is to let:**Solution**

1. Replace the repeated definite integral by one symbol.

Correct option: B**Final answer** $A = \text{integral}$

Q046 MCQ

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:**Solution**

1. Solve $A = 1/3 + A$ after substitution.

Correct option: B**Final answer** $-1/3$

Worked Solutions

Integration: Definite Integrals and Applications

Q047 MCQ

$\frac{d}{dx} \int_a^x f(t)dt$ equals:

Solution

1. Apply the variable-upper-limit theorem.

Correct option: B

Final answer $f x$

Q048 MCQ

When differentiating an integral, the dummy variable t is replaced by:

Solution

1. The derivative evaluates the integrand at the moving limit.

Correct option: B

Final answer

upper-limit variable

Worked Solutions

Integration: Definite Integrals and Applications

Q049 MCQ

If $\int_a^a f(t)dt = 0$, this can determine:

Solution

1. Substitute the equal limits into the original equation.

Correct option: B

Final answer

the unknown lower limit a

Q050 MCQ

For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:

Solution

1. Use the Fundamental Theorem directly.

Correct option: A

Final answer

$2x^2 + 3x - 5$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz WRITTEN 1 Concept Quiz · Written

Find $\int (4x^3 - 6x + 2)dx$.

Solution

1. Apply the power rule term by term.

Final answer

$$x^4 - 3x^2 + 2x + C$$

Quiz WRITTEN 2 Concept Quiz · Written

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

Solution

1. Integrate and use $F(2) = 1$.

Final answer

$$F(x) = x^2 - 5x + 7$$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz WRITTEN 3 Concept Quiz · Written

Evaluate $\int_{-1}^2 (3x^2 - 4x + 1)dx$.

Solution

1. Use an antiderivative and subtract the endpoint values.

Final answer

6

Quiz WRITTEN 4 Concept Quiz · Written

Find $f(x)$ if $f(x) = x^2 + \int_0^1 f(t)dt$.

Solution

1. Let $A = \int_0^1 f(t)dt$; solve $A = 1/3 + A$ after substitution.

Final answer

$f(x) = x^2 - 1/3$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz MCQ 5 [Concept Quiz · MCQ](#)

The derivative of $\int_0^x (3t^2 + 1)dt$ is:

Solution

1. Evaluate the integrand at $t = x$.

Correct option: A

Final answer

$$3x^2 + 1$$

Quiz MCQ 6 [Concept Quiz · MCQ](#)

A primitive of $5x^4$ is:

Solution

1. Divide the power coefficient by the new exponent.

Correct option: A

Final answer

$$x^5 + C$$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz MCQ 7 Concept Quiz · MCQ

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

Solution

1. Integrate and apply $F(0) = 3$.

Correct option: A

Final answer

$$2x^2 + 2x + 3$$

Quiz MCQ 8 Concept Quiz · MCQ

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ equals:

Solution

1. Join adjacent intervals with the same integrand.

Correct option: A

Final answer

integral from 1 to 5

Answer key

Use the question number shown on each page.

- Q001 Integrate term by term and add C to each primitive. Q012 Expand products, integrate, and evaluate the antiderivative at the upper and lower limits. Q020 Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation. Q036 B
- Q002 Expand products or powers, collect like powers, then integrate every term. Q013 Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0. Q021 Let the definite integral be A , substitute the expression for $f(t)$, solve for A , then recover $f(x)$. Q037 C
- Q003 Apply linearity and the power rule; retain the constant of integration. Q014 Use linearity, expand products, then apply the endpoint-substitution rule. Q022 $2x^2 + 3x - 5$. Q038 B
- Q004 Expand each product or square, simplify, and integrate term by term. Q015 Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate. Q023 Replace t by x when differentiating, then use equal limits or the given condition to determine a . Q039 B
- Q005 $F(x) = x^3 + 4x^2 - x + 3$. Q016 Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity. Q024 Apply the variable-limit Fundamental Theorem directly to each integrand. Q040 B
- Q006 Integrate the derivative and use the given point to determine C . Q017 Rewrite each expression until adjacent intervals have matching integrands and compatible limits. Q025 Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a . Q041 B
- Q007 For each item, integrate $F'(x)$, then impose the stated value of F . Q018 Scale to $\int_a^b (x-a)(x-b) dx = -\frac{(b-a)^3}{6}$. Q026 B
- Q008 $f(x) = x^3 - x^2 + x - 19$. Q019 Let $A = \int_0^1 f(t) dt$; then $A = \frac{1}{3} + A$, so $A = -\frac{1}{3}$ and $f(x) = x^2 - \frac{1}{3}$. Q027 B
- Q009 Integrate the given gradient and use the point on the curve to determine the constant. Q028 B
- Q010 Integrate the polynomial gradient and apply the given point condition in each case. Q029 A
- Q011 Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero. Q030 B
- Q031 C
- Q032 A
- Q033 B
- Q034 B
- Q035 B
- Q042 B
- Q043 B
- Q044 B
- Q045 B
- Q046 B
- Q047 B
- Q048 B
- Q049 B
- Q050 A
- Quiz WRITTEN 1 $x^4 - 3x^2 + 2x + C$
- Quiz WRITTEN 2 $F(x) = x^2 - 5x + 7$
- Quiz WRITTEN 3 6
- Quiz WRITTEN 4 $f(x) = x^2 - 1/3$
- Quiz MCQ 5 A
- Quiz MCQ 6 A
- Quiz MCQ 7 A
- Quiz MCQ 8 A

Concept 5 | Applications of Integration to Geometry

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Formula opener

Area

$$A = \int_a^b (y_{top} - y_{bottom}) dx$$

Volume

$$V = \pi \int_a^b y^2 dx$$

Bounds

$$a \leq x \leq b$$

Worked Solutions

Applications of Integration to Geometry

Q001 Written

Find the area bounded by $y = 2x^2 + 1$, the x -axis, $x = -2$, and $x = 1$.

Solution

1. The parabola is above the x -axis on the interval.
2. $S = \int_{-2}^1 (2x^2 + 1) dx = 9$.

Final answer

9

Q002 Written

Find the area bounded by $y = x^2 + x - 6$, the x -axis, $x = -2$, and $x = 1$.

Solution

1. The curve is below the x -axis on $[-2, 1]$.
2. Use $S = -\int_{-2}^1 (x^2 + x - 6) dx = \frac{33}{2}$.

Final answer

 $\frac{33}{2}$

Worked Solutions

Applications of Integration to Geometry

Q003 Written

Find the area bounded by $y = -x^2 + 4$ and the x -axis.**Solution**

1. The intercepts are $x = -2, 2$.
2. Integrate the positive curve from -2 to 2 .

Final answer

$$\frac{32}{3}$$

Q004 Written

Find the area under $y = -x^2 - 2x + 3$ and above the x -axis.**Solution**

1. The roots are -3 and 1 .
2. Integrate $-x^2 - 2x + 3$ between the roots.

Final answer

$$\frac{32}{3}$$

Worked Solutions

Applications of Integration to Geometry

Q005 Written

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.**Solution**

1. Intersections give $x = -3, 1$; the line is upper.
2. Integrate upper minus lower over $[-3, 1]$.

Final answer

$$\frac{32}{3}$$

Q006 Written

Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.**Solution**

1. The intersections are $x = -2, 2$.
2. Use the upper-minus-lower integral $\int_{-2}^2 (8 - 2x^2) dx$.

Final answer

$$\frac{64}{3}$$

Worked Solutions

Applications of Integration to Geometry

Q007 Written

Find the area between $y = -x^2 + 4x$ and $y = x$.**Solution**

1. Intersections are $x = 0, 3$.
2. Integrate $-x^2 + 3x$ from 0 to 3.

Final answer

$$\frac{9}{2}$$

Q008 Written

Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \leq x \leq 3$.**Solution**

1. The curves meet at $x = 2$ within the interval.
2. Split the absolute difference at $x = 2$; the two areas are $80/3$ and $10/3$.

Final answer

$$30$$

Worked Solutions

Applications of Integration to Geometry

Q009 Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.**Solution**

1. Roots are (0,2,3); the curve changes sign at 2.
2. Add the positive area on $[0, 2]$ to the absolute area on $[2, 3]$.

Final answer

$$\frac{37}{12}$$

Q010 Written

Find the area bounded by $y = -x^3 + 4x$ and the x -axis.**Solution**

1. Roots are (-2,0,2); the two lobes have equal area.
2. Double $\int_0^2 (-x^3 + 4x) dx = 4$.

Final answer

$$8$$

Worked Solutions

Applications of Integration to Geometry

Q011 Written

Solve the given cubic area exercises by finding every x -intercept and adding the absolute signed pieces.

Solution

1. Factor the cubic first.
2. Mark the sign on each interval before integrating.

Final answer

Use the numbered solution for each given cubic.

Q012 Written

Evaluate $\int_{-2}^1 |x + 1| dx$.

Solution

1. The sign changes at $x = -1$.
2. Split into two ordinary integrals and add the positive areas.

Final answer

$$\frac{5}{2}$$

Worked Solutions

Applications of Integration to Geometry

Q013 Written

Evaluate $\int_0^3 |x^2 - 4| dx$.**Solution**

1. The root in the interval is $x = 2$.
2. Use $-\int_0^2 (x^2 - 4) dx + \int_2^3 (x^2 - 4) dx$.

Final answer

$$\frac{23}{3}$$

Q014 Written

Evaluate $\int_0^2 |x^2 + 3x - 4| dx$.**Solution**

1. The relevant root is $x = 1$.
2. Reverse the sign on $[0, 1]$, then add the integral on $[1, 2]$.

Final answer

$$5$$

Worked Solutions

Applications of Integration to Geometry

Q015 Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.**Solution**

1. The tangents are $y = -x$ and $y = 3x - 4$, meeting at $x = 1$.
2. Split the area at $x = 1$ and integrate curve minus tangent.

Final answer

$$\frac{3}{2}$$

Q016 Written

Find the area bounded by $y = x^3 + x^2 - 3x + 6$ and its tangent at $(1, 5)$.**Solution**

1. The tangent is $y = 2x + 3$; the intersections are $x = -3, 1$.
2. Factor the difference as $(x - 1)^2(x + 3)$ and integrate.

Final answer

$$\frac{64}{3}$$

Worked Solutions

Applications of Integration to Geometry

Q017 Written

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

Solution

1. The tangent equations are $y = 4x - 4$ and $y = 8x - 16$.
2. Integrate the two squared gaps on $[2, 3]$ and $[3, 4]$.

Final answer

$$\frac{2}{3}$$

Q018 Written

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x -axis, find k .

Solution

1. The total area under the parabola is 3.
2. Set the area cut off by $y = kx$ equal to $3/2$, giving $(2 - k)^3 = 3/4$.

Final answer

$$k = 2 - \sqrt[3]{\frac{3}{4}}$$

Worked Solutions

Applications of Integration to Geometry

Q019 **Written**

Solve the **given** bisection exercises for the given parabolas and lines, using the required area ratio.

Solution

1. Find the total parabola area first.
2. Integrate between the line and curve, then impose the stated ratio.

Final answer

Use the numbered solution for each **given** ratio.

Q020 **MCQ**

When the graph is below the x-axis, area is:

Solution

1. Reverse the sign.

Correct option: B

Final answer

negative of the integral

Worked Solutions

Applications of Integration to Geometry

Q021 MCQ

A bounded area under a curve is found by first checking:**Solution**

1. Draw the graph.

Correct option: A**Final answer**

the graph and sign

Q022 MCQ

The area under $y = -x^2 + 4$ between its roots is:**Solution**

1. Integrate from -2 to 2.

Correct option: C**Final answer**

32/3

Worked Solutions

Applications of Integration to Geometry

Q023 MCQ

Between two curves, the integrand is:**Solution**

1. Identify the upper graph.

Correct option: B

Final answer

upper minus lower

Q024 MCQ

If curves cross inside the interval, area must be:**Solution**

1. The sign of the difference changes.

Correct option: A

Final answer

split at the crossing

Worked Solutions

Applications of Integration to Geometry

Q025 MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Solution

1. Use the intersection limits.

Correct option: B

Final answer

32/3

Q026 MCQ

For cubic area, the sign changes at:

Solution

1. Mark all roots.

Correct option: B

Final answer

each x-intercept

Worked Solutions

Applications of Integration to Geometry

Q027 MCQ

The area for $y = -x^3 + 4x$ is:**Solution**

1. The two lobes are equal.

Correct option: C

Final answer

8

Q028 MCQ

A negative lobe contributes:**Solution**

1. Area is non-negative.

Correct option: B

Final answer

positive absolute area

Worked Solutions

Applications of Integration to Geometry

Q029 MCQ

For an absolute-value integral, first locate:**Solution**

1. Roots set the sign intervals.

Correct option: A**Final answer**

the roots

Q030 MCQ

The value of $\int_0^2 |x^2 + 3x - 4| dx$ is:**Solution**

1. Split at $x = 1$.

Correct option: C**Final answer**

5

Worked Solutions

Applications of Integration to Geometry

Q031 MCQ

If f changes sign at c , split the integral:**Solution**

1. The absolute value changes form.

Correct option: A

Final answer atc

Q032 MCQ

A tangent line at $x=a$ has slope:**Solution**

1. Differentiate the curve.

Correct option: B

Final answer $f'a$

Worked Solutions

Applications of Integration to Geometry

Q033 MCQ

For two tangent lines, split at their:**Solution**

1. The upper tangent changes there.

Correct option: A**Final answer**

intersection x-coordinate

Q034 MCQ

The area in the given parabola-tangent Warm Up is:**Solution**

1. Integrate the two gaps.

Correct option: C**Final answer** $\frac{3}{2}$

Worked Solutions

Applications of Integration to Geometry

Q035 MCQ

Bisection means each part has:**Solution**

1. Set each part to half the total.

Correct option: A**Final answer**

the same area

Q036 MCQ

For $y=2x-x^2$, the total area above the x-axis is:**Solution**

1. Integrate from 0 to 2.

Correct option: C**Final answer**

3

Worked Solutions

Applications of Integration to Geometry

Q037 MCQ

The line $y=kx$ must satisfy the stated:**Solution**

1. Use the given bisection condition.

Correct option: A**Final answer**

area ratio

Q038 MCQ

In a bisection problem, the cut-off area is:**Solution**

1. Set the two parts equal.

Correct option: B**Final answer**

half the total area

Worked Solutions

Applications of Integration to Geometry

Quiz MCQ 1 [Concept Quiz · MCQ](#)

The area under $y = -x^2 + 4$ and above the x -axis is:

Solution

1. Integrate from -2 to 2 .

Correct option: C

Final answer

$$\frac{32}{3}$$

Quiz MCQ 2 [Concept Quiz · MCQ](#)

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Solution

1. Use upper minus lower from -3 to 1 .

Correct option: B

Final answer

$$\frac{32}{3}$$

Worked Solutions

Applications of Integration to Geometry

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Evaluate $\int_{-2}^1 |x + 1| dx$:

Solution

1. Split at $x = -1$.

Correct option: B

Final answer

$$\frac{5}{2}$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

The area enclosed by $y = x^2$ and tangents through $(3, 8)$ is:

Solution

1. Integrate the two squared gaps.

Correct option: C

Final answer

$$\frac{2}{3}$$

Worked Solutions

Applications of Integration to Geometry

Quiz WRITTEN 5 Concept Quiz · Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x -axis.

Solution

1. Use roots (0,2,3) and split at 2.

Final answer

$$\frac{37}{12}$$

Quiz WRITTEN 6 Concept Quiz · Written

Evaluate $\int_0^2 |x^2 + 3x - 4| dx$.

Solution

1. Split at the root $x = 1$.

Final answer

$$5$$

Worked Solutions

Applications of Integration to Geometry

Quiz WRITTEN 7 Concept Quiz · Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Solution

1. Find both tangents and split at their intersection.

Final answer

$$\frac{3}{2}$$

Quiz WRITTEN 8 Concept Quiz · Written

If $y = kx$ bisects the area under $y = 2x - x^2$, find k .

Solution

1. Set the cut-off area equal to half of 3.

Final answer

$$2 - \sqrt[3]{\frac{3}{4}}$$

Answer key

Use the question number shown on each page.

Q001 9

Q002 $\frac{33}{2}$ Q003 $\frac{32}{3}$ Q004 $\frac{32}{3}$ Q005 $\frac{32}{3}$ Q006 $\frac{64}{3}$ Q007 $\frac{9}{2}$

Q008 30

Q009 $\frac{37}{12}$

Q010 8

Q011 Use the numbered solution for each given cubic.

Q012 $\frac{5}{2}$ Q013 $\frac{23}{3}$

Q014 5

Q015 $\frac{3}{2}$ Q016 $\frac{64}{3}$ Q017 $\frac{2}{3}$ Q018 $k = 2 - \sqrt[3]{\frac{3}{4}}$

Q019 Use the numbered solution for each given ratio.

Q020 B

Q021 A

Q022 C

Q023 B

Q024 A

Q025 B

Q026 B

Q027 C

Q028 B

Q029 A

Q030 C

Q031 A

Q032 B

Q033 A

Q034 C

Q035 A

Q036 C

Q037 A

Q038 B

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $\frac{37}{12}$

Quiz WRITTEN 6 5

Quiz WRITTEN 7 $\frac{3}{2}$ Quiz WRITTEN 8 $2 - \sqrt[3]{\frac{3}{4}}$

Concept 1 | Fundamentals of Random Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Expectation

$$E(X) = \sum x p(x)$$

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

Worked Solutions

Fundamentals of Random Variables

Q001 Written

For the sum of two dice, construct the probability distribution and find $P5 \leq X \leq 8$.

Solution

1. List the 36 equally likely pairs, count each sum, and divide each frequency by 36.

Final answer

Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; $P5 \leq X \leq 8 = 20/36 = 5/9$.

Q002 Written

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

Solution

1. Check that all table probabilities add to 1 before evaluating an interval event.

Final answer

Use the sample space or combinations to list every possible X and sum the relevant probabilities.

Worked Solutions

Fundamentals of Random Variables

Q003 Written

Construct both given probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Solution

1. A probability distribution is a table of every possible value of X and its probability.

Final answer

Use binomial or hypergeometric counts, then add probabilities for the requested values.

Q004 Written

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

Solution

1. Build the table, multiply each value by its probability, and add the products.

Final answer

For four fair coins $EX=2$; for the hypergeometric draw $EX=4/3$.

Worked Solutions

Fundamentals of Random Variables

Q005 Written

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Solution

1. Keep the random variable definition unchanged while forming the table.

Final answer

Use $EX = \sum xPX = x$ after constructing each distribution.

Q006 Written

Find EX for all six given random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).

Solution

1. The expected number of successes in repeated identical trials can be checked with n times p .

Final answer

Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.

Worked Solutions

Fundamentals of Random Variables

Q007 Written

For tails in four fair coins, find $E(3X+2)$, $E(-4X+1)$, and EX^2 .

Solution

1. Constants shift the mean; a multiplier scales the mean by the same factor.

Final answer

Use $E(aX + b) = aEX + b$; calculate EX^2 directly from the table.

Q008 Written

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

Solution

1. Do not replace EX^2 by EX^2 ; they are different quantities.

Final answer

Apply linearity for $aX+b$ and use the weighted square sum for X^2 .

Worked Solutions

Fundamentals of Random Variables

Q009 Written

Solve both given Exercise groups of transformed expectations for coins and for balls.**Solution**

1. Write the transformation before substituting the known mean.

Final answer

For every expression use $E(aX + b) = aEX + b$; evaluate powers from the distribution table.

Q010 Written

For black balls drawn from the stated box, find EX and VX .**Solution**

1. The distribution is $PX = 0 = 2/7$, $PX = 1 = 4/7$, $PX = 2 = 1/7$. Hence $EX = 6/7$ and $VX = EX^2 - (EX)^2 = 20/49$.

Final answer

$EX = 6/7$; $VX = 20/49$.

Worked Solutions**Fundamentals of Random Variables****Q011** Written**Find the mean and variance for the two given Try random variables (a numbered card and winning tickets).****Solution**

1. Variance measures spread around the mean, not the mean itself.

Final answerConstruct each probability table, then use $VX = EX^2 - EX^2$.**Q012** Written**Find EX and VX for all five given Exercise random variables.****Solution**

1. For sampling without replacement, use combinations to obtain the exact probabilities.

Final answer

Use the distribution table for each experiment and the variance identity.

Worked Solutions

Fundamentals of Random Variables

Q013 Written

Find the standard deviation of tails when two biased coins have $P(\text{head})=4/7$ and $P(\text{tail})=3/7$.

Solution

1. Find VX first, then take the positive square root.

Final answer

The table gives $VX=24/49$, so $\sigma X=2 \sqrt{6}/7$.

Q014 Written

Find the standard deviation from each given probability table and for white balls drawn from 4 red and 3 white.

Solution

1. Standard deviation has the same units as X and is always non-negative.

Final answer

Use $\sigma X=\sqrt{V(X)}$ after evaluating the variance from the table.

Worked Solutions

Fundamentals of Random Variables

Q015 Written

Find the standard deviation for both given probability-table exercises.

Solution

1. Keep the positive square root only.

Final answer

Compute EX , EX^2 , then $\sigma_X = \sqrt{E(X^2) - E(X)^2}$.

Q016 Written

For the number of winning tickets in the given draw, find VY and σ_Y for $Y=2X+1$ and $Y=-3X-2$.

Solution

1. The constant b changes neither variance nor standard deviation.

Final answer

Use $V(aX + b) = a^2 VX$ and $\sigma(aX+b)=|a|\sigma_X$.

Worked Solutions

Fundamentals of Random Variables

Q017 Written

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X .

Solution

1. Write both transformation rules before substituting numbers.

Final answer

Apply the squared multiplier to VX and the absolute multiplier to σX .

Q018 Written

For a single die, find VY and σY for $Y=X+2$, $Y=3X+1$ and $Y=-2X+4$.

Solution

1. A translation changes the mean but leaves spread unchanged.

Final answer

Use the variance and standard-deviation transformation formulas for each Y .

Worked Solutions

Fundamentals of Random Variables

Q019 Written

Solve the two given transformation problems: a net prize after an entry fee, and finding a, b from $E(aX+b)=7/3$ and $\sigma(aX+b)=3$.

Solution

1. The mean equation determines b after the standard-deviation equation fixes a .

Final answer

Use $E(aX + b) = aEX + b$ and $\sigma(aX+b)=|a|\sigma X$, with $a>0$.

Q020 Written

Solve the two given Try problems involving net winnings and determining a, b for a die transformation.

Solution

1. Entry fees are negative constants; they shift the mean only.

Final answer

Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.

Worked Solutions

Fundamentals of Random Variables

Q021 **Written**

Solve all four given transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Solution

1. Translate the words into $aX+b$ before calculating.

Final answer

Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.

Q022 **MCQ**

A probability distribution must have probabilities that sum to:

Solution

1. Check the table total.

Correct option: B

Final answer

1

Worked Solutions

Fundamentals of Random Variables

Q023 MCQ

For two fair dice, $PX = 2$ is:**Solution**

1. Only the pair (1,1) gives sum 2.

Correct option: B

Final answer

1/36

Q024 MCQ

The number of heads in three coins can be:**Solution**

1. Count heads in the sample space.

Correct option: B

Final answer

0, 1, 2, 3

Worked Solutions

Fundamentals of Random Variables

Q025 MCQ

The expected value is calculated as:**Solution**

1. Multiply each value by its probability.

Correct option: B**Final answer**

$$\sum xPX = x$$

Q026 MCQ

The mean number of tails in four fair coins is:**Solution**

1. Use symmetry or the table.

Correct option: B**Final answer**

2

Worked Solutions

Fundamentals of Random Variables

Q027 MCQ

For a fair die, EX is:**Solution**

1. The values 1 to 6 average to 3.5.

Correct option: B

Final answer

3.5

Q028 MCQ

 $E(aX+b)$ equals:**Solution**

1. Use linearity of expectation.

Correct option: B

Final answer $aEX + b$

Worked Solutions

Fundamentals of Random Variables

Q029 MCQ

In general, EX^2 is:**Solution**

1. Square values before weighting.

Correct option: C**Final answer**

weighted square sum

Q030 MCQ

If $EX=2$, then $E(3X+2)$ is:**Solution**

1. Calculate $3 \times 2 + 2$.

Correct option: C**Final answer**

8

Worked Solutions

Fundamentals of Random Variables

Q031 MCQ

The variance identity is:**Solution**

1. Subtract the squared mean.

Correct option: B

Final answer

$$EX^2 - EX^2$$

Q032 MCQ

Variance can never be:**Solution**

1. It is an expected square.

Correct option: C

Final answer

negative

Worked Solutions

Fundamentals of Random Variables

Q033 MCQ

The first step in a variance problem is to:**Solution**

1. The table supplies EX and EX^2 .

Correct option: A**Final answer**

draw the distribution

Q034 MCQ

Standard deviation is:**Solution**

1. Take the positive square root.

Correct option: B**Final answer** $\sqrt{V(X)}$

Worked Solutions

Fundamentals of Random Variables

Q035 MCQ

Standard deviation has units:**Solution**

1. The square root restores the units.

Correct option: B

Final answer

same units as X

Q036 MCQ

 $V(aX+b)$ equals:**Solution**

1. Constants do not affect spread.

Correct option: B

Final answer

a squared VX

Worked Solutions

Fundamentals of Random Variables

Q037 MCQ

 $\sigma(aX+b)$ equals:**Solution**

1. Standard deviation is non-negative.

Correct option: B**Final answer** $|a| \sigma X$

Q038 MCQ

For $Y=X+3$, the variance is:**Solution**

1. A translation leaves variance unchanged.

Correct option: B**Final answer** VX

Worked Solutions

Fundamentals of Random Variables

Q039 MCQ

A net prize after a fixed fee is modeled as:**Solution**1. The fee is the constant b .

Correct option: B

Final answer

$$aX + b$$

Q040 MCQ

If $a > 0$ and $\sigma(aX + b) = 3$, then a is:**Solution**1. Solve $a \sigma X = 3$.

Correct option: B

Final answer

$$3/\sigma X$$

Worked Solutions

Fundamentals of Random Variables

Q041 MCQ

Changing b in $aX+b$ changes:**Solution**

1. b is a translation.

Correct option: C**Final answer**

the mean only

Q042 MCQ

If $EX=2$ and $EX^2=7$, then VX is:**Solution**

1. Calculate $7 - 2^2$.

Correct option: A**Final answer**

3

Worked Solutions

Fundamentals of Random Variables

Quiz WRITTEN 1 Concept Quiz · Written

Find EX and VX for the number of heads in three fair coins.

Solution

1. Use the binomial distribution.

Final answer

$EX=3/2$ and $VX=3/4$

Quiz WRITTEN 2 Concept Quiz · Written

For $Y=4X-1$ with $VX=2$, find VY and σY .

Solution

1. Use $V(aX + b) = a^2 VX$.

Final answer

$VY=32$ and $\sigma Y=4 \sqrt{2}$

Worked Solutions

Fundamentals of Random Variables

Quiz WRITTEN 3 Concept Quiz · Written

For a fair die, find $E(2X+5)$.

Solution

1. $EX=7/2$, then use linearity.

Final answer

12

Quiz WRITTEN 4 Concept Quiz · Written

If $\sigma X=2$ and $EX=3$, find $\sigma(-5X+7)$.

Solution

1. Multiply standard deviation by $|a|$.

Final answer

10

Worked Solutions

Fundamentals of Random Variables

Quiz MCQ 5 [Concept Quiz · MCQ](#)

If $VX=5$, $V(3X+1)$ is:

Solution

1. Square the multiplier.

Correct option: D

Final answer

45

Quiz MCQ 6 [Concept Quiz · MCQ](#)

For a fair die EX is:

Solution

1. Average 1 through 6.

Correct option: C

Final answer

3.5

Worked Solutions

Fundamentals of Random Variables

Quiz MCQ 7 [Concept Quiz · MCQ](#)

A probability table total must be:

Solution

1. Add all probabilities.

Correct option: B

Final answer

1

Quiz MCQ 8 [Concept Quiz · MCQ](#)

If $\sigma X=4$, $\sigma(X-9)$ is:

Solution

1. Translations do not change spread.

Correct option: B

Final answer

4

Answer key

Use the question number shown on each page.

- | | | | |
|---|---|---|--|
| Q001 Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; $P5 \leq X \leq 8 = 20/36 = 5/9$. | Q009 For every expression use $E(aX + b) = aEX + b$; evaluate powers from the distribution table. | Q019 Use $E(aX + b) = aEX + b$ and $\sigma(aX + b) = a \sigma X$, with $a > 0$. | Q034 B |
| Q002 Use the sample space or combination to list every possible X and sum the relevant probabilities. | Q010 $EX = 6/7$; $VX = 20/49$. | Q020 Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules. | Q035 B |
| Q003 Use binomial or hypergeometric counts, then add probabilities for the requested values. | Q011 Construct each probability table, then use $VX = EX^2 - EX^2$. | Q021 Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule. | Q036 B |
| Q004 For four fair coins $EX = 2$; for the hypergeometric draw $EX = 4/3$. | Q012 Use the distribution table for each experiment and the variance identity. | Q022 B | Q037 B |
| Q005 Use $EX = \sum xPX = x$ after constructing each distribution. | Q013 The table gives $VX = 24/49$, so $\sigma X = 2\sqrt{6/7}$. | Q023 B | Q038 B |
| Q006 Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice. | Q014 Use $\sigma X = \sqrt{V(X)}$ after evaluating the variance from the table. | Q024 B | Q039 B |
| Q007 Use $E(aX + b) = aEX + b$; calculate EX^2 directly from the table. | Q015 Compute EX , EX^2 , then $\sigma X = \sqrt{E(X^2) - E(X)^2}$. | Q025 B | Q040 B |
| Q008 Apply linearity for $aX + b$ and use the weighted square sum for X^2 . | Q016 Use $V(aX + b) = a^2 VX$ and $\sigma(aX + b) = a \sigma X$. | Q026 B | Q041 C |
| | Q017 Apply the squared multiplier to VX and the absolute multiplier to σX . | Q027 B | Q042 A |
| | Q018 Use the variance and standard-deviation transformation formulas for each Y. | Q028 B | Quiz WRITTEN 1 $EX = 3/2$ and $VX = 3/4$ |
| | | Q029 C | Quiz WRITTEN 2 $VY = 32$ and $\sigma Y = 4\sqrt{2}$ |
| | | Q030 C | Quiz WRITTEN 3 12 |
| | | Q031 B | Quiz WRITTEN 4 10 |
| | | Q032 C | Quiz MCQ 5 D |
| | | Q033 A | Quiz MCQ 6 C |
| | | | Quiz MCQ 7 B |
| | | | Quiz MCQ 8 B |

Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Independent

$$\text{Var}(X + Y) = \text{Var } X + \text{Var } Y$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q001 Written

Given the probability distributions of X and Y in the given tables, find $E(X+Y)$.

Solution

1. Find EX and EY separately, then add them.

Final answer

$$EX = 35/3, EY = 9/2, \text{ so } E(X + Y) = 97/6.$$

Q002 Written

Three coins are tossed once; let X , Y and Z be the head indicators. Find $E(X+Y+Z)$.

Solution

1. Each fair head indicator has expected value $1/2$.

Final answer

$$E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q003 Written

For the two given tables with $X=1,2,3$ and $Y=4,5,6$, find $E(X+Y)$.

Solution

1. The expectation of a sum is the sum of the expectations.

Final answer

$$E(X + Y) = EX + EY = 2 + 5 = 7.$$

Q004 Written

For the two bags in the given exercise, find the expected total number of red balls drawn.

Solution

1. Use $E(\text{red drawn}) = \text{sample size} \times \text{the red proportion in each bag}$.

Final answer

Bag A: $6/7$; Bag B: $6/7$; total $E = 12/7$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q005 Written

Three independent subsystems each receive two signals with success probability 0.5. Find $E(X+Y+Z)$.

Solution

1. Add the three binomial means.

Final answer

Each mean is $2(0.5) = 1$, so $E(X + Y + Z) = 3$.

Q006 Written

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z .

Solution

1. Model the score as $aX+bY$, then apply linearity.

Final answer

$Z = 500X + 100Y$, so $EZ = 500 \cdot 1 + 100 \cdot 3/2 = 650$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q007 **Written****Two 500-point coins and two 100-point coins are tossed. Find EZ .****Solution**

1. The expected heads in each two-coin group is 1.

Final answer

$$EZ = 500(1) + 100(1) = 600.$$

Q008 **Written****A die is thrown twice and the score is $Z=3X+5Y$. Find EZ .****Solution**

1. Both die rolls have mean $7/2$.

Final answer

$$EZ = 3(7/2) + 5(7/2) = 28.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q009 Written

For three 100-point coins and one 50-point coin, find the expected score from heads.

Solution

1. Use the expected score of each fair coin and add.

Final answer

$$EZ = 350 + 25 = 175.$$

Q010 Written

In two independent raffles, buy two tickets from a 2/10 winner raffle and two from a 1/6 winner raffle. Find the expected prizes.

Solution

1. Expected winners are sample size times the winning proportion in each raffle.

Final answer

$$EZ = 22/10 + 21/6 = 11/15.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q011 Written

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find $E(XY)$.

Solution

1. Check independence before multiplying expectations.

Final answer

X and Y are independent, so $E(XY) = EXEY = 3/2 \times 7/2 = 21/4$.

Q012 Written

For independent X with probabilities $2/5, 2/5, 1/5$ at 1, 2, 3 and Y uniform on 2, 4, 6, find $E(XY)$.

Solution

1. Compute each mean from its table, then multiply.

Final answer

$EX = 8/5, EY = 4$, so $E(XY) = 32/5$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q013 Written

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find $E(XY)$.

Solution

1. The two experiments are independent.

Final answer

$$E(XY) = EXEY = 3/2 \cdot 3/2 = 9/4.$$

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find $E(XY)$.

Solution

1. Replacement makes the two draws independent.

Final answer

$$EX = EY = 5/2, \text{ so } E(XY) = 25/4.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q015 Written

For independent X and Y with $EX=1$ and $EY=2/7$, find $E(XY)$.

Solution

1. Use the product rule after checking independence.

Final answer

$$E(XY) = EXEY = 2/7.$$

Q016 Written

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. For independent variables, add variances, then take the positive square root.

Final answer

$$VX=2/3, VY=3/4, \text{ so } V(X + Y) = 17/12 \text{ and } \sigma=\sqrt{17/12}.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q017 **Written**

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Solution

1. Each fair die has variance $35/12$.

Final answer

$$V(X + Y) = 35/12 + 35/12 = 35/6; \sigma = \sqrt{35/6}.$$

Q018 **Written**

An independent coin head count X and die score Y are combined. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. Use $V(\text{Bernoulli}(1/2)) = 1/4$ and $V(\text{die}) = 35/12$.

Final answer

$$V = 1/4 + 35/12 = 19/6; \sigma = \sqrt{19/6}.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q019 Written

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Solution

1. Variances of independent Bernoulli counts add.

Final answer

$$V = 2/4 + 3/4 = 5/4.$$

Q020 Written

Independent X is uniform on 1 to 4 and Y on 1 to 6. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. Use the uniform discrete variance for each scale and add.

Final answer

$$V = 5/4 + 35/12 = 25/6; \sigma = 5/\sqrt{6}.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q021 Written

Three independent special dice take values 2,4,6 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.

Solution

1. For a product multiply means; for a sum add variances.

Final answer $EX=4, VX=8/3; E(XYZ) = 64$ and $V(sum) = 8$.

Q022 Written

Three independent dice take values 1,3,5 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.

Solution

1. All three distributions are identical and independent.

Final answer $EX=3, VX=8/3; E(XYZ) = 27$ and $V(sum) = 8$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q023 Written

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Solution

1. Use independence for the product and variance addition for the sum.

Final answer

$$E(\text{product}) = 7/2^3 = 343/8; V(\text{sum}) = 335/12 = 35/4.$$

Q024 Written

Three friends have 5, 4 and 3 independent rolls to score a six. Find $E(XYZ)$ and $V(X+Y+Z)$.

Solution

1. For Binomial($n, 1/6$), use $E = np$ and $V = np(1 - p)$.

Final answer

$$\text{Means are } 5/6, 2/3, 1/2; E(\text{product}) = 5/18 \text{ and } V(\text{sum}) = 5/3.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q025 **Written**

If X, Y and Z are independent head counts from 4, 3 and 2 fair coins, find $E(XYZ)$.

Solution

1. Multiply the three expected head counts.

Final answer

$$E(XYZ) = 23/21=3.$$

Q026 **MCQ**

$E(X+Y)$ equals:

Solution

1. Add the two expectations.

Correct option: B

Final answer

$$EX + EY$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q027 MCQ

For three variables, $E(X+Y+Z)$ equals:**Solution**

1. Extend linearity to three terms.

Correct option: B

Final answer

$$EX + EY + EZ$$

Q028 MCQ

The expected heads in one fair coin toss is:**Solution**

1. Use the Bernoulli mean.

Correct option: B

Final answer

$$1/2$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q029 MCQ

For two signals with $p=0.5$, EX is:**Solution**1. Use np .**Correct option:** B**Final answer**

1

Q030 MCQ

If $EX=2$ and $EY=5$, $E(X+Y)$ is:**Solution**

1. Add the means.

Correct option: B**Final answer**

7

Worked Solutions

Expected Value and Variance of Multiple Variables

Q031 MCQ

 $E(aX+bY)$ equals:**Solution**

1. Apply linearity.

Correct option: A

Final answer

$$aEX + bEY$$

Q032 MCQ

The expected score of one fair 100-point coin is:**Solution**

1. Half the value is expected.

Correct option: B

Final answer

50

Worked Solutions

Expected Value and Variance of Multiple Variables

Q033 MCQ

For a fair die, $E(3X+5Y)$ with two rolls is:**Solution**

1. Use both means $7/2$.

Correct option: C**Final answer**

28

Q034 MCQ

A fixed entry fee changes:**Solution**

1. It is a constant term.

Correct option: B**Final answer**

the mean only

Worked Solutions

Expected Value and Variance of Multiple Variables

Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:

Solution

1. Use sample size times winner proportion.

Correct option: B

Final answer

2/5

Q036 MCQ

$E(XY) = E(X)E(Y)$ requires:

Solution

1. The product rule needs mutual independence.

Correct option: B

Final answer

independence

Worked Solutions

Expected Value and Variance of Multiple Variables

Q037 MCQ

For three fair coins, EX is:**Solution**

1. Use np.

Correct option: C**Final answer** $3/2$

Q038 MCQ

For a fair die, EY is:**Solution**

1. Average 1 through 6.

Correct option: C**Final answer**

3.5

Worked Solutions

Expected Value and Variance of Multiple Variables

Q039 MCQ

With replacement, two card draws are:**Solution**

1. Replacement restores the same distribution and independence.

Correct option: B

Final answer

independent

Q040 MCQ

If $EX=2$ and $EY=3$ for independent variables, $E(XY)$ is:**Solution**

1. Multiply the means.

Correct option: B

Final answer

6

Worked Solutions

Expected Value and Variance of Multiple Variables

Q041 MCQ

For independent X, Y , $V(X+Y)$ equals:**Solution**

1. Add independent variances.

Correct option: B**Final answer**

$$VX + VY$$

Q042 MCQ

The variance of a fair die is:**Solution**

1. Use the standard die variance.

Correct option: B**Final answer**

$$35/12$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q043 MCQ

Sigma(X+Y) is:**Solution**

1. Take the positive root.

Correct option: B

Final answer $\sqrt{V(X + Y)}$

Q044 MCQ

The variance of the sum of two independent fair dice is:**Solution**

1. Double 35/12.

Correct option: B

Final answer $35/6$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q045 MCQ

For independent sums, a constant shift changes:**Solution**

1. Translations do not change spread.

Correct option: C**Final answer**

neither spread measure

Q046 MCQ

For independent X, Y, Z , $E(XYZ)$ equals:**Solution**

1. Multiply the three means.

Correct option: B**Final answer** $E(X)E(Y)E(Z)$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q047 MCQ

For independent X, Y, Z , $V(X+Y+Z)$ equals:**Solution**

1. Add the three variances.

Correct option: B

Final answer

$$VX + VY + VZ$$

Q048 MCQ

For a binomial count with $n=5$ and $p=1/6$, EX is:**Solution**1. Use np .

Correct option: B

Final answer

$$5/6$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q049 MCQ

For three values 2,4,6 equally likely, EX is:**Solution**

1. Average the three values.

Correct option: C**Final answer**

4

Q050 MCQ

For three independent fair dice, $V(\text{sum})$ is:**Solution**

1. Add three copies of $35/12$.

Correct option: C**Final answer** $35/4$

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 1 Concept Quiz · Written

If $EX=4$ and $EY=7$, find $E(3X-2Y)$.

Solution

1. Use linearity: 34-27.

Final answer

-2

Quiz WRITTEN 2 Concept Quiz · Written

For independent variables with $EX=2$ and $EY=5$, find $E(XY)$.

Solution

1. Multiply the means.

Final answer

10

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 3 **Concept Quiz · Written**

Two independent fair dice are thrown. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. Add the two die variances.

Final answer

$$V = 35/6, \sigma = \sqrt{35/6}$$

Quiz WRITTEN 4 **Concept Quiz · Written**

Three independent Binomial counts have $(n,p)=(2,1/2),(3,1/2),(4,1/2)$. Find $V(X+Y+Z)$.

Solution

1. Add $np(1-p)$ for the three counts.

Final answer

$$9/4$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz MCQ 5 [Concept Quiz · MCQ](#)

If $EX=3$ and $EY=4$, $E(2X+Y)$ is:

Solution

1. Compute $2 \times 3 + 4$.

Correct option: B

Final answer

10

Quiz MCQ 6 [Concept Quiz · MCQ](#)

For independent variables $EX=2$, $EY=5$, $E(XY)$ is:

Solution

1. Multiply the means.

Correct option: B

Final answer

10

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz MCQ 7 [Concept Quiz · MCQ](#)

For two independent die rolls, $V(X+Y)$ is:

Solution

1. Add the two variances.

Correct option: B

Final answer

35/6

Quiz MCQ 8 [Concept Quiz · MCQ](#)

For three independent variables, $V(X+Y+Z)$ is:

Solution

1. Extend variance addition.

Correct option: B

Final answer

the sum of variances

Answer key

Use the question number shown on each page.

Q001 $EX = 35/3, EY = 9/2, \text{ so } E(X + Y) = 97/6.$ **Q016** $VX=2/3, VY=3/4, \text{ so } V(X + Y) = 17/12.$ **Q025** $E(XYZ) = 23/21=3.$

Q002 $E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$ and $\sigma = \sqrt{17/12}.$ **Q026** B

Q003 $E(X + Y) = EX + EY = 2 + 5 = 7.$ **Q017** $V(X + Y) = 35/12 + 35/12 = 35/6;$ **Q027** B

Q004 Bag A: $6/7$; Bag B: $6/7$; total $E = 12/7.$ $\sigma = \sqrt{35/6}.$ **Q028** B

Q005 Each mean is $2(0.5) = 1$, so $E(X + Y + Z) = 3.$ **Q018** $V = 1/4 + 35/12 = 19/6;$ $\sigma = \sqrt{19/6}.$ **Q029** B

Q006 $Z = 500X + 100Y, \text{ so } EZ = 5001 + 1003/2 = 6001.$ **Q019** $V = 21/4 + 31/4 = 5/4.$ **Q030** B

Q007 $EZ = 5001 + 1001 = 6001.$ **Q020** $V = 5/4 + 35/12 = 25/6;$ $\sigma = 5/\sqrt{6}.$ **Q031** A

Q008 $EZ = 37/2 + 57/2 = 28.$ **Q021** $EX=4, VX=8/3; E(XYZ) = 64 \text{ and } V(sum) = 8.$ **Q032** B

Q009 $EZ = 350 + 25 = 175.$ **Q022** $EX=3, VX=8/3; E(XYZ) = 27 \text{ and } V(sum) = 8.$ **Q033** C

Q010 $EZ = 22/10 + 21/6 = 11/5.$ **Q023** $E(product) = 7/2^3 = 343/8;$ $V(sum) = 335/12=35/4.$ **Q034** B

Q011 X and Y are independent, so $E(XY) = EXEY=3/27/2=21/4.$ **Q024** Means are $5/6, 2/3, 1/2;$ $E(product) = 5/18 \text{ and } V(sum) = 5/3.$ **Q035** B

Q012 $EX = 8/5, EY = 4, \text{ so } E(XY) = 32/5.$ **Q025** $E(XYZ) = 23/21=3.$ **Q036** B

Q013 $E(XY) = EXEY = 3/21 = 3/2.$ **Q026** B **Q037** C

Q014 $EX = EY = 5/2, \text{ so } E(XY) = 25/4.$ **Q027** B **Q038** C

Q015 $E(XY) = EXEY = 2/7.$ **Q028** B **Q039** B

Q042 B

Q043 B

Q044 B

Q045 C

Q046 B

Q047 B

Q048 B

Q049 C

Q050 C

Quiz WRITTEN 1 -2

Quiz WRITTEN 2 10

Quiz WRITTEN 3 $V = 35/6, \sigma = \sqrt{35/6}$

Quiz WRITTEN 4 $9/4$

Quiz MCQ 5 B

Quiz MCQ 6 B

Quiz MCQ 7 B

Quiz MCQ 8 B